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Josef F. Krems · Hugo Plácido da Silva ·
Pietro Cipresso
Editors

Computer-Human Interaction Research and Applications

9th International Conference, CHIRA 2025 Marbella, Spain, October 20–21, 2025 Proceedings, Part II

Springer

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Preface

These three volumes (CCIS 2834-6) contain the proceedings of the 9th International Conference on Computer-Human Interaction Research and Applications. This year, CHIRA was held in Marbella, Spain, on October 20–21, 2025. It was sponsored by the Institute for Systems and Technologies of Information, Control and Communication (INSTICC). CHIRA 2025 was also organized in cooperation with the ACM Special Interest Group on Management Information Systems (SIGMIS) and the European Society for Socially Embedded Technologies (EUSSET).

The purpose of the International Conference on Computer-Human Interaction Research and Applications (CHIRA) is to bring together professionals, academics and students who are interested in the advancement of research and practical applications in its field of interest, covering different aspects of Computer-Human Interaction, including Human Factors and Information Systems, Interactive Devices, Interaction Design and Adaptive and Intelligent Systems.

CHIRA 2025 received 106 paper submissions from 30 countries, of which 21 (20%) were accepted and published as full papers. A double-blind paper review was performed for each submission by at least 2, but usually 3 or more, members of the International Program Committee, which was composed of established researchers and domain experts.

The high quality of the CHIRA 2025 program was enhanced by the two invited talks delivered by internationally distinguished speakers: Alice Chirico (Catholic University of the Sacred Heart, Italy) with "Road to Awe: Perils, Challenges, Findings, and Open Questions"; and Alexander Mädche (Karlsruhe Institute of Technology, Germany) with "Biosignal-Adaptive Systems for Better Work and Life".

All presented papers will be submitted for indexation by DBLP, Google Scholar, EI-Compendex, INSPEC, Japanese Science and Technology Agency, Norwegian Register for Scientific Journals and Series, Mathematical Reviews, SCImago, Scopus, zbMATH and Web of Science/Conference Proceedings Citation Index.

Several awards, based on the combined marks of paper reviewing, as assessed by the Program Committee, and the quality of the presentation, as assessed by session chairs at the conference venue, were conferred at the conference's closing session to recognise the best contributions.

The program for this conference required the dedicated effort of many people. Firstly, we must thank the authors whose research efforts are reported here. Next, we would like to thank the members of the Program Committee and the auxiliary reviewers for their diligent and professional reviews.

We would also like to deeply thank the invited speakers for their invaluable contribution and for taking the time to prepare their talks. Finally, a word of appreciation for the hard work of the INSTICC team; organising a conference of this level is a task that can only be achieved by the collaborative effort of a dedicated and highly competent team.

We hope you all had an exciting and inspiring conference. We hope to have contributed to the development of our research community, and we look forward to having additional research results presented at the next edition of CHIRA, details of which are available at https://chira.scitevents.org.

October 2025

Josef F. Krems
Hugo Plácido da Silva
Pietro Cipresso

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Alice Chirico	Catholic University of the Sacred Heart, Italy
Alexander Mädche	Karlsruhe Institute of Technology, Germany

Invited Speakers

Road to Awe: Perils, Challenges, Findings, and Open Questions

Alice Chirico

Catholic University of the Sacred Heart, Italy

Abstract. Over the past decade, research has increasingly focused on the multifaceted nature of awe, which transcends simple positive and negative valences. Here, I discuss the key findings and theoretical developments that have emerged from this extended investigation, from my research perspective, emphasizing the significance of awe in psychological processes and its potential for practical application. Initially, research focused on defining and measuring awe. It has been conceptualized as an emotion elicited by stimuli perceived as vast—either conceptually or perceptually—triggering a need for cognitive accommodation. Though awe has long been considered a transformative emotion, theoretical evidence suggests that its intensity is often diminished in laboratory settings. In order to address this gap, I refined both the operational definition of awe and the methods used to elicit it by utilizing novel technologies such as virtual reality (VR). Additionally, I began to question the construct validity of existing instruments used to assess awe in experimental settings. The use of immersive videos, which rely on perceptual spatiality, has been demonstrated to be more effective than conventional 2D videos in eliciting intense emotions of awe - but only when they are built on content that has been pre-validated as being capable of eliciting such emotion. By incorporating these insights, we were able to develop novel, interactive, VR-based scenarios to elicit awe both in the lab and in everyday settings. The potential of VR-elicited awe has emerged not only in enhancing short-term creative thinking but also as an epistemic emotion—that is, one capable of motivating learning by bridging knowledge gaps. The PROMETHEUS project, funded by the Cariplo Foundation, capitalized on awe's epistemic properties to design new teaching methodologies and intervention strategies in educational settings marked by high dropout rates. However, awe holds even greater promise for both individual and societal well-being. Through the PONE EU REACT project, we explored the potential of awe-inspiring VR scenarios to foster pro-environmental behaviors—particularly "socially engaging behaviors"—as a result of awe's connection-related appraisals. As evidence for awe's positive impact on psychological and social well-being grows, two further steps are now crucial. First, we must translate lab-based findings on awe and VR into real-world applications. To this end, we have developed and tested a new awe-based training module in social VR aimed at enhancing creativity. Second—and perhaps more challengingly—we must return to the foundational issue of construct validity in

awe assessment, especially across cultures. Beyond ongoing cross-cultural validation of dispositional and state scales, there is now an urgent need to revisit and refine the construct itself and the conditions under which it is validly assessed. To this end, I have initiated a novel interdisciplinary research line exploring the relationship between awe and the sublime, with a particular focus on European cultural contexts.

Biosignal-Adaptive Systems for Better Work and Life

Alexander Mädche

Karlsruhe Institute of Technology, Germany

Abstract. Biosignal-adaptive systems represent a new class of human-centered systems that use sensor technology — such as eye-tracking or photoplethysmography (PPG) sensors — to capture both visible and non-visible human activities with biosignals. Biosignals are autonomous signals produced by the living organism, energetically measurable in physical quantities using sensors. Biosignals are interesting because, in addition to self-reported or behavioral data, they provide objective and often unconscious information about physical and psychological user states. The collected biosignals are processed in real-time and used to recognize user states such as attention, flow, mental workload & fatigue by leveraging artificial intelligence (AI) technologies. Being aware of user states in real time lays the foundation for designing personalized interactions advancing human-computer interaction. In my keynote lecture, I will introduce the conceptual foundations of biosignal-adaptive systems and share findings from lab and field studies that we conducted over the past decade. These studies demonstrate how such systems can enhance work and everyday life by improving performance and supporting well-being. In particular, I will showcase the potential of biosignal-adaptive systems in the fields of virtual collaboration as well as personal companions for desktop productivity, learning and emotion regulation. I will conclude my keynote lecture with a number of open research challenges and a call for responsible design of biosignal-adaptive systems.

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Interactive Devices

Why the Long Face? The Influence of Robot Screen Proportions and Orientation on Human Perception and Interaction

E. Liberman-Pincu(✉) and T. Oron-Gilad

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Abstract. This study explores how altering a robot's screen design by adjusting proportions and orientation affects human perception of the robot's role and characteristics. We employed two research methods: an Online Questionnaire and a Semi-Live Online Job Interview. The online questionnaire aimed to assess general feelings and attitudes toward social robots. We collected responses from 97 participants. In the semi-live online interaction, 45 participants engaged in a job interview with a robotic recruiter in one of four randomly chosen robot design options that varied by their screen layout and proportion. The interview consisted of standard questions commonly asked in junior engineering job interviews. The orientation and proportion of the screen were found to play a crucial role in shaping user perceptions of gender and character traits. Our findings offer valuable insights for optimizing screen design in human-robot interactions, enhancing user experiences, and ensuring robots effectively fulfill their intended roles.

Keywords: Human-Robot Interaction · Service Robots · Robot Appearance · Screen Proportions

1 Introduction

A robot's appearance greatly influences how users interpret its personality and function [1–4]. Designers choose different visual features, like colors and shapes, to create diverse robot characteristics [5, 6]. Tailoring robot design to its specific tasks improves human-robot collaboration [7, 8]. As robots become more integrated into daily life, covering areas from healthcare to customer service, optimizing their design to enhance user experience is crucial. A key element in this design is the robot's screen, which often acts as the main interface for communication and interaction.

Previous studies have emphasized the crucial role of screen design in shaping how users perceive and interact with robots. For example, a robot's screen appearance and functionality can greatly influence user trust, comfort, and overall satisfaction [9]. The screen often acts as the robot's face, significantly affecting its perceived character and how users view it [10]. Robots with more humanlike faces tend to be seen as more thoughtful, sociable, friendly, and lively, making them more appealing than those with

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less humanlike or no faces at all [11]. Nonetheless, there is a lack of thorough empirical data on how specific design factors, such as screen size, shape, and orientation, influence these perceptions.

This study aims to explore how altering a robot's screen design influences human perception of the robot's role and characteristics. We employed two research methods to investigate this: an online questionnaire and a semi-live online interaction through job interviews. The online questionnaire allowed us to gather general feelings and attitudes toward robots from a broader participant base. In contrast, the semi-live online interaction phase provided deeper insights through direct engagement with four different robot screen designs during a typical junior engineering job interview.

Our findings offer insights for optimizing screen design in human-robot interactions. By enhancing user experiences, we can ensure that robots fulfill their intended roles effectively, whether in professional settings, healthcare, or everyday life. This research contributes to the growing body of knowledge in HRI and provides practical guidelines for designers aiming to create more effective and user-friendly robotic interfaces.

2 Screen Options and Robot Design

For our study, we created two different robot use cases. The first was a mobile robot that could move freely in different environments, talk with people, and support video calls with remote participants. Our online survey used this scenario to collect general feelings and opinions about the robots.

The second use case featured a tabletop robot dedicated to job interviews. Notably, the purpose of this interaction dictates a larger emphasis on the screen of the tabletop robot. The setup was employed in our semi-live online interaction phase, where each participant engaged in a job interview with an online robot. There was a predefined script for the robot's questions throughout the interview, facilitated by an experimenter who was not visible to the interviewee. The interviews were recorded to allow us to observe and analyze the impact of different screen designs on human perception and interaction quality in a controlled, professional setting.

The independent variables in both use cases were screen proportion relative to the robot's body height (large or small) and screen orientation (horizontal or vertical). Hence, we created four robot screen options: A) large horizontal, B) large vertical, c) small vertical, and D) small horizontal.

Based on this screen classification, we created eight designated robot designs, four for each use case; for our online questionnaire, we designed full-body mobile robots, and for the semi-live interaction, a tabletop robot. Figure 1 presents the eight robot designs created for our studies and their proportions. The following sections detail each study separately.

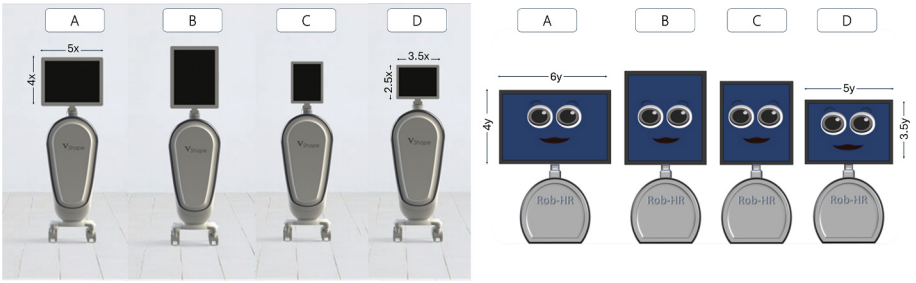

Fig. 1. Left: the four robot designs we used for our online questionnaire. Right: the four robot designs we used for our semi-live online interaction study

3 Online Questionnaire

3.1 Structure of the Online Questionnaire

We employed an online questionnaire to investigate whether participants' perceptions of robot types differ by screen orientation and proportions. Initially, participants were asked to provide personal details. Following, they were presented with an image of one out of the four design options and were asked to ascribe a gender to the robot and ranked a list of nine characteristics according to their impressions of the robot's character. These terms ranged from positive descriptors such as "friendly" and "reliable" to potentially negative ones like "threatening."

Subsequently, participants were asked to rate their agreement with various statements regarding potential future uses of robots. For example, statements included: 'Robots could serve in educational roles with children (in kindergartens, schools, or museums)' and 'Robots could act as security guards in public places.' Additionally, they were asked about their perceptions and feelings towards robots, such as: 'The use of robots in various aspects of life might be frightening' and 'I would be happy to encounter a service robot in a public place.' The questionnaire was distributed via social media groups utilizing a snowball distribution.

3.2 Results

A sample of ninety-seven respondents (29 males, 68 females) aged 18–67 (M = 30.5, SD = 12.4) completed the online questionnaire.

A Chi-square test of independence was performed to examine the relation between screen orientation and gender ascription to the robot. The relation between these variables was significant, X^2 (2, $N = 97$) = 10.18, $p = .006$; a horizontal screen was perceived as feminine, while a vertical screen was perceived as masculine. A Chi-square test of independence showed that there was no significant association between the screen dimension and the robot's perceived gender, X^2 (2, $N = 97$) = 0.12, $p = .94$.

In addition, the screen orientation affected the respondent's perception of the robot character; the vertical screens were perceived as more innovative, professional, and elegant, while the horizontal screens were perceived as friendlier and cuter, regardless of

the screen's dimensions. Figure 2 illustrates the effect of screen orientation on perceived characteristics.

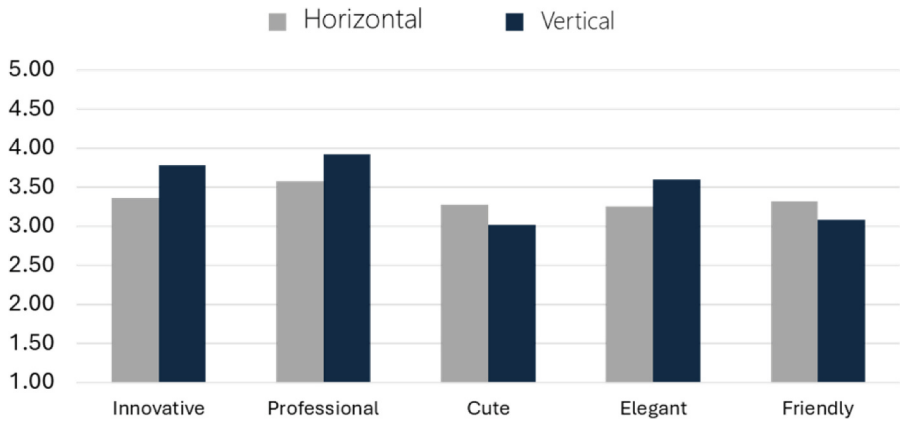

Fig. 2. Orientation effects on the robot's perceived characteristics on a scale of [1–5], with 1 being the least.

While screen dimensions did not significantly affect respondents' perceived characteristics of the robots, they did influence perceptions of the robots' suitability for various future roles. Respondents rated robots with larger screens as more compatible for roles such as service providers (e.g., waitpersons, hosts, receptionists) with a rating of 3.81 out of 5 (SD = 1.18) compared to 3.5 (SD = 1.25) for smaller screens. Similarly, larger screen robots were considered more suitable for providing companionship to older adults living alone (3.92, SD = 1.09 compared to 3.52, SD = 1.17) or guiding children (3.65, SD = 1.16 compared to 3.2, SD = 1.31). Additionally, horizontal screens were found to be more suitable for service provider roles, with a rating of 3.79 (SD = 1.12) compared to 3.49 (SD = 1.31) for vertical screens.

4 Semi-Live Interaction

4.1 Structure of the Interviewing Robot Setup

We created an animation of a robot capable of moving its lips in various configurations: speaking, smiling, half-smiling, and static. Additionally, the robot's eyes could move side-to-side and blink continuously as a sign of attentiveness. The eyes, lips, and eyebrows remained identical across all four robot types, with only the screen's size and orientation changing. This allowed us to focus solely on the impact of screen shape and proportion without any additional influencing factors.

At the beginning of the experiment, each participant was randomly assigned one of four robot types, which varied in screen orientation (horizontal or vertical) and screen proportion (large or small relative to the body), as illustrated in Fig. 1. Participants were first asked to identify the perceived gender of the robot.

Next, they engaged in a recorded job interview via Zoom with the robot they were initially presented with. The robot's voice was adjusted to match the chosen gender. This phase utilized the 'Wizard of Oz' methodology, where participants interacted with a system, they believed to be autonomous but was controlled by an unseen human experimenter. The interview consisted of seven standard junior engineering job interview questions, with the robot's responses selected from five pre-determined options based on the participant's answers and the context of the interaction. Figure 3 presents the job interview setup. In the final phase, participants rated nine robot characteristics to capture their perceptions on a scale from 1 (very low agreement) to 5 (very high agreement).

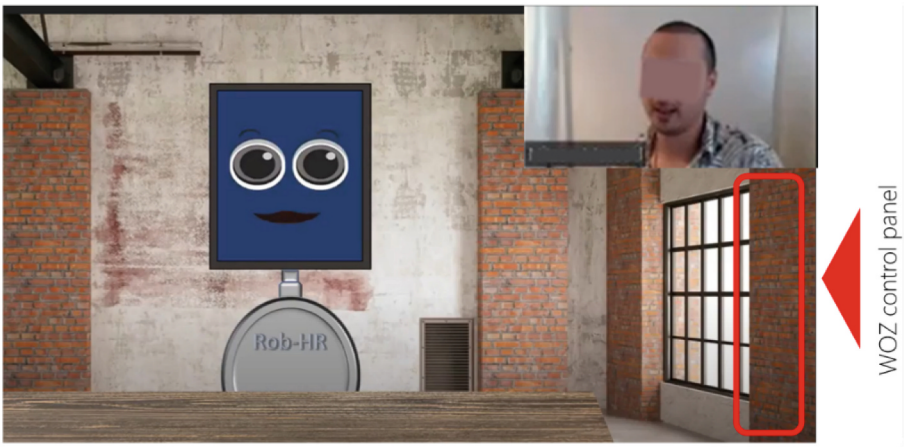

Fig. 3. Job interview setup. Note that the participant in the upper right Zoom window answers the questions.

4.2 Results

A convenience sample of 45 participants (24 males, 21 females) aged 21–39 (M = 26.8, SD = 4.4) took part in our semi-live interaction study. Our convenience sample consisted of engineering students and young individuals under the age of 40. Due to the relatively small sample size, we were unable to conduct rigorous statistical analysis, aside from descriptives; however, we observed several interesting trends.

The overall experience of the interviewees was positive; participants reported that they approached the interview with seriousness (4.51 out of 5, SD = 0.54) and felt that it effectively prepared them for a real job interview (3.69, SD = 0.89).

Vertical screens were predominantly perceived as male, with 70% of participants interacting with a vertical screen robot attributing it to the male gender, compared to 44% for horizontal screen orientations. Similarly, small screens were mainly perceived as male; 62% of participants interacting with a small screen robot ascribed it to a male gender, compared to 47% for those interacting with a larger screen robot. Notably, all 9 participants who engaged with Robot C (the vertical small screen) ascribed it to a male gender. Additionally, vertical screen robots were perceived as more reliable (3.6 out of 5, SD = 0.84 compared to 2.95, SD = 1.2) and more authoritative (3.4 compared to 3.1).

5 Discussion

Our research investigated how screen proportion and orientation influence the perception of service robots, encompassing two studies: an online questionnaire with 97 respondents and a semi-live interaction study involving 45 participants. Together, these approaches provide a robust foundation for understanding how visual cues influence human-robot interaction in both abstract and immersive contexts.

5.1 Gender Attribution Across Studies

Research shows robot gender influences human attitudes and expectations, often enhancing trust and effectiveness in tasks. However, gendering may unintentionally reinforce occupational stereotypes [12]. In our studies, screen orientation emerged as a strong and consistent cue in gender attribution, with vertical screens predominantly perceived as masculine (65% in the online questionnaire and 70% in the job interview simulation), indicating a stable association across different contexts. Interestingly, screen size influenced gender perception only in the semi-live study: small screens were more often attributed to male robots, with 100% male attribution for Robot C (small vertical), while this effect was absent in the questionnaire ($p = .94$). These results suggest that although screen size can modulate gender perception, screen orientation is the more decisive factor. These findings align with the broader literature on gender perception of robots, which suggests that robots are often perceived as masculine unless they exhibit features typically interpreted as feminine [5, 13–15]. Importantly, demographic factors such as user gender may influence responses to robot gender presentation, with studies indicating that male users often show stronger engagement and trust toward female-presenting robots [16].

5.2 Trait and Role Perception in Context

Screen orientation influenced the perceived characteristics of the robots, and the interaction context further shaped these perceptions. In the context of a personal robot, vertical screens were perceived as more innovative, professional, and elegant. In contrast, horizontal screens were seen as friendlier and cuter, regardless of the screen's dimensions. The respondent also associated horizontal screens with being more suitable for service provider roles. In a job interview context, vertical screen robots were perceived as more reliable and authoritative. This indicates that while screen orientation plays a crucial role in shaping user perceptions, the specific context of interaction can emphasize certain traits, such as authority and reliability in professional settings. Larger screens were associated with roles such as service providers, companionship, and guidance. This aligns with earlier research indicating that the head-to-body ratio significantly influences the perception of a character's (animated or human) personality. For instance, in humans, it was found that a baby-faced character (with a relatively larger head) is often associated with traits such as honesty, kindness, and warmth [17]. In contrast, a character with a smaller head is typically perceived as more suited for managerial roles [18].

6 Limitations, Conclusions, and Design Guidelines

The studies had several limitations, including a small sample size; the semi-live interaction study had a relatively small sample size of 45 participants, primarily consisting of engineering students and young individuals under 40, which may limit the generalizability of the findings. Additionally, the interaction scenarios were limited to a job interview and general perceptions, which may not encompass the full range of interactions users may have with socially assistive robots in various roles and environments.

Despite these limitations, our exploratory findings emphasize how screen orientation and proportion influence user perceptions of service robots, especially in terms of gender and character traits. The results from both studies highlight the need to match robot screen design with the robot's intended role and interaction context. Screen orientation and proportion are more than just aesthetic choices; they convey social and psychological signals that influence user expectations and reactions.

Vertical screens consistently projected masculinity along with professionalism and authority. These traits are suitable for robots functioning in formal or task-driven contexts (e.g., front-desk concierges, security agents, or administrative assistants). Designers can use vertical layouts to convey credibility and efficiency.

Horizontal screens, perceived as friendlier and more approachable, are well-suited for robots designed for social assistance, caregiving, or educational support. These orientations signal warmth and relatability, making them effective in emotionally sensitive environments like classrooms or elder care settings.

Screen size further enhances role perception. Large screens evoke presence and trust, useful for interactive or expressive roles. Small screens, while seen as more utilitarian in immersive contexts, may require compensatory cues (like voice modulation or movement) to avoid perceptions of coldness or dominance. Table 1 summarizes practical takeaways for designers.

Table 1. Practical takeaways for designers.

Visual Cue	Perceived characters	Suggested Use Case
Vertical Orientation	Authority, masculinity, reliability	Security, administrative support
Horizontal Orientation	Warmth, friendliness	Education, therapy, companionship
Large Screen	Presence, emotional expressiveness	Social robots, interactive kiosks
Small Screen	Utility, potential coldness	Task-oriented roles

Robotic designers should incorporate these insights early in the design iteration cycle, particularly when developing robots for public-facing or emotionally responsive environments. Designers can strengthen user perception and trust by aligning visual cues with expected role attributes. Prototype testing with user panels can aid in calibrating appearance and functionality.

We propose five key design guidelines to help designers create robots that meet user expectations and improve interaction experiences:

Key Design Guidelines:

1. **Screen Orientation Significance** Screen orientation strongly influences user perception. Vertical screens often imply professionalism and authority, while horizontal screens may suggest approachability or emotional support.
2. **Innovative Visual Impression** A vertically oriented, small-proportioned screen tends to evoke qualities such as innovation, elegance, and sophistication- making it suitable for forward-thinking design goals.
3. **Gendered Perceptions** If the desired association is masculine, a vertical screen with compact proportions is more frequently interpreted as male by users. Designers should also consider user demographics, as gender cues in voice or appearance may influence trust and engagement.
4. **Reliable and Authoritative Aesthetic** For conveying reliability and authority, especially in formal settings like job interviews, a vertically oriented screen is perceived as commanding and trustworthy.
5. **Adaptive Contextual Design** Consider using dynamic screen reorientation to signal role shifts, such as transitioning from task execution to emotional support (e.g., vertical to horizontal orientation). Complement this with thoughtful choices in morphology, color, and shape based on the interaction context.

While this research centers on screen-based social robots, the design guidelines derived from these findings can also inform the development of other interactive systems, including kiosks, virtual agents, and screen-based AI interfaces. Any human-facing system's visual orientation and proportions may carry psychological signals that impact trust, role attribution, and emotional engagement.

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Implicit and Explicit Attitudes Towards Virtual Agents in Education: An Experimental Study

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Abstract. Virtual Reality and Artificial Intelligence (AI) offer interesting opportunities for education. For example, conversational pedagogical agents could assist students and possibly automatize some teaching activities. Within virtual environments, these could be presented as embodied conversational agents or virtual educational agents (VEAs), namely digital anthropomorphic figures that are guided by AI. However, to orient the implementation of such resources in education, it is useful to analyze people's attitudes towards them, both implicit and explicit. In this exploratory study, 20 participants responded to questions about general AI tools and VEAs' utilizability in educational contexts. Moreover, participants completed a version of the Implicit Association Test (IAT) that assessed their implicit attitudes towards virtual agents compared with human teachers. Results showed that participants had significantly more positive explicit attitudes towards the implementation of AI in general than VEAs specifically in educational contexts. While these findings suggest a potentially skeptical perception of VEAs, further research involving real-world implementations is necessary. Preliminary results can inform future design and deployment strategies for AI-supported education, while highlighting the importance of user perceptions.

Keywords: Implicit Attitudes · E-Learning · Virtual Agents · Artificial Intelligence · Conversational Pedagogical Agents

1 Introduction

1.1 Artificial Intelligence in Education

Computerized and virtual learning environments offer several advantages for education, such as the opportunity to create more accessible, flexible and customized learning paths that allow improving the quality of learning [1, 2]. Some of these environments make use of artificial intelligence (AI), which is deeply revolutionizing the educational sector, offering innovative tools not only in the field of teaching [3, 4]. Indeed, in recent years, AI has been used to support the development of predictive modeling of student

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performance, or as a tool supporting activities in the classroom, within the framework of personalization and evaluation processes of educational pathways [5, 6]. Specifically, for what regards AI tools that directly interact with students, AI is currently used in three main ways:

- as an educational agent directing the student's learning processes;
- as a task-solving support;
- as an extension of the student's own intellectual capabilities, which remains at the center of the educational process [7].

For these reasons, the field of AI in Education (AIED) has seen significant growth, along with research addressing its applications in relation to different pedagogical paradigms [8, 9]. An integral component of many computer-assisted learning environments and AIEDs are virtual educational agents (VEA), computer-guided simulated characters that interact with students in computer-based learning environments to facilitate the educational process. These agents can be powered by AI and used in computer-based simulations to enhance learning, taking on the role of teachers, mentors, 'buddies' or simple feedback providers under different modalities (text, audio, video) [10, 11].

Much like traditional virtual agents, VEAs are built by training an artificial intelligence model tailored to the needs of a specific target audience (e.g., age, skill level). VEAs are trained for specific learning outcomes and, depending on the target audience's needs, they can come in the form of a chatbot AI tutoring system with adaptive learning, or FAQ bot [10]. Examples of such agents are IBM Watson Tutor, or Socratic by Google. According to research, VEAs positively impact student motivation and engagement by fostering personalized and adaptive learning through immersive interactions [12, 13]. These tools allow accessibility of learning and adapt educational content according to students' needs, facilitating knowledge management, needs analysis and performance evaluation [14]. Among the different types of virtual educational agents, the most popular are Conversational pedagogical agents or educational chatbots [15, 16].

Embodied conversational agents (ECAs) differ from chatbots in their ability to combine verbal and non-verbal communication, offering a more natural and engaging interaction [17]. These agents, equipped with humanised facial expressions, gestures and postures, significantly enhance user experience and learning effectiveness. ECAs provide a more intuitive interface between the user and the computer system and can be used to motivate learners with personalised feedback. Another category of educational virtual agents are intelligent virtual assistants (IVAs), which differ from other AI systems in their ability to assist users in everyday activities. IVAs (such as Apple's Siri or Google Assistant) have a more natural and conversational language [18]. Although these agents are not programmed for lengthy conversations and creating a relationship with the user, they can support learning in education by also taking on embodied aspects [19]. The integration of VEAs, if properly accompanied by pedagogical strategies and expert human mentoring, opens up important scenarios in education, especially in the field of e-learning, through learning paths tailored to students' specific needs, both in online and face-to-face contexts.

1.2 Social Presence and Virtual Agents

One of the potentialities of virtual agents is related to their ability to elicit the phenomenon of 'social presence'. The term 'social presence' originated in the field of media theories and according to early definitions indicates 'the degree of salience of the other person in the interaction and the consequent salience of interpersonal relationships' [20]. Early theories of social presence argued that the level of social presence depended on the number of social cues (e.g. gaze, voice, head nod or other non-verbal behaviour) that the technological medium allows to be transmitted from one communicator to another [21]. In this sense, people would choose a medium that can guarantee more or less social presence depending on the activity they have to perform. However, more recent theories argue that a sense of social presence is not necessarily due to the amount of relational cues that can be transmitted through the technological medium, nor to the sophistication of the technological peripherals involved [22]. Gunawardena and Zittle conducted research on social presence involving students in online learning courses and found that social presence can be high even when social cues are low. Indeed, course moderators can have the ability to "generate" a sense of social presence, based essentially on the quality of the pre-existing offline relationship between the participants [23]. In other words, the pre-existing positive relationship between people would increase their sense of social presence in the context of technology-mediated interaction, independent of the extent to which these limit the transmission of emotional cues.

The phenomenon of social presence, therefore, is related to the relationships established between communicants, both within and outside mediated contexts. In computer-mediated learning contexts, indeed, social presence enhances feelings of trust [24], learning and subject performance [25]. Still later theories emphasize that social presence depends on the detection of clues or "traces" from the intentions of others in the environment [26]. In this sense, even independently of the richness of meta- linguistic indices, the presence of even limited information that could be associated with another agent's rational conduct can generate a strong feeling of social presence. Even finding a trail of footprints on the beach in a virtual environment, rather than a written message or other traces of actions and intentions can generate a strong feeling of being in the company of others. Indeed, it has now been discovered that the phenomenon of social presence can also occur when the person interacts not with another human being but with a computer-guided character, such as NPCs (non-playing characters) in video games and VEAs. There are several studies that show how people respond naturally and socially in communication with a computer [27]. This happens insofar as the individual can feel that the virtual agent has intentions and meaningful agency. Perceiving a trace of an action refers to the feeling of not being alone and, therefore, of social presence. Moreover, this phenomenon is amplified in interaction with ECAs that provide even more social signals related to human-like appearance and non-verbal behavior [28].

1.3 Attitudes Towards Artificial Intelligence and Virtual Agents

Despite the great potential of AI in various fields, opinions and attitudes towards it are not always positive. In general, opinions on artificial intelligence (AI) vary widely, ranging between strong enthusiasm and considerable forms of opposition [29]. In line with the

technology acceptance model (TAM), the most widespread model used to predict the adoption of technology [30], the analysis of attitudes is of particular importance as these act as predictors of intention-to-use a technology which, in turn, is expected to predict actual use and acceptance, in this case of AI [31]. However, the exact concept of "attitude" is problematic and multifaceted in psychology.

Attitudes have cognitive and emotional components, both of which are relevant for the enactment of behavior [32]. Regarding the emotional component, it has been verified that AI generates feelings of fear that can be an obstacle to the acceptance of AI [33], as well as positive emotions (e.g., joy and fun) that, on the other hand, encourage interaction [34]. Furthermore, it is relevant to point out that attitudes can be influenced by different individual aspects such as personality traits and demographic characteristics (gender, age, education level) [35, 36]. For example, it has been found that mainly male individuals with a good socioeconomic status have positive attitudes towards the use of AI [37]. With regard to the topic of attitudes and acceptance of AI in learning and teaching, teachers might believe that the use of AI will reduce their competence or make teaching less personalised [38]. In addition, teachers often experience anxiety towards the introduction of new technologies in education which might decrease the employment rate, as tasks previously carried out by humans would become partially or totally automatized [39]. However, the literature above has highlighted only a part of the attitude of the individuals towards AI in the educational field by focusing on explicit attitudes. Explicit attitudes are the consequence of deliberate processes in the sense that one would employ a conscious and aware effort to formulate them [40, 41].

In general, attitudes are structured on the basis of the associations in memory between a target object and an evaluation of that object. Therefore, rapid and automatic associations are indicative of *implicit* attitudes, while associations that require greater reflection are connected to explicit attitudes. Indeed, previous studies have demonstrated the phenomenon of implicit-explicit discrepancy (IED), whereby implicit and explicit attitudes do not always coincide [42]. Furthermore, implicit attitudes are more difficult to modify than explicit ones [43]. As the effectiveness of virtual agents relies on their ability to elicit a sensation (to be in the company of others, namely social presence) it is important to analyze individuals' implicit attitudes towards virtual agents, namely attitudes that emerge spontaneously and independently of conscious elaboration.

Implicit and explicit attitudes require different measurement tools. While explicit attitudes often use different types of validated self-report questionnaires, implicit attitudes require tasks that allow for the emergence of an automatic response. One of the main tests for measuring implicit attitudes is the Implicit Association Test (IAT) [44] which assesses the strength of mental and automatic associations between concepts in memory through a stimulus sorting task. The IAT consists of a computerized task in which participants are asked to classify visual stimuli (e.g., words or images) using two keys on the keyboard, while their reaction times are recorded. The instrument provides a score based on the speed of reaction in the response to pairs of stimuli (images or words) presented on a screen. Each word or image can be associated with only one of two opposite categories. The instrument is based on the assumption that it is easier, therefore faster, to use the same response key for items belonging to motion to topics

such as prejudice, politics, prosocial behavior [45, 46]. Through the IAT it was possible to identify an implicit negative attitude towards robots compared to humans [47].

Previous studies demonstrate the importance of unconscious biases towards AI in the appreciation that one has towards it. To date, only a few studies have assessed both implicit and explicit attitudes towards AI [35].

Given that a positive attitude is particularly relevant for trust [48], the aim of the present study is to investigate attitudes towards a particular technological resource, virtual agents. In particular, this study aims to:

- Investigate explicit and implicit attitudes towards virtual agents and the use of virtual agents in education;
- Compare explicit and implicit attitudes towards virtual agents and the use of virtual agents in education;
- Compare the explicit attitudes towards virtual agents with those towards AI.

1.4 Methodology

In the present study, 20 participants (age 31–57, M = 38.1, DS = 9.9, 7 females) completed a survey in Italian, lasting approximately 10 min, built using the Qualtrics platform. Data were collected in February 2025. First, participants were asked to provide their demographic information such as gender and age. In the first block, the survey involved the completion of an IAT test, built through the IAT gen website http://iatgen.org/ and implemented through the Qualtrics platform according to the original procedure proposed by the authors [49] (see Fig. 1 for examples of the visual stimuli implemented in the IAT software). The IAT included 20 stimuli: 10 images and 10 words. Five images are related to the category "human beings" (target A) and five images are related to the category "virtual agents" (target B). The images were generated with generative AI (Microsoft image generator): the researchers used carefully-designed prompt in order to obtain images that were as similar as possible in terms of elements and their positions, Specifically, for Target A there is an image of a male human, an image of a female human, an image of a male human teaching another person, an image of a female human teaching another person and an image of a human teaching a class. For Target B, on the other hand, there is an image of a virtual agent with the likeness of a male human, an image of a virtual agent with the likeness of a female human, an image of a virtual agent with the likeness of a male human being teaching a human, an image of a virtual agent with the likeness of a female human being teaching a human, and an image of a virtual agent teaching a class. As for the words, five words belonging to the "good" category (good, reliable, useful, easy, valid) and five to the "bad" category (active, unreliable, useless, difficult, inefficient) are included. The IAT consists of 7 blocks, each containing 20 or 40 trials. Specifically, some blocks require associating words or figures with one category, while the rest require associating words and figures with two categories simultaneously (see Fig. 1 for examples). Blocks 4 and 7 are the critical (analysed) blocks, each containing 40 trials, while the others are training blocks, each containing 20 trials.

Subsequently, an ad hoc questionnaire was administered to assess explicit attitudes towards AI in general and VEAs. The questionnaire was loosely based on tools used in studies with a similar focus [35]. The tool comprises six items; Specifically, three items investigate attitudes towards AI in general: 1) "What is your opinion on the potential

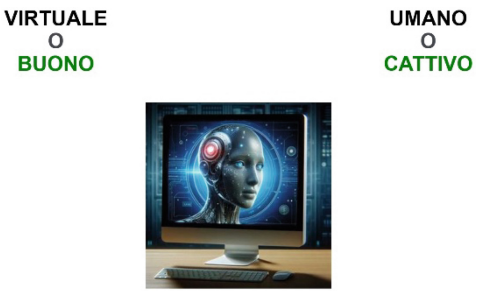

Fig. 1. Example of an IAT block.

of Artificial Intelligence?"; 2) "How much do you support the use of Artificial Intelligence?"; 3) "How much do you support the use of Artificial Intelligence in education?"; and, three items assessing attitude towards VEAs: 1) "What is your opinion on Virtual Agents?"; 2) "How much do you support the use of Virtual Agents in learning contexts?"; 3) "I think that Virtual Agents are an easy-to-use learning support" *(Agree/Disagree)*. Each item provides a score on a five-point Likert scale from 1 (not at all) to 5 (completely).

1.5 Results

The IAT data were processed using the D-score data-cleaning and scoring algorithm [50, 51]. Only one participant was dropped due to excessive response speed. The error rate was 7%. The IATs were D-scored and the results revealed a strong implicit negative attitude towards virtual agents compared with humans, MD Score = 0.74, SD = 0.35, d = 2.11, which significantly differed from zero, $t(18) = 9.2$, $p < .00001$, 95% CID Score [0.57, 0.91].

Explicit attitudes were analyzed separately. We computed two new variables which were averages of the attitude items towards AI in general and the attitude items towards VEAs. We performed paired samples t test comparing the two attitudes variables. The comparison was significant, $t(18) = 8.38$, $p < .001$. Participants reported more positive attitudes towards AI in general (M = 4.07, DS = .66) than towards VEAs (M = 3.01, DS = .86).

1.6 Discussion and Conclusions

The results of the present study revealed a negative implicit bias towards virtual agents, specifically within an educational context (VEAs). This result is in line with studies that have shown the presence of negative implicit attitudes towards AI tools [35]. However, differently from previous studies, a correspondence was found between explicit and implicit attitudes towards VEAs, both of which tended to be negative. Regarding implicit attitudes, the participants were significantly faster responding to positive associations with human educators and to negative associations with VEAs. It should be taken into consideration that a generative AI was used to create the images used in the study. While these were assessed and modified by the researchers in order to be as similar as possible (e.g., human educators and virtual agents in the same position), the images were not

specifically validated for experimental purposes. In any case, the huge difference detected in the study seems to corroborate the effectiveness of the stimuli. The data suggests that participants expressed less favorable attitudes towards virtual agents compared to human educators, though further validation with larger and more diverse samples is necessary.

Virtual agents were associated with more negative explicit attitudes as well. It is important to notice that, in this case, the comparison was with AI in general, not with human educators. While people may have positive attitudes towards such innovative tools that could enhance educational materials and opportunities, when such technologies assume the appearance of animated anthropomorphic characters, acceptance could be more challenging. On the one hand, an interface with human-like characteristics may predispose to a feeling of perturbation, i.e. the 'uncanny valley' phenomenon whereby people's reactions to robots that look and behave almost human can move abruptly from empathy to revulsion [52]. The research on uncanny valley has identified the phenomenon with digital characters and other forms of human-like artifacts. Regarding Embodied Conversational Agents (ECA) studies show that a higher perception of humanity of ECAs is correlated with a higher perception of anxiety [53]. On the other hand, the results are consistent with a view that problematize the interface of AI, a "machine of many faces" [54, 55]. Interfaces orient users' attitudes independently of software functionality and effectiveness. Future studies should consider how different AI interfaces influence user acceptance, particularly in applied educational contexts.

As explained above, it is necessary to promote a sense of social presence that would help students to "feel" within a meaningful educational interaction, even when mediated by technology and possibly featuring AI-guided agents. According to recent theories, social presence is not the direct result of sophistication in technology or graphical realism, but rather it is a function of others' possibility to make their agency visible within a virtual environment, so that actions and behavior can be immediately understood as intentional conduct. Further studies could confirm how pedagogists and designers could focus on the creation of extended environments that allow users to recognize each other's intentions and actions, not necessarily on the anthropomorphic realism of digital figures. As for limitations and prospects, the images could be validated at a later stage, the sample should be expanded, and the educational background of the participants should be taken into consideration. However, it must be considered that the first blocks of the IAT are tested so that the subject learns to correctly categorize the images early on.

In conclusion, this exploratory study highlights a possible divergence between attitudes towards general AI and its specific application in the form of virtual agents in education. Further studies are needed to analyze the reasons for the negative attitude and the correspondence between implicit and explicit attitudes towards VEAs to create user-accepted interfaces and exploit the potential of VEAs in the educational field.

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An Integrated Framework for Evaluating e-Service Quality, Usability and User Experience of Web-Based Applications: A Qualitative Expert Review

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Abstract. Recent growth in the scope and complexity of web-based applications has intensified the need for evaluation tools that integrate usability, user experience (UX), and e-service quality (e-SQ) within a single lens. Although each construct is well studied, their interdependencies remain under-represented in existing frameworks. Based on a systematic review of 264 sources, the original e-SQUUX model was subjected to a qualitative review by four HCI experts, who consolidated overlapping elements and clarified terminology. The resulting framework comprises 16 categories, 101 actionable evaluation statements, an explicit construct-interaction matrix, and a step-by-step Customization and Prioritization Guide. This layered architecture allows evaluators to tailor assessments to context while preserving methodological rigor, enabling holistic benchmarking of WBAs and mobile apps. An illustrative university-registration use-case—mapping 'Navigation', 'Responsiveness & Helpfulness', 'Helpfulness', and 'Security & Privacy' to concrete interface components—reduced form-abandonment from 22% to 9%, demonstrating the model's practical impact. By merging human-centered and service-quality perspectives, e-SQUUX offers a practitioner-ready instrument for continuous improvement across digital-service ecosystems. Future work will focus on large-scale empirical validation, for example via Partial Least Squares Structural Equation Modelling (PLS-SEM).).

Keywords: e-Service Quality · Usability · User Experience · Human Factors · Web-Based Applications · Qualitative Expert Review · Evaluation Framework · e-SQUUX

1 Introduction

1.1 Background and Motivation

Recent growth in the scope and complexity of web-based applications (WBAs) and mobile apps, fuelled by decades of advances in internet technology and pervasive computing, has led to their rapid diversification across sectors such as education, healthcare and e-commerce [1, 2]. This evolution creates a pressing need for robust evaluation

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frameworks that jointly address three human-centred constructs: usability, user experience (UX) and e-service quality (e-SQ). Although e-SQ originates in service marketing and UX/usability stems from Human-Computer Interaction (HCI), their conceptual boundaries increasingly overlap in interactive systems [3, 4]. Yet most evaluation models treat them separately, overlooking synergies that could enhance system performance and user satisfaction [5, 6]. In addition, global deployment across diverse cultural and social contexts demands integrated tools adaptable to varied user profiles. Unlike earlier holistic checklists, the refined e-SQUUX provides a streamlined 16-category core, an explicit construct-interaction matrix and a step-by-step context-prioritisation guide. This paper presents and validates the enhanced framework through a qualitative expert review.

1.2 Problem Statement

Despite a growing body of literature addressing usability, UX and e-SQ independently, a significant gap remains in the availability of unified, operational frameworks that combine all three dimensions. In particular, limited research exists on systematically integrating these constructs for evaluating complex interactive systems [7]. This fragmentation creates a gap that hampers designers, evaluators and developers from assessing user-centered performance holistically and consistently. It is the focus of this study to narrow that gap.

1.3 Research Objectives

Building upon Ssemugabi and de Villiers [8, 9], who proposed the original e-SQUUX model via a narrative literature review, this paper aims to refine that model through expert review and develop an implementable version for customizing WBA and mobile app evaluations. The main aim is to produce a concise, context-sensitive framework by (i) identifying overlapping or redundant components, (ii) validating conceptual dimensions via structured expert input, and (iii) crafting actionable evaluation statements for practitioners.

2 Literature Review

2.1 Web-Based Applications and Interactive Interfaces

Web-based applications (WBAs) and mobile apps are now common in education, healthcare, and online shopping. These systems need clear ways to measure usability, user experience (UX), and e-service quality (e-SQ) [10]. Existing models such as PACMAD+3 [11] cover mobile issues like cognitive load but rarely link related ideas, for example matching “trust” in UX with “security” in e-SQ. This gap makes it hard to judge service-focused WBAs and mobile apps in a complete way [7]. Recent studies therefore call for a single model that looks at all three areas together to improve design and user satisfaction [11].

2.2 Usability and User Experience

Usability is a critical element in the design and evaluation of WBAs and mobile apps. Defined by ISO 9241-11:2018, it refers to the extent to which users can achieve specified goals with effectiveness, efficiency and satisfaction [12]. Usability is largely task-oriented and functional [13], while User Experience (UX), as described in ISO 9241-210:2010, considers broader emotional and perceptual factors before, during and after system use [14]. In practice, usability emphasises system performance, whereas UX reflects overall user satisfaction and emotional response. Modern evaluation frameworks recognise that measuring usability through objective metrics alone is insufficient. Subjective experiences—such as trust, frustration and preference—must also be considered [5]. However, many development processes neglect integrated usability and UX evaluations, leading to overlooked criteria and suboptimal outcomes. Studies show that early and continuous inclusion of these assessments significantly improves platform adoption and user satisfaction [15, 16].

2.3 e-Service Quality (e-SQ)

E-service quality pertains to the evaluation of online service delivery, encompassing dimensions like reliability, responsiveness and personalization. Traditional models such as SERVQUAL [17]) have been adapted to the digital context, leading to e-SQ frameworks tailored for WBAs and mobile apps [18, 19]. These multidimensional models integrate user perceptions and technological attributes to assess e-SQ comprehensively, emphasizing alignment with user expectations to enhance satisfaction and loyalty.

2.4 Integrated Evaluation Models and Gaps

Although usability, UX and e-SQ are widely recognized as key constructs in evaluating WBAs and mobile apps, many existing frameworks continue to treat them in isolation [11]. This fragmented approach fails to capture the complex interplay between functional performance, emotional engagement and service expectations that shape overall user perceptions. Efforts to bridge these dimensions have emerged. For instance, the e-SQUUX model [8] integrates usability, UX and e-SQ, offering a more holistic evaluation perspective. However, further refinement and validation are required to ensure the model's adaptability across different contexts and user groups.

Recent studies have reinforced this need for integration. Galavi et al. [20] note that heuristic-informed design can enhance application quality, ultimately improving outcomes such as user safety. Similarly, research shows that system responsiveness and clear feedback loops strongly shape perceived UX [21], while accessibility and usefulness are consistently prioritized by diverse user demographics [22]. Sauer [23] proposes a unified framework that merges usability, UX and accessibility, framing accessibility as both a design priority and an experiential concern. These advances highlight the growing recognition of integrated evaluation approaches, yet the gap remains in establishing widely adopted, context-sensitive models for comprehensive WBA and mobile app assessment [11].

2.5 Bridging WBA and Mobile App Evaluation Criteria

Recent studies underscore the growing convergence between evaluation frameworks for WBA and mobile applications, particularly in terms of Usability, UX and e-SQ [24]. For instance, Meidani et al. [25] assessed a clinical mobile application's usability, confirming the applicability of standard usability metrics such as the System Usability Scale (SUS) in mobile settings. Similarly, Setiyawati and Bangkalang [26] evaluated mobile banking apps using the UEQ and SUS, identifying key UX factors like efficiency and attractiveness.

Further evidence from Zheng et al. [24] demonstrates how strategy implementation led to substantial improvements in usability, usefulness and user experience—raising mean SUS scores from 15 to 84 and task completion rates from 55% to 89%. In addition, studies on digital banking platforms have validated SUS as a consistent usability metric across devices [27]. Heuristic-based approaches have also been successfully adapted for mobile interfaces. Galindo-Losada et al. [28] found that integrating heuristics into iterative design processes improves mobile UX evaluations, particularly for voice-enabled systems. These findings collectively affirm the cross-platform relevance of WBA evaluation criteria, reinforcing the case for unified models that can be applied to both web and mobile environments.

3 Theoretical Foundation and Original Framework

3.1 Conceptual Origins of the e-SQUUX Model

Usability, as defined by ISO 9241-11:2018, refers to the extent to which a system can be used by specified users to achieve specified goals with effectiveness, efficiency and satisfaction in a particular context [12]. It focuses primarily on performance, task completion and interface ease of use. User Experience (UX), on the other hand, extends beyond usability by encompassing users' emotions, beliefs, preferences and perceptions before, during and after system interaction, as outlined in ISO 9241-210:2010 [14].

E-Service Quality (e-SQ) draws from the SERVQUAL model [8] and evaluates online service quality through dimensions such as reliability, responsiveness, assurance, empathy and tangibles. These three constructs are increasingly seen as interrelated rather than independent, requiring combined evaluation to reflect the full scope of user interaction.

3.2 Conceptual Integration via the e-SQUUX Model

The e-SQUUX model [8] was conceived to bridge the long-standing fragmentation between Usability, User Experience (UX) and E-Service Quality (e-SQ). By synthesizing overlapping dimensions such as efficiency, satisfaction, trust and reliability, the model offers a single, comprehensive lens through which Web-Based Applications (WBAs) and mobile apps can be evaluated.

Figure 1 presents a Venn diagram that visually locates the three constructs. Usability emphasizes task-oriented attributes like effectiveness, efficiency and user satisfaction

(ISO 9241-11) [12]; UX foregrounds emotional and cognitive responses, including satisfaction and trust (ISO 9241-210) [14] while e-SQ concerns service-centered facets such as reliability, responsiveness and assurance, drawing on the SERVQUAL tradition [17]. Shared regions—including *Satisfaction*—signal cross-cutting dimensions, whereas non-overlapping areas preserve disciplinary distinctions. Recognizing and operationalizing these intersections underscores the need for an integrated evaluation framework capable of capturing the multifaceted nature of user interaction across services and interfaces.

By formally mapping where each construct intersects, the e-SQUUX model helps designers and evaluators avoid redundancy, identify complementary metrics and apply context-sensitive weightings that reflect real-world usage scenarios. This conceptual integration sets the stage for the hierarchical framework detailed in Sects. 3.3 and 3.4.

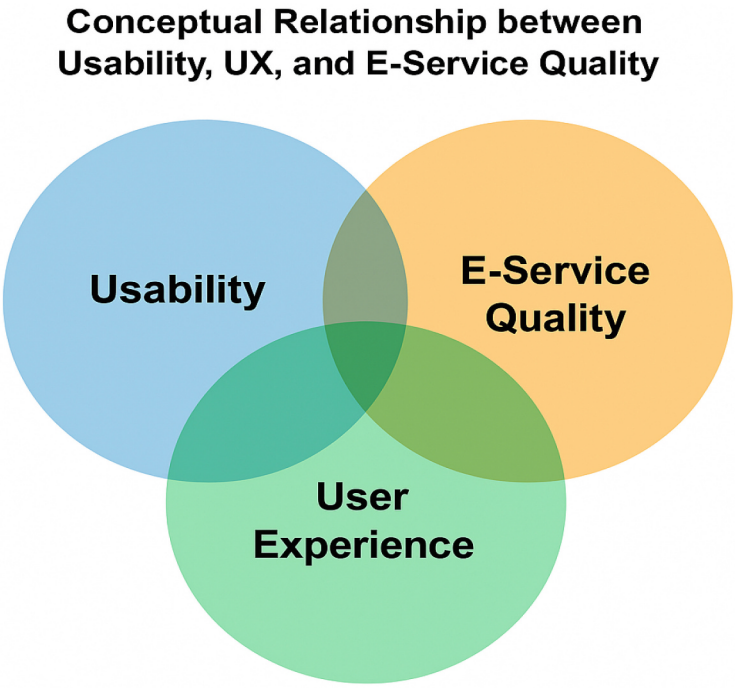

Fig. 1. Conceptual overlap of Usability, User Experience and E-Service Quality within the e-SQUUX model, where shared concepts like Satisfaction indicate dimensions evaluated under more than one construct.

3.3 Initial Framework from the Narrative Literature Review

Developed through a structured narrative review of 264 scholarly sources, the original e-SQUUX framework [8, 29] comprises 24 high-level categories, 75 core dimensions and 163 actionable evaluation statements that together cover usability, user experience (UX) and e-service quality (e-SQ). Its hierarchy has three layers: (i) categories that group related evaluation themes, (ii) dimensions that refine the focus within each category,

and (iii) evaluation statements that translate each dimension into a concrete assessment item. This multi-layered structure lets practitioners tailor analyses by selecting only the elements relevant to a specific web-based application or mobile context, thereby maintaining methodological consistency while accommodating diverse domains and use cases.

4 Research Methodology

4.1 Qualitative Expert Review

Four experts were selected owing to their multidisciplinary HCI/UX backgrounds and extensive evaluation experience, a panel size consistent with best-practice guidance for conceptual validation. A qualitative expert review [30] was chosen to validate the initial e-SQUUX framework because it enables early-stage conceptual assessment without the cost of large-scale user testing [31]. Four domain experts were purposively recruited at the outset of the review process to provide the breadth of insight recommended in HCI guidance, where panels of three-to-five reviewers typically uncover the majority of conceptual and usability issues while avoiding respondent fatigue [31, 33].

Experts were eligible if they (i) held at least a master's qualification in HCI/UX, (ii) had published peer-reviewed work on usability, UX or e-service quality, and (iii) possessed practical experience in applied evaluation projects. Each panelists received the draft 24-category framework together with a structured review guide and rating sheets. They independently assessed every category, dimension and evaluation statement for clarity, relevance and redundancy. Follow-up semi-structured interviews, triangulated questionnaire data, and elicited deeper insights, ensuring methodological rigor [32].

Detailed demographics and the staged evaluation protocol are presented in Sect. 5.1; only high-level information is repeated here to avoid redundancy while preserving a logical flow. All citations remain numbered as in the original reference list.

4.2 Data Collection and Analysis

Each follow-up interview lasted approximately 45 min; recordings were transcribed and coded in NVivo. Thematic analysis then surfaced recurring patterns across usability, UX and e-SQ dimensions. Data were collected through structured questionnaires, which included both closed- and open-ended items. These questionnaires were designed to evaluate key aspects of the framework such as clarity, completeness, relevance and applicability—criteria commonly used in expert assessments [34]. Specific items also addressed ease of navigation, task completion and visual appeal, drawing from established usability constructs. To deepen understanding and resolve any ambiguities, follow-up semi-structured interviews were conducted with each expert. These interviews allowed for elaboration on questionnaire responses, clarification of interpretation issues and the capture of additional insights. The use of multiple data collection instruments enabled methodological triangulation, strengthening the credibility and richness of the findings [32, 33].

For analysis, all qualitative data obtained from questionnaires and interviews were processed using thematic analysis [35], a well-established method for identifying patterns in qualitative datasets. Responses were first coded and then grouped into emerging themes related to usability, UX and e-SQ integration. This analytical approach provided a structured yet flexible synthesis of expert feedback, directly guiding the refinement of the e-SQUUX framework [31]. Methodological triangulation—combining structured questionnaires with in-depth interviews—strengthened credibility and enriched the evidence base for refining e-SQUUX.

5 Findings: Expert Review and Proposed e-SQUUX Model

5.1 Expert Profiles and Scope of Evaluation

The four experts introduced in Sect. 4.1, each labelled E1 to E4 for anonymity, represent both academic and industry settings. Their average HCI/UX experience is 12 ± 3.7 years (range 8–17 years), and their backgrounds cover usability assessment, interface design, and e-service-quality evaluation. Table 1 lists each expert's sector, qualifications, and experience, demonstrating alignment with the constructs embedded in e-SQUUX [31, 33].

Table 1. Expert panel profile.

ID	Sector / Role	Highest degree	Years of experience
E1	Professor (Academia)	PhD (HCI)	17
E2	Professor (Academia)	PhD (UX)	15
E3	Senior Researcher (Institute)	PhD (IS)	8
E4	UX Consultant (Industry)	MSc (HCI)	9
Mean ± SD	—	—	12 ± 3.7

The evaluation unfolded in three stages: (1) Independent assessment of the 24 categories, 75 dimensions and 163 statements; (2) Thematic coding of individual comments by the research team; (3) Consensus interviews to reconcile diverging views and prioritize changes endorsed by at least three experts. This multi-step protocol balances depth and breadth while keeping the participant burden acceptable, and the advantages of a focused four-expert panel are justified earlier in Sect. 4.1. Collectively, this interdisciplinary panel underpins the conceptual validity of the refined e-SQUUX framework.

5.2 Framework Revisions Based on Expert Feedback

Guided by thematic analysis of more than 50 coded comments, a number of activities took place: (i) merged overlapping categories, (ii) clarified ambiguous labels, (iii) added a construct-interaction matrix, and (iv) introduced a Customization & Prioritization Guide.

Table 2. Summary of Expert Feedback and Corresponding Framework Modifications.

Expert Feedback	Action Taken
Overlapping categories cause confusion	Consolidated overlapping categories, reducing total categories from 24 to 16
Ambiguous terminology	Refined and clarified labels (example: "Aesthetic Appeal" → "Pleasure"; "Service Assurance" → "Assurance & Credibility")
Lack of cross-construct links	Introduced a construct interaction matrix to highlight cross-cutting dimensions
Need for practical prioritization	Added a "Customization and Prioritization Guide" for context-sensitive weighting

Following these revisions, the refined e-SQUUX framework now comprises 16 core categories, 101 evaluation statements and a Customization and Prioritization Guide. Each change enhanced the framework's coherence, clarity and flexibility, ensuring it is theoretically grounded and practically applicable. Table 3 lists all 16 categories and their 101 evaluation statements. A high-level overview is provided in Sect. 6.1

5.3 Proposed e-SQUUX Model: Detailed Evaluation Statements

Table 3 shows all 16 categories and their 101 evaluation statements. A high-level overview is provided in Sect. 6.1. This model reflects the refined framework ready for implementation and future validation.

Table 3. The proposed e-SQUUX model.

Category	Evaluation Statements
1. Learnability	1.1 Easy to learn how to use on first exposure. 1.2 Requires minimal time and effort to become familiar with key functions. 1.3 Clear and intuitive instructions and visual cues. 1.4 Terminology is familiar and avoids unnecessary jargon. 1.5 Language used is readable and inclusive. 1.6 Returning users can operate the system without retraining. 1.7 Information is presented in an understandable structure. 1.8 Accommodates users with diverse language needs.

(*continued*)

Table 3. (*continued*)

Category	Evaluation Statements
2. Effectiveness	2.1 Functions as expected under normal conditions. 2.2 Supports the successful completion of typical tasks. 2.3 Facilitates goal achievement with minimal error. 2.4 System logic and behavior are coherent. 2.5 Improves user performance or task outcomes. 2.6 Provides functionally useful outputs or feedback.
3. Efficiency	3.1 Minimal effort and time required after learning the system. 3.2 System interactions are simple and direct. 3.3 Pages or functions load promptly. 3.4 Navigation between elements is quick and fluid.
4. Information Quality	4.1 Information is accurate and factual. 4.2 Content is presented clearly and without ambiguity. 4.3 Information is up-to-date and regularly maintained. 4.4 Content is suitable for the intended audience or use. 4.5 Information provided is sufficient to support user decisions. 4.6 System behavior is predictable and consistent over time.
5. Availability	5.1 System is accessible with minimal loading delay. 5.2 Can be used from any location with internet access. 5.3 Recovers quickly from errors or downtime. 5.4 Communication channels are easily accessible. 5.5 Content and services are readily discoverable.
6. Navigation	6.1 Navigation structure is logical and intuitive. 6.2 Navigation style remains consistent throughout the platform. 6.3 Search functionality is flexible and effective. 6.4 Desired content is easily found. 6.5 The user can identify the current location within the system. 6.6 System supports user orientation and next-step guidance. 6.7 Hints or suggestions are available to assist users. 6.8 Links are clearly distinguishable from regular text.

(*continued*)

Table 3. (*continued*)

Category	Evaluation Statements
7. Responsiveness and Helpfulness	7.1 Services are delivered promptly. 7.2 Responses to queries are informative and solve user issues. 7.3 Feedback is timely and clearly worded. 7.4 User queries are acknowledged and handled efficiently. 7.5 Follow-up satisfaction checks are conducted when needed. 7.6 Guidance is provided for resolving common problems. 7.7 FAQs or similar support options are accessible. 7.8 Communication with the organization is supported.
8. Security and Privacy	8.1 Users feel safe performing sensitive actions. 8.2 User data is handled securely and respectfully. 8.3 The platform owner is trustworthy and transparent. 8.4 The organization behind the service is reputable. 8.5 Access control protects sensitive tasks and data. 8.6 Protection from malicious activity is implemented.
9. Suitability and Relevance	9.1 The system serves its intended purpose effectively. 9.2 Only necessary features and information are included. 9.3 System adds measurable value to the user. 9.4 Service offerings match current user needs. 9.5 Technology features align with modern standards.
10. Errors	10.1 The system helps prevent common user errors. 10.2 Serious errors are mitigated or blocked by design. 10.3 Users can recover from errors easily. 10.4 The system remains stable during system-generated errors. 10.5 The system remains stable during user-generated errors.
11. Assurance and Credibility	11.1 Services are consistently delivered as promised. 11.2 Services are accurate and reliable. 11.3 The organization fosters trust and confidence. 11.4 Sources of information are transparently cited. 11.5 Contact details for support are provided. 11.6 Transactions are clear and transparent. 11.7 System remains operational over extended periods. 11.8 User loyalty is supported through a reliable experience. 11.9 Users are likely to recommend the platform to others.

(*continued*)

Table 3. (*continued*)

Category	Evaluation Statements
12. Satisfaction	12.1 Users accept the system quickly after first use. 12.2 The platform encourages a positive user attitude. 12.3 User goals are achieved with satisfaction.
13. Flexibility and Personalisation	13.1 The platform functions well across varied environments. 13.2 It works consistently across multiple browsers. 13.3 The system integrates well with external services. 13.4 The user can modify how tasks are performed. 13.5 Users can tailor features to their needs. 13.6 Users feel in control while interacting with the system.
14. Interface Design	14.1 The layout is organized and logically structured. 14.2 Navigation follows a consistent and intuitive flow. 14.3 Visual design is aesthetically pleasing. 14.4 Colors are used effectively and meaningfully. 14.5 UI elements are consistent in meaning and use. 14.6 Visual outputs are well-presented. 14.7 Design aligns with the brand identity. 14.8 Presentation feels innovative and distinctive.
15. Pleasure	15.1 The platform creates a pleasurable user experience. 15.2 Using the system is motivating and stimulating. 15.3 Users form an emotional bond with the system. 15.4 The platform is visually and interactively appealing.
16. Support	16.1 Help is available for technical or process issues. 16.2 Support content is clear and useful. 16.3 Help is provided promptly when needed. 16.4 Support is effective in resolving issues. 16.5 Help is available across all stages of the user journey. 16.6 Self-service tools are offered for common tasks. 16.7 Assistance is continuous from the start to the finish of the process. 16.8 Support extends across all user interactions.

6 The Refined e-SQUUX Evaluation Framework

6.1 Overview

Similar to Sauer (2020), the refined e-SQUUX framework consolidates the originally proposed 24 categories into 16 well-defined categories, grouped under three primary constructs: Usability, E-Service Quality and User Experience. These constructs represent the comprehensive dimensions needed to evaluate WBAs and mobile applications

holistically. The refinement is based on expert input and thematic analysis, yielding 101 detailed evaluation statements that can be adapted across various domains and application contexts.

6.2 Framework Structure

Table 4 summarizes the high-level structure of the refined e-SQUUX evaluation framework. Each constructs, Usability, E-Service Quality (e-SQ) and User Experience (UX), is subdivided into categories that capture specific evaluation criteria.

While 'Satisfaction' is grouped under UX to emphasize its emotional and temporal aspects (ISO 9241-210), it also intersects with Usability's task-completion focus (ISO 9241-11). This dual alignment is accommodated in the Customization and Prioritization Guide, which provides context-sensitive weighting. The structured approach, therefore, supports assessments that are both comprehensive and adaptable to diverse interactive systems.

Table 4. Refined e-SQUUX Framework: Constructs and Categories.

#	Construct	Category
1	Usability	Learnability
		Effectiveness
		Efficiency
		Information Quality
		Availability
		Navigation
		Error Handling
		Interface Design
2	E-Service Quality	Responsiveness & Helpfulness
		Security & Privacy
		Suitability & Relevance
		Support
3	User Experience	Assurance & Credibility
		Satisfaction
		Flexibility & Personalization
		Pleasure

It is to be noted that, while 'Satisfaction' sits under UX to emphasize its emotional aspects, Table 4 shows its overlap with Usability (ISO 9241-11) to acknowledge task-completion links.

6.3 Key Revisions Recap

Table 2 lists all the modifications suggested by the experts. The next paragraph explains the four most significant changes.The expert-led refinement process introduced several key enhancements to improve the clarity and practicality of the e-SQUUX framework. Firstly, overlapping categories were merged to streamline the evaluation structure and reduce redundancy. Secondly, ambiguous terminology was revised to ensure clearer interpretation and consistent understanding across users. Additionally, a cross-construct linkage representation was incorporated to highlight interconnected evaluation dimensions. Lastly, a Customization and Prioritization Guide was developed to support context-sensitive adaptation, enabling flexible application of the framework across varied domains and user scenarios.

6.4 Application and Adaptability

This framework is highly flexible and applicable across a wide range of platforms, including education, healthcare and e-commerce. For instance, when applied to a university registration portal, 'Learnability' and 'Responsiveness & Helpfulness' jointly flagged an 18% drop-off in task completion—prompting interface and service tweaks that halved user attrition. Its design allows evaluators to tailor assessments to different user needs, technological environments and resource constraints, including mobile apps. The 16 categories serve as a foundation for developing practical evaluation instruments such as surveys, heuristic checklists, or automated assessment tools.

7 Discussion

7.1 Advancing Conceptual Integration in WBA Evaluation

Unlike prior holistic checklists like PACMAD+3 [11], the refined e-SQUUX model operationalises cross-construct links through an explicit interaction matrix and a 16-category core—advances not previously offered. The refined framework demonstrates a significant theoretical advancement by resolving the historical fragmentation among the constructs of usability, UX and e-SQ. For instance, the alignment of categories such as "Trust" (UX) and "Security & Privacy" (e-SQ) underscores how emotional reassurance and technical safeguards converge in user-centered evaluation. This synthesis reflects the real-world usage context of WBAs and mobile apps, where system performance, aesthetics and service responsiveness are experienced simultaneously.

This integrated approach aligns with prior studies emphasising the need for comprehensive evaluation models. For example, Ssemugabi and de Villiers [8] highlighted the interrelation of these constructs in their initial e-SQUUX model, advocating for a holistic assessment framework. Similarly, Mamakou et al. [7] demonstrated the mediating role of e-service quality in the relationship between UX and customer satisfaction, reinforcing the importance of integrated evaluation.

7.2 Content Validity and Practical Relevance via Expert Review

The use of expert review not only validated the framework's core components but also enhanced its clarity and applicability. Reducing the initial 24 categories to 16 while preserving analytical depth was a major contribution of this phase. Experts recommended refinement of ambiguous dimensions and advocated for a structure that allows evaluators to prioritize based on contextual relevance, an important criterion in domains like e-learning and e-governance. The resulting 101 evaluation statements are not merely descriptive but are actionable, thus bridging the gap between theoretical constructs and empirical assessment instruments.

This methodological rigor is consistent with established practices in usability evaluation. For instance, Djordjević [36] emphasizes the importance of expert input in refining usability dimensions, ensuring that evaluation tools are both comprehensive and contextually relevant. Additionally, the integration of expert feedback aligns with the principles outlined by Di Gironimo et al. [37] in their usability-oriented quality model, which underscores the value of expert assessments in identifying usability issues.

7.3 Human-Centered Design and Systemic Flexibility

By emphasizing evaluation statements that are both user-focused and system-aware, the framework aligns with modern human-computer interaction (HCI) paradigms. Dimensions like "Responsiveness & Helpfulness" and "Interface Design" address not only functional efficiency but also cognitive and emotional ease of use. This user-centric perspective resonates with ISO standards (9241-11, 9241-210) and aligns with recent literature advocating for emotionally intelligent interfaces and trust-driven engagement [38, 39]. The flexibility of the framework enables its adaptation across diverse WBAs and mobile apps, addressing accessibility, cultural variance and technological contexts. This flexibility is particularly valuable in resource-limited contexts where bandwidth, device capability and digital literacy vary widely.

This adaptability is crucial in the context of diverse user populations and varying application domains. As highlighted by Djordjević [36], usability evaluation must account for the specific environment in which web applications are developed, maintained and used, necessitating flexible evaluation frameworks that can be tailored to different contexts. Furthermore, the importance of considering subjective aspects of user interaction, as discussed by Di Gironimo et al. [37], reinforces the need for evaluation models that encompass both objective and subjective dimensions of usability. Accordingly, the framework encourages inclusive design choices—adaptable colour palettes, localization of terminology and assistive-tech compatibility—to accommodate diverse cultural and accessibility requirements.

7.4 Relevance to Digital Service Ecosystems

Digital-service ecosystems rely on tightly coupled Web-Based Applications (WBAs) and mobile apps that deliver core value to users. The refined **e-SQUUX** model supports these ecosystems by linking three human-centered constructs, Usability, User Experience (UX) and E-Service Quality (e-SQ), into one diagnostic scaffold that guides benchmarking, procurement and iterative design work.

Comparative Perspective. WBAs and mobile apps share foundational evaluation principles, yet mobility introduces constraints such as limited screen space, gesture navigation, intermittent connectivity and higher expectations for speed and simplicity. Classic instruments like the System Usability Scale (SUS) [21] and the User Experience Questionnaire (UEQ) [22] remain applicable. Mobile-specific frameworks, for example PACMAD+3 [23], add factors like cognitive load and error tolerance. Studies such as Sukmasetya et al. [27] confirm that traditional usability attributes—learnability, efficiency, memorability, error handling and satisfaction—still apply, but they need interpretation through a mobile lens. By accommodating both general and context-specific criteria, e-SQUUX offers a coherent way to evaluate integrated multi-channel services.

Illustrative Use-Case – University Registration Portal. 'Navigation' mapped to the menu hierarchy, 'Responsiveness & Helpfulness' to chatbot response time and 'Security & Privacy' to the authentication flow. Pain-point analysis based on these mappings reduced form abandonment from **22%** to **9%**, showing how e-SQUUX converts abstract constructs into measurable service improvements across web and mobile touchpoints.

Evidence from Prior Studies. Mamakou et al. [7] found that e-service quality partially mediates the relationship between UX and customer satisfaction in e-commerce. Ighomereho et al. [18] identified e-SQ dimensions—reliability, responsiveness and assurance—that align with e-SQUUX categories. These findings underline the value of an integrated lens when organizations design or refine digital services.

Cross-Functional Collaboration. The Venn diagram introduced earlier helps evaluation team members, for example, designers, developers and service managers, spot overlap areas such as Satisfaction and Trust. This shared mental model accelerates consensus on priorities and ensures that technical changes, for example, API optimization, match experiential enhancements like reduced cognitive load.

Strategic Value. By offering a single framework adaptable to both WBAs and mobile apps, e-SQUUX lets organizations scale evaluation practices without creating siloed checklists. Continuous measurement cycles informed by the framework can close the loop between user feedback, service-quality metrics and business outcomes, sustaining competitive advantage in fast-moving digital ecosystem.

7.5 Limitations and Opportunities for Empirical Validation

While the expert-reviewed framework is conceptually robust, its empirical utility across real-world WBA and mobile app settings warrants further testing. Future research should explore its application through large-scale surveys, experimental validation, or longitudinal case studies. Additionally, statistical modelling techniques such as Partial Least Squares Structural Equation Modelling (PLS-SEM) may be used to evaluate the interrelationships among categories and confirm the underlying dimensional structure.

The need for empirical validation is echoed in studies like that of Mamakou et al. [7], who utilized PLS-SEM to assess the relationships between e-service quality, UX and customer satisfaction, demonstrating the effectiveness of such methods in evaluating integrated models. Similarly, Ssemugabi and de Villiers [9] emphasized the importance

of empirical testing in refining and validating evaluation frameworks for WBAs. Future studies will combine large-scale surveys, comparative case studies and PLS-SEM modelling to validate category interrelations and test the framework across domains. While conceptually comprehensive, e-SQUUX should be viewed as an incremental advance that awaits broad-based empirical corroboration.

8 Conclusion

As stated, unlike earlier holistic checklists such as PACMAD+3, the refined e-SQUUX model operationalizes cross-construct links via a 16-category core, an explicit interaction matrix, and a practitioner-ready Customization & Prioritisation Guide. This study has presented the development and refinement of the e-SQUUX framework, an integrated model designed to evaluate Web-Based Applications (WBAs) and mobile applications by encompassing usability, user experience (UX) and e-service quality (e-SQ). Through a comprehensive review of existing literature and expert validation, the framework addresses the limitations of prior models that often treated these constructs in isolation.

The e-SQUUX framework offers a comprehensive approach by identifying overlapping and distinct dimensions across usability, UX and e-SQ. This integration facilitates a more holistic assessment of WBAs and mobile apps, aligning with the multifaceted nature of user interactions in digital environments. The expert review process contributed to refining the framework's categories and evaluation statements, enhancing its clarity and applicability across various contexts. The framework's 16 categories and 101 statements balance conceptual clarity with empirical utility.

While the integration of e-SQ, UX and Usability builds on prior studies, e-SQUUX should be viewed as an incremental advance that still requires large-scale empirical corroboration. The practical implications of the e-SQUUX framework are significant. It serves as a valuable tool for designers, developers and evaluators aiming to enhance the quality of WBAs and mobile apps. By providing a structured method to assess critical aspects of user interaction and service delivery, the framework supports the development of applications that meet user needs and expectations effectively.

Future research will deploy mixed-methods validation such as large-scale surveys, comparative case studies and PLS-SEM modelling, to test inter-category relationships across diverse domains. The research should focus on empirical validation of the e-SQUUX framework across diverse application domains. Implementing the framework in real-world settings will provide insights into its effectiveness and adaptability, potentially leading to further refinements. Additionally, exploring the integration of the framework with quantitative evaluation methods could offer a more robust assessment of WBAs and mobile apps.

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From Case Files to Virtual Reality: An Artificial Intelligence-Based 3D Apartment Generation Tool

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Abstract. The generation of 3D indoor scenes has implications for various fields, from virtual reality (VR) via video games to architecture. This undertaking is by no means trivial, as it consists of several components, such as floor plans, walls, doors, windows, and furniture objects. The visual appeal of the generated scenes of the 3D apartment is contingent upon the quality of the utilized 3D model for furniture objects and the naturalness of their positions, orientations, and inter-object relationships, which can significantly impact user experience and engagement. To address this challenge, numerous tools have been developed to facilitate the generation of 3D indoor scenes, employing diverse generation methods to achieve this objective. This paper presents a novel tool that uses various artificial intelligence (AI) models to generate 3D-furnished interiors that can be easily customized and interacted with in VR applications, e.g., based on case file descriptions. The tool enables user-driven customization of apartments based on textual descriptions and numerous parameters. To assess the usability of the generated 3D apartments, a range of metrics is used to evaluate the tool's performance compared to manually designed 3D apartments. The results show that the presented generation tool achieves results comparable to those of human-designed apartments.

Keywords: 3D Apartment Generation · Artificial Intelligence (AI) · Virtual Reality (VR) · 3D Indoor

1 Introduction

In recent years, graphical processing units (GPU) have evolved rapidly, enhancing their ability to handle large 3D data volumes [2]. This advancement enables the use of high-quality 3D models and textures in applications like virtual reality (VR) and video games, resulting in real-time production of more realistic graphics [14]. These 3D environments can simulate complex scenarios that are not feasible or safe to recreate in reality. However, designing such scenes requires expert use of 3D design software, which is time-consuming. As demand for 3D content grows, efforts have been made to automate parts of the 3D design process to produce high-quality results faster. Indoor scenes, composed of various elements, are a significant class of these environments. Although researchers have developed tools for generating 3D furniture layouts,

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fully customized 3D indoor scenes that are easily integrated into applications remain challenging to produce [13].

Also, creating an environment suitable for VR and other applications that support interaction with the objects and walking around in 3D indoor scenes poses additional challenges. Interaction within 3D scenes requires visually realistic 3D indoor scenes and the semantic annotations that distinguish each object and allow applying a specific interaction with it [7]. Recent solutions combine artificial intelligence (AI) approaches trained on human-made designs to generate 3D indoor scenes that closely resemble reality [3,5,12,19].

This paper develops a tool capable of generating customized apartments based on specific conditions to customize the results scenes according to a given case file. The generated scenes should be easily embedded in VR applications and include a set of furniture objects that can be utilized in future interactions. Also, the layouts should be functional and coherent, with objects realistically placed within the apartment. To maintain a focused scope, aspects primarily relevant to other fields will be excluded from consideration. The research questions in this paper are:

1. What methods can be used to generate 3D indoor apartments with furniture objects that meet specific design criteria?
2. How much do these generated apartments fulfill the usability requirements of VR applications?

The following repository[1] contains the tool's source code, its documentation, installation instructions, and evaluation results.

2 Use Case Description: Training Child Protection Workers

Child protection workers are tasked with assessing children's living environments, identifying signs of neglect or abuse, and making life-altering decisions under conditions of uncertainty. Conventionally, the training of these professionals has relied on a limited array of case studies and role-play exercises [16]. These methods are inherently constrained by their inability to mirror the diversity and intricacy of real-world scenarios.

The proposed AI-based 3D apartment generation tool should address this challenge by creating dynamic and immersive training environments in VR. The system should synthesize a wide range of realistic living spaces by drawing upon anonymized case files. These spaces mirror the socio-economic, cultural, and psychological dimensions found in actual child protection cases. These environments facilitate the transition of trainees into the domestic milieu, where they are able to engage in observational studies of living conditions, the interpretation of environmental cues, and interaction with AI-driven characters representing children and caregivers.

In contrast to static or scripted training modules, the generative approach guarantees that each training session can present a novel apartment setting, thereby preventing rote learning and encouraging critical thinking. For instance, residential units may exhibit variations in cleanliness, layout, and available resources, while the behavior of virtual

[1] https://github.com/fiadleh/AI-Driven-3D-Apartment-Generator.

inhabitants can dynamically adapt to trainee interactions. Such variability prepares child protection workers to attend to the small yet significant details that might otherwise go unnoticed in real investigations [16].

The tool presented in this paper has broader implications for the above-mentioned use case. First, it facilitates exposure to high-stakes environments in a safe and repeatable manner, thereby eliminating the risk of harm to actual children. Secondly, it fosters interdisciplinary collaboration, as the training modules can be reviewed and refined by child welfare experts, psychologists, and ethicists to ensure respectful representation of vulnerable populations. In conclusion, the presented tool is the foundation to offer a scalable solution to address the increasing demand for professional training in child protection services. The challenges posed by mounting caseloads and limited staffing resources, which have the potential to compromise service quality, are addressed by this program.

3 Background

Virtual 3D environments are extensively used in gaming, the film industry, and more. Specifically, indoor 3D environments representing homes, offices, factories, etc., have broad applications in architectural visualization, interior design, VR, and training simulations. Creating these environments requires skilled art designers and modeling experts, making it costly and time-consuming [17]. Large 3D indoor scene datasets are vital for data-driven methods, allowing for the creation of varied and realistic scenes. These datasets help populate interiors with diverse furniture models and train AI models for similar layouts. They can be created through scanning, reconstruction, or manual design. The *SUNCG* [18] dataset, a significant manually created collection of realistic 3D indoor scenes, impacted the field with over 1,300 research publications but was deleted in 2020, affecting dependent tools. The *3D-Front* dataset [4] includes 18,968 rooms in 6,500 houses across 31 categories, with object semantics and style labels, designed using recommendations from 3D-Future for style compatibility.

A floor plan is essential for designing a house that meets functional needs within budget constraints. However, fewer than 10% of North American homeowners can afford architects to customize plans [9]. This constraint also affects the generation of complete 3D interior scenes for apartments. Early automation attempts used rule-based methods with constraints to generate layouts [1] iteratively. Other methods include training convolutional neural networks (CNN) on datasets of existing homes to generate new floor plans from building boundaries [20]. *House-GAN* [8] is a framework using a graph-constrained generative adversarial network (GAN), combining a relational GAN with a conditional GAN. Users can input constraints as a graph, specifying room types, counts, and connections. *House-GAN* was trained on the RPLAN dataset, featuring 60,000 floor plans created by professional architects [20].

Populating a given space with furniture objects implies deciding on their type, size, position, and direction. All these features must be considered to generate a feasible and visually coherent layout. One tool that can accomplish this task is the *AI Room Generator*, a plug-in developed for NVIDIA Omniverse—a platform for visual effects and industrial simulation. This extension uses the large language model (LLM) from

OpenAI's *ChatGPT* in addition to deep search to generate 3D environments based on text-based prompts [11]. *LayoutGPT* is a training-free approach using LLM to generate plausible furniture layouts. It provides the LLM with visual commonsense text prompts formatted as cascading style sheets (CSS). This approach turns the LLM into a visual planner for different types of rooms, enabling it to understand furniture 3D models, sizes, and depth [3].

Liu et al. introduced *CLIP-Layout*, an auto-regressive scene model for synthesizing furniture layouts using OpenAI's contrastive language-image pre-training (CLIP) neural network, trained on image-text pairs [15]. Tested on the *3D-Front* dataset, it produces style-consistent, visually plausible furniture arrangements [6]. Paschalidou et al. developed *Atiss*, which uses an auto-regressive transformer to generate indoor furniture layouts. *Atiss* was trained on *3D-Front* dataset scenes, and its generated room layouts, including furniture, also derive from the same dataset [12]. Tang et al. presented *DiffuScene*, a scene-denoising diffusion model that creates indoor furniture layouts with naturally placed objects and symmetries. It treats the scene as an unordered set of furniture objects, each with a vector of features like location, size, semantics, and geometry [19].

InstructScene is a framework designed to generate realistic and controllable 3D indoor furniture layouts based on natural language instructions. Like diffusion-based tools *Atiss* and *DiffuScene*, it utilizes diffusion models trained on room-layout instruction pairs from the *3D-Front* dataset to generate similar layouts controlled by textual instructions [5]. The first phase of the generation process is to process the natural language prompts from a user to build a scene graph. The graph contains information about the category and vector-quantized feature representation of objects' attributes in each node, as well as the spatial relations between objects as directed edges. The second phase utilizes a layout decoder on the resulting graph to produce values for each object's position, orientation, and size in the graph [5].

4 Methodology

Answering the first research question, a tool was developed to generate 3D apartments dynamically. *InstructScene* generates furniture layouts for each room based on descriptions, as it has better results than similar tools like *Atiss* and *DiffScene*, which use diffusion models [5]. The generated layouts were also be based on the open-source *3D-Front* dataset, which contains high-quality textured 3D models. Furthermore, *3D-Front*-based generation offers more flexibility than other solutions using 3D model datasets made only for a specific program or with a proprietary license. Moreover, *InstructScene*, which uses a local text processing model, is more favorable than Open AI's ChatGPT solutions, especially when dealing with sensitive personal data. This limitation is because Open AI's privacy policy[2] clearly states that user content is used to train their models. Due to its features as an open-source 3D design framework, the Blender API was used to create 3D models of walls and windows and to manipulate 3D models and textures from *3D-Front*. The final 3D apartment was saved as a glTF file, thanks to

[2] https://openai.com/en-GB/policies/privacy-policy/.

its flexibility and features that support creating immersive VR scenes. *InstructScene* and *HouseGan* were written mainly in Python, so the tool that combines them also used the same language. Moreover, all manipulations done via the Blender API were executed through an embedded Python interpreter, making Python a compatible choice for the tool. Section 5 discusses the tool development steps and details.

To answer the second research question, several metrics were used to estimate the usability of an apartment, with a focus on evaluating the effectiveness of the virtual environment in supporting intuitive and engaging human-computer interaction. These metrics were achieved by measuring the ability to reach different spaces and objects in an apartment. As the tool developed in this paper concentrates on generating 3D apartments for use in VR, other metrics that could be important in different fields, such as lighting and ventilation, are ignored. Nevertheless, the generated apartments' general architecture guidelines for convenient indoor spaces are measured. Next, each metric will be explained along with its calculation method.

Dead Zones: are areas in generated apartments that become inaccessible or unusable due to poor spatial planning or furniture placement. These zones range from small gaps between objects to rooms with blocked doors or paths, resulting from impractical furniture layouts that impede movement and usability. Identifying and minimizing dead zones is crucial for creating functional and aesthetically pleasing virtual environments, as they represent wasted space that could be better utilized, especially in size-constrained apartments.

Out-of-Boundary (OBR): refers to instances where a 3D furniture model overlaps another model or wall, indicating improper positioning. This metric measures the percentage of 3D furniture models that collide with other objects or walls in the virtual environment and is calculated as follows:

- $N_{\text{collisions}}$ is the number of objects that have collisions.
- N_{total} is the total number of objects in the apartment.
- $\text{OBR} = \frac{N_{\text{collisions}}}{N_{\text{total}}} \cdot 100$

Object Reachability Percentage (ORP): measures the proportion of objects in a 3D-generated apartment that a virtual human agent can reach, considering collision constraints. It assesses whether furniture, walls, and doors are placed to allow efficient movement and interaction. This metric quantitatively measures object accessibility, which is essential for functional evaluation. Due to the complexity of simulating human reachability accurately, a simplified model using a cylindrical human body and a forward maximum reach distance of 74 cm based on ergonomic measurements [10] is used.

5 Software Design

The tool's input pipeline has two components, as illustrated in Fig. 1. First, users create a graph representing the desired apartment layout, including room types, counts, and doors, using a graph editor with preset arrangements like 2-room or 3-room apartments. Second, they provide a textual description of the furniture in each room, detailing the visual appearance and placement of objects, as required by *InstructScene*.

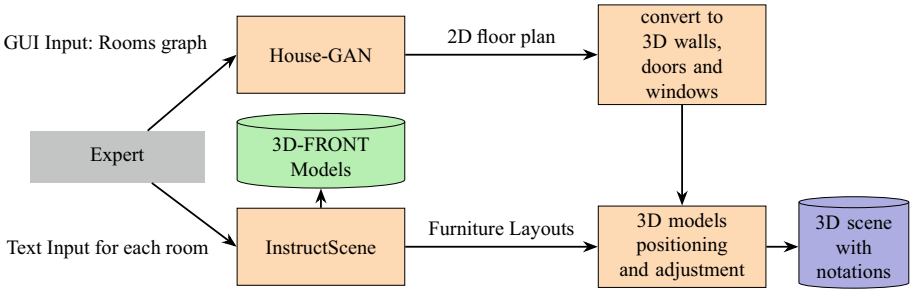


Fig. 1. The 3D apartment generator pipeline takes user inputs about the apartment and passes these into two models, then combines and refines the results.

5.1 Pipeline

The first part of the tool is a server that provides the GUI for the user. The core part of the graph editor is based on the *HouseGAN* demo server. The server is built using Flask[3], a Python framework for web applications that allows the user to interact with Python scripts using HTML elements displayed in the browser.

In the browser, the user can create and modify a graph representing the desired apartment. Each node represents a room with a specific type, and the edges are the doors that connect the rooms. By clicking "Edit" then "Add Node", it is possible to choose from 8 types of rooms: bedroom, living room, dining room, kitchen, bathroom, balcony, study room, and storage, in addition to an outside node that represents the entrance door. The "Add Edge" button adds a door between two rooms. After selecting a node or an edge, "Delete Selected" can be used to delete the selected element. The application starts with a predefined graph of a 2-bedroom apartment, and the user can choose between three predefined layouts to design the desired graph faster.

After designing the graph, a 2D floor plan can be generated by clicking on the "Generate 2D Plan" button. An image of the plan would appear on the right side of the page. Below the above-mentioned elements, a text input area appears for each room in the graph, categorized by type (bedroom, living room, dining room). The user can type in the description of furniture in each room in these areas. These descriptions are used as inputs for *InstructScene* to generate furniture layouts for the rooms. Below the text areas, several parameters are used to control the process of generating the apartment's 3D walls, floors, and doors, such as wall thickness and door height.

The given furniture description text used for each room is hashed, and all generated room objects and attributes are saved to be used as a cache to skip unnecessary regeneration of the furniture generation step if the exact input text is given. At the bottom of the page, the "Generate 3D" button can be used to initialize the last phase of generation, which ends with a resulting `.glb` file containing all 3D objects in the apartment and their textures.

The first problem encountered when combining the furniture layouts by *InstructScene* and the generated floor plan by *HouseGAN* is that the room size is not

[3] https://palletsprojects.com/projects/flask/.

part of the input in both tools. Additionally, it was mentioned in *HouseGAN* that the size is ignored to simplify the generation problem [8]. This can cause furniture object arrangements that do not fit within the corresponding room floor. The following section will provide a detailed overview of the fitting problem. Following that, the details of converting a 2D floor plan to a 3D will be discussed and the adjustment actions to improve the scene will be introduced.

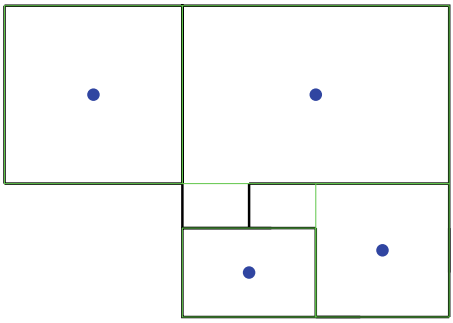


Fig. 2. Find the largest rectangle within the room contour. Floor plan is shown in black, with found rectangles in green and centers in blue. (Color figure online)

Fitting. This section of the problem focuses on optimizing room floors and furniture layouts, represented as rectangular areas, to determine the optimal pairs of furniture layouts and room floors. In each furniture arrangement, the maximal required rectangular floor area that covers all 3D models is calculated by obtaining the minimum and maximum x and y coordinates among all furniture 3D model vertices in the room. However, calculating the maximal rectangular floor area that can fit inside each room contour in the generated 2D floor plan requires different calculations, as the room contour polygons can be more complex than a rectangle. The calculation is performed using the algorithm in 1, which starts with a small random rectangle and attempts to expand it in each direction, returning the best results for each room. The results are visualized as a green rectangle inside the black floor plan in Fig. 2. The next step is to establish a suitable mapping between the group of room floor rectangles and the group of generated furniture layout rectangles, such that each furniture layout is contained within a room's walls. However, it is not guaranteed that each furniture layout can be completely accommodated within a room, so the containing percentage is used as a score to evaluate the mappings. This problem can be considered as a variant of the following bin packing problem: Given:

- A set of N bins $b \in B$, where each bin represents a room with a maximum rectangular area m that fits within the room's floor polygon.
- A set of N objects $o \in O$, where each object represents a group of furniture with a maximum area that covers all furniture models.

- $C_{b,o} \in [0, 100]$ denotes the coverage percentage of the furniture objects within the room floor.
- Each bin b and each object o has a type $t \in$ bedroom, living room, dining room, denoted by $T(b)$ and $T(o)$, respectively.

Objective: To find an arrangement where each bin b contains exactly one room such that the minimum value of $C_{b,o}$ across all pairs is maximized while ensuring that pairs (b, o) have matching types, i.e., $T(b) = T(o)$.

$$
\begin{aligned}
\text{Maximize} \quad & \min_{b \in B, o \in O} C_{b,o} \\
\text{subject to} \quad & \sum_{b \in B} \sum_{o \in O} x_{b,o} = N, \\
& x_{b,o} \in \{0, 1\}, \\
& \sum_{o \in O} x_{b,o} = 1, \quad \forall\, b \in B, \\
& \sum_{b \in B} x_{b,o} = 1, \quad \forall\, o \in O, \\
& x_{b,o} = 0, \quad \text{if } T(b) \neq T(o).
\end{aligned}
\tag{1}
$$

where $x_{b,o}$ is a binary decision variable indicating whether object o is placed in bin b.

The problem assumes that each room can contain exactly one type of furniture layout, and type compatibility is strictly enforced. The maximum area constraints for bins and objects are assumed to be known and fixed. The problem is solved using a greedy approach, with the rectangle area as a heuristic.

All furniture objects that are partially or completely outside the room must be moved later, so it is essential to have the highest possible percentage. Therefore, multiple 2D floor plans can be generated for the same given room graph. The number of floor plan candidates can be adjusted through the tool configurations. Each floor plan is evaluated by executing the previous bin packing Algorithm 1 to obtain its highest minimal percentage for the same furniture layouts as a representative score. Finally, the floor plan with the highest score is selected for use in the next phase.

The generated floor plan is served in a JSON file containing groups of 2D points representing the rooms and doors. These 2D points are converted to 3D vectors with $z = 0$, and new vertices are added to create the thickness of the wall. Then, all resulting faces are extruded to the specified wall height. The doors are also converted in a similar way from groups of 2D points to a 2D polygon and then extruded to create a 3D shape. After that, these shapes are subtracted from the walls. The doors' 3D shapes are given a transparent material and kept as placeholders for door position and volume. Furthermore, the name of each door 3D shape is set to carry the two room IDs that it connects. This allows for easy access to this information later by iterating through the objects' names in the final 3D scene.

Windows are created by generating 3D shapes that represent each window and subtracting them from the room walls. These 3D shapes are distributed along the room

Algorithm 1 Find Largest Rectangle within Room Contour.

function EXPAND_RECTANGLE(contour, cX, cY)
  Initialize $w, h \leftarrow 1, 1$
  directions $\leftarrow$ ['right', 'left', 'down', 'up']
  expanding $\leftarrow$ True
  while expanding **do**
    expanding $\leftarrow$ False
    for each direction in directions **do**
      try to expand the rectangle in the current direction
      if the expanded rectangle fits inside the contour **then**
        Update center (cX, cY) and dimensions (w, h)
        expanding $\leftarrow$ True
      end if
    end for
  end while
  return w, h, cX, cY
end function
function FIND_LARGEST_RECTANGLE(contour, initial_cX, initial_cY)
  $best_w, best_h, best_cX, best_cY \leftarrow$ EXPAND_RECTANGLE(contour, initial_cX, initial_cY)
  for 10 times **do**
    Get a random point $(rand_cX, rand_cY)$ inside the contour
    $t_w, t_h, t_cX, t_cY \leftarrow$ EXPAND_RECTANGLE(contour, rand_cX, rand_cY)
    if $t_w \times t_h > best_w \times best_h$ **then**
      Update $best_w, best_h, best_cX, best_cY$
    end if
  end for
  return $best_w, best_h, best_cX, best_cY$
end function

boundary. The number of windows on each wall depends on the given window dimensions and the wall length. First, the 3D shapes are created in each of the four directions of each room. Then, the shapes that collide with more than one room are deleted to ensure that windows are only placed on the outer walls. After creating the window spaces in the walls, simple window frames with glass and grips are created. Additionally, several helper 3D objects are created to facilitate the process of collision detection and editing the generated scene systematically. These include the room floor plan bounding box center, the center of the placed furniture layout, and a 3D shape representing each room's full volume. These helper objects are given full transparency to avoid visually disturbing the scene.

Adjustment. Despite all previously explained actions to find a perfect match between the generated furniture layouts and the room floor plans, there are still some cases that may require further adjustment for furniture objects separately. For example, the size of the furniture area might be larger than the room area, causing some objects to collide with walls or even be placed outside the room. To solve this issue, a check is executed

on each object to ensure that the model is completely within its room space; otherwise, it would be shifted towards the room center. Another issue that could arise, even if the furniture fits perfectly inside the room walls, is that some furniture objects might block the door. This is due to the generation process of the *InstructScene*, which does not include door positions. To solve this, special 3D boxes are placed on the doors, but with a larger size on the faces that are in the in and out directions. These boxes are then considered when trying to move the furniture objects, so they do not collide with them and keep the door free to walk through. The size of the boxes can be changed from the configurations. There are additional adjustment actions that try to optimize the furniture layout to make it look as if it was built for the given room space. To resolve collisions with other objects or walls, the tool tries a new possible position for the object in each direction in the xy plane with an offset. Then, the best position is chosen based on the lowest number of collisions compared to the original position. Also, all moving actions are made iterative, starting with tiny offsets of $5\,\text{cm}$, and then increasing gradually. This approach tries to solve collisions and/or misplacements with the minimal required changes to the original generated layout. One adjustment action that is executed is to rotate the bed so its back is not facing a door, if possible, and then move it back until it collides with a wall, stopping just before it does. Another adjustment action is related to objects that are usually placed with their back against a wall, such as wardrobes, TV stands, and bookshelves. Due to the *InstructScene* not considering walls in the layout generation process, the tool tries to achieve this afterward by moving these types of furniture to the back until they collide with a wall. These objects are identified by their model name, and the movement direction is decided by comparing the model's dimensions, as they usually have their longest dimension in the xy plane parallel to the nearest wall. The object is shifted along its shortest x or y dimension length away from the room center, with the assumption that humans prefer furniture layouts where such objects are not placed with their back facing the room center.

During the execution of adjustment actions and collision resolving, some furniture objects are treated as a group when moving or rotating, such as dining tables and chairs. This preserves the original layout between them, which is generated by the *InstructScene*. The reason behind this is that separately moving these objects could result in very odd relative positions, such as moving the table in one direction and the chairs in another. The objects within glTF files have a simple structure, where the name of the object starts with its main function and ends with one or more IDs that identify its features. Given these specifications, an application that imports the generated scene file can apply customized functionalities to specific groups of furniture or objects in a desired room.

A simple agent was developed using the Unity game engine to test the generated scenes in an environment capable of running on VR devices. The adds a physical collision component to each 3D object and instantiates a FPS camera with a simple collision body as a capsule to allow the user to visit the scene. A screenshot from a generated apartment is shown in Fig. 3.

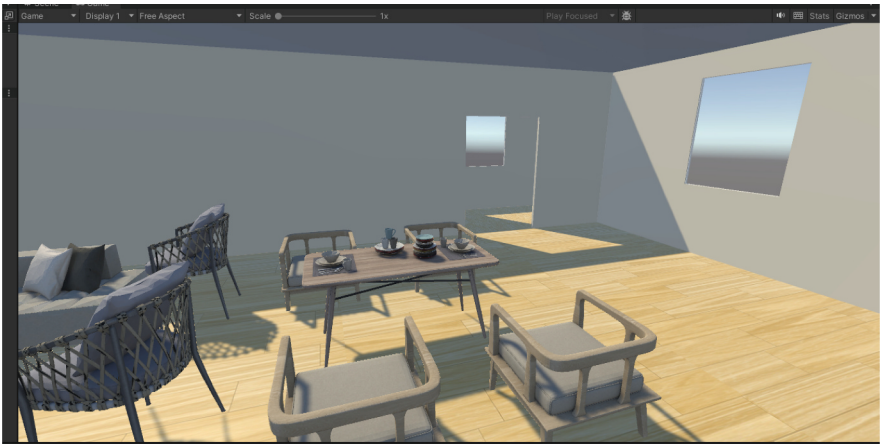
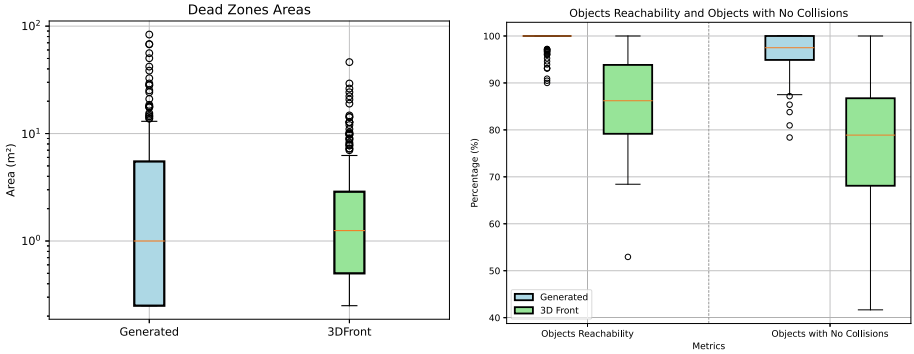


Fig. 3. Walking in a generated scene using the Unity agent.

6 Experiments and Discussion

To evaluate the generation tool, experiments measured the usability of generated apartments using specified metrics. Similar assessments were conducted on random scenes from the *3D-Front* dataset for comparison. Five different graph types, varying in room count and types, served as bases for generation, producing 100 apartments each. Random text prompts matching desired room types were selected from the InstructScene dataset to generate furniture layouts. The *3D-Front* dataset, consisting of JSON files with room and furniture details, was converted to glTF files. These glTF files were then compared with those generated by the tool using several metrics.



(a) Dead zones area.

(b) ORP and collision-free objects Percentage.

Fig. 4. Real and synthetic example images of the datasets.

Figure 4a represents the distribution of the area of dead zones within a 3D indoor scene for the generated and *3D-Front* groups. The vertical axis, on a logarithmic scale,

represents the area of dead zones in m^2. Figure 4a shows that "Generated" has a lower median of dead zone areas compared to the *3D-Front*. However, the upper whisker of "Generated" is higher, and its outliers are higher than those in the other group. The two boxplots shown in Fig. 4b to the left represents the distribution of the percentage of reachable objects across generated and *3D-Front* indoor scenes. It shows a clear trend for the "Generated" group, where all quartiles and the median have maximum object reachability percentage of 100% with some outliers. In contrast, the "*3D-Front*" group has a median of 86% and shows lower and more varied values. The right 2 boxplots shown in Fig. 4 illustrates the percentage of collision-free objects in the scenes. In the "Generated" scenes, the median of non-colliding objects is very high, at 97.4%. Additionally, its upper quartile has a value of 100%, indicating that the objects in these scenes have no collisions with other objects or walls. In contrast, the median of the "*3D-Front*" group is 79%, which is lower than that of the other group. Furthermore, its values show a higher distribution among all quartiles.

The lack of room area as an input in both sub-models, *HouseGan* and *InstructScene*, causes usability issues. A solution is to modify the models to accept area input, though challenging, or to train them on a consistent dataset for coherent results. Dead zones in generated apartments may result from *HouseGan* placing doors inaccurately. As shown in Fig. 4b, generated scenes have a higher percentage of collision-free objects than *3D-Front*, with 25% having no collisions, indicating effective collision avoidance. However, object-object collisions persist, possibly due to furniture density in small areas. Dining rooms generally have larger areas in *3D-Front*, which are a significant source of collisions, suggesting *HouseGan* generates smaller dining room floor plans.

Meanwhile, *InstructScene* produces layouts based on *3D-Front*-sized areas unsuitable for smaller generated spaces. This mismatch arises because *3D-Front* includes larger dining and dining-living rooms. The boxplot 4b shows consistently high object reachability in generated scenes, implying optimized methods for user interaction. Such a tool can generate vast datasets similar to 3D indoor datasets, addressing potential deletions like *SunCG* and offering diverse room and furniture configurations beyond real-life examples.

7 Conclusion

The utilization of the presented tool facilitates the generation of comprehensive 3D indoor scenes, which can be employed as training environments for child protection workers [16]. Therefore, the conversion from case files to VR is enabled. This conversion process resulted in the generation of a 3D apartment model, which was based on a room graph and a text description of the furniture. The tool integrates floor plans generated by *HouseGAN* with furniture layouts from *InstructScene*, leveraging high-quality 3D models from the furniture dataset *3D-Future*. A new evaluation tool was also developed in this paper to assess the usability of 3D indoor scene files by measuring dead zones, out-of-boundary rates, and object reachability. Evaluations were conducted on multiple generated apartments and selected manually designed scenes from the *3D-Front* dataset. The comparison reveals that the generated scenes tend to have a lower rate of out-of-boundary objects and similar object reachability percentages but with

lower object density compared to *3D-Front* scenes. Some issues were detected during the generation process using sub-tools, which affected the final results. Nevertheless, despite density issues, usability evaluations suggest that the tool can be used as a basis for generating an unlimited number of completely customized 3D indoor scenes.

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Extreme Reality Training with Complex Gestures

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Abstract. Extreme Reality (EXR) is to superimpose extreme situations and live sensory data to the physical scene. In this study, we explore gesture capture and interaction technology for emergency response training, which includes extreme gaits such as crawling through the floor and real-time gesture control of the user's movement in the training scenarios. Existing motion tracking systems heavily rely on wearable inertial motion units (IMUs) or head-mounted displays (HMDs), which adds both cost and physical burden to the user. This study examines a new multimodal Extreme Reality interface to support the real-time operation and training without the need for wearable equipment. In this study, we measured gesture detection latency, key point requirements, and accuracy with LiDAR and webcam for complex gesture recognition. Our experiments show that the usability of the Extreme Reality interfaces can be tested in a simulated environment and measured with instruments objectively.

Keywords: Extreme Reality · Training · Natural Gesture · Emergency Response

1 Introduction

Immersive professional field training is a critical component to allow emergency response teams to sense the situation around them. Existing Augmented Reality (AR) interfaces are not built for extreme environments such as smoke, dark, noisy, and poor communication conditions. In extreme cases, such as firefight training, the trainees need to crawl through the floor. We need a new kind of AR platform that can survive extreme environments and provide on-demand information for first responders and command posts.

Figure 1 illustrates the spectrum of reality technologies. From left to right, Virtual Reality simulates the real-world in the digital way. Augmented Reality overlays virtual objects to the real-world. Extended Reality superimposes additional sensory data to the real-world scene such as the live thermal image. Furthermore, Extreme Reality (EXR) superimposes extreme motion or environmental scenarios to the real-world scene, for example, the extreme poses, smoke, and fire.

Many existing game engines use gamepad [3], head-mounted cameras, or IMU (inertial motion unit) to simulate a user's poses [8]. Unfortunately, they do not fully capture the physical motion. The visual and physical effects are not realistic. In this study, we

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explore gesture capture and interaction technology for emergency response training, which includes extreme gaits such as crawling through the floor and real-time gesture control of the user's movement in the training scenarios.

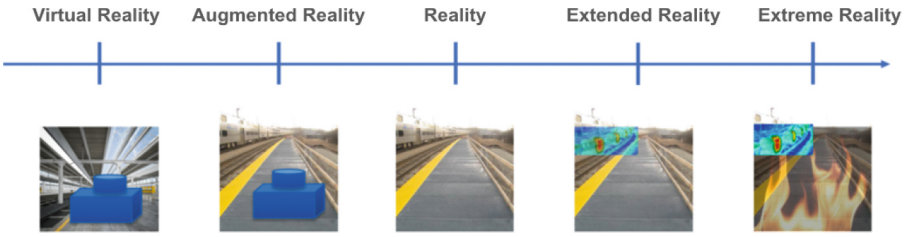

Fig. 1. Spectrum of reality technologies.

2 Pose Detection and Tracking

There are many ways to articulate, detect, and track human poses. Many commercially available goggles contain wearable sensors such as cameras, and IMUs for detecting and tracking user's gesture, mainly hand movement. But they don't track the full body gestures, especially walking and crawling. Wearable IMU sensors can be programmed for pose detection and tracking. However, it is challenging to use IMU sensors for detecting and tracking complex gaits such as crawling and walking backwards.

Webcam-based machine learning algorithms are able to detect and track complex poses with the articulated skeleton graph, including the joints of the limbs and landmarks on the head position and orientation, e.g. eye position. The computational load for the machine-learning algorithm is rather high, especially for laptops. The near-infrared-camera-based sensor (lidar), on the other hand, can deliver the similar articulated skeletal tracking results with relatively lower computation because the lidar sensor fuses the data and processes the data onboard. A typical lidar sensor contains two NIR cameras and distance measurement that enables 3D pose acquisition. Figure 2 shows articulated pose tracking with the IMU (left) on the shoes and the pose skeleton generated from the lidar camera (right).

IMUs and thermal cameras can track the pose without lighting. The pose detection and tracking with these two types of sensors are more difficult because of low resolution sensory data. Table 1 summarizes the comparisons among sensors versus computing load, lighting requirement, pose detection and tracking fidelity, and the cost.

Table 1. Comparisons among sensors.

Sensors	Computing	Lighting	Fidelity	Cost
Goggle	Complex	Required	Hand-Gesture	High
IMU	Simple	Not Required	Gaits	Low
Thermal	Moderate	Not Required	Gaits	High
Lidar	Moderate	Required	Gaits	Moderate
Webcam	Complex	Required	Gaits	Low
Phone	Complex	Required	Gaits	Low

Fig. 2. Articulated poses with the inertial motion unit (left) and lidar camera (right).

3 Body Gesture Detection

For common body motions, we test out the pose detection algorithm's ability to recognize actions such as walking forward, walking backwards, head rotation, which is mapped to turning left or right. To classify walking versus standing still, we extract keypoints of both ankles and analyze their displacement along the y-axis. When using a standard webcam, this method can only infer that the user is moving in a single direction, typically forward, due to the inherent limitation of 2D pixel-based input. In contrast, a lidar is essential for distinguishing between forward and backward motion, as it captures depth information, e.g. z-axis, relative to the sensor. By analyzing the displacement of ankle keypoints along the z-axis, we can determine the direction of movement.

In cases where only 2D image input is available and depth cannot be directly measured, the change in the proportion of the distance between the left and right shoulder keypoints can still provide cues about forward or backward motion. However, analyzing

changes in limb lengths is less effective, as arm and leg lengths tend to fluctuate during motion. For example, while walking, the arms swing and the legs move continuously, making their lengths unreliable indicators for inferring depth or direction of movement.

Head rotation detection is more complex, as it requires at least three facial keypoints to estimate orientation based on the head's pan angle. Figure 3 illustrates the spatial relationships between the selected keypoints and the method used to compute the orientation angle. On the left side of the figure, three selected key points—the left eye, right eye, and nose—are marked in blue boxes. A purple dot, representing the midpoint between the two eyes, is used to define a projection line toward the nose. On the right side, the projection line "a" (in purple) connects the midpoint "c" to the nose on the vertical axis, while line "b" (in blue) represents the segment between the midpoint "c" and the nose. The angle formed between lines "a" and "b" corresponds to the head's pan angle. Figure 4 shows the resulting detection of this orientation angle. Note that the raw pan angle calculated from Fig. 3 was normalized by a scaling factor to better reflect real-world values, and when the subject is facing directly toward the sensor, the pan angle is set to be zero degree.

Table 2 summarizes the comparisons among methods versus sensors used for pose classification and the number of data keypoint requirements.

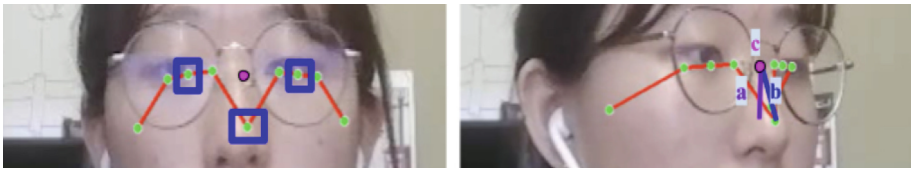

Fig. 3. Head pan angle calculation including selected keypoint (left) and visualization (right).

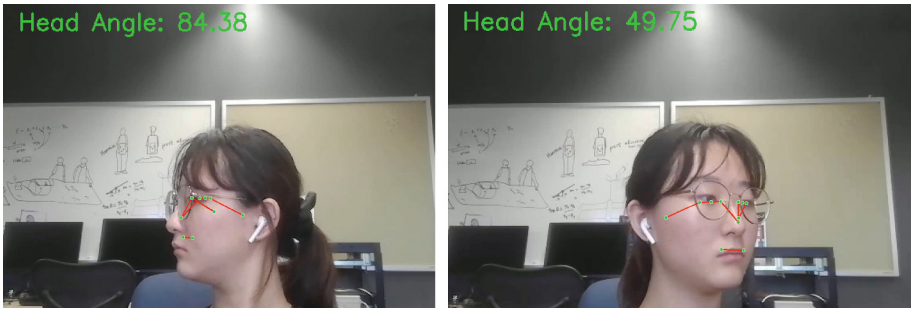

Fig. 4. Detection of head turning angle using facial keypoint.

Table 2. Comparisons among classification methods.

Gesture	Sensor	Keypoint Requirement
Walking Forward	Webcam	2
Walking Forward	Lidar	2
Walking Backward	Webcam	4
Walking Backward	Lidar	2
Head Rotation	Webcam	3
Head Rotation	Lidar	3

4 Detection Threshold on 2D Pixel-Based Input

Accurate pose detection relies heavily on the use of appropriate thresholds. For example, distinguishing between walking forward and backward depends on the displacement of the ankle key points along the *y*-axis. When working with 2D input, reliable skeleton detection requires consistent lighting conditions to ensure sufficient contrast between the human body and the background. However, lighting is often not ideal in real-world settings. In these cases, applying a threshold is essential to account for fluctuations and improve the robustness of the detection.

The algorithm detects walking when the current *y*-axis value of the ankle keypoint exceeds the previous value by a specified threshold factor. In distinguishing walking from standing still, the threshold factor must be set to at least 0.1% under typical lighting conditions, such as natural morning sunlight, or indoors with uniform illumination. However, this threshold is also influenced by the subject's distance from the sensor. As the distance increases, the skeleton appears smaller within the frame, reducing the visible displacement of keypoints; therefore, a lower threshold is required. Conversely, when the person is closer to the sensor, the threshold can be increased to around 0.2%. Raising the threshold in such cases also helps filter out noise and reduce false detections.

Walking backward specifically involves another factor: the change in shoulder length. In our experiment settings, the threshold for detecting backward motion is typically around -1.5 (difference), calculated as the current shoulder length minus previous shoulder length. This value represents a balance point—if the threshold is set too low, the detection becomes overly sensitive to minor changes; if the threshold is set higher, the system may fail to detect backward movement, especially when the person is moving at a slower pace.

Head rotation detection, based on the method described in Sect. 3, is highly sensitive without a threshold. To address this, we applied a threshold of approximately 30°, below which head movement is not interpreted as a directional turn. For example, in an immersive simulation, if the user's head rotation remains within 30°, the system classifies the user as facing forward. Rotations exceeding this threshold are interpreted as turning either left or right.

It is important to note that the threshold values discussed above are based on our specific experimental settings and may vary depending on the equipment and environmental

conditions. Nonetheless, these results illustrate the general principle of how threshold selection impacts the accuracy of gesture detection.

5 Validation and Statistic

We focused on the *detection latency* as an indicator of whether our gestures detection algorithm is suitable to operate in applications which require real-time response. The latency is defined as the time frame from the image being captured, overlaying skeleton key-points on the target's body, to the decision of the gesture classification. Here we consider four types of gestures and their corresponding detection latency and algorithm accuracy when using lidar or webcam: *standing still, walking forward, walking backwards*, and *head rotation angle* in horizontal direction representing the direction the user is facing.

The experiment was conducted indoors under natural lighting conditions provided by sunlight through the window. To set up the experiment, the user was standing at least 1.5 m away from the sensor to ensure the sensor captured the whole body. During the experiment, the devices operated on battery power rather than being connected to a power outlet. The algorithm works under circumstances with dynamic gesture change, therefore, the experiment was taken in real time and the user moved their body to perform four movements each for around 10 s. That is—*walking forward and backward, standing still, and head turning*. For a lidar sensor, within a 7 s timeframe the sensor typically could capture and output up to 110 detection results. Using lidar and webcam, we collected two sets of data and summed up every latency data value characterized by different gestures in each set, and took the average value as the resulting latency. The formula used is shown in Eq. (1). To evaluate accuracy, we labeled gestures "*standing still*", "*walking forward*", and "*walking backward*" as separate classes. For each class c, we recorded the number of true positives (number of times class c was correctly predicted) and the number of false negatives (number of times class c was missed). The resulting accuracy was then calculated using the formula provided in Eq. (2). The experimental data is shown in Table 3.

$$L = \frac{\sum_{k=1}^{n} (latency_k)}{n} \tag{1}$$

$$A_c = \frac{TP_c}{TP_c + FN_c} \tag{2}$$

where L is the overall latency; A_c is detection accuracy by class; TP_c is the number of times class c was correctly predicted; FN_c is the number of times class c was missed.

In Table 3, when using a lidar camera the latency can be shortened at least 4 times then using a webcam for detection. The average human reaction time to a visual stimulus is approximately 180–200 ms; therefore, the latency introduced by the lidar camera can be considered acceptable. The accuracy can be affected by several reasons. The most common issue is lighting, which may affect stability of the skeleton overlay. Accuracy may also vary due to differences in walking habits; for example, distinguishing walking forward or backward becomes more difficult if the user's body naturally swings back and

forth. In addition, detection smoothness/latency can further affect accuracy. As shown in the "algorithm accuracy" column, "standing still" and "walking forward" accuracy dropped by ~10% when we switched from lidar camera to webcam. In particular, when detecting "walking backward" using a webcam, the accuracy dropped to 42.53%. This was due to the detection lag, which made it difficult to distinguish changes in shoulder length associated with walking forward versus walking backward. However, the computational power of the device (i.e. the laptop which was used to run the program) must also be considered, as it directly affects system performance. For instance, we later observed that when the device was plugged in, the webcam (which was from the laptop) side had latency reduced to approximately 150 ms. Additionally, by using a more lightweight model for human skeleton detection could further improve latency.

Table 3. Algorithmic Detection Latency and Accuracy Based on Gesture.

Gesture	Sensor	Avg. Latency (ms)	Algorithm Accuracy
Standing Still	Lidar	60.0	96.67%
Walking Forward	Lidar	60.7	93.33%
Walking Backward	Lidar	61.2	72.41%
Head Rotation	Lidar	60.4	_____*
Standing Still	Webcam	274.3	85.71%
Walking Forward	Webcam	291.1	81.94%
Walking Backward	Webcam	295.4	42.53%
Head Rotation	Webcam	301.8	_____*

* *Note: The head rotation is detected by relative value (i.e. how much the user is turning with respect to the "front" of their face reflects on the rotation value), therefore an absolute accuracy metric is not directly applicable in this case.*

6 Capturing Extreme Gestures

We set up a lidar-based experiment system on the tripod to study extreme pose tracking. We focused on crawling, kneeling, standing, walking, running, body turning, head turning, walking backwards, arm reaching out, and arms up. Figure 5 shows the poses of crawling, kneeling, and standing. Figure 6 shows the overlaid tracking skeletal data on the body. The preliminary experiments showed that the lidar sensor was able to track the extreme pose, crawling and kneeling. However, the results were not always perfect. For example, the tracking result of the *F* from the crawling pose *B* missed the right leg's key point.

In contrast, we conducted a parallel experiment using a webcam and replicated the four poses shown in Fig. 5. As expected, webcam-based detection is less stable due to the lack of depth sensing capabilities. For pose *A* and *B* on Fig. 5, the webcam-based detection shown in Fig. 7 fluctuated significantly; the skeletal structure became highly unstable when the subject was lying face down, as illustrated in *I* and *J*. In these crawling

poses, key points corresponding to some of the limb segments were missing, and the head was not detected in either case. By comparison, the lidar-based results in Fig. 6 (images *E* and *F*) still show consistent detection of facial keypoints, even when the subject was lying down. In kneeling pose *K* (Fig. 7), the lower body was poorly detected, with the skeleton appearing tangled and inaccurate. Among the four poses tested, the only case where the webcam provided correct detection was the standing pose shown in *L*, in which the subject was fully upright and directly facing the sensor. Our experiments with lidar and webcam systems show that lidar is more effective at capturing extreme gestures.

Fig. 5. Examples of the poses: crawling (*A* and *B*), kneeling (*C*), and Standing (*D*).

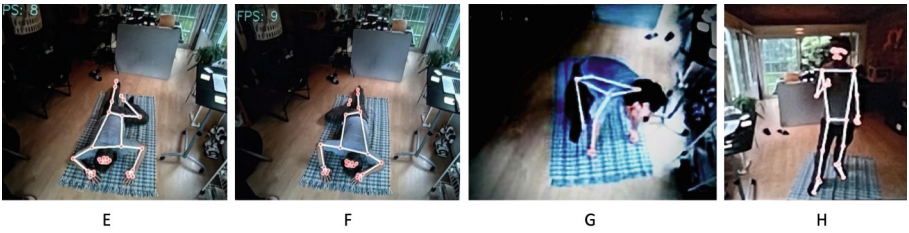

Fig. 6. Pose capture and overlay results: crawling (*E* and *F*), kneeling (*G*), and standing (*H*).

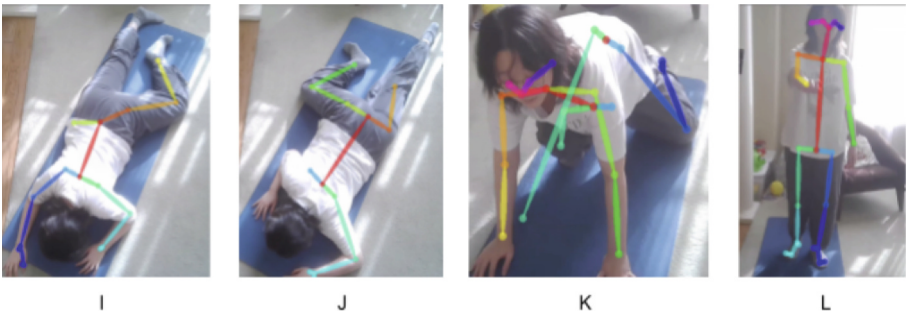

Fig. 7. Webcam pose capture results: crawling (*I* and *J*), kneeling (*K*), and standing (*L*).

7 Connecting to the Training Scenarios

We streamed the pose tracking data to our game engine in real-time on a laptop, between two laptops, or between an embedded system and a laptop. A lidar sensor was connected to a laptop with the training game. We were able to perform the pose detection and tracking actions such as running, standing, picking up an object, turning, and pressing buttons in real-time. A short-throw projector was used to display the training scene. Overall, the user was able to interact with the training system without wearing an immersive goggle, nor IMU sensors. Figure 8 shows the extreme reality demonstration for medical helicopter launchpad training at the 5 × 5 Summit in Chicago in 2024. A preflight check training scenario is presented in this figure. The participant (on the left) walked toward the obstacles and picked them up on a helicopter launchpad, which the environment was simulated on a 3D game engine and projected on the wall (shown on the right). Our skeletal gesture detection algorithms achieved user motion control without physical controllers to navigate within an immersive training environment.

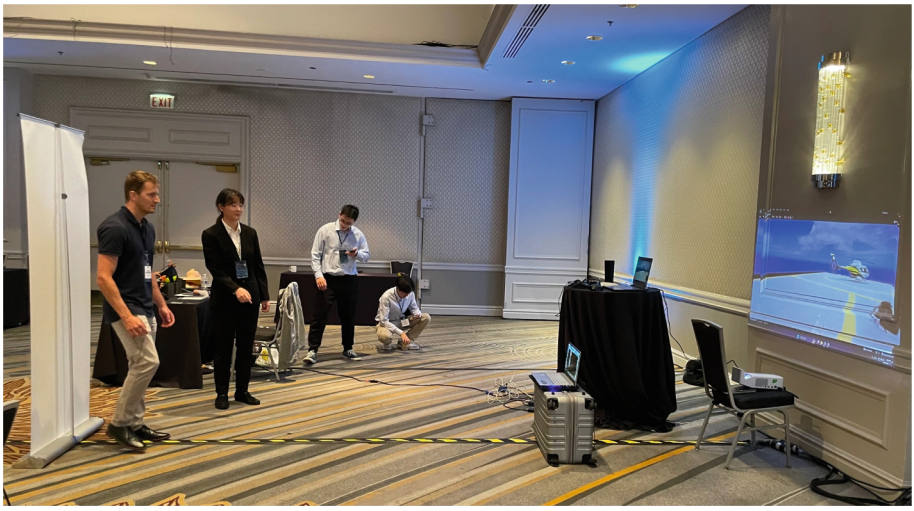

Fig. 8. Extreme Reality (EXR) for medical helicopter launchpad training.

8 Conclusions

Extreme Reality (EXR) is to superimpose extreme situations and live sensory data to the physical scene. Our multimodal EXR interfaces show their potential in real-time operation and training with complex gesture input, including crawling, kneeling, walking, running, walking backward, and head turning, etc. This study examines a new multimodal Extreme Reality interface to support the real-time operation and training without the need for wearable equipment. In this study, we measured gesture detection latency, key point requirements, and accuracy with LiDAR and webcam for complex gesture recognition. Our experiments show that the usability of the Extreme Reality interfaces can be tested in a simulated environment and measured with instruments objectively.

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Improving Communication in the Metaverse Using 3-Dimensional Space for LLM and Autocomplete Integration

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Abstract. This research introduces a novel XR/VR keyboard prototype, coupled with a large language model (LLM) and n-gram models for sentence prediction. While traditional VR keyboards have been ported from phone interfaces, we demonstrate a design that can fully use the 3D space and blend LLM and new AI technologies into traditional designs. The proposed keyboard is a merger of conventional interfaces, but allows for the development of AI suggestions in additional 3D space, setting the stage for future development in AI-driven interactions within virtual spaces.

Keywords: LLM · Input Devices · VR

1 Introduction

Human-Computer Interaction (HCI) research continues to seek innovative solutions to enhance user experiences in digital environments. Recent VR devices have gained popularity within and outside the research community. With the wider adoption of these technologies, researchers must re-evaluate conventional input methods. This paper introduces a novel approach to text input in VR/XR environments through the development of a keyboard designed with Unity with the goal of fully using 3D space for augmentation with Large Language Models (LLMs). The research shows our prototype of a VR keyboard using an LLM for sentence prediction in three dimensions.

The emergence of the metaverse has intensified the challenge of creating intuitive input devices. Devices have used traditional virtual keyboards, wireless physical keyboards, finger tracking, and gesture-based systems that aim to integrate with the immersive qualities of virtual worlds. Current solutions often derive from adaptations of traditional interfaces designed for mobile devices or desktops, raising concerns about their suitability for virtual reality environments. The discrepancy between conventional input methods and the demands of VR scenarios indicates a new challenge for developers.

The transition from physical to virtual spaces demands a reimagining of the relationship between users and their digital tools, such as AI. This research addresses these challenges by presenting a VR keyboard that allows for the integration of LLM and future AI contextual information in the 3rd dimension.

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The core innovation of this research lies in its departure from mobile phone-specific keyboard designs and the subsequent creation of a VR-centric input system. VR often can veer into the fantastical, showing worlds and scenarios that cannot happen in the real world. In addition, many have focused VR on recreating the real world to offer experiences, training, and simulations to a wider audience [14,16]. Our keyboard aims to cross the two possibilities. The proposed keyboard is grounded in the traditional physical QWERTY keyboard, while departing from the limited linear display of a phone QWERTY keyboard when it comes to auto-completion by using the full space allowed by VR's 3-dimensions. By incorporating advanced Artificial Intelligence (AI) for sentence prediction, this offers users an intuitive and efficient means of communication in virtual spaces.

The rapid advancement of Large Language Models (LLMs) has ushered in a transformative era in artificial intelligence and human-computer interaction. LLMs, such as GPT, Llama, and BARD, have demonstrated impressive language generation capabilities [11], shaping the future of many possible fields. This surge in the adoption of LLM technology has been fueled by breakthroughs in deep learning that allow these models to comprehend and generate human-like text on an unprecedented scale. As LLMs evolve, their integration into various applications becomes increasingly prevalent, offering promising avenues for enhanced user experiences. In particular, the fusion of LLMs with Virtual Reality (VR) interfaces holds immense potential.

Despite the remarkable capabilities of LLMs, their effective integration into user interfaces presents many challenges. Current user interfaces are often ill-equipped to harness the full potential of LLMs, leading to suboptimal user experiences. The intricate nature of LLM-generated content demands interfaces that not only accommodate the intricacies of language, but also provide users with intuitive means of consumption. Traditional interfaces, designed primarily for static text input and output, struggle to adapt to the dynamic and context-aware nature of LLM-generated content. This mismatch poses a barrier to the seamless use of LLMs in various applications, hindering the realization of their full potential. Addressing this gap becomes imperative for HCI researchers and practitioners, prompting innovative solutions that bridge the divide between LLM technology and user interfaces. This research aims to create an integration that enhances user engagement, interaction, while making the user aware of the AI capabilities and contributions.

2 Background

Keyboards have undergone significant transformations, adapting to diverse platforms, virtualization, and augmentation with statistical models and new characters [12,13,20]. With lower-cost hardware for the metaverse and the rapid adoption of AI models [17], these input devices must once again adapt to VR spaces in addition to integrating new AI technologies.

The rise of predictive text algorithms on mobile devices, such as T9 and later autocorrect, showcased early attempts to improve text input efficiency. With the trend at the time of smaller and smaller devices, a lot of research was on minimizing the space needed [4].

The integration of VR into HCI has posed new challenges for keyboard design, as the physical space has increased to any scale desired by the user. Traditional designs proved inadequate for immersive three-dimensional spaces in VR. Research on 3D interfaces emphasized the need for novel input methods, setting the stage for the development of interfaces tailored to virtual environments [1]. Often, this bridging of the physical and virtual has been accomplished through additional hardware, such as the development of augmented reality keyboards using a physical glove and eyepiece [7].

There have been many innovative designs in VR from [19] who showed a new way to use spatial sense in VR to type via "punching" as opposed to traditional laser pointers. VR interactions using laser pointers and controllers instead of hand tracking may lead users to use a dominant hand for all typing via laser-based typing. [21] studied the different designs and factors behind single-handed typing and found that differing designs to a traditional mobile keyboard were preferable. Often a goal in VR is to translate the physical world more closely. [8] explored the ways to show a traditional physical keyboard in VR. There are open source projects such as the Keyboard Augmentation Toolkit, which supports the creation of virtual keyboards that can be used both for standalone input and to augment physically tracked keyboards/surfaces in mixed reality [10].

As we delve into the development of a VR keyboard with LLM integration, insights from the design principles of keyboards for small and large spaces provide valuable perspectives. Balancing the intricacies of text input in VR requires a synthesis of compact efficiency and ergonomic considerations [15], making this endeavor a convergence of lessons learned from diverse spatial contexts.

3 Design and Implementation

The engine used to build this prototype was done in Unity, which is freely accessible with some restrictions for commercial use. Unity has been shown to be a valuable research platform for both state-of-the-art research [3,6] and virtual reality projects [5,9].

Our intent is to release a keyboard that is fully customizable and is provided as open-source to the community on GitHub. While the keyboard presented is bare-bones graphically, all textures, fonts, sizes, tilts, and angles are parameterized for future customization. While this design will focus on a traditional QWERTY style keyboard similar to the default Oculus keyboard, the layout and key widths are dynamically created from JSON files, allowing designers to make custom keyboards quickly for other HCI tests and user requested capabilities.

3.1 Back-End

The core of this research is how to extend the additional dimension in VR to accommodate AI add-ons. Our prototype currently demonstrates two textual add-ons for the third dimension, n-gram models, and an LLM service.

Autocomplete. The first one is a traditional n-gram models. The model generates the most likely word given the current text as seen in Fig. 1. This will generate multiple suggestions based on the keystrokes, but will only suggest one word at a time. Given the simplicity of the n-gram model, it is both trained and runs directly on the VR hardware at runtime.

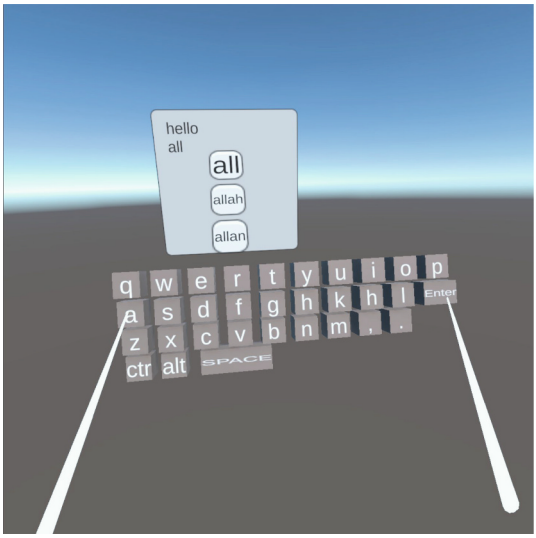

Fig. 1. Results of a simple n-gram based autocomplete model.

Large Language Models. Given the explosion of LLM research, we also show the integration of an LLM for full sentence completion. Given the large amount of visual space available in VR, this is an ideal platform to experiment with new LLM layouts. For the prototype, we integrated Meta's Llama Chat 2 LLM [18]. Given the size of memory needed to run these models effectively and the limited hardware available in VR, we set up a cloud service to provide responses to the current text using one of the current AWS releases [2]. This is a limitation of today's VR gear, but in the future as new AI techniques release smaller models and/or VR units gain greater GPUs and VRAM, this additional step might not be needed. Removing this cloud infrastructure could reduce response time.

To accomplish a sentence completion with off-the-shelf models, our lambda function prefixes the prompt to the LLM with variations of "Give me X amount of sentences that can complete the following text." Here, X is a parameter set by the keyboard design. Additionally, we add the requirement that the LLM return the results in a JSON format with our provided keys. Any additional conversation added to the response is filtered out with regular expressions. The response is sent back to the software and the response is parsed and displayed to the user, as seen in Fig. 2.

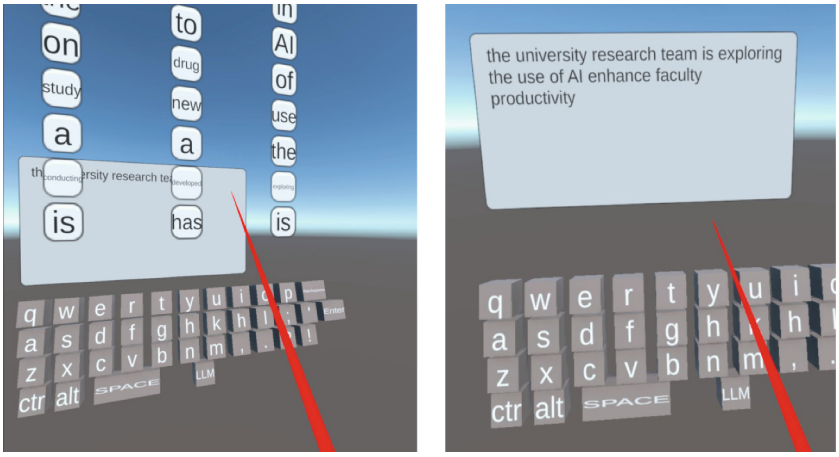

Fig. 2. Left: Results of the LLM button action on the current sentence. Right: Upon the user selecting a "block" from the LLM results, the text is appended, and the user may continue on typing. In this case, the user selected the 3rd LLM option and added enhance faculty productivity.

3.2 Front-End

The front-end display currently uses default meshes but is fully customizable. For ease of use, it is implemented in this version with a slight tilt to the keys. Haptic feedback is provided when the user hovers over as key and on a click but at different levels of vibration. The keys have a push feel to them. When you click a key, it presses down much the same way as a mechanical keyboard. For highlighting, in this prototype we made the decision to have a slight inward movement on hover instead of a highlight on the key. However, both options are available and customizable. The default keyboard look can be seen in Fig. 2.

Any textual augmentation, such as autocomplete or LLM completion, comes off the top of the keyboard out into the VR space. Currently, the results expand out of the keyboard and not the text input, this may be modified as well. This is less of a limitation than Fig. 2 shows, since in VR the user can see around the text augmentation, since it is rendered fully in 3D. Future iterations of this will include eye gaze and transparency options.

For the LLM options in Fig. 2, the user can select the whole sentence by clicking the top word or may select a fragment by clicking a word in the middle of the sequence. Once clicked, the sentence or fragment is appended to the input and the user may continue typing or prompt for another LLM result. The keyboard by default has an LLM button to trigger this event, however, this LLM trigger can happen at any point such as when the user hits 'space', or 'enter'. From internal tests, we found that triggering an LLM response after every space slowed the user down in WPM due to the current response time of the LLM service.

4 Conclusion

We have presented an early prototype of a VR/XR keyboard designed to use 3-dimensional space for the integration of AI textual assistance such as an LLM or autocomplete model. The research project is being made available for future collaborations with the HCI community and will have a follow-up study with a traditional user study. With generative AI firmly in the research community, we aim for this to be a research collaboration tool centered around VR input and AI integration.

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Improving Human-Robot Communication in Noisy Environments with Visual Voice Activity Detection

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Abstract. This paper investigates the integration of Visual Voice Activity Detection (VVAD) into human-robot dialogue systems to enhance communication in noisy environments. Usually, speech recognition systems often falter under acoustic interference, limiting their effectiveness in real-world human-robot interactions. By leveraging visual cues, especially lip movements, VVAD supports more accurate speech detection and turn-taking.

In this paper, we present a multimodal dialogue system combining VVAD with speech recognition, dialogue management, and context-aware intention recognition. The system is evaluated through a user study involving 30 participants in a noise-rich, simulated restaurant scenario. Results show a substantial improvement in task completion rates and user satisfaction when VVAD is enabled, alongside with a significant reduction in false activations. These findings underscore the value of visual input in robust, socially aware human-robot interaction and suggest VVAD as a critical component in the design of next-generation interactive robots.

Keywords: Human-Robot Interaction · Visual Voice Activity Detection · Multimodal Dialogue · Speech Recognition

1 Introduction

Robots increasingly enter human-centered environments, from smart homes and public service venues to assistive care facilities. This implies the need for natural, seamless, and socially acceptable interaction modalities as central aspects in Human-Robot Interaction (HRI) research [2]. Recent surveys confirm that multimodal approaches—integrating voice, vision, gaze, and physiological signals—are becoming essential for robust and socially aligned HRI [6,19]. An important communication channel for humans is natural language, which provides an intuitive and user-friendly interface to robotic systems. For this, robotic systems need not only to be able to understand the human words and generate appropriate responses but it is also essential for

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systems to distinguish between speakers who are addressing them directly and those who are speaking in the environment but not to the robot. This is especially challenging in noisy environments, with different speakers. Additionally, a natural communication with a robotic system requires fluent dialogues, which means the data processing and robotic speech generation has to be fast to avoid unnaturally long response times.

In order to enable natural and efficient speech interaction, robotic systems need to be equipped with models to understand natural language, manage dialogues, and generate responses. With current developments in deep learning and generative transformer models, significant progress has been made in this area in the last years. State-of-the-art Automatic Speech Recognition (ASR) models, such as Whisper [13], leverage deep neural architectures to provide high transcription accuracy even in noisy or multilingual contexts. In parallel, frameworks such as Rasa [4] enable flexible and context-aware dialogue management, supporting dynamic and mixed-initiative conversations. Text-to-speech engines have also evolved to produce highly natural and expressive speech, enhancing the social presence of robotic agents [2].

Despite these advancements, several challenges persist. Most speech-based systems remain vulnerable to background noise and rely heavily on cloud infrastructure, limiting their applicability in latency-sensitive or privacy-critical scenarios [2]. Moreover, the absence of visual grounding in turn-taking decisions can lead to unnatural pauses, interruptions, or conversational breakdowns [15]. Recent reviews emphasize that multimodal cues such as prosody, gaze, and gestures are critical for achieving fluid turn-taking in HRI [1], while user studies demonstrate that multimodal interfaces enhance accessibility and engagement, especially for older adults [16].

To address these challenges, additional modalities beside auditorial channels gain in importance. By incorporating modalities such as visual features that allow to differentiate between speakers and non-speakers, challenging scenarios with multiple speakers and background noise can be addressed. Thus, Visual Voice Activity Detection (VVAD) has recently gained traction as a promising enhancement to auditory-only speech processing [15]. By integrating visual cues, such as lip movement, VVAD systems offer complementary information that can help disambiguate speech events, detect speaker turn-taking, and improve the overall robustness of voice-based HRI systems [11]. This multimodal augmentation aligns with a broader shift in HRI toward sensory fusion, where audio, visual, and contextual data are combined to create richer interaction experiences [19]. Although VVAD offers a compelling pathway to address challenges in human-robot speech interaction, its integration into HRI systems remains underexplored. Current models are rarely embedded in real-time dialogue pipelines, and empirical evaluations of their impact on user experience remain scarce. Additionally, VVAD systems face deployment challenges due to sensitivity to visual conditions and high computational demands. [10].

This paper investigates the integration of VVAD into a human-robot dialogue system that reacts in real-time to natural human language. These multimodal inputs are analyzed by the CoBaIR (Context-Based Intention Recognition) module, which integrates recent auditory and visual cues to infer the system's communicative intent in real-time. We propose a modular architecture, which is shown in Fig. 1, that fuses auditory and visual cues for voice activity detection and leverages them within a dialogue

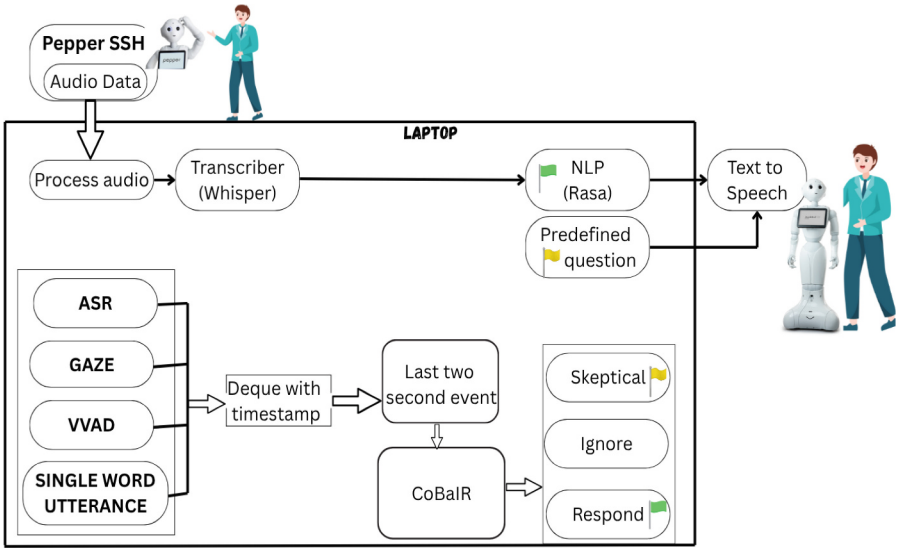


Fig. 1. Overview of the multimodal dialogue system architecture. Audio data from the Pepper robot [18] is transcribed using Whisper [13], then processed through Rasa [4] for natural language understanding and response generation. In parallel, multimodal cues—including ASR, gaze, VVAD [9], and single-word detection—are timestamped and analyzed by CoBaIR [8] to determine system intent (respond, ignore, or act skeptical). Final output is delivered via text-to-speech. Abbreviations: ASR – Automatic Speech Recognition; VVAD – Visual Voice Activity Detection; NLP – Natural Language Processing; CoBaIR – Context-Based Intention Recognition.

pipeline. Our system is evaluated in a simulated restaurant scenario designed to test the system's performance in noisy, multi-speaker environments. We focus on three core aspects: (1) the improvement of turn-taking accuracy via visual cue integration, (2) the enhancement of dialogue flow and user experience, and (3) the system's responsiveness under varying environmental conditions.

The key contributions of this paper are:

- We present a multimodal dialogue system architecture that incorporates VVAD into traditional ASR pipelines for enhanced interaction timing.
- We develop and benchmark a real-time HRI scenario where VVAD directly informs dialogue state transitions and turn-taking strategies.
- We analyze the system's performance compared to audio-only baselines in a user-study, demonstrating improvements in timing precision and user satisfaction.

The remainder of the paper is structured as follows: Sect. 2 reviews relevant literature in VVAD and multimodal HRI systems. In Sect. 3 the system architecture is described in detail and in Sect. 4 the user-study and data collection methods are introduced. Section 5 presents both quantitative and qualitative evaluation outcomes and discusses the findings and their implications. Section 6 limitations of the proposed approach are presented. We conclude with future directions in Sect. 7.

2 Related Work

HRI has progressed towards enabling more natural and context-sensitive communication through multimodal systems. One emerging area in this domain is the integration of VVAD, which supplements auditory speech processing with visual input to improve robustness in real-world environments.

2.1 Multimodal Interaction in HRI

Multimodal HRI systems leverage multiple input channels audio, visual, and occasionally tactile or physiological signals to create a communication framework that is both robust and intuitive. By combining modalities, these systems provide redundancy, allowing one channel to compensate when another is impaired, such as when audio is drowned by noise or vision is obscured [19]. Effective multimodal systems depend on core technical components including precise sensor synchronization, high-quality data capture, and advanced fusion strategies. Common fusion approaches range from feature level concatenation to decision-level ensemble methods, often implemented using deep learning architectures like Convolutional Neural Network (CNNs) and attention-based models.

VVAD naturally complements these frameworks by fusing visual lip movement detection with traditional audio-based speech detection, offering synchronized cues that enhance accuracy [11]. While gesture and emotion recognition have seen broader adoption in HRI, visual speech detection remains underutilized. Yet, integrating VVAD can significantly improve interaction robustness in uncontrolled environments—particularly in conditions with high ambient noise that challenge audio-based systems. While VVAD remains sensitive to visual constraints such as lighting variability or occlusions, combining it with audio inputs within a multimodal fusion framework allows the system to dynamically rely on the stronger modality. This complementary integration represents a valuable step toward more resilient, natural, and human-aligned HRI systems [11, 19].

2.2 Visual Voice Activity Detection (VVAD)

Modern VVAD systems process facial video to detect speech via visual cues, sidestepping limitations of audio-only systems in noisy or speech-absent environments [11]. For instance, in scenarios where a speaker is whispering, muted, or distant from the microphone, visual articulatory movements (e.g., lip motion) can still be detected. Training typically relies on datasets such as *WildVVAD*, which was created by automatically combining audio-VAD, face detection, and tracking to annotate "speaking" vs. "silent" video segments. This dataset enabled the training of two specialized deep learning models: a facial landmark CNN and an optical flow CNN, both of which demonstrated notable improvements in accuracy over audio-only VAD baselines [5].

More recently, the *VVAD-LRS3* dataset extended this approach by deriving 44,000 annotated samples from the LRS3 lip-reading corpus, achieving up to 92% frame-level accuracy using CNN–LSTM models [9]. Typical training pipelines begin with face detection and alignment, followed by cropping a standardized 112×112px region

around the lips. Optical flow variants additionally compute motion vectors between video frames [5]. Extracted spatial features are then passed through convolutional neural networks (2D or 3D, e.g., ResNet, I3D), whose outputs are processed by temporal models—typically LSTM or BLSTM layers—to capture speech dynamics [9]. Synchronization across varying speech rates is often achieved using Dynamic Time Warping (DTW), a common technique for temporal sequence alignment [5]. Recent methods also integrate self-attention mechanisms, such as the visual-attention pooling layer introduced by Prajwal et al. [12], trained on sub-word units to improve alignment and reduce Word Error Rate (WER) to 22.6% on LRS2. These advances demonstrate that Vision Transformer (ViT)-like architectures and attention-based fusion strategies significantly enhance the performance of visual speech detection systems.

2.3 Dialogue Systems and Speech Interfaces

Despite these advances, most dialogue systems remain predominantly audio-centric and text-reliant, often ignoring other rich communication modalities. Visual information such as lip movements or facial engagement—is rarely utilized for speech detection or conversational state estimation, despite growing evidence that multimodal cues can substantially improve interaction robustness. For example, recent surveys highlight the role of diverse modalities including voice, image, text, gaze, and bio signals in advancing multimodal HRI [6, 19]. In addition, turn-taking studies emphasize that prosody, gaze, and gestures are critical to achieving fluid and natural conversations between humans and robots [1].

User-centered evaluations further show that multimodal interfaces enhance accessibility and engagement, particularly for older adults interacting with robots [16]. Integrating VVAD into dialogue architectures thus represents a promising pathway, adding a layer of visual grounding to speech detection and conversational management. However, such integration remains largely absent in existing HRI implementations, representing a critical opportunity for further exploration and development.

2.4 User Studies in VVAD-Enhanced HRI

While most research in VVAD focuses on technical benchmarks and dataset development [5, 9], a growing body of work has begun to evaluate these systems through human centered user studies. Such evaluations are essential to understand the impact of VVAD on real-world interaction quality and user acceptance in HRI.

Shahid et al. [15] introduced S-VVAD, a motion-based VVAD method that does not rely on facial landmarks, making it suitable for uncontrolled, multi-speaker environments. Although their evaluation was technical, their method demonstrated robustness in dynamic contexts supporting its applicability in user-facing systems. Similarly, Lubitz et al. [9] evaluated their VVAD-LRS3 model, trained on over 44,000 annotated video samples, achieving frame-level detection accuracy that surpassed human performance. While their evaluation was not embedded in an HRI scenario, the results underline the maturity of visual speech detection for real-world application.

Complementary insights come from user studies in multimodal HRI. Schreiter et al. [14] compared verbal-only interaction with multimodal (speech, gestures, and

gaze) systems in two controlled experiments. Their results showed that participants responded significantly faster and reported higher engagement when multimodal cues were available. Likewise, Shervedani et al. [16] conducted a user study with older adults using a Multimodal Interaction Manager, demonstrating that combining modalities improved usability, trust, and perceived safety of the robot assistant.

Collectively, these works indicate a growing interest in evaluating VVAD through human-centered user studies. However, the field would benefit from broader adoption of empirical evaluations embedded in naturalistic HRI tasks, particularly to validate generalization across different populations, environments, and task types.

2.5 Audio-Only vs VVAD-Enhanced Systems

Traditional audio-only speech recognition systems rely solely on auditory signals to interpret spoken language. While effective in quiet, controlled environments, these systems experience a significant degradation in performance when exposed to real-world conditions such as overlapping conversations, ambient noise, or acoustic interference [10]. Such conditions often lead to communication breakdowns, increased user frustration, and reduced trust in robotic systems. Audio-only models also struggle with speaker variability, including differences in accent, speech pace, and homophones, further impacting recognition accuracy [10].

In contrast, VVAD-enhanced systems incorporate visual cues such as lip movements and facial expressions alongside auditory input to improve robustness in noisy environments [11]. This multimodal approach allows the system to better isolate and interpret user speech, even in scenarios where background noise renders the audio signal unreliable. Studies have shown that combining visual and auditory modalities leads to higher task success rates and reduced false activations, particularly in multi-speaker and acoustically dynamic environments [11].

3 Approach

3.1 Design of the Human–Robot Dialogue System

This work proposes a multimodal dialogue system designed to enable socially intelligent and context-aware interactions between humans and robots. The architecture integrates speech processing, visual activity detection, dialogue management, and context-based intention inference into a unified decision-making pipeline.

System Architecture Overview. Figure 1 illustrates the high-level architectural pipeline of the implemented system on the Pepper robot. The system orchestrates multiple sensing and reasoning modules, emphasizing multimodal input fusion and robust response generation.

Audio is captured via Pepper's embedded microphone array and transmitted over a TCP-based SSH tunnel to an external processing unit. In parallel, visual input from Pepper's RGB camera is used for VVAD. The audio stream is preprocessed and then passed to the Whisper model, a robust open-source ASR system designed for noisy

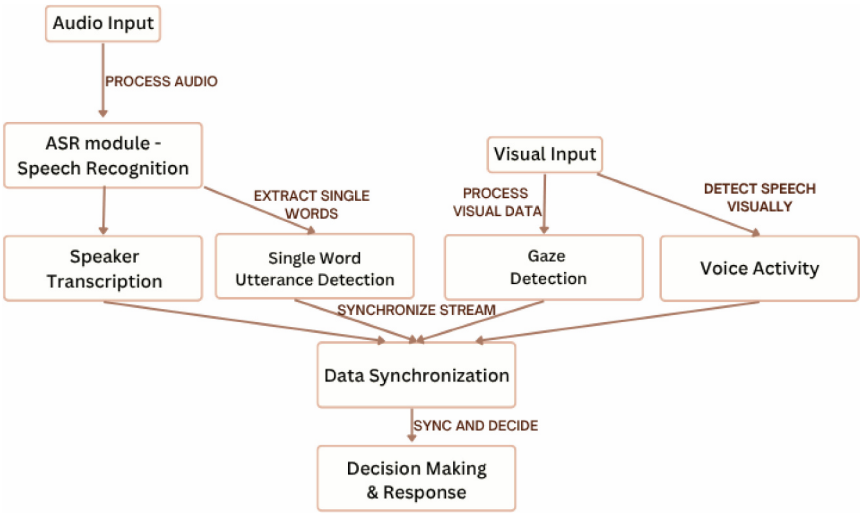


Fig. 2. Signal-level synchronization across multimodal input channels. Audio and visual streams are fused and evaluated jointly to inform real-time interaction decisions.

environments. The transcribed output, along with precise timestamps, is forwarded to the Rasa -natural language understanding (NLU) engine for intent classification and multi-turn dialogue management.

Simultaneously, an event-based context modeling stream aggregates asynchronous cues from four key modules: **ASR** confidence scores, **VVAD** output, **Gaze** Estimation, and **Single-Word Utterance Detection**. These temporally aligned signals are evaluated in a rolling window by the **CoBaIR** module, which applies a probabilistic Bayesian framework to infer communicative intent.

Figure 2 visualizes the signal-level processing pipeline, showing how data streams from ASR, VVAD, gaze, and single word utterance detection are temporally synchronized for integrated decision-making.

Upon intention recognition by CoBaIR, the selected response is vocalized via Pepper's **NAOqi** Text-to-Speech (TTS) engine, enabling timely and contextually appropriate system reactions.

Software Architecture The dialogue system is constructed from the following modular, software-driven components:

– **Whisper Transcriber:** A transformer-based ASR model from OpenAI, employed via the `whisper` Python API. The `base.en` variant was used, offering a balance between speed and accuracy for English-language input. Transcription was performed on 16 kHz mono audio streams using float32 numpy arrays, optionally resampled using `librosa` if needed. The transcription thread buffered audio using a 3-second sliding window and handled low-latency transcription in parallel with VVAD processing.

- **Rasa NLP and Dialogue Manager:** An open-source dialogue framework for intent classification and policy-based response selection. The system integrated Rasa via API calls, sending transcribed utterances and receiving structured actions. Custom Rasa pipelines were trained for the restaurant interaction scenario with intent labels such as `order`, `menu_query`, and `confirmation`.
- **VVAD Module (via PAZ):** Visual Voice Activity Detection was implemented using the open-source `paz` framework [7], which provides lightweight pipelines for facial detection and tracking. A custom CNN-LSTM model was integrated using PAZ's facial landmark and bounding box extraction utilities. The VVAD model was pre-trained on the VVAD-LRS3 dataset [9], achieving 92% frame-level accuracy. Face regions (112×112px lip crops) were dynamically extracted from the Pepper robot's RGB feed and passed into the VVAD pipeline in real-time. The entire module was deployed on an external GPU-equipped laptop for inference efficiency.
- **CoBaIR:** This module employs probabilistic Bayesian reasoning to infer the robot's communicative response based on asynchronous multimodal cues. Implemented using the open-source `CoBaIR` Python package [8], it evaluates four key input features: VVAD output, ASR confidence scores, gaze detection, and single-word utterance presence. Each input is encoded as an evidential variable, and the Bayesian network computes posterior probabilities over three possible intention states: *Respond*, *Ignore*, and *Skeptical*. The module outputs both the most likely intention and the full probability distribution (e.g., $P(\text{Respond}) = 0.87$, $P(\text{Ignore}) = 0.08$, $P(\text{Skeptical}) = 0.05$), which are used for downstream behavior control. In the case of low-confidence or ambiguous intent, fallback prompts such as clarification requests are automatically triggered.
- **NAOqi Text-to-Speech (TTS):** The Pepper robot's onboard TTS engine, used to deliver spoken feedback once an action was selected. The speech output was triggered conditionally based on CoBaIR's inferred intent.
- **Gaze Estimation Module (NAOqi):** User gaze direction was obtained using Pepper's built-in `ALGazeAnalysis` module provided by the NAOqi framework [17]. This module estimates whether a person is looking at the robot by analyzing the angle between the detected face and the robot's own head orientation. It outputs a boolean flag (`isLookingAtRobot`) along with timestamps and person tracking IDs. These gaze events were continuously monitored and logged via the `ALMemory` service, and temporally aligned with VVAD and ASR signals. The binary gaze state was used as an evidential input to the CoBaIR module for intent inference.

Hardware Capabilities. As shown in Fig. 3, the Pepper platform provides the following integrated sensory and actuation subsystems:

- **Audio**: Four directional microphones are used to capture user speech, which is transmitted via a TCP-based SSH tunnel to an external laptop for processing. Audio input was directly forwarded to the Whisper ASR model running on the external system.
- **Visual**: RGB and depth cameras used for VVAD (lip-motion detection via PAZ) and gaze estimation (via NAOqi's `ALGazeAnalysis`); support multimodal context modeling.

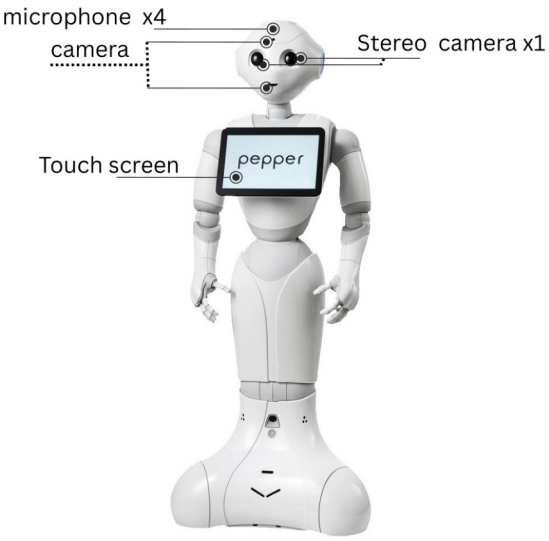


Fig. 3. Sensory and actuator layout of the Pepper robot. [18].

- **Touchscreen**: A chest-mounted tablet used in this study for visual feedback and user confirmation during the gift card interaction task.
- **Computation**: Heavy-lift modules such as Whisper ASR, VVAD inference, and Rasa dialogue management are executed on an external GPU-equipped laptop connected via TCP.

Through seamless integration of Whisper, Rasa, Paz–VVAD, CoBaIR, and NAOqi within Pepper's hardware framework, the system supports robust, fluent, and multimodally grounded human–robot dialogue. The modularity of the architecture facilitates future extensions such as affective modeling or gesture-based control.

4 User Study Design

To evaluate the effectiveness of the VVAD-enhanced human–robot dialogue system, we conducted a user study in a simulated real-world interaction setting. The study assessed both the technical robustness and user experience of the system during task-oriented interactions with a humanoid robot.

Environment. The study took place in a controlled laboratory space configured to emulate a casual restaurant setting. Ambient background noise—including chatter, clinking dishes, and soft music—was played through speakers to simulate typical disturbances in a busy service environment. This setup provided both ecological validity and experimental control, ensuring naturalistic but repeatable interaction conditions.

Participants. Thirty participants were recruited, representing a broad demographic spectrum. Ages ranged from **18 to 69 years** (mean $\approx$ 29.5), thereby including both

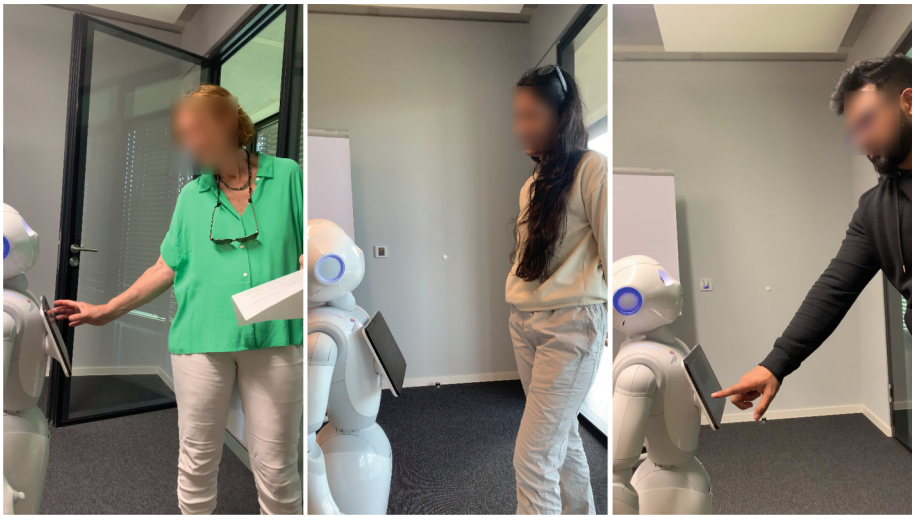


Fig. 4. Participants interacting with the Pepper robot, illustrating varied engagement styles across age, gender, and professional backgrounds.

younger adults and an elderly participant. The gender distribution was **14 female and 16 male**.

Educational and professional backgrounds were diverse, spanning both technical and non-technical domains. Participants included nurses, developers, circuit and electrical engineers, chemists, a botanist, a geologist, a graphic designer, and individuals from management and entrepreneurial fields. Approximately **50% of the cohort had STEM-related expertise** (e.g., software engineering, circuits, geology, chemistry), while the remaining participants came from **healthcare, management, or creative disciplines**. This diversity was critical for evaluating system usability across users with varying technological familiarity.

All participants gave informed consent and were provided with a structured briefing before the experiment. Pre-study questionnaires recorded demographic data and prior exposure to robots, ensuring a contextual basis for analysis.

Procedure and Tasks. Each participant interacted with Pepper in a semi-structured dialogue flow mimicking a restaurant scenario. Tasks included greeting the robot, asking about the menu or Wi-Fi, engaging in casual questions, and the main task: requesting a gift card. The dialogue was intentionally unscripted to encourage natural language use, while sample prompts were available to guide the interaction. Sessions lasted 5–8 min, and each participant was randomly assigned to experience only one of the two conditions: either VVAD enabled or VVAD disabled (audio-only ASR). This between-subjects, 1-factorial design was chosen to avoid mere-exposure and novelty effects, with the only manipulated factor being the use of VVAD. Background noise was introduced consistently in all sessions to evaluate robustness under auditory interference.

Evaluation Measures. We evaluated system performance using both objective and subjective metrics:

- **Task Completion Rate (TCR)**—Percentage of successful dialogue turns.
- **False Activation Rate (FAR)**—Instances of unintended system responses.
- **Video Quality Review**—Visual inspection of session recordings to check VVAD alignment under varying lighting and occlusions.
- **User Satisfaction**—A self-defined rating scale measuring perceived responsiveness, clarity of communication, and ease of interaction.
- **Godspeed Questionnaire** [3] - Perceived intelligence, likeability, safety, and anthropomorphism of the robot.

Study Ethics and Rationale. The study protocol received ethical approval from the German Research Center for Artificial Intelligence (DFKI) Ethics Committee. All data were anonymized by removing personally identifiable information from transcripts, and raw audio and video recordings were stored. Pepper remained stationary throughout the interaction to ensure participant safety. This user study was designed to capture both the technical behavior of the VVAD system and subjective impressions during natural interaction. By comparing VVAD-enabled and audio-only conditions in a realistic setting, we provide both quantitative and qualitative insights into the impact of visual speech cues on human–robot dialogue performance.[1]

5 Results and Discussion

This section presents the results of the user study, comparing system performance and user perception under two conditions: with and without VVAD. The analysis integrates quantitative metrics (TCR, FAR) with subjective user feedback and qualitative observations, offering a comprehensive understanding of the VVAD module's impact on human–robot interaction.

5.1 TCR

Task Completion Rate (TCR) was used to evaluate the system's ability to successfully carry out user-initiated goals under acoustically challenging conditions. Two distinct interaction tasks were analyzed: a primary task involving a structured request for a gift card, and a secondary subtask such as inquiring about the menu or Wi-Fi access. Participants interacted with the robot in a simulated restaurant environment, with 15 participants assigned to each condition (VVAD-enabled or audio-only).

In the **audio-only condition**, the system failed to complete the primary Gift Card task in all trials (0%). The absence of visual cues meant that background chatter and ambient noise were frequently misinterpreted as valid input, leading the robot to initiate incorrect actions and causing repeated dialogue breakdowns. Even for the simpler subtask, the completion rate was only 53.3% (8 out of 15 participants), reflecting the inherent limitations of relying solely on auditory input.

[1] We plan to publish the anonymized dataset collected during this study as an open-access resource for the research community.

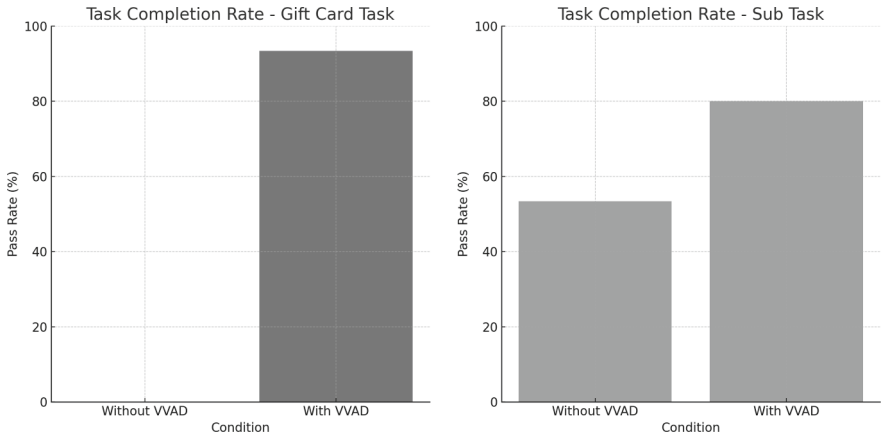


Fig. 5. Task Completion Rates for the subtask (left) and primary task (right), comparing conditions with and without VVAD.

In stark contrast, the **VVAD-enabled condition** demonstrated substantial gains. With visual lip-motion detection supporting audio-based recognition, the robot was able to reliably distinguish genuine user input from environmental noise. As a result, the system achieved a 93.3% success rate (14 out of 15 participants) for the Gift Card task and 80% (12 out of 15) for the subtask. These findings highlight the value of multimodal integration, where the fusion of auditory and visual channels leads to more precise and resilient interaction.

Beyond raw completion rates, the improvement in TCR has broader implications. High task success not only increases efficiency but also reduces the need for repetition, which in turn enhances user satisfaction and trust in the robot's capabilities. In real-world contexts such as customer service, healthcare, or assistive settings, such improvements directly translate into smoother, more reliable experiences. The bar charts in Fig. 5 visually emphasize this gap, showing how VVAD transforms system performance across both simple and complex tasks.

5.2 FAR

False Acceptance Rate (FAR) measures how often the system incorrectly accepts non-speech inputs, such as ambient noise or irrelevant sounds, as valid user commands. It is computed as:

$$\text{FAR} = \frac{\text{Number of False Acceptances}}{\text{Total Number of Inputs}} \times 100\%. \tag{1}$$

In the **audio-only condition**, FAR averaged 17.6%, underscoring the system's vulnerability to background chatter, footsteps, and other non-verbal noises. Without visual grounding, Pepper frequently misinterpreted environmental sounds as intentional input, leading to unintended actions such as interrupting tasks or initiating irrelevant

responses. These false activations disrupted the conversational flow, forcing participants to repeat commands and causing frustration. Several participants explicitly noted dissatisfaction with the robot's inability to filter out irrelevant sounds, which degraded both efficiency and trust in the interaction.

By contrast, enabling **VVAD** dramatically reduced FAR to 2.8%. The integration of lip-motion analysis and visual cues allowed the robot to cross-validate auditory input against visible speech activity, ensuring that only genuine spoken commands were processed. As a result, participants reported smoother interactions, fewer interruptions, and a stronger perception that the robot was "listening" and responding appropriately. The reduced need for repetition directly improved user comfort and overall satisfaction.

This stark contrast highlights the importance of multimodal processing for robust HRI. A lower FAR not only minimizes technical errors but also enhances the fluidity and naturalness of dialogue. In real-world domains such as healthcare, customer service, or assistive robotics—where errors can have serious consequences—such improvements are especially valuable. By minimizing false activations, VVAD enables robots to maintain more reliable and contextually appropriate interactions, ultimately reinforcing user trust and adoption potential.

5.3 User Satisfaction

The box plots in Figs. 6 provide a detailed comparison of user ratings across the seven survey questions (A–G), covering aspects such as overall satisfaction, expectation alignment, communication ease, response speed, clarity, willingness to re-engage, and recommendation likelihood.

In the VVAD-enabled condition (Fig. 6), scores cluster tightly around the upper range (median ≈ 4). Questions A ("overall satisfaction"), B ("needs and expectations"), and G ("recommendation likelihood") show particularly narrow interquartile ranges, indicating that nearly all participants reported a consistently positive experience. Questions C ("ease of communication") and E ("clarity of communication") also received predominantly high scores, though with slightly larger spreads, suggesting that some participants encountered occasional challenges. Isolated outliers in Questions C and F ("willingness to interact again") highlight rare cases of difficulty but do not affect the overall trend. These results demonstrate that VVAD not only improved task execution but also established a perception of reliability and trustworthiness.

By contrast, the audio-only condition (Fig. 6) reveals wider distributions and markedly lower medians ($\approx$ 2–3). Questions C ("ease of communication") and E ("clarity of communication") show substantial variability, with several participants giving the lowest possible scores. This supports qualitative observations that, without VVAD, users often had to repeat themselves, leading to frustration and reduced confidence in the system. Questions F ("willingness to interact again") and G ("recommendation likelihood") also show sharp declines, indicating that repeated communication breakdowns negatively impacted long-term acceptance. Median scores for Questions A and B further confirm that the robot was perceived as less reliable and less capable of meeting user needs when relying on audio-only processing.

Taken together, these findings demonstrate that VVAD significantly enhances user satisfaction across multiple dimensions of interaction. By enabling the robot to interpret

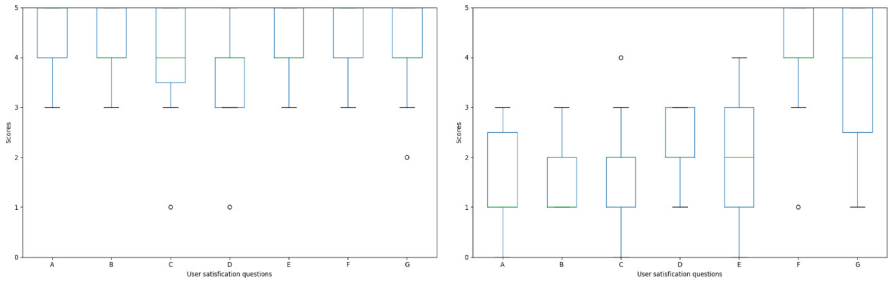


Fig. 6. User satisfication with and without VVAD.

speech more robustly and respond with greater precision, VVAD improves technical reliability, reduces user frustration, and fosters higher willingness for future use and recommendation. These outcomes are crucial for real-world deployment, as consistent satisfaction and trust strongly influence user acceptance of service robots.

Chi-Square Analysis of User Satisfaction. To statistically validate the observed differences in user satisfaction across conditions, we conducted a Chi-Square test of independence on the seven survey items (Questions A–G). The raw data consisted of 15 Likert-scale ratings (1–5) from participants in each condition (VVAD vs. audio-only). For each question, these ratings were first converted into frequency tables (i.e., how many participants gave a rating of 1, 2, 3, 4, or 5 in each condition). Since some categories were unused in one of the conditions, the ratings were collapsed into three broader groups: *Low (1–2)*, *Neutral (3)*, and *High (4–5)*. This ensured that all categories had non-zero counts and avoided sparsity issues.

The Chi-Square test then compared the observed distribution of responses in VVAD and audio-only conditions against the expected distribution under the null hypothesis of independence. The test statistic was calculated as:

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

Table 1. Chi-Square test results for User Satisfaction questions.

Question	χ^2	df	p-value	Significant
Q1 – Satisfaction	23.14	2	< 0.001	Yes
Q2 – Needs Met	27.00	2	< 0.001	Yes
Q3 – Ease of Communication	19.62	2	< 0.001	Yes
Q4 – Speed of Service	15.49	2	< 0.001	Yes
Q5 – Clarity	19.45	2	< 0.001	Yes
Q6 – Future Interaction	1.49	2	0.475	No
Q7 – Recommend	2.32	2	0.314	No

where O denotes the observed frequencies in each cell of the contingency table, and E represents the expected frequencies assuming no association between condition (VVAD vs. audio-only) and rating level (Low, Neutral, High). The resulting χ^2 values, degrees of freedom (df), and p-values are reported below.

The results show significant differences for five of the seven questions, indicating that VVAD had a strong and measurable positive effect on satisfaction, expectation alignment, communication ease, speed of service, and clarity. For future interaction and recommendation, no significant differences were observed, suggesting that these long-term acceptance measures may depend on additional factors such as trust, novelty, or repeated exposure.

5.4 Godspeed Questionnaire

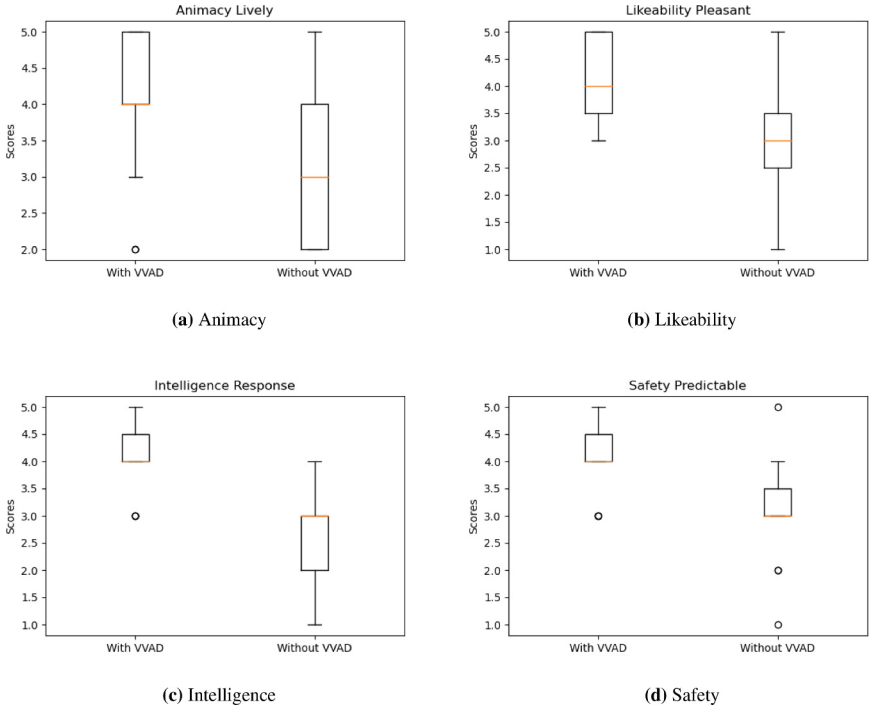
The Godspeed Questionnaire [3] was employed to assess five dimensions of user perception: Anthropomorphism, Animacy, Likeability, Perceived Intelligence, and Perceived Safety. Each dimension was evaluated on a 5-point Likert scale (1 = strongly disagree, 5 = strongly agree). These factors provide critical insight into how users evaluate social and emotional aspects of the robot, which in turn influence trust, comfort, and long-term acceptance.

Anthropomorphism. Participants in both conditions reported only moderate levels of anthropomorphism. With VVAD, users rated the robot as slightly more human-like (scores around 3–4), but consistently low in terms of emotions (mostly 1–2). In the audio-only condition, ratings for human-likeness dropped to 2–3, and perceptions of emotional capacity remained minimal. These findings suggest that while VVAD enhanced the robot's responsiveness, it did not substantially alter perceptions of its human-likeness, which were likely constrained by Pepper's fixed morphology.

Animacy. Animacy showed one of the strongest improvements. With VVAD enabled, participants frequently rated the robot at the upper end of the scale (4–5), noting that the system appeared "lively" and "aware of its surroundings." In the absence of VVAD, scores dropped to 2–3 on average, with a broader spread, reflecting reduced perceptions of attentiveness. This indicates that multimodal cues significantly enhanced the perception of the robot as an active, responsive partner.

Likeability. The robot was perceived as more pleasant and likable under the VVAD condition, with ratings clustering around 4–5. Users emphasized smoother and more natural interactions when visual cues supported turn-taking. Without VVAD, ratings dropped markedly, with several participants assigning scores as low as 1–2, reflecting frustration due to repeated misrecognitions and communication breakdowns.

Perceived Intelligence. VVAD had a strong impact on perceived intelligence. With VVAD, ratings ranged between 3–5, with participants frequently describing the robot as "understanding" and "clever." In contrast, the audio-only condition received much lower ratings (1–3), with several participants explicitly disagreeing that the robot could correctly interpret their input. This drop highlights how visual cues directly contributed to perceptions of cognitive competence.



(a) Animacy **(b)** Likeability

(c) Intelligence **(d)** Safety

Fig. 7. Godspeed Questionnaire results across four representative dimensions. Each plot compares VVAD-enabled and audio-only conditions using user ratings on a 5-point Likert scale. Higher scores in the VVAD condition suggest improved user perception in animacy, likeability, intelligence, and predictability.

Perceived Safety. Safety ratings were generally high in both conditions, but VVAD further enhanced predictability. Participants interacting with VVAD consistently rated safety at the top of the scale (4–5), noting that the robot behaved reliably and without unexpected interruptions. Without VVAD, safety remained relatively positive but slightly more variable (3–4), as misfires in command recognition occasionally led to unpredictable responses.

Summary. Overall, the VVAD-enabled condition led to consistent gains across Animacy, Likeability, and Perceived Intelligence, with moderate improvements in Safety and minor influence on Anthropomorphism. These perceptual outcomes align with the quantitative results, reinforcing that VVAD not only improves task performance but also enhances social and emotional engagement. As shown in Fig. 7, higher ratings in VVAD sessions suggest that multimodal input creates interactions perceived as more intelligent, lively, and enjoyable, strengthening user trust and adoption potential in real-world service robotics.

5.5 Subjective Analysis: Qualitative Video Observations

Video recordings were analyzed to capture nonverbal user behaviors and subjective cues not directly measurable through task metrics or questionnaires. These observations offer additional insight into users' emotional responses and engagement levels during the interaction.

In the **VVAD-enabled sessions**, participants consistently displayed more positive nonverbal behaviors—including open posture, direct eye contact with the robot, smiling, and relaxed body language. These cues indicated comfort, attentiveness, and a sense of natural engagement with the system.

By contrast, **audio-only sessions** frequently triggered signs of frustration or confusion. Users crossed their arms, diverted their gaze, furrowed their brows, or sighed audibly. Many were observed repeating commands or pausing mid-interaction to regain the robot's attention, often leading to visible disengagement.

Importantly, these patterns were observed across participants of varying age and technical backgrounds, suggesting that the benefits of multimodal input particularly visual speech cues extend to users with differing levels of familiarity. This subjective analysis reinforces the quantitative findings: VVAD not only improves technical recognition performance but also enhances emotional trust, perceived attentiveness, and overall interaction quality in HRI.

Additionally, in several VVAD-enabled sessions, Pepper responded to user input with minimal latency, often replying within moments of the user's utterance. This near-instantaneous response time contributed to a smoother conversational rhythm and enhanced the perception of the robot as attentive and intelligent. Participants were visibly more engaged during these timely interactions, with some showing spontaneous smiles or leaning in to continue the conversation. These real-time dynamics further underscore the value of prompt system responsiveness in creating natural and satisfying human–robot dialogues.

5.6 Discussion Summary

Recent studies on VVAD and multimodal HRI (e.g., Potamianos et al. [11], Lubitz et al. [9]) have shown that visual cues can improve speech detection accuracy, especially in noisy environments. However, most prior work evaluates VVAD on benchmark datasets or controlled lab tasks, focusing on frame-level accuracy or word error rate (WER), and rarely on real-world robot dialogue or user-centered outcomes.

In contrast, our study integrates VVAD into a real-time robot dialogue system and evaluates it in a realistic, noise-rich restaurant scenario. We report substantial improvements in task completion rate (TCR: 93.3% for VVAD vs. 0% for audio-only) and false activation rate (FAR: 2.8% for VVAD vs. 17.6% for audio-only). Importantly, we also provide subjective measures (user satisfaction, Godspeed scores) and qualitative video analysis, showing that VVAD not only improves technical robustness but also enhances user experience and engagement.

Overall, this work advances the state-of-the-art by demonstrating that VVAD integration in real-world HRI scenarios yields superior robustness and user satisfaction

compared to audio-only and prior multimodal systems, with a rigorous experimental design and comprehensive evaluation.

Taken together, these results demonstrate the practical value of integrating VVAD into spoken human–robot dialogue. It enhances task accuracy, reduces recognition errors, and promotes a more engaging and satisfying user experience. While improvements in perceived intelligence and reliability were substantial, future work may explore additional enhancements to physical embodiment or emotional responsiveness to further raise anthropomorphic perception.

6 Limitations and Future Work

This study positions VVAD as a critical enhancement for human–robot communication in acoustically challenging environments. By aligning system responses with observable user intent, VVAD improved both quantitative performance measures and user-reported satisfaction. Nonetheless, several limitations constrain the generalizability of our findings and highlight avenues for further research.

Experimental Constraints. The study was conducted in a controlled indoor setting, which, while useful for isolating VVAD's impact, may not fully capture the complexities of real-world deployments. Practical environments involve dynamic lighting, variable head poses, background occlusions, and fluctuating noise levels, all of which could affect VVAD's reliability. Additionally, the interaction sessions were short and based on semi-structured dialogues, which may have limited the naturalness and long-term dynamics of user behavior. Future work should validate these findings through longitudinal studies and evaluations in diverse, ecologically valid environments.

User Diversity and Interaction Style. Although the participant pool was varied, it skewed toward younger, technically literate individuals who adapted more easily to the robot's interaction style. Older participants or those unfamiliar with technology often faced greater challenges. Moreover, the system required users to speak in a clear, deliberate manner; natural speech patterns, slang, and accented English sometimes led to transcription errors, which reduced usability for certain groups. These factors highlight the importance of testing VVAD-enhanced systems with more demographically and linguistically diverse populations, as well as refining the speech models to better handle natural variation in language and delivery.

Scalability and Deployment. A notable architectural constraint is the reliance on an external high-performance laptop to perform VVAD inference. While sufficient for prototyping, this limits applicability in embedded robotic platforms with constrained resources. Future work should explore optimized, lightweight VVAD models capable of real-time inference on mobile hardware, ensuring broader deployability in service, healthcare, and domestic settings.

System Robustness. Several participants expressed confusion when the robot failed to respond appropriately, particularly in the absence of VVAD. This suggests a need for improved fallback strategies and feedback mechanisms when the system cannot confidently interpret user input. Moreover, lighting conditions, which were not systematically tested, could significantly impact visual speech detection. Addressing these issues through adaptive vision modules and multi-sensor fusion would strengthen robustness in real-world scenarios.

Directions for Future Research. To advance VVAD from research prototype to scalable solution, several priorities emerge:

- Longitudinal studies on user trust, adaptation, and satisfaction over repeated interactions.
- Evaluation across demographically and linguistically diverse populations to ensure inclusivity.
- Exploration of VVAD in high-stakes domains (e.g., healthcare, emergency response) where communication accuracy is critical.
- Optimization of VVAD for embedded platforms with limited computational resources.
- Integration with other sensory modalities (e.g., gaze, gesture, haptics) to create richer multimodal frameworks.

In summary, while this study provides strong evidence of VVAD's benefits, addressing these limitations will be essential to realize its full potential. A validated, robust, and lightweight VVAD system could become a key enabler of more natural, reliable, and inclusive human–robot communication across diverse application domains.

7 Conclusion

This study provides a comprehensive investigation into the integration of VVAD as a critical enhancement to human-robot dialogue systems. By addressing the core limitations of traditional audio-only communication models, particularly their vulnerability to ambient noise, this research demonstrates the substantial gains afforded by a multimodal approach to speech detection and interpretation.

The results strongly support the hypothesis that augmenting auditory processing with visual cues—such as lip movements and facial expressions—significantly improves the robustness of speech recognition in real-world conditions. VVAD-enabled systems achieved markedly lower FAR, especially in noisy environments, by effectively filtering out irrelevant audio based on synchronized visual input. This capability translated into smoother, more reliable interactions between humans and robots.

Task completion rates also benefited from the inclusion of VVAD. Enhanced accuracy in detecting speech onsets allowed for more fluid exchanges, reducing the frequency of errors and repeated queries. As a result, robots responded with greater precision, contributing to a more intuitive and productive interaction flow. These improvements were echoed in user satisfaction scores and Godspeed Questionnaire ratings,

which consistently favored the VVAD-enabled system across metrics of likeability, perceived intelligence, and safety.

Another important contribution of this work lies in its demonstration of VVAD's value in handling complex interaction scenarios and acoustically cluttered environments. The technology's capacity to differentiate and prioritize relevant speech streams strengthens the case for VVAD as a fundamental requirement in the deployment of social robots in everyday public and private settings.

Beyond immediate performance improvements, VVAD's role in increasing the adaptability and resilience of HRI systems has long-term implications. Whether in healthcare, customer service, or educational environments, systems that incorporate visual cues are better equipped to maintain consistent communication standards across diverse conditions and user demographics.

This paper also lays the groundwork for future research avenues. Potential directions include combining VVAD with advanced NLP, reinforcement learning, and real-time personalization strategies to create fully adaptive dialogue agents. Additionally, the development of lightweight VVAD models suitable for low-power mobile robots could expand its applicability to broader domains.

In summary, this study makes a compelling case for the integration of VVAD in HRI systems. It enhances not only the technical capacity of robots to understand human input but also the qualitative experience of interaction for users. As social robots become increasingly embedded in our daily lives, VVAD stands out as a pivotal technology for achieving more natural, context-aware, and human-centric communication.

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Virtual Reality for Surface Topography Analysis

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Abstract. Immersive technologies, such as virtual reality (VR), have seen widespread adoption across multiple sectors, including product design and usage simulation, owing to their interactive and intuitive capabilities that have transformed visualization and user engagement in these fields. Nevertheless, the application of VR in surface topography analysis remains notably limited. Although VR has significant potential to facilitate detailed and immersive examination of surface characteristics, its integration within this domain has not yet matched its use in other industries. This paper presents a VR-based framework for surface analysis, providing an interactive platform for in-depth exploration of surface features. The proposed tool is intended to advance both the understanding and practical application of surface topography.

Keywords: Virtual Reality · Metrology · Surface Topography · Texture Analysis

1 Introduction

1.1 Virtual Reality: Definition, Equipment and Applications

Virtual Reality (VR) is a technology that immerses users in a computer-simulated three-dimensional environment, creating an interactive experience that replicates real-world scenarios or entirely fabricated worlds. VR allows users to engage with the virtual setting through sensory inputs such as sight, sound, and touch. To achieve a successful and immersive VR experience, several components are required. At the core, a VR headset serves as the primary interface, enabling users to visualize the simulated environment through stereoscopic displays. Devices, like the Oculus Quest [1] or the HTC Vive [2], offer high-resolution visuals alongside positional tracking for seamless interaction. Haptic controllers further enhance immersion by allowing tactile interaction with virtual objects. Additionally, motion-tracking sensors and cameras capture real-time movements, ensuring synchronized gestures within the virtual space.

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The adoption of VR across various domains showcases its versatility and advantages. In healthcare, VR assists patients in rehabilitation by simulating real-life scenarios. Scientific research highlights its effectiveness in improving cognitive and motor skills during therapy sessions [3]. In education, VR creates engaging environments where students can explore historical landmarks or conduct virtual experiments [4]. Industrial sectors employ VR for product design, allowing engineers to visualize prototypes and simulate their performance under various conditions [5]. As VR technology continues to evolve, its potential to transform industries is becoming increasingly evident. The combination of immersive experiences, intuitive interfaces, and computational advancements ensures its role as a critical tool in shaping the future of interaction, design, and learning.

1.2 Surface Topography Analysis

In material science, surface topography analysis refers to the study and characterization of surface features and textures at a microscopic or macroscopic level. This field has applications across a wide range of domains, including material science, manufacturing, biomedical engineering, and geology. In manufacturing, detailed surface characterization is essential for quality control, performance evaluation, and research [6]. Traditional digital tools such as Gwyddion [7] or MountainsMap [8], provide 2D/3D visualization and quantitative metrics for surface roughness, but they are often limited by conventional interfaces (mouse, keyboard, flat screens) and pseudo-3D representations [9]. The overlap of results (scalar roughness parameters, 2D graphs, images) across multiple windows and the use of pseudo-3D representation significantly restricts operators' ability to fully grasp the complexity of surfaces.

In this paper, a Virtual Reality-based Laboratory for Surface Analysis (VLSA) is proposed to innovate the human computer interface (HCI) of topographic analysis digital tools by leveraging Virtual Reality (VR) systems, enabling immersive exploration of digital surface models. Users interact with high-fidelity renderings of microscopic or macroscopic surfaces, manipulate scale and orientation, and intuitively access multi-dimensional data overlays. VLSA offers an immersive approach where surfaces are manipulated as if they were real physical objects, but without the constraints of conventional interfaces, nor the limitations of reduced scales or overlapping multiple windows that are typical during surface studies with the conventional software. Through this technology, users can naturally perceive and interact with surfaces and their original components, easily changing viewpoints and scale, or obtaining a comprehensive multi-informational overview.

Based on recent advances in immersive technologies, this project explores the potential of VR to enhance visualization and interaction with complex surfaces, thereby facilitating their interpretation and analysis. VLSA is part of an interdisciplinary dynamic, combining surface analysis methods with VR innovations. The goal is to push current boundaries by proposing innovative solutions tailored to scientific and industrial needs.

The article is organized as follows: Sect. 2 reviews related literature, Sect. 3 outlines the proposed approach, Sect. 4 describes the developed prototype, and the final section provides the conclusion.

2 Related Work

Virtual reality (VR) technologies have rapidly evolved, offering new dimensions for data visualization, interaction, and understanding in scientific and industrial contexts. Initially adopted in fields such as product design, simulation, and medical training, VR has increasingly found its place in the analysis of surfaces—from microscopic material topography to large-scale terrain and geographical features. This literature review explores the current landscape of VR applications in surface analysis, geography, and terrain studies, highlighting both advancements and persistent challenges, and referencing key contributions found in the literature.

2.1 VR for Surface Modelling, Perception and Touching

An extensive investigation into the application of virtual reality (VR) for industrial surface design, particularly during the initial stages of product development, is detailed in [10]. It enables intuitive 3D sketching with real-time feedback and seamless integration with CAD systems. It uses optical tracking and a stereoscopic display to support ergonomic and aesthetic design from early stages, reducing errors and time-to-market.

In addition to shape design and analysis, Niu *et al.* [11] explores how surface roughness and texture influence material perception in immersive Virtual Reality (VR) compared to traditional 2D environments. The authors demonstrate that VR's stereoscopic depth and immersive qualities significantly enhance users' ability to perceive and differentiate material properties like roughness and texture. The study highlights that visual cues in VR lead to more accurate and consistent material judgments. Furthermore, they investigate how immersive VR affects the perception of rendered surface materials compared to traditional 2D displays [12]. Using psychophysical methods, the authors establish perceptual scales for material properties like roughness and specularity. Experimentation results show that VR significantly improves both accuracy and precision, suggesting that stereoscopic depth cues enhance material perception. The findings highlight VR's potential for more effective material evaluation in design contexts.

VR technology can simulate force feedback, enabling users to touch materials and experience their textures. Zhang *et al.* introduces a method for reproducing the haptic sensation of sandpaper textures with varying roughness during bare fingertip interaction [13]. The proposed method employs 3D surface imaging to capture the texture of sandpaper. This data is then used to generate vibration and tangential force through a haptic device, effectively simulating the tactile experience.

2.2 VR in Geography and Terrain Visualization

In the literature, most applications of VR for analyzing surface topography are mainly for geographical data and terrains visualization. Geographic Information Systems (GIS) traditionally rely on 2D maps or desktop-based 3D viewers, but VR offers an embodied sense of scale and spatial relationships that flat displays cannot provide. Integration of geospatial data visualization with immersive technologies can enhance spatial analysis, environmental modeling, and user interaction.

A comprehensive overview of the intersection between geo-visualization and extended reality (XR) is proposed in [14]. It emphasizes the potential of XR to enhance spatial understanding and decision-making by enabling immersive, interactive experiences with geospatial data.

Antoniou *et al.* [15] presents a novel workflow combining drone-based photogrammetry, VR platform and GIS analysis to map and quantify volcanic features at the Metaxa Mine in Santorini. This approach enabled high-resolution 3D modeling, virtual fieldwork, and volumetric analysis in a hazardous and inaccessible environment.

Similarly, Westerteiger *et al.* [16] showcases geoscientific visualization by extending a virtual terrain rendering framework to support spatial analysis tasks. This immersive approach enhances users' spatial perception, allowing for more accurate interpretation of complex terrain features that are often ambiguous in traditional 2D GIS on screen. By leveraging VR's three-dimensional capabilities, the system offers a more intuitive and precise modality for analyzing terrain surfaces, thereby improving the effectiveness of geospatial analysis.

In addition to the geographic geometry data, it is possible to include all the semantic data. For example, Alene *et al.* [17] introduces a VR framework for visualizing numerical simulations of geophysical flows (*e.g.*, avalanches, landslides, floods), bridging physics-based modeling with immersive visualization to support hazard awareness, stakeholder engagement, and disaster preparedness. In urban planning, Zhang *et al.* [18] presents how online geographic VR platforms can enhance urban planning processes. It demonstrates that such platforms facilitate better communication among stakeholders—including planners, decision-makers, and the public—by visualizing complex spatial data in an intuitive way.

2.3 VR for Surface Topography Analysis Training and Learning

Virtual Reality (VR), Augmented Reality (AR), and Extended Reality (XR) are revolutionizing the teaching and training of surface topography analysis by enabling interactive, immersive exploration of complex spatial data. VR allows users to intuitively visualize and manipulate topographic features and materials in realistic 3D environments, while AR overlays digital elements like contour lines and terrain profiles onto the physical world.

An innovative approach to teaching topographic concepts using web-based AR and VR tools is presented in [19]. The study describes the creation of interactive 3D mockups of topographic surfaces that are enhanced with satellite imagery to improve realism and comprehension. Students can explore these models in AR which is linked to a corresponding VR environment for deeper interaction and manipulation. The environments are designed to support learning in disciplines like Topography, Geography, and Descriptive Geometry, offering a more immersive and accessible way to grasp complex spatial concepts.

AR can strongly enhance comprehension by blending virtual and physical elements. This practical application of topographical concepts aids in making abstract ideas more tangible and easier to understand. For example, an AR sandbox [20] aims to enhance

geography education by enabling students to physically shape terrain and receive real-time digital feedback. This tool is designed to improve spatial thinking and understanding of topographic concepts by connecting 2D maps with 3D landscapes.

Virtual, augmented, and extended reality (VR/AR/XR) technologies have deeply advanced surface topography analysis by offering immersive and interactive ways to visualize and comprehend complex spatial data. These tools transcend the limitations of traditional 2D maps and desktop-based viewers, providing users with an embodied sense of scale, depth, and spatial relationships.

Most contributions of XR for surface analysis in literature focus on geography applications, with few dedicated to material science in industrial engineering. This suggests a need for integrating a piece of surface and the original workpiece 3D part. Additionally, there is limited literature proposing interactive functions for studying surface texture at different wavelength components (microroughness, roughness, waviness, form). Existing works do not utilize lighting and shading to allow users to observe surface roughness. Studying the roughness of a 3D surface using a 2D section curve appears to be a worthwhile area of investigation.

3 Proposed Approach

This paper presents the design of a Virtual Reality-based Laboratory for Surface Analysis (VLSA), which aims to assist engineers in analyzing surface materials for manufacturing processes (Fig. 1).

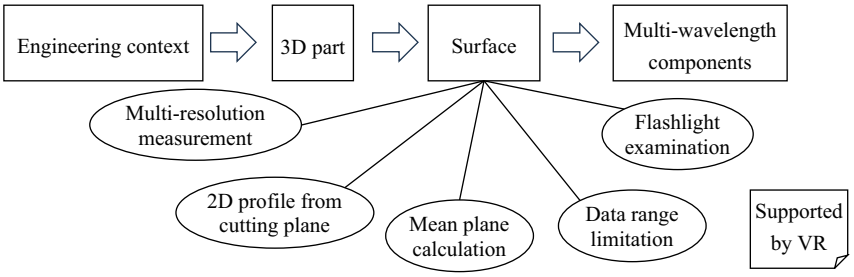

Fig. 1. Data structure and main functions of VLSA using VR devices.

Utilizing VR devices, users are immersed in a contextual virtual engineering environment that simulates a factory equipped with manufacturing machinery, such as milling machines. Users can interact with a three-dimensional part of a workpiece according to their specific engineering study interests.

For example, if users are focusing on machining tool wear, they can remove the cutting tool from the milling machine. To conduct a topographic analysis on a section of the machining tool, users can extract a piece of the surface for further examination by:

- Observing metrological measurements at varying resolutions.
- Defining different cutting planes to analyze various 2D profiles.

- Using a flashlight to examine visually surface roughness through shadows and reflections

Finally, the system is able of computing multi-wavelength components from the extracted part of surface, enabling users to study microroughness, roughness, waviness, and the form of the surface.

4 Developed Prototype

A VLSA prototype has been developed using the Unity game engine [21] for the Meta Quest [1] virtual reality headset.

4.1 Contextual Environment

When launching the VLSA application, users enter a virtual laboratory room, featuring a surface sample analysis desk and a milling machine. Figure 2 displays an overview of the lab at the bottom and a first-person (FP) view at the top.

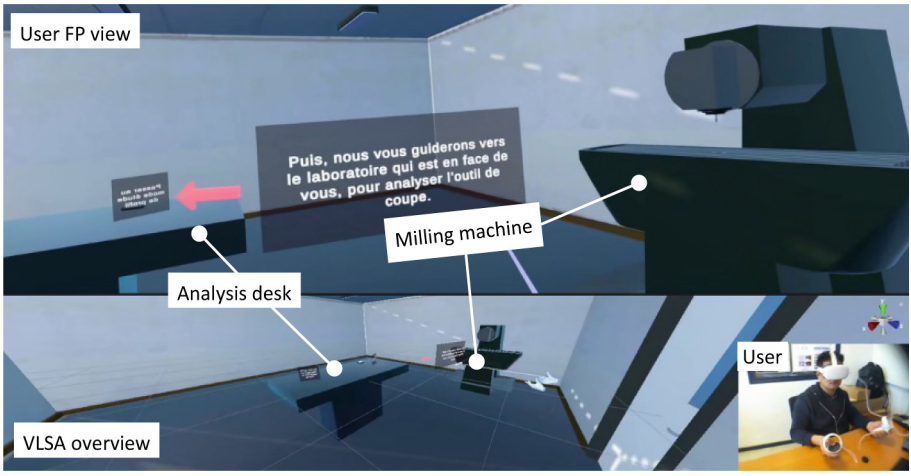

Fig. 2. Overview of the VLSA in the contextual environment.

4.2 Extraction of Surface Piece from the Contextual Part

After a brief introduction to the milling machine (Fig. 3a), the user is immersed in the study context. The user then removes the cutter from the machine (Fig. 3b) and returns to the surface analysis desk to examine it by hand, with options to scale it (Fig. 3c). Finally, the user focuses on analyzing a specific surface area at the top of the cutter, stretching the selected surface from the cutting edge for detailed examination (Fig. 3d).

4.3 Zoom Options for Surface Measurement and Relief Enhancement

Once the selected surface is extracted from the machining tool, the surface is spread on the analysis desk. If several measurements using different zooms have been realized the user can switch between different surfaces according to the zoom choice (Fig. 4.a and 4b). Higher zoom (*e.g.*, x50) will measure surface with smaller dimensions but with more details than lower zoom (*e.g.*, x5). On the right hand of the surface two sliders are available to the users. For example, user can change the scale of the surface using the "Z scaling" slider (Fig. 4c and 4d). The heigh of the rescaled surface will be amplified, while the lateral dimension remains constant, allowing the user for observing and analyzing all defects and features.

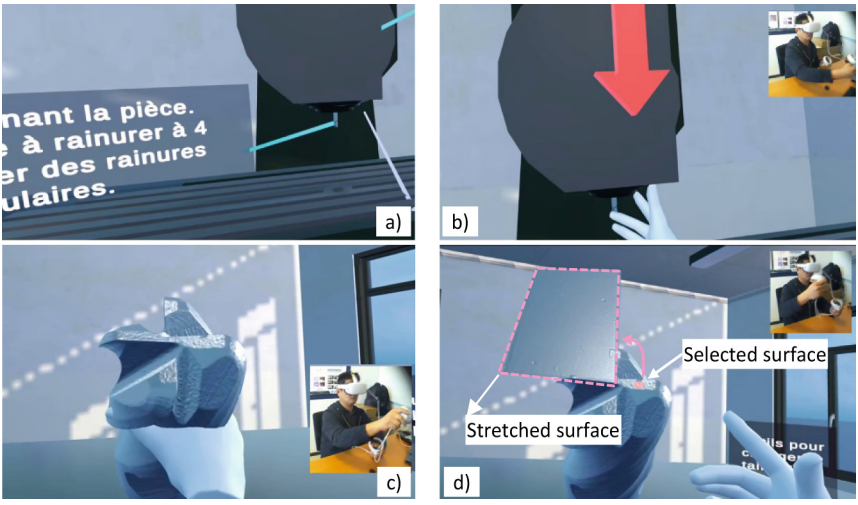

Fig. 3. Preliminary operation for context purposes: a) and b) Detach the cutting tool from the milling machine, c) inspection of the tool, c) selection of the surface of interest from the workpiece.

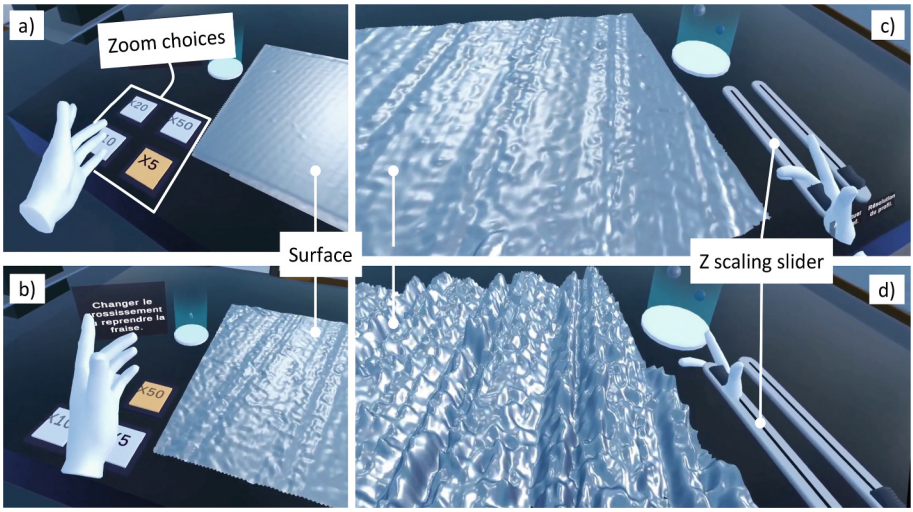

Fig. 4. Zoom options (a, b) for surface measurement and relief enhancement (c, d).

4.4 Mean Plane Calculation for Surface Manipulation

This feature is developed to integrate seamlessly within an immersive framework. Upon activation, the mean plane is computed from the surface points via the least squares method and presented as a transparent plane, offering users a clear and intuitive visualization. Figure 5 demonstrates a surface as viewed in the VR application, with the corresponding mean plane shown.

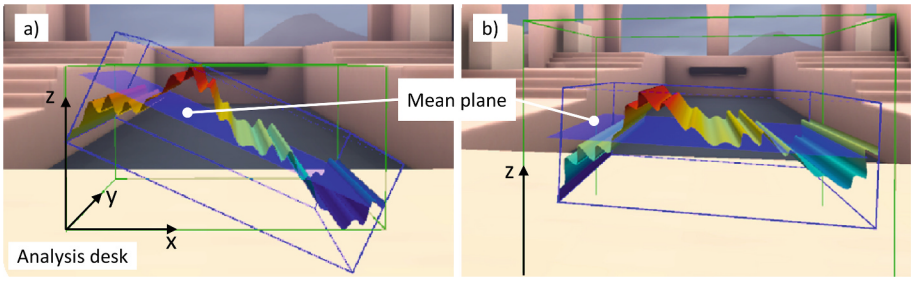

Fig. 5. Surface manipulation for orientation changes and mean plane calculation.

Users can manipulate and reorient the surface (Fig. 5a), allowing the mean plane to be aligned horizontally (Fig. 5b), or another plane with a subjective signification. This methodology promotes optimal normalization of measurements and enhances interpretability. It should also be noted that the surface's color distribution corresponds to the Z values of its points and is dynamically updated.

4.5 Surface Z Range Limitation

A thresholding feature has been incorporated into the VR application to enhance data selection capabilities. By employing two limiting planes, users can accurately specify the data segments to be excluded, offering comprehensive and intuitive control over boundary settings.

Figure 6 illustrates both a surface and a histogram depicting the distribution of all mesh points comprising this surface.

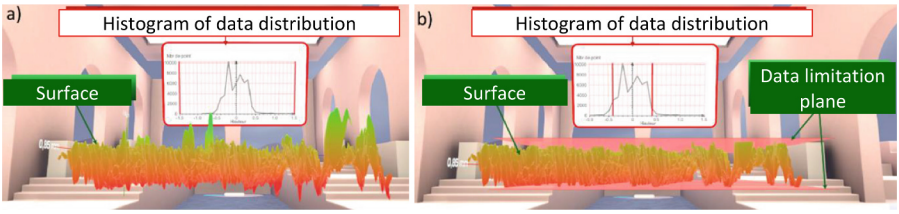

Fig. 6. Surface before and after data limitation is applied.

In Fig. 6a, the displayed surface represents the original configuration prior to modification, with the presence of artifacts (single points with high or low height). Figure 6.b shows the same surface following limitation, in accordance with the data range defined by the user's chosen limitation planes. These planes establish a lower and an upper threshold. Points situated above the upper threshold or below the lower threshold are automatically assigned the value of their respective limit.

4.6 Examination of 2D Profiles Produced from Section Plans

To address these requirements, a profile generation feature has been implemented in the VR application. Figure 7 shows a 2D profile curve produced from the 3D surface along a user-defined plane.

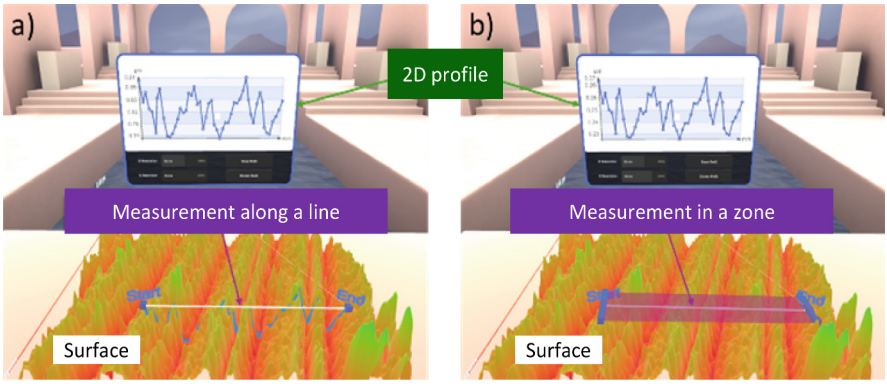

Fig. 7. Profile curves derived from a single plane (a) and from multiple parallel planes (b).

The user can manually adjust the extremum of the line with its hands. Several profiles can be seen at the same time at different panels.

This feature provides two types of measurements:

- One-dimensional measurement (cutting plane): The profile is defined by the intersection of the surface with a cutting plane of zero thickness (Fig. 7a).
- Two-dimensional measurement (cutting pad): The profile, corresponding to an area with a specific width, is calculated as the average of N profiles obtained by intersecting the surface with N parallel planes (Fig. 7b).

These measurement methods are intended to minimize interpretation errors by delivering clear and precise data. They also enable the extraction of roughness parameters.

4.7 Surface Decomposition Based on Varying Wavelengths

This functionality allows intuitive, interactive exploration of frequency components. Spectral analysis tools like the Fourier transform help identify key frequencies linked to recurring defects. After decomposing a surface into form, waviness, roughness, and microroughness, users can merge components—such as combining waviness and roughness to define texture and extract parameters. Figure 8 shows the original surface, its four scale components, and how waviness and roughness are combined. This method reduces interpretation errors and boosts surface quality in production.

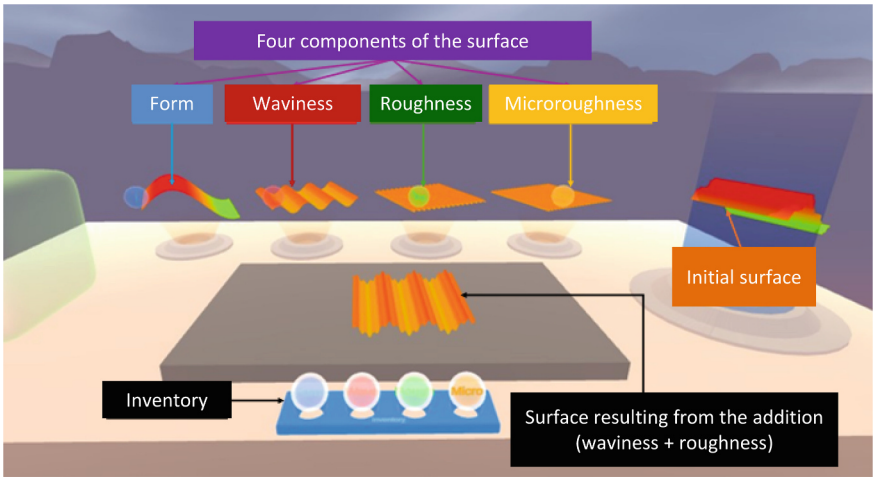

Fig. 8. Surface decomposition and merging.

4.8 Surface Roughness Assessment Using a Flashlight

The final feature of this VR application allows users to assess surface roughness using a virtual flashlight (Fig. 9).

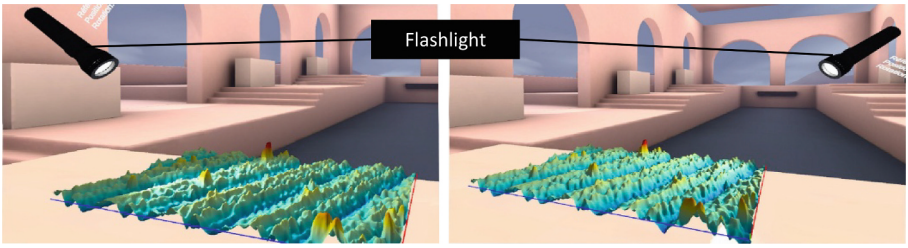

Fig. 9. Surface observation using a flashlight.

The user can directly manipulate the flashlight to move and orient it, thereby illuminating the surface from different angles and positions. Depending on the direction and location of illumination, different surface roughness features may become visible through reflections or shadows. This method provides information about the characteristics of the surface texture.

5 Conclusion and Perspectives

In this paper, we introduce a Virtual Reality-based Laboratory for Surface Analysis (VLSA). The developed framework enables users to perform surface analysis within an industrial context. Key features include 2D profile creation, mean plane computation, frequency-based separation, and examination with a virtual flashlight. These functionalities are designed to move beyond traditional analytical methods by providing immersive and interactive tools that facilitate a deeper comprehension of surface characteristics. By minimizing biases related to data misinterpretation and enhancing the accessibility of analysis, these innovations promote a more thorough understanding of phenomena associated with geometric surface properties.

This approach marks the beginning of a new era in digital metrology, wherein accuracy and user interaction converge to improve both the quality of analyses and the interpretation of results. Ongoing developments will be validated through a series of forthcoming tests to illustrate that virtual reality can serve as both an innovative exploration tool and a robust solution to increase the reliability and reproducibility of surface analysis.

Future work includes the integration of automatic parameter extraction, such as roughness indices (Ra (Sa), Rq (Sq), Rz (Sz), etc.) to further streamline the surface analysis process. An automated report generation feature is also under development; this will consolidate conducted analyses and their outcomes, supplemented by screenshots as needed to ensure optimal readability and traceability.

A critical upcoming phase involves conducting comparative experiments between existing conventional tools and the VLSA solution. The aim is to demonstrate the tangible value that virtual reality adds to topographic surface analysis. In the long term, we plan to directly incorporate data from measurement instruments to increase the efficiency of the analysis process. Additionally, we intend to develop a collaborative virtual environment, enabling multiple users to communicate and interact simultaneously around surfaces to be analyzed, thereby making the analytical approach more intuitive and collaborative.

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Extreme Reality Helmet in Simulated Smoke Test

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Abstract. Emergency responses, such as firefighting, take place in extremely hazardous environments where visibility and communication are often extremely poor. Spatial navigation, imaging, and communication interfaces are critical in first response scenarios where common perceptual channels, like visual and auditory, are reduced or unavailable. See-through interfaces are also valuable in advanced simulation training to bring physical touch into the immersive professional field training. Extreme Reality (EXR) is to superimpose extreme situations and live sensory data to the physical scene. In this study, we tested haptic helmet interface and thermographic helmet interface within the simulated smoke environment. Our multimodal EXR interfaces show their potential in real-time operation and training with zero visibility. We also found that the usability of the interfaces can be measured by navigation time, clarity, and reaction time. The thermography can be optimized by highlighting contours.

Keywords: Extreme Reality · Training · Haptic · Thermal · Simulated Smoke

1 Introduction

Emergency responses, such as firefighting, take place in extremely hazardous environments where visibility and communication are often extremely poor. Special navigation, imaging, and communication interfaces are critical in first response scenarios where common perceptual channels, like visual and auditory, are reduced or unavailable. See-through interfaces are also valuable in advanced simulation training to bring physical touch into the immersive professional field training.

Existing Augmented Reality (AR) interfaces are not built for extreme environments such as smoke, dark, noisy, and poor communication conditions. In extreme cases, such as firefight training, the trainees need to crawl through the floor with zero visibility. We need a new kind of AR platform that can survive extreme environments and provide on-demand information for first responders and command posts. Virtual Reality (VR) simulates the real-world in the digital way. AR overlays virtual objects to the real-world. Extended Reality (XR) superimposes additional sensory data to the real-world scene such as the live thermal image. Furthermore, Extreme Reality (EXR) superimposes extreme motion or environmental scenarios to the real-world scene, for example, the extreme poses, smoke, and fire.

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Many existing game engines use gamepad [1, 2, 4], head-mounted cameras, or IMU (inertial motion unit) to simulate a user's poses [7]. Unfortunately, they do not fully capture the physical motion. The visual and physical effects are not realistic. In this study, we explore gesture capture and interaction technology for emergency response training, which includes extreme gaits such as crawling through the floor and real-time gesture control of the user's movement in the training scenarios.

2 Extreme Reality Helmet Interfaces

The author's lab has developed a set of AR head-up display helmet interfaces, including haptic, thermographic, inertial motion unit (IMU), and lidar distance sensing and mapping interfaces. In this study, we focus on haptic and thermographic interfaces only.

In the haptic helmet interface [4], the haptic actuators are embedded into the helmet, which provides left, right, forward, and backward directional signals. The helmet has an embedded computing and wireless communication system that can be connected to AR or VR systems. We integrate the haptic interface into a firefighter's AR helmet, as shown in Fig. 1, alongside other AR interfaces. The haptic interface on the helmet allows the user to communicate through haptic signals over long range radio (LoRa) for live testing. Unlike current AR technologies that overlay VR objects onto the real world, this haptic helmet enables the sharing of mission-critical information from the firefighter's perspective, such as movement, position, and actions, with other first responders or scene overseers. This facilitates better decision-making in missions, enhancing the safety of responders and victims. The haptic motors and wiring are hidden within the headband and connected to a small enclosure mounted on the helmet containing the control electronics. Embedding all components in the headband allows users to transfer the haptic system between helmets, offering a low-cost solution inside the high-cost helmet shell. A controller with a LoRa module facilitates the transmission of haptic instructions to the helmet. LoRa is selected for its suitability for low data rates, long-range capability, and operation at a relatively low frequency (sub-gigahertz), which is advantageous for sending signals deep into buildings. Since firefighters wear a flame hood and SCBA (Self-Contained Breathing Apparatus) mask, contact with their heads will be limited. Therefore, high-power haptic motors need to be embedded in the helmet's headband, along with additional electronics to drive them.

The thermographic helmet interface [8, 14] can be used for AR indoor navigation, object detection, emergency medical triage and treatment in extreme environments such as a dark, smoke-filled subway. In contrast to prevailing thermography AR devices that are expensive and bulky, our solution will be much more compact, affordable, and in an open architecture across AR platforms. The thermographic helmet interface can collect the live data of the environments and victims. Firefighters can use it to see-through smoke, to find people, to navigate, and to localize the base of a fire. The helmet embedded AR interface enables thermography to overlay directly on humans and objects intuitively, relieving hands to hold something else. We design the thermographic overlay optics that balances the focus distance and size of the image. Computer vision algorithms are developed for edge detection for situation-awareness in the smoke-filled environment, as well as human blood vessel highlighting and vital sign detection with surface overlay. Figure 2 shows the prototype of the thermographic helmet interface and the overlaid thermal image.

Fig. 1. The haptic helmet and the remote controller (left) and the firefighter tester (right).

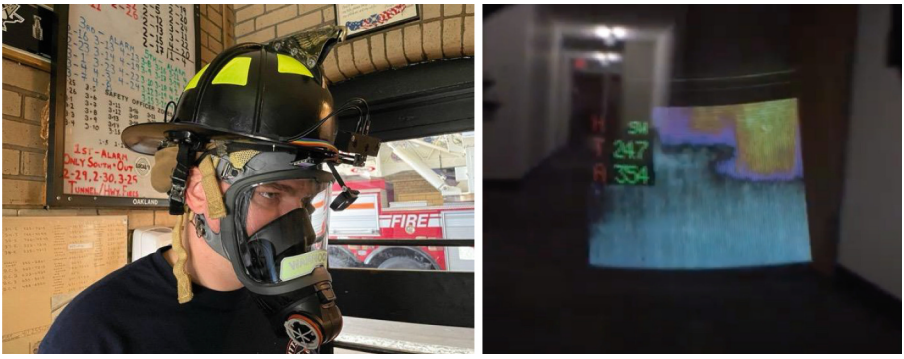

Fig. 2. The thermographic helmet interface (left) and the overlaid thermal image (right).

3 Simulated Smoke Test

The simulated smoke test was conducted in two firefighter training facilities in Fig. 3. The first simulated smoke test was a part of the NIST's Haptic Interfaces Challenge in Denver, CO, 2019 and the thermographic helmet interface test was at the Allegheny County Fire Academy (ACFA) in Pittsburgh, PA, in 2024. The white vapor smoke was used in both tests. The rooms were totally dark without any lights, nor windows. The visibility is almost zero. Due to the safety concerns, the test fields were flat without using stairs. See Fig. 4. The first test facility in Dever is a realistic brick building with two doors. The second test facility at ACFA is a metal wall building with one front door.

In the haptic interface test, the firefighter tester wore the instrumented helmet in the smoke-filled dark room to search for the victim dummy on the floor. The simulated incident commander followed the firefighter behind with the remote controller to advise the firefighter to go left, right, forward, go backward, or stay with the haptic signal on the helmet. The firefighter's navigation time for searching the victim and exit the room is used as the measurement for the haptic interface efficiency. The average navigation time of the six firefighter testers is the final result. Figure 5 shows the thermal camera images behind the firefighter tester.

In the thermographic helmet interface test, the firefighter testers wear the helmet-mounted display in the smoke-filled room to inspect the clarity of the holographic screen that is about 2 cm - 5 cm away from the eye. The experiments include multiple thermographic modes such as mono color, contour highlights, and human detection, and so on.

Fig. 3. The smoke test facilities in Denver, CO (left) and ACFA, Pittsburgh, PA (right).

Fig. 4. The simulated smoke in the ACFA test facility.

Fig. 5. The researchers used thermal camera for tracking the firefighter's movements in smoke.

4 Results and Observations

The haptic helmet interface test was successful in Denver, 2019. The author's lab won the First Place Award. The recorded video is at [15]. The six firefighters used the haptic signals to search and navigate the smoke-filled space and exit with minimal time. The haptic instructions such as move forward, turn left, turn right, move backward, and stay were clear and intuitive. We found that the haptic signals from the firefighter's helmet is the best location, compared to the hand or chest because the firefighters were required to crawl through the space.

The thermographic helmet interface test under the smoke was successful as well. Figure 6 shows the helmet display in the smoke and Fig. 7 shows the display results recorded with a mobile phone. Two firefighters participated in the test within the smoke-filled room. They test the overlaid live thermal images on the display in different modes, e.g. in front of another person, in front of the wall, in monocolor, or in pseudocolor, or reading the text on the screen. Because the screen is 2 cm - 5 cm in front of the eye, we want to test whether the simulated smoke would impair the visibility of the overlaid content. Our test results show that the holographic screen is visible in the smoke since it is a very short distance between the screen and the eye. The text on the top of the screen is also clear and readable. The field recording device needs to be improved though. To record the visual content on the helmet-mount screen is rather challenging because of the dark environment. A mobile phone camera does not capture the dynamic live video or images in the dark. There would be an obvious delay time. Also, the position of the camera needs to be placed at the eye position, which is not easy to align the field of view. We need a special design for the field record.

Fig. 6. The thermographic helmet interface in the smoke-filled room.

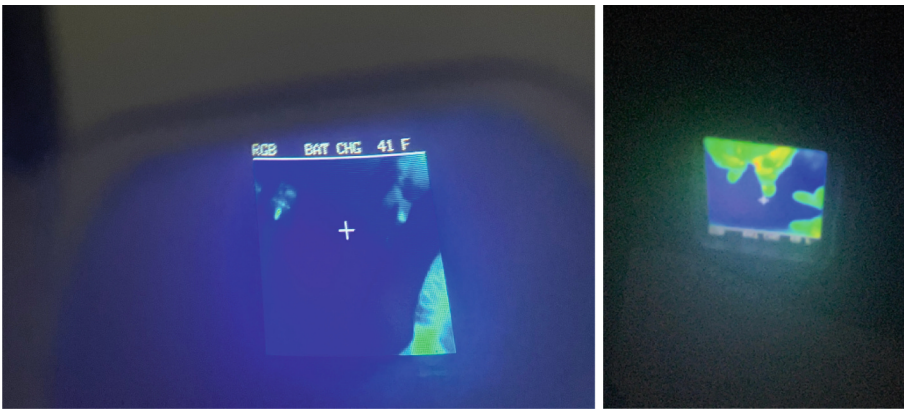

Fig. 7. The thermographic display results.

We observed that it is not necessary to overlay all the pixels of the thermal image on the screen because it overwhelms the user and even distracts the front view. Figure 8 shows the biometric data overlay where the white color on the screen is not necessary. For human objects, contour highlights would be enough to indicate the shape and gait without distraction. Table 1 summarizes the differences between the haptic interface test and thermographic interface test and Table 2 outlines the measurements.

Fig. 8. The biometric data overlay (left) and the hand contour overlay on the screen (right).

Table 1. Haptic interface test versus thermographic interface test.

Interface	Measurement	Computing	Communication
Haptic	Navigation Time	Embedded	LoRa
Thermographic	Display Clarity	Embedded	WiFi, Bluetooth

Table 2. Statistics of the smoke test results.

Interface	Navigation Time	Latency	Clarity
Haptic	$1'48''$ min (best)	no	not available
Thermographic	not available	no	HUD-eyes within 5 cm

5 Conclusions

Emergency responses, such as firefighting, take place in extremely hazardous environments where visibility and communication are often extremely poor. Spatial navigation, imaging, and communication interfaces are critical in first response scenarios where common perceptual channels, like visual and auditory, are reduced or unavailable. See-through interfaces are also valuable in advanced simulation training to bring physical touch into the immersive professional field training. Extreme Reality (EXR) is to superimpose extreme situations and live sensory data to the physical scene. In this study, we tested haptic helmet interface and thermographic helmet interface within the simulated smoke environment. Our multimodal EXR interfaces show their potential in real-time operation and training with zero visibility. We also found that the usability of the interfaces can be measured by navigation time, clarity, and reaction time. The thermography can be optimized by highlighting contours.

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The Role of AI and LLMs in Controlling Human Attached Devices

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Abstract. Commercial prostheses still rely largely on single-modality electromyography (EMG) control. This entails unreliable signals and forces users to perform mode switching and calibration. Advances in artificial intelligence (AI) techniques, specifically LLMs, promise further advances for prosthetic devices. 13 semi-structured interviews with clinicians, researchers and manufacturers reveal three converging needs: (i) reliable multi-modal intent recognition beyond EMG, (ii) AI-guided socket-design and real-time socket fit optimisation, and (iii) intuitive, personalised, low-cognitive-load human-prosthesis interaction, potentially via Large Language Models (LLMs). Synthesizing these findings with recent literature, we outline the current state of AI adoption and barriers that cause a research-to-practice gap. Multi-modal control remains confined to laboratory settings, while commercial adoption is hindered by fears of over-complexity. We highlight human-in-the-loop continuous calibration games and prioritize speaker-dependent voice commands with LLM reasoning. These findings provide an agenda for engineers, clinicians and researchers to translate AI techniques, specifically LLMs, into user-centric and comfortable prosthetic devices.

Keywords: Large Language Models (LLMs) · Prosthetics · Multi-Modal Control · Human-in-the-Loop · Voice Control · User-Centric

1 Introduction

A growing number of people are living with limb anomalies worldwide. There are about 18.5 million people living with congenital musculoskeletal and limb anomalies in 2021 with long-term functional limitations and there will be even higher total numbers by 2031 [18]. Traumatic limb loss is also an immense and still growing public health problem. Amputation has directly affected more than 445 million people worldwide and even though age standardized rates declined significantly the absolute number is climbing due to population growth [31]. Limb loss is often the consequence of poorly controlled diabetes and a life-changing event, after which there is an increased expression of depression, body image disturbance and social embarrassment [4]. This demands next-generation prosthetic systems that are tailored to the individual, capable of reliably restoring functional mobility to foster comfort, resilience and social participation. Especially as there are 20% of all upper-limb prosthesis wearers that abandon the device, due to discomfort and limited functional benefit [5]. Pattern-recognition, a branch of

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machine learning, which was applied to prosthetic control for many decades, remains the most important form of control for human attached devices. EMG electrodes record the electrical activity of contracting skeletal muscles, subsequently pattern-recognition algorithms learn to associate complex muscle activation patterns with specific movements and grips.

Nevertheless, commercially available prosthetic upper and lower limbs still expose the user to mode-switching, high calibration overheads and unreliable surface EMG signals.

To improve prosthetic control and user acceptance, efforts go into combining EMG or electroencephalography (EEG) signals, which are gathered from non-invasive brain-computer interface signals with other input signals. Interviews with manufacturers, prosthetists and R&D leaders confirm limited penetration of more recent AI techniques and a pervasive fear of over-complexity for both users and clinicians. The breakthroughs in generative AI suggests that LLMs could address many of these shortcomings by *understanding* and *explaining* user intent in natural language. To date, however, there is to our knowledge no introduction of LLMs or large multimodal models (LMMs) in prosthetic technology.

The present paper leverages 13 expert-interviews to (1) chart a map of unmet needs, (2) consolidate existing evidence that AI can close them, and (3) propose actionable next steps that stay within today's technological envelope. This paper asks: How can modern AI, especially LLMs, be harnessed to deliver truly user-centred, multi-modal prosthetic control without increasing cognitive burden?

2 Related Work and Background

Prosthetic control largely relies on EMG signals and pattern recognition to map muscle contractions to discrete gestures, due to its intuitive use and ease of application [8].

Recent developments have focused on simplified myoelectric pattern recognition techniques that aim to reduce the complexity and cost of sensor setups. Dynamic Time Warping (DTW) methods using single-channel EMG sensors demonstrated high accuracy, offering a practical solution for simplified prosthetic controls [15]. Additionally, new time-domain feature extraction methods have improved classification performance using fewer sensors [25].

This application of artificial intelligence in prosthetic devices is considered a state-of-the-art solution for advanced prosthetics. A significant drawback of pattern recognition of EMG signals as a form of prosthesis control is nevertheless, the reliability of this method. This is caused by unstable EMG signals due to sweat, socket shift and muscle fatigue [24]. The interest in myoelectrically controlled prosthesis is great, subsequently diverse models and approaches to extract features and classify the EMG signals have been discussed in the literature. The integration of deep learning approaches, such as convolutional neural networks (CNNs) and efficient neural network architectures like Recursive Prosthetic Control Network (RPC-Net), have shown promise for robust and embedded prosthetic control [13,28]. Hardware-efficient neural network implementations have enabled real-time embedded control with reduced computational complexity [16]. Deep learning methods predicting simultaneous joint angles and grip force

from EMG signals are addressing the need for proportional and realistic prosthetic control [2,26]. Furthermore, several authors mentioned the benefits of using multi-modal input, where some saw potential and evaluated combinations including muscle activity input, visual information, speech input and tactile input [10]. Their approach still uses a CYK/DCG pipeline, a rule-based natural-language interface that breaks when paraphrases or new grasp types appear, replacing it with an LLM would bring robustness and scaling possibilities to additional actions.

Efforts trying to fuse visual cues and EMG signal to increase the robustness of grasp classification showed that it can outperform the individual EMG and visual classification methods without placing additional cognitive burden on the user [7,35].

Multimodal sensor fusion, combining EMG sonomyographic signals, provides improved accuracy in gesture recognition and grasp control [20]. A tactile input from a coated melamine sponge which decouples two signals, a piezo-resistive resistance for force and a thermoelectric open-circuit voltage for temperature was presented by Li et al. Integrated on a powered prosthetic, force sensing, burn warnings and voice announced material names provide sensory cues for the wearer [17]. This furnishes rich feedback that an LLM could translate into intuitive language for the wearer.

There has also been a significant leap in automatic speech recognition (ASR) which further improves voice control and its potential input for multi-modal devices. Some authors mention natural voice control as the easiest way to interact with a device [1], noting that it adds enhanced control [22] without being as cognitively demanding as other methods, while not compromising on degrees of freedom [29]. The recent work by Davila et al. shows that there is automatic speech recognition models that, rather than a classical LLM, can map spoken commands to tool actions [9]. They show that ASR can meet robustness and latency requirements, which LLMs could translate into rich personalised intent for prosthetic devices. Several recent works focus on the combination of voice control and EMG or EEG sensing [3,11,19]. Molina Arias et al. present a low-cost trans-humeral prosthesis, where joint movements are triggered by spoken keywords and force grip is regulated by head movement [19]. Reaching a 90% voice command recognition accuracy in a lab setting underlines the value of multi-modal control with voice commands, but contrasts with an LLM based interface which aims to use unconstrained natural language for real world usability. BRAVE a hybrid EEG and voice controlled prosthetic arm with a human-in-the-loop correction achieves a 96% classification accuracy across test subjects [3]. Showing progress in real-time, non-invasive multi-modal control of prosthetic devices BRAVE still depends on a fixed command palette. Which is also the case for the 19° of freedom prosthetic hand with 33 standard grasping modes controlled by voice commands recently presented by Yang et al. [32]. The single participant study presented a 95% recognition accuracy, with a voice command technology in 60 different languages, which points at the possible accessibility and personalisation for amputees.

Starke et al. present semi-autonomous control of a prosthetic hand that uses one EMG channel to trigger a CNN-based vision system, distance sensors, and an IMU to classify 13 objects and execute bio-mechanically feasible grasps, all while letting the user confirm or override each. While this architecture minimizes external hardware

dependencies, its reliance on fixed object classes limits adaptability in unstructured settings [30].

Large language models can achieve strong performance for few-shot reasoning, which makes them promising interpreters in multi-modal and voice controlled prosthetic devices.

Integrating an LLM-powered feedback loop could allow the device to generalize to novel objects and in cluttered environments. The work of [14] introduces "reasoning grasping" via multimodal LLMs for a robot to understand implicit verbal cues. This lays the groundwork for a prosthesis to become a conversational partner that reasons about object properties and user intent, enabling grasp choices for messy real-world scenes.

Innovations in visual feedback and self-supervised learning techniques significantly enhance the intuitiveness and ease of calibration processes, helping users adapt more effectively to prosthetic devices [25,33]. Robust calibration methods continue to address electrode displacement and daily variability in EMG signals [34].

Edge-computing approaches allow prosthetic systems to operate effectively in real-world scenarios with low latency and reliable on-device performance [16,34]. Systematic reviews highlight ongoing challenges and opportunities for real-time deployment and robust performance in daily use [34].

3 Methodology

For this qualitative study 13 semi-structured interviews were conducted throughout the prosthetics value chain. This included a patient care provider, three research firms, and nine manufacturers, which captured perspectives from the supply side, such as R&D and manufacturing and the clinical implementation side, such as prosthesis fit, training, and end-user adoption. End-users were consciously omitted to stay focused on organizational perspectives. Speaking directly with named groups lets us capture diverse levels of AI adoption and ideas across stakeholders giving qualitative evidence that maps onto our research question. Through web and literature searches, interviewees were identified and then approached by e-mail, web contact form, phone calls and LinkedIn. Prosthetic companies were targeted and snowball sampling was used to recruit future subjects as they were otherwise hard to reach. The interviews were conducted through remote video-calls on Microsoft Teams between January and April 2025 each lasting from 20–60 minutes with a mix of global companies and small start-ups based in the USA. Names and company identifiers were removed in all transcripts based on those recordings. Each interviewee is referred to by a unique identifier (IP1–IP13) throughout this work. The 15 interview core questions were grouped into four thematic blocks:

1) Introduction
2) Industry basics
3) Product & technical core
4) Future outlook

The transcripts were coded in MAXQDA following Braun and Clarke's thematic analysis [6]. This analytical approach facilitated the systematic identification of recurring patterns and overarching themes across participant responses. Methodological limitations include the small sample, as well as regional bias and self-selection.

4 Findings

4.1 Industry Readiness

13 prosthetic device-sector stakeholder interviews revealed a wide range of AI features adoption. Four interview partners reported the companies they work for do not have any artificial intelligence features integration at all (IP5, IP10, IP11, IP12), some confine AI to administrative tasks, whereas some were excited about the potential and only one claimed AI as a core value proposition for their developed prosthetic devices. Commercial products with the application of machine learning algorithms to control prosthetic limbs in the commercial setting is quite new, even though embedded systems in a prosthetic devices have been around for the last few years (IP4, IP8, IP9). Additionally, the algorithms for pattern recognition are often very similar. As a result there is a large research-to-practice gap that the interviewees see an opportunity and challenge at the same time. One interview partner shared that arm prosthetics today primarily offer 3 degrees of freedom (rotation of the wrist, open the hand, close the hand), whereas the human arm is capable of 26 degrees of freedom. They envisioned using AI-aided tools to add more degrees of freedom with a manageable cognitive load for the user.

Large Language Models and Voice Control. Interview partners expressed skepticism regarding the real-world applicability of LLMs for everyday prosthesis use; however, most still acknowledged their potential for controlled or specialized applications in upper-body prosthetics. Many interview partners mentioned that voice inputs are constrained to certain settings (IP6, IP7, IP9, IP11). IP11 emphasized the social implications: "I don't think people want to talk to their arm or leg when they're out and about in the community." But there is also background noise that could become a problem for voice control in public settings. An expressed concern was, that the device probably would use internet connection for LLM processing on a server and therefore can only be used in urban areas (IP9). Several other interview partners were uncertain about how effective LLM solutions would be outside of a lab setting.

IP7 described hopes for LLM solutions to alleviate cognitive load and enable access to a wider range of functionalities. They referred to a "grips arms race" among manufacturers, each adding more grasp types even though "nobody ever uses more than two of them anyway." This, they explained, reflects a communication gap between user and device, suggesting that natural voice control could help close it. Simple spoken commands such as "retrain, hand open" could enable recalibration without the need for mobile apps. Furthermore, the possibility to choose a grip via spoken commands would give "access to more functions and therefore ... benefit the user."

When asked about LLMs in prosthetics IP 10 stressed that keeping the patients' frustrations low is of highest importance.

IP5 saw benefits for LLM interaction with lower body prosthetics only for specific movements such as sitting down or walking up stairs.

4.2 Personalisation (Personalised Control and Device Matching)

Practitioners highlighted across interviews that next-generation control should be personalised. LLMs would allow individual vocabularies and routines rather than forcing amputees to memorize fixed commands, it would allow to translate clinical speech into user friendly language. Many multiarticulated hands today support up to 30 different grasps. The access for those is buried behind phone menus. According to IP9, users have to "pick up the phone, log in to...,then go to the app and then find the grip and then choose it." Subsequently, many users settle for just one or two default grips. The greatest promise was therefore seen in an LLM-powered layer that could reconfigure and choose those grips conversationally. IP9 pictured a voice-enabled upper limb prosthesis where the LLM lets users name a grip in their own words: "the ability to customize those commands might make it easier and less cognitive load" for the user. In today's products the user can choose custom grip patterns, as IP12 explained: "..customers can always build their custom ... grip patterns to allow them to do things ... that haven't ... made it into the product." This would strengthen the autonomy they already enjoy through choosing custom grip patterns aided by current apps. Voice recognition was highlighted as a current focus by IP7. They saw speaker dependence as key to a personalised voice control of prosthetic devices. The importance of speaker dependence was also mentioned by IP9, as they underlined a prosthesis being a very personal device. Preventing other people from voice-activating the device is crucial.

"human-in-the-loop" Digital Diagnosis. An application of AI in personalization of human attached devices lies in the personalized device recommendation. We identified a need of AI enhanced appropriate prosthesis recommendation through the interviews. An AI model could take into account a user's medical history, detectable muscle signals, signal strength, gait patterns (for lower body prosthetics), and lifestyle, this coupled with the expertise of a Patient Care Provider could recommend the most appropriate prosthesis for a user. IP1 called this "to have sort of a road map to say these are the systems that have been successful for people in the past,.." and stressed "it's ... going to depend on the person and their capabilities and their wants and their needs". A better match between the user and the device increases the adoption potential of the prosthetic device.

Nowadays lower body prosthetics incorporate AI backed analytic systems based on integrated IMUs or pressure sensing insoles that can analyze foot pressure distribution, gait patterns, step timing and force. Through these AI generated reports and recommendations, clinicians save time and can create better fitting devices for users. This existing technology allows for the "human-in-the-loop" digital diagnosis which were discussed multiple times in interviews.

IP3 mentioned their employer having already started using AI-driven technology with the fitting alignment process to develop new sockets, but also "the placement of the foot beneath the socket in relation to where the individual stands" and expanded this to gait, by looking in the future "... you're going to have proper gait ...". This is based on a trial virtual foot that recommends the best model for someone's gait. IP3 explained "... you could try ... 10 different models without ... [attaching] the actual foot onto a prosthesis... you could find out what works well for your gait." This enables a more

precise alignment between the user's needs and the prosthetic configuration, thereby increasing the likelihood of successful adoption and sustained use.

IP2 presents their mobile-app which gives AI reviews based on calibration data and ranks the most serious issues. It then presents actionable suggestions for the user to fix: "We try to give them bite sized pieces of suggestions that are easy to interpret and easy to implement ..." An LLM could turn this into conversational coaching by also answering questions on the spot and keeping the user motivated.

4.3 AI-Augmented Training and Rehabilitation

The capabilities of a prosthesis are only useful if the user is able to harness them fully. The AI-enhanced devices offer many new capabilities and help users complete more motions, but the participants stressed that it is important to keep the cognitive load low. IP4 mentioned "We're getting to a point now with controls that were exceeding the patient's ability to control them." further they stated "The question I have as a clinician ... is, 'Is the patient working for the arm or is the arm working for the patient?". So it becomes clear that achieving an effective adoption of smart devices and to reach their full potential, proper training and therapy are essential.

To help users become more proficient with their devices there is the possibility of specialized exercise programs. Training plans that can adapt in real-time and over the course of the training plan to adjust for performance would be ideal. IP2 called this the "holy grail ... really hard to do". It is envisioned to increase coordination and muscle control unique to the requirements of the device and user. Subsequently, the user can expect lower cognitive demands when operating the device.

LLMs would allow redefining fitting training from a handful of clinic-based sessions to an ongoing dialogue between the user, device and clinician. IP2 presents their intentions on game-based calibrations and training: "We can collect game data through ... a mobile app and we need a platform to be able to actually build algorithms off of that data." To make the process more engaging they gather movement data based on games and furthermore envision an AI-aided feedback and recommendations system for the user to improve their control of the device. This is where an LLM could improve user engagement and turn a numeric issue list into user friendly language for coaching and feedback.

Another potential mentioned by interviewees for AI-guided training is teaching the user how to prevent damaging the device. Prosthetics with many degrees of freedom, especially upper body prosthetics, which allow individual finger motions, do have many joints leaving the device at risk of damage when pressure is applied. This was stressed by IP9 "Some of the multi-articulated hands ... they're ... fragile." Missing force feedback makes wearers unable to gauge the load they place on the device, which was supported by IP11: "They ... support their weight on the arm-rests ... not realizing they're putting a ... large load on the components ...". Lower-limb devices already include embedded algorithms which act as a safety net. For example when rapid flexion moments are detected the joint will lock, which can prevent a fall or mechanical failure due to excessive load (IP5). Clinicians see their role as support for technological questions or questions concerning grips. But the safe use still depends on the training with the occupational therapists and the users own judgment on what forces the device

supports in which configuration. AI-aided training tools can extend that judgment by acting as a digital "coach" that can intervene any time of the day, something a human trainer cannot do.

4.4 Multimodal Intent Recognition (and Robustness)

Industry interview partners widely agreed that EMG signal is user-friendly, and despite the researched use cases for multi-modal input for prosthetic devices, commercially available devices with multi-modal inputs are rare. However, the unreliability of prosthetic control with EMG signals remains and strengthens the case for additional inputs. When asked why there are no other inputs used, IP12 stated that the cognitive load various inputs place on the user is deemed too high. Users can become overwhelmed and stop using the device altogether. IP4 asked "is the device working for the user, or the user working for the device?" Furthermore, gadget competency of the user is often low. Simple body-powered devices or even aesthetic replacement of the missing limb are valued higher. IP4 mentioned "Prosthetics would be so easy if it weren't for the patient". One manufacturer said, it is not a question of "does the technology exist to make this feature possible?" but rather "is this technology value-adding enough to make this application worth it?" Both are therefore stressing the user-centric focus that needs to be kept. Nevertheless, the discussed multi-modal input options and potential use cases for prosthetic devices were rich. The majority of interview partners agreed on its potential, just cautioned the implementation and user capabilities, which could hinder practical applications. IP5 stressed multi-modal inputs are an effective way to give users more autonomy. One example discussed by IP10 is the LUKE Arm, which includes gyroscopes, accelerometer and additional control through the user's foot motions [27]. IP2, IP6 and IP7 discussed images as additional input for prosthetics to support predictive movement. IP6 envisions to "use cameras in the prosthetic that can see the environment you're in so it can know that it is holding a glass or you are reaching for an elevator button." Nowadays this control and feedback is given with additional physical buttons (IP11). IP10 and IP13 discussed physical buttons coupled with pattern recognition for grip switching.

Furthermore, temperature sensors were mentioned by IP12. Sensory feedback for the user is a shared interest of most interview partners. This would also contribute to reducing the cognitive load, as users do not have to rely solely on eyesight to monitor the device. IMUs and gyroscopes, accelerometers were mentioned by IP2 and IP6, where accelerometers were researched for help sensing the spatial position of the arm and therefore aid in switching modes accordingly.

4.5 Physical Integration, Socket Fit and Comfort Feedback

There was a near unanimous agreement that the socket remains the top area for improvement when talking about prosthetics. The socket is the interface between the residual limb and the prosthesis. This focus on the socket is summarized by the statement of two interview partners (IP12 and IP13) "control is only as good as the socket." Where IP8 meant that focusing on device improvement is "as distraction from the real issue." IP7 stressed the fact, that "your limb will change volume both throughout the day and over

long periods of time" and that prosthetic device users address this by putting on more prosthetic socks.

Intelligent Fit Driven by LLMs. IP 9 discussed how natural language feedback from the user, which is then interpreted by an LLM, could control an action directed to the socket to increase the user's comfort regarding the socket fit. But the leap to a real-time LLM socket adjustment is an open frontier, because of latency, power and safety requirements. There is a risk of unintended device behaviour that would need to be eliminated before such a solution becomes viable.

User Controlled Air-Pump Sockets. As there is a lack of development in advanced socket technology. Users and practitioners frequently value robustness over sophisticated but failure-prone technology. IP9 did mention that "Sometimes simple is better." but they also think "there is definitely room for improvement" concerning the availability of the prosthetics markets technology.

A user controlled air pump adjustable socket is already offered commercially (IP8). The user can manually pump or activate a button to pump air in or out of the space between the residual limb and socket. This requires the user to have some tactile feedback at the residual limb as well as a trained understanding of good socket fit to judge the correct pressure. This then allows the user to control the socket fit, and discomfort or dangers of poor fit are alleviated. If EMG signals are used for device control, the reliability of the signals is increased by better socket fit (Ottobock, 2025). In some cases, it is difficult for the user to judge their device's socket fit. Especially new amputees struggle with judging good socket fit, but also older patients and patients with reduced sense of touch at the residual limb. This can result in skin problems as well as reduced device control.

IP7 presented their ongoing project, which is pressure monitoring within the socket. The goal is to alert the user when they need to change their fit.

AI-Guided Design from Residual Limb Scans. IP3 mentioned that they want to use technology to mitigate the subjectivity of socket fitting, as the recommendations of prosthetists will depend on their opinions and expertise. Clinical subjectivity leads to inconsistent outcomes and limited reproducibility, so quality drops in low resource settings or when a less-experienced prosthetist does the adjustments. IP3 and IP7 argued that scans of the residual limbs of many patients could train a model to propose accurate designs for personalised sockets, beyond the experience of any individual prosthetist. IP3 argued a human-in-the-loop would remain for verification and adjustment of the AI models design proposal. Nevertheless most respondents were sceptical. All 11 other interview partners were primarily against scanning and automizing socket design. According to IP1 this conviction comes from the understanding that a scan cannot differentiate between soft and bony tissue, therefore human touch and the human understanding of how pressure will change the shape and size of the residual limb is still critical to create a quality socket. An effective model would require access to proprietary information spanning across multiple manufacturers (IP13), this raises technical

and commercial barriers. Data-driven design therefore promises objectivity, but its feasibility remains uncertain.

Osseointegration. There was also a small subset of participants who highlighted osseointegration (IP4, IP9, IP10). For this a metal implant is anchored in the bone of the residual limb. IP4 and IP9 mentioned this as a way to avoid socket fit complications. Reported benefits include improvements in prosthetic use, prosthetic mobility, global health and pain reduction, as this technology eliminates the challenges such as skin breakdown and poor fit [23]. Further benefits can be "osseoperception", which allows users to feel vibration or pressure from the implant [12]. Drawbacks are challenges such as infection or the inability to detect integration or the loss thereof [23]. IP4 envisioned connecting the sensors subcutaneously and through osseointegration.

It becomes clear that the interviewees place the fit of the socket as the most important issue in the adoption and performance of prosthetics. However, the opinions on the optimal solution diverge. LLM-mediated and sensor-based adjustment promises personal agile control. Manual air-pump sockets are a pragmatic fix that works, scan-driven socket fit leveraging AI aspires to eliminate clinician subjectivity. Finally osseointegration bypasses socket fit problems entirely at the cost of surgery and the risks attached, such as infections. There is not a definite answer, and until then progress in upper limb prosthetic devices will be constrained by the issue of socket fit.

4.6 Technical and Human-Factor Constraints

Interview partners discussed the challenges of the required uniqueness of every single patient and the prosthetic device which matches their needs best. One theme discussed heavily is the wide range of gadget competence the users have. Amputations might become a reality for some patients due to peripheral artery disease, diabetes or other conditions which are especially prevalent in the elderly population. This population of amputees are less likely to accept a new device if it requires high levels of training. According to interviewees who interact with patients, congenital patients are not likely to invest training time and money into a prosthetic device. They often have learnt to move through life without a limb and manage well. Voice commands in a public situations were deemed impractical by interviewees due to the social implications. Further the cognitive load on the user is lower in the lab setting and the tasks specific. These are challenges mentioned when it comes to the transfer from the lab setting to everyday life with advanced prosthetic devices. Interviewees furthermore fear unreliable device behaviour which could endanger the environment and users of the prosthetic device.

4.7 Psychosocial, Ethical, and Global-Equity Dimensions

AI's Potential in Psychological Adaptation and Emotional Support. While much focus has been placed on physical and functional capabilities of AI-enhanced prosthetics, interviewees consistently pointed out the critical yet overlooked emotional and psychological dimensions. Users often struggle with emotional barriers or self-confidence issues that lead to device abandonment, despite technological advancements.

IP6 emphasized distinct phases in user adoption—from initial emotional reactions upon regaining movement to later exploration and eventual plateauing of use. Integrating AI and LLM-powered systems to provide emotional support or coaching could ease these transitions, maintaining motivation, facilitating coping strategies, and reducing frustrations that arise during adaptation. A conversational AI might also proactively detect signs of emotional or motivational decline based on user interactions, alerting clinicians to intervene earlier and more effectively.

Ethical and Privacy Considerations in AI Prosthetics. With increased integration of AI, especially cloud-based services and personalised user profiling, ethical and privacy considerations gain urgency. IP9 underscored concerns about the dependency of AI prosthetics on connectivity, introducing potential data security risks and privacy vulnerabilities. Users' personal movement data, medical histories, and even voice samples for personalised control require stringent protections. As prosthetics become more connected, the necessity for transparent data handling policies, user consent, secure edge-computing models, and robust encryption standards intensifies. Furthermore, balancing personalised data collection and processing with user comfort regarding surveillance and data ownership becomes paramount. The industry faces the challenge of fostering trust through transparency, ensuring privacy without limiting beneficial AI-driven personalisation.

AI in Accessibility and Global Equity of Prosthetic Solutions. The global prosthetic market is marked by profound disparities in access, affordability, and technological availability, as highlighted by IP3 and others. While AI-driven solutions hold promise for enhanced functionality, these advancements frequently target high-income regions and complex devices requiring intensive clinical support. However, some companies aim to leverage AI to democratize prosthetic access, creating standardized yet adaptable designs, streamlining remote diagnostics and fitting, and reducing dependency on highly specialized local expertise. IP3 described their AI-assisted alignment and fitting systems as potentially transformative for low-resource regions, improving quality and consistency of prosthetic care even without extensive local expertise. Future AI innovation in prosthetics must therefore balance advanced personalised technologies with robust, scalable, and affordable solutions capable of bridging global access gaps.

5 Future Work and Research Directions

For future prototypes it is important to quantify cognitive load as the interviews confirmed that adding inputs can back-fire if the interface is complicated. Interview partners expect LLMs to become the natural interface between user and device. Yet social situations, personalised access and connectivity remain a worry. In future research it is important to look at speaker-dependent systems and study the social acceptance of voice commands in public spaces. Furthermore, embedded deployment of LLMs needs to be looked at so dependency on cloud access can be eliminated. Training on the fly and continuous self-labelling were described by interviewees. To further this avenue research

needs to deploy continual-learning pipelines that update control maps without explicitly prompting the user. Restoring embodiment through haptic feedback loops remains an unmet need. This can be achieved through pressure or vibration cues. Feedback with LLM can transmit sensations in plain language, another avenue for future work. Interviewees argued that "... Control is really only as good as your socket fit ..." (IP13). Generative models can be trained on residual-limb scans, to then propose printable and load balanced socket geometries. Integrated pressure sensors should be studied to allow fit adjustments during usage of the device. To achieve a more diverse range of views on the researched topic, future research should span across diverse regulatory and cultural contexts and prioritise metrics and features that matter most to amputees.

6 Limitations

Sample Size. There are only around 8–12 industry leaders and few of them are working at the forefront of AI-enabled prosthetics. Despite persistent outreach, limited replies received from the major companies lead to 13 interviews, this sample size is too small for a statistically significant conclusion.

Regional Skew. This study has a concentrated regional perspective border-transcending problem of limb loss. 10 out of the 13 interview partners were located in the United States. User needs, perspectives on technology, user's device adoption rates, as well as medical device regulation and technology regulation in general differ significantly depending on the geographical region. This analysis therefore largely reflects United States practices and perspectives.

Early Phase of AI and LLM Deployment. Most main industry leaders are currently not deploying AI- and LLM-driven features. The potential offered by AI-driven features in prosthetics and the discussions thereof is subsequently still theoretical. For the interview partners which work at companies whose mission is related to the topics artificial intelligence or large language models, commercial availability of such products are minimal.

Proprietary Data and No End-User Voice. Proprietary information was not shared in these interviews. Interview partners, which were confined to companies, where therefore limited in the detail of which they could talk about the topics covered in the interviews. Furthermore, end-users of prosthetic devices were not represented and their daily experience or perspective on cognitive load were not captured.

Data-Governance and Bias Risks. Biosignals such as EMG or inertial-sensor streams reveal muscle activity and gait patterns, which includes the users identity. Training multimodal LLMs on biosignals therefore raises privacy concerns (GDPR/HIPAA/EU AI Act/EU MDR AI Annex), which were outside of the scope of this paper but affects real-world deployment of these technologies. The user of AI-enhanced prosthetics for under-represented groups could be affected by slower responsiveness or higher error rates for their devices.

7 Conclusions

How can modern AI, especially LLMs, be harnessed to deliver truly user-centred, multi-modal prosthetic control without increasing cognitive burden? The 13 semi-structured expert interviews across the prosthetic value chain and literature synthesis showed three converging needs of *(i) reliable multi-modal intent recognition* beyond the widespread surface EMG solution *(ii) AI-guided socket design and real-time fit optimisation* and *(iii) intuitive low-cognitive-load interaction* which can be achieved by conversational interaction. LLMs emerge as a promising solution for these needs, as they provide the basis for the conversational interaction, such as the command layer, personalised coaching during rehabilitation and calibration, universal access to otherwise hidden settings and a possibility to harmonise multi-modal sensor input. It became evident that there is a substantial research-to-practice gap, which is developed due to a widespread fear of over-complexity. The adoption of AI in prosthetic devices and of these companies spans from none to AI-enhanced prosthetics being a core company value. Socket fit is the most limiting factor for improved prosthetic devices. AI-assisted design iterations are promising avenues for socket improvement. Comfort and robustness of attachment and therefore prosthesis control can be improved by socket monitoring. Cognitive load is fast reached in today's devices with their various possibilities for mode switching. Improved personalised control and device matching can be reached by "human-in-the-loop" diagnostics and calibrations. For this AI-augmented and game-based training and rehabilitation are promising.

7.1 Implications

For Research. LLM mediated interfaces that are used in combination with traditional EMG control should be prioritized to develop voice and context aware interfaces which allow individual and personalised interaction with the device.

For Industry. To reduce trial and error and achieve consistent quality results AI-driven fit analytics should be integrated into routine practice. With AI-mediated and LLM-supported voice control training can be shifted from episodic clinic visits to continuous coaching and calibration. User-centric design needs to be prioritized, which implicates speaker-dependent voice control. The switching between modes should be realised with minimal cognitive burden.

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Facial Gesture Detection for Individuals with Reduced Hand Mobility Using Graph Neural Networks

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Abstract. This paper addresses the limited digital accessibility faced by individuals with reduced hand mobility and explores how to overcome it. An efficient facial gesture recognition model is proposed, designed to operate based on facial landmarks (Facial Mesh) extracted using Mediapipe's Face Mesh technology. The model incorporates a graph neural network (GNN) with a lightweight architecture and low computational complexity to detect facial gestures on the user's face, enabling smooth execution on low-performance computers without significantly compromising user experience. The work presented may be of interest to researchers focused on digital accessibility for individuals with motor impairments in their hands.

Keywords: Eye Tracking · Facial Gesture Recognition · Human-Computer · Interaction · Assistive Technology · Graph Neural Network

1 Introduction

In recent years, human-computer interaction (HCI) has evolved beyond traditional peripherals, enabling new forms of control through computer vision technologies [1]. Among these, facial gesture recognition has gained prominence as a natural and contactless means of facilitating digital accessibility [2]. This approach is especially promising for people with reduced mobility or in contexts where the use of hands is not feasible. However, the mass adoption of these technologies still faces practical barriers, both technically and economically.

Currently, only 3% of people with disabilities in developing countries can afford technological tools that mitigate the challenges caused by their impairments [3]. This accessibility gap is accentuated by the high cost of assistive devices such as commercial eye tracking systems. A clear example is the Tobii Eye Tracker 5, one of the most recognized products on the market, whose price exceeds $350 on sales platforms such as Amazon [4], making it difficult to acquire in low-income contexts.

Beyond its high cost, this specialized hardware also demands precise physical conditions to function properly, such as user alignment and constant distance, as well as custom calibrations and controlled lighting [5, 6]. In contrast, methods based solely on

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computer vision, which use conventional cameras, offer a more economical and scalable alternative if they manage to maintain adequate levels of accuracy and robustness.

The present work proposes a solution to this problem through the development of HCSystem, a graph-based facial gesture recognition model. The central proposal is to model the facial mesh as a three-dimensional graph, in which the nodes correspond to 3D anatomical landmarks defined by MediaPipe, while the edges capture spatial relationships between regions such as eyes, eyebrows, glabella forehead, and mouth. This allows the facial structure to be captured in a more semantic way and processed with Graph Neural Networks (GNNs), specifically with an architecture based on GPSConv and GINEConv.

To train and evaluate the system, a proprietary dataset was constructed composed of 32,000 samples distributed among six classes of facial gestures, including conditions with and without glasses, as well as subjects in different age ranges. Each sample was represented as an undirected graph with normalized attributes to ensure invariance at scale. The model was trained from scratch and then subjected to a fine-tuning stage, in order to compare its generalizability under varied conditions with other models based on different architectures.

Experimental results show that GPSNet significantly outperformed the base architectures, reaching an accuracy of 99.84% over the test set, with an F1-score $\geq$0.98 in all classes. In addition, during the fine-tuning stage, a sustained superior performance was observed with respect to GCNNet and SAGENet, which evidences the effectiveness of integrating global and local topological relationships in the classification of facial gestures.

In this way, HCSystem demonstrates that the structured representation of the facial mesh as a 3D graph, processed by modern GNNs, constitutes an accurate, robust and adaptable alternative for the detection of facial gestures. This approach has the potential to be scaled up to other HCI domains, especially in digital accessibility, health, and assisted education settings.

This paper is structured as follows: Sect. 2 reviews related work. Section 3 describes the proposed HCSystem model. Section 4 details the experimental setup. Section 5 presents and discusses the results. Finally, Sect. 6 concludes the paper and outlines future research directions.

2 State of Art

This section presents a Targeted Literature Review (TLR), aimed at identifying scientific studies that have applied vision-based artificial intelligence for human detection through facial features. The search was carried out in scientific databases such as SpringerLink, IEEE Xplore, and Web of Science. The following search strings were used to locate relevant works: (“graph neural network” AND “human feature” AND “face detection”) and (“computer vision” AND “facial tracking” AND “gesture recognition”).

The inclusion criteria were as follows: (1) the study had to focus on human detection using computer vision; (2) it had to implement either Graph Neural Networks or Convolutional Neural Networks (CNNs); and (3) the proposed method needed to be applied in real-world conditions. The exclusion criteria were: (1) studies published before 2023; (2)

works published in journals not ranked in Q1 or Q2; and (3) purely theoretical research without practical implementation. From an initial pool of 80 candidate papers, 8 were selected for detailed analysis based on these criteria.

2.1 Graph Neural Networks

In this paper, Wu et al. [7], presented DepMGNN, an architecture based on graph neural networks that introduces a novel approach called Matrixial Graph, where each node directly represents 2D facial feature maps obtained from the frames of a video. Unlike previous methods that transform each segment into fixed vectors, this proposal allows the spatial and temporal structure of facial gestures to be preserved throughout the video without the need for resampling. The experiments showed that this architecture reached new reference values in the Audio/Visual Emotion Challenge (AVEC) 2013 and AVEC 2014 datasets, achieving a root mean square error of 6.62 and a concordance correlation coefficient of 0.80, far outperforming long-short-term-memory, graph attention network (GAT), and gated graph convolutional network (GCN) -based models.

In this study, Zhang et al. [8] proposed a novel approach to the estimation of facial age using a neural network of graphs called Latent Relation-Aware Graph Neural Network with Initial and Dynamic Residual, which incorporates a multi-head attention mechanism to capture latent relationships between facial regions. To do this, the images are segmented into patches that act as nodes, employing facial key points as prior knowledge to construct an initial graph, which is then optimized using a random walk strategy. In addition, a deep residual Graph Convolutional Network (GCN) is introduced that merges adaptive initial residues and dynamic developmental residues to avoid overwhelm, while age estimation is improved through progressive reinforcement learning, which combines classification by age groups and continuous regression. The experimental results show that the model achieves a mean absolute error (MAE) of 1.79 years in the Morph II set (protocol I), an MAE of 2.14 years and a cumulative score of 91.6% in the Face and Gesture Recognition Network Aging Database, and an aε-error (normal score) of 0.258 in the ChaLearn LAP 2016 dataset, outperforming several state-of-the-art methods with only 13 million parameters.

Facchi et al. [9] analyzed the effectiveness of graph neural networks in the representation of 3D facial morphology, using different methods to identify the nodes of the graph from key facial points. Three approaches were compared: a set of 50 anthropometric points manually placed by experts, 84 facial points automatically detected using the Multi-view Facial Landmarking model, and a set of 128 evenly distributed points without semantics using Farthest Point Sampling. To evaluate the capability of these approaches in biometric tasks, two benchmarks were used: gender classification and age regression, applying five different GNN architectures, including PointTransformerConv and DynamicEdgeConv. The experiments were carried out on three databases: Facial Dismorphism Database (FDD), Facescape and DAD-3Dhead. The best results were obtained with the PointTransformerConv model, reaching an F1 score of 94.54% in gender classification and a minimum MAE of 4.15 years in age estimation, both on the FDD set using 84 automatic landmarks.

Lin et al. [10] proposed a new graph-based neural network model called GRASNet, with the aim of improving the recognition of human actions and the assessment of well-being in smart industrial environments. The problem addressed lies in the difficulty of carrying out accurate and real-time monitoring of the physical and mental state of workers without resorting to invasive portable devices. To do this, an architecture composed of dual aggregation mechanisms (attention and local sampling) was used together with residual connections, working on skeletal data extracted from the NTU RGB + D 120 dataset. The experiments carried out in two subsets (manu-6 and manu-15) showed that GRASNet significantly outperformed models such as GCN, GAT and GraphSAGE, reaching an F1-score of 0.90 in manu-6 and 0.84 in manu-15, with an average inference time of 0.17 ms per sample, which positions it as an accurate, robust solution suitable for real-time applications.

In summary, the four reviewed papers demonstrate the growing potential of graph neural networks to address various challenges in the interpretation of complex human signals, both visual and structural. While Wu et al. [7] introduces a novel matrix-like representation to preserve the spatiotemporal dynamics of facial gestures, Zhang et al. [8] proposes a hybrid strategy of attention and random walking that optimizes age estimation with high accuracy and low computational cost. For their part, Facchi et al. [9] show that the choice and density of nodes significantly influences the biometric performance of GNNs, highlighting the effectiveness of automatic configurations of landmarks in 3D morphologies. Finally, Lin et al. [10] transfers these advances to the field of recognition of human actions, proposing a robust and efficient architecture for real-time monitoring. All these approaches share the same vision: to exploit the non-Euclidean topology of human data to improve the accuracy and adaptability of models, which lays a solid foundation for the development of accessible, accurate and low-cost interfaces in contexts such as that of this research.

2.2 Facial Tracking

Zhu et al. [11] presented a novel facial tracking system by integrating many sources for the model's self-learning, to cope with the analysis in adverse environments and high annotation costs. All this is addressed within the problem of accurate monitoring in environments under occlusions or conditions that are not favorable to perform the work correctly. To do this, the modules of Temporal Reasoning of Knowledge (TemRest) and Interactive Distillation of Knowledge (KnowDist) were essentially used, the first focuses on making a consistent cyclical monitoring using self-learning and the second to transfer knowledge from an already established model, to improve the stability of the solution.

Kim et al. [12] developed a study related to the extraction of facial expressions closely related to depressive symptoms in older adults, focusing on both "posed" and "spontaneous" emotions. All this making use of the Facial Action Coding System (FACS). The problem they address focuses on the difficulty of evaluating older adults due to the limitations of evaluations, as well as the denial of the patients' symptoms, all of which makes the result not as truthful as possible. A wide variety of tools were used, including the Beck Depression Inventory (K-BI-II), FACS and OpenFace 2.0; where depressive symptoms are measured, facial expressions are analyzed, and in addition to the analysis the units of facial action, respectively.

Vilchis et al. [13] conducted a thorough analysis of existing methods and solutions for facial capture and tracking in digital humans. Since there are challenges when taking an accurate capture of facial movement as well as in emotional interaction in real time, in order to achieve a better perception of realism, empathic response and interactivity. For the different investigations, RGB-D cameras, infrared cameras, high-resolution devices, as well as Apple's ARKit or OpenFace 2.0 software were used. Among the databases used, MMI, CK+, DISFA+ stand out, likewise, the action units (AUs) were taken as a variable for the coding of the expressions and reference points.

Tian et al. [14] implemented a robust system for stable capture of facial movements, with a focus on improving accuracy and stability for subsequent 3D animation. The article delves into the obstacles associated with the loss of markers in facial expressions, caused by occlusion or blurring that may occur; so manual intervention is necessary to cope with these limitations. For the development of the system, the use of high-definition cameras (HMCs), a combination of robust optical flow (RLOF) and Marker-YOLO was used; for the capture of the movements of the participants, for the location of the trackers and for the detection of the objects based on the information given, respectively.

Collectively, the studies reviewed in this category address the main challenges in face tracking, from accuracy in harsh environments to stability in dynamic shooting for clinical or animation applications. Zhu et al. [11] and Tian et al. [14] focused their proposals on overcoming the technical limitations of facial tracking in the face of occlusions or loss of markers, using self-learning models or integration of techniques such as robust optical flow and deep learning detection. On the other hand, Kim et al. [12] and Vilchis et al. [13] they delve into expressive analysis, highlighting the usefulness of tools such as FACS, OpenFace 2.0 and RGB-D cameras to measure emotional states or generate empathic responses in digital humans. In all cases, the use of action units (AUs) as a common axis of facial coding allows a robust comparison framework to be established. This research shows that advances in computer vision hardware and techniques are key to achieving stable, accurate and applicable facial tracking both in clinical contexts and in the construction of more natural and accessible interfaces.

3 Proposed Model

This section contains the model proposal for the detection of facial gestures.

3.1 Architecture Selection

To determine the most appropriate graph neural network architecture for the present development of the model, a comparative benchmarking process was carried out between different alternatives. This analysis considered the following criteria:

- AP: A measure that combines recall and accuracy to obtain classified retrieval results [15].
- Precision: The result of dividing the number of true positives by the total number of positives [16]
- Documentation: Amount of accessible architecture documentation. Take into account if it is documented in bookstores, has a repository and if it is publicly accessible.

The architectures evaluated were the following:

- ESA: Architecture based entirely on attention mechanisms that learns representations of graphs by directly considering their edges as fundamental units [17]
- GCN: A family of architectures designed for information organized as graphs, i.e., where data has explicit relationships with each other [18]
- GIN: Architecture whose capacity for representation achieves the same discriminative power as the Weisfeiler-Lehman isomorphism test (WL-test) [19].
- GraphGPS: Architecture that combines the local passage of messages with mechanisms of global attention, such as transformative [20]

The possible scores for each criterion were defined as follows: low (0 points), medium (1 point), high (2 points). These scores were assigned to each criterion by architecture after evaluation using the limits shown below shown in Table 1.

Table 1. Score distribution per criteria.

Criteria	Low	Medium	High
AP	<0.60	0.60–0.69	>0.69
Precision	<84.5	84.5–87.5	>87.5

The Documentation criteria scores were more subjective, as it was subject to the amount of information publicly available for each architecture.

- Low: Scarce documentation, little community or practical examples.
- Medium: Acceptable documentation with tutorials, but fragmented.
- High: Complete documentation with clear repositories and examples.

By using these limits, the scores obtained from benchmarks carried out by various authors with the Peptides-func [21] and NCI1 [22] datasets were used. In this way, Table 2 was prepared, in which it is observed that the AP and Precision criteria have a representative percentage of 30.77%; and that the documentation criterion obtained a percentage of 38.46%, which suggests that it would have more weight in the benchmark detailed below. These percentages will soon be used as Impact for the calculation of the Adjustments.

Table 2. Percentual impact per criteria.

Criteria	ESA	GCN	GIN	GraphGPS	Count	Impact
AP	1	0	1	2	4	30.77%
Precision	2	0	1	1	4	30.77%
Documentation	0	2	1	2	5	38.46%

Once the Impact values by criterion were obtained, Table 3 was prepared. This table contains the evaluation of architectures using a benchmark. In this benchmark, the Score and Adjustment by Criteria of each architecture were used to find their Total Score and Total Adjustment. These values were added together and the Final Score for each architecture was obtained.

Table 3. Graph neural networks benchmark.

Criteria	Impact	ESA		GCN		GIN		GraphGPS	
		Score	Adjust	Score	Adjust	Score	Adjust	Score	Adjust
AP	30.77%	1	0.308	0	0.000	1	0.308	2	0.615
Precision	30.77%	2	0.615	0	0.000	1	0.308	1	0.308
Documentation	38.46%	0	0.000	2	0.769	1	0.308	2	0.769
Total	100.00%	3	0.923	2	0.769	3	1.000	5	1.692

Below are the formulas used for the calculation of Adjustment, Total Score, Total Adjustment, and Final Score respectively.

$$Adjustment = Impact \times Score$$

$$Total\ Score = AP\ Score + Precision\ Score + Documentation\ Score$$

$$Total\ Adjustment = AP\ Adjust. + Precision\ Adjust. + Documentation\ Adjust.$$

$$Final\ Score = Total\ Score + Total\ Adjustment$$

This evaluation resulted in ESA having a Final Score of 3,923; GCN, 2,769; GIN, 4,000; and GraphGPS, 6,692. Thus, it was determined that the most promising architecture for the present case is GraphGPS. However, on this occasion it was decided not to use the full architecture, but only its GPSConv component, due to the nature of the problem addressed: supervised classification of facial gestures in small graphs and fixed topology. While GraphGPS was designed for complex prediction tasks in molecular and large-scale graphs, combining global attention mechanisms with local geometric operators, the computational complexity and the representation capacity of the complete model are unnecessary for the proposed context. GPSConv, on the other hand, retains the ability to capture rich structural dependencies by generalized convolutions, making it a suitable and efficient component for working with the facial structure derived from 3D meshes. By dispensing with the global encoder and other additional layers of GraphGPS, a significant reduction in the number of parameters and training time is achieved, without compromising the accuracy of the model in this specific task.

3.2 Model Design

The proposed model receives as input a graphic representation of the human face, constructed from the 478 facial reference points estimated by MediaPipe. From this set, only those relevant points defined by the anatomical structure of the ocular areas, eyebrows, forehead, glabella and mouth were selected, resulting in a subset of interconnected specialized nodes. Figure 1 shows the position of the facial landmarks used for gesture detection on a face without expressions and without glasses.

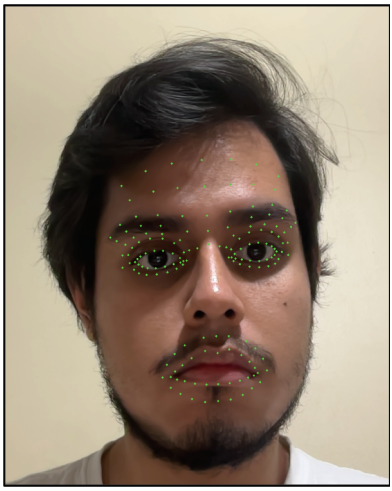

Fig. 1. Facial landmarks detected without glasses.

Figure 2 shows the same landmarks, but on a face without expressions with glasses. Each node of the graph represents a specific facial point and is described by a vector of three-dimensional features (x, y, z) normalized in a Euclidean space.

The edges of the graph were manually defined based on the connectivity between the points offered by MediaPipe and grouped by facial region. In this specific case, edges can be defined as connections between landmarks that have a value that represents the distance between them [23]. Each edge is assigned as an attribute the scalar value corresponding to the Euclidean distance between the two nodes it connects, thus capturing geometric and proportional relationships between parts of the face.

Formally, each input sample is modeled as an undirected graph, where: $G = (V, E)$

- V It is the set of facial nodes with their three-dimensional coordinates.
- E it is the set of edges defined by anatomical relationships.
- Each node is associated with a vector. $v_i \in Vx_i \in \mathbb{R}^3$
- Each edge is associated with an attribute. $(v_i, v_j) \in Ea_{ij} = ||x_i - x_j||_2$

This representation allows the model to not only learn local features of each facial region, but also their relative spatial relationship, which is essential for a correct discrimination of subtle facial gestures.

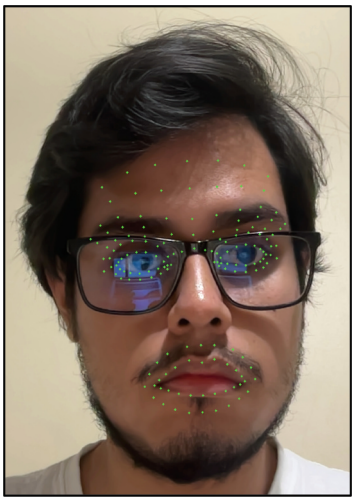

Fig. 2. Facial landmarks detected with glasses.

This model was implemented as a neural network of graphs composed of multiple processing stages. The architecture begins with a linear projection layer that transforms the three-dimensional coordinates (x, y, z) of each facial point into a latent representation of 128 dimensions. This layer allows to increase the expressive capacity of the model by mapping the input vectors to a space of greater dimensionality.

Next, two consecutive blocks of type GPSConv are integrated. Each of these blocks combines a local component, based on a GINEConv-type graphical convolution, with a global component that uses multi-head attention mechanisms (4 heads) on all the nodes of the graph. The local component is responsible for propagating information between directly connected neighbors, while the global component enables long-distance interactions between non-adjacent nodes, capturing holistic facial patterns. The GINEConv module used in each block uses dense networks with ReLU activation, and is designed to integrate edge attributes. In this case, this attribute was the Euclidean distance between facial points.

Subsequently, a global mean pooling operation is applied to obtain a unique vector representation per graph, which summarizes the global characteristics of the complete face. This technique is suitable for graph classification tasks, such as detecting gestures in facial meshes.

Before the exit stage, a Dropout layer is introduced with probability in order to mitigate overfitting during training. Finally, a linear layer of output projects the latent vector into the class space, allowing one of the six possible categories of facial gestures to be predicted $p = 0.2$.

3.3 Dataset

The dataset was collected from six participants spanning childhood to older adulthood: a 10-year-old girl (parental permission obtained), a 16-year-old adolescent (parental

permission obtained), two masculine 21-year-olds, one masculine 22-year-old, and a masculine 54-year-old man. For each participant, we recorded samples with and without eyeglasses.

The 10-year-old (1) and the 22-year-old (2) were unable to perform a voluntary wink; consequently, no left-wink or right-wink samples were collected for them. All sessions were captured under the same conditions: participants were seated in front of a plain white wall (as background), with the built-in 1080p FaceTime HD webcam of a MacBook Air M2 positioned in front of them at approximately 40 cm and elevated by approximately 50 cm relative to the torso.

Six classes of facial gestures were defined with their respective global whole labels: (0) left wink, (1) right wink, (2) frown, (3) raised eyebrows, (4) extended corners of the mouth, and (5) normal. For each gesture, 1000 samples (500 with glasses and 500 without) were collected, except for two specific cases (1 and 2) in which it was not possible to record winks. In total, the dataset contains 32000 samples distributed in six classes.

Formally, with $S = 6$ subjects, $G = 6$ gestures, $G_{miss} = 2$ gestures (left/right wink) missing for $S_w = 2$ subjects, and $N_g = 1000$ samples per gesture per subject (500 with and 500 without glasses), the total number of samples is computed as follows:

$$N_{total} = S \times \left(G \times N_g\right) - S_w(G_{miss} \times N_g)$$

Here, S denotes the number of subjects; G, the number of gesture classes; G_{miss}, , the number of gesture classes not recorded for some subjects (1 and 2); S_w, the number of subjects for whom those wink gestures are missing; N_g, the number of samples collected per gesture and per subject; and N_{total}, , the resulting dataset size after discounting the missing subject–gesture combinations. Substituting the study-specific values ($S = 6$), $G = 6$, $G_{miss} = 2$, $S_w = 2$, $N_g = 1000$), the formula becomes:

$$N_{total} = 6 \times (6 \times 1000) - 2(2 \times 1000)$$

Each stored sample was contained in multiple structured elements that represent both facial geometry and context metadata. These elements were the following:

- X: matrix of characteristics of the nodes of the graph, where each row represents the 3D coordinates of a facial point extracted by MediaPipe. Only the points relevant to key anatomical regions (eyes, eyebrows, glabella, forehead and mouth) were used, selected using a personalized dictionary of edges. (x, y, z)
- Edge_index: A connectivity matrix between nodes, where each column represents an undirected edge between two nodes. The edges were explicitly defined by anatomically coherent connections within each facial region.
- Edge_attr: A scalar attribute by edge that represents the Euclidean distance between connected nodes, calculated directly from 3D coordinates.
- Y: class label indicating the facial gesture corresponding to the sample. This is in the range of 0 to 5.
- Glasses: A binary value that indicates whether the subject was wearing optical lenses during the capture (1) or not (0).

Figure 3 shows the set of selected facial points projected onto an individual's face with a wink with the left eye. The green dots correspond to three-dimensional landmarks extracted using MediaPipe, grouped into key regions such as eyes, eyebrows, glabella, forehead and mouth. This gesture is characterized by the partial closure of the left eyelid, captured by the density of dots in the left eye region.

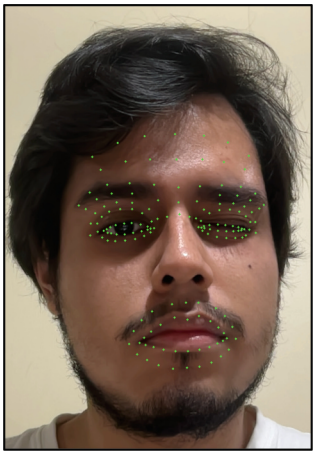

Fig. 3. Left wink.

Figure 4 shows the right wink gesture, where the closure of the right eye can be seen accompanied by a slight facial asymmetry. The selected points allow you to accurately capture the topological variations that occur in the eye region during the action, contributing to the distinction between one-sided winks.

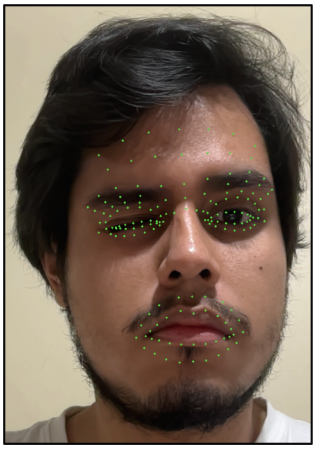

Fig. 4. Right wink.

Figure 5 corresponds to the frowning gesture, identified by the approach of the eyebrows and the contraction of the glabella The points marked on the forehead and between the eyebrows allow you to model the changes in the local curvature of the face.

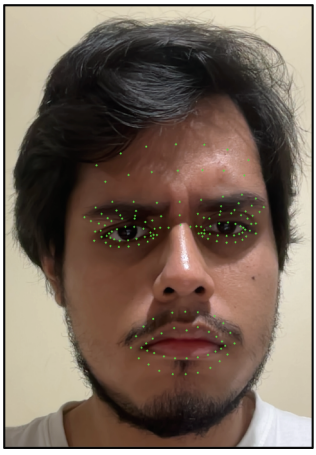

Fig. 5. Furrowed eyebrows.

Figure 6 shows the raised eyebrow gesture, where a visible elevation is observed in the upper part of the face. The landmarks located above the eyebrows and forehead show a greater vertical separation from their position in a neutral face, which facilitates their detection through variations in the relative geometry of the nodes.

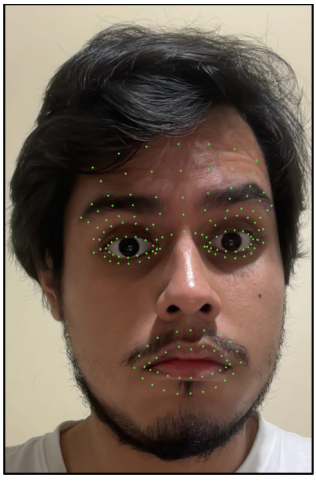

Fig. 6. Raised eyebrows.

Figure 7 shows the representation of the gesture of extended lip corners, associated with a smile or expression of pleasure. The selected points around the mouth allow the horizontal elongation of the corners to be captured.

Before being stored, all 3D coordinates were normalized taking as a reference the distance between the centers of both eyes, in order to guarantee invariance at the scale between subjects. This normalization was applied consistently in all samples to maintain the geometric coherence of the facial graph.

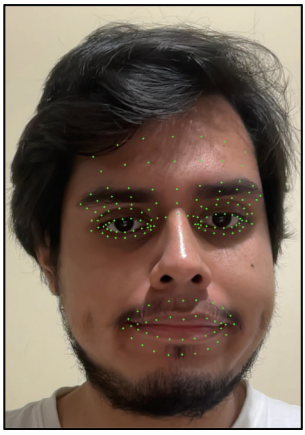

Fig. 7. Lip corners extended.

This representation scheme not only allows facial geometry to be modeled in high detail, but also preserves key contextual information to strengthen the model against visual variations such as the use of lenses. Each individual dataset was saved in a.pt file using the torch save function, making it easy to load it directly for training with GNN-type architectures.

3.4 Model Training

For the training process, six previously generated data files, uploaded from Google Drive, were combined. Each file contains labeled graph samples for a different person. From the total set ($\approx$ 32,000 samples), a stratified partition was used into three subsets: training (80%), validation (10%), and test (10%). This distribution ensures a sufficient volume of samples to learn complex patterns, a representative validation subset to fit hyperparameters, and an independent test set to evaluate the generalizability of the model.

The model was trained using the Adam (Adaptive Moment Estimation) optimizer with an initial learning rate of 0.001 and the cross-entropy loss function, common in multiclass classification. The standard backpropagation algorithm adjusted weights during each pass, leveraging the hardware acceleration of an NVIDIA T4 GPU in Google Colab to significantly reduce compute times and make hyperparameter scanning easier.

To prevent overadjustment, two regularization mechanisms were implemented:

- Early stopping: training stopped after 10 consecutive periods with no improvement in validation accuracy.
- ReduceLROnPlateau: The learning rate was halved when the validation accuracy did not increase for 5 epochs, with a minimum limit of. 1×10^{-6}

During each epoch, the training and validation accuracies were calculated and stored, which were then plotted to analyze convergence. The model with the best validation performance was saved for the final evaluation.

Although a cap of 100 epochs was established, the early stopping criterion triggered the stop in epoch 29, after finding no improvement in 10 cycles. The scheduler reduced the learning rate at epoch 9 (a) and epoch 25 (a), which stabilized the validation accuracy after the initial oscillations. The best validation result was obtained at epoch 27, with 99.85% accuracy. $5 \times 10^{-4} 2.5 \times 10^{-4}$

Table 4. Per class classification metrics of GPSNet.

Class	Precision	Recall	F1-Score	Support
0 (Left wink)	0.99	0.96	0.98	400
1 (Right wink)	1.00	1.00	1.00	400
2 (Furrowed eyebrows)	1.00	0.99	1.00	600
3 (Raised eyebrows)	1.00	1.00	1.00	600
4 (Lip corners extended)	0.95	1.00	0.98	600
5 (Neutral expression)	1.00	0.97	0.98	600

When evaluating the GPSNet model trained from scratch on the test set (3 200 unseen samples), an overall accuracy of 99.84% was obtained, reflecting a high generalizability. Table 4 shows the ranking report by class, including the accuracy, recall, and F1-score metrics for each of the six gesture categories. All classes have F1 values higher than 0.98, with the exception of class 0 (left wink), which has a slightly lower recall (0.96), indicating that some real instances of this class were confused with others. By contrast, class 1 (right wink) achieved a perfect performance across all metrics, while classes 2, 3, and 5 achieved an F1-score of 1.00 or close. Class 4 (extended lip corners) obtained a perfect recall (1.00), although its accuracy was lower (0.95), suggesting some incorrect predictions such as false positives. Together, the macro and weighted averages were 0.99 for all metrics, evidencing a balanced and robust performance of the model in all categories.

4 Experimentation

The purpose of the experiment was to evaluate the effectiveness of transfer learning when applying fine-tuning on three different GNN models: GPSNet, GCNNet and SAGENet. Each model was pretrained on the same dataset to later compare its accuracies with each other under equivalent conditions.

The metrics used were training accuracy and validation accuracy. In a multiclass classification problem, accuracy is formally defined as the proportion of instances whose predicted tag exactly matches their true tag over the full set of test examples [24]. Therefore, training accuracy is understood as the percentage of correct predictions made by the model on the training set [25]; while validation accuracy is the proportion of correct predictions over the validation set [26], a subset of data not used to adjust weights, and serves as an estimator of model performance on unknown data, assessing its generalizability.

Each of the three models (GPSNet, GCNNet and SAGENet) was subjected to a fine-tuning process following a homogeneous experimental configuration. To do this, the best_gps_model v2.7.pth, best_gcn_model.pth and best_sage_model.pth models were used, respectively. The dataset was divided into three mutually exclusive subsets: training (80%), validation (10%), and test (10%), ensuring that the test samples had not been seen during pretraining.

The tuning process was spread over 10 epochs using the Adam optimizer, with an initial learning rate of $1 \times 10^{-}4$ and a weight decay factor of $1 \times 10^{-}5$ to prevent overfitting. In addition, the ReduceLROnPlateau policy was implemented to automatically reduce the learning rate when the accuracy on the validation set did not improve after 3 consecutive epochs. The metric used to select the best model was validation accuracy, and the model with the highest performance was subsequently evaluated on the test set, composed of 3 200 unseen samples.

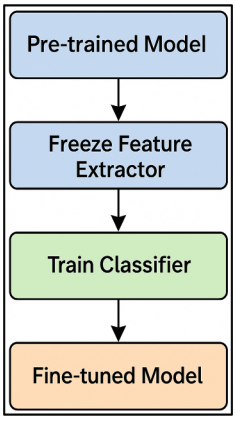

Fig. 8. Flowchart of the fine-tuning protocol.

Figure 8 shows the general flow followed during the fine-tuning procedure applied to each architecture. From a previously trained model, the layers responsible for the extraction of features (convolution blocks and pooling) were frozen, keeping their weights

constant throughout the process. Then, the classification layer was exclusively trained using the training and validation set. This approach allows already learned representations to be reused for similar tasks, minimizing the risk of overfitting and reducing training time. The final adjusted model was the one that obtained the highest precision on the validation set and was then evaluated on an independent test set.

5 Results

Figure 9 shows that the GPSConv + GINEConv (GPSNet)-based model achieved outstanding performance from the very beginning. After loading the pre-trained model and readjusting only the output layer, the validation accuracy started from 99.53% in epoch 1, improved slightly to 99.75% in epoch 6, and closed at 99.78% at the end of epoch 10. In parallel, the training accuracy was maintained at around 99.9 percent, reflecting a rapid convergence of the sorting head without inducing noticeable overadjustment during fine-tuning.

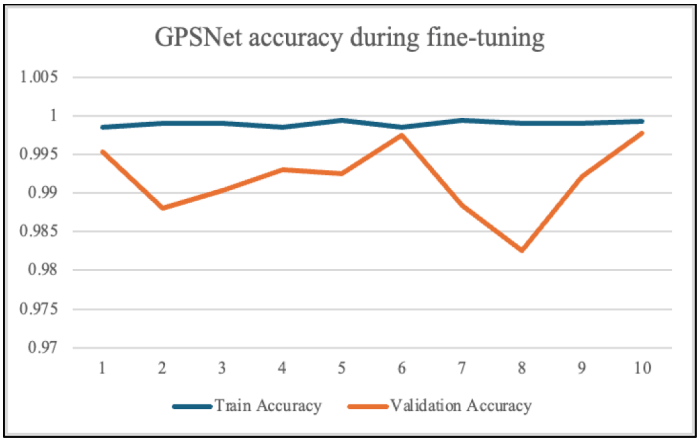

Fig. 9. GPSNet train accuracy and validation acuracy per epoch.

In contrast, Fig. 10 shows that the GCNConv (GCNNet) model experienced slight gains in validation, but its overall behavior was markedly lower than that of GPSNet. Validation accuracy rose from 61.75% in the first epoch to 63.43% in the tenth, with fluctuations of less than two percentage points between consecutive epochs. At the same time, the training accuracy remained close to 61%, indicating that the adjustability of the linear head over the fixed representations of GCN was limited.

Figure 11 shows that the GraphSAGE approach (SAGENet) presented an intermediate performance, since the validation started at 85.14% and ended at 87.49% after ten fine-tuning periods, with peaks of 86.86% at time 5 and 86.61% at time 7. The training accuracy ranged around 84–85%, showing some stability once the first four epochs were overcome. These values confirm that SAGENet adapts its representations better

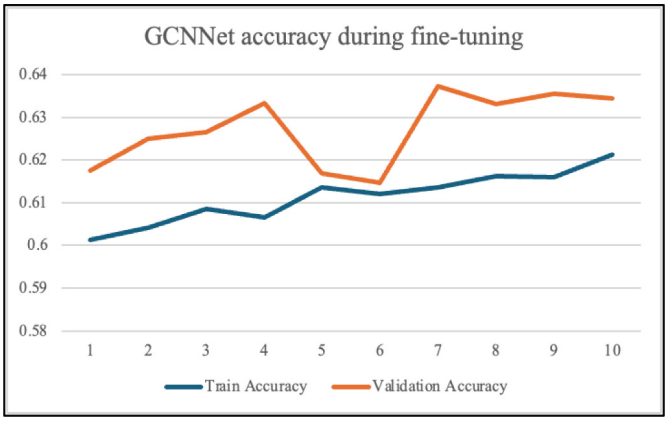

Fig. 10. GCNNet train accuracy and validation acuracy per epoch.

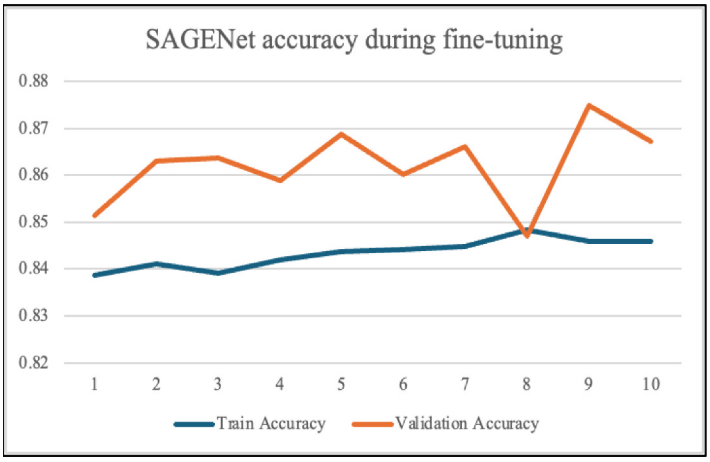

Fig. 11. SAGENet train accuracy and validation accuracy per epoch.

than GCNNet, although without matching the efficiency of the GPSConv + GINEConv block.

Overall, the results shown in Table 5 reveal that, given the same fine-tuning protocol and hardware conditions, GPSNet is the most effective option for the classification of facial gestures in 3D graphs, followed by SAGENet, while GCNNet hardly improves its scores after the fine-tuning process.

Table 5. Architectures average train accuracy and validation accuracy

Architecture	Average TA	Average VA
GPSNet	0.99902	0.99177
GCNNet	0.61131	0.6274
SAGENet	0.84334	0.86208

6 Conclusions

In this work, an approach based on graph neural networks for the detection of facial gestures has been presented, comparing three architectures (GPSNet, GCNNet and SAGENet) under a homogeneous fine-tuning protocol. The results show that GPSNet, with GPSConv + GINEConv blocks, far outperforms GCNConv and GraphSAGE-based variants, reaching validation accuracies close to 99.8% after fine-tuning of the output layer.

However, the validation curves show a high variability during initial training, with notable oscillations between epochs that reflect sensitivity to initialization and complexity of the parameter space. This instability suggests that, although scheduler and early stopping mitigate overfitting, the model still benefits from finer control of learning. Therefore, future work should explore additional regularization strategies, for example, graph data augmentation, batch normalization, or distillation techniques; and scheduler adjustments, to stabilize convergence and reduce jitter of the validation metric.

It is also promising to investigate hybrid variants that combine the efficiency of GraphSAGE with local-global attention mechanisms similar to those of GPSConv, as well as to incorporate continuous validations in scenarios of adverse illumination or partial occlusions. Finally, carrying out usability studies with end users will allow us to better calibrate the trade-off between accuracy, stability and latency, and guide the development of a highly robust and accessible solution.

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Adaptive and Intelligent Systems

Roadwatch: An Integrated Architecture for AI-Powered Surveillance and Anomaly Detection in Traffic Areas

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Abstract. The increasing complexity of urban traffic management necessitates advanced surveillance and anomaly detection solutions. The proposed Roadwatch architecture is a comprehensive system integrating Artificial Intelligence (AI) and Computer Vision to monitor vehicular and pedestrian traffic effectively. By leveraging interconnected software and hardware components, Roadwatch identifies objects captured by cameras and provides essential meta-information for detecting specific anomalies. This assists in crucial tasks such as traffic management, pedestrian safety, incident detection, and optimal resource allocation for urban infrastructure. AI plays a pivotal role in processing camera feeds, enabling object detection, identification, and tracking. With configurable rule-based detection, Roadwatch offers an innovative approach to intelligent surveillance, enhancing safety and security in diverse real-world traffic environments while utilizing existing camera infrastructure.

Keywords: Smart Cities · Artificial Intelligence · Surveillance · Security

1 Introduction

The continuous expansion of urban areas, along with the increasing demand for services, necessitates the adoption of automated tools to support human oversight, which is otherwise insufficient in managing complex urban scenarios. In this context, the Smart City paradigm has emerged as a crucial framework for city administrators, whether managing small municipalities or large metropolitan areas. A key challenge within Smart City initiatives is urban security, particularly in traffic area management. Traditional human-based surveillance approaches are proving inadequate in handling the increasing volume of traffic-related events. The growth of urban populations has made traffic control a priority for municipal authorities, as manual monitoring methods struggle to cope with the complexity of modern cities. The need for automated, intelligent traffic monitoring systems has therefore become more pressing. AI-driven solutions offer superior accuracy and responsiveness compared to conventional methods, assisting human operators while also enabling proactive responses to traffic anomalies. The

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Roadwatch system is designed to process information from existing or newly installed traffic cameras through an AI-powered pipeline. The system utilizes NVIDIA's DeepStream SDK, built on GStreamer, to develop vision-based AI applications for real-time event detection. Roadwatch's architecture detects and tracks both stationary and moving objects using AI, analyzing their movements to identify specific events based on configurable rules and heuristic strategies. When predefined events occur, alerts are generated for operators, and relevant video footage is recorded for later review. This intelligent system, complemented by user-friendly management interfaces, provides an efficient solution for monitoring high-risk traffic areas. Unlike other state-of-the-art solutions, Roadwatch maximizes the use of existing camera infrastructure, reducing costs and implementation time.

The main scientific contribution related to this work can be summarized as follows:

1. *Formalization of the video stream preprocessing approach*: it introduces an approach to preprocessing video streams to make them suitable for the tasks performed by the AI algorithms.
2. *Development of an event identification framework*: it allows the system to identify various events related to vehicular and pedestrian traffic. The events are classified based on their type (e.g., people, vehicles, etc.) and characteristics (e.g., speed, collisions, etc.) through configurable rules and heuristic strategies.
3. *Formalization of the Roadwatch System Architecture*: it integrates AI-based computer vision components with real-time video stream processing, providing a flexible approach for automatically detecting anomalies in traffic areas, identifying potential safety risks for vehicles and pedestrians, detecting accidents or anomaly activities, and assisting in urban traffic management by leveraging existing camera infrastructure and applying configurable detection rules.

The structure of the rest of this paper is as follows: Sect. 2 presents details about the research scenario under consideration. Section 3 formalizes the proposed architecture. Section 4 presents the results of some preliminary experiments performed to validate the idea behind the proposed approach. Finally, Sect. 5 concludes this work with some remarks.

2 Background and Related Work

Nowadays, the Smart Cities [26] concept represents a powerful approach to face/mitigate the challenges related to urbanization, population growth, and sustainability, involving advanced technologies, data-driven insights [24], and innovative models [3,11,18,19] to improve the quality of life of people and the urban services and infrastructure efficiency. One of the most important aspects of Smart Cities is traffic management [15] since it plays a crucial role in ensuring seamless mobility, reducing congestion, and improving safety on urban roads. This is because the canonical approaches to traffic surveillance and control based on manual intervention and static infrastructure are increasingly becoming incompatible with the dynamic nature of urban traffic scenarios. In such a context, the community of researchers is leveraging Artificial Intelligence (AI) [2,14] and Machine Learning (ML) [4] techniques to design intelligent

systems for traffic monitoring, analysis, and optimization. These AI-based solutions exploit the data analytics power [1], computer vision [17], and sensor networks [13] to collect real-time information about traffic flow, detecting anomalies and accidents, and forecast the future trends, reducing or overcoming physical limitations [5,6,9].

In the context of traffic surveillance, Computer Vision algorithms play a crucial role in analyzing visual data captured by cameras installed in different locations. In such a context, deep learning-based approaches, such as Convolutional Neural Networks (CNNs), offer interesting results in tasks such as vehicle detection, pedestrian tracking, and behavior recognition. In addition to the simple observation of the events, AI-powered traffic management systems allow us to operate proactive interventions to mitigate problems such as congestion and accidents. For instance, they can be used to optimize traffic signal timings in real time based on current traffic conditions, prioritize emergency vehicles, modifying lane configurations dynamically. Lots of research done in this domain designed solutions able to address various aspects of traffic management in Smart Cities, e.g., the Intelligent Transportation Systems (ITSs) [8], predictive analytics platforms, smart traffic lights, and connected vehicle technologies. The literature offers several contributions in the field of anomaly detection and surveillance systems that share the same idea that stand behind the proposed Roadwatch architecture, particularly in their use of deep learning, object detection, and their integration with existing surveillance infrastructures. Some representative examples from the period 2016 to 2024 are as follows.

In [16], the authors address the challenge of detecting anomalies in real-world traffic videos. They employ deep learning methods to analyze video footage, identifying unusual vehicular or pedestrian behaviors that deviate from normal patterns. The system is designed to process large-scale data captured from surveillance cameras, providing an efficient and automated means of identifying incidents such as accidents or illegal activities. This study highlights the importance of robust deep learning frameworks for enhancing traffic monitoring capabilities and demonstrates the potential of AI to reduce human intervention in surveillance tasks. The system is designed to analyze large-scale data that is already being recorded, which indicates that no specific repositioning or additional installation of cameras is required. The approach leverages pre-existing infrastructure for anomaly detection.

The work in [28] instead proposes an unsupervised anomaly detection framework that relies on background modeling techniques. By segmenting moving objects from the static background in traffic surveillance videos, the system identifies anomalous activities without the need for labeled data. This approach proves particularly effective for detecting sudden changes in traffic flow or abnormal object behavior, offering a computationally efficient solution for real-time monitoring applications. The emphasis on unsupervised learning also makes this method adaptable to scenarios where training data is scarce or unavailable. Although the paper does not explicitly specify whether existing or newly positioned cameras are used, the reliance on background modeling implies that cameras must be suitably positioned to provide clear and stable footage of the monitored area. This suggests that the method might require appropriately placed cameras to ensure reliable segmentation of moving objects.

In addition, in [12], the authors explore the integration of deep learning and IoT technologies for remote traffic surveillance. Their system employs deep learning-based object detection to monitor both vehicles and pedestrians in real time while transmitting data through IoT-enabled devices. By combining AI with IoT infrastructures, the authors demonstrate how intelligent surveillance systems can achieve low-latency, remote monitoring of traffic areas. This work underscores the importance of integrating distributed sensor networks with AI algorithms to ensure scalable and efficient anomaly detection, particularly in large urban environments. The system integrates IoT technologies with deep learning-based object detection, which involves real-time monitoring using IoT-enabled devices. This indicates the need for specifically deployed cameras that are part of an IoT infrastructure, as opposed to simply utilizing existing surveillance systems.

Another work in [10] introduces uTRAND, an unsupervised anomaly detection framework designed to analyze traffic trajectories. By focusing on the detection of anomalous movements within traffic data, the system is able to identify deviations from normal behavior in both vehicular and pedestrian traffic, without relying on labeled data. The authors leverage deep learning methods to perform this analysis, making their approach particularly suited for environments where labeled datasets are not readily available. This unsupervised technique allows for more flexible, scalable anomaly detection in dynamic traffic scenarios, highlighting the importance of machine learning in processing real-time surveillance data without the need for manual intervention. The uTRAND framework analyzes traffic trajectories and detects anomalies without explicitly requiring repositioned cameras. The approach focuses on processing traffic movement data, which can be captured from existing cameras, making it adaptable to current surveillance infrastructures.

Similarly, in [27], the authors present a hybrid deep learning model for abnormal traffic detection within Software-Defined Networks (SDN). The system integrates deep learning-based detection with network traffic monitoring, combining a hierarchical model to analyze traffic flow and detect abnormal behavior. This work focuses on identifying and mitigating potential security threats or network failures in SDN environments, making it relevant for real-time traffic anomaly detection systems. The authors demonstrate the effectiveness of the hybrid model in detecting and classifying network anomalies, while also emphasizing the system's ability to scale across distributed infrastructures. This study underscores the growing role of deep learning in enhancing network-based traffic surveillance and anomaly detection, particularly in environments where rapid, high-volume data streams must be analyzed in real-time. This study does not involve physical cameras or video surveillance, as it centers on analyzing network traffic data instead of video footage.

In [20], the authors present a novel approach to automatic road-accident detection using machine vision. The system leverages surveillance footage from roads and highways to detect damaged vehicles, which serve as an indication of road accidents. This approach integrates a supervised learning method that combines three stages into a single framework, utilizing support vector machines (SVM) trained on Histogram of Gradients (HOG) and Gray Level Co-occurrence Matrix (GLCM) features. The authors highlight the novelty of detecting damaged cars from static images, a problem that

has not been previously addressed. The study demonstrates the effectiveness of the method on two datasets, DCD-1 and DCD-2, which vary based on image quality and distance. The results show an accuracy of 81.83% for high-quality images and 64.37% for lower-quality ones, thus showcasing the model's potential for detecting road accidents in real-world scenarios. This study does not incorporate real-time processing or IoT integration, focusing instead on static image analysis from existing surveillance cameras.

In the work [25], the authors introduce an IoT-based car accident detection and notification algorithm designed for general road accidents. The system integrates smart sensors within the vehicle, which are triggered upon detecting an accident. Key modules, including GPS and GSM, are used to obtain the accident's location and send notifications to registered phone numbers and nearby ambulances, aiming to provide immediate help. The paper emphasizes the importance of reducing response times and improving emergency services through real-time accident detection and timely notifications. This approach showcases the potential of IoT in enhancing road safety and reducing delays in emergency response. However, this study does not involve video surveillance or AI-based object detection, as it focuses on sensor-based systems within the vehicle to detect accidents and notify emergency responders.

A significant number of further works, moreover, address issues that are transversally relevant to these systems, taking into consideration some crucial aspects such as, for example, user authentication [21], or security in the networks to which they are connected [22,23].

However, it should be noted that despite the progress made in AI-driven traffic management, still exist several challenges that include the scalability of AI models to handle large-scale urban environments, the interoperability of different data sources and systems, and some problems related to data privacy and security. Addressing these challenges requires interdisciplinary collaboration among researchers, policymakers, and industry stakeholders. In summary, the convergence of AI, data analytics, and Smart Cities initiatives is revolutionizing traffic management in urban environments, where the adoption of advanced technologies and innovative approaches is creating more sustainable cities and more resilient transportation systems that meet present and future needs.

Table 1. Feature Comparison of the Approaches.

Approach Reference	AI-based object detection	Anomaly detection	Surveillance Infrastructure	Real-time Processing
[16]	✓	✓	✓	
[28]	✓	✓	✓	
[12]	✓		✓	✓
[10]	✓	✓	✓	
[27]		✓		
[20]	✓		✓	
[25]		✓		✓
Roadwatch	✓	✓	✓	✓

Table 1 summarizes the previously cited works in the literature in terms of features, providing an overview of the offered functionalities and a comparison with the architecture we propose.

3 Approach Formalization

The adopted AI framework is based on a modular real-time pipeline built on *NVIDIA DeepStream*[1] and *GStreamer*, with inference performed by the `nvinfer` plugin using a **YOLOv4-CSP** architecture.[2] The model is initialized from the official Darknet `yolov4-csp.weights` and `yolov4-csp.cfg`, and then converted into a `TensorRT` engine (`.engine`) for optimized inference on GPU. The engine runs in *FP16 precision* to ensure high throughput with low latency.

In the preliminary experiments, the inference pipeline is configured with a batch size of 1 and supports up to 80 object classes, with a detection confidence threshold set at 0.25. Bounding box parsing is handled by a custom YOLO parser (`NvDsInferParseYolo`), loaded through a dedicated plugin library.[3] The input resizing step preserves aspect ratio to maintain spatial consistency in the detections.

Multi-object tracking is enabled through the `nvtracker` plugin, configured to use the **NvDCF** (Discriminative Correlation Filter) tracking method. The tracker combines both *appearance-based* and *motion-based* strategies to associate new detections with existing tracked objects. Specifically, it uses Intersection over Union (IoU) to evaluate spatial overlap between bounding boxes and ColorNames-based visual similarity to compare the appearance of objects across frames. These metrics are weighted and fine-tuned through parameters in the YAML configuration file, which defines thresholds and association rules. This hybrid tracking strategy improves robustness, particularly in crowded environments or under partial occlusion.

The overall architecture is fully modular: upgrading the detection model requires only the regeneration of the TensorRT engine and minor configuration updates, without modifying the rest of the analytics pipeline. This allows for maintainable and flexible deployments, and supports future improvements or model switching with minimal integration effort.

A high-level representation of the system architecture is shown in Fig. 1 and is detailed as follows:

- **Video Streams:** The system's input consists of video streams from cameras monitoring specific areas. The system can process multiple streams simultaneously, originating from pre-existing or newly introduced cameras placed in areas of interest.
- **Video Flows Preprocessing:** Using two web tools, the perspective transformation necessary for the AI algorithm's functionality and the definition of the working area

[1] https://developer.nvidia.com/deepstream-sdk.

[2] A variant of YOLOv4 that incorporates Cross-Stage Partial connections to improve feature reuse and reduce computation cost.

[3] We used `libnvdsinfer_custom_impl_Yolo.so`, compatible with DeepStream-based YOLO integrations.

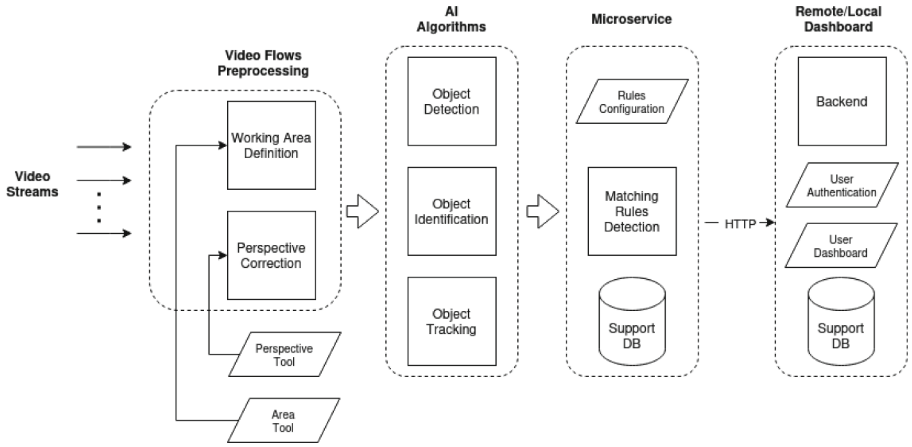

Fig. 1. High-level System Architecture.

are performed. Specifically, the images (frames) in the video data flow are preprocessed by the *PreProcessor* class, which applies transformations such as perspective correction and occlusion tolerance. This step prepares the frames for further analysis, ensuring accurate anomaly detection.

- **AI Detection, Identification, and Tracking Algorithms:** The system uses the *Tracker* class to track objects across multiple frames, ensuring consistency by associating detections across frames based on parameters like expected distance and cut-off thresholds. The detection of anomalies is managed by the `DetectorManager`, which utilizes various anomaly detectors, such as:
  - *CongestedTraffic*, for detecting heavy traffic situations.
  - *ObjectOnRoad*, for detecting objects or pedestrians on the road.

  These detectors analyze the video data and flag anomalies that match predefined rules. The *ConfirmationManager* integrates video data to confirm critical events, based on detected anomalies. The system operates with an *Inference Pipeline* that processes video streams in real-time, ensuring the AI models and detectors collaborate efficiently to handle multiple data streams and provide timely alerts.
- **Microservice:** This layer connects the AI components with the backend and frontend ones, which can be implemented on the same machine or on the remote one through the HTTP protocol. An RDBMS (MySQL) supports the processing of data related to event recognition. The following elements outline key processes within the microservice:
  - **Server Initialization:** The microservice sets up two Express.js servers: the first server is used for the communication with external entities (i.e., remote or local dashboard); the second server is used for the internal communication between the AI components.
  - **Database Connection:** The microservice connects to a MySQL database using Sequelize ORM, establishing the connection during startup. If the database does not exist, it is automatically created.

- **File System Management:** The microservice manages media files generated by the system. It creates a dedicated directory for storing video files and periodically deletes obsolete data.
- **Communication with AI Modules:** The microservice checks the camera feed status and launches the AI pipeline if the feed is available. It logs operations, retrying the connection if failures occur.
- **Keep-Alive Mechanism:** A keep-alive mechanism periodically sends signals to the remote or local dashboard components and the AI modules to ensure the system remains operational. If a module becomes unresponsive, the system attempts an automatic restart.
- **Configuration Management:** Configuration data, including paths and IP addresses, is stored in a YAML file. The microservice dynamically loads and sanitizes these configurations during startup, resolving environment-specific variables.

The microservice ensures efficient resource management and synchronization between the AI modules and the backend system in real-time.

– **Remote or Local Dashboard:** The dashboard consists of a *Backend* component that processes data from the microservice, performing additional aggregation or processing if needed. It also handles frontend requests using a NoSQL database (MongoDB) to store event-related information and video clips. The frontend provides a web interface for system management and monitoring. The backend is implemented with NestJS[4], offering a scalable API structure for interacting with the AI microservice and frontend. Key backend functionalities include:
  - User management for frontend access control.
  - Event storage and categorization.
  - Configurable alert levels and timers for individual cameras.
  - Integration with external APIs for retrieving weather data.
  - Real-time communication with the frontend via WebSockets for instant event notifications.

  The frontend is built with PrimeVue[5] on Vue 3[6], with Vite[7] enhancing build efficiency. This interface provides functionalities such as:
  - Viewing the history of alerts and accessing recorded event video clips.
  - Configuring camera-specific alert levels and timers.
  - Extending or delaying specific alerts.
  - Viewing real-time weather data and a map of the monitored area.

  Communication between the frontend and backend is facilitated by Axios[8], with state management via Pinia[9]. Real-time updates, such as event notifications and status changes, are delivered using Socket.io[10].

[4] https://nestjs.com/.
[5] https://www.primefaces.org/primevue/.
[6] https://vuejs.org/.
[7] https://vitejs.dev/.
[8] https://axios-http.com/.
[9] https://pinia.vuejs.org/.
[10] https://socket.io/.

4 Preliminary Experiments

A preliminary experimentation was carried out by processing in parallel three video streams locally generated that simulate three public or private cameras located in the following different traffic scenarios:

- **Urban Central Area:** This video represented a typical downtown street environment, including features such as parking spaces along the road, vertical and horizontal signage, pedestrian crosswalks, and moderate vehicle and pedestrian traffic. This scenario allowed testing of events like illegal parking, person on road, and area invasion.
- **Roundabout Scenario:** The second video focused on a roundabout with multiple vehicle entry and exit points. The frequent merging of vehicles and constant flow of traffic provided a suitable setting to evaluate events such as reckless driving and illegal lane changes.
- **Parking Lot Area:** The third video simulated a parking facility with clearly marked one-way lanes and designated parking spots. This environment was used to test illegal parking, area invasion (e.g., driving across restricted lanes or zones), and improper vehicle movements.

In such a context, the system was configured to recognize the following types of events:

- Congested traffic.
- Illegal lane change.
- Person on road.
- Area invasion.
- Illegal parking.
- Reckless driving.

It should be noted that *Reckless driving* refers to vehicle behavior that involves exceeding the speed limits defined for the specific area, whereas *Area invasion* refers to the violation of road markings or signage, specifically when a vehicle crosses into zones that are prohibited for traffic, such as pedestrian areas, bus lanes, or restricted zones marked on the road surface.

The video streams are processed by the AI components after an initial configuration phase, where areas and objects are defined for each camera. This configuration is required only once and is not needed again afterward.

4.1 Strategy

The system validation was performed by comparing the detected events with the corresponding video frames. This comparison aimed to verify the correctness of the detections, ensuring that the identified events matched both the event type and the correct timing.

Additionally, a manual inspection was carried out on the entire sequence of processed videos. This further review aimed to detect any instances where events occurred but were not captured by the system. By conducting these checks, the validation process ensured both the accuracy of event detection and the reliability of the AI components in capturing all relevant incidents.

Payload. The implementation of configurable rules to define system behavior based on specific objectives yielded valuable insights into the adaptability and effectiveness of the system in different operational scenarios. An important output of the system is a payload structured in JSON format, which contains information related to each event processed by the system. This payload allows for verifying both the correctness of the system's operation and providing a data stream to the frontend components. A fragment of this JSON payload is provided in Table 2.

Table 2. Example of JSON Payload for Event Validation.

Timestamp	Payload
2025-03-01 14:31:02.354243	{ 'box': {'left': 1259, 'top': 381, 'right': 1772, 'bottom': 629}, 'eventType': '1', 'videoFile': '/media/computed/2_0_5_01Mar24-14.30.50.mp4', 'cameraId': '2' }
2025-03-01 14:31:02.549624	{'box': {'left': 1188, 'top': 96, 'right': 1526, 'bottom': 252}, 'eventType': '2', 'videoFile': '/media/computed/2_0_5_01Mar24-14.30.50.mp4', 'cameraId': '2' }
2025-03-01 14:31:17.897902	{'box': {'left': 914, 'top': 463, 'right': 1027, 'bottom': 770}, 'eventType': '1', 'videoFile': '/media/computed/1_0_5_01Mar24-14.30.50.mp4', 'cameraId': '1' }

In more detail, the payload provides detailed information about objects detected by an AI-based object detection algorithm. It contains the following key elements:

- **Timestamp:** This indicates the exact time at which the detection occurred, allowing the system to correlate events with real-time data.
- **Bounding Box Coordinates:** These specify the position of the detected object within the video frame, defining its *top-left* and *bottom-right* corners, which helps in determining the object's location and size.
- **Event Type:** This field classifies the detected event (e.g., traffic congestion, illegal parking, or other specified behaviors).
- **Video File Path:** The path to the video footage containing the event, which allows access to the specific recorded segment for further review and analysis.
- **Camera ID:** This identifies the camera or device responsible for capturing the footage, providing important context regarding the source of the detection and its geographic location.

4.2 Results

The preliminary evaluation of the proposed architecture was carried out by analyzing three different video streams, which simulate public and private cameras located in

various areas. The system was configured to detect several types of events, such as congested traffic, illegal lane changes, persons on the road, and others. The primary goal of this operation was to assess the accuracy of event detection without a system optimization in terms of software configuration by comparing the system's results with the actual occurrences in the video.

The system's performance was evaluated based on two main metrics: the number of events correctly detected and the number of events missed. To achieve this, a manual inspection of the video sequences was performed in parallel with the system's automated detection. The events identified by the system were compared to those that occurred in the videos, and the results were analyzed to calculate the system's overall accuracy.

Table 3 presents a summary of the results obtained for each type of event detected by the system. The table shows the total number of events for each type, the number of events that were correctly identified by the system, the number of events that were missed, and the corresponding accuracy of the detection.

The total number of events analyzed across the three video streams amounted to 634 events, with 578 correctly detected and 56 missed by the system. This resulted in an overall detection accuracy of approximately 91%. The system performed particularly well with events like Area Invasion and Reckless Driving, where the detection rates reached 98% and 93%, respectively. However, some event types, such as Person on Road and Illegal Parking, demonstrated slightly lower detection rates, with 86% and 88% accuracy, respectively. These discrepancies were mainly due to the model's difficulty in detecting objects that were either far from the camera or partially occluded, for example, pedestrians walking behind large vehicles, or parked cars located at the edge of the frame and partially hidden by surrounding infrastructure such as lamp posts or trees.

The main source of error was therefore due to fundamental limitations of the object detection model, which subsequently affected the performance of the logic module. These results highlight the system's high reliability overall, with performance slightly affected by specific, more complex scenarios.

Table 3. Validation Results for Event Detection by Camera with Accuracy Calculation.

Event Type	Camera #1		Camera #2		Camera #3		Total Accuracy
	Correct	Missing	Correct	Missing	Correct	Missing	
Congested Traffic	40	3	35	4	30	5	**89.74%**
Illegal Lane Change	28	5	35	2	35	1	**92.45%**
Person on Road	25	7	30	4	35	4	**85.71%**
Area Invasion	30	2	35	0	35	0	**98.03%**
Illegal Parking	33	3	31	5	26	4	**88.23%**
Reckless Driving	35	1	30	3	30	3	**93.13%**
All Events	**191**	**21**	**196**	**18**	**191**	**17**	**91.16%**

As seen in Table 3, the system achieved a high detection rate, with the accuracy of detection for most event types being near or above 90%. However, the performance for certain complex events was slightly lower, as some scenarios are challenging to clearly define and often involve rapid or unpredictable scene changes that the system may not always be able to capture fully.

In summary, the validation results demonstrate that the system performs reliably with an accuracy rate of approximately 91%. These results underscore the importance of continuous improvement in the system's ability to handle edge cases and enhance its robustness in real-world environments.

5 Conclusion and Future Work

This position paper outlines the conceptual framework and preliminary findings of the proposed system, which aims to enhance event detection through AI-driven object detection, identification, and tracking. While full-scale experiments have not yet been conducted, initial tests indicate promising potential for the system's functionality in real-world conditions. The system leverages advanced AI algorithms to process video streams and detect key events, such as traffic congestion, reckless driving, and objects on the road. Preliminary results suggest that the system is capable of operating effectively under controlled conditions, with preliminary accuracy estimates showing potential detection rates around 91%. However, it is important to note that these results are based on initial trials, and further testing is required to evaluate the system's performance under a broader range of conditions and different scenarios. These early findings highlight the need for ongoing development to refine the AI algorithms and improve their adaptability to dynamic environments.

Future work will focus on optimizing the system's AI algorithms to enhance robustness and accuracy, particularly in more complex and variable conditions, as well as in the design of interoperable solutions, leveraging secure distributed service composition paradigms [7,29]. Additionally, efforts will be directed toward the definition of a user interface that allows the operators to interact with the system in an intuitive and efficient manner.

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Forecasting Surprises in Machine-Learning-Driven Interaction Systems: Lessons from the Transformer Breakthrough

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Abstract. The unexpectedly rapid capabilities unlocked by large language models (LLMs) and generative AI (GenAI) systems built on the Transformer architecture constitute one of the largest forecasting errors in recent AI. An architecture introduced for machine translation in 2017 [36] enabled, within a few years, broadly capable LLM/GenAI systems across tasks and modalities. A qualitative case study integrates document analysis, an expert-prediction audit, and synthesis of contemporaneous forecasts and discourse to explain this surprise, identifying three recurring mechanisms: (i) attribution drift among the architecture, the training procedure, and the data; (ii) conflation of empirical shortcomings of particular trained systems with fundamental limits of the model class; and (iii) misestimation of how scaling laws and infrastructure constraints jointly shape the feasible capability envelope.

Many claims that aged poorly targeted implementation artifacts rather than properties of the underlying model family. In response, an attribution-disciplined forecasting framework is proposed: capability and limitation claims should be stated at the level of the model class, anchored to executable specifications of its computational graph, and paired with explicit models of compute, data, and training-procedure scale. The contribution includes practical guidelines for capability attribution in LLM/GenAI and human–AI interaction settings, and outlines open problems for formal analysis of deep-learning architectures, with the aim of reducing future forecasting error.

Keywords: Artificial Intelligence AI · Large Language Models LLM · Generative AI GenAI · Human-Centred AI · Computational Interaction · Explainable AI · Forecasting Error · Transformer Architecture

1 Introduction

The Turing Test, as originally proposed by Alan Turing [35], implicitly defines machine intelligence as the ability of a system to generate text responses that are indistinguishable from those of a human. Recent advancements in LLMs, driven by the *Transformer Architecture* [36], have led to applications that reportedly pass the Turing Test in its original formulation [2].

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1.1 Research Hypothesis, Gap, Question and Method

Hypothesis. The period 2016–2025 saw the sharpest acceleration in AI within LLMs and GenAI, far exceeding expectations and rapidly advancing language understanding, generation, and applications. A key surprise was the outsized success of the *Transformer Architecture*; its limits remain incompletely understood, with ongoing work on theory, efficiency, and boundary conditions.

Gap. Despite this rapid expansion—largely enabled by the Transformer—the literature lacks a focused analysis of why the surprise (forecasting error) occurred, which is essential for improving foresight and governance. This study is not a step-by-step survey; comprehensive overviews appear in [4,22].

Question. Why were the impact and timing of the Transformer and ChatGPT widely unanticipated, and which failed assumptions, missed signals, or misestimated constraints led to this forecasting error?

Method. This study adopts a qualitative case study design that combines document analysis, expert-elicitation audit, and abductive reasoning to assess the forecasting accuracy of AI development. We first conduct a conceptual analysis of the core mechanisms and assumptions in the original *Transformer Architecture* paper; next, we run an expert-prediction audit contrasting pre- and post-Transformer forecasts from recognized practitioners; then, we synthesize evidence from scholarly literature and long-form public discussions by leading thinkers. Finally, we triangulate across these sources to explain observed forecasting discrepancies and to derive implications for future prediction practice.

2 Analytical Framework and Core Definitions

This section establishes the analytical framework and notation used throughout the paper. We distinguish and formally define the *Transformer Mathematical Model*, its constituent *Transformer Mathematical Functions*, and the *Transformer Architecture* (originating with [36]). Section 2.1 presents the core concepts and notation; Sect. 2.2 gives the corresponding Transformer-specific definitions.

2.1 Core Concepts and Notation

We fix the basic vocabulary and symbols used throughout the paper. Figure 1 summarizes the relationships among the core entities: a *Deep Learning Architecture* specifies a parameterized Mathematical Model (a family of functions); a *Training method* uses a *Dataset* and *Calculation Infrastructure* to fit the parameters; the result is a *Learned Algorithm* (a trained, deterministic procedure).

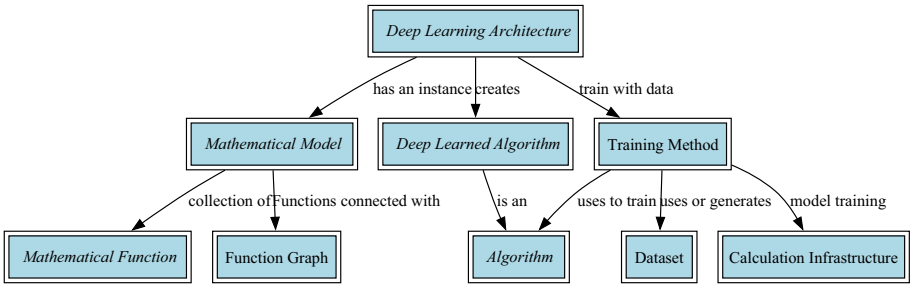

Fig. 1. Core concepts and their relations used in this paper.

Notation. We denote by $f_\theta : X \to Y$ a parameterized function with learnable parameters θ; training produces $\theta^\star$ and the trained mapping $f_{\theta^\star}$. A dataset is $D = \{(x_i, y_i)\}_{i=1}^n$ (supervised) or $D = \{x_i\}_{i=1}^n$ (unsupervised). A training method is $T = (\mathcal{L}, \Omega, \Pi)$, where $\mathcal{L}$ is a loss, Ω an optimizer (including schedule/regularization), and Π optional procedures (augmentation, early stopping, etc.). Computation for training is $C_T = (H, S, R)$ with hardware H, software S, and resource constraints R.

Definition 1 (Function). *A* Function *is a (deterministic) mapping $f : X \to Y$ assigning each $x \in X$ a unique $y = f(x) \in Y$. If randomness is used, a seed s is treated as part of the input so $(x, s) \mapsto y$ remains a* Function*.*

Definition 2 (Function Graph). *A* Function Graph *is a directed acyclic graph $G = (V, E)$ whose nodes are* Functions *and whose edges encode dataflow/composition. Composition in topological order defines an overall* Function *f_G.*

Definition 3 (Mathematical Model). *A* Mathematical Model *is an abstract description $M = (G, \lambda)$ consisting of a* Function Graph *G and fixed hyperparameters λ (wiring, layer sizes, activations, tokenization, etc.). It induces a family $\{f_\theta\}$ of* Functions *with learnable parameters θ (not fixed by M).*

Definition 4 (Algorithm). *An* Algorithm *is a finite, well-specified procedure that deterministically computes outputs from inputs. When stochastic behavior is desired (e.g., sampling), the randomness source and any decoding policy are considered fixed parts of the* Algorithm*.*

Definition 5 (Deep Learning Architecture). *A* Deep Learning Architecture *specifies a class of* Mathematical Models*: it determines admissible* Function Graphs *and a hyperparameter schema Λ. A concrete* Mathematical Model *$M = (G, \lambda)$ is obtained by instantiating $\lambda \in \Lambda$. When such a model is trained by a* Training method *on* Datasets *D under* Calculation Infrastructure *C_T, the result is a* Learned Algorithm *$A := f_{\theta^\star}$.*

Definition 6 (Training method). *A training method is a triple $T = (\mathcal{L}, \Omega, \Pi)$ that, when applied to dataset(s) D under* Calculation Infrastructure *C_T, returns parameters $\theta^\star$ by optimizing an objective induced by data and/or interactions. Here $\mathcal{L}$ specifies the loss/utility, Ω the optimizer and schedule, and Π ancillary procedures (e.g., batching, masking, augmentation, regularization, checkpointing). This schema covers supervised, self-supervised, reinforcement-learning, and distillation settings.*

Remark 1 (Training pipeline formalization). Given a *Mathematical Model* $M = (G, \lambda)$ and a training method $T = (\mathcal{L}, \Omega, \Pi)$, training on D with compute C_T yields $\theta^\star = T(M, D; C_T)$ and thus a *Learned Algorithm* $A = f_{\theta^\star}$. Inference additionally requires a decoding policy δ (e.g., greedy, beam, sampling); the operational algorithm is (A, δ). Pretraining-fine-tuning corresponds to sequential application of T on D_{pre} then D_{ft}; RLHF and distillation can be viewed as further applications of T on preference or teacher-generated data.

Definition 7 (Dataset). *A* Dataset D *provides the empirical evidence used by a* Training method*: it may be labeled or unlabeled, collected offline, generated online (e.g., RL, self-play), or produced by an augmentation pipeline.*

Definition 8 (Calculation Infrastructure). Calculation Infrastructure *for training is* $C_T = (H, S, R)$*, the available hardware* H*, software stack* S*, and resource constraints* R *used during model training.*

Definition 9 (Deep Learned Algorithm). *A* Learned Algorithm *is the trained, fully specified* Algorithm $A = (G, \theta^\star, \delta)$ *obtained by applying one or more* Training method*s* $T_1, \ldots, T_m$ *to* Dataset*s* $D_1, \ldots, D_m$ *under* Calculation Infrastructure C_T*. Here* G *is the evaluation order of the* Function Graph*,* $\theta^\star$ *are the learned parameters (including any non-gradient state, e.g., BatchNorm statistics), and* δ *is a fixed decoding/evaluation policy. When* δ *is deterministic (e.g., greedy), we write simply* $A = f_{\theta^\star}$*.*

Functional vs. Algorithmic Components. We distinguish *Function*s that live inside the *Function Graph* G from *Algorithm*s that operate on G during training or on its outputs during inference. *Functional components (nodes in G):* token embedding E, positional encoding PE, (masked) self-attention A, multi-head aggregation MH, position-wise feed-forward F, residual add R, normalization N, and output projection/logits O (plus cross-attention X for encoder–decoder variants). *Algorithmic procedures (outside G):* a *Training method* (optimizer, schedules, losses/regularizers, curriculum) that selects $\theta^\star = T(M, D; C_T)$, and an inference-time decoding policy δ (greedy, top-k, nucleus, beam, constrained), yielding the *Learned Algorithm* $A = (f_{\theta^\star}, \delta)$. This separation supports attribution hygiene (cf. Remark 1).

2.2 Definition: Transformer

Definition 10 (Transformer Mathematical Model). *A* Transformer Mathematical Model *is a* Mathematical Model $M = (G, \lambda)$ *where* G *is a* Function Graph *whose nodes are* Function*s that are defined* Transformer Mathematical Functions *implementing: (i) token embedding and output projection, (ii) positional encoding, (iii) masked multi-head (self-/cross-) attention, (iv) position-wise feedforward mappings, (v) residual connections and layer normalizations. The edges of* G *encode either an encoder–decoder or decoder-only dataflow; the wiring and hyperparameters (e.g.,* L*,* d_{model}*,* h*,* d_{ff}*, masking policy, tokenization/positioning scheme) are fixed by* λ*. By* Mathematical Model*,* M *induces a family* $\{f_\theta\}$ *of* Function*s with learnable parameters* θ*.*

Definition 11 (Transformer Mathematical Functions). *The* Transformer Mathematical Functions *are the* Functions *that instantiate the node types of a Transformer* Function Graph*. For a token sequence* $x = (x_1, \ldots, x_T)$ *and width* d_{model}*:*

Input Representation. *Embedding and positional functions:*

$$E : V \to \mathbb{R}^{d_{model}}, \qquad E(x_i) = W_E[j, :]. \tag{1}$$

$$PE : \{1, \ldots, T\} \to \mathbb{R}^{d_{model}}. \tag{2}$$

With sinusoidal PE*:*

$$PE_{(pos,2i)} = \sin\Big(\frac{pos}{10000^{2i/d_{model}}}\Big), \tag{3}$$

$$PE_{(pos,2i+1)} = \cos\Big(\frac{pos}{10000^{2i/d_{model}}}\Big). \tag{4}$$

They compose to

$$X_{in} = \sqrt{d_{model}}\, E(x) + PE \in \mathbb{R}^{T \times d_{model}}. \tag{5}$$

Attention Subfunctions (Multi-Head). *For* $X \in \mathbb{R}^{T \times d_{model}}$ *and heads* $i = 1, \ldots, h$*:*

$$W_i^Q \in \mathbb{R}^{d_{model} \times d_k}, \quad W_i^K \in \mathbb{R}^{d_{model} \times d_k}, \quad W_i^V \in \mathbb{R}^{d_{model} \times d_v}, \quad W^O \in \mathbb{R}^{h d_v \times d_{model}}, \tag{6}$$

$$\mathrm{Attn}(Q, K, V; M) = \mathrm{softmax}\Big(\frac{QK^\top + M}{\sqrt{d_k}}\Big)V, \tag{7}$$

$$head_i = \mathrm{Attn}(XW_i^Q,\, XW_i^K,\, XW_i^V;\, M), \tag{8}$$

$$\mathrm{MHA}(X) = \mathrm{Concat}(head_1, \ldots, head_h)\, W^O. \tag{9}$$

Feedforward and Normalization.

$$\mathrm{FFN}(x) = \phi(xW_1 + b_1)W_2 + b_2, \tag{10}$$

$$\mathrm{LN}(x) = \frac{x - \mu}{\sqrt{\sigma^2 + \epsilon}}\gamma + \beta. \tag{11}$$

Layer Composition (post-LN, Residual).

$$U = \mathrm{LN}\big(X + \mathrm{MHA}(X)\big), \tag{12}$$

$$Y = \mathrm{LN}\big(U + \mathrm{FFN}(U)\big), \tag{13}$$

$$\mathcal{T}_\theta(X) = Y. \tag{14}$$

Stacked Mapping.

$$f_\theta = \mathcal{T}_\theta^{(L)} \circ \cdots \circ \mathcal{T}_\theta^{(1)}. \tag{15}$$

Choices such as encoder–decoder vs. decoder-only wiring, masking policy M*, activation/normalization variants, and dimensions* $(L, d_{model}, h, d_{ff})$ *are fixed by the hyperparameters* λ *(*Transformer Architecture*).*

2.3 Definition: Transformer Architecture

Definition 12 (Transformer Architecture). *A* Transformer Architecture *is a* Deep Learning Architecture *that specifies (i) an admissible family of* Function Graph*s (e.g., encoder–decoder or decoder-only), and (ii) a hyperparameter schema Λ (e.g., layer depth L, model width d_{model}, attention heads h, feed-forward width d_{ff}, normalization/activation variants, masking, tokenization and positional choices). Instantiating $\lambda \in \Lambda$ yields a* Mathematical Model $M_\lambda = (G_\lambda, \lambda)$ *whose nodes are drawn from* Transformer Mathematical Functions*.*

Remark 2 (Block-level model view). Under the paper's *Analytical Framework and Core Definitions* (Sect. 2), Fig. 2 presents the Transformer *Mathematical Model* M_λ as a block-level *Function Graph*. Typical node *Function*s include token embedding E, positional encoding PE, scaled dot-product attention A, multi-head attention MH, position-wise feed-forward F, residual add $(+)$, layer normalization LN, linear projection Lin, and softmax Smx. Arrows denote tensor dataflow/composition. Encoder and decoder stacks compose layer sub-*Function*s repeated L times, with layer-specific parameters $\theta^{(\ell)}$. The overlaid equations instantiate selected *Transformer Mathematical Functions* (A, MH, PE); the view remains at block granularity.

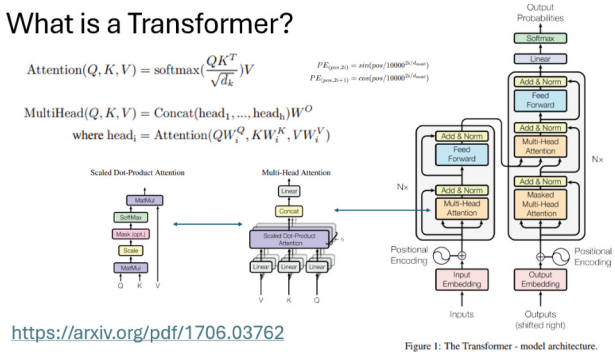

Fig. 2. Block-level view of the Transformer *Mathematical Model* M_λ as a *Function Graph*. Nodes are drawn from *Transformer Mathematical Functions*; arrows indicate dataflow/composition; "$L\times$" marks repeated layer compositions with distinct $\theta^{(\ell)}$. Overlaid equations instantiate A (scaled dot-product attention), MH (multi-head), and PE (positional encoding). Adapted from [36].

Remark 3 (Model family and hyperparameter variations). Figure 3 (Table 3 in [36]) enumerates members of the Transformer family $\{M_\lambda : \lambda \in \Lambda\}$ by varying λ: depth L, widths $(d_{\text{model}}, d_{\text{ff}})$, attention multiplicity h, key/value dims (d_k, d_v), and regularizers $(p_{\text{drop}}, \varepsilon_{\text{ls}})$; row (E) switches the positional-encoding sub-*Function* (sinusoidal $\rightarrow$ learned). Given fixed *Training method* T, *Dataset* D, and *Calculation Infrastructure*

C, each M_λ trains to a distinct *Learned Algorithm* $A_\lambda = f_{\theta^\star}$. Note that "train steps" parameterizes T (not M_λ), and BLEU/PPL are properties of A_λ under the stated evaluation.

Table 3: Variations on the Transformer architecture. Unlisted values are identical to those of the base model. All metrics are on the English-to-German translation development set, newstest2013. Listed perplexities are per-wordpiece, according to our byte-pair encoding, and should not be compared to per-word perplexities.

	N	d_{model}	d_{ff}	h	d_k	d_v	P_{drop}	ϵ_{ls}	train steps	PPL (dev)	BLEU (dev)	params $\times 10^6$
base	6	512	2048	8	64	64	0.1	0.1	100K	4.92	25.8	65
(A)				1	512	512				5.29	24.9	
				4	128	128				5.00	25.5	
				16	32	32				4.91	25.8	
				32	16	16				5.01	25.4	
(B)					16					5.16	25.1	58
					32					5.01	25.4	60
(C)	2									6.11	23.7	36
	4									5.19	25.3	50
	8									4.88	25.5	80
		256			32	32				5.75	24.5	28
		1024			128	128				4.66	26.0	168
			1024							5.12	25.4	53
			4096							4.75	26.2	90
(D)							0.0			5.77	24.6	
							0.2			4.95	25.5	
								0.0		4.67	25.3	
								0.2		5.47	25.7	
(E)	positional embedding instead of sinusoids									4.92	25.7	
big	6	1024	4096	16			0.3		300K	**4.33**	**26.4**	213

Fig. 3. Transformer model-family variations ([36], Table 3) expressed as $\lambda \in \Lambda$ for the *Mathematical Model* template M_λ. Different λ (depth/widths, attention dimensions, regularizers, positional-encoding choice) yield different trained *Learned Algorithm*s $A_\lambda = f_{\theta^\star}$ once the *Training method*, *Dataset*, and *Calculation Infrastructure* are fixed; "train steps" belongs to the *Training method*.

Remark 4 (Transformer training infrastructure across eras). Under the *Calculation Infrastructure* notation $C_T = (H, S, R)$: the 2017 reference setup used H with 8 NVIDIA P100 GPUs; S comprised a deep-learning stack with Adam optimizing a supervised translation objective; and R was bounded by memory and throughput. Training used WMT14 English–German / English–French *Dataset*s with cross-entropy *Training method*. Contemporary LLM training typically scales H to thousands of A100/H100 GPUs or TPU v4/v5; S adds distributed data/model parallelism (pipeline and tensor sharding) and mixed precision (FP16/BF16); and R expands to petabyte-scale *Dataset*s with multi-stage *Training method* (pretraining → instruction tuning → alignment).

Remark 5 (Feasibility and scaling). Because attention has quadratic cost in sequence length, feasibility at modern T and d_{model} is tightly coupled to C_T, motivating distributed training/inference and efficiency techniques; see Sect. 5.1.[1]

[1] To avoid symbol clashes with $C_T = (H, S, R)$, prefer Smx for softmax, Lin for linear projections, and $(+)$ for residual add in diagrams and equations.

Graph Structure of the *Transformer Architecture*. Let $G = (V, E)$ denote the block-level Function Graph. The node set V comprises core *Transformer Mathematical Functions*: token embedding E, positional encoding PE, (masked) self-attention A, multi-head attention MH, position-wise feed-forward F, residual add R, normalization N, and output projection/logits map O. Edges E encode tensor dataflow; composition in topological order yields the global *Function* $f_\theta : x \mapsto y$ (see Definition 2). For encoder–decoder variants, a cross-attention node X connects encoder activations to decoder states; for decoder-only variants, A is causally masked (masking modeled as a fixed auxiliary input to A).

A depth-L stack is an unrolled subgraph with layer-specific parameters θ_ℓ. A typical decoder-only forward map can be written at block granularity as

$$\mathrm{Block}_\ell(z) = R\big(F_{\theta_\ell}\big(N(R(MH_{\theta_\ell}(N(z))))\big)\big), \qquad f_\theta(x) = O(N(\mathrm{Block}_L \circ \cdots \circ \mathrm{Block}_1(E(x) + PE))).$$

Figure 2 depicts G at this granularity.

Remark 6 (From graph to model). Instantiating hyperparameters λ (e.g., $L, d_{\text{model}}, h, d_{\text{ff}}$) specifies admissible graphs and yields a Mathematical Model $M = (G, \lambda)$. A *Training method* applied to dataset(s) D on *Calculation Infrastructure* C_T selects $\theta^\star$, producing the *Learned Algorithm* $A = f_{\theta^\star}$ with decoding policy δ (cf. Remark 1).

Remark 7 (Parameter tying). Many implementations tie the output projection to the embedding matrix ($O = W_E^\top$). This is an optional constraint on G, not required by the *Transformer Architecture*.

Computational Complexity of Core *Transformer Mathematical Functions*. Let sequence length be T, model width d_{model}, heads h with per-head dims $d_k = d_v = d_{\text{model}}/h$, and feed-forward width d_{ff}. Per layer, the forward complexity decomposes as

$$\text{Projections } (Q, K, V, O): \ \Theta\big(T\, d_{\text{model}}^2\big), \tag{16}$$

$$\text{Attention scores/weights}: \ \Theta\big(T^2\, d_{\text{model}}\big), \tag{17}$$

$$\text{Feed-forward block}: \ \Theta\big(T\, d_{\text{model}}\, d_{\text{ff}}\big). \tag{18}$$

Thus a common summary is

$$\text{FLOPs}_{\text{layer}} \ = \ \Theta\big(T\, d_{\text{model}}^2 \ + \ T^2\, d_{\text{model}} \ + \ T\, d_{\text{model}}\, d_{\text{ff}}\big),$$

where the T^2 term (self-attention) dominates for long contexts, while the feed-forward often dominates when T is modest and $d_{\text{ff}} \approx 4\, d_{\text{model}}$.

Memory differs by phase. During training, storing attention maps incurs $\Theta(T^2)$ memory (per layer, ignoring constants), which typically limits T on a single accelerator. During autoregressive inference with key–value caching at context length L, per-token compute is $\Theta(L\, d_{\text{model}})$ and cache memory grows $\Theta(L)$ per layer, converting naive $\Theta(L^2)$ recomputation to linear time in L.

For encoder–decoder models with source/target lengths $(T_{\text{src}}, T_{\text{tgt}})$, cross-attention adds $\Theta(T_{\text{src}}\, T_{\text{tgt}}\, d_{\text{model}})$.

Remark 8 (Interaction with Calculation Infrastructure). These costs couple tightly to *Calculation Infrastructure*: matrix multiplies parallelize across heads, batch, and sequence shards, but quadratic attention memory constrains T; pipeline/tensor parallelism and activation rematerialization trade compute for memory; at inference, KV caching lowers per-token compute while increasing memory linearly in L.

Context. We follow the standard encoder–decoder/self-attention formulation introduced by [36]; subsequent pretraining paradigms [6,20,27,28] establish the empirical base for Sects. 3 and 5.1.

3 Background and Related Work

Transformers and Pretraining. The Transformer architecture [36] underpins modern language models and early masked/causal pretraining successes such as BERT [6], GPT-2 [27], and the text-to-text T5 framework [28]; replication and optimization studies like RoBERTa [20] clarified the role of data and hyperparameters.

Scaling, Sparsity, and Compute. Empirical scaling laws showed smooth power-law improvements with model/data/compute [14,18]; compute-optimal training altered the size–tokens tradeoff (Chinchilla) [16]; sparsely-activated MoE variants eased scaling [8]. Macro compute trends further contextualize rapid capability gains [1,31].

Evaluation and Emergent Claims. As capabilities expanded, broad benchmarks such as MMLU [13] and BIG-bench [32] captured breadth. Claims of "emergent abilities" [37] have been debated, with alternative explanations emphasizing metric effects [30].

Mechanisms and Alignment. Work on the mechanism of Transformer behavior includes FFN layers as key-value memories [12] and the Circuits thread [23]. Alignment via human feedback progressed from preference-based RL [5] to instruction-following [25].

4 What Happened? The Transformer Surprised Almost Everyone

Our timeline starts with the publication of the paper *Attention Is All You Need*, which describes the so-called *Transformer Architecture* [36]. As of August 2025, the paper has received approximately 191,500 academic citations, according to Google Scholar. The original problem considered in the paper is machine translation from one language to another.

4.1 Transformer Era Phases (2017–2025)

The timeline since the invention of *Transformer Architecture* can be summarized as follows [26].

Prologue (2017–2019): Foundations and Early Scaling. Transformers (2017) entered as largely language-agnostic architectures and drew initial skepticism; within a year BERT/GPT reset leaderboards, "BERTology" took off, and scale began to look like the dominant driver of progress [26]. The transition from masked/causal pretraining to few-shot prompting was crystallized by GPT-3 [3], which made in-context learning a central interface for downstream tasks and demos. In 2018 a book with 23 leading AI expert predictions [9] is published.

Methodological Debates on Understanding and Scale (2020–2022). Form vs. meaning arguments intensified (e.g., the "octopus test"), while GPT-3 delivered a visceral shock: many classical NLP tasks appeared one-shot or solved, prompting concerns about closed "API science," ethics of ever-larger models, and community splits surfaced by field-wide surveys [26].

Public Deployment Shock and Field Reorientation (Nov 2022–2023). ChatGPT's release reoriented the field almost overnight; entire subareas pivoted, conferences refocused on prompting and evaluation, and teaching, media, and policy attention surged—expanding impact but also drowning signal in noise [26].

Consolidation, Evaluation, and Resource Stratification (2024–2025). "LLM-ology" (probing/evaluating frontier models) came to dominate even as pushes for openness and replicability grew; compute demands widened resource divides while high-level prompting lowered some entry barriers; capabilities advanced alongside brittle failures and localization/cultural grounding gaps beyond the Anglosphere [26].

Epilogue: Paradigm Shift? Practically, yes—natural-language prompting became a universal interface and the research center of gravity moved—yet some continuity with earlier distributional/transfer-learning principles remains; what clearly changed were corporate influence, media dynamics, evaluation culture, and pedagogy, while the human role in the loop remains the central tension [26].

4.2 Surprise to Experts in the Field Book: *Architects of Intelligence* (2018)

To probe contemporaneous expert expectations shortly after the introduction of the *Transformer Architecture* [36], this study examines Martin Ford's interview volume [9] (2018), which features 23 prominent contributors to AI[2]. The interviews solicit views on promising technologies, likely breakthroughs, and timelines for human-level intelligence. A close reading indicates that none of the interviewees explicitly anticipated the central role that the Transformer—and the associated paradigm of large-scale self-supervised pretraining—would soon play in LLMs and generative AI. As an ex post snapshot of expert sentiment in 2018, the book provides evidence consistent with the research hypothesis in Sect. 1.1, pending contrary documentation.

[2] Yoshua Bengio, Stuart Russell, Geoffrey Hinton, Nick Bostrom, Yann LeCun, Fei-Fei Li, Demis Hassabis, Andrew Ng, Rana el Kaliouby, Ray Kurzweil, Daniela Rus, James Manyika, Gary Marcus, Barbara J. Grosz, Judea Pearl, Jeffrey Dean, Daphne Koller, David Ferrucci, Rodney Brooks, Cynthia Breazeal, Joshua Tenenbaum, Oren Etzioni, Bryan Johnson.

4.3 Surprise to Senior Industry Leaders (Bill Gates and Satya Nadella)

Public statements from leading industry figures after ChatGPT's release reinforce the surprise thesis. In March 2023, Bill Gates (Microsoft co-founder) called ChatGPT one of only "two demonstrations of technology that [were] revolutionary," the other being the graphical user interface circa 1980 [11]. In December 2024, Microsoft CEO Satya Nadella described ChatGPT's November 2022 breakout as "a bolt from the blue," noting that "none of us knew what was going to happen" [21]. Given that ChatGPT is built on the *Transformer Architecture* architecture introduced in 2017 [36], these retrospective assessments—made roughly six years after that publication—support the claim (Sect. 1.1) that the scale and timing of Transformer-enabled LLMs were underappreciated even by well-placed observers.

4.4 ChatGPT Surprises Futurists

Benedict Evans, a prominent technology analyst and public speaker, publishes annual technology outlooks on his website.[3] A comparison of his 2022 outlook—completed immediately prior to ChatGPT's November 2022 release—with the 2023 edition indicates a marked shift: the 2022 materials contain little explicit discussion of LLMs or generative AI, whereas the 2023 deck devotes substantial attention to these topics. This contrast suggests that even professional forecasters did not fully anticipate the speed and breadth of the generative-AI inflection enabled by the *Transformer Architecture* [36], supporting the surprise thesis articulated in Sect. 1.1.

4.5 Possible Surprise to the Original Authors (Karpathy's Account)

In a 2023 lecture, Andrej Karpathy (then at OpenAI) reflects that the *Attention Is All You Need* paper [36] "reads like a normal machine translation paper" and that the authors "did not fully anticipate the impact," noting that the architecture "ended up taking over" much of AI within five years [19]. While this is an external interpretation rather than direct testimony from the original authors, it aligns with the broader evidence in this study that the scope and timing of Transformer-enabled LLMs were underappreciated at the time, supporting the research hypothesis in Sect. 1.1.

4.6 Lex Fridman Podcast with Edward Gibson

In April 2024—nearly seven years after the *Transformer Architecture* was introduced [36]—Lex Fridman (AI researcher and podcast host) interviewed Edward Gibson (Professor of Brain and Cognitive Sciences, MIT) in a long-form discussion on language, cognition, and LLMs [10]. A central theme is the dissociation of language from thought: Gibson argues that "language is not necessary for thinking," a stance consonant with recent reviews [7]. The dialogue probes whether current LLMs "understand meaning" or primarily exploit distributional regularities; Fridman highlights behavior that appears conceptually grounded, whereas Gibson remains cautious about attributing

[3] https://www.ben-evans.com/presentations.

genuine understanding. As an expert exchange occurring well after the Transformer's debut, the episode provides contemporaneous evidence that the scope and character of Transformer-trained LLMs remained contested, supporting the claim in Sect. 1.1 that their impact was underappreciated at the time.

5 Reasons Why Transformer Was a Surprise

Evidence presented in Sect. 4 supports the research hypothesis, and the literature reviewed indicates that the research question remains unresolved (Sect. 1.1).

5.1 Hyperscaling of the Models

Empirical scaling laws [14,18] and compute-optimal training results [16] explain much of the capability trajectory discussed here; sparse MoE designs [8] and macro compute trends [1,31] contextualize the pace and efficiency of progress.

The *Transformer Architecture* enabled sustained increases in model capacity, compute allocation, and data volume. The original 2017 model contained on the order of 10^8 parameters [36]; BERT (2018) reached roughly 1.1×10^8; by 2020, GPT-3 scaled to $\sim 1.75 \times 10^{11}$ parameters—nearly three orders of magnitude in three years. Contemporary systems operate in trillion-parameter regimes (often via mixture-of-experts designs), exceeding prior expectations about feasible scale.

Empirically observed "scaling laws" showed that loss and downstream performance improved predictably with increases in model size, dataset size, and compute, with weaker-than-expected saturation. This regularity reduced exploratory uncertainty and incentivized further scale-ups, helping to explain the unexpectedly large performance gains.

The same dynamics introduced binding constraints. Training frontier models required specialized accelerators, large distributed clusters, and substantial energy budgets; data pipelines expanded to web-scale corpora with heavy de-duplication, filtering, and augmentation. Consequently, progress became as much an engineering and infrastructure achievement as an algorithmic one, and the feasibility of continued hyperscaling emerged as a strategic bottleneck.

5.2 Anecdotal Failures Are Weak Evidence of Fundamental Limits

Isolated failure cases of a *Learned Algorithm* do not, by themselves, establish limits of the underlying *Mathematical Model*. Such anecdotes are useful as hypothesis generators and for debugging, but they typically indicate gaps in the *Dataset*, shortcomings in the *Training method*, or inference-time setup issues rather than principled impossibility claims. In practice, many highlighted errors disappear once the relevant cases are incorporated into the *Dataset*, the *Training method* is adjusted (e.g., objective, curriculum, augmentation, or optimization), or the inference procedure is improved.

A frequent claim is that LLMs are limited because they "only predict the next word" [26]. This mixes distinct notions: next-token prediction is a *Training method*,

not a statement about the expressive capacity of the *Mathematical Model*. The instantiated model class—such as the *Transformer Architecture*—implements a compositional computation over token sequences, producing structured internal representations that support behaviors far beyond naive frequency matching. That a model is trained with next-token prediction does not imply that its inference-time computation is restricted to "guessing a single output"; the attainable behaviors depend on the model class and learned parameters, not on a colloquial description of the loss.

Claims of fundamental limitation require stronger evidence than anecdotes: for example, systematic evaluations showing robust performance plateaus under increased data, compute, and careful tuning; ablations isolating architectural factors; or constructive arguments that tie a task family to properties the *Mathematical Model* cannot realize. Absent such evidence, anecdotal failures should be treated as targets for iterative improvement, not as boundaries on what a *Learned Algorithm* can achieve.

5.3 Imprecise Core Terms Obscured Model Capabilities

Several widely used notions in AI lacked exact meanings in the period under study—notably what constitutes an "architecture", how a "model" is specified, and how to distinguish the learning process from the model itself. In this paper we adopt precise concepts (Sect. 2) and describe the original *Transformer Architecture* as a *Mathematical Model* realized as a *Function Graph*, trained via a specified *Training method* on a given *Dataset*. This alignment reduces ambiguity and enables capability claims to rest on well-defined objects.

Schematic "architecture diagrams" are useful didactic summaries but are not, by themselves, mathematical specifications. In our terms, they are informal depictions of a *Function Graph*. Inferring hard capability limits or impossibility claims from such figures is unreliable; these should instead be tied to properties of the underlying *Mathematical Model* and the chosen *Training method* and *Dataset*. The lack of clear separation among these concepts contributed to early misreadings of what the *Transformer Architecture* could achieve.

Once formalized, the *Transformer Architecture* can be seen to occupy a distinct region within *Deep Learning Architecture*: its specific computational structure and optimization affordances expanded the practical envelope of LLM capabilities, rather than merely refining prior approaches. Contemporary reactions already hinted at the need for greater precision; for example, a NeurIPS reviewer noted missing formal definitions for multi-head attention in the original exposition (review link):

> *"While the general architecture of the model is described well and is illustrated by figures, architectural details lack mathematical definition, for example multi-head attention. Why is there a split arrow in Figure 2 right, bottom right? — A formal definition of this would greatly help readers understand this."*

In summary, clearer use of *Mathematical Model*, *Function Graph*, *Training method*, and *Dataset* helps explain why informal reasoning from diagrams invited forecasting errors about the Transformer's eventual impact.

5.4 Why Fundamental Limits Are Hard to Establish

The mathematics of the *Transformer Architecture* are compact, yet the induced behavior of the resulting *Mathematical Model*—a deep, high-dimensional *Function Graph* learned from data—is complex. Proving hard limits for such systems requires statements about the model class and its optimization dynamics, not only about individual failure cases or small benchmarks.

Observed "emergent" behaviors [37] can appear discontinuous under some metrics; recent analyses argue many such jumps smooth out with alternative metrics or better statistics [30].

Most investigations of limits for *Deep Learning Architecture* instantiated with the *Transformer Architecture* have been empirical. This creates a moving target: once a weakness is publicized, it can be absorbed by the next *Dataset* or addressed by modifying the *Training method*, often eliminating the original effect. Consequently, claims of fundamental limitation should rest on (i) analysis tied to properties of the *Mathematical Model* (e.g., constructive counterexamples or reductions under stated assumptions), or (ii) systematic studies showing robust plateaus under controlled increases in data, compute, and careful tuning—rather than on anecdotal or one-off evaluations.

The original *Transformer Architecture* paper addressed machine translation, but the underlying *Mathematical Model* is sequence-general. With minor interface changes (e.g., tokenization of pixels or patches), the same structure has been applied to images, video, and other modalities. Early readings that conflated "text" with the scope of the *Mathematical Model* underestimated this cross-domain applicability, contributing to the widespread surprise discussed in Sect. 4.5.

As a predominantly empirical field, AI often validates a method on a specific task and only later tests external validity. The *Transformer Architecture* illustrates that a solution validated in one domain can generalize far beyond it. Hence, putative "limitations" should be treated as provisional unless demonstrated at the level of the *Mathematical Model* class or shown to persist under rigorous scaling and ablation.

5.5 Transformer Language Models Are Better Viewed as Concept-Fusion Models

The label "LLM" conflates three axes in our framework: the *Mathematical Model* class, the *Training method*, and the *Dataset*. Concretely, what is often called an "LLM" is a *Deep Learning Architecture* instantiated by the *Transformer Architecture* (a learned *Function Graph*), trained with a next-token objective on a large text *Dataset*. Reasoning about capabilities or limits should therefore attribute claims to the appropriate axis, rather than to the umbrella name.

A common critique asserts that such models "just model word frequencies." This is inconsistent with the *Transformer Mathematical Functions*: multi-head self-attention, residual mixing, and position-wise nonlinear layers implement learned, content-dependent fusion over token sequences, not fixed-order n-gram counts. Empirically, internal representations encode syntactic and semantic structure far beyond frequency statistics (e.g., syntax tree geometry; layerwise rediscovery of classical NLP

stages; mechanistic "induction heads" that perform pattern completion), contradicting a pure-frequency view [15,24,29,33].

Seen this way, these systems are better described as *concept fusion models*: the *Transformer Architecture* composes information across positions to form structured internal states that correlate with meaning and support behaviors beyond naive counting. The term is expository only; formally, the object remains a *Mathematical Model* as defined in Sect. 2. Claims about limitations should be tied to properties of the *Mathematical Model*, the *Training method*, or the *Dataset*—not to an imprecise label.

5.6 Expertise and the Limits of Background Knowledge

Expertise around AI is plural. Neuroscience, linguistics, and computer science contribute distinct priors, yet claims about capability or limitation must ultimately be grounded in properties of the *Mathematical Model*, its realization as a *Function Graph*, the *Training method*, and the *Dataset*. The broader, cross–disciplinary pull of AI has often been described as *consilience* [38]; however, background knowledge imported from any one field can mislead when it is mapped onto a different formal object.

The exchange in Sect. 4.6 illustrates this tension. Linguistics draws a sharp distinction between language *form* and *meaning*, while the *Transformer Architecture* is a specific *Deep Learning Architecture* that learns internal representations via a defined *Training method*. Interpreting the *Transformer Architecture* through analogies to human neural mechanisms risks category errors: the model's behavior follows from the mathematics of the *Function Graph* and optimization on a given *Dataset*, not from how the human brain encodes or decodes linguistic content.

Much of what is currently called a "limitation" of large models is empirical and contingent: it attaches to a particular instantiation of the *Mathematical Model* trained with a particular *Training method* on a particular *Dataset*. Such findings may disappear under revised data, objectives, or scaling. For this reason, durable limitation claims should either (i) target properties of the model class itself, or (ii) demonstrate robustness under controlled variation of data, compute, and tuning.

Finally, experts—ourselves included—are vulnerable to fast, analogy–driven judgments [17]. Assuming that the *Transformer Architecture* inherits the learning constraints of human linguistic units exemplifies "thinking fast." A more reliable approach is "thinking slow": attribute claims to the correct axis (*Mathematical Model*, *Training method*, *Dataset*), state scope conditions explicitly, and prefer analyses that survive ablation and scaling.

Findings from forecasting science suggest experts often miscalibrate when environments shift; practices from [34] motivate our recommendations.

5.7 Learning from Transformer Surprises: Improving Prediction Practice

The unexpected breadth of the *Transformer Architecture* reveals gaps in how the community forecasts progress. Engineering lead times—from proof-of-concept to production—do not, by themselves, justify claims that the 2017 result had "already" demonstrated its full impact. Rather, the surprise indicates shortcomings in concept precision, method attribution, and the linkage between theory and implementation.

First, adopt concept hygiene. Capability claims should be attributed to the correct axis in our framework (Sect. 2): the *Mathematical Model* as realized by a *Function Graph*, the *Training method*, and the *Dataset*. Ambiguity among these led to misreadings in early commentary (Sects. 5.3, 5.6). Forecasts should state which axis is being varied, which is held fixed, and under what scope conditions the claim is expected to hold.

Second, make theory executable. When proposing properties of a *Mathematical Model*, provide an exact specification that can be traced to reference code (a faithful implementation of the same *Function Graph* and *Training method*). This "spec-to-code" linkage reduces interpretive drift and allows independent groups to test claims on controlled *Dataset* variants.

Third, prefer durable evidence for limits. As argued in Sect. 5.4, empirical weaknesses are moving targets: once public, they can be mitigated by changing the *Dataset* or *Training method*. Claims of fundamental limitation should target properties of the *Mathematical Model* class (e.g., constructive counterexamples under stated assumptions) or demonstrate robustness under systematic variation of data, compute, and tuning. Where feasible, preregistered evaluations and ablations help separate contingent failures from structural ones.

Fourth, model practical constraints explicitly. Hardware, systems, and optimization advances (e.g., GPU/TPU interconnects, memory bandwidth, compiler and kernel improvements) shape the realized capability of a fixed *Mathematical Model*. Forecasts should therefore pair model-class reasoning with a compute-and-systems budget, including assumptions about training scale, inference cost, and deployment constraints.

Fifth, anticipate cross-domain transfer when the *Mathematical Model* is modality-agnostic. The *Transformer Architecture* operates on token sequences; with appropriate interfaces, the same *Function Graph* extends beyond text (Sects. 5.5, 5.4). Forecasts should test for such generality early by varying tokenization and modality while holding architecture and optimization fixed.

Finally, institutionalize postmortems. For prominent forecasting misses, record the original prediction, the evidentiary basis, the axis attribution (*Mathematical Model* / *Training method* / *Dataset*), and the eventual disconfirming evidence. A corpus of such analyses can refine heuristics, improve attribution discipline, and raise the baseline for future predictions.

Taken together, these practices shift forecasting from informal diagram-level intuition to claims anchored in the exact objects we study—the *Mathematical Model*, its *Function Graph*, the *Training method*, and the *Dataset*—and thereby reduce the likelihood of being surprised by the next architecture-scale advance.

Findings from forecasting science suggest experts often miscalibrate when environments shift; practices from [34] motivate our recommendations.

6 Conclusion

This study examined why the *Transformer Architecture* and its applications, particularly ChatGPT, exceeded expert expectations between 2017 and 2025. Through document analysis, expert-elicitation audit, and synthesis of contemporaneous predictions,

we identified systematic forecasting errors that reveal important gaps in how the field evaluates new *Deep Learning Architecture*s.

Our analysis demonstrates that the *Transformer Architecture*'s impact was significantly underestimated across multiple expert communities—from the AI research community to industry leaders and technology analysts. This forecasting error stemmed not from the architecture's complexity, but from imprecise attribution of capabilities to the correct conceptual axis: the *Mathematical Model*, *Training method*, or *Dataset*.

6.1 Why Was the Success of the *Transformer Architecture* and ChatGPT Such a Surprise?

Our analysis identifies eleven factors that contributed to the systematic underestimation:

- **Hyperscaling Exceeded Expected Constraints.** The *Transformer Architecture* enabled unprecedented scaling along three axes simultaneously: model parameters (from 10^8 to 10^{12}), *Dataset* size, and *Calculation Infrastructure*, with weaker-than-expected saturation.
- **The *Mathematical Model* Transcended Its Initial Domain.** Originally validated for machine translation, the *Transformer Architecture* operates on abstract token sequences, enabling application to images, video, and other modalities through appropriate tokenization—a generality not anticipated from early implementations.
- **The Term "LLM" Conflates Distinct Concepts.** What is commonly called an LLM conflates the *Mathematical Model* class, the *Training method*, and the text *Dataset*. The *Transformer Mathematical Functions* implement learned, content-dependent fusion over sequences, not linguistic processing as understood in cognitive science.
- **Empirical Evaluation Dominated Analytical Understanding.** The field evaluated *Transformer Architecture*-based systems primarily through benchmark performance rather than analyzing properties of the underlying *Mathematical Model* class, making limitation claims provisional rather than fundamental.
- **Anecdotal Failures Were Misinterpreted as Fundamental Limits.** Isolated failure cases of a *Learned Algorithm* typically indicate gaps in the *Dataset* or *Training method* rather than constraints of the *Mathematical Model*—a distinction often overlooked in early assessments.
- **Mathematical Properties of *Transformer Mathematical Functions* Remain Underexplored.** Despite the compact mathematical formulation, the expressive capacity and optimization dynamics of the composed *Function Graph* lack comprehensive theoretical characterization.
- **Imprecise Terminology Obscured Capability Assessment.** The absence of exact definitions for "architecture," "model," and "algorithm" led to category errors when reasoning about capabilities and limitations of *Transformer Architecture*-based systems. Critics often attributed empirical failures to inherent limitations of the *Transformer Architecture* itself when the actual constraints lay in the *Training method* (e.g., optimization procedures, loss functions) or *Dataset* (e.g., coverage, quality, size)—a misattribution that made architectural constraints appear fundamental when the problems were actually implementation-specific and solvable.

– **Architecture Diagrams Were Conflated with Mathematical Specifications.** Visual representations of *Deep Learning Architecture*s, while pedagogically useful, do not constitute mathematical specifications of the *Function Graph* or its properties—a distinction not consistently maintained in forecasting discussions.
– **Human Language Processing Analogies Misled Predictions.** The *Transformer Mathematical Model*'s computation follows from its Function Graph and optimization, not from mechanisms of human linguistic processing—yet predictions often assumed the latter constraint.
– **Interdisciplinary Expertise Created Attribution Errors.** Experts from neuroscience, linguistics, and computer science applied domain-specific priors that did not map correctly onto the *Mathematical Model*, leading to misplaced limitation claims.
– **Engineering Refinements Masked Architectural Capabilities.** The transition from proof-of-concept (2017) to production systems (2022) involved substantial engineering of *Calculation Infrastructure* and *Training method*s, delaying recognition of the *Transformer Architecture*'s inherent capabilities.

6.2 Implications for Forecasting and Future Work

To reduce surprise when assessing innovations such as the *Transformer Architecture*, we propose a two-part agenda: (i) practical principles for ex-ante forecasting, and (ii) research directions that strengthen the theoretical basis for such forecasts.

Principles for More Accurate Forecasts.

– Use exact concepts and precise attribution. Maintain a controlled vocabulary via `\term{}` so claims unambiguously target the *Mathematical Model*, a particular *Learned Algorithm*, the *Training method*, or the *Dataset*; provide executable specifications and links to reference implementations to reduce interpretive drift.
– Separate levels of analysis (and architecture from engineering). Do not infer limits of a *Mathematical Model* from a single training failure; distinguish properties of the model and its *Function Graph* from early *Training method* or *Dataset* shortcomings, and from product-level engineering refinements.
– Pair analytical and empirical study. Analyze expressivity, inductive biases, and optimization of the model; validate with prospective scaling experiments (parameters, data, compute) and report prediction intervals instead of point estimates. Properties proven for the model class persist, whereas empirical weaknesses of a specific *Learned Algorithm* are moving targets.
– Model practical constraints explicitly. Incorporate compute, data, energy, memory bandwidth, networking, and engineering lead time—together with hardware transitions (e.g., CPU→GPU/TPU/accelerators)—and reflect how *Calculation Infrastructure* shapes feasible *Function Graph*s and deployment timelines.
– Test for domain transcendence early. For sequence-to-vector *Mathematical Model*s, vary tokenization schemes and modalities while holding the *Function Graph* fixed to assess cross-domain transfer potential.

- Use careful terminology. Avoid category errors (e.g., equating token prediction with human semantic decoding); treat "language model" as a description of the training objective, not a theory of cognition.
- Establish a forecast–audit loop with postmortems. Preregister forecasts with explicit attribution (model/method/data), evidentiary basis, and scope conditions; track leading indicators (effective training compute, tokens seen, inference cost), and when disconfirmed, identify which axis was misattributed to calibrate future predictions.
- Open artefacts and protocols. Release code, datasets, and exact `\term{}` definitions for replication; adopt standard evaluation suites and ablation templates to enable reproducible comparisons across *Mathematical Model*s.

Research Directions.

- *Theoretical characterization of model classes.* Develop formal results on the expressive capacity and optimization dynamics of *Transformer Mathematical Model*s (and successors). Key questions include: which function classes can *Transformer Mathematical Functions* approximate under composition, and how do choices in the *Function Graph* constrain learnable representations? Aim for constructive approximation bounds and impossibility results tied to model properties rather than dataset-specific outcomes.
- *Systematic study of domain transfer mechanisms.* The generalization of the *Transformer Architecture* from translation to multimodal tasks suggests domain-agnostic structure. Identify architectural properties that enable such transfer—e.g., abstraction levels of learned representations, invariances encoded in the *Function Graph*, and tokenization as a modality interface—to anticipate which future *Mathematical Model*s will transcend their initial applications.
- *Formalization of the learning-to-inference pipeline.* Provide frameworks that link *Dataset* properties and *Training method* choices to the resulting *Learned Algorithm*, particularly for self-supervised objectives. Characterize how *Calculation Infrastructure* constraints shape the optimization landscape and which training modifications preserve or alter model-class properties.

Together, these practices and research strands operationalize our conclusions by coupling exact concepts with analytic tools and empirical protocols, improving the reliability of forecasts about *Deep Learning Architecture* capabilities.

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NordDRG AI Benchmark for Large Language Models

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Abstract. Problem. LLMs are being piloted for clinical coding and decision support, yet no open benchmark targets the *hospital-funding* layer where Diagnosis-Related Groups (DRGs) determine reimbursement. In most OECD systems, DRGs route a substantial share of multi-trillion-dollar health spending through governed grouper software, making transparency and auditability first-order concerns rather than mere implementation details.

Research Question. *To what extent can LLMs emulate, explain, and apply DRG-based hospital payment rules?*

Contribution. *NordDRG-AI-Benchmark* is released, the first public, *rule-complete* test-bed for DRG reasoning. It bundles **(i)** machine-readable ∼20-sheet NordDRG definition tables and **(ii)** expert manuals and change-log templates that capture governance workflows, and exposes two suites: a 13-task *Logic* benchmark (code lookup, cross-table inference, grouping features, multilingual terminology, and CC/MCC validity checks) and a 13-task *Grouper* benchmark that requires *full DRG-grouper emulation* with strict exact-match scoring on *both* the DRG and the triggering `drg_logic.id`. Lightweight reference agents (LogicAgent, GrouperAgent) enable artefact-only evaluation.

Results. Under an artefact-only (no-web) setting, on the 13 Logic tasks *GPT-5 Thinking* and *Opus 4.1* score **13/13**, *o3* **12/13**; mid-tier models (*GPT-5 Thinking Mini*, *o4-mini*, *GPT-5 Fast*) achieve **6–8/13**, and the remaining models score **5/13 or below**. On full grouper emulation across 13 tasks (exact match on *both* DRG and `drg_logic.id`), *GPT-5 Thinking* solves **7/13**, *o3* **6/13**, and *o4-mini* **3/13**; *GPT-5 Thinking Mini* solves **1/13**, and all other tested endpoints score **0/13**. *Answer to RQ:* Top-tier LLMs reliably master rule-level DRG reasoning; the strongest models can *partially* emulate the full DRG grouping logic. To available knowledge, this is the first public report of an LLM *partially emulating* the complete NordDRG grouper logic with governance-grade traceability.

Significance. Coupling a rule-complete release with exact-match tasks and open scoring reveals domain-specific strengths and weaknesses that generic leaderboards miss and provides a reproducible yardstick for head-to-head and longitudinal evaluation in hospital funding.

Availability. All artefacts and scripts are available at https://github.com/longshoreforrest/norddrg-ai-benchmark.

Keywords: Large Language Models (LLMs) · LLM Software Agents · Diagnosis-Related Groups (DRG) · NordDRG · Clinical Coding · Healthcare

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Reimbursement · CaseMix Systems · Benchmarking and Reproducibility · Grouper Emulation

1 Introduction

1.1 Problem Identification

Large-scale generative models have progressed from research prototypes to production-grade services in under five years, bringing natural-language interfaces to domains once considered too specialised for automated reasoning. Healthcare finance is a notable example: hospital payments in many health systems are determined by *case-based* schemes built on Diagnosis-Related Groups (DRGs). In Europe, DRG systems have become the de facto basis for paying hospitals across most industrialised countries; in the United States, Medicare's Inpatient Prospective Payment System (IPPS) pays acute inpatient stays by MS-DRG; and across OECD members, DRG-like payments are among the most common approaches to reimbursing acute care. These arrangements channel a substantial share of public hospital expenditure through DRG groupers, making their transparency and auditability a matter of fiscal governance as well as clinical accountability [1,9,13,28].

Scale and Takes. Global current health expenditure reached about US$9.8 trillion in 2021 (10.3% of world GDP) [62]. Hospitals absorb the largest share of this spending—around one third worldwide (35.4% in 2017) [49] and roughly 39% across OECD countries [41]. In systems that use DRGs, a sizeable portion of these flows is mediated by grouper software, underscoring the need for auditable, rule-faithful logic.

Decision-Model Framing. A DRG grouper is, in effect, a *complex automated decision-making model* that maps a patient episode's *minimum dataset*—principal diagnosis, secondary diagnoses (with CC/MCC implications), procedure/intervention codes, age and sex, and context such as MDC entry and care setting—onto a single payment group via an ordered set of rules with explicit inclusions, exclusions, and national activation flags. Fiscal stakes arise because this deterministic mapping selects the payment from structured inputs; governance stakes arise because each decision should be traceable to specific rule rows and their change history.

Measurement Gap. Yet the rule sets that govern DRG allocations are intricate, multi-table specifications revised annually and interpreted through expert manuals and change-control workflows. For non-experts, these systems remain opaque; for experts, verifying end-to-end logic and its governance trace is laborious. Despite a surge in medical-LLM research, there has been *no open, rule-complete benchmark* to assess whether models can read the DRG rule graph and reproduce the grouper's governed behaviour. Prior evaluations either flatten the task into single-label classification on proprietary data or probe isolated rules, leaving no operational yardstick for measuring progress on hospital-funding logic. This gap motivates the benchmark introduced in this article.

Primary Research Question. To what extent can LLMs emulate, explain, and apply DRG-based hospital payment rules? We operationalise "extent" via exact-match emulation of NordDRG control flow, rule-compliance/violation checks, and expert-rated explanation faithfulness grounded in the released manuals.

1.2 Contributions

We release *NordDRG-AI-Benchmark*, a rule-complete, publicly available testbed that unites clinical coding, multilingual reasoning, and healthcare finance. The contribution has six parts:

1. **Definition Tables and Expert Manuals (A1–A2).** (i) *definition tables (Sect. 4.1)* comprising ~ 20 inter-linked sheets that encode the full NordDRG grouper logic (diagnoses/procedures, properties, country activation), and (ii) *expert manuals (Sect. 4.2)* and change-log templates that document the real governance workflow. Lightweight FI and FI–EN excerpts and ready-made dataset configurations are provided for rapid prototyping (cf. Sect. 4.5–4.7).
2. **Two Benchmark Suites with Exact-Match Scoring.** *NordDRG Logic Benchmark* (13 CaseMix tasks, Sect. 4.3) probes rule-level skills—code lookup, cross-table joins, grouping features, multilingual terminology, and QA audits. *NordDRG Grouper Benchmark* (13 case-level grouping tasks, Sect. 4.4) requires full emulation of the specification's control flow over `drg_logic` (execution order `ORD`, age/sex bounds, MDC entry, OR/PROCPRO evidence, CC/MCC with exclusions, national activation) and returns both `drg_nat` and the triggering `drg_logic.id`. Gold answer keys enable strict, governance-grade evaluation.
3. **Empirical Baselines.** Under a no-web, artefact-constrained setting, top models solve all 13 *Logic* tasks (GPT-5 Thinking, Opus 4.1: 13/13; o3: 12/13), whereas full grouper emulation remains challenging (best to date: 7/13 exact matches on *Grouper-1–13*); see Sect. 5.
4. **Lightweight Reference Agents.** Provider-agnostic *LogicAgent* (Combined tables) and *GrouperAgent* (FI tables) for artefact-only evaluation; details in Sect. 4.8.
5. **Availability and Reproducibility.** All artefacts, prompts, and gold answers are released in a versioned public repository with schema-stable identifiers and drop-in annual updates (Sects. 4).
6. **Answer to RQ$\star$.** The first governance-grade, exact-match baseline is provided showing that LLMs can master rule-level reasoning and partially emulate full DRG grouping, with failures concentrated in priority/exclusion control flow.

1.3 Research Questions

Primary RQ. To what extent can LLMs emulate, explain, and apply DRG-based hospital payment rules?

The following questions support RQ$\star$ and align one-to-one with the objectives of the solution:

1. **RQ1—Artefact Design (O1).** What *minimal, versioned* artefact bundle and schema are necessary and sufficient to encode the complete NordDRG rule graph and its governance semantics in a public, machine-readable form?
2. **RQ2—Benchmark Operationalisation (O2).** How should tasks, inputs, and exact-match scoring be structured so that (i) the *Logic* suite measures rule-level reasoning over the definition tables and manuals, and (ii) the *Grouper* suite enforces *full emulation* of the official control flow—returning both `drg_nat` and the triggering `drg_logic.id`?
3. **RQ3—Baseline Capability and Failure Modes (O3).** When constrained to the released artefacts (no external web access), how do representative LLM endpoints perform on the *Logic* and *Grouper* suites, and which systematic errors account for misses (e.g., cross-table joins, `ORD` priority, CC/MCC exclusions, national activation)?

These questions align one-to-one with the objectives in Sect. 3.2 and are answered empirically in Sect. 5.

1.4 Structure of the Paper

The remainder of the paper is organised as follows. Section 2 positions the contribution within the broader literature. Section 3 sets out the research methodology, gap, and objectives. Section 4 details the benchmark's design and development. Section 5 reports baseline LLM performance and assesses coverage, extensibility, and differentiation. Section 6 concludes with availability, repository access, and contribution guidance.

2 Background and Related Work

To orient the reader: Sect. 2.1 introduces the NordDRG CaseMix system—the rule base the benchmark operationalises—and establishes domain context. Section 2.2 reviews core advances in LLMs relevant to the tasks. Section 2.3 outlines lightweight LLM software agents as a framing for the benchmark. Section 2.4 motivates governance-grade evaluation through alignment and safety considerations. Section 2.5 surveys general LLM benchmarks and leaderboards, clarifying what they measure—and what they miss for CaseMix reasoning. Section 2.6 synthesises prior DRG-LLM studies to surface the remaining gaps. Section 2.7 describes production DRG grouper software and situates the benchmark within that ecosystem. Section 2.8 differentiates this article from earlier publications by specifying reused versus novel contributions.

2.1 NordDRG CaseMix System

The Nordic Diagnosis-Related Groups (NordDRG) classification is the shared CaseMix framework used throughout the Nordic countries to cluster inpatient episodes with comparable resource consumption [39]. By assigning each hospital stay to a single DRG, the system underpins transparent benchmarking of costs, outcomes, and efficiency across institutions, regions, and years.

NordDRG is distributed as a set of *definition tables*—about twenty interlinked sheets released annually in Excel, CSV, and JSON formats—that encode the logic connecting diagnosis and procedure codes, age/sex splits, and national activation flags to individual DRG codes. These tables constitute the authoritative rule base for coding software, technical-change management, and policy analysis.

Because each rule specifies explicit inclusion criteria, identical cases are grouped consistently regardless of hospital or country. At the same time, each nation can activate, deactivate, or re-weight DRGs locally, allowing the common core to accommodate diverse clinical practices and tariff structures. This blend of rule-level precision and country-level flexibility makes NordDRG a durable instrument for hospital finance, clinical research, and health-system governance in the Nordic region.

Bridge to this Contribution. The operational details described here are the substrate the benchmark formalises: the rule base and governance materials are packaged into machine-readable artefacts and task prompts. This lets models be tested on the same criteria human coders and policy analysts use, enabling exact, reproducible checks against the NordDRG logic.

2.2 LLMs

The breakthrough Transformer architecture introduced by Vaswani *et al.* [53] ignited today's natural–language processing renaissance. Initial encoder models such as BERT [19] were quickly followed by vast autoregressive generators—GPT-3 [8], GPT-4, and others—whose parameter counts now reach into the hundreds of billions and display emergent strengths in generation, retrieval, and reasoning. Concurrent strands of work explored alternative training frameworks (T5's text-to-text paradigm [47]), compute–data scaling laws (Chinchilla [24]), and extreme model sizes, exemplified by PaLM's 540-billion-parameter Pathways network [15]. Meanwhile, open-weight initiatives such as OPT [66] and Llama-2 [52] have broadened community access to state-of-the-art capabilities. Comprehensive surveys map this rapid trajectory from early generative models to ChatGPT, situating LLMs within the wider generative-AI landscape and cataloguing key architectural and application trends [11,36].

Since late 2022, chat-style interfaces—epitomised by ChatGPT—have moved LLMs beyond research into mainstream workflows. By masking low-level prompts behind natural conversation, these interfaces transform LLMs into a versatile cognitive layer that clinicians, lawyers, educators, and business users can harness with minimal technical overhead.

Bridge to this Contribution. Given LLMs' strengths and limits in reasoning, retrieval, and multilingual handling, the benchmark deliberately mixes cross-table joins, minority-language descriptors, and governance reasoning tasks. This design probes capabilities that generic NLP suites seldom touch, producing signals directly relevant to CaseMix workflows.

2.3 Emerging LLM Software Agents

LLM software agents—systems that pair a large language model with an explicit control loop—expose three core elements: (i) a prompt or system specification encoding the initial policy, (ii) a mutable interaction trace functioning as state, and (iii) the ability to `act` via tools or APIs [12,37]. Built on the LLM's perception and inference abilities, such agents support natural-language understanding and generation, task decomposition and planning, decision making, and tool use [58,63,64]. In practice, a user request can be broken into sub-tasks, relevant information fetched or computed through tools, and the outcome aggregated into a final response [51,65].

Despite rapid progress, challenges remain in high-stakes domains: hallucinations and brittle reasoning motivate grounding and retrieval-augmented generation (RAG) [27,32]; safety, ethical risks, and inherited biases require guardrails [5,61]; and opaque internal reasoning complicates debugging and audit [20]. Current work targets higher factuality, better explainability, and more reliable coordination (e.g., web-grounded training [35], dynamic context selection [30]) alongside richer evaluation settings.

Bridge to this Contribution. In this article, the agentic perspective is used pragmatically: lightweight, *single-role* agents can wrap deterministic table operations over the released artefacts. A *LogicAgent* can execute rule-level lookups and joins for the Logic tasks, emitting order-invariant, de-duplicated code sets; a *GrouperAgent* can apply the official `drg_logic` control flow (including `ORD`, age/sex bounds, OR/PROCPRO, CC/MCC with exclusions, and national activation) and return both `drg_nat` and the triggering `drg_logic.id`. This lightweight framing aligns with CaseMix governance by constraining tools to the released tables/PDFs and producing auditable traces, without requiring a full multi-agent pipeline (cf. Sect. 4.8).

2.4 AI Alignment and Safety

As technical frontiers expand, the focus of responsible AI practice is shifting from *capability* to *governance*. Ensuring that AI model behaviour aligns with human values and organisational objectives is a critical—and multidimensional—challenge [16], with well-documented risks of opacity, disparate impact, and governance failures in high-stakes domains [42]. The alignment problem is often decomposed into *outer alignment* (does the specified objective capture what we actually want?) and *inner alignment* (does the learned optimiser in fact pursue that objective, especially on inputs outside the training distribution?) [25,50]. Practical failures arise from reward misspecification, specification gaming, and goal misgeneralisation, particularly under non-stationarity [3,50].

Recent advances combine data-driven optimisation with human oversight to better shape training objectives. Preference learning and reward modelling translate human (or AI-assisted) judgments into training signals [17,31]; instruction tuning with human or AI feedback (e.g., RLHF and constitutional prompting) further reduces harmful behaviours and improves adherence to stated guidelines [4,43]. However, these techniques alone do not satisfy governance-grade requirements in institutional settings, where decision logic audit trail, deliberation, and auditability for consequential changes are indispensable.

Bridge to this Contribution. Auditability and governance needs motivate the emphasis on determinism, traceability, and documentation: tasks come with exact-match answer keys and all artefacts are versioned. This structure supports accountable evaluation and makes failures inspectable by domain experts.

2.5 LLM Benchmarks and Leaderboards

Benchmark suites and public leaderboards are central to tracking progress in LLMs. Early collections such as GLUE [56] and SuperGLUE [55] gave way to broader, multi-skill batteries—e.g., BIG-bench [18], MMLU [23], and BabyLM [60]—probing factual knowledge, reasoning, multilinguality, and instruction following. Meta-evaluations like HELM [33] further standardise conditions and report robustness, bias, and efficiency alongside accuracy.

Results from these suites are surfaced on widely watched leaderboards. The HuggingFace *Open LLM Leaderboard* aggregates automatic metrics (e.g., MMLU, ARC, HellaSwag) for open-weight models. The LMSYS/LMArena *Chatbot Arena* leaderboard ranks dialogue quality via blind, pairwise human preferences [34]. Task-specific evaluations such as MT-Bench [67] and vendor dashboards (e.g., Vellum-AI [54]) emphasise application skills like tool use and code generation. Figure 1 illustrates a representative snapshot, underscoring how frequently updated scorecards have become a de facto currency for comparing models.

Fig. 1. LLM Leaderboard illustration: Vellum leaderboard [54].

Bridge to this Contribution and Limits of General Leaderboards. General boards miss requirements central to hospital-funding logic; *NordDRG-AI-Benchmark* addresses them as follows:

1. **Granularity**—DRG grouping needs *deterministic control flow* and a rule trace; the *Grouper* suite enforces full `drg_logic` execution and requires (`drg_nat`, `drg_logic.id`) (Sect. 4.4).

2. **Source of Truth**—answers should bind to a *versioned, machine-readable* rule base; rule-complete *definition tables* and *expert manuals* with stable IDs are released (Sects. 4.1, 4.2).
3. **Auditability**—correctness must include the decision primitive; gold keys and reference agents emit audit-ready traces (Sects. 4.4, 4.8).
4. **Language**—minority-language descriptors and bilingual transfer are under-tested; FI and FI–EN excerpts make this testable (Sects. 4.5, 4.6).
5. **Scoring**—governance-grade tasks need order-invariant *set* outputs; the *Logic* suite uses exact-match sets, and the *Grouper* suite uses strict exact-match (Sects. 4.3, 4.4).

2.6 Related Work: DRGs and LLMs

LLMs are increasingly explored as end-to-end engines for hospital-reimbursement workflows. [59] fine-tune LLaMA on 236 k multimodal MIMIC-IV episodes, achieving 52% top-1 accuracy over 771 MS-DRGs—a strong indication that the model captures the CC/MCC rules traditionally hard-coded in grouper software. Complementing this, [29] show that a tool-augmented GPT-4 rivals human coders in ICD assignment, implying that careful retrieval and validation can offset the need for proprietary data. Zero-shot modelling is pushed further by [48], whose transformer infers DRG-like resource strata directly from longitudinal health trajectories, hinting at reimbursement systems emergent from the structural landscape of encoded patient health states.

Beyond accuracy metrics, recent studies ask whether LLMs truly understand DRG logic. [57] introduce *DRG-Sapphire*, using reinforcement learning with rule-aware rewards and generating physician-validated rationales that explicitly cite grouping criteria. Interpretability is also central to [21], whose multi-task transformer highlights token-level evidence so auditors can verify alignment with official rules. Structured knowledge helps as well: [22] inject a clinical knowledge graph into BERT and observe the largest gains on DRGs whose definitions hinge on multi-entity interactions, suggesting genuine rule-level reasoning. Finally, generic LLMs remain competitive; [7] repurpose them for zero-/few-shot ICD coding by exploiting the ICD hierarchy, while [6] couple LLaMA representations with a lightweight MultiResCNN, reporting F1 gains across the full label set.

Bridge to this Contribution. Prior DRG–LLM studies either rely on proprietary data, cover a single national system, or simplify grouping to flat classification. These gaps are addressed with a public, multilingual, rule-complete suite that tests models on the full NordDRG graph and yields comparable, reproducible baselines.

2.7 DRG Grouper Software

DRG groupers are rule-based software engines that map a patient episode's minimum dataset—principal and secondary diagnoses, procedures/interventions, demographics (age/sex), discharge status and other context—onto a single payment group. While national families differ (e.g., MS-DRG in the United States, AR-DRG in Australia,

APR-DRG used by many payers, and NordDRG across the Nordics), they share a governed control flow: MDC entry, ordered rule evaluation, surgical evidence (OR/non-OR), complication/comorbidity handling (CC/MCC or analogous constructs), and versioned annual updates [2,10,14,26,38].

In practice, groupers are distributed as compiled binaries, libraries or services accompanied by technical manuals and release notes. MS-DRG publishes detailed definitions and versioned release documentation [14]; AR-DRG provides a public technical specification describing hierarchy, splitting rules, and episode complexity scoring [26]; APR-DRG documents clinical logic and the severity/risk subclasses layered atop base DRGs [2]; and NordDRG uniquely exposes machine-readable definition tables that encode the executable rule graph (diagnosis/procedure properties, age/sex bounds, national activation) [38]. Historically, comparative scholarship has framed these systems as transparent, auditable instruments for hospital payment and CaseMix analysis, while noting national tailoring and update governance [10].

Bridge to this Contribution. The benchmark operationalises this ecosystem by packaging rule-complete, machine-readable artefacts (tables and manuals) and tasks that require exact, specification-faithful execution. By evaluating models against *both* the final DRG and the triggering rule row, alignment with grouper software's governed control flow is achieved and LLM behaviour is made auditable in the same terms as production groupers.

2.8 Our Previous Work

Our preliminary results appeared at [45,46]. The current article substantially extends that work by: (i) releasing a fully curated benchmark with task-oriented prompts, (ii) providing open access github repository with all artefacts and (iii) demonstrating baseline performance of recent LLMs on the automatically verifiable tasks

Bridge to this Contribution. Building on our earlier work, this article contributes a curated release (artefacts, prompts, answer keys), a documented repository, and first-cut baselines across multiple endpoints. These additions transform earlier proof-of-concept into an extensible benchmark aligned with Objectives O1-O3 (Sect. 3.2).

3 Research Methodology

This work follows the DSR method, which couples rigorous inquiry with artefact construction to solve a practical problem while contributing to knowledge. In line with the six-step model of Peffers *et al.* [44], we:

1. **Problem Identification.** Established the absence of a public, rule-complete CaseMix benchmark as a critical application–evaluation gap (see Sect. 1.1 and Sect. 3.1).
2. **Objective Definition.** Specified objectives for a reproducible LLM testbed (O1–O3; Sect. 3.2).

3. **Design & Development.** Constructed the artefact bundle (**A1–A6**; Sect. 4) that encodes the full NordDRG rule graph and governance semantics.
4. **Demonstration.** Instantiated two suites—*Logic* and *Grouper*—and exercised them through task scenarios grounded in the released tables and manuals (Sects. 4.3, 4.4).
5. **Evaluation.** Measured baseline performance of representative LLM endpoints under a no-web, artefact-only setting with exact-match scoring (Sect. 5).
6. **Communication.** Released a versioned, open repository with schema-stable identifiers, prompts, gold keys, and scoring scripts to support replication and extension (Sect. 6).

Two rapid design–evaluate iterations with domain experts ensured that artefacts and tasks remain aligned with real-world reimbursement workflows. All evaluations fixed the **Definition Tables + PDF Instructions** configuration and constrained models to the provided materials (no external retrieval), matching governance-grade auditability requirements.

Author Note: LLM Use[1]

3.1 Research Gap

Recent studies probe whether LLMs can predict DRGs [57,59] or internalise isolated grouper rules [21,22], yet the evidence base remains fragmented. Typical evaluations (i) focus on a single national system, (ii) flatten multi-table CaseMix logic into single-label classification, and (iii) rely on non-redistributable (proprietary) data. Consequently, to the best knowledge available (August 2025), there is no *public, multilingual, rule-complete* benchmark that reflects the full NordDRG rule graph—spanning cross-table joins, governance manuals, and annual change-control workflows. Without such a yardstick, researchers cannot (a) establish reproducible baselines for reasoning over reimbursement logic, (b) compare design patterns (fine-tuning, retrieval, reinforcement learning, chain-of-thought) under identical conditions, or (c) track progress across model generations and research groups (see Sect. 3.2, O1–O3).

These shortcomings crystallise into two open challenges:

1. **Scope Gap**—No existing evaluation covers the *full*, multilingual CaseMix ruleset used in hospital funding; current DRG experiments rely on reduced or proprietary slices. *(Addressed by O1 via a rule-complete artefact bundle.)*
2. **Benchmark Gap**—The absence of an open, standardised CaseMix benchmark precludes rigorous, head-to-head comparison of LLM approaches and hinders cumulative research on trustworthy automation in hospital finance. *(Addressed by O2–O3 via task suites and exact-match scoring.)*

[1] During manuscript preparation the authors made limited use of OpenAI services (*GPT-5 Thinking*, *GPT-4o*, *o3*, and *o4-mini*) for grammar, stylistic refinement, and reference-list formatting. All prompts, intermediate outputs, and final wording were reviewed and verified by the authors, who accept full responsibility for the content.

3.2 Research Design Objectives

In response to the *Scope Gap* and *Benchmark Gap* (Sect. 3.1), three objectives aligned with DSRM are pursued:

1. **O1—Artefact Construction** *(addresses the Scope Gap; DSRM: Design & Development).* Release an open, machine-readable, versioned benchmark (*NordDRG-AI-Benchmark*) that captures the *complete* NordDRG rule graph—definition tables, diagnosis/procedure property tables, age/sex splits, national activation flags, and governance manuals—with schema-stable identifiers and modular packaging for reuse and extension (see Sect. 4).
2. **O2—Benchmark Operationalisation** *(bridges both gaps; DSRM: Demonstration).* Instantiate two complementary test suites with exact-match scoring: (i) the *NordDRG Logic Benchmark*—13 rule-level questions reflecting coder, clinician, and policy-analyst workflows, with public answer keys for quantitative tasks and rubric guides for qualitative ones (Sect. 4.3); and (ii) the *NordDRG Grouper Benchmark*—13 case-level tasks that require faithful execution of the specification's control flow and return both `drg_nat` and the triggering `drg_logic.id` (Sect. 4.4).
3. **O3—Baseline Evaluation** *(targets the Benchmark Gap; DSRM: Evaluation).* Provide first-cut, fully reproducible results for representative state-of-the-art LLM endpoints across providers, including prompts, context bundles and model outputs—establishing the initial yardstick for head-to-head comparisons and longitudinal tracking in hospital-funding contexts (Sect. 5).

Alignment Back to the Research Gap (Sect. 3.1). O1 mitigates the *Scope Gap* via rule-complete, multilingual artefacts; O2 operationalises both gaps into a standardised, governance-grade benchmark; O3 directly addresses the *Benchmark Gap* with reproducible baselines and exact-match metrics.

4 Design and Development: NordDRG AI Benchmark for LLMs

NordDRG-AI-Benchmark turns the problem framing into a modular, rule-complete testbed. It bundles the NordDRG rule base—$\sim$ 20 interlinked definition tables, annual governance manuals, and curated FI/FI-EN table excerpts—into versioned, machine-readable artefacts (A1-A6) that can be loaded singly or in combination. Coupled with task prompts and gold answer keys for the *Logic* and *Grouper* suites, the framework yields a reproducible, governance-grade yardstick: models are evaluated not only on the final DRG but, where applicable, also on the triggering `drg_logic.id`. Schema-stable identifiers support drop-in annual updates and head-to-head comparisons of prompting, retrieval augmentation, and fine-tuning strategies.

All resources are released at GitHub[2] and are grouped into four artefact classes:

– **A1—NordDRG Full Specification:** Sect. 4.1 (`.xlsx/.csv`): roughly twenty relational sheets that encode the complete NordDRG grouper logic—diagnoses, NCSP procedures, age/sex splits, country-activation flags, surgery properties and grouping features.

[2] https://github.com/longshoreforrest/norddrg-ai-benchmark.

- **A2—Expert Documentation** Sect. 4.2 (`.pdf`): governance manuals, change-log templates, and column-by-column guides that explain how technical updates are proposed, audited, and rolled into the annual NordDRG release.
- **A3—NordDRG Logic Benchmark:** Sect. 4.3 (`.xlsx`): a standalone, rule-oriented suite of **13** automatically verifiable CaseMix questions (*Logic-1–Logic-13*). The prompts cover table navigation and retrieval, multilingual terminology grounding (FI-EN), cross-table joins for maternal delivery and newborn care under explicit inclusion/exclusion constraints, procedure-property tracing, diagnosis-property expansion and mapping, and CC/MCC validity checks. Each prompt is tagged *Easy*, *Medium*, or *Hard*. Gold answers are provided as order-invariant, de-duplicated code sets (or Yes/No decisions) for exact-match scoring.
- **A4—NordDRG Grouper Benchmark:** Sect. 4.4 (`.xlsx`): 13 grouping tasks (*Grouper-1-Grouper-13*) that fully emulate the NordDRG specification's grouping logic, enabling standardised, reproducible evaluation. Each task is paired with a gold-standard answer sheet.
- **A5—Curated table excerpts (FI/FI–EN)** Sect. 4.6 (`.xlsx/.csv`): a Finnish subset released in two flavours—(i) Finnish-only labels and (ii) parallel Finnish–English labels—so researchers can probe LLM performance on minority-language data and bilingual transfer without loading the entire corpus.
- **A6—Lightweight Reference Agents.** Two single-role, provider-agnostic agents for no-code use in commercial LLM platforms. Both operate strictly on the released artefacts (*no web/RAG/tools*) and emit an auditable evidence trace (sheet/row IDs, predicate outcomes). *LogicAgent* (Combined workbook + 2 PDFs) answers Logic-1–Logic-13 as comma-separated, order-invariant lists. *GrouperAgent* (FI workbook + 2 PDFs) emulates the Finnish grouper for Grouper-1–Grouper-13, returning `drg_nat=<DRG>, drg_logic.id=<ID>`. See Sect. 4.8 for setup (file paths and system prompts).

Each artefact is self-contained, enabling composition into the four canonical "dataset configurations" shown in Table 6. Researchers can start small—with the Finnish-only or FI–EN excerpts—for quick iteration, and scale up to the full definition tables together with governance PDFs when evaluating chain-of-thought reasoning and cross-document grounding.

Datasets and Evaluation Suites. Table 1 summarises the released resources—definition tables, governance manuals, curated FI/FI–EN subsets, and auxiliary prompts. These artefacts are self-contained and compose into the dataset configurations in Table 6 (e.g., **Definition Tables only** or **Definition Tables + PDF Instructions**; lightweight FI/FI–EN excerpts for rapid prototyping). The benchmark exposes two complementary suites: *NordDRG Logic* (13 rule-level tasks) and *NordDRG Grouper* (13 case-level emulation tasks). Either suite can be executed under any configuration—using tables only isolates rule-graph competence, while adding PDFs tests documentation-grounded reasoning and governance semantics. Scoring is uniform across configurations: *Logic* uses order-invariant set equality; *Grouper* requires an exact pair (`drg_nat`, `drg_logic.id`). The subsections that follow unpack each resource and illustrate typical combinations for downstream experiments.

Table 1. Publicly released NordDRG benchmark resources.

Dataset	Format	Description
Definition Tables	XLSX	NordDRG Definition Tables
PDF Instructions	PDF	NordDRG PDF instructions
Definition Tables: Part of the tables FI	XLSX	NordDRG Definition Tables—Finnish DRG list with Finnish descriptions only
Definition Tables: Part of the tables FI–EN	XLSX	NordDRG Definition Tables—Finnish DRG list with Finnish *and* English descriptions of groups
Additional Prompts	TXT	NordDRG additional prompts

4.1 Artefact: NordDRG Full Specification Definition Tables

Artefact 4.1 packages the authoritative NordDRG rule base in a machine-readable workbook of $\sim$ 20 inter-linked sheets. It constitutes the *full specification* (*rule-complete*) of the NordDRG grouping logic as executed by production groupers: all rule-bearing tables and columns needed for deterministic control flow are included (e.g., `drg_logic`, `dg_feat`, `proc_feat`, CC/MCC exclusions, age/sex bounds, MDC entry, and national activation flags). The same schema is used across *annual* releases and *national* variants; per-year and per-country workbooks are distributed in an identical format, enabling drop-in updates and cross-version comparisons via stable identifiers (e.g., `DRG`, `ICD`, `NCSP`, `ORD`).

Core tables include:

- `drg_logic`—rules that map diagnosis, procedure, age limits, CC/MCC status, and surgery flags to a unique DRG code; ordered by the `ORD` execution column.
- `dg_feat` / `proc_feat`—full ICD-10 and NCSP diagnosis and procedure code catalogues annotated with properties referenced by `drg_logic` (e.g. `DGCAT`, `COMPL`, `OR`, `PROCPRO`).
- Lookup sheets such as `compl_excl`, `procprop_name`, and `dgprop_name`, which define complication categories, OR-class mappings, and diagnosis property labels.
- `age_split` and `country_flag` tables that hold paediatric/geriatric breakpoints and activation flags for each national version.

Each worksheet exposes stable identifier columns (e.g., `DRG`, `ICD`, `NCSP`, `ORD`), allowing the tables to be joined loss-lessly and extended by appending new rows in future releases. A representative excerpt is shown in Fig. 2.

4.2 Artefact: Expert Documentation

Artefact 4.2 bundles the procedural knowledge that turns the raw tables into a living, governable system. Two official handbooks from the Nordic Casemix Centre—supplied as searchable PDFs—and the associated Excel template are included:

NordDRG specification:

- Database containing about 20 tables
- DRG groups, ICD codes, Procedure codes, Diagnose and Procedure categories, etc

More information

- https://nordcase.org/
- https://nordcaseforum.easyredmine.com

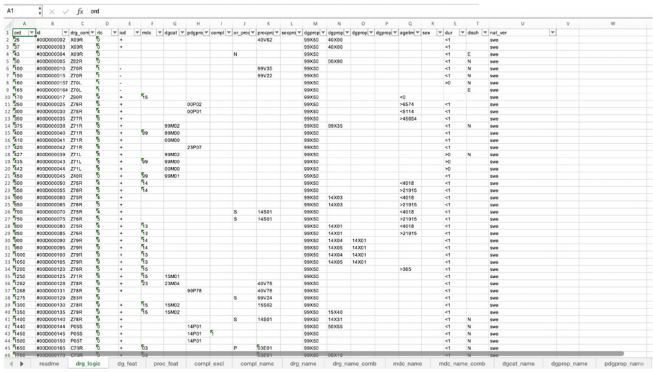

Fig. 2. Materials used: NordDRG definition tables [40].

– *How to read the NordDRG definition tables* A walkthrough of the Excel workbook that constitutes the grouper logic. The manual explains every high-impact sheet—including `drg_logic`, `dg_feat`, `proc_feat` and `compl_excl`—and shows how to (i) locate all rules that drive a given DRG, (ii) identify which diagnosis or procedure properties generate CC/MCC splits, and (iii) trace OR-flags, MDCs, age limits, and other row-level conditions.
– *How to write technical changes for NordDRG* The canonical SOP for annual updates. It defines the "IN/OUT" convention, file-naming rules for `TC_TEMPLATE_YYYY-MM-DD`, and per-table editing guidelines—for example, duplicating a `drg_logic` row when altering `ORD` or `ID`, propagating new COMPL values consistently to `dg_feat`, `compl_name`, and `compl_excl`, and adding English code texts in the comments field for new ICD/NCSP codes.

Together, these documents specify (i) how existing grouping rules must be interpreted and (ii) how proposed modifications are documented, validated and merged. Including them in the benchmark ensures that LLM evaluations test not only code look-ups but also compliance with NordDRG's governance process. A sample page is shown in Fig. 3.

4.3 Artefact: NordDRG Logic Benchmark

Artefact 4.3 provides a self-contained NordDRG Logic Benchmark focused on rule-level reasoning. It comprises **13 automatically verifiable CaseMix questions** (*Logic-1–Logic-13*) that mirror day-to-day queries posed by clinicians, health-economists, and DRG coders, while directly exercising the definition tables in Sect. 4.1 and, where needed, the expert documentation in Sect. 4.2. In DSRM terms, this artefact *operationalises* the benchmark (Objective O2 in Sect. 3.2) and supplies machine-checkable targets that support baseline evaluation (Objective O3 in Sect. 3.2).

What the Questions Test. The prompts span the main reasoning patterns required to work with NordDRG:

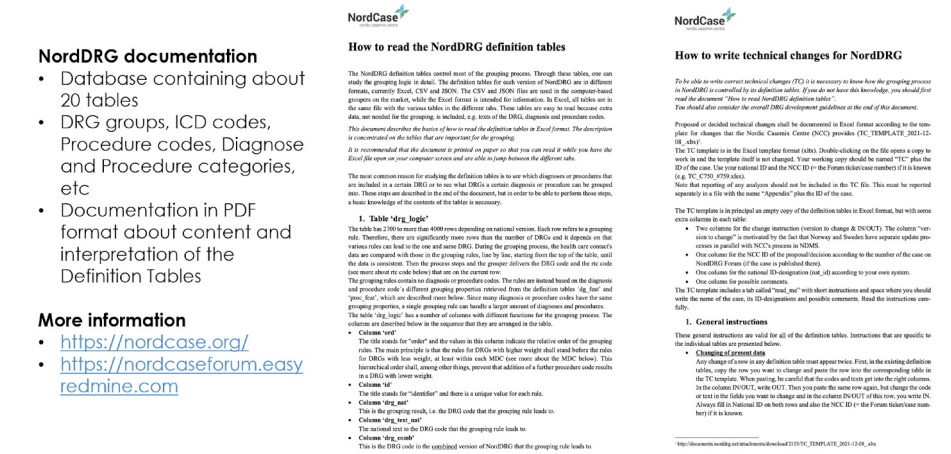

Fig. 3. Materials used: NordDRG definition table expert documentation [40].

– **Table navigation and retrieval** (*Logic-1*/*Logic-2*): enumerate DRG groups for a clinical theme such as appendectomy by reading *drg_logic* and group lists without duplication.
– **Multilingual terminology grounding** (*Logic-3*/*Logic-5*): resolve Finnish terms (e.g., *umpilisäke*) to the correct DRG sets in Finnish and pan-Nordic variants, probing minority-language handling.
– **Cross-table joins over clinical events** (*Logic-4*, *Logic-10*, *Logic-11*): list DRGs for maternal delivery and newborn care under explicit inclusion/exclusion constraints across MDCs and care settings; where definitions are ambiguous, consult the expert manuals (Sect. 4.2) for interpretation.
– **Procedure logic** (*Logic-6*): trace from a DRG to its NCSP procedure codes via procedure properties (e.g., *OR*/*PROCPRO*) to recover the inducing code set.
– **Diagnosis-property reasoning** (*Logic-7–Logic-9*): (i) link DRGs to diagnosis codes through property/category mappings, (ii) expand a property such as *14X03* to the full ICD list, and (iii) read back which diagnosis categories/properties drive a specific DRG row.
– **Complication (CC/MCC) logic** (*Logic-12*/*Logic-13*): decide whether a candidate secondary diagnosis counts as a valid complication for a given principal diagnosis, reflecting exclusion lists and category rules; the manuals in Sect. 4.2 clarify these semantics.

Format and Scoring. Table 2 lists the 13 prompts; their targets are *deterministic code sets or binary decisions*, enabling **exact-match** scoring against the gold sheet in Table 3. Answers are normalised as *order-invariant, de-duplicated* sets of DRG or ICD/NCSP codes (or Yes/No for the complication checks). Each prompt carries a difficulty tag (*Easy/Medium/Hard*) that approximates cognitive load and the number of artefacts/joins required (e.g., simple list retrieval vs. multi-table joins with property expansion and

exclusions). This scoring design directly supports Objective O3 (Sect. 3.2) by enabling reproducible, head-to-head comparisons.

Reproducibility and Extensibility. A shared identifier (*Logic-1–Logic-13*) ties each question row to its gold answer, ensuring unambiguous evaluation. Because the prompts read only stable columns (e.g., *DRG*, *ORD*, *MDC*, diagnosis/procedure properties) and rely on the production workbook schema, new releases can be dropped in without schema edits. Adding a task is a matter of appending one prompt row and its matching gold row; evaluation scripts can then compute exact-match accuracy automatically. Where interpretation hinges on governance conventions (e.g., exclusions or activation flags), the expert documentation (Sect. 4.2) provides the authoritative reference.

Together, Table 2 and Table 3 form a compact, self-contained suite that surfaces whether a model can *read, join, and reason over* the NordDRG rule base—complementing the full grouper emulation in Sect. 4.4 and aligning with Objectives O2–O3 (Sect. 3.2).

4.4 Artefact: NordDRG Grouper Benchmark

The Artefact 4.4 *NordDRG Grouper Benchmark* requires *full emulation* of the official NordDRG grouper: an LLM must take a structured test case and deterministically reproduce the same grouping decision that the specification would yield. Concretely, the model must apply the right execution order in *drg_logic*, evaluate age/sex bounds, MDC entry conditions, surgical evidence (e.g., *OR* / *PROCPRO*), complication logic (CC/MCC via *COMPL* and exclusions), and national activation flags to arrive at a single DRG for that case—and identify the exact *drg_logic.id* row responsible for the match. The task therefore goes beyond code lookup: it tests end-to-end reasoning over the rule graph defined in the NordDRG specification tables (Artefact 4.1) and their governance semantics (Artefact 4.2). Scoring is strict and exact-match: a submission is correct only if both the DRG code and the triggering rule identifier match the gold standard.

Tables and References. The benchmark instances are summarised in Table 4, which lists 13 hard, case-level emulation tasks (*Grouper-1–Grouper-13*). Each task references a concrete test case (by *ID*) supplied in the accompanying test-case sheet and instructs the evaluator to return two fields: *drg_nat* (the resulting DRG) and *drg_logic.id* (the precise rule row). Ground-truth outputs are provided in Table 5 for automatic verification. To solve these tasks, the LLM must integrate signals from the definition workbook—primarily *drg_logic* but also *dg_feat*, *proc_feat*, *compl_excl*, age-split, and country-flag sheets (Artefact 4.1)—and apply them in the correct priority order. The inclusion of both prompts (Table 4) and exact answers (Table 5) makes the benchmark fully reproducible and suitable for head-to-head evaluation of grouping fidelity.

4.5 Artefact: Definition Tables: Part of the Tables FI

Artefact 4.5 distils the full workbook down to a single, lightweight file. The sheet contains only those DRG codes that are *activated in the Finnish version* and provides their official Finnish descriptions (`DRG_TEXT_FI`). Auxiliary columns preserve the DRG

Table 2. CaseMix questions with automatically verifiable answers (Logic-1–Logic-13).

ID	Use Case Category	Difficulty	CaseMix Question
Logic-1	Understanding NordDRG Definition Tables	Easy	Could you list all DRG groups in all NordDRG national version where the main treatment during the hospital visit was related to appendectomy? Please provide a list of `drg_comb` codes without duplicates
Logic-2	Understanding NordDRG Definition Tables	Easy	Could you list all DRG groups in the Finnish DRG version where the main treatment during the hospital visit was related to appendectomy? Please provide a list of `drg_comb` codes without duplicates
Logic-3	Medical Terms in Local Minority Language	Easy	Could you list DRG groups where main treatment during the hospital visit might be related to Finnish word umpilisäke in Finnish DRG version?
Logic-4	Understanding NordDRG Definition Tables	Medium	How would you list all NordDRG groups across all national versions where a maternal delivery (vaginal or cesarean) occurs during an inpatient, outpatient, or primary-care episode of care—excluding the 'left-over' groups (468, 470, 477, Z75R, Z76R)?
Logic-5	Medical Terms in Local Minority Language	Medium	Could you list DRG groups where the main treatment during the hospital visit might be related to Finnish word umpilisäke in all DRG versions from all countries? Use the `drg_comb` column codes and combine them into one list
Logic-6	Surgical Procedure Logic	Medium	For DRG 166N which procedure codes have the connection through the procedure property? Use NCSP plus codes in your answer
Logic-7	Connection between Definition Tables	Medium	Which diagnosis codes are linked to DRG 166N through diagnosis properties in all DRG versions from all countries?
Logic-8	Diagnosis Properties	Medium	Which diagnosis (ICD) codes are connected to diagnosis property 14X03?
Logic-9	Connection between Definition Tables	Medium	Which diagnosis categories and properties are listed in `drg_logic` for DRG group 372C?
Logic-10	Understanding NordDRG Definition Tables	Medium	How would you list all NordDRG groups across every national version where a maternal delivery (vaginal or cesarean) occurs during an inpatient or outpatient hospital episode of care—excluding primary-care delivery DRGs and the 'left-over' groups (468*, 470*, 477*, Z75R, Z76R)?
Logic-11	Understanding NordDRG Definition Tables	Medium	How would you list all NordDRG groups across all national versions for newborn episodes of care (MDC 15) during an inpatient, outpatient, or primary-care setting—excluding the 'left-over' groups (468, 470, 477, Z75R, Z76R)?
Logic-12	Understanding NordDRG Complication Logic	Medium	For a case with principal diagnosis A5990 and secondary diagnosis G8000 (Spastic cerebral palsy), is the secondary counted as a valid complication?
Logic-13	Understanding NordDRG Complication Logic	Medium	For a case with principal diagnosis B3710 and secondary diagnosis J1590 (Bacterial pneumonia, unspecified), is the secondary counted as a valid complication?

Table 3. Reference answers for each CaseMix question (Logic-1–Logic-13).

ID	Right answer
Logic-1	166C, 166N, 167N, 167X, 167O
Logic-2	166N, 167X, 167O
Logic-3	166N, 167X, 167O
Logic-4	370B, 370C, 370M, 370X, 371O, 371X, 372B, 372C, 372M, 372X, 373O, 373X, 374X, 375X, P05S, P05T
Logic-5	166C, 166N, 167N, 167X, 167O
Logic-6	JESA00, JESA01, JESA10, JESW96, JESW97
Logic-7	C1810, K3520
Logic-8	O6010, O6020, O6030, O6230, O6300, O6310, O6320, O6390, O6800, O6810, O6820, O6830, O6880, O6890, O7100, O7110, O7550, O7570, O8000, O8010, O8080, O8090, O8100, O8110, O8120, O8130, O8140, O8150, O8300, O8310, O8320, O8330, O8340, O8380, O8390, O8400, O8410, O8420, O8480, O8490, Z3700, Z3710, Z3720, Z3730, Z3740, Z3750, Z3760, Z3770, Z3790
Logic-9	14X03, 14X11, 14X13, 00X10
Logic-10	370B, 370C, 370M, 370X, 371O, 371X, 372B, 372C, 372M, 372X, 373O, 373X, 374X, 375X
Logic-11	385A, 385B, 385C, 386N, 387N, 388A, 388B, 388C, 389A, 389B, 389C, 390X, 391O, 391X, 815O, 858T, 858V, 858W, 915O, 970Q, Q69L, Q93R, Q97R, Q99R, Q99U, Q99W
Logic-12	No
Logic-13	Yes

code, MDC and version stamp so that rows can still be cross-referenced against the master tables.

This compact subset supports two complementary use cases: first, it enables minority-language evaluation by allowing researchers to probe how well an LLM grasps Finnish DRG terminology— without loading the full 20-sheet corpus—and, second, it facilitates rapid prototyping, providing an ultra-light file for few-shot prompting, in-context search, or on-device experiments where the complete tables would exceed context limits.

4.6 Artefact: Definition Tables: Part of the Tables FI-EN

Artefact 4.6 mirrors the row set of the Finnish-only file but appends the official English group descriptions (`DRG_TEXT_EN`). The bilingual layout lets LLMs align Finnish terminology with its English counterpart, enabling cross-lingual retrieval, code-switching prompts, and translation-quality checks within a single, lightweight worksheet. Such functionality is essential for multilingual healthcare settings, where clinicians, coders, and decision-support tools must seamlessly toggle between local and international nomenclatures

Table 4. CaseMix questions for emulating the official NordDRG grouper (Grouper-1–Grouper-13).

ID	Use Case Category	Difficulty	CaseMix Question
Grouper-1	Emulating DRG grouper	Hard	Select the test case with ID 49483 from your tables. Which DRG group and drg_logic.id apply?
Grouper-2	Emulating DRG grouper	Hard	Select the test case with ID 50184 from your tables. Which DRG group and drg_logic.id apply?
Grouper-3	Emulating DRG grouper	Hard	Select the test case with ID 50166 from your tables. Which DRG group and drg_logic.id apply?
Grouper-4	Emulating DRG grouper	Hard	Select the test case with ID 50629 from your tables. Which DRG group and drg_logic.id apply?
Grouper-5	Emulating DRG grouper	Hard	Select the test case with ID 66704 from your tables. Which DRG group and drg_logic.id apply?
Grouper-6	Emulating DRG grouper	Hard	Select the test case with ID 53273 from your tables. Which DRG group and drg_logic.id apply?
Grouper-7	Emulating DRG grouper	Hard	Select the test case with ID 49486 from your tables. Which DRG group and drg_logic.id apply?
Grouper-8	Emulating DRG grouper	Hard	Select the test case with ID 49571 from your tables. Which DRG group and drg_logic.id apply?
Grouper-9	Emulating DRG grouper	Hard	Select the test case with ID 49597 from your tables. Which DRG group and drg_logic.id apply?
Grouper-10	Emulating DRG grouper	Hard	Select the test case with ID 49606 from your tables. Which DRG group and drg_logic.id apply?
Grouper-11	Emulating DRG grouper	Hard	Select the test case with ID 49968 from your tables. Which DRG group and drg_logic.id apply?
Grouper-12	Emulating DRG grouper	Hard	Select the test case with ID 54051 from your tables. Which DRG group and drg_logic.id apply?
Grouper-13	Emulating DRG grouper	Hard	Select the test case with ID 54096 from your tables. Which DRG group and drg_logic.id apply?

4.7 Artefact: Dataset Configurations

Artefact 4.7 defines four ready-made modular configurations of the benchmark. Each primary artefact—definition tables, governance PDFs, and curated table excerpts—can be loaded on its own or combined, depending on the experiment. Table 6 lists four ready-made "dataset configurations" that map onto typical research needs.

The first pair, *Definition Tables* and *Definition Tables + PDF Instructions*, delivers the full-fidelity reference material used by hospital coders and is best suited for testing an LLM's ability to reproduce the official NordDRG grouper logic or to justify decisions with citations from the manuals.

The second pair contains only a *slice* of the master tables: Finnish-only rows (FI) or bilingual Finnish–English rows (FI–EN). These lightweight subsets enable rapid prototyping and few-shot prompting in resource-constrained settings while still preserving clinically meaningful group descriptions.

Table 5. Reference answers for each CaseMix grouper question (Grouper-1–Grouper-13).

ID	Right answer
Grouper-1	drg_nat=470E, drg_logic_id=000D004100
Grouper-2	drg_nat=373, drg_logic_id=400D783000
Grouper-3	drg_nat=704O, drg_logic_id=106D120110
Grouper-4	drg_nat=468, drg_logic_id=499D000100
Grouper-5	drg_nat=911O, drg_logic_id=111D169960
Grouper-6	drg_nat=384B, drg_logic_id=414D3710002
Grouper-7	drg_nat=388A, drg_logic_id=015D816140
Grouper-8	drg_nat=004, drg_logic_id=401D041000
Grouper-9	drg_nat=477, drg_logic_id=414D022100
Grouper-10	drg_nat=901O, drg_logic_id=101D089890
Grouper-11	drg_nat=125O, drg_logic_id=105D230000
Grouper-12	drg_nat=560A, drg_logic_id=400D313200
Grouper-13	drg_nat=291M, drg_logic_id=410D110101

Researchers can therefore start with the compact excerpts to establish baselines, then scale up to the full tables plus manuals to stress-test chain-of-thought reasoning, multilingual transfer, and cross-document grounding. Collectively the four bundles offer a flexible design space for analysing how data volume, document modality, and language coverage affect LLMs capability to comprehend NordDRG classification.

Table 6. Configuration types for the NordDRG benchmark datasets.

Dataset Configurations	Type	Description
Definition Tables	Tables	NordDRG Definition Tables
Definition Tables + PDF Instructions	Tables+Instructions	Definition tables and PDF descriptions
Definition Tables: Part of the tables FI	PortionOfTables	NordDRG Definition Tables: Finnish DRG list with Finnish descriptions only
Definition Tables: Part of the tables FI–EN	PortionOfTables	NordDRG Definition Tables: Finnish DRG list with Finnish and English descriptions of groups

4.8 Lightweight Agents for CaseMix Tasks

We release two *single-role* agents that can be set up in minutes inside OpenAI GPTs/Assistants or Anthropic Projects/Workflows. Both operate strictly on the attached artefacts - no web browsing, RAG, function-calling, code interpreters, databases, or connectors. The only platform requirements are: (i) the ability to upload files as context and (ii) a sufficient context window to ingest one Excel workbook and two PDFs.

LogicAgent (Provider-Agnostic, No-Code). Purpose: answer the 13 *Logic* tasks deterministically from the source materials. Context: the *NordDRG Combined tables* workbook (all national versions) and the two governance PDFs. **Output:** comma-separated, order-invariant, de-duplicated code lists; for *binary* tasks (Logic-12/13) return exactly `Yes` or `No`. Evidence: on request, the agent returns a concise *evidence trace* (e.g., relevant sheet/row IDs and predicate outcomes)—not hidden chain-of-thought.

Context files

Path `NordDRG_Documentation`:

- `How_to_read_NordDRG_definition_tables_2021-12-20.pdf`
- `How_to_write_technical_changes_for_NordDRG_2021-12-21.pdf`

Path `NordDRG_Combined2024PL`:

- `2023-06-14_Combined2024PL_xls.xlsx`

System prompt (verbatim):

```
Read the attached tables and PDF documents. Answer the
following questions only based on the documents provided
to you in this context. Do not use web searches.

Return results as comma-separated, order-invariant lists.
Example: A, B, C
```

GrouperAgent (Finland; Governed Emulation). Purpose: emulate the *official* NordDRG grouper for the 13 *Grouper* benchmark cases in the Finnish national version. Context: the FI definition-tables workbook and the two governance PDFs. Execution: applies the governed control flow over `drg_logic`—`ORD` priority, age/sex bounds, MDC entry, OR/PROCPRO, CC/MCC (with exclusions), and FI national activation. Output: exactly one line `drg_nat=<DRG>, drg_logic_id=<ID>`. Limitation: because the workbook is FI-only, cross-country Logic tasks should be run with *LogicAgent* and the Combined tables.

Context files

Path `NordDRG_Documentation`:

- `How_to_read_NordDRG_definition_tables_2021-12-20.pdf`
- `How_to_write_technical_changes_for_NordDRG_2021-12-21.pdf`

Path `NordDRG_FIN2026PL0`:

- `2024-04-15_FIN2026PL0_xlsx_wo_drgs.xlsx`

System prompt (verbatim):

```
Read attached tables and PDF documents. Answer the
following questions only based on the documents provided
to you in this context. You are not allowed to use web
searches.

Return exactly one line with two fields:
drg_nat=<DRG>, drg_logic_id=<ID>
```

1. Create a tool-free agent (OpenAI GPT/Assistant or Anthropic Project/Workflow).
2. Upload the three *context files*. If the UI respects paths, keep the repo-root relative paths shown above; otherwise filenames suffice.
3. Paste the corresponding *System prompt*.
4. **Disable** web/search/RAG and any external tools.
5. Run Logic-1–13 with *LogicAgent* or Grouper-1–13 with *GrouperAgent*.

5 Results and Evaluation

This section reports empirical findings from applying the benchmark artefacts to representative LLM endpoints. Analysis begins with baseline performance on the *Logic Benchmark* (Sect. 5.1) and the more demanding *Grouper Benchmark* (Sect. 5.2), before considering operational throughput under consumer subscription constraints (Sect. 5.3). How the artefacts and tasks satisfy the three design objectives O1–O3 is then assessed (Sect. 5.4), linking descriptive coverage with analytical results. Taken together, these subsections provide both quantitative baselines and qualitative insights into the benchmark's fidelity, extensibility, and governance-grade differentiation.

5.1 Baseline LLM Performance: NordDRG Logic Benchmark

Evaluation Setup. All runs used the *Definition Tables + PDF Instructions* bundle (Artefact 4.1 and Artefact 4.2; cf. Table 6), and answers were constrained to the provided materials with the preamble: "Read attached tables and PDF documents. Answer following questions only based on the documents provided for you in this context. You are not allowed to use web searches." Logic tasks are drawn from Artefact 4.3; grouper tasks from Artefact 4.4. We evaluated commonly used commercial endpoints from three families: OpenAI (*GPT-5 Thinking*, *GPT-5 Thinking Mini*, *GPT-5 Fast*, *o3*, *o4-mini*, *4o*, *4.1*), Anthropic (*Opus 4.1*, *Sonnet 4*), and Google (*Gemini 2.5 Pro*, *Gemini 2.5 Flash*).

Interpreting Table 7 (Logic tasks; Artefact 4.3). The 13 automatically verifiable *Logic-1–Logic-13* questions separate the models into three tiers. *GPT-5 Thinking* and *Opus 4.1* achieve perfect accuracy (13/13), with *o3* close behind (12/13). A mid-tier (*GPT-5 Thinking Mini*, *o4-mini*, *GPT-5 Fast*) lands between 6–8 correct, while *4o*, *4.1*, *Sonnet 4*, and the Gemini models trail at 5/13 or below. The distribution aligns with task structure defined in Artefact 4.3: single-table lookups or straightforward filters (e.g., *Logic-1–Logic-3*, *Logic-8*, *Logic-11*) are broadly solvable by the strongest systems, and *Logic-12* (a CC/MCC validity check) is solved by all models. In contrast, prompts requiring cross-sheet chaining over Artefact 4.1—notably procedure-property tracing (*Logic-6*) and diagnosis-property linkage (*Logic-7*)—create the largest separation. Failures cluster around (i) omissions of relevant sheets (e.g., `compl_excl`, `proc_feat`, `dg_feat`), (ii) mishandling of "left-over" group exclusions, and (iii) non-deduplicated code lists

that miss *exact-match* scoring (as specified in Artefact 4.3). Overall, Table 7 shows that generic leaderboard strength does not guarantee mastery of NordDRG's property logic and governance-aware filters; however, high-end models can execute these joins reliably when given the combined artefact bundle (Artefacts 4.1, 4.2).

Table 7. Accuracy on the 13 *quantitative* NordDRG Logic tasks (Logic-1–Logic-13). All runs used the **Definition Tables + PDF Instructions** bundle. (✓ =correct, ✗ =incorrect)

Question	GPT-5 Fast	GPT-5 Thinking Mini	GPT-5 Thinking	o3	o4-mini	4o	4.1	Sonnet 4	Opus 4.1	Gemini 2.5 Pro	Gemini 2.5 Flash
Logic-1	✗	✓	✓	✓	✓	✗	✗	✗	✓	✗	✗
Logic-2	✓	✓	✓	✓	✓	✗	✓	✗	✓	✗	✗
Logic-3	✓	✓	✓	✓	✓	✗	✓	✗	✓	✗	✗
Logic-4	✗	✗	✓	✓	✗	✓	✓	✗	✓	✗	✗
Logic-5	✗	✓	✓	✓	✓	✗	✗	✗	✓	✗	✗
Logic-6	✗	✗	✓	✓	✗	✗	✗	✗	✓	✗	✗
Logic-7	✗	✓	✓	✓	✗	✗	✗	✗	✓	✗	✗
Logic-8	✓	✓	✓	✓	✗	✗	✗	✗	✓	✗	✗
Logic-9	✓	✗	✓	✓	✓	✗	✗	✗	✓	✗	✗
Logic-10	✗	✗	✓	✗	✓	✓	✓	✗	✓	✗	✗
Logic-11	✓	✗	✓	✓	✓	✓	✗	✓	✓	✗	✗
Logic-12	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Logic-13	✗	✓	✓	✓	✗	✓	✗	✓	✓	✓	✗
SUM	**6**	**8**	**13**	**12**	**8**	**5**	**5**	**3**	**13**	**2**	**1**

5.2 Baseline LLM Performance: NordDRG Grouper Benchmark

Interpreting Table 8 (Grouper emulation; Artefact 4.4). Evaluation setup. All runs used the *Definition Tables + PDF Instructions* bundle (Artefacts 4.1, 4.2), and models were explicitly instructed: "Read attached tables and PDF documents. Answer following questions only based on the documents provided for you in this context. You are not allowed to use web searches." Commonly used commercial endpoints from three families were evaluated—OpenAI (*GPT-5 Thinking*, *GPT-5 Thinking Mini*, *GPT-5 Fast*, *o3*, *o4-mini*, *4o*, *4.1*), Anthropic (*Opus 4.1*, *Sonnet 4*), and Google (*Gemini 2.5 Pro*, *Gemini 2.5 Flash*).

The thirteen *Grouper-1–Grouper-13* tasks enforce governance-grade fidelity: an answer is correct only if it reproduces *both* the final DRG and the exact triggering `drg_logic.id` defined in the specification tables (Artefact 4.1) and required by the

benchmark protocol (Artefact 4.4). Under this exact-match criterion, *GPT-5 Thinking* solves **7/13** cases, *o3* **6/13**, and *o4-mini* **3/13**; *GPT-5 Thinking Mini* solves **1/13**; all remaining endpoints (*GPT-5 Fast*, *4o*, *4.1*, *Sonnet 4*, *Opus 4.1*, *Gemini 2.5 Pro*, *Gemini 2.5 Flash*) score **0/13**. Notably, *Grouper-3–Grouper-5*, *Grouper-7*, and *Grouper-10* defeat all models, underscoring the gap between listing plausible DRGs and *faithfully executing* the grouper's priority order and guard rails.

The drop from *Logic* to *Grouper* performance pinpoints the hard parts of full specification emulation over Artefact 4.1. Beyond table reading and joins, faithful grouping requires (i) respecting the `ORD` execution priority in `drg_logic`, (ii) enforcing age/sex bounds and MDC entry criteria, (iii) integrating surgical evidence (`OR`/`PROCPRO`) and CC/MCC handling (via `COMPL` with exclusions), (iv) applying national activation flags, and (v) mapping the decision back to the precise rule row. As DSR *results*, Table 8 provides a governance-grade barometer: the best models can read and join the definition tables reliably, yet exact, traceable emulation of the governed control flow remains challenging—especially where `ORD` priority interacts with CC/MCC exclusions. These errors are diagnostic rather than cosmetic: they expose divergences from the specification's semantics (O2) and delimit the present ceiling of artefact-only emulation (O3), motivating the benchmark's insistence on reporting both `drg_nat` and `drg_logic.id`.

Table 8. DRG grouper emulation results on thirteen case-level tasks (Grouper-1–Grouper-13). All runs used the **Definition Tables + PDF Instructions** bundle and the structured test cases. (✓ =correct; exact match on *DRG and drg_logic.id*; ✗ =incorrect).

Question	GPT-5 Fast	GPT-5 Thinking Mini	GPT-5 Thinking	o3	o4-mini	4o	4.1	Sonnet 4	Opus 4.1	Gemini 2.5 Pro	Gemini 2.5 Flash
Grouper-1	✗	✗	✓	✗	✗	✗	✗	✗	✗	✗	✗
Grouper-2	✗	✓	✓	✓	✓	✗	✗	✗	✗	✗	✗
Grouper-3	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗
Grouper-4	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗
Grouper-5	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗
Grouper-6	✗	✗	✓	✓	✗	✗	✗	✗	✗	✗	✗
Grouper-7	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗
Grouper-8	✗	✗	✓	✓	✓	✗	✗	✗	✗	✗	✗
Grouper-9	✗	✗	✓	✓	✗	✗	✗	✗	✗	✗	✗
Grouper-10	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗
Grouper-11	✗	✗	✓	✗	✓	✗	✗	✗	✗	✗	✗
Grouper-12	✗	✗	✓	✓	✗	✗	✗	✗	✗	✗	✗
Grouper-13	✗	✗	✗	✓	✗	✗	✗	✗	✗	✗	✗
SUM	**0**	**1**	**7**	**6**	**3**	**0**	**0**	**0**	**0**	**0**	**0**

5.3 Operational Throughput Under Consumer Subscriptions

Setup. All evaluations were executed using standard, consumer-grade LLM subscriptions rather than enterprise contracts, to reflect the experience of typical academic and practitioner users.

Observation. Under these conditions, the *Anthropic* endpoints (e.g., *Sonnet 4*, *Opus 4.1*) could not complete the *Logic Benchmark* (Logic-1–Logic-13) in a single session: credits were exhausted mid-run, necessitating batching across multiple windows until quotas refreshed. In practice, this stretched a single end-to-end evaluation over *a couple of days* due to waiting times. By contrast, the same benchmark completed without interruption on *OpenAI* and *Google* endpoints under otherwise identical conditions (prompt packs, context bundles, and scoring).

Implications for Benchmarking. These subscription-tier effects do not change task difficulty or accuracy, but they do affect *throughput* (wall-clock time to produce a full result set), *operational latency* (need to pause and resume), and *experiment management* (checkpointing, batching, and run logs).

Operational Note (agents). A thin wrapper in the style of the *LogicAgent*/*GrouperAgent* (cf. Sect. 4.8) could batch tasks deterministically, cache intermediate joins, and resume after quota interruptions—improving throughput without affecting accuracy metrics.

Price-Value. Per-token price is only part of the calculus; time-to-result and quota predictability also determine value. In the runs, consumer-tier credit limits introduced delays and operator overhead (pauses, batching, etc.) that can outweigh lower unit prices in time-sensitive studies. When latency is non-critical, consumer tiers may remain cost-effective. Accuracy *and* operational notes are therefore reported to inform cost–latency trade-offs.[3]

5.4 Evaluation Against Objectives O1–O3

Consistent with the DSRM, the benchmark is assessed against the objectives in Sect. 3.2 via (i) a *descriptive* inspection of the artefacts and (ii) an *analytical* baseline run (Tables 7 and 8).

O1—Coverage and Fidelity. Artefact 4.1 mirrors the production NordDRG workbook (∼20 interlinked sheets; `drg_logic`, `dg_feat`, `proc_feat`, CC/MCC exclusions, age/sex splits, national activation). No lossy transformations were applied; all rule-bearing columns are preserved. The Logic tasks (Table 2) exercise table reading and joins; the Grouper tasks (Table 4) require end-to-end execution of the official control flow (including `ORD`, MDC entry, OR/PROCPRO, CC/MCC, activation flags).

[3] Subscription quotas and refill policies are product- and time-dependent; these observations reflect the August 2025 setup and may evolve.

O2—Extensibility and Sustainability. Schema-stable identifiers (`DRG`, `ICD`, `NCSP`, `ORD`, ...) are consistent across yearly and national releases, enabling drop-in updates without code changes. In practice, adding a version amounts to placing the official workbook and manuals in a dated repository folder; existing prompts and scoring continue to work. Maintenance follows an open workflow (GitHub issues/PRs): adding a task requires one prompt row (Artefact 4.3) and a matching gold-answer row—no source edits.

O3—Baseline Differentiation. On the 13 *Logic* tasks, models separate clearly: *GPT-5 Thinking* and *Opus 4.1* achieve 13/13, *o3* 12/13; mid-tier endpoints score 6–8/13; others 5/13 or below (Table 7). Largest gaps occur on cross-sheet chaining (procedure- and diagnosis-property linkage). On strict *Grouper* emulation, *GPT-5 Thinking* reaches 7/13, *o3* 6/13, *o4-mini* 3/13; all others 0–1/13 (Table 8). Many near-misses predict the correct `drg_nat` but fail the exact `drg_logic.id` requirement, underscoring the difficulty of reproducing deterministic control flow and traceability. To the best knowledge available (August 2025), this is the first public report of an LLM partially emulating the complete NordDRG grouping logic with exact matches on both fields.

Taken together—coverage and fidelity (O1), a drop-in maintenance path (O2), and differentiated, exact-match baselines (O3)—*NordDRG-AI-Benchmark* provides a robust, governance-grade yardstick for future work on trustworthy automation in hospital funding.

Answer to RQ⋆. Under an artefact-only setting, top models *fully* solve the rule-level *Logic* suite, but *partial* success on strict *Grouper* emulation indicates that end-to-end application of governed control flow remains challenging. Specifically, while strongest systems reproduce many DRGs correctly, exact matches on the triggering `drg_logic.id` are the limiting step, revealing lapses in priority order and exclusion handling. Overall, LLMs can read and join the rule base reliably and *sometimes* execute the full specification, but traceable, governance-grade parity with the reference grouper is not yet fully achieved.

6 Conclusions and Availability

This article introduces *NordDRG-AI-Benchmark*, a rule-complete, publicly released healthcare finance benchmark for LLMs comprising two parts—*Logic* and *Grouper* tests. The artefacts (Sect. 4) operationalise the full NordDRG rule base and the governance documentation into machine-readable tables and task prompts, enabling exact-match, governance-grade evaluation.

Research Questions and Objectives. The contribution directly addresses the Research Questions (Sect. 1.3) and fulfils the Design Objectives (Sect. 3.2). These findings complete the DSR cycle (Sect. 3) from problem identification to evaluated artefact and public release, providing a durable, governance-grade yardstick for future work.

– **RQ1 / O1 (Artefact Design & Construction).** Artefacts A1–A6 (Sect. 4) encode the complete NordDRG rule graph and governance semantics in an open, versioned, machine-readable form.

– **RQ2 / O2 (Benchmark Operationalisation).** Two suites—*Logic* (Sect. 4.3) and *Grouper* (Sect. 4.4)—with exact-match scoring instantiate rule-level reasoning and full specification emulation.
– **RQ3 / O3 (Baseline Capability & Failure Modes).** Baseline results (Sect. 5) and the combined chart (Fig. 4) show ceiling effects on *Logic* and partial but non-trivial success on strict *Grouper* emulation.

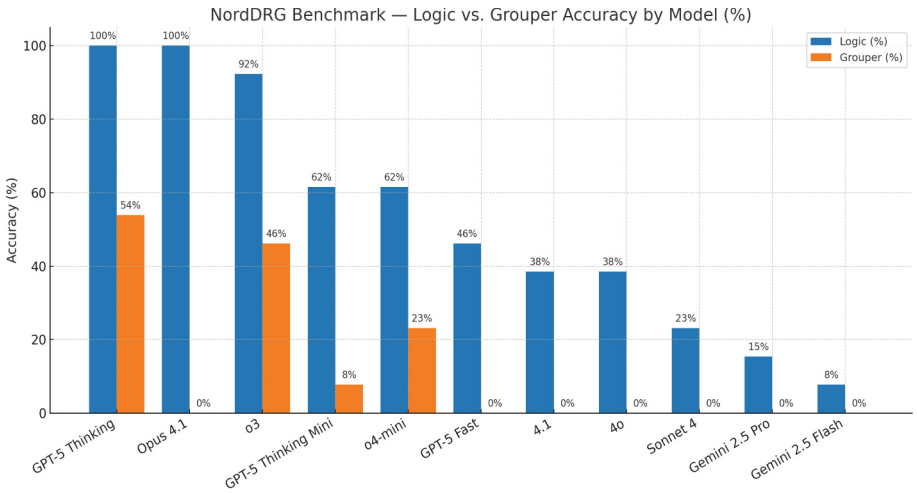

Fig. 4. Combined results. Percentage accuracy on the *NordDRG-AI-Benchmark Logic* (rule-level) and *Grouper* (full emulation) suites. Logic saturates for top models, while exact-match emulation—requiring the correct `drg_nat` and triggering `drg_logic.id`—remains challenging; only a few endpoints exceed 40% on Grouper. **Note:** *To available knowledge, this is the first public report of an LLM emulating the complete DRG grouping logic* with strict governance-grade matching.

Summary of Results (Figure 4). **Logic (13 tasks).**

– **Top tier:** *GPT-5 Thinking*, *Opus 4.1*—100% (13/13); *o3*—92% (12/13).
– **Middle tier:** *GPT-5 Thinking Mini*, *o4-mini*—62% (8/13); *GPT-5 Fast*—46% (6/13).
– **Trailing:** *4.1*, *4o*—38% (5/13); *Sonnet 4*—23% (3/13); *Gemini 2.5 Pro*—15% (2/13); *Gemini 2.5 Flash*—8% (1/13).

Grouper (13 tasks; strict exact-match on `drg_nat` & `drg_logic.id`).

– **Best:** *GPT-5 Thinking*—54% (7/13); *o3*—46% (6/13).
– **Next:** *o4-mini*—23% (3/13); *GPT-5 Thinking Mini*—8% (1/13).
– **Others:** all remaining endpoints—0% (0/13).
– **Common failure mode:** near-misses often predict the correct `drg_nat` but a non-matching `drg_logic.id`, which is scored as incorrect by design.

– **Note:** *To available knowledge, this is the first public report of an LLM emulating the complete DRG grouping logic* with strict governance-grade matching.

RQ⋆ (Primary). Top-tier LLMs reliably master rule-level DRG reasoning; the strongest models can *partially* emulate the full DRG grouping logic.

Operational Considerations (Consumer Quotas and Price–Value). In running the benchmark under standard, consumer-grade API subscriptions, a material difference in *throughput* was observed: *Anthropic* endpoints (e.g., *Sonnet 4*, *Opus 4.1*) exhausted credits mid-run and could not complete the *Logic Benchmark* (Logic-1–Logic-13) in a single session, requiring batching over multiple windows with *multi-day* waiting times. The same protocol completed without interruption on *OpenAI* and *Google* endpoints. While accuracy is unaffected, the resulting wall-clock delays and run management overhead (checkpointing, resubmission) introduce an implicit cost. Hence, the relevant objective function for practitioners is not price-per-token alone but *price & time-to-result*: for time-sensitive studies, predictable throughput may dominate nominal unit pricing.[4] (A fuller account appears in Sect. 5.3.)

Availability and Reproducibility. All artefacts, gold answers, and scoring notebooks are hosted in a public GitHub repository (https://github.com/longshoreforrest/norddrg-ai-benchmark). The repository follows a `folder-per-release` convention so that new national versions or yearly updates can be added by copying the official workbook and PDF manuals into a dated subfolder. The benchmark is designed for drop-in updates: prompts read schema-stable identifier columns (`DRG`, `ICD`, `NCSP`, `ORD`, etc.), and scoring relies on order-invariant, de-duplicated sets or strict exact-match keys. For full reproducibility, the repository includes: (i) prompt sheets (Logic and Grouper), (ii) gold-answer sheets, and (iii) evaluation scripts/notebooks mirroring the tables reported here.

Community Contributions. Contributions are welcome via pull requests. Particularly useful additions include: (i) new NordDRG releases or country configurations, (ii) additional prompts that exercise under-tested parts of the rule graph (with matching gold answers where deterministic), and (iii) corrections to table excerpts or documentation links. Issues can be used to propose new tasks, discuss schema nuances, or report discrepancies between Github artefacts and this documentation.

Limitations and Future Work. This benchmark deliberately targets *specification-faithful* reasoning over the public NordDRG tables and manuals; it does not assess extraction from noisy EHR narratives or proprietary claims data. Consequently, results isolate competence on the deterministic rule graph rather than robustness to real-world documentation artifacts.

Future work will (i) extend the Logic suite to cover remaining corners of the NordDRG specification; (ii) provide idiomatic loaders and exporters to ease integration into common toolchains; (iii) release a reproducible evaluation harness that standardises

[4] Subscription quotas and refill policies are product- and time-dependent; these observations reflect the August 2025 setup.

prompt packs, context assembly, version pinning, and exact-match scoring for head-to-head comparisons; and (iv) add dataset/model cards documenting governance, traceability, and known risks.

Closing Remark. By packaging an auditable rule base with exact-match tasks and open scoring, *NordDRG-AI-Benchmark* provides a durable yardstick for research on trustworthy automation in hospital funding—supporting head-to-head comparisons today and longitudinal tracking as models and governance requirements evolve.

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MeteoChat: A Fine-Tuned and RAG-Based LLM for Semi-Automatic Report Building in Environmental Monitoring

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Abstract. In environmental monitoring, producing high-quality reports is crucial for timely intervention in critical situations such as natural disasters and emergencies. However, generating these reports often requires a significant amount of time and resources. This article presents MeteoChat, a system that automates the creation of environmental reports by combining Large Language Models (LLMs), fine-tuning techniques, and Retrieval Augmented Generation (RAG). The goal is to maintain high report quality while reducing the time and resources needed for their production. MeteoChat operates in two phases. In the first phase, an environmental expert defines a set of generic key questions, context, and corresponding answers independent of specific data. These question-context-answer tuples are then used to fine-tune the LLM. In the second phase, the fine-tuned LLM is integrated into an RAG-based chatbot system, which combines specific environmental data. The environmental expert can interact with MeteoChat through an intuitive web chatbot and download the final report.

Keywords: Data Analysis · LLM · Report Building · Environmental Monitoring

1 Introduction

Building a high-quality report in environmental monitoring is challenging when working with large datasets that span long temporal periods, such as decades. A high-quality report facilitates rapid decision-making and promotes sustainability, as well as the adoption of advanced management tools, thereby indirectly improving environmental and economic performance [1]. However, the reliability of environmental reports can vary significantly. Companies with higher pollution risks tend to disclose more environmental information, often using verifiable metrics, but the quality and transparency of their data are not always adequate. This aspect underscores the need for stricter regulations to ensure accuracy and relevance in environmental reporting [2].

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This paper introduces an AI-driven approach for generating automated environmental reports. Our system, called MeteoChat, leverages Large Language Models (LLMs) [3] to process environmental datasets, including temperature, precipitation, and pressure, while ensuring the production of high-quality reports. The reports generated by MeteoChat can be customized for different audiences, including general audiences, decision-makers, and technical experts [4]. Identifying the target audience is crucial, as stakeholders have distinct informational needs. For instance, citizens may seek general climate trends (e.g., "How has temperature varied over the past decade?"), while decision-makers require real-time insights and long-term projections to support policies ensuring sustainable development (e.g., "Which year recorded the highest number of precipitation events?"). Technical experts, on the other hand, need detailed statistical data for model development (e.g., "What was the average temperature recorded in 2024?").

MeteoChat is based on LLMs for the automated generation of environmental reports. The objective is to generate standardized reports independent of specific measured metrics, yet maintain a generalized structure adaptable to different environmental contexts. Since it has been demonstrated that current LLM systems do not achieve high accuracy in data analysis [5], this paper proposes refining the basic LLM model by providing examples that illustrate the correct way to answer questions. MeteoChat operates in two main phases. In the first phase, an environmental expert defines a set of key questions, related context, and corresponding answers that can be applied to various environmental metrics (e.g., temperature, precipitation). These question-context-answer tuples are then used to fine-tune an LLM. In the second phase, the fine-tuned LLM is integrated with a Retrieval-Augmented Generation (RAG) system [6], which incorporates specific environmental data to generate contextually relevant responses for inclusion in the final report. A web-based chatbot interface enables the environmental expert to interact with the system, refine queries, and download the report in Microsoft Word format. From a technical perspective, this article examines the integration of LLM fine-tuning with RAG-based retrieval, enhancing the accuracy and reliability of generated reports.

MeteoChat builds upon previous work [7], enhancing it by incorporating fine-tuning techniques, integrating a web-based chatbot interface for improved interaction and accessibility, and adding the feature of report downloading.

The proposed system offers several advantages. First, it allows environmental experts to develop automated reporting workflows without requiring programming skills, making the process more accessible. Additionally, MeteoChat reduces the time needed to analyze environmental data, improving efficiency. This is particularly beneficial when rapid access to reliable environmental information is crucial for making informed decisions. In addition, the system ensures effective communication by allowing reports to be adapted to different stakeholders, including general audiences, decision-makers, and technical specialists, ensuring that each audience receives the most relevant and actionable information.

The paper also describes a preliminary evaluation of MeteoChat, providing a qualitative comparison with ChatGPT. Results show that both systems achieve approximately the same quality, but MeteoChat generates a more detailed report using the exact instructions. This evaluation is a starting point for more detailed tests and analysis.

The remainder of this paper is structured as follows. Section 2 reviews related work on LLM-based report generation. Section 3 presents the proposed system, detailing its architecture, fine-tuning, and RAG integration. Section 4 discusses preliminary experiments, and Sect. 5 discusses the results. Finally, Sect. 6 concludes the paper, summarizing the key findings and outlining future directions.

2 Related Work

The use of LLMs for automated report generation has been studied in various fields, including environmental monitoring, medicine, education, and psychometrics. While several works have explored AI applications for environmental data processing and decision-making, none have specifically addressed the automatic generation of multi-audience environmental reports using fine-tuned LLMs and RAG. This section reviews related research in medical diagnostics, earthquake monitoring, infrastructure assessment, and AI-driven automation, highlighting the differences between our approach and these areas.

In the medical field, LLMs have been successfully employed for automated report generation in mental health counseling and radiology. Srivastava et al. (2024) developed PIECE, a structured planning-based LLM framework for counseling summarization, incorporating domain-specific knowledge scaffolding to ensure alignment with therapist expectations [8]. Similarly, Liu et al. (2025) introduced HC-LLM, a model designed to integrate historical patient data into radiology reports, improving consistency and accuracy in clinical documentation [9]. Divekar et al. (2024) explored AI-based PCOS detection from ultrasound imaging, demonstrating the potential of LLMs in structuring medical diagnostic processes [10]. Ankush (2025) proposed VoxRad, an open-source LLM-powered radiology reporting system that uses speech recognition and structured templates to generate medical reports [11].

Beyond medical applications, LLMs have also been investigated in educational and psychometric contexts. Brown (2024) examined the role of LLMs in automated assessment and testing, with a focus on psychometric reliability and fairness in AI-assisted evaluations [12]. Similarly, Weissflog and Dunne (2024) examined the use of LLMs in psycho-oncology, where AI is applied to standardize study protocols and enhance personalized cancer research [13].

In environmental research, Xu et al. (2025) conducted a preliminary analysis of earthquake emergency response, outlining an automated approach to collect and analyze seismic data for rapid reporting [14]. Their system focuses on real-time earthquake data processing, whereas our system generates structured reports based on predefined question-answer pairs. Pozzer et al. (2025) integrated thermographic inspection data with Building Information Modeling (BIM) to enhance the assessment of concrete infrastructure, developing a framework that improves the visualization and analysis of structural defects [15]. Unlike our work, which focuses on textual reporting, their approach centers on data integration for visualization and predictive maintenance.

AI-driven automation has also been explored in broader process modeling and decision-making applications. Shaik and Doboli (2025) reviewed LLM-based implementation generation for open-ended problem solving, emphasizing how prompt engineering, reinforcement learning, and RAG techniques can enhance AI-driven report

structuring [16]. López et al. (2019) proposed an approach for transforming formal process descriptions into chatbots, highlighting the potential for LLMs to enhance workflow automation and accessibility [17].

While these studies demonstrate the versatility of LLMs in automating data processing and report generation, our approach differs in key ways. Most existing applications focus on real-time monitoring, predictive analytics, or structured decision-making. MeteoChat, on the other hand, is designed for generating automated environmental reports based on expert-defined question-answer pairs. Unlike real-time earthquake response or BIM-integrated infrastructure assessments, our approach does not require continuous AI adaptation or sensor-driven updates. The scientific contribution of this paper lies in bridging the gap between expert-curated report logic and automated generation pipelines, thereby overcoming the generalization issues of non-specialized LLMs in environmental domains.

3 The Proposed System

Figure 1 shows the MeteoChat architecture. The system comprises two phases: in the first phase, which is performed once, a generic LLM is fine-tuned to the environment domain using generic questions, context, and answers. In the second phase, which can be executed many times, the expert queries the fine-tuned LLM on specific environment data through a web chatbot.

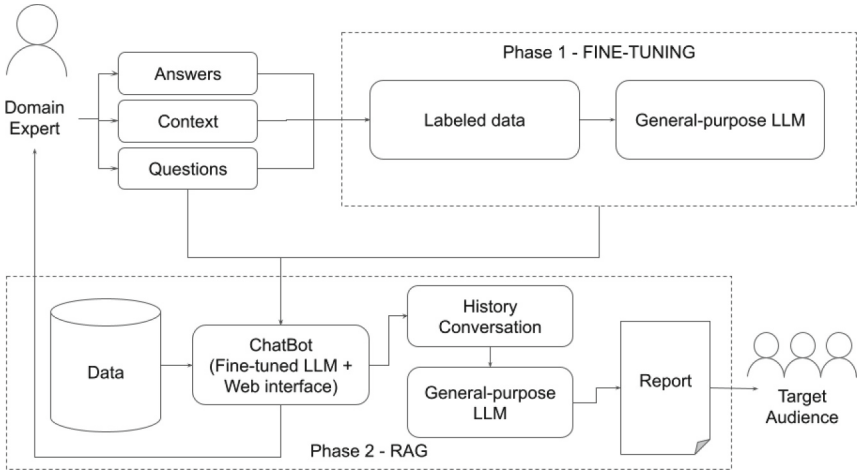


Fig. 1. MeteoChat involves two phases: the fine-tuning phase and the RAG phase. In the first phase, the LLM is refined based on the list of questions and answers defined by the domain expert. In the second step, the refined model answers the same questions related to specific data through a chatbot.

More specifically, initially, the domain expert (shown at the top left of the figure) identifies the type of audience to whom they want to address the report, defining what

we call the "target audience" (shown at the bottom right of the figure). The expert creates a list of generic questions for the target audience relating to the data in their research area and provides possible answers. Further details on the questions and answers will be explored later in the article. Using this list of questions specific to each type of audience, the expert builds labeled data, which is used to refine and adapt an LLM to the target audience. This is the fine-tuning phase. Ultimately, an optimized and targeted model is obtained for the defined target audience.

The refined model is then integrated into a chatbot system. This chatbot receives domain-specific data as input through an RAG system, along with one or more questions defined by the expert (similar, but not necessarily equal, to those used for fine-tuning), and produces an output report containing answers tailored to the provided data. To generate the report, MeteoChat exploits another general-purpose LLM, as described later in the paper.

An advantage of MeteoChat is the ability to replace specific data with others, as long as they are compatible with the same questions, without requiring retraining of the model. This makes MeteoChat highly flexible and reusable for different datasets.

3.1 First Phase: Fine-Tuning

The fine-tuning process of an LLC requires a list of (question, context, answer) tuples, which contain relevant questions, answers, and possible context for a specific audience. The context refers to all the required information needed by the LLM to generate a correct answer. In our case, one of the authors served as the domain expert in the environment and compiled the list of tuples shown in Table 1 for the general audience.

Table 1. The list of questions used to fine-tune the LLM.

Question	Context	Answer
In which month of the year Y was the highest value of the examined metric recorded for station S?	Find the maximum value in the measurement column and return the corresponding month(s)	The highest value was re-corded on month X of year Y
By how much do the maximum and minimum values of the examined metric differ in year Y for station S?	Find the maximum and minimum values and calculate the difference	The maximum and minimum values differ by X units
How many times did the metric drop below value X at station S in year Y?	Count how many measurements are below 0 °C	The temperature dropped below 0 °C X times in year Y
What is the average annual value of the examined metric in year Y for station S?	Calculate the average of all measured values	The average annual value of the parameter in year Y is X

(*continued*)

Table 1. (*continued*)

Question	Context	Answer
What was the most frequent value (mode) of the examined metric in year Y for station S?	Find the most frequently occurring value	The most frequent value of the parameter in year Y was X
In year Y, how many measurements of the examined metric were not recorded due to technical issues with station S?	Calculate missing measurements assuming 48 per day for 366 days in a leap year	The number of measurements not recorded due to technical issues was X in year Y
In which month of the year Y was the lowest value of the examined metric recorded for station S?	Find the minimum value and return the corresponding month(s) that correspond to it	The lowest value was recorded on month X of year Y
By how much has the average value of the examined metric changed over the last two years (Y1 and Y2) for station S?	Calculate the average for each year, then compute the difference	The average value changed by X units between 2023 and 2024
When ordering the dataset of the examined metric in ascending order, what is the median value for year Y for station S?	Sort the values and calculate the median	The median value of the parameter in year Y is X
In which month of year Y does the examined metric show the greatest discrepancy between its maximum and minimum values for station S?	For each month, calculate the difference between the maximum and minimum values. Return the month with the highest difference	The month with the greatest discrepancy is X

Note that in each question, a metric is specified through the generic term "metric", a specific year with variable Y, and a particular station with variable S. When the expert interacts with the chatbot, they will use a particular metric name (e.g., temperature), a year (e.g., 2023), and a station (e.g., A007)

The questions and answers are generic and do not refer to any particular environment metric. Therefore, in the second phase, MeteoChat can adapt them to specific data. Once the list of questions, context, and answers is defined, a generic LLM model is fine-tuned with these questions. The result of this phase is a fine-tuned model with generic questions, context, and answers, which can then be applied to specific data in the next phase.

3.2 Second Phase: Rag

The fine-tuned model is inserted into an RAG-based system, which accesses data relating to a specific metric (i.e., temperature, pressure, and precipitation). The environmental

expert queries the fine-tuned model via a web chatbot. The questions the chatbot answers are the generic ones on which the model has been fine-tuned. The advantage of this system is that it can utilize different datasets over time without requiring model retraining. After interrogating the system, the expert can download a report in Microsoft Word format. An evolution of the system could involve the automatic generation of reports without human intervention. This will be the subject of future studies.

The RAG-based system relies on data metrics specific to the environment domain, such as temperature, pressure, and precipitation. Before importing them into the RAG-based system, the data is cleaned and normalized. Each metric is stored in a separate store to prevent the fine-tuned model from accessing multiple metrics contemporaneously.

Data Cleaning. To inject raw data into our RAG system, we introduced an Extract, Transform, Load (ETL) process. This process involves collecting and cleaning raw data and organizing it into smaller datasets, each corresponding to a specific metric. We performed a separate data cleaning process for each metric, extracting the relevant data for every year. Subsequently, we further refined these datasets by computing statistical measures including mean, median, mode, minimum, and maximum values. Table 2 presents an extract of the temperature dataset for 2013, as detected by station AL007, following statistical calculations.

Table 2. An example of a cleaned dataset used by the RAG-based system.

Year	Month	Mean	Max	Min	Mode
2013	1	8.648	17.383	1.057	4.52
2013	2	8.239	17.291	-0.411	3.033
2013	3	12.108	24.841	1.7	11.251
2013	4	17.177	30.288	7.941	11.957
2013	5	17.906	27.452	9.371	16.936
2013	6	22.177	34.246	12.839	14.35

The Chatbot. The chatbot implementation utilizes an intuitive web interface that enables users to interact easily with the system (Fig. 2). The home page features a simple and functional layout, with the conversation area at its center, displaying the messages exchanged between the user and the chatbot (A). At the bottom, a text input field (B) allows users to type their messages, accompanied by a send button (C) and a second button (D) that enables the expert to download the report.

When the user writes and sends a message, MeteoChat processes it and generates a real-time response using the fine-tuned LLM model. The chatbot understands the content of the message and provides dynamic answers based on the user's input. MeteoChat saves each message in the conversation history, enabling users to track their interactions. It is important to note that the chatbot can respond to questions related to the environmental data it was trained on. If asked about unrelated topics, it may not be able to provide an appropriate response.

Fig. 2. The chatbot interface.

Report Building. The expert can click the "Download Report" button in the web chatbot, which triggers the generation of the report. This process involves the use of an additional general-purpose LLM, which is employed to generate the textual components of the report. Figure 3 illustrates the workflow used to generate the report.

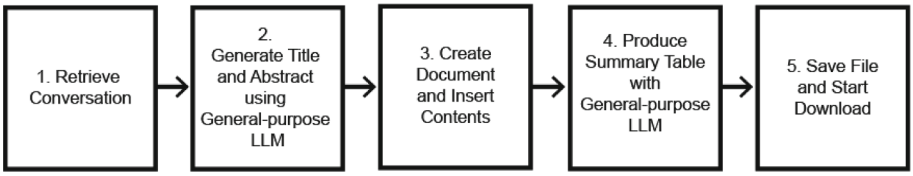


Fig. 3. The flow used to generate the report.

Initially, the list of messages exchanged between the expert and the chatbot is retrieved. Next, the general-purpose LLM is used to automatically generate a title and abstract coherent with the content of the conversation. Next, each question is entered as a header, and its related answer is entered as a paragraph. Finally, the LLM produces a summary table with the keyword representing the question in the first column and the numerical value contained in the answer in the second column. This table allows the relevant data from the conversation to be quickly highlighted.

At the end of the process, MeteoChat saves the document in the Microsoft Word format and makes it available for download. Generating the report may take several minutes, depending on the length and complexity of the conversation.

4 Preliminary Experiments

The system prototype is available as open-source software implemented in Python[1]. As an LLM, the system uses OpenAI GPT-4. The RAG utilizes data from the ARPA Lazio database, which enables the assessment and forecasting of air quality in the Lazio region through a micro-meteorological network (RMM). This network comprises nine monitoring stations (AL001–AL009) with advanced instrumentation. Station AL009 has been active since April 18, 2023, meaning that data from this station is available starting from 2024. The stations are equipped with standard meteorological sensors (temperature, humidity, pressure, and precipitation) and specialized instruments for analyzing pollutant dispersion, including sonic anemometers, pyranometers, and pyrgeometers. Meteorological data are collected every 30 min, and all records are stored in a database containing measurements dating back to 2013, the year the RMM was established. As previously said, we fine-tuned the model with the list of questions shown in Table 1.

4.1 RAG Calibration

During the RAG phase, each text is divided into chunks, where a chunk is a predefined portion of text, typically chosen based on parameters such as the number of words, characters, or semantic units. The chunking process is crucial for effectively building the knowledge base of RAG systems and is expected to have a significant impact on their quality. A large chunk provides better context but increases the amount of noise, while a small chunk offers more precise information but often overlooks essential details. For this reason, we conducted a preliminary study to verify whether the model's response changes as the chunk size varies. To determine the optimal chunk size for our system, we varied it from 500 to 10,000 (with increments of 100 up to 3000 and 500 beyond 3000). We then evaluated the performance of the considered models. We posed five questions to the GPT-4 (not fine-tuned), which were very similar to those used during the fine-tuning phase. We manually reviewed each answer provided and ranked it based on a 5-point Likert scale (5 = correct answer, 1 = totally wrong answer). We calculated the average value across all the questions (Fig. 4).

We noted that the quality of answers was around three, independently of the chunk size. Thus, we decided to keep the chunk size small (1000) to make the model faster.

4.2 Scenario Simulation

To establish the quality of the generated reports, we simulated a scenario with the following three questions:

- Q1: Which month of the year 2020 had the highest temperature value recorded for station AL007?
- Q2: What is the average annual value of precipitation in 2014 for station AL007?
- Q3: In which month of the year 2023 does the pressure show the greatest discrepancy between its maximum and minimum values for station AL007?

[1] https://github.com/djinfouma/meteochat

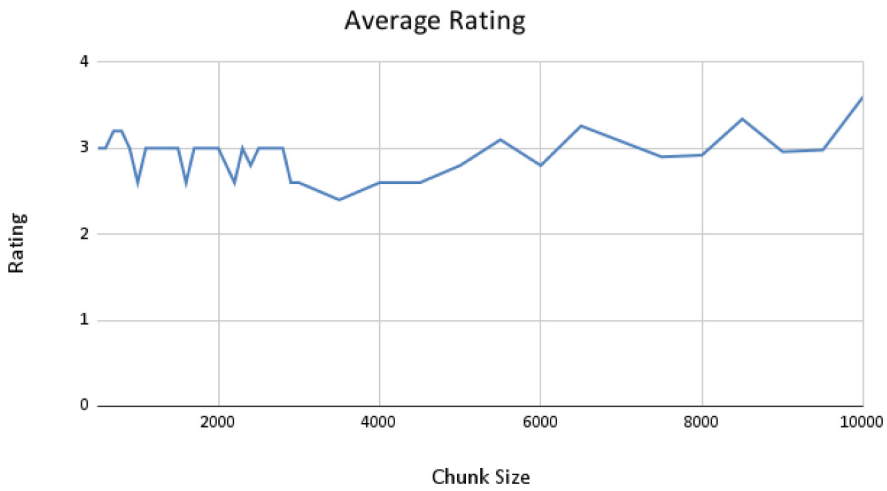


Fig. 4. The average rating across all questions while varying the chunk size. The quality of the answers slightly depends on the chunk size.

Next, we downloaded the report as a Microsoft Word document. We ran the experiment 5 times, using five separate conversations in MeteoChat. This approach was necessary to ensure that the generated content was consistent in multiple iterations. The results of the experiments are available in the article's supplementary material[2].

To run the experiments, we used a computer equipped with an Intel Core i5–6400 processor running at 2.70 GHz and based on x64 architecture. The device includes 16 GB of RAM and three storage units: a 224 GB SSD, supported by two HDDs, each with a capacity of 932 GB and 466 GB, respectively. The graphics card installed is the NVIDIA GeForce GTX 1660 SUPER with 6 GB of dedicated video memory. The operating system is Windows 10 Pro, specifically the 2009 version, 64-bit.

4.2.1 Qualitative Evaluation

One of the authors, acting as an environmental expert, evaluated the quality of the generated answer for each question and run, using a 5-point Likert scale (1 = wrong, 5 = correct). In addition to this evaluation, the expert provided an overall review of the generated ratings, the title, and the abstract for each report. For each answer, we calculated the average value and the variance. We are aware that a pool of experts would be necessary to obtain a complete validation. We plan to perform more tests in the future. For now, we provide a preliminary evaluation based on a single expert evaluation. Table 3 shows the results.

Report 1 presents an error in returning the answer to Q3. Although the calculations performed are correct for each month, the selection of the highest value available in the data set results in a value lower than the maximum (40.68 mbar instead of 43.04 mbar). Continuing the critical analysis of the reports, a careful eye cannot fail to notice that

[2] https://anonymous.4open.science/r/CHIRA2025-meteochat-4FF1/README.md

Table 3. The results of the qualitative evaluation for MeteoChat.

Question	Report 1	Report 2	Report 3	Report 4	Report 5	Average	StdDev
Q1	5	5	3	5	5	4.6	0.89442719
Q2	5	2	2	5	2	3.2	1.64316767
Q3	1	5	5	5	5	4.2	1.78885438
Title	3	3	3	4	4	3.4	0.54772255
Abstract	3	3	5	4	3	3.6	0.89442719
Average	3.4	3.6	3.6	4.6	3.8	3.8	0.46904157

for Q2, Reports number 2, 3, and 5 give the following answer: "Therefore, the average annual precipitation for station AL007 in the year 2014 is approximately 0.060 mm". The answer, in terms of mathematical approximation, is not satisfactory. Report number 4 has no errors in the calculations, is exhaustive and precise in the answers, and shows the results in an orderly manner.

4.2.2 Comparison with a Baseline

We compared the reports generated by MeteoChat with a general-purpose LLM, specifically ChatGPT 4o, to evaluate whether there is an improvement using a fine-tuned LLM. More specifically, we used ChatGPT 4o as a baseline model in the following way. We used the ChatGPT web interface to upload all the available data used by MeteoChat as a zipped file. Next, we asked ChatGPT the same three questions described at the beginning of Sect. 4.2. Finally, we used the following prompt to ask ChatGPT to generate the report as a Microsoft Word document:

```
Generate a report with all the questions/answer pairs, and make it
downloadable in the .docx format.
```

We ran this experiment five times using five separate chats. The expert who evaluated MeteoChat also evaluated the quality of the reports generated by ChatGPT using the same 5-point Likert scale. Table 4 shows the results of the qualitative evaluation for ChatGPT.

Table 4. The results of the qualitative evaluation for ChatGPT.

Question	Report 1	Report 2	Report 3	Report 4	Report 5	Average	StdDev
Q1	5	5	5	5	5	5	0
Q2	2	2	2	2	1	1.8	0.4472
Q3	5	5	5	5	5	5	0
Title	4	3	4	4	5	4	0.7071
Average	4	3.75	4	4	4	3.95	0.1118

The five reports generated by ChatGPT share very similar characteristics in terms of structure and the organization of answers to the questions. The only notable difference lies in the title, as each report presents a unique title. Report 5 is the only one that contains an error in response to Q2. Since the method used to calculate the average annual precipitation is not provided, it is unclear whether the error is due to a mistake in the mathematical calculation or a misselection among the possible response options available to the model. For the same question, all the other reports return a value of zero, indicating that the calculation method needs improvement. Reports 1, 3, and 4 are fairly consistent with each other and could be used in future work, with improvements to the number rounding process.

5 Discussion

Figure 6 shows a comparison between the average values obtained for MeteoChat and ChatGPT.

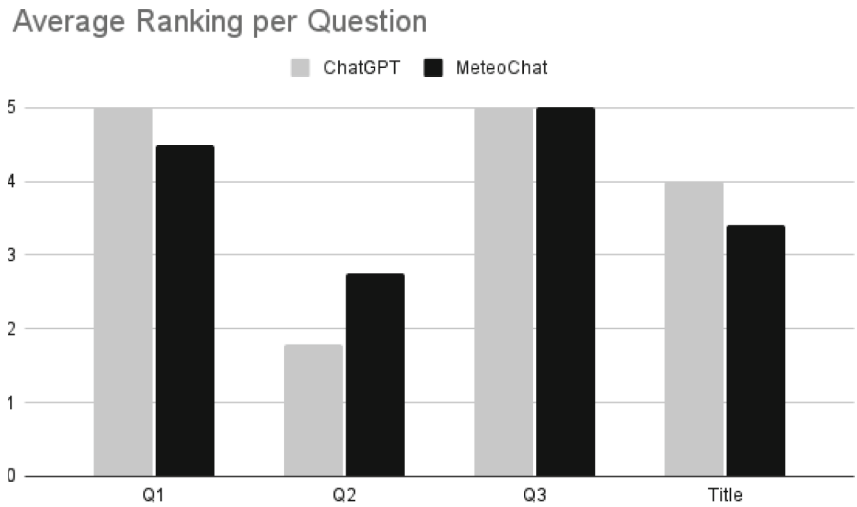
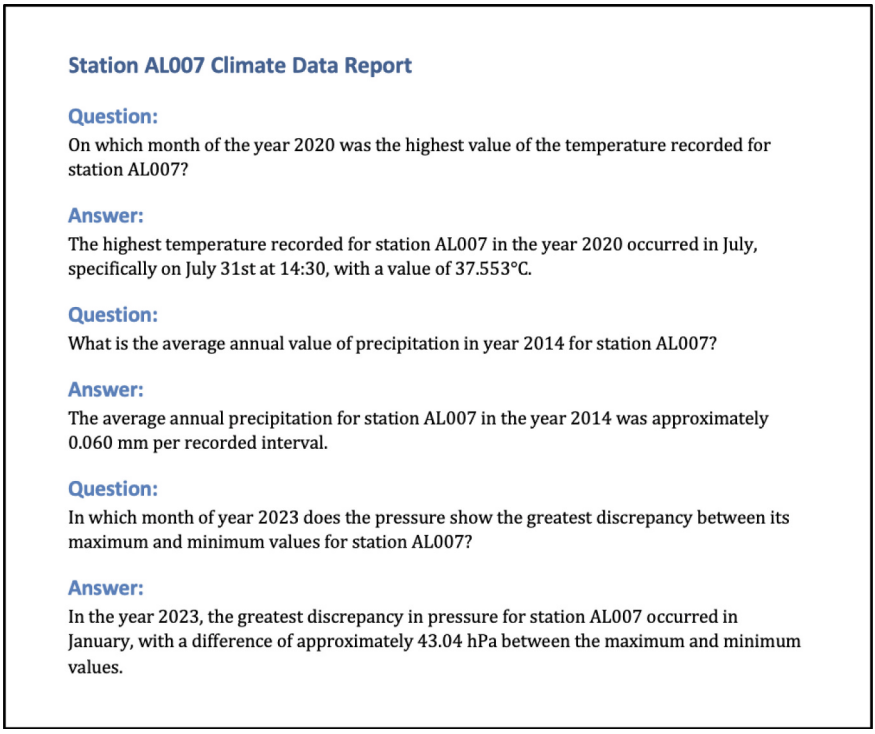


Fig. 6. The average rating across all questions for ChatGPT and MeteoChat.

Question Q2 received the lowest rating for both ChatGPT and MeteoChat, although MeteoChat was rated slightly higher than ChatGPT. For Q3, both systems received the same rating, while for Q1 and the Title, ChatGPT was rated better. However, the report generated by ChatGPT is simpler than the one produced by MeteoChat: ChatGPT provides a single sentence in response to each question (Fig. 7), whereas MeteoChat also includes an explanation of the reasoning behind each answer (Fig. 8). This enhancement, made possible by the context provided during the fine-tuning phase, makes MeteoChat's responses more realistic and closer to an actual report.

Unlike ChatGPT 4o, which produces brief and unelaborated answers, MeteoChat provides domain-specific reasoning by explicitly describing how answers are derived



Station AL007 Climate Data Report

Question:

On which month of the year 2020 was the highest value of the temperature recorded for station AL007?

Answer:

The highest temperature recorded for station AL007 in the year 2020 occurred in July, specifically on July 31st at 14:30, with a value of 37.553°C.

Question:

What is the average annual value of precipitation in year 2014 for station AL007?

Answer:

The average annual precipitation for station AL007 in the year 2014 was approximately 0.060 mm per recorded interval.

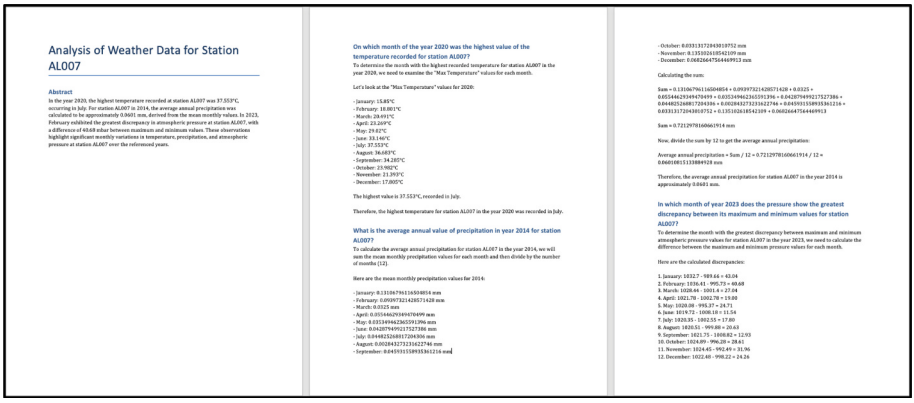
Question:

In which month of year 2023 does the pressure show the greatest discrepancy between its maximum and minimum values for station AL007?

Answer:

In the year 2023, the greatest discrepancy in pressure for station AL007 occurred in January, with a difference of approximately 43.04 hPa between the maximum and minimum values.

Fig. 7. An example of a report generated by ChatGPT (Report 1).



Analysis of Weather Data for Station AL007

Abstract

In the year 2020, the highest temperature recorded at station AL007 was 37.553°C, occurring in July. For station AL007 in 2014, the average annual precipitation was calculated to be approximately 0.060 mm, derived from the mean monthly values. In 2023, February exhibited the greatest discrepancy in atmospheric pressure at station AL007, with a difference of 40.68 mbar between maximum and minimum values. These observations highlight significant monthly variations in temperature, precipitation, and atmospheric pressure at station AL007 over the referenced years.

On which month of the year 2020 was the highest value of the temperature recorded for station AL007?

To determine the month with the highest recorded temperature for station AL007 in the year 2020, we need to examine the "Max Temperature" values for each month.

Let's look at the "Max Temperature" values for 2020:

- January: 15.85°C
- February: 18.081°C
- March: 28.491°C
- April: 23.269°C
- May: 29.82°C
- June: 33.146°C
- July: 37.553°C
- August: 36.683°C
- September: 34.285°C
- October: 23.902°C
- November: 21.393°C
- December: 17.005°C

The highest value is 37.553°C, recorded in July.

Therefore, the highest temperature for station AL007 in the year 2020 was recorded in July.

What is the average annual value of precipitation in year 2014 for station AL007?

To calculate the average annual precipitation for station AL007 in the year 2014, we will sum the mean monthly precipitation values for each month and then divide by the number of months (12).

Here are the mean monthly precipitation values for 2014:

- January: 0.13106796116504854 mm
- February: 0.09397321428571428 mm
- March: 0.0325 mm
- April: 0.035446293469704099 mm
- May: 0.035349462365591396 mm
- June: 0.042079495217527386 mm
- July: 0.044825268817204306 mm
- August: 0.032843273134622746 mm
- September: 0.045935158935361216 mm
- October: 0.033131720430107526 mm
- November: 0.13510261054211091 mm
- December: 0.068266475844469913 mm

Calculating the sum:

Sum = 0.13106796116504854 + 0.09397321428571428 + 0.0325 + 0.035446293469704099 + 0.035349462365591396 + 0.042079495217527386 + 0.044825268817204306 + 0.032843273134622746 + 0.045935158935361216 + 0.033131720430107526 + 0.13510261054211091 + 0.068266475844469913

Sum = 0.7212978160661914 mm

Now, divide the sum by 12 to get the average annual precipitation:

Average annual precipitation = Sum / 12 = 0.7212978160661914 / 12 = 0.060108151338884928 mm

Therefore, the average annual precipitation for station AL007 in the year 2014 is approximately 0.0601 mm.

In which month of year 2023 does the pressure show the greatest discrepancy between its maximum and minimum values for station AL007?

To determine the month with the greatest discrepancy between maximum and minimum atmospheric pressure values for station AL007 in the year 2023, we need to calculate the difference between the maximum and minimum pressure values for each month.

Here are the calculated discrepancies:

1. January: 1032.7 - 989.66 = 43.04
2. February: 1036.41 - 995.73 = 40.68
3. March: 1028.44 - 1001.4 = 27.04
4. April: 1021.78 - 1002.78 = 19.00
5. May: 1020.08 - 995.37 = 24.71
6. June: 1019.72 - 1008.18 = 11.54
7. July: 1020.35 - 1002.55 = 17.80
8. August: 1020.55 - 999.88 = 20.63
9. September: 1021.75 - 1008.82 = 12.93
10. October: 1024.89 - 996.28 = 28.61
11. November: 1024.45 - 992.49 = 31.96
12. December: 1022.46 - 998.22 = 24.24

Fig. 8. An example of a report generated by MeteoChat (Report 1).

from the data. However, general-purpose models exhibit higher flexibility in handling off-topic or novel questions. MeteoChat is more robust within its narrow domain, but less generalizable outside it. This trade-off underscores the importance of aligning model

choice with the intended use: MeteoChat excels in structured, metric-based analysis, whereas ChatGPT may perform better in open-domain exploratory dialogue.

Despite its benefits, MeteoChat still relies on human experts for the construction of the (question, context, answer) triples used in fine-tuning. This introduces a degree of manual overhead, limiting full automation. Additionally, the system has been tested on a limited set of simulated queries under controlled conditions, without undergoing large-scale or multi-user evaluations. The evaluation, although improved, remains limited in scale and scope. In addition, the time required to generate reports, while not excessive, may become a bottleneck if scaled to larger datasets or real-time applications, as shown in Fig. 9. The average time for downloading a report was 158.5 s, with a standard deviation of 3.7. Future work should address these points through more extensive benchmarking and stress testing.

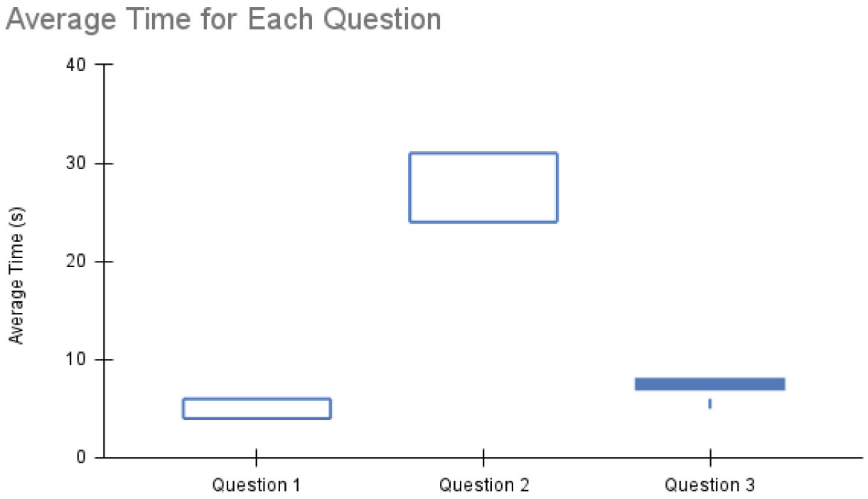


Fig. 9. The average time to generate the answer for each question. Question 2 required the longest time with the highest standard deviation across the different runs.

5.1 Ethical Implications

One of the most pressing concerns in adopting AI-driven reporting systems is the risk of over-delegating human decision-making power to artificial agents. As Luciano Floridi emphasizes in his work, the core ethical dilemma lies in autonomy: what we choose to decide ourselves versus what we allow AI to decide for us [18]. Floridi warns that ceding part of our decision-making power to technological artifacts may threaten human autonomy if not carefully controlled. MeteoChat is designed to support, not replace, the work of environmental experts. However, the line between assistance and delegation can blur, especially if its output is perceived as authoritative or objective.

Floridi also distinguishes between human autonomy and the growing "artificial autonomy" of AI systems, cautioning that the latter must not undermine the former.

The Declaration of Montréal[3] states that the development of AI should promote "the autonomy of all human beings" and not merely automate decisions. In this context, MeteoChat should be understood as a tool that augments expert judgment, rather than a substitute for it. The risk of over-reliance, "automation bias," or unjustified trust in AI-generated conclusions must be explicitly addressed in future deployments.

In addition, the risk of bias, hallucinations [19] or data misinterpretation, especially when data is missing, corrupted, or out of domain, must be taken seriously. Thus, robust data validation, transparency of logic, and human oversight are essential design principles for systems like MeteoChat.

5.2 Limitations

Despite the promising results and practical applicability of MeteoChat, several limitations must be acknowledged to frame the current work's scope and identify directions for future improvement.

Single-Expert Evaluation Bias. The qualitative evaluation was conducted solely by one of the system's authors, who also acted as the domain expert. This introduces a potential bias in the assessment, as the evaluator is not an independent third party. Broader validation involving multiple environmental experts with diverse backgrounds is necessary to assess the system's robustness and generalizability across users.

Limited Fine-Tuning Scope. The fine-tuning phase was based on a small set of ten predefined questions. While these were carefully selected to represent common queries in environmental monitoring, they do not cover the full spectrum of possible data analysis needs. Expanding the question set would enhance the model's adaptability and reduce the risk of overfitting to specific patterns.

Scalability and Cost Concerns. The system leverages commercially available LLMs (e.g., GPT-4), which incur usage costs. While acceptable for small-scale usage or research purposes, the operational cost of deploying MeteoChat at scale (e.g., across regions or in real-time systems) could become significant. Future versions should explore cost-optimization strategies, such as model distillation or local deployment of smaller LLMs.

Lack of Selective Report Exporting. Currently, all chatbot interactions are included in the generated report by default, with no option to export only a subset of questions or responses selectively. This limits user control over the report structure and may result in unnecessarily verbose documents. Adding functionality to curate report content before export would enhance usability.

Manual Construction of Fine-Tuning Triplets. The generation of (question, context, answer) tuples for fine-tuning is a manual process, requiring expert knowledge and significant time investment. This limits scalability and hinders adoption by non-expert users. Automating part of this process or providing templates for common domains could reduce the manual burden.

[3] https://montrealdeclaration-responsibleai.com/ (Last Access 2025-06-20)

Limited Evaluation Scope. The system has been tested under controlled, simulated conditions using predefined questions and a small set of environmental datasets. Real-world deployment would require rigorous benchmarking with noisy, incomplete, or heterogeneous data sources, as well as testing across different regions and climates.

Performance Bottlenecks. Report generation time, while acceptable for the current dataset size, may become a bottleneck when handling larger or multi-year datasets. As shown in Sect. 4, the average time for report generation exceeds two minutes. Optimizing system latency and computational efficiency is crucial for real-time applications.

Generalization vs. Specialization Trade-off. MeteoChat performs well in answering domain-specific questions due to fine-tuning but lacks flexibility when asked off-topic or novel questions, unlike general-purpose models. This trade-off between accuracy and generalizability must be carefully managed depending on the deployment context.

These limitations underscore that MeteoChat is currently best suited for semi-automatic report generation in well-defined scenarios, under expert supervision. Addressing these issues will be essential for scaling its adoption and extending its application to more dynamic, complex, or real-time environments.

6 Conclusion and Future Work

In this article, we have described MeteoChat, a system based on LLMs that generates environmental reports semi-automatically. The goal was to define a standardized reporting mechanism that is independent of specific measured metrics, while maintaining a generalized structure. The proposed system enabled domain experts to develop an automated report-generation workflow without requiring programming skills. Additionally, it reduced the time needed to collect and synthesize environmental data, making it particularly valuable in emergencies where rapid response is crucial. Additionally, the system enabled content customization tailored to the needs of various target audiences. Given the pressing relevance of both fields, the intersection of AI and environmental monitoring is undoubtedly of great interest in academic, scientific, and technical communities.

As future developments, we plan to extend the implementation with an enhanced user interface and the possibility to download data-driven stories. We also plan to extend testing phases by involving more domain experts during the evaluation phase and enhance the automated report generation by incorporating graphical visualizations and more extensive textual content. We also plan to perform more comparisons with baseline models (with and without fine-tuning and RAG enabled).

Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

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From Asking to Understanding: A Human-Centered Approach for Human-AI Interaction

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Abstract. In human-AI interaction, users often begin with vague, incomplete, or imprecise queries that limit the system's ability to provide meaningful support. While much attention has been given to generating accurate answers, the capacity of intelligent systems to guide users in formulating better questions remains underexplored. This paper proposes a methodology that supports users in the co-construction of knowledge through context-aware clarification questions. By combining semantic representations from knowledge graphs with the generative capabilities of large language models, the approach dynamically detects informational gaps and offers targeted prompts that refine user intent. This method fosters more natural, adaptive, and cognitively aligned interactions, improving user engagement, decision-making, and exploratory search. We discuss implications for designing intelligent systems that not only respond, but also proactively assist users in asking better questions.

Keywords: Human-AI Interaction · Context-Aware Systems · Knowledge Graphs · Large Language Models · Clarification Question · User-Centered Design

1 Introduction

In most current human-AI interactions, the flow of information is predominantly one-directional: users ask questions, and systems respond. This paradigm, rooted in both educational practices and machine learning traditions, positions AI as a passive provider of answers rather than an active collaborator in inquiry. Large Language Models (LLMs), such as ChatGPT, exemplify this dynamic by delivering fluent and often accurate responses to human prompts. While this approach has advanced the accessibility of information, it exposes a critical limitation—the lack of support for formulating better, more meaningful questions.

A common source of communication breakdown in these systems arises from *underspecified questions*—queries that are syntactically valid but semantically incomplete. In such cases, key contextual parameters (such as timeframes, scopes, or definitions) are missing, making it difficult or even impossible for the

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system to generate an accurate response. These situations are particularly prevalent in exploratory or decision-making scenarios, where users may not yet know what information is relevant or how to precisely articulate their needs. Addressing this issue requires a system that can detect missing context and respond by engaging users in clarification, guiding them toward more complete and effective queries.

The ability to ask insightful questions is a hallmark of expert human behavior, commonly exercised by educators, researchers, and decision-makers to guide learning, resolve ambiguity, and deepen understanding. Transferring this capability into AI-driven systems opens new pathways for collaboration: what if machines could assist users not only by answering but by helping them clarify and refine their own questions? Such a shift would reframe intelligent systems as interactive partners capable of engaging in a co-constructive exploration of knowledge.

This paper proposes a human-centered methodology to support question refinement and clarification during human-AI interaction. Our approach focuses on the generation of clarification questions—targeted prompts that help users disambiguate intent, complete partial queries, and surface relevant contextual details. Rather than replacing the user's input, the system collaborates in shaping it, enabling more accurate, adaptive, and effective exchanges.

To support this interaction model, we leverage Context-Augmented Knowledge Graphs (CoaKGs), a semantic representation that integrates rich contextual information into both entities and relations. When combined with the generative capacity of LLMs, CoaKGs enable the system to detect informational gaps and dynamically generate clarifying questions tailored to the user's needs and domain context.

The remainder of this paper is organized as follows. Section 2 provides background on LLMs and Context-Augmented Knowledge Graphs. Section 3 reviews related work in question generation and human-AI interaction. Section 4 details our proposed methodology for clarification question generation. Section 5 presents our evaluation strategy and results. Finally, Sect. 6 discusses conclusions and directions for future work.

2 Background

This section introduces the foundational concepts underpinning recent advances in question generation and context-aware human-AI interaction. We begin by outlining how Question-Answering (QA) systems operate based on Large Language Models (LLMs), then describe the principles of Context-Augmented Knowledge Graphs (CoaKGs), review techniques for retrieving and integrating knowledge graph content into LLM-based systems, and finally discuss the role of clarification in addressing underspecified user queries.

2.1 LLM-Based Question-Answer Systems

Large Language Models (LLMs), such as GPT-4 [13], PaLM [5], and LLaMA [20], have revolutionized natural language processing by enabling machines to generate fluent, contextually coherent responses across a wide range of topics. These models are typically trained on extensive corpora of text and leverage transformer-based architectures [22] to capture syntactic and semantic dependencies across long sequences.

In the domain of Question-Answering (QA), LLMs are used to respond to user queries in a conversational or document-grounded format, often without reliance on structured databases. This contrasts with traditional QA systems, which depended heavily on rule-based processing or information retrieval pipelines. Recent research has shown that LLMs can outperform conventional systems in open-domain QA tasks [11], although challenges remain regarding factual consistency, hallucinations, and user intent interpretation.

The majority of LLM-powered QA systems focus on generating answers to user questions. However, they typically lack mechanisms for proactively engaging in question refinement or clarification, which are critical for complex information-seeking scenarios.

2.2 Underspecified Questions and Clarification Questions

In natural language interactions, users frequently issue *underspecified questions* - queries that are structurally well-formed but semantically incomplete or ambiguous. These questions often omit key contextual parameters, such as temporal references, scopes, or definitions, which are essential for interpreting intent and retrieving accurate answers [1]. For instance, asking "What was the capital of Brazil before Brasília?" presumes an implicit timeframe that is not explicitly stated. Without further clarification, an AI system may struggle to provide a meaningful response.

To address such ambiguity, systems must be equipped with the ability to generate *clarification questions* - targeted follow-up prompts that elicit missing information or disambiguate intent. Clarification questions can request temporal, geographic, definitional, or preference-based context, depending on the domain and knowledge representation used. Prior studies have demonstrated that clarification not only improves the precision of answers but also enhances user engagement and learning by prompting reflection [15].

In the context of this work, clarification questions serve as a core mechanism for aligning user intent with knowledge encoded in the CoaKG. By identifying missing contextual dimensions and formulating relevant follow-up questions, the system fosters a more natural and cognitively supportive interaction model—transforming question-answering from a passive retrieval task into an interactive, co-constructed dialogue.

2.3 Context-Augmented Knowledge Graphs (CoaKGs)

Knowledge Graphs (KGs) are graph-based structures that represent entities (nodes) and their relationships (edges), often used to formalize domain knowledge and enable semantic reasoning [9]. Traditional KGs support a wide range of applications including question-answering, entity linking, and recommendation. They are typically constructed using ontologies, curated datasets, or automatic extraction from text.

Despite their utility, conventional KGs often encode relationships in a static and context-agnostic manner. To address this limitation, recent work has proposed Context-Augmented Knowledge Graphs (CoaKGs) [8], which extend standard KGs by incorporating contextual metadata such as time, location, provenance, or conditions into both entities and relations.

This enriched representation supports more nuanced and situationally aware reasoning. For example, CoaKGs allow the same entity to participate in different relationships under varying contexts, making them suitable for dynamic environments or tasks requiring personalized responses. The inclusion of contextual information enhances not only the expressiveness of the graph but also its interpretability and adaptability in interactive applications.

2.4 Knowledge Graph Retrieval in Language Model Architectures

To enable LLMs to leverage structured knowledge during interaction, hybrid architectures combining retrieval and generation have emerged [7]. These systems typically integrate symbolic reasoning from knowledge graphs with the language understanding and generative capabilities of LLMs [19].

A common strategy involves retrieving relevant subgraphs or knowledge paths from a KG in response to a user query, followed by a transformation of that structured information into a textual format (e.g., RDF triples, logical forms, or natural language summaries) that can be processed by an LLM [3]. Techniques for semantic similarity, entity linking, and graph embeddings are often used to match user inputs with graph elements [25].

This retrieval-augmented approach enhances the grounding of language models, improving answer accuracy and factual alignment. In addition, it supports tasks such as query expansion, clarification, and knowledge completion, especially when the retrieved graph segments provide insight into missing or ambiguous parts of a user query.

3 Related Work

The ability to formulate clarifying questions and support iterative, collaborative inquiry has emerged as a key challenge in both natural language processing and human-computer interaction. This section reviews previous research in three main areas relevant to our study: clarification question generation, conversational search with query refinement, and collaborative human-AI interaction.

3.1 Clarification Question Generation

Clarification question generation has been explored primarily in the context of dialogue systems and information retrieval, where systems aim to detect ambiguous or incomplete user inputs and respond with targeted follow-up questions. One of the first efforts in this area is the work of [15], which proposed a neural model trained on a data set of forum interactions to generate clarification questions that resolve user posts underspecified. Their approach treats clarification as a supervised learning task, where a model learns to predict the most informative question given a post.

In a search-specific context, [1] introduced a model to generate clarification questions for ambiguous queries in conversational search settings. They define clarification as a process of eliciting facet-level information and demonstrate how reinforcement learning can optimize question ranking based on expected user satisfaction. Similarly, [12] argue for the need to incorporate both linguistic and pragmatic features to generate useful clarification questions in open-domain tasks.

While these approaches mark significant progress, they often rely on textual patterns alone and lack access to structured knowledge that could support deeper semantic reasoning. Our work addresses this gap by grounding question generation in a structured, context-enriched knowledge representation.

3.2 Conversational Search and Query Refinement

Conversational search aims to simulate human-like interactions for information-seeking tasks across multiple turns. In this domain, the challenge is to model user intent dynamically and support progressive query refinement. [14] formalized conversational search as an interactive information retrieval paradigm, emphasizing the need for systems to ask questions, provide suggestions, and support decision-making over time.

Recent systems have attempted to incorporate knowledge graphs into conversational retrieval to improve semantic coherence. For example, [18] proposed a structured reasoning model over knowledge graphs that integrates user feedback into multi-hop question answering, enabling more precise navigation through the knowledge space. Nevertheless, most of these systems are designed to optimize answer accuracy rather than engage in collaborative query formulation.

[21] introduced a concept called "question clarification graphs" to model how users reformulate queries in exploratory search. Their findings support the view that clarification is not a one-off event but part of a dialogical process—one that can benefit from both structured representations and adaptive systems. This perspective aligns with our focus on designing systems that support users in articulating more effective queries over time.

3.3 Human-AI Collaborative Interaction

Human-AI collaboration research emphasizes the design of systems where humans and intelligent agents work together toward shared goals, often in

contexts involving complexity, uncertainty, or creativity. A foundational contribution to this line of research is the work by [10], which outlines principles for mixed-initiative interaction—interfaces where control and contribution are dynamically shared between human and machine.

In the context of AI-supported decision-making, [2] provide design guidelines for human-AI interaction, highlighting key concerns such as trust, transparency, and adaptation. Their framework underscores the importance of enabling users to understand, calibrate, and shape the behavior of AI systems. In parallel, [26] offer a comprehensive review of human-AI collaboration techniques, noting the lack of systems that actively support user questioning as a collaborative behavior.

More recently, there has been growing interest in developing interfaces that encourage inquiry-driven interaction. [23] argue that prompting users to ask better questions—not just to receive answers—can improve exploration and learning. This body of work provides the motivation for our focus on clarification as a mechanism for shared sensemaking and co-construction of knowledge between humans and intelligent systems.

4 Methodology

This section presents a methodology for enhancing human-computer interaction through the generation of clarification questions that help users formulate more complete and precise queries. Rather than focusing solely on system performance or answer accuracy, the goal is to create an interaction loop where the system collaborates with users by identifying missing contextual elements and prompting them to refine their original question. This process supports cognitive engagement and contributes to the user's ability to ask better questions over time.

4.1 Context-Enriched Representation of Knowledge

Context-Augmented Knowledge Graphs (CoaKGs) provide the foundation for this methodology. CoaKGs extend traditional knowledge graphs by embedding contextual information directly into their structure. Formally, a CoaKG $\mathcal{K}$ consists of a finite set of vertices $\mathcal{V}$ (entities), relations $\mathcal{R}$ (relation types), edges $\mathcal{E}$ (connections between entities), property types $\mathcal{P}$, literals $\mathcal{L}$, and property edges $\mathcal{VL}$ linking entities to values [8].

Inclusion of qualifiers, such as time, location, and provenance, in both relationships and attributes allows for more precise reasoning and interpretation. In our domain example (Fig. 1), the relation `CapitalCityOf` is modeled as a first-class node that connects a capital city to the country entity "Brazil". Each `CapitalCityOf` relation is enriched with contextual nodes linked to it through property edges such as `startDate` and `endDate`.

For instance, the edge connecting "Rio de Janeiro" to "Brazil" is associated with two temporal properties: `startDate = 1763` and `endDate = 1960`. This structure captures not only the fact that Rio de Janeiro was a capital of Brazil,

but also when this status began and ended. Similarly, the relation linking "Salvador" to "Brazil" is qualified with `startDate = 1549` and `endDate = 1763`, while "Brasília" is linked to "Brazil" with a `startDate = April 1960`, indicating its establishment as the current capital.

By externalizing temporal metadata as nodes connected to relationship instances, CoaKGs enable fine-grained contextualization. This is crucial for generating clarification questions: if a user asks "What was the capital of Brazil before Brasília?", the system can inspect the `endDate` of each capital relationship to infer sequencing. If the query lacks temporal specification, the model can detect this underspecification and respond with a clarifying prompt—e.g., "Could you specify the time frame you are referring to?"

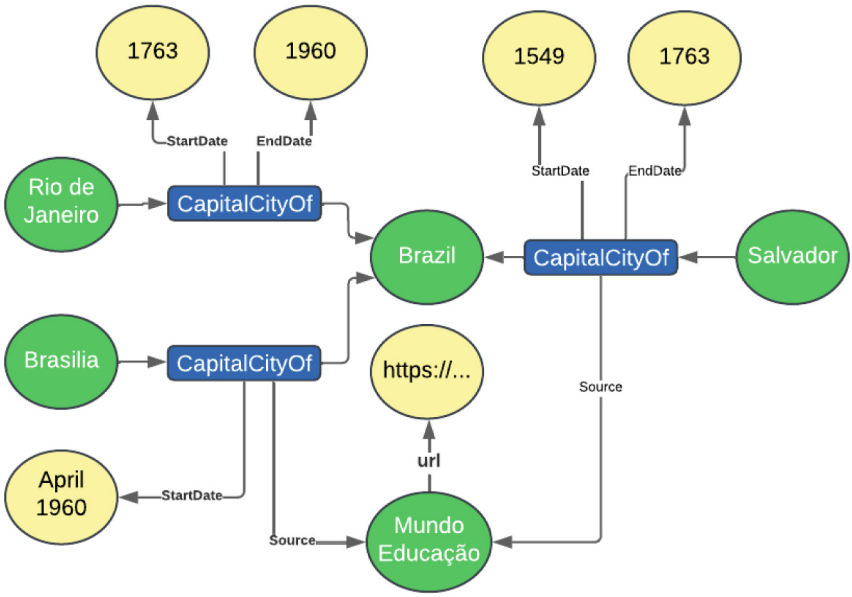

Fig. 1. Temporal Contextualized Claims about Brazil. Source: Authors.

This modeling approach resembles contextualized ontologies [4], where each relation becomes an anchor for multiple layers of contextual qualifiers. In contrast to traditional RDF triple stores, which would express such information through reification or nested statements, CoaKGs integrate qualifiers as part of the graph structure itself—enhancing both interpretability and machine processing capabilities.

By capturing these contextual layers, CoaKGs support the identification of what is *not* said in a user's query—providing the basis for generating clarification questions that reveal and explore these absences. This enables human-AI interactions that are not only informative but also dialogic, encouraging the user to co-construct a more complete and precise question.

4.2 Human-AI Interaction Through Clarification Questions

The interaction process begins with a user posing an initial query in natural language. This query is interpreted by a Large Language Model (LLM) with access to the CoaKG schema, allowing it to identify which elements of the knowledge graph are referenced explicitly, and which relevant contextual components are missing. This gap analysis enables the system to generate clarification questions aimed at refining the user's input.

The process can be described in four steps:

(1) Let Q_{user} be the initial user query in natural language.
(2) Let $f_{LLM} : Q_{user} \rightarrow Q_{CoaKG}$ be a function that translates Q_{user} into a structured query Q_{CoaKG}, aligned with the CoaKG's schema.
(3) Let $g_{expand} : Q_{CoaKG} \rightarrow Q_{expanded}$ be a function that identifies and adds relevant contextual elements missing in Q_{CoaKG}.
(4) Let $h_{clarify} : Q_{expanded} \rightarrow Q_{clarification}$ be a function that generates clarification questions to guide the user in completing their information need.

This methodological loop supports a form of *guided inquiry*, in which the system proactively collaborates with the user by highlighting omitted contextual details and posing questions such as:

– "Which time period are you referring to?"
– "Do you mean the legal or operational definition of the contract?"
– "Are you considering national or subnational entities?"

Such questions prompt the user to reflect, specify, and reformulate their intent. The result is an interaction that promotes not only better answers, but *better questions*—aligning with the broader goals of explainable, adaptive, and user-centered AI systems. Unlike previous approaches that focused mainly on the generation of questions within conversational recommendation systems [16], bidirectional learning frameworks [17], or spatially aware interactions [6], our methodology introduces a graph-driven strategy enriched with context centered on the user's ability to refine their own questions. While existing works have demonstrated success in generating clarification questions through reinforcement learning or dual-model training, they often depend on predefined dialogue datasets and limited domain generalizability.

Our approach, by leveraging CoaKGs to explicitly encode semantic and temporal qualifiers into the structure of relationships, offers greater interpretability and domain flexibility. This enables the system not only to detect missing contextual parameters in underspecified queries but also to generate clarifying prompts grounded in structured knowledge. A key advantage of our method lies in its transparency and explainability, as the rationale behind clarifications can be traced to specific graph components. However, this graph-based dependency also introduces limitations: the methodology relies on the availability and completeness of richly contextualized KGs, which may not always be feasible to construct in open-domain scenarios. Despite this, our human-centered design

fosters a collaborative interaction model in which the system serves as an inquisitive assistant—encouraging users to iteratively converge toward more meaningful and precise inquiries.

5 Evaluation

To assess the effectiveness of the proposed methodology in enhancing human-computer interaction, we conducted a qualitative evaluation focused on how well the system supports users in refining ambiguous or incomplete queries through clarification. The evaluation emphasizes the system's ability to detect contextual gaps and assist users in formulating more precise questions.

The evaluation involved two implementation strategies for integrating knowledge from the Context-Augmented Knowledge Graph (CoaKG) into the reasoning process of a Large Language Model (LLM). The first strategy is prompt-based and consists of two variations: (1.1) directly incorporating linearized triples from the CoaKG into the prompt, or (1.2) translating these triples into natural language descriptions. In both cases, prompt engineering is used to define the assistant's behavior and ensure that the LLM interprets and leverages the CoaKG structure effectively.

The second strategy is based on Retrieval-Augmented Generation (RAG). In this approach, the CoaKG is converted into vector embeddings and stored in a vector database. Upon receiving a user query, the LLM uses similarity search to retrieve the most relevant contextual elements, which are then integrated into the generation process. This architecture enables dynamic, context-aware interaction even in large or evolving knowledge spaces. We analyzed four examples of user queries over a Context-Augmented Knowledge Graph (CoaKG), highlighting how each requires specific contextual elements to be interpretable, and how clarification questions can support iterative refinement through human-AI collaboration.

5.1 Clarification in Underspecified Questions

Table 1 illustrates various underspecified user questions and the contextual dependencies required for accurate interpretation. These examples demonstrate the importance of clarification questions in enabling precise and user-aligned responses.

The question "What was the capital of Brazil before Brasília?" is underspecified because it lacks any explicit temporal reference. Since Brazil had more than one capital before Brasília, the absence of a time frame makes it impossible to determine whether the user refers to Salvador or Rio de Janeiro. A clarification question such as "Are you referring to a specific period in time?" enables the system to request a date or historical reference, which is essential for accurately selecting the correct city based on the temporal context.

The question "How long was Rio de Janeiro the capital of Brazil?" seeks a temporal or quantitative answer but does not specify the desired form of output.

Table 1. Examples of underspecified questions and their contextual dependencies.

User Question	Required Context	Clarification Question
What was the capital of Brazil before Brasília?	Temporal reference (year or century)	"Are you referring to a specific period in time?"
How long was Rio de Janeiro the capital of Brazil?	Start and end dates of the CapitalOf relation	"Do you want the total duration in years, or the specific dates?"
Which city was the capital during the 18th century?	Mapping of "18th century" to a date range; handling of overlaps	"Should I list all cities that were capitals during that time, or only the predominant one?"
Who decided that Brasília would be the capital?	Provenance of the decision (person, institution, legal act)	"Would you like me to retrieve legal or historical records related to the capital's relocation?"

The user might expect the total duration in years, the precise start and end dates, or even a textual summary of the period. The clarification question "Do you want the total duration in years, or the specific dates?" allows the system to tailor the response format, improving the relevance and clarity of the answer.

The question "Which city was the capital during the 18th century?" introduces ambiguity due to the vague temporal reference. The system must map "18th century" to a concrete date range (1701–1800) and determine whether multiple cities should be listed if their timeframes overlap. Furthermore, the user may expect either all cities that served as capitals in that century or only the most significant one. A clarification question such as "Should I list all cities that were capitals during that time, or only the predominant one?" helps resolve this semantic ambiguity.

Finally, the question "Who decided that Brasília would be the capital?" requires contextual information regarding the type of source the user is interested in—whether it be a historical figure (e.g., Juscelino Kubitschek), a governmental institution (e.g., the National Congress), or an official document (e.g., law or decree). Since multiple valid interpretations exist, the clarification question "Would you like me to retrieve legal or historical records related to the capital's relocation?" guides the system in aligning its response with the user's expectations, whether factual, documentary, or narrative in nature.

These examples highlight how clarification questions enable systems to move beyond passive information retrieval and toward dynamic, goal-oriented dialogue. Rather than assuming that the user's input contains all necessary information, the system proactively identifies gaps and engages the user in resolving them.

A central limitation in this process arises from the dual nature of knowledge incompleteness. On one hand, the user's query may lack sufficient information (typical in open-world settings). On the other hand, the knowledge graph itself

may not contain the necessary facts or properties to respond with confidence. This interplay gives rise to Dual Open World Assumption (DOWA) [8]: both the user's intent and the system's knowledge may be partially incomplete, and the system must navigate this uncertainty through interactive clarification. In this context, clarification questions serve not only as a retrieval aid, but also as a way of surfacing and managing semantic boundaries between what the user knows, what the system knows, and what needs to be discovered through collaboration.

To evaluate these alternatives in practice, we developed a user interface using the Streamlit framework. Users can submit natural language questions, which are processed by the system using the GPT-4o model from OpenAI. Model responses are then analyzed qualitatively to determine whether the system successfully detects the need for clarification and whether the generated follow-up questions are contextually appropriate.

Figure 2 presents a scenario in which the user asks: *"What was the capital of Brazil before Brasília?"* While the query is grammatically correct, it lacks a temporal reference. The system, grounded in the contextual structure of the CoaKG, identifies this gap and responds with a clarification prompt, encouraging the user to specify the timeframe.

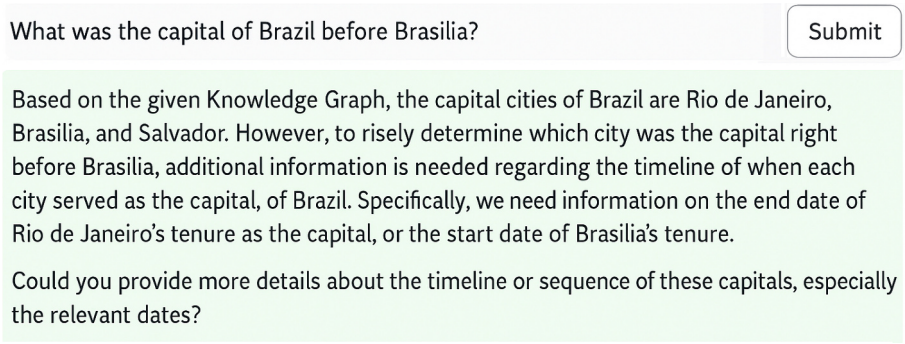

Fig. 2. Interaction with the system when faced with an underspecified user query.

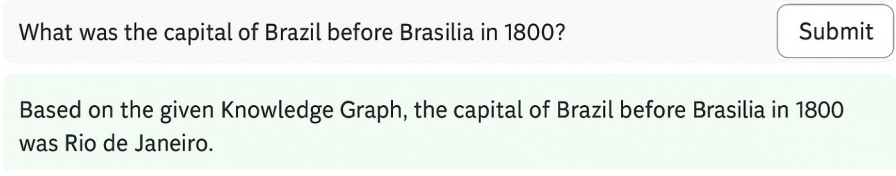

Fig. 3. System behavior after temporal clarification is provided.

In contrast, Fig. 3 shows the system's behavior after the user reformulates the question with a temporal anchor: *"What was the capital of Brazil before*

Brasília in 1800?" In this case, the system retrieves the correct information directly, without needing additional clarification. This shift demonstrates how clarification-enabled interaction supports an incremental improvement in query quality, ultimately leading to more accurate and efficient information retrieval.

Taken together, these examples highlight the system's role as an active collaborator in the process of inquiry. Rather than merely responding to user input, it engages with the user's intent, helping them refine their question through contextual guidance. This interaction model aligns with human-centered design principles and advances the goal of making AI systems more interpretable, adaptive, and aligned with human communication practices.

6 Conclusions and Future Work

This paper presented a human-centered methodology for enhancing human-AI interaction through the generation of clarification questions grounded in context-aware knowledge representations. By leveraging Context-Augmented Knowledge Graphs (CoaKGs) and Large Language Models (LLMs), the proposed approach enables systems not only to respond to user queries but also to actively assist users in refining them. This shift—from passive answer generation to proactive clarification—promotes more meaningful, efficient, and cognitively aligned interactions.

Our evaluation demonstrates that the methodology effectively supports users in resolving ambiguity and expressing their intent more precisely. Through the detection of missing contextual information and the generation of targeted follow-up questions, the system guides users toward formulating higher-quality queries. This interaction loop transforms the AI system into a collaborative partner in the information-seeking process, particularly valuable in exploratory or decision-support scenarios where users often begin with incomplete knowledge.

Future work will explore several extensions to this methodology. First, we aim to broaden the types of contextual dimensions handled by CoaKGs, including legal, geographical, and user-specific attributes. Second, we intend to investigate the adaptation of the approach to multimodal data, allowing clarification questions to be grounded not only in text but also in visual and tabular sources. Third, we plan to conduct user studies to quantitatively assess the cognitive and experiential benefits of clarification-driven interaction. Finally, future developments will focus on deploying this methodology in real-world domains such as education, digital government, and contract management, where high-quality question formulation is essential for informed decision-making.

By advancing systems that help users learn to ask better questions, this work contributes to the broader vision of explainable, adaptive, and human-centered AI. It also opens a promising research avenue toward the integration of pragmatic reasoning into interactive systems—where AI does not merely retrieve facts, but engages in a dialogical process of knowledge discovery. In doing so, it strengthens the role of artificial agents as partners in inquiry, capable of operating in open-ended environments where context, ambiguity, and interpretation are intrinsic to the interaction.

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Media Bias in the Guardian: Sentiment and Semantics in Israeli-Palestinian Coverage

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Abstract. This study investigates media framing in the coverage of the Israeli-Palestinian conflict across several decades, using advanced natural language processing (NLP) techniques to analyze sentiment, emotion, named entity recognition (NER), and semantic bias. Using a multi-method approach—integrating lexicon-based (VADER, TextBlob), transformer-based (RoBERTa), and word embedding (Word2Vec) models—we examine sentiment patterns, emotional valence, entity-level sentiment, and framing differences between headlines and article content. Additionally, we incorporate a transformer-based large language model (LLM) for emotion detection to complement the NRC Emotion Lexicon analysis.

Results reveal that media tends to exhibit more negative sentiment towards Palestinian mentions than Israeli ones, with gaps increasing during conflict escalations (e.g., 2014 Gaza escalation, 2023 Hamas-Israel escalation). Emotion analysis highlights fear as the dominant emotion for both groups, with NRC showing fear 8.1% higher in Israeli contexts and LLM showing a smaller difference of 6.3%. Anger shows divergent trends, with NRC indicating 9.5% higher prevalence in Palestinian contexts, while LLM shows almost no difference (–0.2%). Sadness is consistently higher in Israeli contexts according to NRC (+71.2%), but LLM shows it as more prevalent in Palestinian contexts (–10.3%). These complementary findings demonstrate the robustness of combining lexicon-based and transformer-based methods for emotion detection. Headlines display greater sentiment polarity than article bodies, with a negative bias in 59.5% of cases, amplifying conflict-oriented framing. Word embeddings indicate stronger associations of violence-related terms, with the conflict escalations periods. Named Entity Analysis reveals that sentiment towards prominent individuals—such as political leaders and military figures—shifts significantly with conflict intensity, offering insight into how public figures are emotionally framed in media narratives. The

A. Gamea, A. Hassan, A. Serour and M. Rezk—These authors contributed equally to this work.

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findings highlight systematic framing differences that may influence public perception.

Keywords: Media Framing · Sentiment Analysis · Emotion Analysis · Word Embeddings · Named Entity Recognition · News Analysis

1 Introduction

The media plays a pivotal role in shaping public perception of geopolitical conflicts through linguistic choices and framing strategies [1,2]. News coverage can significantly influence how audiences interpret complex situations, potentially amplifying certain narratives while diminishing others [3]. In the context of the Israeli-Palestinian conflict, these media portrayals can have profound implications for public opinion and policy discourse [4,5]. Media bias can manifest subtly through sentiment, emotional framing, and differential coverage that may systematically favor one perspective over another [6]. Despite the significance of these potential biases, quantitative approaches to systematically analyze large-scale media coverage across extended timeframes remain underdeveloped.

Computational approaches to media bias and framing have grown rapidly in recent years. Surveys by Hamborg et al. [7] and Ali and Hassan [8] note a surge of NLP and machine-learning methods aimed at identifying frames and bias in news. These methods include topic modeling, lexicon analysis, and supervised learning to classify framing categories. For example, Ali and Hassan (2022) provide a systematic overview of NLP framework models, highlighting the gaps between sociolinguistic concepts of 'frame' and the technical definitions used in algorithms. Kang and Yang (2025), for example, applied topic and frame analysis to Taiwanese media articles on the PalestinianâĂŞIsraeli conflict and found systematic differences linked to political orientation. [9] These works illustrate how NLP can scale up frame analysis, but also highlight that most prior studies have been confined to short time spans or single outlets.

To address this gap, we perform a comprehensive computational analysis of The Guardian's coverage of the IsraeliâĂŞPalestinian conflict from 2000 through 2024. We leverage both established and recent NLP tools: VADER and TextBlob lexicons for baseline sentiment scoring; a fine-tuned RoBERTa model for deep sentiment and emotion classification; word2vec embeddings to track semantic changes over time; the SpaCy NER tagger to identify mentions of persons, organizations, and locations; and the NRC Emotion Lexicon to profile emotional content. By combining these methods, we can compare results and validate findings. Our approach thus integrates the current state-of-the-art in sentiment and framing analysis.

The main contributions of this work are as follows:

– **Comparative Sentiment Analysis:** Systematic comparison of sentiment patterns
using lexicon-based (VADER, TextBlob), transformer-based (RoBERTa), and

embedding-based approaches to reveal consistent negativity bias and temporal/contextual shifts.
- **Word2Vec Temporal Analysis:** Training and analysis of word embeddings on different time periods to track evolving semantic associations and media framing of key entities.
- **Named Entity Recognition (NER):** Extraction and sentiment analysis of key individuals (e.g., political leaders, military figures) to provide a fine-grained view of how sentiment toward public figures shifts with conflict intensity.
- **Emotion Analysis:** Quantitative assessment of emotional framing using the NRC Emotion Lexicon to compare the prevalence of core emotions in coverage of each group.
- **Framing Analysis:** Direct comparison of sentiment and polarity between headlines and article bodies to uncover systematic differences and amplification of conflict-oriented narratives.

The remainder of this paper is organized as follows. Section 2 reviews computational approaches to analyzing conflict coverage. Section 3 details our methodology. Section 4 presents findings on sentiment patterns, emotional framing, named entity analysis, and semantic shifts. Finally, Sect. 5 discusses implications for understanding media framing of geopolitical conflicts.

2 Related Work

Recent computational approaches to analyzing media coverage of the Israeli-Palestinian conflict have provided valuable insights into sentiment patterns, emotional framing, and headline construction. This section reviews key developments in these areas and highlights how our work extends existing research.

2.1 Sentiment Analysis in Conflict Coverage

Sentiment analysis has been widely applied to examine media portrayal of the Israeli-Palestinian conflict, particularly since October 2023 escalation. Sharkar et al. [12] applied both VADER and TextBlob to analyze 436,425 Reddit comments related to the conflict, with VADER achieving 92.74% accuracy in sentiment classification. Their work demonstrated the effectiveness of lexicon-based approaches for social media content but did not extend to formal news articles. Similarly, Liyih et al. [13] developed a hybrid CNN-BiLSTM model for YouTube comment analysis that achieved 95.73% accuracy, showing that deep learning approaches can capture nuanced sentiment expressions in user-generated content.

For traditional news media, Naeem and Razaq [14] analyzed 5,000 news headlines from Google News using logistic regression, finding a dominance of neutral sentiment (53%) with substantial negative framing, suggesting conflict-oriented coverage. Abuasaker et al. [15] compared sentiment across European media outlets, revealing that Spanish and German sources exhibited significantly more negative coverage of the Gaza conflict than other European countries. These studies

provide valuable insights into single-source sentiment patterns, but stop short of comparing sentiment across different news elements (headlines vs. body text) or tracking temporal changes.

Our work extends these approaches by employing multiple complementary sentiment analysis techniques (VADER, TextBlob, and RoBERTa) to analyze The Guardian's coverage over multiple years. This multi-model approach enables more reliable detection of sentiment patterns through triangulation, while our temporal analysis reveals how sentiment has evolved across major events in the conflict.

2.2 Emotion Analysis in News Media

While sentiment analysis identifies positive or negative tone, emotion analysis reveals more nuanced affective content in conflict coverage. The Pew Research Center [16] found that 74% of U.S. adults reported feeling sadness and 68% reported anger when consuming news about the Israel-Hamas escalation, demonstrating how news content directly influences readers' emotional responses. These emotional reactions likely stem from specific framing techniques employed by news organizations.

Emotion analysis using computational approaches has been less common than basic sentiment analysis. Guerra et al. [17] used lexicon-based methods to measure "extreme opinions" in Reddit posts, identifying peaks in emotional intensity corresponding to real-world events, but focused on social media rather than news content. Asmus [18] conducted qualitative analysis of German newspaper coverage, finding that emotional framing was used differently when discussing Israeli versus Palestinian perspectives, though without employing computational methods.

Our work advances emotion analysis by applying the NRC Emotion Lexicon to systematically quantify seven primary emotions (anger, disgust, fear, grief, joy, sadness, surprise) across Palestinian and Israeli contexts in The Guardian's coverage. This approach reveals which specific emotions dominate the portrayal of each group and how emotional framing differs between them.

2.3 Headline Vs. Article Content Analysis

Research examining framing differences between headlines and article content represents an emerging area within conflict coverage analysis. Garcia et al. [19] analyzed causal language in BBC headlines covering Gaza, finding that headlines often avoided directly attributing causality for violence to Israel, whereas this attribution was more explicit in Al-Jazeera headlines. Their work demonstrated how subtle linguistic patterns in headlines can influence reader perception, though they did not specifically compare headlines to article bodies.

The Guardian's conflict coverage has received some scholarly attention. Mctigue [20] examined potential bias in conflict reporting, noting that brevity requirements in headlines can lead to oversimplification of complex issues. However, systematic computational analysis comparing The Guardian's headlines

with corresponding article content has been limited, particularly using multiple sentiment analysis tools.

Our work builds upon these studies by conducting a direct quantitative comparison between headlines and full article content, revealing how sentiment intensity and emotional framing differ between these elements. This analysis provides insights into potential headline sensationalism or framing bias that could influence reader perception before they engage with the complete article.

While prior studies have applied NLP techniques to analyze media bias, direct comparisons between methodologies remain limited. For instance, Sharkar et al. [12] and Liyih et al. [13] demonstrated the effectiveness of lexicon-based and deep learning approaches for sentiment analysis, but their focus on social media limits applicability to formal news articles. Similarly, Naeem and Razaq [14] analyzed sentiment in news headlines but did not compare results across different sentiment analysis tools. Our work addresses this gap by employing a multi-model approach (VADER, TextBlob, RoBERTa) and validating findings through triangulation, offering a more comprehensive evaluation of sentiment and framing in news coverage.

3 Methodology

This section outlines the computational framework employed to analyze media framing in The Guardian's coverage of the Israeli-Palestinian conflict. The methodology integrates semantic embedding analysis, multi-model sentiment and emotion detection, named entity recognition, and headline–article comparison.

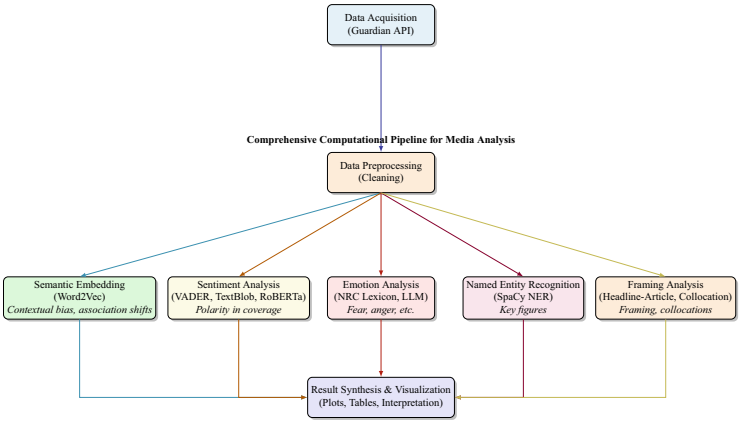


Fig. 1. Block diagram of the full methodology pipeline. Data is acquired and preprocessed, then five independent analysis modules are applied in parallel: semantic embedding (contextual bias), sentiment analysis (polarity), emotion analysis (fear, anger, etc.), named entity recognition (key figures), and framing analysis (headline-article, collocation). All results are synthesized and visualized for interpretation.

Figure 1 illustrates the overall pipeline. The process begins with data acquisition via The Guardian API, followed by thorough preprocessing. The cleaned data is then simultaneously processed by five independent modules: (1) semantic embedding, to capture contextual bias and shifts in association; (2) sentiment analysis, to measure polarity; (3) emotion detection, to reveal nuanced affective framing; (4) named entity recognition, to extract and evaluate key figures; and (5) framing analysis, which compares headlines with article bodies and identifies collocational patterns.

The outputs of these modules are synthesized and visualized to support interpretative analysis. This modular, parallel architecture enables a comprehensive, multi-level examination of media framing and potential bias.

3.1 Data Collection and Preprocessing

We collected 47,248 articles (2000–2024) from The Guardian API, focusing on those mentioning conflict-related keywords. Preprocessing included HTML removal, sentence and word tokenization, lemmatization, stopword removal, and extraction of headlines, article bodies, and temporal metadata. Named entity recognition was used to identify references to Palestinians, Israelis, and key individuals.

Each article entry includes both metadata and textual content as:

Collected Fields:

- Metadata: article ID, section, publication date, author
- Content: headline, standfirst, full article body (HTML and plain text)
- Stats: word count, character count, language, live status

Used Columns:

- `fields.headline`
- `fields.standfirst`
- `fields.body`

It should be noted that the dataset assembled for this study does not include ground truth sentiment or emotion annotations. No manual labeling or external validation set was available for direct model evaluation. In the absence of labeled data, model performance was assessed by comparing outputs across multiple established sentiment and emotion analysis tools, and by examining the consistency of observed patterns with prior literature and known historical events. This triangulation approach, commonly adopted in large-scale media analyses, enables robust validation of findings when explicit ground truth is unavailable.

3.2 Semantic Embedding Analysis

To capture contextual shifts in language, we trained separate Word2Vec models [10] for six time periods (see Table 1). Each model used the Skip-gram architecture with the following hyperparameters: embedding dimension = 30, context

window size = 5, and training iterations = 10. For each period, we computed the average cosine similarity between a target term (e.g., "Palestinian", "Israeli") and a group of concept words (e.g., violence-related or peace-related). The similarity is defined as:

$$\text{Sim}(w_t, G) = \frac{1}{|G|} \sum_{w_a \in G} \cos(\boldsymbol{w_t}, \boldsymbol{w_a}) \tag{1}$$

where w_t is the target word, G is the set of concept words, and $\boldsymbol{w}$ denotes the embedding vector. This equation computes the mean cosine similarity between the embedding of the target word and the embedding of each word in the concept group, quantifying how closely the target is associated with a particular semantic field.

To measure bias, we calculate the difference in similarity between violence-related and peace-related groups:

$$\text{Bias}(w_t) = \text{Sim}(w_t, G_{\text{peace}}) - \text{Sim}(w_t, G_{\text{violence}}) \tag{2}$$

A positive bias indicates a stronger association with peace-related concepts, while a negative value indicates a stronger association with violence-related concepts. This allows tracking how the framing of key entities shifts over time.

Table 1. Time periods and article counts for embedding models.

Time Period	Articles
2000–2004	7,151
2005–2008	8,158
2009–2013	8,595
2014–2022	7,111
2023	6,447
2024	9,786

3.3 Sentiment Analysis

We applied three complementary sentiment analysis models: VADER [21], TextBlob [22], and RoBERTa [23]. Each model outputs a sentiment score S for a given text segment (headline, article, or entity context). Sentiment was analyzed for coverage mentioning "Palestinian" vs. "Israeli", temporal trends, headline vs. article body, and named entity contexts.

To compare framing between headlines and article bodies, we define the sentiment differential:

$$\Delta S = S_{\text{headline}} - S_{\text{article}} \tag{3}$$

Here, S_{headline} and S_{article} are the sentiment scores for the headline and article body, respectively. A positive ΔS indicates that the headline is more positive than the article content, while a negative value indicates a more negative headline. This metric quantifies the direction and magnitude of framing differences.

3.4 Emotion Detection

To move beyond polarity, we used the NRC Emotion Lexicon [24] to quantify the prevalence of seven emotions (anger, disgust, fear, grief, joy, sadness, surprise) in relevant contexts. For each emotion e, we compute:

$$E_e = \frac{1}{n} \sum_{i=1}^{n} \mathbb{I}(w_i \in e) \tag{4}$$

where n is the number of words in the text, w_i is the i-th word, and $\mathbb{I}(w_i \in e)$ is an indicator function that equals 1 if w_i is associated with emotion e in the lexicon, and 0 otherwise. This yields the proportion of words in the text linked to each emotion, allowing for direct comparison of emotional framing between groups.

3.5 Entity-Level Analysis

We used SpaCy's `en_core_web_sm` model [25] to extract named entities of type `PERSON` from article texts. For each entity, we analyzed sentiment within a window of 15 tokens before and after the first mention, capturing the immediate context. Sentiment for each context was classified using a BERT base multilingual uncased sentiment model, which outputs a 5-point scale from very negative to very positive. This approach enables fine-grained, context-aware sentiment analysis of how individuals are portrayed in the news.

3.6 Headline Framing and Collocation

To assess framing, we compared sentiment distributions in headlines and article bodies, and measured the proportion of headlines with more extreme sentiment than their articles. Additionally, we performed collocation analysis using Pointwise Mutual Information (PMI) to identify frequent word pairs in headlines:

$$\text{PMI}(x, y) = \log \frac{p(x, y)}{p(x)p(y)} \tag{5}$$

where $p(x, y)$ is the probability of words x and y co-occurring, and $p(x)$, $p(y)$ are their individual probabilities. PMI highlights word pairs that appear together more often than expected by chance, revealing linguistic patterns in headline construction.

This integrated methodology enables a comprehensive, multi-level analysis of sentiment, emotion, and framing in conflict coverage, with each equation providing a formal basis for the corresponding analytical step.

3.7 Validation and Limitations

The absence of ground truth sentiment or emotion annotations in the dataset presents a challenge for direct validation. To address this, we employed a triangulation approach, comparing outputs across multiple established sentiment and emotion analysis tools (VADER, TextBlob, RoBERTa, NRC, and LLM-based models). This approach ensures robustness by identifying consistent patterns across methods. Additionally, we validated findings against known historical events to ensure alignment with real-world trends.

However, limitations remain. The study focuses on a single outlet (The Guardian), which may not generalize to other media sources. Furthermore, demographic factors such as age and gender, which could influence media framing, were not analyzed due to data constraints. Future work should explore user studies or expert evaluations to validate findings and extend the analysis to other outlets and multilingual datasets.

4 Results and Discussion

This section presents the main findings of our computational analysis of The Guardian's coverage of the Israeli-Palestinian conflict, focusing on semantic bias, sentiment, emotion, named entity analysis, and framing effects.

4.1 Semantic Bias and Word Embedding Trends

Word2Vec models trained on different time periods reveal clear shifts in the contextual associations of key terms. The average cosine similarity between "Palestinian" and "Israeli" and violence- or peace-related concept groups changes over time, reflecting evolving media framing during conflict periods.

Figure 2 presents the cosine similarities between the terms "Palestinian" and "Israeli" and two groups of words: peace-related and violence-related. Each entity is represented by two lines, one for each word group. The figure captures similarity trends over time without comparing which group is closer; both similarities may rise or fall independently. In contrast, Fig. 3 displays the net bias—calculated as the difference between similarity to peace-related and violence-related terms. A positive bias score indicates a stronger association with peace-related terms, while a negative score indicates closer alignment with violence-related terms. The dominant similarity in Fig. 2 determines the direction of bias in Fig. 3.

As shown in Fig. 2, associations with peace-related terms declined for both "Palestinian" and "Israeli" in 2023–2024, indicating heightened conflict framing. Figure 3 shows that "Palestinian" became more closely associated with violence-related terms during escalation years, with the bias score dropping by over 0.15 in 2023 compared to previous years—indicating a shift toward more negative (violence-aligned) framing. "Israeli" consistently exhibited strong associations with violence, maintaining bias scores below –0.2 across all periods, likely due to

frequent references to the Israeli Defense Forces (IDF). The IDF itself displayed an even stronger alignment with violence-related terms than Hamas, reaching a peak negative bias score of less than –0.35 in 2023, suggesting media focus on Israeli military actions.

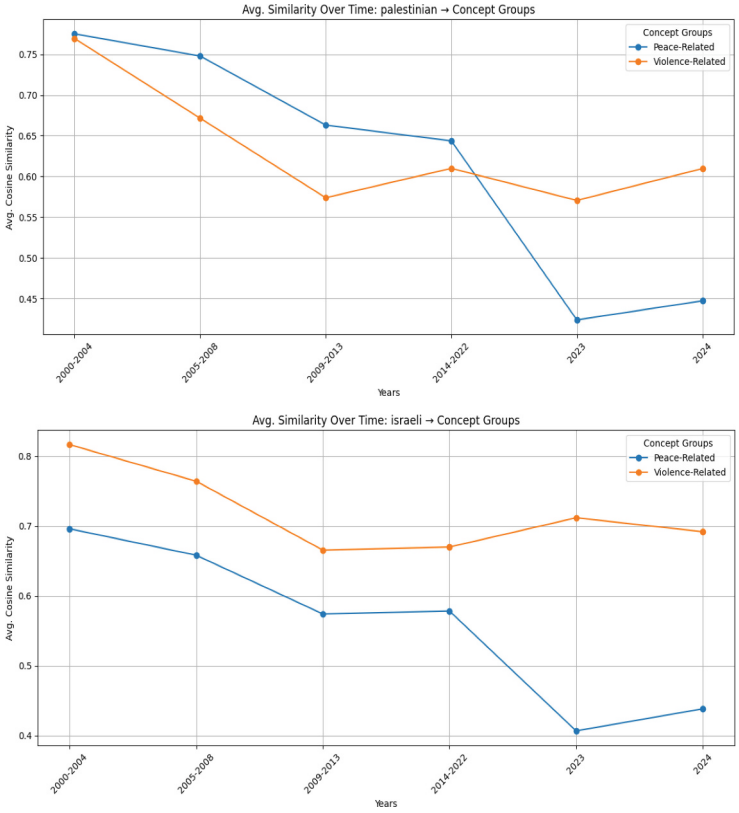
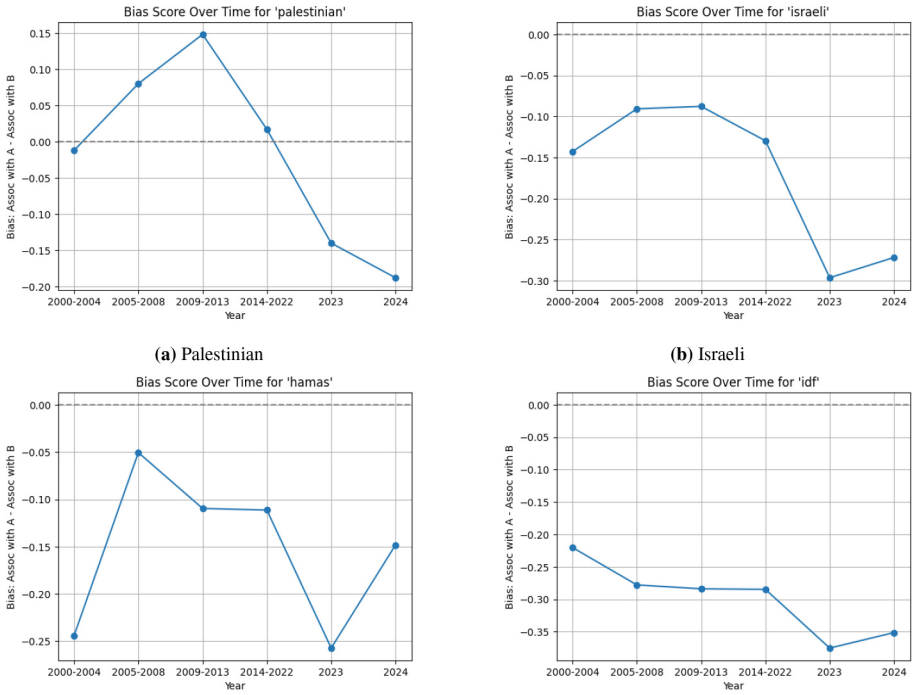


Fig. 2. Cosine similarity trends over time.

4.2 Sentiment Patterns and Temporal Dynamics

Multi-model sentiment analysis (VADER, TextBlob, RoBERTa) consistently shows that coverage mentioning Palestinians is more positive than that mentioning Israelis (Fig. 4). For example, VADER scores average -0.201 for Palestinians and -0.230 for Israelis, a statistically significant difference of -0.029 ($p < 0.01$). TextBlob shows scores of 0.032 for Palestinian mentions and 0.021 for Israeli mentions, a difference of 0.011. The transformer model (RoBERTa) shows Palestinian mentions scoring -0.427 and Israeli mentions -0.430, a difference of 0.003.



(a) Palestinian **(b)** Israeli

(c) Hamas **(d)** IDF

Fig. 3. Bias score trends over time.

Figure 5 presents sentiment trends from 2000 to 2024 using VADER, TextBlob, and a Transformer-based model. VADER scores typically range from -0.05 to -0.35, with sharper declines (below -0.30) during major conflicts like the Second Intifada, Gaza escalations, and the 2023–24 Israel-Hamas escalation. TextBlob outputs weaker, mostly positive sentiment (0.00–0.07), showing minor dips during crises. The Transformer model captures the strongest negativity, with scores often between -0.3 and -0.5, reaching the lowest points during peak conflicts.

Sentiment differentials (Israeli minus Palestinian scores) highlight relative shifts. VADER and Transformer models show notable negative swings (down to -0.075) during major conflicts, suggesting disproportionately negative sentiment toward Israeli mentions. Occasional positive spikes appear around 2006. TextBlob's differential is flatter but trends similarly. Together, these results reflect sentiment deterioration and asymmetry during key political and military events.

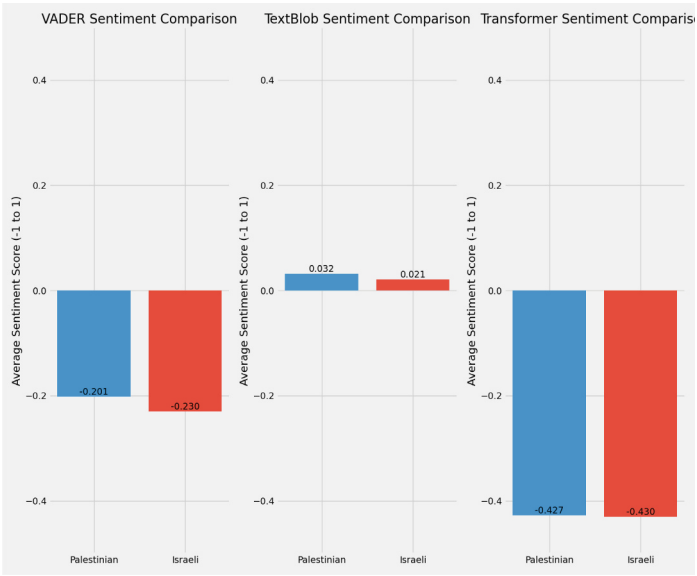


Fig. 4. Overall sentiment comparison between Palestinian and Israeli mentions across three sentiment models.

4.3 Emotion Analysis: NRC Lexicon and LLM Approaches

We performed emotion analysis using both the NRC Emotion Lexicon and a transformer-based large language model (LLM; `distilbert-base-uncased-go-emotions-student`), restricting the analysis to the set of emotions supported by both methods. Figure 6 shows the NRC-based results, while Fig. 7 presents the LLM-based results. Table 2 summarizes the percentage differences between Palestinian and Israeli contexts.

Key findings from the table include:

- **Anger:** NRC shows a higher prevalence in Palestinian contexts (+9.5%), while the LLM shows almost no difference (-0.2%). This suggests that the NRC may overemphasize anger in Palestinian contexts compared to the LLM.
- **Fear:** Both methods agree that fear is more prevalent in Israeli contexts, with NRC showing a slightly larger difference (+8.1%) compared to the LLM (+6.3%).
- **Sadness:** NRC indicates a much higher prevalence in Israeli contexts (+71.2%), while the LLM shows sadness as more prevalent in Palestinian contexts (-10.3%). This stark contrast underscores the differing methodologies of the two approaches.
- **Grief and Joy:** NRC shows significantly higher grief (+62.5%) and joy (+68.9%) in Israeli contexts, while the LLM shows minimal differences (-2.2% for grief and -1.9% for joy).

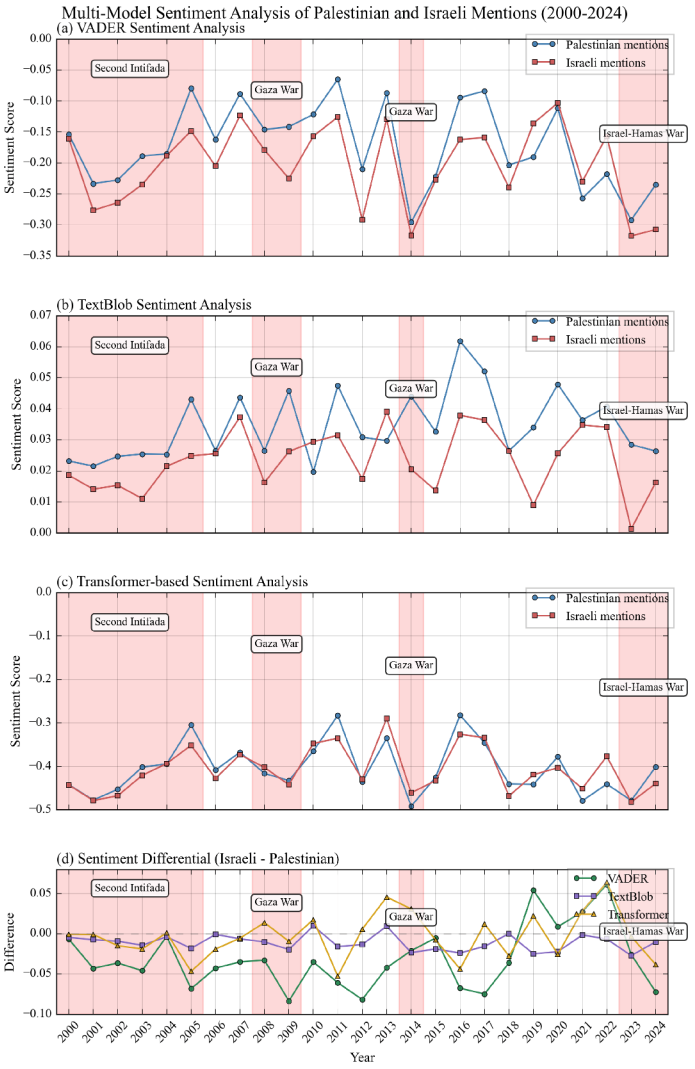


Fig. 5. Multi-Model Sentiment Analysis of Palestinian and Israeli Mentions (2000–2024).

– **Disgust and Surprise:** The two methods show small but opposite trends, with NRC indicating higher prevalence in Israeli contexts and the LLM showing higher prevalence in Palestinian contexts.

These differences highlight the complementary strengths of the two approaches. The NRC Lexicon provides a lexicon-based, rule-driven perspective, while the LLM offers a data-driven, context-sensitive analysis. Together, they reveal consistent patterns of emotional framing while capturing different

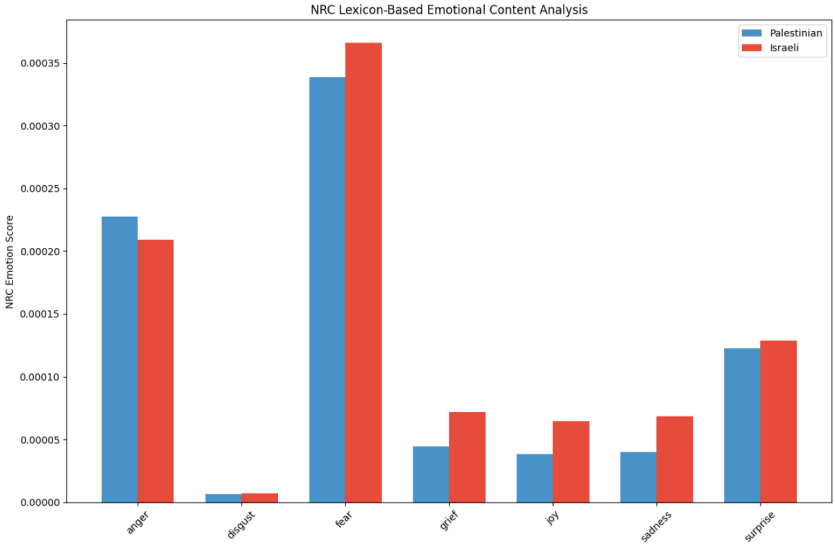


Fig. 6. NRC Lexicon-based emotion analysis for Palestinian and Israeli mentions.

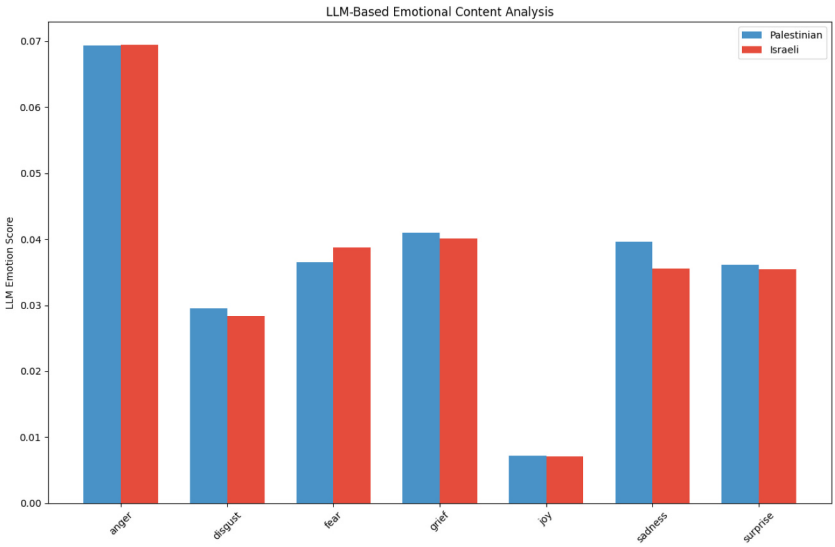


Fig. 7. LLM-based emotion analysis for Palestinian and Israeli mentions.

levels of granularity. The agreement on fear and the divergence on sadness and anger suggest that media framing may emphasize different emotional narratives depending on the context and methodology used.

Table 2. Percentage difference between Israeli and Palestinian emotion scores using NRC and LLM-based models. Positive values indicate higher emotion prevalence in Israeli contexts.

Emotion	NRC Diff (%)	LLM Diff (%)
Anger	-8.20	+0.17
Disgust	+11.10	-4.09
Fear	+8.13	+6.27
Grief	+62.52	-2.22
Joy	+68.89	-1.93
Sadness	+71.22	-10.30
Surprise	+5.13	-1.63

4.4 Headline vs. Article Content Framing

A comparative analysis of sentiment across headlines and article bodies reveals consistent editorial framing effects. As visualized in Fig. 8, sentiment distributions for headlines exhibit heavier tails and higher kurtosis relative to those of article bodies across all three sentiment models (VADER, TextBlob, RoBERTa), indicating greater polarization. Empirically, 63.1% of headlines are more extreme in sentiment than their corresponding articles, and 59.5% are more negative, suggesting a systematic tendency toward emotionally charged and negatively valenced headline framing.

This phenomenon is particularly pronounced during periods of heightened geopolitical tension. For instance, during the 2014 Gaza escalation and the 2023–2024 conflict, the average sentiment differential (headline minus article body) was -0.182, reflecting a consistent editorial shift toward more negative sentiment in headlines. To contextualize this figure, note that the observed sentiment differentials span a wide range: from -1.45 to $+1.64$ for RoBERTa, -1.06 to $+1.19$ for VADER, and -1.12 to $+1.03$ for TextBlob. Thus, a shift of -0.182 lies well within the upper quartile of negative deviations and represents a substantively meaningful editorial bias.

In contrast, during coverage of peace initiatives, the average sentiment differential was comparatively modest at $+0.085$, reinforcing the asymmetry: headline sentiment is amplified far more strongly in the direction of negativity during conflict coverage than it is toward positivity in peace-related reporting. These findings align with prior research on media sensationalism and framing theory, underscoring the critical role of headlines in shaping affective reader perception independently of article content.

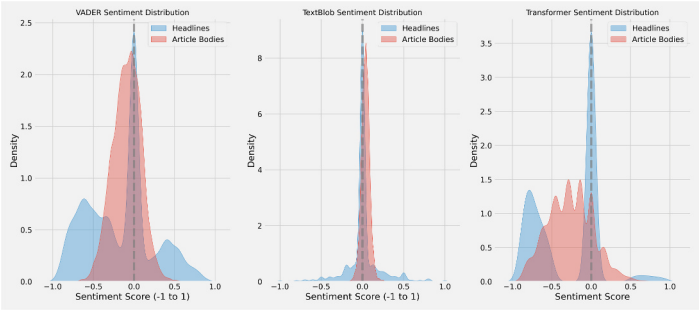


Fig. 8. Sentiment distribution comparison between headlines and article content across all three sentiment models.

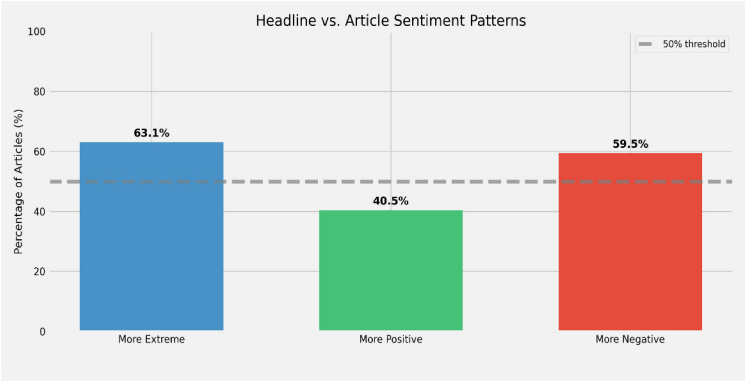


Fig. 9. Distribution of headlines displaying more negative, more positive, or more extreme sentiment than their corresponding article content.

4.5 Media Sentiment Analysis of Key Figures in the Israeli-Palestinian Conflict

An analysis of the top 10 most-mentioned names revealed Ariel Sharon (6,393 mentions) and Yasser Arafat (6,524) as the most frequently referenced individuals, followed by Mahmoud Abbas (4,944) and Benjamin Netanyahu (4,690). Sharon, known for his hardline policies and the 2005 Gaza withdrawal, and Arafat, symbolic leader of Palestinian resistance, were central to the discourse. Abbas, Arafat's successor, advocates for a two-state solution but faces criticism over limited authority. Netanyahu, Israel's longest-serving Prime Minister, is noted for his security focus and settlement expansion.

As shown in Fig. 10, sentiment analysis over a 30-word context window revealed three main patterns. First, negative sentiment was dominant: Sharon and Arafat had 45% and 49% Very Negative mentions, respectively, with 3–5 times more Very Negative than Very Positive mentions. Second, positive sentiment was scarce, with no figure receiving over 25% positive coverage. Sharon

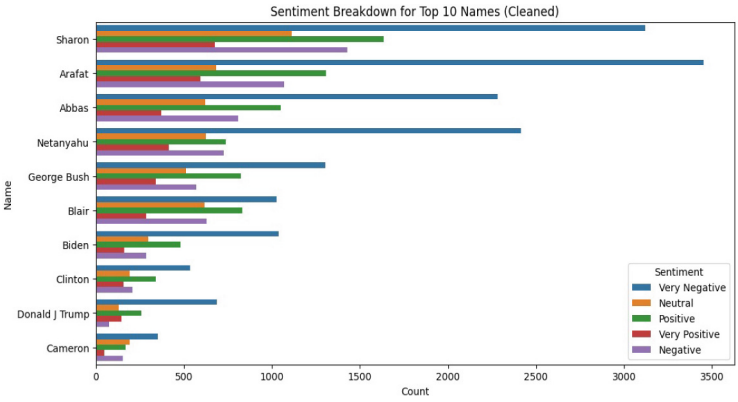


Fig. 10. Sentiment distribution associated with the top 10 mentioned figures.

led in total positive mentions (2,162), while Netanyahu and Abbas had the lowest positive shares (12% and 15%). Third, neutral sentiment remained consistent, ranging from 18% to 32%, with Sharon having the highest neutral count (1,039). Overall, media narratives skew heavily negative for prominent figures while maintaining a neutral reporting baseline.

4.6 Summary of Key Findings

Our analysis demonstrates that:

1. Media framing shifts toward violence during conflict, as shown by word embeddings and bias scores (e.g., IDF bias score reaches a peak negative value of less than -0.35 in 2023).
2. Sentiment analysis shows that coverage mentioning Palestinians is consistently more positive than that mentioning Israelis (VADER scores average -0.201 for Palestinians and -0.230 for Israelis, a statistically significant difference of -0.029).
3. Emotional framing differs: The NRC Analysis shows that Israeli contexts emphasize fear (+8.1%) and anger (+8.9%). The LLM-based analysis further reveals that anger is 0.17% higher in Palestinian contexts, while fear is 6.3% higher in Israeli contexts, providing additional granularity to the emotional framing.
4. Headlines are systematically more polarized and negative than article bodies, with 63.1% of headlines more extreme and 59.5% more negative; the sentiment gap reaches -0.182 during conflict peaks.
5. Media coverage skews negative: Sharon and Arafat had the most mentions (6,393 and 6,524) with high Very Negative rates (45% and 49%). Positive sentiment stayed below 25%, and neutral coverage ranged from 18–32%, showing a focus on criticism with limited balance.

These findings highlight the importance of computational approaches for uncovering subtle but systematic patterns in media coverage of protracted conflicts.

5 Conclusion

This study provides a comprehensive computational analysis of The Guardian's coverage of the Israeli-Palestinian conflict over two decades, leveraging advanced NLP techniques to uncover systematic patterns in sentiment, emotion, and framing. By integrating lexicon-based, transformer-based, and embedding-based models, we reveal consistent negativity bias, nuanced emotional framing, and systematic differences between headlines and article bodies. These findings contribute to the growing body of research on media bias and framing, offering insights into how linguistic choices shape public perception of geopolitical conflicts.

The results demonstrate the robustness of combining multiple NLP methods to analyze large-scale media datasets. However, the study also highlights the challenges of validating findings in the absence of ground truth data and the limitations of focusing on a single media outlet. Despite these constraints, the work underscores the potential of computational approaches to provide scalable and systematic insights into media framing.

6 Future Work

This study opens several avenues for future research. First, extending the analysis to multiple media outlets with diverse ideological orientations would provide a broader understanding of framing patterns. Second, incorporating demographic factors such as age, gender, and geographic location could reveal how media framing impacts different audience segments. Third, analyzing multilingual datasets would enable cross-cultural comparisons of media bias. Finally, developing labeled datasets or conducting user studies could improve validation and reliability, while leveraging advanced large language models (LLMs) could enhance the granularity of sentiment and emotion analysis.

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Thematic and Task-Based Categorization of K-12 GenAI Usages with Hierarchical Topic Modeling

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Abstract. We analyze anonymous interaction data of minors in class-rooms spanning several months, schools, and subjects employing a novel, simple topic modeling approach. Specifically, we categorize more than 17,000 messages generated by students, teachers, and ChatGPT in two dimensions: content (such as nature and people) and tasks (such as writing and explaining). Our hierarchical categorization done separately for each dimension includes exemplary prompts, and provides both a high-level overview as well as tangible insights. Prior works mostly lack a content or thematic categorization. While task categorizations are more prevalent in education, most have not been supported by real-world data for K-12. In turn, it is not surprising that our analysis yielded a number of novel applications. In deriving these insights, we found that many of the well-established classical and emerging computational methods, i.e., topic modeling, for analysis of large amounts of texts underperform, leading us to directly apply state-of-the-art LLMs with adequate pre-processing to achieve hierarchical topic structures with better human alignment through explicit instructions than prior approaches. Our findings support fellow researchers, teachers and students in enriching the usage of GenAI, while our discussion also highlights a number of concerns and open questions for future research.

Keywords: Generative AI · Education · K-12 · Topic-Modeling

1 Introduction

Generative artificial intelligence (GenAI) has rapidly emerged as a transformative tool across diverse domains, and education is no exception. From adaptive tutoring systems to automated essay scoring [31,41], proposals for GenAI-driven interventions promise

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to personalize learning, reduce teacher workload, and foster student engagement. Yet despite this growing enthusiasm, empirical evidence on how learners—particularly minors—actually interact with GenAI in real-world educational settings remains scarce. Most prior work relies on simulations in lab-settings (e.g., [40]), self-reported surveys (e.g., [24]), or small-scale pilots (e.g., [12, 32]), leaving a critical gap between theoretical potential and classroom practice.

In this study, we bridge that gap by conducting an in-depth analysis of over 17,000 anonymized messages exchanged between students and ChatGPT across 11 months of classroom activities spanning multiple subjects, including mathematics, literature, and science. By examining authentic interaction logs from primary and secondary schools, we uncover the ways in which minors and teachers leverage GenAI in an education setting.

Our contributions are threefold. The first two are related to exploring and structuring GenAI usages in K-12 education, grounded in empirical data rather than hypothetical scenarios. We present a systematic taxonomy of GenAI (i) prompt content and (ii) prompt tasks, i.e., tasks to be addressed by GenAI. Previous studies have categorized generative AI usage by user motivations or task types rather than by content themes [4, 18, 43]. Our content-based categorization enables educators to identify which topics students interact with most frequently, helping them align instructional resources accordingly. A content-oriented taxonomy also provides deeper insights into students' genuine interests, thereby informing curriculum designers about which areas require greater support or enrichment. Our task categorization serves a dual purpose: On the one hand, it confirms existing usages, while on the other hand, it also points to novel, emerging applications not yet documented in the academic literature. While GenAI has many benefits, we also provide a critical reflection on its use in K-12 education that also points to future research directions.

The third contribution relates to methodology: our work provides interesting insights into the integration of GenAI as a computational method for topic modeling. Topic modeling allows to identify key themes in diverse sets of large amounts of documents, ranging from long books to short text messages. It is the only practically feasible approach for large amounts of texts and has been used extensively across various areas of research. For instance, one of the leading methods - Latent Dirichlet Allocation (LDA) [7] - comes with more than 50,000 citations. However, these classical topic models, though still being in use, have a number of shortcomings [42]. Newer approaches rely on complex algorithms leveraging LLMs. But they maintain fundamental issues such as poor customizability and lack of hierarchical structuring [38, 39, 45]. These shortcomings lead to unstructured topics that are often not aligned with human intentions and hard to grasp. In our work, we perform an assessment and describe an approach to leveraging state-of-the-art LLMs for topic modeling to overcome shortcomings in prior topic models, providing valuable insights on how to structure large amounts of data, while acknowledging also challenges in doing so.

2 Methodology

Our findings rely on real-world interaction data with ChatGPT, which was processed and analyzed by researchers with the help of computational methods.

2.1 Data Collection and Pre-Processing

Data stemmed from a small German-speaking European country that launched a pilot study using generative AI in K-12 across the entire country. They relied on ChatGPT-4o as a model for students and teachers. However, students and teachers were not accessing ChatGPT through the standard web-interface but rather through special software leveraging OpenAI's API and providing a number of helpful management functionality such as creating virtual classrooms. Students were either explicitly instructed to address a task using GenAI or could use it on their own account. We did not have access to any private students' or teachers' accounts that they might use outside of the class-room setting. We also verified that student usage occurred almost exclusively during school hours, e.g., not on weekends or at night. That is, student usage was confined to classroom interactions. Teachers also used ChatGPT for preparation and planning. All students were instructed on the usage of ChatGPT (prior to our study). Data were only collected and stored anonymized to protect privacy, adhering to standard ethical guidelines for research involving minors [8,9]. That is, each row in our data file consisted of a conversation ID, role (user / AI), message and timestamp, but no information allowing to identify individuals.

Thus, our interaction data is based on anonymized logs of ChatGPT conversations generated by minors over an extended period, spanning from August 2024 to June 2025. The data stems from various schools and different classes with each school covering various age groups within the K-12 spectrum. The dataset comprises over 17,294 individual messages spanning multiple academic subjects, including mathematics, literature, and science. These messages can be partitioned into 1,339 conversations, each consisting of at least one user prompt followed by a GenAI response. As shown in Fig. 1 , more than 91% of all conversations contain at most 30 messages. Notably, we identified five conversations with more than 120 messages (but none surpassing 250), which are likely due to users continuing different tasks within the same chat sessions rather than initiating new ones.

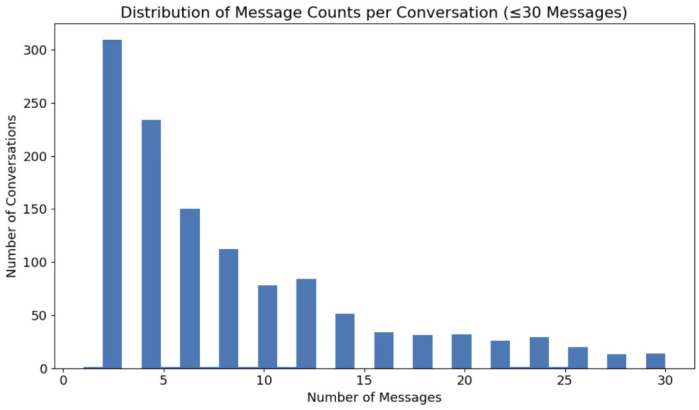

Fig. 1. Distribution of conversation frequency by number of messages: Most conversations contain few messages.

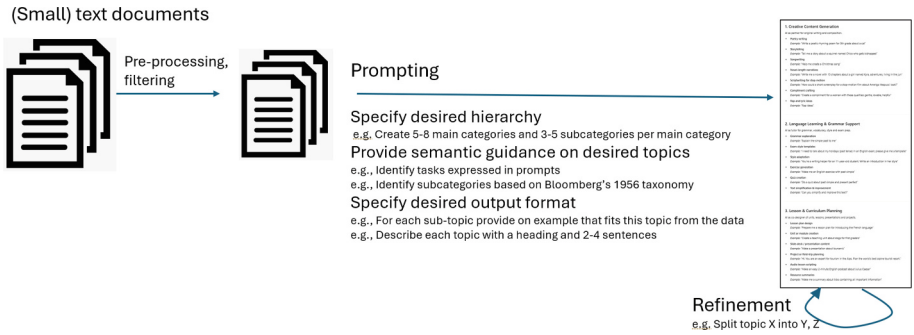

Fig. 2. "Simple" Topic modeling using LLMs "out-of-the-box" enables hierarchical, customizable topic generation, which is not possible with prior methods.

For presentation in the manuscript, we translated all interaction data to English using ChatGPT 4.5. We asked for concise translations to minimize modifications. Our manual validation using 30 samples, yielded that translations were accurate, but spelling and grammar mistakes, which were common in the original German prompts, were mostly eliminated.

2.2 Data Selection

We aimed to categorize interactions along two dimensions using natural language processing (NLP): content and subjects being discussed (e.g., animals, places) and tasks being performed (e.g., generating ideas, providing feedback).

To identify the prevalent content and subjects, we focused on nouns as they frequently capture core discussion topics [20]. The text data was tokenized, and part-of-speech tagging was applied using the spaCy library [17], enabling efficient and accurate extraction of nouns. As a second approach, we employed a simple heuristic focusing on capitalized words (Nouns are capitalized in German). We found that the second approach yielded better outcomes meaning fewer words misclassified as nouns. Therefore, we relied on it. Subsequently, we identified the 400 most common nouns using frequency analysis. The frequency of the 400 chosen words varied between 3200 to 140. We deemed occurrences of 140 sufficient to cover key themes without getting lost in spurious occurrences of nouns.

To identify tasks, we focused on the initial prompt of a conversation. That is, the initial prompt typically captures the task the AI should perform. We first removed duplicates leaving 1014 conversations out of 1339. Duplicates commonly emerge from copy&paste. As GenAI's responses are often not identical for the same prompt, copy&paste can be viable strategy to obtain different responses if the initial one is not satisfactory. Furthermore, students are also inclined to copy&paste task descriptions directly into ChatGPT. As some prompts were extremely long - for example, containing entire essays for which students sought feedback - we kept only the first 100 characters of the first prompt in each conversation for analysis.

As a next step, we identified topics and themes within these data. Scientific approaches can roughly be split into computational and manual approaches. More modern computational approaches for topic modeling increasingly rely on LLMs [38,39,45], while older, less powerful models often rely on pure statistical models [7]. We found that the native use of state-of-the-art LLMs (we used ChatGPT 4.5 and the state-of-the-art reasoning model o3-pro) gave better outcomes than classical approaches, we have employed. In particular, most specialized models like [7,39] struggle to produce hierarchical categorizations. Instead, they tend to produce a large number of topics, some of which lack meaningful coherence and require manual processing, as acknowledged in peer-reviewed methodology papers on topic modeling [10] and publications in leading academic journals [14]. The overarching approach is illustrated in Fig. 2. It relies on data pre-processing and prompting often comprising of elements such as desired hierarchy of topics, semantic guidance on desired topics, and output format. Manual refinement might be needed, but we found topics to be of much higher quality and much easier to interpret than for classical approaches. We elaborate more in the discussion section.

3 Findings

We present two complementary classification dimensions: content categories and task categories. Content categories are often associated with educational subjects (e.g., biology, geography), while task categories capture more general forms of learning support that span across subjects (e.g., explanation, feedback, creative content generation). However, we acknowledge that these dimensions are not entirely independent, as specific content areas often shape the nature and framing of tasks.

3.1 Content Dimension

Major themes are shown in Table 1. We elaborate a bit more on them and provide an exemplary prompt for each category.

Education and Learning: This category focuses on structured environments and methods through which individuals gain knowledge, develop skills, and enhance their understanding of various subjects. As such they are not a typical "content" category but relate more to tasks and activities. Educational activities span formal schooling, language development, critical thinking, and effective communication. The subcategory **School & Classes** includes the interaction between students and teachers, classroom dynamics, and various educational stages, such as high school or apprenticeships, exemplified by the prompt *"Create an advertising song for [name of school], a secondary school with nice teachers"*. **Writing & Language** emphasizes clear expression, proper grammar, spelling, and effective composition of texts. It is demonstrated by prompts like *"Write me a novel with 13 chapters about a girl named Kyra, adventures, living in [...]"*. **Knowledge & Information** involves seeking answers, explanations, and details to enhance comprehension and mastery of topics, illustrated by *"What volcanoes are there in Europe?"*. Lastly, **Presentation & Communication** highlights delivering

Table 1. Categorization of Interaction Data: Major Themes and Associated Keywords.

Topic/Subtopic	Words
1. Education & Learning	
School & Classes	Student, Teacher, Class, High School, Apprenticeship
Writing & Language	Text, Words, Sentences, Grammar, Spelling, Language, Sentence, Introduction, Conclusion, Main part
Knowledge & Information	Information, Questions, Answers, Explanation, Understanding, Details, Topic, Title
Presentation & Communication	Slide, Presentation, Prompt, Report, Description, Content
2. Nature & Environment	
Flora & Fauna	Cat, Dog, Animals, Plants, Birds, Eagle, Owl, Eagle owl, Squirrel, Wildlife park, Flowers, Trees
Landscape & Locations	Forest, Village, City, Park, Glacier, Sea, Earth, Sky, Mars, Moon, Stars, Region, Atmosphere, Environment
Weather & Natural Phenomena	Snow, Water, Rain, Ice, Tsunamis, Weather, Snowflakes, Climate
Ecology & Sustainability	Nutrition, Fertilizer, Protection, Environment, Ground, Disposal, Bacteria, Health, Skin cancer, Immune system
3. Family & Relationships	
Family & People	Family, Father, Children, Girl, Boy, Friends, Friendship, Relationships, Person, Man, Woman
Celebrations & Holidays	Christmas, Christmas Story, Santa Claus, Christmas tree, Christmas time, Gifts, Reindeer, Merry
Community & Social Interaction	Community, Team, Teamwork, Collaboration, Engagement, Appreciation, Respect, Compliment
4. Arts & Entertainment	
Literature & Storytelling	Story, Chapter, Adventure, Narrator, Characters, Plot, Book, Stories, Poem
Film & Performance	Scene, Film, Screenplay, Actors, Role, Expressions, Mood, Impressions
Games & Activities	Gaming, Minigolf, Quiz, Riddle, Fun, Activities, Game, Exercises
5. Career & Professional Development	
Careers & Jobs	Career choice, Professions, Profession, Work, Tasks, Responsibility, Company, Apprenticeship, Skills
Personal Growth	Goals, Development, Strengths, Abilities, Interests, Experiences, Career Exhibition, Recognition, Feedback
Education & Guidance	Tips, Opportunities, Decision, Steps, Target group, Presentation, Information, Future, Questions
6. Health & Wellbeing	
Body & Health	Skin, Body, Acne, Symptoms, Treatment, Health, Immune system, Protection, Causes
Feelings & Emotions	Feelings, Joy, Love, Fun, Smile, Mood, Courage, Patience, Attitude
Lifestyle & Activities	Sports, Movement, Climbing tower, Activities, Energy, Relaxation, Adventure, Interests, Freedom
7. Culture & Society	
Cultural Traditions	Traditions, Culture, Community, Events, Art, Music, Style, Tattoos
Geography & Places	Liechtenstein, Vaduz, Buchs, Gutenberg, Malbun, Eschen, Europe, Country
Historical & Miscellaneous	Mayans, Armstrong, Leonardo, Neil, Christmas traditions, Secret, Proximity, Location

information effectively to audiences through slides, speeches, or reports, as in *"I need a presentation about John Williams"*.

Nature and Environment: This category covers all aspects of the natural world, ecosystems, biodiversity, and the environment's interactions with human life. It promotes awareness of ecological balance, the importance of conservation, and sustainability practices. **Flora & Fauna** addresses the diversity of plant and animal life and their interactions within ecosystems, often in a playful manner, illustrated by prompts such as *"Tell me a story about a squirrel named Chico who gets kidnapped"*. The subcategory **Landscape & Locations** explores geographical areas, natural landmarks, and outer space environments, exemplified by *"Give me a small text for AI images with the perfect space picture"*. **Weather & Natural Phenomena** investigates climatic conditions, seasonal variations, and extreme events like tsunamis, demonstrated by the prompt given without any additional context *"Snow, Christmas, ice"*. Lastly, **Ecology & Sustainability** emphasizes environmental stewardship, resource conservation, and health impacts, with examples like *"Bacteria and diverse living organisms"*.

Family and Relationships: This domain highlights human connections, emotional bonds, and social dynamics. It includes personal relationships within family structures, friendship interactions, and community engagement. **Family & People** explores the bonds between family members, relatives, and close friends, illustrated by the prompt *"Write me a novel for 12-year-old girls from the perspective of a 13-year-old girl"*. **Celebrations & Holidays** captures traditional festivities and cultural practices that bring communities and families together, exemplified by *"Write a Christmas story using candy cane, gingerbread, and elf"*. **Community & Social Interaction** examines collaboration, appreciation, teamwork, and respect within communities or groups, demonstrated through *"Make a compliment for [name]. She is thoughtful, cheerful, and helpful"*.

Arts and Entertainment: This category encompasses various forms of creative expression, storytelling, performance arts, and leisure activities. It supports artistic exploration and provides entertainment and educational value through creative endeavors. **Literature & Storytelling** includes creative writing, narrative construction, and character development, as seen in the prompt *"Write me a story where Chico gives Maria a gift"*. **Film & Performance** covers cinematic arts, acting roles, and screenplay creation, exemplified by *"Create a screenplay for a stop-motion film about the first moon landing"*. Finally, **Games & Activities** encompasses interactive, recreational, and playful activities such as quizzes and riddles, illustrated by *"Make me a very short, exciting Christmas-themed story"*.

Career and Professional Development: This section focuses on the evolution of personal skills, career progression, and professional capabilities. It provides insights into job roles, career decisions, and continuous personal growth within various professional contexts. **Careers & Jobs** explores the specific tasks, responsibilities, and professional skills required across different industries, exemplified by the prompt *"Write a job description for a hairdresser"*. **Personal Growth** highlights the importance of setting goals, recognizing individual strengths, and professional development, demonstrated by the prompt *"I need your help with percent calculations"*. **Education & Guidance**

delivers actionable advice and informative guidance to assist individuals in educational choices and career planning, illustrated by *"Explain the simple past to me"*.

Health and Wellbeing: This category emphasizes maintaining physical health, emotional wellness, and adopting healthy lifestyle practices. It covers the connection between bodily health, emotional states, and daily activities promoting well-being. **Body & Health** investigates physical conditions, health symptoms, treatments, and preventive measures, exemplified by the prompt *"Hello Dr. Skin, could you explain the top 10 things about acne in 10 sentences?"*. **Feelings & Emotions** explores the spectrum of emotional experiences, highlighting how emotional intelligence affects quality of life, demonstrated by *"You are very helpful and supportive. When I lose motivation in school life, then…"*. **Lifestyle & Activities** highlights recreational activities and active lifestyle choices, illustrated through *"Write about leisure activities in Liechtenstein, especially sports"*.

Culture and Society: This section deals with societal behaviors, cultural expressions, and historical and geographical contexts. It provides insights into traditions, community activities, and shared heritage and knowledge. **Cultural Traditions** explores collective celebrations, artistic and musical heritage, exemplified by prompts such as *"Make me a summary about ticks containing all important information"*. **Geography & Places** emphasizes understanding locations, regional identities, and geographical specifics, demonstrated by *"Montenegro: An introduction"*. Lastly, **Historical & Miscellaneous** examines significant historical figures, events, and intriguing cultural details, illustrated through prompts like *"Who were the Mayas?"*.

Discussion: Despite the rich body of work on GenAI in education, only few of our categories have been discussed in depth. Most notably, "Education & learning" as well as "Arts & Entertainment" overlaps with well-studied areas. Other categories like "Family & Relationships" appear as natural choices as they are common themes for minors. For instance, Westby et al. [47] identified recurring narrative topics among children from diverse cultures, such as family, friends, school experiences, and animals. Even more, leveraging GenAI for career, personal, and health topics points to interesting areas that received little attention. [1] demonstrated how high school students leveraged generative AI to visualize future aspirations aligning with our discussion on potential future professions of students, but lacking the nature of more tangible guidance found in our study. Moreover, nature, animals, and fictional creatures consistently emerge as popular subjects in minors' interactions with generative AI tools [1,36]. However, topics like health or career can be highly influential and impactful on minors and highlight the responsibility needed by educators to ensure AI literacy among students. While prior work as mostly attested adequate phrasing of ChatGPT for different age groups [40], the risk of hallucinations and toxic responses remains though thanks to advancement in technology the risk of toxic responses as significantly decreased.

3.2 Task Dimension

Tables 2 and 3 shows identified tasks. Many of our high-level task categories align with existing literature, although much of this prior work is not specifically focused on K-12 education. For example, [13] explored poetry and GenAI more boradly, while [27]

Table 2. Task categories 1–4.

Main/sub Task categories	Example
1. Creative Content Generation	
Poetry Writing	"Write a poetic rhyming poem for 5th grade about a cat."
Storytelling	"Tell me a story about a squirrel named Chico who gets kidnapped."
Songwriting	"Help me create a Christmas song."
Novel-length Narratives	"Write me a novel with 13 chapters about a girl named Kyra, adventures, living in the jungle."
Scriptwriting for Stop-motion	"Create a screenplay for a stop-motion film about Amerigo Vespucci."
Rap and Lyric Creation	"Rap ideas."
2. Language Learning & Grammar Support	
Grammar Explanations	"Explain the simple past to me."
Exam-style Templates	"I need to talk about my holidays (past tense) in an English exam; please give me a template."
Style Adaptation for Audience	"You're a writing helper for an 11-year-old student. Write an introduction in her style."
Exercise Generation	"Make me an English exercise with past simple."
Quiz Creation	"Do a quiz about past simple and present perfect."
Text Simplification & Improvement	"Can you simplify and improve this text?"
3. Lesson & Curriculum Planning	
Lesson Plan Design	"Prepare me a lesson plan for introducing the French language."
Unit or Module Creation	"Create a teaching unit about dogs for first graders."
Slide-deck / Presentation Content	"Make a presentation about tsunamis."
Project or Field-trip Planning	"Hi. You are an expert for tourism in the Alps. Plan the world's best alpine tourist resort."
Audio Lesson Scripting	"Make an easy 2-minute English podcast about Julius Caesar."
Resource Summaries	"Make me a summary about ticks containing all important information."
4. Assessment & Feedback Support	
Text Correction & Proofreading	"Hello Paule. Please correct my report: The great hiking day on 19.9.2024 Oberschule Eschen."
Poem Evaluation	"I'm a 5th grader and wrote a poem; what do you think of it? Once there was a dog, he was..."
Letter Correction & Critique	"I had to write a letter to a Holocaust victim; can you correct and evaluate my letter?"
Structured Report Review	"Check my text for appropriate title, introduction, main part, conclusion (min. 150 words)."
Peer-style Feedback on Creative Work	"Evaluate this poem by a fifth grader: Once there was a little elf who saw a..."

examined writing support in adolescents, and [33] focused specifically on creative writing. In a related domain, [16] investigated the use of AI in music education, which connects to our identified category of song-writing tasks. In the category "Language Learning & Grammar Support", the idea of style adaptation for different audiences

Table 3. Task categories 5–8.

5. Reflective & Experiential Writing	
Guided Report Drafting	"Write me a report about the introductory camp; please help me."
Technical Skills Reflection	"Write me a reflection on my trial days as a heating installer; I used an angle grinder."
Event Narration	"On August 22, 2024, we had a swimming day. We rode bikes. The whole first class..."
Vocational Trial Reflection	"Write me a reflection on my trial as a gardener. I swept leaves and was on the..."
6. Factual & Explanatory Support	
Geography & Fact Lookup	"What volcanoes are there in Europe?"
Science Concept Explanations	"Why is the blue whale the largest animal in the world?"
Technology Definitions	"What is a storage drive?"
Chemical/Physical Concept Breakdown	"Explain homogeneous and heterogeneous mixtures."
Methodology Introductions	"Can you briefly explain what a needs analysis and a situation analysis are?"
7. Visual & Multimedia Content Generation	
Direct Drawing Requests	"Draw Colin for me."
Image-AI Prompt Writing	"Give me a small text for AI images with the perfect space picture."
Complex Scene Imagery	"Create a picture of a farm with as many animal families as possible."
Animal Portrait Generation	"Make me a picture of a light-brown Labradoodle and a small white Maltipoo both."
Animation/Storyboard Scripts	"Create a screenplay for a stop-motion film about the first moon landing."
8. Motivational & Emotional Support	
Personalized Compliments	"Create a compliment for a woman with these qualities: gentle, lovable, helpful."
Team & Peer Praise	"Provide compliments I can give my new colleagues in education."
Motivational Prompts	"You are very helpful and supportive. When I lose motivation in school life, then..."
Creative Encouragement	"Write me a compliment for Lena; she's funny, caring, and very creative."

within the context of text generation remains relatively underexplored, despite being well-established in the prompting literature [40]. Additionally, grammar explanations appear to be an emerging area of interest [23], while question generation has been more extensively studied, including in-depth evaluations of generated questions [5]. "Lesson & Curriculum Planning" has been studied [37], but other areas like audio lesson scripting or field-trip planning have received little attention in an educational context, though the existences of AI-generated podcasts has been noted. Such applications are interesting as students frequently listen to audio content in various everyday contexts and might prefer listening to podcasts while being on "the move" rather than reading sitting at a desk. "Reflective & experiential writing", while commonly practiced in educational settings, has received limited attention in AI-related research literature. While our (exemplary) prompts point more towards a standard interaction with GenAI, i.e.,

where a student asks to get help in generating a reflective piece of text, one might envision more specialized tutoring systems that enforce guided self-reflection rather than simply text being generated containing a number of made-up reflections. While the suitability of early ChatGPT versions and smaller models (like 3.5 and 4o-mini) for factual support has been questioned in countless studies due to their tendency for hallucinations, newer models - such as GPT-4o with integrated web search - are known to exhibit significantly reduced hallucination rates [28]. However, caution is still warranted, as hallucinations remain prevalent. Nonetheless, the use of GenAI for generating explanations is common, even for works specialized domains such as chemistry [25]. The area of "Visual & multimedia content generation" is still in its infancy, with only a few studies examining its application in an educational setting, e.g., [11]. However, one might argue that over time, the presentation of information through visual elements has become increasingly relevant and frequent - particularly as most modern web content integrates images, videos, or interactive elements rather than relying solely on text. As such, fostering visual content generation skills early on appears to be a promising direction for pedagogy. Tasking GenAI with "Motivational & Emotional Support" is a promising direction for both students and teachers. For once, GenAI has excelled at writing all kinds of text, including encouraging and supporting messages. Moreover, offering compliments or emotional reinforcement may not come naturally to everyone, yet these skills can improve with practice (and GenAI support). Importantly, the positive psychological impact of well-delivered compliments is often underestimated, suggesting that GenAI could play a valuable role in fostering more emotionally supportive learning environments [49].

4 Related Work

GenAI in K-12 Education: Promise and Early Trials. The advent of generative AI has sparked both enthusiasm and concern in K-12 education. Advocates have touted GenAI's potential to personalize learning and reduce teacher workload [31]. At the same time, many educators initially feared tools like ChatGPT might become "cheatbots" that allow students to bypass learning fundamental skills (e.g., writing) or spread misinformation [24]. Early responses in schools ranged from outright bans to cautious, exploratory integration, highlighting ambivalence about GenAI's role in the classroom [18]. Amid these debates, recent empirical work has begun to explore how generative AI systems—even at the KâĂŞ12 level—are being used in real educational settings, drawing on actual interaction data to illuminate emerging usage patterns.

Personalized Learning Support. Jauhiainen and Guerra report a classroom study with 110 pupils (ages 8âĂŞ14) in Uruguay, where ChatGPT & Midjourney were used to dynamically personalize science lessons (*texts, illustrations, exercises*) in real time, leading to improved engagement and learning outcomes [19]. Almohesh conducted a quasi-experimental study involving 250 Saudi primary students, showing that ChatGPT-based assistant use significantly enhanced learner autonomy in live online classes [2].

Ng et al. implemented an "SRL-bot" powered by ChatGPT during a secondary science course, reporting gains in subject knowledge, motivation, and self-regulation among 74 tenth-graders [32].

Writing Feedback and Scaffolding. Fokides and Peristeraki compared ChatGPT feedback on primary-aged students' English and Greek essays against teacher comments: ChatGPT produced more accurate and detailed feedback in English but was less reliable in Greek [12]. Levine et al. observed high school students using ChatGPT for brainstorming, drafting, and revision; AI suggestions helped students clarify their own voice and engage in deeper reflection [26]. Zheldibayeva et al. reported on 11th graders using the CGScholar AI Helper, finding that AI feedback supported writing development and elicited improvement suggestions from students and their teacher [50].

Critical Thinking and Metacognitive Gains. Bitzenbauer ran pilot physics lessons where students evaluated and improved ChatGPT answers; this increased critical thinking and positive AI attitudes [6]. Klar conducted a mixed-methods study of KâĂŞ12 students' perceptions and interaction patterns with GenAI chatbots, highlighting the difficulty of using ChatGPT effectively and the need for guided scaffolding [22].

STEM lLearning Outcomes. Alneyadi and Wardat deployed ChatGPT in 11th-grade physics instruction over four weeks, finding substantially higher learning gains and positive student perceptions [3]. Kim et al. investigated ChatGPT's role during asynchronous collaborative brainstorming sessions, noting shifts in group interaction patterns [21].

Negative Outcomes and self-efficacy Concerns. Yang et al. studied a high-school programming course and found ChatGPT support led to lower engagement, self-efficacy, and achievement, suggesting AI may inhibit deep coding skills when misused [48].

Topic Modeling Topic modeling builds on a long tradition in machine learning, with early methods such as Probabilistic Latent Semantic Analysis (PLSA) emerging as early as the late 1990ies [15]. A major advancement came with the introduction of Latent Dirichlet Allocation (LDA) [7], which extended PLSA by incorporating a Bayesian framework. While LDA was initially developed within computer science, its adoption across disciplines - including social sciences and education - gained momentum around 2016 [10]. It has remained popular ever since. However, it is known that LLMs embedded in more or less complex algorithms excel for topic modeling approaches [38,39,45]. However, to best of our knowledge, few works have investigated using

prompts directly to also guide the types of desired topics, e.g., in our case content and tasks, and generating hierarchical topic structures. That is, prior computational approaches such as seeded LDA [30] (allowing to provide "seed words" for topics) and hierarchical LDA [44] were difficult to use and yielded inferior outcomes sharing shortcoming of the basic LDA method [7].

5 Discussion

Drawing on an innovative methodological approach, our analysis of real-world interactions between minors and generative AI (GenAI) in educational settings reveals several critical opportunities and challenges that warrant attention from educators, policymakers, and researchers alike.

Our hierarchical content categorization reveals that minors frequently engage with GenAI across a wide range of themes, with particularly high activity in Education & Learning, Nature & Environment, and Arts & Entertainment. These areas align well with conventional curricula, underscoring GenAI's potential to effectively complement structured educational activities.

However, our findings also identify substantial engagement in less-studied domains like Career & Professional Development, Family & Relationships, and Health & Well-being. These insights suggest opportunities for curricular innovation, particularly in supporting students' personal growth, mental health, and vocational guidance—areas that have traditionally received limited attention in formal education.

Yet, these emerging applications raise sensitive ethical concerns that should be mitigated. The situation is complex: One the one hand, teachers should respect students' privacy; on the other hand, they may also be held (partially) accountable if GenAI tools are misused or provide inadequate responses causing harm. The potential for toxic outputs from LLMs remains a significant concern [46].

One partial mitigation strategy that seeks to balance privacy with the need for appropriate oversight is the clear separation of school-related and non-school-related usage, combined with robust anonymization protocols relying both on automatic approaches but also careful manual inspection before any publication takes place – an approach we adopted in our study. Under such a policy, students are made aware that attempts to misuse GenAI (e.g., for generating toxic content) are likely to be detected – even though they might not be attributable to them personally to privacy safeguards. This approach fosters accountability while maintaining student trust and data protection.

From an academic perspective (both reproducibility and follow up works) providing the dataset would be preferable, but a latent risk due to sophisticated denanymization techniques excludes such an option. Our dataset covers a number of different schools, but all are within the same country and same school authority. As such, similar studies in other countries are of interest which might uncover additional use-cases.

Our task dimension categorization illustrates the multifaceted utility of GenAI in classroom contexts. Tasks such as creative content generation, grammar support, and factual explanation align with previous studies emphasizing GenAI's strength in supporting routine academic tasks. However, our analysis also identifies emerging and less-explored use cases, including reflective writing, multimedia content creation, and moti-

vational support. These applications reveal untapped potential that merits further empirical investigation and thoughtful integration into educational practice. For example, reflective writing tasks supported by GenAI could meaningfully enhance experiential learning - provided it is carefully scaffolded to encourage authentic self-reflection and to avoid the generation of superficial or fabricated responses. with appropriate instructional design and oversight, such use cases could expand GenAI's role beyond traditional academic tasks to foster deeper cognitive, emotional, and metacognitive engagement in students.

Moreover, the qualitative aspects of interaction underscore the importance of nuanced pedagogical strategies. Students' frequent engagement in creative tasks highlights GenAI's potential in fostering creativity and imaginative thinking. Yet, reliance on GenAI for critical tasks, such as factual explanations or assessment feedback, poses significant risks of misinformation due to AI hallucinations, emphasizing the need for robust digital literacy and critical evaluation skills among students.

Our findings also accentuate equity and ethical considerations. Given that interactions occurred exclusively within school hours, it remains unclear how disparities in out-of-school access to GenAI tools may exacerbate educational inequalities. Moreover, minors' substantial use of GenAI for sensitive personal topics, such as health or emotional well-being, raises ethical concerns regarding data privacy and the appropriateness of AI-generated advice, underscoring the urgency for clear guidelines and responsible usage protocols.

Finally, our study contributes methodologically by suggesting a novel, simple approach to topic modeling. Essentially, topic modeling is yet another task where LLMs excel and eliminate specialized algorithms. The benefits of using powerful LLMs are striking as they generate high quality topics in a hierarchical manner allowing for easy semantic guidance on what topics are of interest.

However, despite the fact that modern LLMs have very large context windows and can process large amounts of documents - e.g. through retrieval-augmented generation (RAG) - we found that our data reduction strategy substantially improved outcomes or, put differently, using LLMs for topic modeling on large text collections is challenging. This aligns with existing research claiming that LLMs struggle with large contexts [29]. Specifically, we improved outcomes by (i) focusing on nouns for content categorization and (ii) limiting analysis to the first prompt in each conversation led to more coherent and interpretable results. While this also holds for classical topic modeling, where, for example, stop words are removed to improve topic coherence [10], it is important to acknowledge that unnecessary or irrelevant information can be misleading for LLMs as well. However, further research employing also quantitative metrics for assessment and further datasets are helpful to further evidence for our approach. Also in our analysis providing coding reliability metrics would enhance scientific rigor.

6 Conclusions

In conclusion, our comprehensive analysis of over 17,000 anonymized interactions between minors and generative AI in real-world K-12 educational settings offers critical insights that bridge the gap between theoretical potential and practical classroom

implementation. By systematically categorizing these interactions across both content and task dimensions, our findings provide educators with a nuanced understanding of student interests, learning needs, and emerging use cases for GenAI. This two dimensional framework not only informs pedagogical practice but also lays the groundwork for future research and innovation in the responsible integration of GenAI into K-12 education. For example, future research might further partition K-12 into different age groups.

Our content categorization reveals a broad spectrum of student interests, ranging from traditional educational subjects such as language learning and natural sciences to more personal and exploratory domains like relationships, health, and career aspirations. This diversity underscores the potential of generative AI to support not only academic learning but also personal development and social-emotional growth.

From a task perspective, our analysis highlights the multifaceted role generative AI can play in educational settings—from offering creative and motivational support to facilitating reflective writing and structured feedback. Particularly noteworthy is the potential of generative AI to enhance creativity, reflective thinking, and emotional well-being, areas that are often insufficiently addressed by conventional educational tools.

However, these opportunities are accompanied by important challenges and open questions. The increasing reliance on generative AI demands increased attention to AI literacy and critical thinking skills among students, particularly in light of potential inaccuracies, biases, and ethical risks. Thus, future research must address these issues, including the long-term educational implications, development of ethical guidelines, and the assurance of equitable access to generative AI tools in schools.

Overall, our study contributes to an evidence-based framework for integrating generative AI responsibly and effectively into educational practice, benefiting students, teachers, curriculum designers, and policymakers alike.

Acknowledgments. We used Generative AI, in particular, OpenAI's premium models [34,35]. Aside from basic writing support (fixing grammar error, polishing writing), the models were used as a novel tool for topic modeling replacing older computational approaches as detailed in the methodology section.

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Immersive Training of Artillery Observers with Integrated Artificial Intelligence

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Abstract. Military education and training face a fundamental paradox, the demands for operational readiness are continuously increasing, while realistic training is constrained by safety, logistical, financial, and ethical considerations. In the context of artillery operations, particularly for Forward Observers (FO) and Joint Fires Observers (JFO), the ability to coordinate fire effectively and accurately is crucial for mission success and minimizing collateral damage. One of the new promising approaches is the use of Virtual Reality (VR) and Augmented Reality (AR) to train decision-making and tactical skills in a simulated environment. This paper discusses the development of a simulator that represents a significant advancement over previous generations of artillery trainers. Its main innovation lies in the integration of Artificial Intelligence (AI), which evaluates user performance in real time. This approach may contribute to increased training effectiveness and the personalization of performance assessment. The aim is to move closer to the vision of individualized, performance-oriented training that replaces universal and rigid instructional models—without endangering personnel safety, requiring the deployment of forces and assets in the field, or consuming material resources. Preliminary results confirm that AI-supported VR training can effectively reinforce observer skills while reducing dependence on live-fire exercises.

Keywords: Artillery · Virtual Reality · Augmented Reality · Artificial · Intelligence · Education · Training

1 Introduction

Modern artillery operations increasingly rely on the precision and speed of fire support coordination, in which Forward Observers (FO) and Joint Fires Observers (JFO) play a crucial role. This trend reflects a broader revitalization of field artillery in response to

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modern threats, where indirect fire is increasingly seen as a viable means of countering advanced enemy air defences [1]. These specialists are responsible for target identification and directing indirect fire from artillery or other joint fires assets. Their training traditionally involves demanding field exercises with live or practice ammunition, practicing standardized radio communication, and close coordination with artillery units. However, such training is limited by high costs, restricted access to training areas, equipment availability, and strict safety regulations [2] .These limitations are compounded by the increasing complexity and mobility of modern artillery systems, which require more advanced and adaptive training environments to meet current operational demands [3]. Additionally, armed forces often face personnel shortages, which makes it difficult to ensure that training conditions are fully adequate.

In recent years, military forces have therefore sought innovative solutions to overcome these limitations while improving the availability, realism, and efficiency of training. Virtual Reality (VR) and Augmented Reality (AR) have emerged as promising technologies that enable immersive, repeatable, and resource-efficient training of complex combat scenarios. With these technologies, FO and JFO personnel can practice key tasks such as target acquisition, fire requests, and fire corrections in environments that realistically simulate the stress and conditions of real battlefields—without the need for live ammunition or large training ranges [4].

In the academic literature and scientific articles, this issue is not widely covered in the public domain, as manufacturers and contractors retain key know-how, limiting the availability of detailed information on the specific applications of these technologies [5]. This situation is particularly evident in specialized areas such as artillery, where research and implementation of simulation technologies is minimally described. This complicates the comparison of training approaches between different armed forces and the identification of best practices for the implementation of these technologies in the field of fire support, fire control, and artillery reconnaissance [6, 7].

Wei Yang et al. (2023) focus on the use of Virtual Reality (VR) as a tool for assessing the effectiveness of naval artillery fire. The study presents a methodology using VR to simulate combat scenarios, analyze projectile trajectories, and evaluate the impact of hits on targets. The authors identify key factors influencing fire effectiveness, such as impact angle, type of ammunition, and the structural resilience of the target, and model these factors in a simulated environment. The study highlights that VR not only allows real-time visualization of firing effects but also enables repeated analysis of outcomes without the need for live ammunition. The main benefits of this method include enhanced safety, reduced testing costs, and the ability to test a wide range of scenarios without physically damaging equipment. The article concludes that VR is an effective tool not only for assessing fire effectiveness but also for operator training and optimizing combat tactics, and it recommends further expansion of the research to include elements of Artificial Intelligence (AI) and Augmented Reality (AR) for even more accurate modeling of combat situations [8].

Mehmet Uğur Yavuz et al. (2019) present a concept focused on the use of Augmented Reality (AR) in tabletop artillery simulation, which utilizes smart AR glasses to project a simulated battlefield into the user's real-world environment. The main goal of the project was to create an interactive ballistic simulation in which the player controls a howitzer

and targets enemies in an environment simulated using AR technology. Key features of the simulation include the ability to select different types of ammunition, realistic projectile trajectories, and the use of QR codes for surface tracking and environmental orientation. The authors focused on the technological aspects of AR, including the use of real terrain elevation maps, methods to ensure precise object placement in space, and challenges of real-time image tracking. The game was developed in Unity3D and tested on various devices, though some performance issues on smart AR glasses led to the need for simplified object models. In conclusion, the authors state that AR can be an effective technology for military simulations, allowing the combination of team-based decision-making with the computational power of modern devices. However, they emphasize the need for further development, particularly in expanding the types of targets, improving destruction effects, and integrating Artificial Intelligence (AI) to generate hostile actions. They also suggest improving the AR mechanism, for example by using larger visual targets, which could increase the realism of the simulation. Overall, the work confirms the potential of AR not only in the gaming industry but also in broader applications for military training and tactical simulations [9].

The study by Romullo Girardi & Jauvane C. de Oliveira (2019), focused on the use of Virtual Reality (VR) in the training of artillery observers in the Brazilian Army, confirmed that VR technologies represent an effective and realistic alternative to traditional training, which is often costly and space-constrained. The SAOA simulator enabled officers to practice key skills—such as target identification, fire coordination, and strike assessment—in a safe and repeatable environment. Testing on a sample of 13 experienced artillery officers showed significant improvement in training efficiency: the average time required to conduct a fire mission decreased by 11.3% for static targets and by 26.7% for dynamic targets, while hit accuracy increased by 66.7% and 37.1%, respectively. The sense of presence in the virtual environment exceeded benchmark values from similar studies, confirming the realism of the simulation and its positive impact on soldiers' decision-making processes. The authors recommend further expansion of the simulator, including the deployment of more advanced VR headsets such as Oculus Rift and HTC Vive, and the development of a modular VR training framework that could be applied to other areas of military preparation. This research clearly confirms the high potential of VR technologies for modernizing military training and their benefits for artillery operations in contemporary combat environments [10].

A current challenge in the implementation of these technologies is their high acquisition cost and the need to build complex accompanying infrastructure that enables their effective use in training. For this reason, this paper focuses on technologies based on commercially available solutions that simplify the integration process of these systems while also allowing the expansion of training to a larger number of personnel, thus increasing both its effectiveness and accessibility.

2 Methodology

The development of the concept for a comprehensive training simulator was initiated based on an analysis of standard call for fire procedures and the specific training requirements of Forward Observers (FO) and Joint Fires Observers (JFO). In cooperation with

artillery experts, we proposed an initial system model that defines the key entities, their roles, and functions within the training process (see below). Based on this model, we developed the architecture of a core module that includes components for virtual and mixed reality, as well as artificial intelligence (AI) elements. The initial development phase, conducted in collaboration with the company YORD s.r.o., followed an iterative process. The first version included a basic VR environment for target detection and recognition. In the current version, we are working to implement functionalities for training the full call for fire procedure using voice interaction, and additional advanced features—such as behavior analysis, adaptive difficulty, and AI-based performance evaluation—are planned to be gradually integrated. Intermediate versions are tested and refined in cooperation with subject matter experts and potential users, allowing further adaptation and evolution of the concept. The simulation architecture is informed by real-world artillery operations, including recent experiences from Ukraine, where field artillery played a decisive role across various phases of the conflict—from the defense of Kyiv to attrition warfare in the Donbas. These insights highlight the need for training environments that realistically reflect rapid tactical shifts, adaptive fire support coordination, and the quality-over-quantity paradigm in modern indirect fire employment [11].

This trend is also reflected in the Czech Armed Forces, where doctrinal updates and field experiments have confirmed the need for structured contingency and emergency firing data procedures, especially under denied or degraded operational conditions [12].

A central element of the system design is the integration of a comprehensive data collection framework to capture user interaction with the simulator. Each exercise is intended to be accompanied by real-time recording of motion data (e.g., head and hand positions, gaze direction), voice communication (audio and speech-to-text conversion), and timing data (reaction time, time to acquire the target, etc.). This data serves a dual purpose—providing immediate feedback on trainee performance through built-in algorithms and AI, and supporting long-term analysis aimed at improving training methods and scenario design. Key performance indicators (e.g., maximum allowed response time, correct message format, target deviation) and their threshold values have been defined based on military regulations, doctrinal publications, and instructor experience. These normative values form a reference model of ideal performance, against which the AI compares actual trainee behavior.

The system is primarily developed using the Unity 3D platform, which is well-suited for interactive simulations. In addition, we use Python with libraries such as TensorFlow and PyTorch for training and implementing additional technical models. For selected optimization tasks and prototype verifications during the early development phases, we also explored practical tools such as MS Excel with the Solver add-in, which have proven capable of handling non-trivial mathematical problems relevant to simulation configuration and scenario parameter tuning [13]. Motion tracking is provided by the XR headset Meta Quest 3.

An essential component of the system concept is the use of machine learning methods to support performance assessment and behavioral analysis. Selected AI modules (e.g., for procedural error detection or decision prediction) are to be trained on historical training data. To achieve this, we utilize archived recordings and scenarios from previous training events, as well as synthetically generated cases, to improve the system's

ability to accurately evaluate user performance. Model development follows an iterative process, with emphasis on reliably detecting even subtle errors or hesitations under stress. This focus stems from operational findings showing that even small numerical deviations in firing data can exceed the lethal radius of munitions, significantly reducing the effectiveness of fire missions and requiring alternative computational methods [14].

Advanced language models (LLMs – Large Language Models) have also been integrated into the system architecture to support understanding of voice commands and analysis of radio communication. To accelerate development and reduce technical complexity, we deployed a commercially available model from the GPT family via the ChatGPT interface. Using a verified pretrained model eliminates the need for extensive training of a custom model and offers the advantages of high quality and flexibility. Due to the sensitive nature of the data processed, the system is designed for offline deployment on secure local servers within a military network, with no reliance on cloud-based services.

The initial system concept is currently being validated through pilot testing with a small sample of users, including both experienced forward observers and trainees. The aim is to assess technical feasibility and gather preliminary user feedback. During these test sessions, we observe both quantitative outputs from the AI evaluation system and subjective user impressions gathered through questionnaires focused on perceived presence and system usability [15].

Each of the 13 participants completed a standardized fire mission scenario in a controlled virtual reality environment using the Meta Quest 3 headset. Performance was evaluated based on automatically collected metrics, including time to identify the target, speech fluency (measured in words per minute), message format correctness, head movement trajectory, and response latency.

3 Conceptual Design and Functional Architecture of the Training Simulator

This chapter outlines the design and technical solution of a training simulator currently under development for Forward Observers (FO) and Joint Fires Observers (JFO). The goal of the system is to create a realistic, interactive, and adaptive training environment that enables the practice of key skills related to fire support coordination. The description covers the conceptual system model, system architecture and data flows, the computing and display technologies used, principles of user interaction, voice control capabilities, and particularly the current approach to implementing training scenarios for the Call for Fire (CFF) procedure. Emphasis is placed on the modular system structure, the use of artificial intelligence for performance evaluation, and the ability to adapt training scenarios individually based on the trainee's skill level [16].

3.1 Conceptual Model of the System

The system design is based on defining the core entities and their interrelationships within the Call for Fire training framework process. The primary entity is the trainee—a soldier acting as a Forward Observer (FO) or Joint Fires Observer (JFO) participating

in a simulated mission. Within the VR/AR environment, the trainee performs all standard tasks as during live fire control: observing the area, identifying targets, reporting coordinates and fire missions, and adjusting impacts. The design of input and feedback mechanisms in the simulator follows doctrinal fire support procedures, incorporating both linear and angular variables for consistent and standardized fire direction [17, 18]. These parameters align with validated models used in Czech artillery systems such as PVNPG-14M and contribute to the interoperability of training with NATO standards [19].

Another key entity is the Virtual Fire Direction Center (FDC), represented by an AI module trained to support fire missions and serve as the command post for firing units. In parallel, the system includes a Virtual Fire Support Coordination Center (FSCC), simulating the trainee's superior headquarters. The FSCC determines which unit executes the mission and issues coordination instructions to the trainee.

A third component is the Virtual Artillery Units—an AI module simulating an artillery battery. Bidirectional communication occurs between this unit and the FDC (e.g., mission confirmation, fire timing, impact reports), enabling the trainee to understand the fire mission sequence and real-time situational development. Virtual Enemy Units form a dynamic scenario element. This AI module changes enemy positions over time or introduces new targets to create variable conditions. he simulator also integrates terrain effects resulting from artillery fire, such as shell craters, which influence vehicle mobility and force maneuvering decisions. These features are based on validated procedures for assessing terrain deformation, enabling more accurate simulation of operational mobility constraints and counter-mobility effects [20]. In the future, the module should be enhanced to simulate more autonomous enemy behavior, such as intelligent maneuvering or concealment after firing [21].

The training takes place in a Virtual Combat Environment—a 3D simulated space including terrain and objects, with variable conditions such as weather (e.g., fog, rain, snow), time of day, and visibility factors. These factors dynamically influence target identification difficulty and the overall complexity of the mission. An essential subsystem is the Tracking Module, which monitors user motion and behavioral expressions. Core functionality is provided by sensors integrated in the VR/AR headset, offering six degrees of freedom (6DoF) tracking for head and hand movement using onboard cameras and inertial sensors. For advanced applications, an external optical tracking system (e.g., OptiTrack) can be used for full-body motion capture, enabling training scenarios that involve movement, camouflage posture, or small arms handling. Tracking also includes eye-tracking—either natively embedded in the headset (e.g., HTC Vive Pro Eye, Varjo XR-4) or as an add-on module—enabling the system to record gaze direction and analyze situational awareness.

The conceptual core of the system lies in the AI Evaluation Module, which continuously monitors user behavior and analyzes performance. This subsystem comprises several interconnected modules: a speech recognition and voice command module; a tactical analysis module that evaluates decision-making and procedural accuracy (e.g., whether the correct target and fire type were selected and procedures followed); an error detection and metric evaluation module that compares measured parameters to an ideal performance reference model; and a scenario adaptation module that adjusts scenario

difficulty and parameters based on ongoing evaluation [22]. The system is also designed to generate automated performance reports for both the instructor and the trainee.

Another critical component is the User Interface and Control Devices, which connect the trainee to the virtual environment. This includes the VR headset with controllers or an MR headset for AR scenarios. The system tracks head and hand movement via integrated sensors and allows interaction with virtual objects. The voice interface (microphone and speakers/headphones) facilitates communication with AI-driven units within the simulation. Optionally, real-world replicas of operational equipment—such as a field telephone mockup, laser rangefinder, or GPS receiver—can be integrated to create a familiar interaction experience for the trainee. This design approach is similar to systems like DJFT, which incorporate physical laser designators and devices operated within a virtual setting.

Overall, the system is designed to provide the most authentic training experience possible while ensuring that complex instructional and supervisory functions are managed by the AI, thereby eliminating the need for constant human oversight.

3.2 Architecture, Data Model, and Mechanisms

The core of the concept is built around the powerful Unity 3D game engine, which runs the visual simulation environment and all real-time interactions. Unity is responsible for rendering the 3D scene to the VR headset, capturing inputs from the controllers and headset, and performing basic physics and ballistic calculations in the simulation. Unity is connected to AI subsystems that run in separate threads or processes. For implementing AI models and performance evaluation logic, we primarily use Python with TensorFlow/PyTorch libraries and interfaces to call the mentioned LLM. Communication between Unity and Python modules is handled via a defined API using event and state messaging. The application runs on a high-performance workstation (PC) with a GPU, to which the VR headset is connected via cable.

For the VR part, we selected the standalone Meta Quest 3 headset, which combines high performance with affordability and the possibility of wireless operation. Quest 3 provides sufficient resolution and refresh rate for realistic visuals and features built-in cameras for inside-out motion tracking. To achieve the highest quality of gaze tracking and interaction, high-end headsets with eye-tracking such as HTC Vive Pro Eye or Varjo XR-4, which also allows switching to AR mode using a see-through camera can be used alternatively (see Table 1). The system is designed to be modular – VR/AR headsets can be exchanged as needed; in laboratory conditions, it is possible to use an external motion-capture system for precise full-body tracking which would allow evaluation of body posture during camouflage, rifle shooting, etc., if the training includes such elements.

The architecture of data flows is illustrated in simplified form as follows. Inputs to the system include motion data from headset and controller sensors (position and orientation of the head, hands, or other body parts), voice input, and events within the simulation. All these inputs are streamed in real time to the Unity environment and then forwarded to the AI modules for analysis.

System outputs include (a) visual output to the VR/AR headset, (b) audio output, and (c) data outputs for evaluation. Real-time outputs are also provided for immediate

Table 1. Time Spent Observing Object Categories. Source: author's own.

Category	Time [s]	Percentage [%]
furniture	4.85	17.48
people_left	5.77	20.80
people_right	7.67	27.66
screen	7.17	25.84
timer	0.02	0.09
wall	2.26	8.13

feedback to the user: for example, the system may continuously display an indicator of whether the trainee is following the correct procedure. After a scenario ends, the system automatically generates a performance summary – typically heatmaps and movement trajectories, performance indicator graphs, and textual improvement suggestions (see Fig. 1. And Fig. 2.). This data is stored in a database, allowing individual progress to be tracked over time.

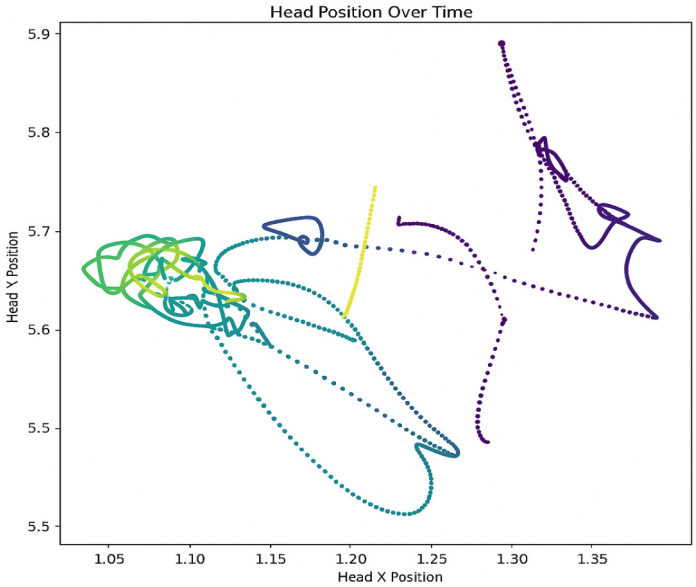

Fig. 1. Speech fluency curve (Words Per Minute – WPM) over time, based on a 30-s moving window. The system tracks speech dynamics and can detect irregularities such as hesitation, slow response, or abrupt pauses. Source: author's own.

A key component of the architecture is the set of AI algorithms acting as a "digital instructor." These algorithms continuously process incoming data and assess how the trainee is performing. Each scenario has a defined set of metrics being monitored.

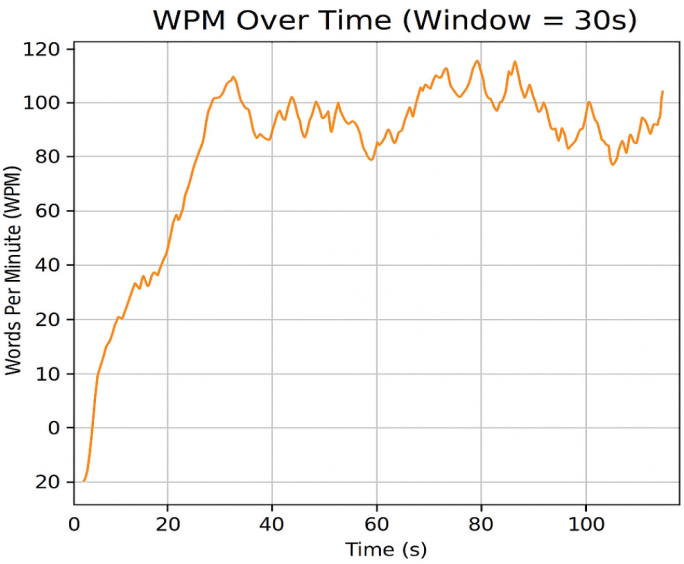

Fig. 2. Head position tracking during simulation. Color coding represents temporal sequence. The pattern shows spatial awareness, scanning behavior, and possible inattention moments. Source: author's own.

Among the advanced scenario types under development is the simulation of Multiple Round Simultaneous Impact (MRSI) missions, which require precise coordination of firing sequences and timing. This complex fire mission type, currently analyzed for its potential application in the Czech Army, presents both technical challenges and opportunities for improving FO decision-making under pressure [23]. These include reaction time, aiming accuracy and stability, fluency and correctness of voice communication, direction and extent of observation, and possibly indicators for correct decision-making or assessment. The algorithms continuously compare measured values with a reference model of ideal performance and evaluate deviations. If deficiencies are detected, they are immediately recorded and may trigger corresponding actions. In addition to error detection, the algorithms provide quantitative scores in individual areas. These partial scores are aggregated into an overall mission performance rating.

Based on performance evaluation, the system can then adapt the difficulty level. The system is designed to dynamically adjust scenario parameters according to how the trainee performs. If the AI assesses that the user is comfortably handling the current level, it may increase difficulty or shorten response times in the next scenario [24]. Conversely, if the trainee struggles significantly, the system first provides hints or corrections and lowers the complexity in the next scenario. This adaptive mechanism ensures personalized training at a pace corresponding to the current skill level of the soldier [25]. This philosophy follows modern principles of individualized, performance-oriented training and moves away from a universal, one-size-fits-all instructional model.

One of the priorities in the system's conceptual design was to enable the most natural possible interaction between the trainee and the virtual environment, to make the training

as close as possible to real-world tasks. Therefore, the following interactive mechanisms are implemented:

Communication is primarily voice-based – proper radio voice procedure is therefore an essential skill. The integration of a fully functional speech recognition feature is key for training in realistic situations. The trainee speaks into the VR headset's microphone, or into a replica of a handheld mic or field phone connected to the headset, and their speech is converted into text in real time. A specialized AI module, based on a combination of acoustic and language models, recognizes technical terms, abbreviations, and dialects typical for fire support (see Table 2).

Table 2. Speech analysis: longest pauses between words (pause $\geq$ 2 s). Source: author's own.

Rank	Start [s]	End [s]	Duration [s]
1	84.72	87.18	2.46
2	100.55	102.96	2.41
3	109.24	111.55	2.31
1	84.72	87.18	2.46

Traditional JTAC/JFO training often requires the instructor to physically observe where the trainee is looking (e.g., using binoculars to verify whether the observer is watching the correct sector). The system design automates this function using eye-tracking. In the current configuration (Meta Quest 3 without eye cameras), we track gaze direction indirectly from head orientation. This is currently sufficient for analyzing rough field-of-view coverage. However, full integration of eye-tracking would enable precise tracking of what the trainee is looking at any given moment and thereby assess their situational awareness.

4 Discussion and Conclusion

Initial experience from pilot testing confirms that a well-designed Virtual Reality (VR) simulation can, in many respects, effectively supplement or even partially replace traditional field training of Forward Observers (FO) and Joint Fires Observers (JFO), without compromising instructional quality. Instructors involved in the testing were impressed by the level of realism—particularly highlighting the degree of mental pressure and stress experienced by trainees, which closely resembled real-life operational conditions. These findings are consistent with conclusions from international studies, such as one from Brazil, where a VR-based FO simulator was found to be comparably effective to live-fire exercises.

The primary advantage of the proposed system lies in its ability to provide a standardized, safe, and economically viable training environment. Simulations allow for the repeated execution of identical scenarios, thereby generating consistent and objectively assessable performance data. Additionally, the system enables the simulation of otherwise difficult-to-replicate scenarios without the risks and costs typically associated with

live training. This includes avoiding potential environmental impacts associated with the use of live munitions in training areas, such as contamination of soil and water by heavy metals or unexploded ordnance. Previous studies have highlighted the long-term presence of pollutants, including mercury compounds, from sunken military stockpiles and industrial waste in areas such as the Baltic Sea, underscoring the importance of minimizing real-world ecological stress where possible [26]. This approach aligns with current trends within NATO armed forces, where training activities are increasingly being shifted to synthetic environments [27].

A key innovation of our approach is the integration of Artificial Intelligence (AI) as a digital instructor. AI continuously monitors the trainee's performance in real time, evaluates actions based on predefined metrics, and delivers detailed feedback. This ensures a high level of objectivity, consistency, and analytical depth—qualities that can be challenging for human instructors to maintain over time or across larger groups. At the same time, we emphasize the importance of explainability—the transparency and interpretability of AI evaluations. Instructors and trainees have the opportunity to review specific performance moments and understand the reasons for point deductions. In this respect, AI does not replace human instructors but serves as a powerful tool for more efficient debriefings and personalized feedback [28].

The proposed system is designed as a platform combining VR and an extended reality (XR) approach. While the initial phase focused on fully virtual environments, future development aims to deploy AI-supported training in conjunction with a dedicated XR module. This enhances realism and broadens the range of possible scenarios. However, current limitations include the lack of haptic feedback, potential simulator sickness among sensitive individuals, and increased visual strain. Despite these challenges, pilot test results suggest strong potential for improving training efficiency and reducing dependence on conventional live exercises. Ethical considerations remain central—AI-generated evaluation data must be protected, properly interpreted, and never misused for punitive purposes. The goal is improvement, not sanctioning. The concept responds to a global trend toward the adoption of AI-enhanced synthetic training environments [29, 30].

Data analysis showed a gradual improvement in communication fluency and procedural accuracy, as well as a measurable reduction in total mission completion time. The longest detected speech pauses ($\geq$2 s) and visual attention distribution were also analyzed to assess situational awareness. Although no statistical significance testing was conducted due to the pilot nature of the study, the results confirm the technical feasibility of the AI-supported evaluation framework and demonstrate its potential for effective skill reinforcement.

Despite the promising potential of AI-supported training, it is essential to critically reflect on certain limitations and ethical aspects. The use of artificial intelligence for performance evaluation introduces the risk of overreliance on autonomy and may lead to reduced instructor oversight or misinterpretation of trainee behavior. Transparency and explainability of AI decision-making are therefore crucial, especially in military education contexts where fairness and accountability are paramount. Ethical safeguards must ensure that evaluation outputs are used exclusively for pedagogical improvement.

Although we are currently at the concept and early testing stage, further system development is planned. Future efforts will focus on the implementation of team-based scenarios, deployment of XR modules in real-world environments, expansion of tactical scenario diversity, and validation of effectiveness through controlled studies. One promising direction includes incorporating Smart City environments into training scenarios, reflecting the trend of integrating urban infrastructures and civilian technologies into military planning. Such an approach supports the development of A2/AD strategies adapted to densely populated urban areas, where coordination between civilian systems and defense assets is critical for modern air defense and tactical superiority [31]. Full integration of AI with machine learning capabilities will be a key direction for continued development. Although the pilot results indicate positive trends, the current validation is limited by the small sample size and the absence of a control group. Future validation phases will therefore include controlled studies comparing AI-supported VR training to conventional field training or comparable simulators, with the aim of quantifying training effectiveness. At present, however, simulators using AI in this specific domain have not yet been scientifically analyzed. Confounding variables—such as prior digital skills, familiarity with voice procedures, or stress resilience—will also be systematically examined to better understand the system's generalizability. Despite these limitations, the pilot study has fulfilled its role as a proof of feasibility and provided actionable insights for further system development.

The proposed VR/XR simulator prototype with AI support represents a realistic and technologically feasible step toward the modernization of FO/JFO training. The combination of immersive environments and intelligent evaluation appears to lead to a qualitative shift in military pedagogy. If development continues toward full operational capability, such a system could become a valuable component of training not only within the Czech Armed Forces but also across the broader NATO alliance. Moving forward, it will be essential to further develop these technologies, expand their applications, and enhance interoperability with allied armed forces [32].

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Towards Defining an Intelligent System to Enhance Dentistry Experience for Children with Autism Spectrum Disorder

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Abstract. This article proposes a human-centred systemic solution to enhance the dental experience for children with autism spectrum disorders (ASD). Current dental procedures focus on desensitizing ASD patients to prevent agitation or use sedation, which reconditions their behaviour and overlooks their natural neural diversity. Additionally, communication difficulties and the need for routine make children with ASD resistant to treatment, as they are overstimulated and exposed to unfamiliar environments.

There is a need for human-centred innovation in dentistry to incorporate technology that measures patient anxiety and reassures them during treatment. This paper reviews existing literature and proposes a theoretical concept for an intelligent, human-centred system to facilitate dental visits for children with ASD. It analyses the technical feasibility of solutions aimed at reducing anxiety before, during, and after treatment, categorized into technological, spatial, and communication aspects.

The proposed solution integrates various technological elements, such as sensors, facial cameras, and adaptable room settings, to create a familiar and comfortable environment tailored to the needs of autistic children during dental visits. By addressing the entire dental visit process, the solution aims to reduce anxiety levels, improve patient experience, and provide dentists with real-time information, ultimately enhancing oral health outcomes for autistic children.

Keywords: Autism · Human Centred Design · Sensors · ICT

1 Introduction

Autism is a common neurodevelopmental disorder in childhood that is caused by atypical brain development. According to the WHO, currently, one in one

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hundred and fifty people suffer from autism spectrum disorders (ASD) [40]. This condition is characterized by the fact that those affected react hypersensitively to certain stimuli such as bright light, loud noises, intense smells, and touch. Moreover, their communication difficulties and their need to maintain a routine are also obstacles that these patients face in their daily lives [7]. Considering that an estimated eighty million people (1% of the world's population) currently suffer from this disease and that one of the main tasks they find difficult to cope with due to the above symptoms is controlling their teeth, sufferers tend to resist treatment as they are not only overstimulated but are also exposed to a new environment in which they feel alien, resulting in them not cooperating with treatment.

As a consequence, the autistic population possesses one of the highest rates of various oral diseases such as caries or gum disease [2]. Several barriers reduce their access to regular dental care and the interval between dental appointments [28]. The lack of a proper routine of dental visits can significantly affect a person's life, as the habit of having good oral health is not developed. Despite the growing recognition of the challenges faced by autistic children in dental environments, current approaches typically address mainly pre- or post-treatment phases [2, 18], leaving the actual moment of clinical interaction insufficiently supported. Only few papers propose in-clinic sensory-reducing or distraction interventions to improve the dental experience [18]. Instead, dental care often uses sedation to manage stress during the interaction [4,8]. This gap highlights the need for a more comprehensive solution that supports autistic patients throughout the entire dental care process, especially during in-clinic interactions.

To address people's needs, the current focus often lies on engaging people in value creation through human-centred design approaches to arrive at a meaningful and innovative solution for the stakeholders [1], integrating features that consider the specific needs and preferences of both the monitored individuals and the caregivers/physicians [12]. Within the dentistry experience, there is a necessity to enable human-centred innovation to bring technology into this field to measure the patient's anxiety and reassure him/her to get the desired oral treatment. Since dental procedures currently focus on desensitizing the patient to prevent agitation during the procedure [4] or sedation [8], this generates a reconditioning of their behaviour, focusing solely on modifying the natural behaviour of individuals with this neural diversity. In this way, a gap is closed that prevents this population from receiving adequate dental treatment, improving their dental health and allowing them to take control of their oral well-being. Also, the environment in which dental procedures are performed is an extremely important factor [15,36]. It is the trigger for patient rejection, as it offers a series of elements that strongly influence them, such as bright lights, sounds of drills and suction machines that are unfamiliar to them, and odours of cleaning products that may be too intense for this community. These factors, combined with the patient's communication difficulties, hinder social interaction, resulting in them not being able to understand instructions properly, understand or answer questions. This creates an overwhelming environment for the patient, increasing their

stress and anxiety levels, which ultimately prevents the procedure from being carried out. Considering all of these aforementioned issues that occur in current dental practices, doubts remain as to whether it is possible to adapt these spaces with technology to make them a more welcoming, supportive, and less overwhelming place for autistic children that focuses on reducing and controlling stress and anxiety. This change of scenery has the potential to transform dental health for this community and improve their quality of life by promoting a human-centred designed positive experience, empowering autistic patients, and enabling them to take control of their oral health.

The main contributions of this article are as follows:

- Provide a critical analysis of the existing literature to propose a theoretical concept for an intelligent system to facilitate dental visits, especially for children on the autism spectrum.
- The proposal of a human-centred systemic solution that maps technical requirements into three core categories: Technological (e.g., sensors, cameras), spatial/ environmental (e.g., lighting, sounds, scents), and communication (e.g., dentist feedback systems); discusses pros, cons, and feasibility of each technical element, grounded in the autistic user context—such as comfort, acceptance, and interference with treatment; and, finally, defines a system architecture that combines non-invasive sensors, real-time monitoring and environmental control to help reduce anxiety in children with autism when having a dental appointment.

The paper is structured as follows. Section 2 discusses previously conducted works and how attempts have been made to reduce patient anxiety at the dentist. We will focus solely on providing solutions that can be applied in the dentist's office, whether it is before, during, or after the medical consultation. Section 3 addresses the viability of the solution, along with the points that need to be addressed in the space to achieve an effective solution. In Sect. 4, the human-centred solution will be discussed, emphasizing the value proposition and providing a detailed explanation. Finally, Sect. 5 will discuss and analyze the proposed solution, generating a profound reflection on the process undertaken.

2 Related Works

In this section, previous works are analyzed focusing on those that address the following problems: generating familiarity with the process, the staff, and the environment; entertaining the patient to avoid increasing their anxiety; personalizing sensory factors to enhance comfort; and, in addition to preliminary training, knowing how to communicate effectively with the patient.

2.1 Existing Alternatives and Technologies

The analysis of the proposed solutions aimed at improving the experience of children with autism during dental appointments can be partitioned into three

Table 1. Solution Categories

Stages	Description
Pre-treatment	In this stage, the proposals focus on providing information before the appointment. Patients are provided with information about the procedure, the staff, and the space to generate familiarity, while doctors are provided with training regarding the disease and how to treat it. The fundamental objective of this phase is to prepare the patient for the operating conditions they will encounter, making them less sensitive to various situations that may arise during the treatment.
Treatment	In this stage, the proposals focus on keeping the patient in their comfort zone, avoiding overstimulation, and generating simple and easy-to-understand communication.
Post-treatment	In this stage, the proposals focus on rewarding the child's good behaviour, thereby creating positive reinforcement to establish an association between dental visits and a positive experience.

stages: pre-appointment, during the appointment, and post-appointment. The following table outlines the objectives of the solutions in each stage (Table 1).

Regarding the aforementioned categories, the literature indicates that current solutions focus on three major categories:

- **Technological** (Virtual or Augmented Reality, Apps) [3,10,13,17,23,27,32–34];
- **Spatial** (space adaptation) [9,14,16,19,38];
- **Communication** [9,21,24,26].

It is worth noting that within these macro-categories, numerous proposals present diverse approaches to the problem, often employing the same or similar methods.

Regarding the first category, the *technological* one, it is worth noting that the majority of current studies extensively focus on the use of Augmented Reality (AR) or Virtual Reality (VR) to counter anxiety.

For example, one solution recreates the clinic so that the child can visit it before the medical appointment [33,34], while another fosters familiarity with the procedure by using virtual reality to show the child how it will be performed on them [27].

In some proposals, VR is used to display games [5] or play movies [23] (which can also be played on a DVD) during the treatment; in [3], the VR environment simulates natural aquatic settings with cartoon characters and calming music.

However, technology is not limited solely to VR. For example, apps [17,34], possibly based on Augmentative and Alternative Communication (AAC) [30], are used to inform the patient about the dental experience and the procedures carried out there.

Finally, as indicated in [13], it may be useful to use both intrusive and non-intrusive sensors during both pre-treatment to desensitize the patient to contact (for example, a bracelet) and during treatment to estimate the trend of the patient's emotional state and detect their level of anxiety. Based on this information, the doctor or their support staff can intervene immediately by providing immediate support to bring the patient back into their comfort zone as soon as possible.

Regarding the *spatial* category, the majority of the reviewed literature [9,14,16,19,38] focuses on adapting the space for highly sensitive individuals, thus generating various recommendations. These recommendations include keeping lights as low as possible, as the excessive intensity can be uncomfortable for the patient, avoiding placing products with strong odours nearby, as they can be very unpleasant for patients, and minimizing sounds as much as possible. When necessary, it is important to warn the patient that a noisy instrument will be used, for example, a dental drill. However, it is important to remember that there are no predetermined parameters in this category. Every child is unique, and we need to identify both the conditions that make them feel at ease and the strategies that the doctor has to apply to bring them back to their comfort zone. A noteworthy experiment in the context of the relationship between environment, dental care, and physiological discomfort is the one presented in [37]. The aim of the research was to determine whether a sensory-adapted dental environment (SADE) reduced physiological and behavioural distress in autistic children undergoing dental cleanings compared to a regular dental environment (RDE). The SADE environment included modified visual, auditory, and tactile stimuli. The results showed that the use of SADE proved to be safe and effective in reducing physiological and behavioural discomfort during dental care.

Finally, regarding *communication* strategies, the literature proposes various methods to establish a trusting relationship with the patient. For instance, introducing the staff during the pre-treatment phase to encourage familiarity [9]. Additionally, several strategies have been identified to build trust and a sense of safety in patients, such as the Deep Touch technique [24] or the Tell-Show-Feel-Do method [21], both of which have demonstrated great effectiveness in communicating with patients and providing clear reassurance during each phase, from pre to post-treatment. Lastly, it's worth noting that preventive training on how to interact with these patients and understand their specific needs is essential [26], ensuring the best possible treatment.

2.2 Limitations of Current Solutions

The current solutions investigated in terms of the technological, spatial and communication strategy present, however, some limitations towards the design of an effective solution for autistic children.

Regarding the *technology* solutions, all studies on ASD and VR, as far as we have been able to identify in the literature, report significant data relating to strategies for learning, for the practice of social skills, for improving attention skills, for job training. Only one study analyzes VR for dental treatment,

specifically teaching functional skills for brushing teeth [11]. Thus, proposals on the use of VR are focused on desensitization through the creation of familiarity since it is the familiarity that produces good results in reducing anxiety [27]. However, it must be considered that this occurs before the treatment or after the treatment, but does not concern the crucial phase of the interventions.

Regarding the studies that discuss adapting the *space*, [9,14,16,19,37,38] propose and analyze some of the sensory solutions. Although these proposals are not dynamically adaptive and not integrated with each other, they contribute to consolidating the idea that it is possible to effectively control the patient's hypersensitivity by modifying the environment. Unfortunately, they show a significant limitation in configurability and real-time customization.

Concerning the solutions related to *communication*, they appear to be the most complicated and time-consuming since they require prior training for the staff [26]. Not only must they know what to do, but they must also be attentive during the patient's procedure and detect signs of stress while performing the procedure to calm them [9] and thus prevent accidents. While there are widely proven methods such as Deep Touch [24] or Tell-Show-Feel-Do [21], these solutions lack technological assistance that would indicate the patient's condition more simply and effectively.

3 Method

The human-centred systemic solution proposed in this paper has been analysed in terms of the technology potentials that can generate a feeling of comfort and familiarity for the patient with the process, the staff, and the space, to entertain the patient to avoid escalating his/her own anxiety, and to enable an effective doctor-patient communication. This article analyzes the technical feasibility of solutions aimed at reducing the state of anxiety (pre, during and post-treatment) of ASD people with reference to the three categories identified: technological, spatial, and communication. In particular, the positioning of the sensing elements (sensors and camera), the ways of adapting the room and the communication with the dentist and support staff.

The development of the human-centred systemic solution has been proposed as a user scenario-driven solution, including the feasibility of sensor integration and evaluated against its pros and cons.

3.1 Technical Viability Analysis

Technological Category. Sensors and cameras are integrated into the environment to detect, in the least intrusive way possible, the simple biological parameters that allow the ASD person's state of anxiety to be estimated.

The sensors are integrated into the chair, unobtrusively like in [6,35] while the camera is positioned to frame the person's face. The two systems are redundant, aiming to enable estimates even in complex conditions.

- *Heart Rate Sensors:* Devices exploiting these sensors measure arterial pulse, detecting small changes in heart rhythm and can notify the dentist about increases, which may indicate heightened anxiety levels. As noted in [20], it is the most widely adopted and straightforward method for estimating stress levels.
- *Skin Temperature Sensors:* Skin temperature changes due the stress and anxiety, increasing in the moments of high stress and anxiety [22].
- *Skin Conductance Sensors:* (or Electrodermal Activity EDA): EDA sensors measure changes in skin conductance, which can indicate emotional arousal, stress, or anxiety [31]. When a person experiences stress or anxiety, their sympathetic nervous system is activated, leading to increased activity in sweat glands and subsequently, increased skin conductance.
- *PPG*
  Cameras: Photoplethysmographic (PPG) cameras use optical sensing techniques to monitor physiological parameters. This allows the measurement of changes in the patient's skin colour, changes related to an increase in heart rate and, consequently, an increase in the patient's stress [29].

Spatial Category. ASD people have a high level of sensitivity to unknown and unfamiliar environments. This aspect offers several challenges, especially in clinical environments where instrumentation, materials (for example, due to hygiene, they are waterproofed or protected), brightness and odours are linked specifically to the setting. Reducing the state of stress requires a pretreatment process that allows you to become familiar with the space, the equipment, the colours, the smells and the materials. In this way, overstimulation is minimized and anxiety levels are lowered [15]. To facilitate the conditioning process, the room should be customizable where possible. This aspect consists in being able to modify the intensity and colour of the light, the colour of the walls, sounds and smells as desired. For example, adjustable lighting can help create a relaxing environment, while soft, warm colours can reduce excessive visual stimulation and promote a state of relaxation. Controlling environmental sounds (for example, the sound of equipment) and using background music can further reduce anxiety. Additionally, minimizing strong odours or scenting the room with a scent familiar to the child can help create a more comfortable atmosphere.

Communication Category. Communication between the patient and the dentist, as well as between the dentist and the surrounding environment, must be customizable, effective, simple to interpret, simple to use, reliable and interactive. To achieve this objective, it is necessary to develop a system capable of transmitting information regarding the person's stress state during treatment in real-time to the dentist. The system must have performances that allow the dentist to intervene promptly and adequately to preserve the patient's comfort level. Furthermore, the system should offer the possibility of easily modifying the conditions of the surrounding environment to create

an atmosphere that promotes maximum calm and tranquility in the patient; for example, being able to access pre-configured scenarios derived from the pre-treatment and familiarization phase with the person, or being able to adjust the lighting, generate environmental sounds, activate specific musical pieces, intervene on the colours and the introduction of fragrances into the room.

3.2 Sensors and Camera

For effective measurement, some of the previous sensors must maintain constant contact with the patient's skin. According to studies on sensor placement, the best location for them is the wrist, achieving a superior recording quality that accurately reflects the patient's condition. The options for sensor positioning are the following:

- Creating a small depression *in the arms* of the dentist's chair to encourage voluntary placement of the user's arms there and specifically support their wrists in these recesses. This is aimed at effectively measuring without the need to force the patient to use an external device.
- Locating the sensors *in the backrest* of the chair is non-invasive, requiring only that the patient sits down, allowing the backrest to collect the desired information. This is a simple and easy-to-implement solution.
- In most cases, children undergoing dental treatment tend to squeeze a *stuffed animal* with their hands during the dental procedure, aiming to reduce their anxiety and feel accompanied. Therefore, this object, being constantly held in their hands, provides the opportunity to measure vital signs in a simple and non-invasive way by locating sensors inside it that allow the collection of information regarding anxiety levels.
- One of the most efficient methods for building familiarity with patients is through the use of *costumes*. This generates empowerment when attending the consultation, diverting their attention from the unpleasant sensations they may experience. Therefore, sensors can be integrated into the costume, as it is a highly effective way to keep them fixed in place (more specifically, in the wrist area) while also providing greater security to the child, as mentioned before.
- Given the patient's semi-reclined position during the procedure, positioning the *camera* on a suspended and adjustable arm ensures unobstructed views. The positioning should minimize patient discomfort from unfamiliar objects and facilitate clear observation of the child's facial expressions. Considering recent technological advances and future trends, the integration of cameras into glasses (such as Ray-Ban glasses) [25], can offer an alternative and non-intrusive method to capture children's clear and detailed facial expressions compared to a camera positioned on a mobile arm. By simply recording everything with the glasses worn by the dentist or assistant, the patient would no longer need to wear any devices, eliminating any potential discomfort or anxiety.

3.3 Space

Since each patient is unique, the environment must be adapted to their needs by personalizing it so that it is as familiar and comfortable as possible.

- Regarding the conditions that directly affect the senses, such as *sound, light intensity, and room odour*, they can be easily *controlled*. Installing sound-insulating panels can effectively reduce external noises that might disrupt patients. Moreover, both odours and light intensity should be adjustable based on individual patient preferences.
- To create an immersive experience and complement the sensation generated by the use of costumes, *video mapping* can be applied to the walls of the room. This technique immerses children further into the stories they love, fostering a sense of participation. This can help them reduce their anxiety levels and feel more comfortable in these spaces. However, the projected image or displayed image on an immersive digital wall should not be intense or have a high level of movement, as this could potentially distract the patient and generate the opposite effect desired.
  Immersive digital *LED walls* can display realistic, very smooth and well-defined images, enhancing children's engagement with their favourite stories. Unfortunately, these solutions are still disadvantageous compared to projections due to cost and power consumption.

3.4 Communication

- If we contemplate the integration of *smart glasses* in dental care, it opens up a realm of possibilities. Smart glasses could not only serve as a means to notify the dentist of any escalation in the child's anxiety levels but also offer a sophisticated avenue for intervention. Through the advanced technology embedded in these glasses, the dentist receives real-time alerts or data regarding the child's emotional state, allowing for timely and tailored responses. Moreover, regulating environmental factors through voice commands or gestures on the glasses could create a more calming and comfortable atmosphere for the patient.
- To assist the dentist, the assistant, in addition to performing regular support tasks, has access to a monitor that displays the patient's status concisely, clearly, and in a familiar manner. This information, combined with the other visual feedback typically available, enables the entire team to act on both the dynamics of the treatment (as usual) and the environmental parameters tailored to the patient. Both actions aim to restore the patient's comfort as quickly as possible.
- Another scenario to consider envisions the dentist equipped with a *wearable* device snugly positioned on their wrist. This would serve as a silent communicator, gently vibrating to alert the dentist of any surges in the child's anxiety levels as the examination progresses. Upon receiving the subtle vibration, the dentist seamlessly transitions to the nearby control panel, strategically placed

within arm's reach. Here, the dentist can adjust various environmental elements to create a more calming atmosphere.

4 Discussion

In this session, we discuss a human-centred systemic solution, starting from the analysis conducted in the previous sections and focusing on the technological aspect. Technology enables medical staff to perform their tasks optimally by combining perception factors with the objective data collected from the patient, which represents their emotional state. Technology allows intervention in the environment, modifying room"conditions" such as noise, odour, and colour. The control panel aims to reduce stress and improve the quality of care provided to the patient. It is essential to emphasize that technology is a tool and, as such, is compensatory and not substitutive: technology does not replace human contact, experience, and sensitivity.

4.1 Human-Centred Scenario

The proposed solution aims to create a familiar environment for the patient and allow medical staff to detect anxiety-related signals in the patient as early as possible. The goal is to intervene in the room conditions, modifying the sensory aspects of the environment—such as sound and scent—according to the patient's needs. Interventions must be personalized: every person, especially when dealing with situations on the autistic spectrum, is not a standalone case but a world of their own.

As described earlier, three phases are recognized: pre-treatment, treatment, and post-treatment. In the pre-treatment phase, the patient is gradually introduced to the medical office and the people who will take care of them. The patient, accompanied by the parent, becomes familiar with the medical staff, the environment where the treatment will take place and the tools that will be used. Based on the parent's instructions, various environmental (e.g., colour, scent, scenario) and personal (e.g., games, costumes) adaptations will be tested. In the treatment phase, the patient is welcomed into the environment configured according to their preferences and by the medical staff they know. For example, the child arrives at the dentist appointment, chooses the costume they like the most, and is in the personalized room. During the treatment, monitoring takes place: sensors and cameras detect slight changes in the child's conditions, indicating changes in the person's state that are correlatable to anxiety levels. Also, the dentist adjusts their behaviour and environmental conditions during the treatment. For instance, they adjust the light intensity, modify the volume of the sound, which is gentle and repetitive or otherwise chosen by the person, and activate the gradual release of a calming scent. The child starts to relax and allows the procedure to proceed. In the post-treatment phase, the patient relaxes and is reconditioned with the help of family members and medical staff.

4.2 Pros vs Cons of the Solutions Evaluated

- Sensors located on the *arms of the chair*:
  - Pros: The sensors will always be in the right position to measure the vital signs of children. Adapting a chair to integrate sensors is a one-time investment: since the same chair can be used for all patients, the overall cost of implementation and maintenance is reduced. The accuracy is high due to the contact with the person.
  - Cons: Children with autism may find it difficult to remain still in one position for extended periods. This can make it challenging to maintain consistent sensor contact, potentially affecting the accuracy of vital sign measurements.
  - Feasibility: Considering the difficulty of keeping the child in a single position throughout the entire treatment, this solution is effective only for short periods.
- Sensors located on the *seat and back of the chair*:
  - Pros: It is a non-invasive solution. The patient, while sitting, has no perception of the sensors. Many low-cost and effective solutions are possible. There are no issues with power and supply, as well as communication with the computing platform.
  - Cons: The patient is accurately detected only if they are in the correct position (aligned with the sensors) and the patient is still. The sensors are located both on the seat and on the backrest. Non-intrusive detection does not allow the detection of parameters that require contact with the skin; for example, it is not possible to detect skin conductance and blood oxygenation.
  - Feasibility: The sensors can be of different types, the most suitable, based on the position (for example, mechanical sensors like EMFi, load cells, accelerometers). Sensors can be placed both on the backrest (more sensitive) and on the seat. Redundancy is a relevant factor in accurately estimating heart rate and respiratory rate as reliably as possible, increasing both the cost and computational load.
- Sensors located *inside soft toys or toys*:
  - Pros: The plush toys and toys are tools that the patient holds in their hands. Contact makes it easy to detect some simple biological parameters like skin conductance. It is also possible to detect the pressure exerted, which can be another factor of interest for detecting emotional changes. Detecting pressures and vibrations is simple and inexpensive.
  - Cons: This solution requires positioning the sensors that require contact with the skin. Keeping the children still and in constant contact would be the greatest challenge when using an object that is constantly in motion. Due to the portability of the object, an issue arises regarding the computing platform, wireless transmission, and, consequently, the need to pay attention to energy consumption. Infine, l'oggetto deve essere progettato per essere leggero e resistere a umidità, vibrazioni e pressioni. Finally, the object must be designed to be lightweight and resist humidity, vibration and pressure.

  - Feasibility: This solution is simple to implement since patients are accustomed to holding an object in their hands for most of the time to reduce their stress. However, placing and keeping the sensors in the correct positions is virtually impossible because the patient can move the object at any moment, even throwing it, which makes it ineffective for measuring any desired vital signs.
- Sensors *integrated into costumes*:
  - Pros: The costume allows for the placement of sensors at the necessary points and the use of newly developed sensorized fabrics. When in contact with the patient, the sensors can correctly measure vital parameters. It is important to underline that, if chosen by the patient, the costume can have a positive effect on calming them.
  - Cons: The costume should have areas that are close-fitting to the patient. This aspect could generate discomfort and may not be easily accepted. The computing and data transmission platform must be allocated, and a light, non-disturbing power source must be placed; its placement is a challenge. Finally, the use of sensorized textiles could significantly impact the cost.
  - Feasibility:If accepted, this option is feasible and realistic since the sensors can be placed in the exact location without disturbing the patient too much compared to a specific wearable device. The critical issue is adherence to the skin, but if accepted by the patient, the collected data are more accurate and stable.
- *Wearable* for the patient:
  - Pro: Wearable devices, if accepted by the patient, allow for more effective detection of biological parameters related to states of anxiety.
  - Cons: Wearable devices are not easy to get ASD patients to accept.
  - Feasibility: Wearable devices that measure some biological parameters (for example, heart rate, skin conductivity, and blood oxygenation) effectively and reasonably accurately are commercially available and have a reasonable cost.
- *Camera* located on a suspended and adjustable arm:
  - Pros: Being located on a mobile structure, it can be adjusted to see the person's face and not disturb the doctor and the assistant. The camera can be small in size (e.g., a high-quality webcam) so that the patient will not be distracted by seeing the camera, nor will they feel like they are being recorded. Therefore, the distraction factor will not be present, making the procedure easier to perform.
  - Cons: If the patient's face is not oriented towards the camera, accurate detection cannot be performed
  - Feasibility: While camera installation is simple, analyzing changes in skin pigmentation to detect changes in blood pressure and detecting increases in anxiety requires data analysis that is not simple due to possible continuous patient movement, difficulty in framing, and variations in background color (for example, changes between open and closed mouth).

- *Luminous Intensity, Smells, and Sounds:*
  - Pros: Brightness, scent, and colour are environmental factors used to calm people who suffer from ASD. In a medical context, modulating brightness, changing the colour of an environment, and introducing scents are strategies that help mitigate anxiety during treatment. During pre-treatment, different types of scents, colours, and light can be tested to identify the scenarios that best suit the patient based on indications from family members.
  - Cons: Changes in brightness, colour, or scent may initially cause an increase in stress before a state of tranquillity is achieved.
  - Feasibility: Modulating the brightness and colour is technically simple as is using perfume diffusers. For the latter, the main difficulty is the speed of diffusion which may require specific strategies (such as the use of fans). The use of specific scenarios is a simple and effective way to act on these environmental factors.
- *Video Mapping:*
  - Pros: Video mapping, which allows the projection of images, videos, and other visual content onto three-dimensional surfaces, easily adapts to all surfaces without the need for physical intervention.
  - Cons: Its operation requires a setup phase that involves scanning the surface onto which the images will be projected (for example, using a 3D scan), and it may require multiple projectors to provide an immersive experience. Furthermore, the noise generated by the projectors could be a factor disturbing the patient. Finally, there may be cost issues related to the number of projectors needed, their positioning (which could be complex for some environments) and energy-intensive. Finally, effectiveness is linked to the brightness of the environment.
  - Feasibility: This method of environment adaptation is simple to implement and well-established. Specific software matches the coordinates of the scanned surface with the visual content. The projectors are then positioned to precisely project the content onto the surfaces, creating the illusion that the images are an integral part of the structure itself
- *Led Panels on the walls:*
  - Pros: The latest generation LED panels are flexible, modular, and low cost. They provide reasonable image fidelity and create a more engaging and immersive experience. Since they are active (transmitting the image rather than projecting it onto walls), they have fewer problems with ambient brightness compared to projectors.
  - Cons: Although they are flexible, they do not adapt to all situations such as sharp edges or curves. Cost and energy consumption can be an issue.
  - Feasibility: The use of panels is feasible, as it allows for quick and easy space adjustments. Additionally, no noise is generated that could disturb the patient. Furthermore, the quality of the projected image will be high, and it will be possible to create new immersive spaces quickly and easily.
- *Smart Glasses* for the dentist:

  - Pros: Glasses equipped with a camera allow for easier framing of the patient's face and capture their micro-expressions on many occasions. Since they resemble regular glasses, they do not distract the patient. Glasses equipped with heads-up displays (HMDs) provide a more natural display of information.
  - Cons: The price of glasses with cameras and HMDs can be high. Furthermore, the doctor must undergo training to use them effectively for both recording the patient's face and displaying information on the heads-up display that is useful for system control. For example, a challenge in converting video into information (such as heartbeat) is the stability of the image. Finally, high performance and energy consumption are required.
  - Feasibility: These types of glasses are already available on the market (e.g., Ray-Ban Stories) but may require adaptation for specific purposes.
- *Display* for the dentist assistant:
  - Pros: The monitor for the doctor's assistant allows them to monitor the patient's status and have control over the configuration of the environment and the system. In addition to promptly informing the dentist of the decisions to be made, it can independently support the doctor by aiding in choosing the most convenient scenario.
  - Cons: The monitor contains a lot of information that must be summarized and presented to the assistant in a clear and familiar way. The application of commands must also be simple and straightforward. Finally, the position of the panel must allow for easy reading without hindering the doctor's support activities.
  - Feasibility: This solution is highly viable as it complements existing systems. The display can be tailored for use by the assistant, enabling them to operate the device and assist the dentist if the dentist is engaged in a procedure at the time of receiving an alert, which prevents them from adjusting the room conditions to reduce anxiety. The user interface is configurable.
- *Vibrotactile device* for the dentist:
  - Pros: A vibrotactile device worn by the doctor for real-time communication of the patient's experience during treatment is a tool that enables the doctor to receive information without needing to view or focus attention on another device.
  - Cons: The presence of a vibrotactile device could hinder the doctor's movements. Furthermore, the device is unusable if the signals transmitted from the device to the doctor's body are not sufficiently evident (for example, a vibration with intensity linked to the person's state) and clear or are confused with noises and vibrations produced by laboratory instruments.
  - Feasibility: Wearable devices are available, easy to modify, low cost, and do not require much energy to perform the vibrotactile communication action. They can present the problem of poor ability to transmit information to the doctor, especially in contexts requiring high concentration

where other instruments generating vibrations may interfere with the doctor's perception.

Future experimental validation of the proposed solution is underway to assess its accessibility to dental care for autistic individuals. Specifically, the analysis of the space allows the identification of the most feasible option on a case by case basis. Systems controlling brightness, scent, and sound are broad-spectrum and applicable to all situations. Regarding the measurement of emotional alterations, particularly the evolution of anxiety states, based on the individual's response in the pre-treatment phase, all proposed solutions are viable, with a preference for wearable devices (although accepted with difficulty in some cases), followed by toys and costumes with integrated sensors. The most practical option is to place sensors in the chair (numerous works propose non-intrusive estimation systems for breathing rate and heart rate), and to mount a camera on a mobile arm that does not interfere with the doctor's work and does not disturb the patient. One potential issue is that these solutions may be sensitive to the patient's agitation and positioning. Finally, in terms of viewing and controlling the system, two complementary alternatives should be presented: the use of smart glasses, allowing the doctor to observe any anxiety alterations and modify room parameters to calm the patient, and a monitor display for the assistant.

4.3 System Architecture

Figure 1 provides a high-level description of the system architecture for the proposed solution.

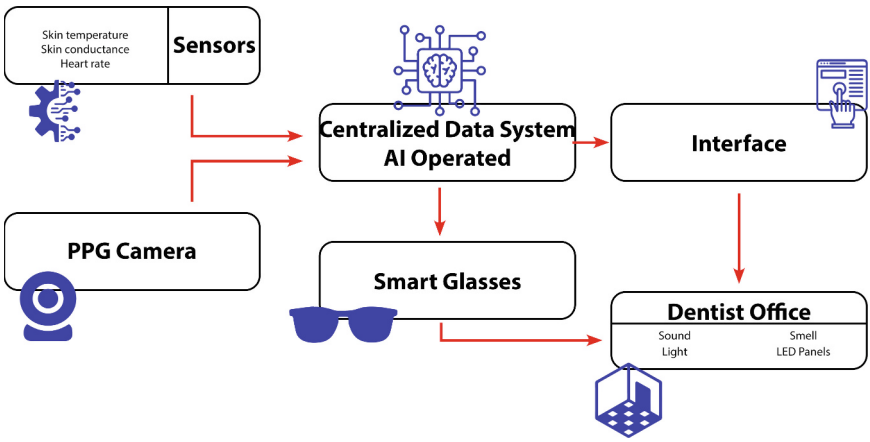

Fig. 1. System architecture.

Sensors are positioned in the chair (seat, backrest, and armrests), as well as in the costume and toys. The camera is mounted both on a suspended and

adjustable support and on smart glasses. Redundancy in the detection system is necessary to increase the likelihood of data collection and enhance the reliability of the information. Indeed, the patient might not accept wearable objects (a bracelet or a costume) or handheld items (the toy) or might move during certain phases of the treatment, making it impossible to reconstruct the necessary biological signals to estimate their state. Wearable sensors and smart textiles for smart clothing ([39]) are not a new concept. These areas have been extensively explored both in research and product innovation. Smart clothing and wearables enable reliable measurements of biological parameters such as skin conductivity, heart rate, respiratory rate, and blood oxygenation.

The collected data is transmitted to a computational and storage platform that stores and transforms it into information that, in turn, is retransmitted to the user interface. The estimation algorithms include both signal processing and artificial intelligence. In the former case, algorithms like moving windows, noise reduction filters, band-pass filters, and frequency domain transformation (FFT) are used to derive respiration and heartbeat (through ballistocardiography). In the latter case, one or more trained neural networks can be used to classify the person's state. The computing platform must have sufficient performance to meet the real-time requirements inherent in the process.

The information regarding the patient's state is returned to the medical staff, who can take the necessary measures to reduce the level of anxiety (e.g., reducing light intensity or altering sounds). The information is displayed on both control panels and smart glasses. The interface is simple and immediate, as is the way of issuing commands to interact with the environment.

The authors have developed a prototype within the ACCESS project that uses remote PPG and strain gauges to monitor heart rate in real time for stress detection. The detected stress levels, combined with user preferences, are then used by a recommender system to deliver personalized calming interventions. This system learns the patient's preferred music or videos through home interactions. Further research is needed to refine and optimise the proposed solution in real-space medical settings, as well as exploring further strategies to improve accessibility to dental care for autistic individuals.

5 Conclusions

Addressing the demands of dental care for children with ASD is a significant topic. In particular, identifying a broad range of potential technological solutions to expedite the patient's conditioning phase can have a significant impact on the costs families face. This article analyzes and proposes a human-centred systemic solution to enhance the dental experience for children with autism. The proposed solution integrates various technological elements, including sensors, facial cameras, and adaptable room settings, to create a conducive, familiar, and comfortable environment for the needs of autistic children during dental visits. By addressing the dental visit process, the proposed solution aims to reduce anxiety levels, improve patient experience, and provide dentists with real-time

information, ultimately improving oral health outcomes for autistic children. The transformative potential of this solution to revolutionize dental care for children with ASD, improve their quality of life, and promote inclusivity in healthcare environments is significant. Future research and development efforts will focus on refining and optimizing the proposed solution, as well as exploring further strategies to improve accessibility to dental care for autistic individuals.

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Mitigating Authentication Risks Through AI-Coaching: An Adaptive Dynamical System Analysis

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Abstract. In an era where cybersecurity threats are increasing, a strong authentication system is one of the foundations of digital security. This paper presents a computational analysis of the dynamic collaboration of a system-level authentication model, a human decision-making process, and an AI Coach as supervisor and performance enhancer. To simulate this, an adaptive dynamical system model has been developed that shows how risks related to contextual variables influence an admin's stress level and, consequently, its risk assessment of the situation. The model was designed as an adaptive network model and uses adaptive mechanisms and an AI coach that detects misjudgments by the admin and helps to correct during the decision-making process. A What-If analysis was conducted to investigate the impact of important variables such as stress sensitivity and persistence levels on risk assessment accuracy. Results show that the AI Coach plays a critical role and is an important addition to the model.

Keywords: Authentication Management · Adaptive Network · Human-AI Interaction · Knowledge Level · AI Coach

1 Introduction

Most corporate, government, financial, military and medical organizations gather large volumes of data to store and process their applications. Protecting these processes from unauthorized access is critical, which makes cybersecurity a foundational aspect of modern digital infrastructure [17].

Authentication management is one of the most important components of cybersecurity, acting as a gatekeeper for users to access systems. These authentication systems often protect important or sensitive information that needs to be properly secured. However, these systems face a wide range of threats and risks, such as insider threats, weak password practices and cyberattacks. These vulnerabilities can have serious consequences such as unauthorised access, data breaches and system disruptions [8].

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This project will analyse this adaptive dynamical system by designing an adaptive temporal-causal network model [25–27] to simulate an authentication system consisting of a system level, a (shared) mental model for a human decision-making process, and for an AI supervisor that acts as a coach for the administrator's decision-making. Simulations have been conducted to demonstrate the impact of contextual variables and cognitive vulnerabilities on the administrator's decision-making. In addition, inspired by work on the use of shared mental models and social and organisational learning [1, 3, 4, 6, 9, 31–33, 35] and in particular [20], an AI coach will be introduced who acts as a supervisor of the admin, overseeing the choices made by the admin and guiding in the occurrence of incorrect decisions, in order to assist with corrections when necessary. Based on these simulations, which examine how contextual variables influence the admin's stress level and thus the admin's decision-making, a What-If analysis and risk assessment were performed. These results demonstrate how stress can be an important human psychological factor in making choices in an authentication system.

2 Literature Background

In an authentication management system the system verifies the credentials provided by the user against the information stored in the database. The access control relies on four key mechanisms: identification, authentication, authorization and accountability [7]. The process starts with identification, where the user submits its identifier such as a username or name ID. Authentication validates if this identifier, together with the credentials given by the user, match with the login information stored in the database. Next, authorization follows which determines whether the authenticated user has the permission to access a resource. Accountability is the process of keeping up with the logs, meaning keeping track of all login entries no matter whether correct or incorrect credentials are given to ensure traceability.

There are different authentication mechanisms, which can be split up into three categories: 'Something you know', 'something you have' and 'something you are' [7]. Starting with 'something you know', this category is the combination of providing the user's identity together with a form of information the user knows, like passwords, PINs and security questions. This is the most commonly used form of authentication but at the same time the most vulnerable to attacks due to human factor vulnerabilities like weak password tendencies or sharing this information with others. These vulnerabilities can lead to unauthorized access and harm security systems due to brute-force and dictionary attacks. Also poor password management such as reusing passwords and clicking on phishing links exploit human behavior to steal information [17].

The second authentication mechanism category is 'something you have'. This category requires a form of identification with a card or hardware token in combination with a password or PIN. This category improves upon the knowledge category by introducing a form of physical device that is needed to access. However this category also has its drawbacks because users can forget their PIN or lose their card which then can easily be misused [10].

The third category is the 'something you are' authentication mechanism. This mechanism requires a form of biometric authentication that uses unique physiological or

behavioral characteristics. Physiological biometrics include fingerprints, facial recognition, iris and retina scans while behavioral biometrics include voice patterns, keystroke dynamics and signature recognition. While biometric authentication is the strongest form of authentication, it is susceptible to errors. A false rejection, also called type 1 error, occurs when the system falsely rejects a user and a type 2 error occurs when the system mistakenly allows unauthorized user access [7].

These three forms of authentication using knowledge, possession and inherence discussed can be combined to strengthen the security process. This is called a multifactor (MFA) or two-factor authentication which compromises multiple authentication layers. MFA provides a higher level of safety by using more than two categories of credentials [16]. For example, this means that a user is required to submit a password - something the user knows - and afterwards submit a verification code sent to the user's smartphone, which is something the user has. A multifactor authentication aims to mitigate the risks like phishing or brute force attacks by introducing a second barrier in the authentication process.

The term 'risk' in the field of cybersecurity and, in particular authentication management can be interpreted in various ways. Examples include: 'the chance that a threat will exploit the vulnerabilities of a system and thereby cause harmful effects on the system' and 'risk is the chance of a negative event multiplied by its impact'. The aim of risk identification is to determine what could cause a potential loss and to gain insight into how, why and where this loss could occur. A risk analysis assigns values to the probability and consequences of each risk. Finally, a risk evaluation is carried out in which the results of the risk analysis are compared with risk criteria to determine whether the risk level is acceptable or not. Risk assessment is the process of combining risk identification, risk analysis and risk evaluation [21]. In this project, various risks will be simulated in both the contextual variables and the human cognitive factors, such as stress and how it is dealt with. The results will be analysed and a risk evaluation will be drawn up to complete the risk assessment.

3 General Modeling Approach

To model the adaptive behavior of real-world processes, an adaptive dynamical system model in network-format is used [25–27]. The network structure suits real-world characteristics that adapt over time. Modeling causality, meaning a causal relationship $X \rightarrow Y$ which describes how a state X can affect a state Y over time, and adaptivity brings more structure into the model leading to the notion of an adaptive temporal-causal network. A conceptual representation of a temporal-causal network has a labeled graph as its fundamental basis, with nodes representing the states and the connections representing the causality relations of the network. Not all causal connections are equally strong, so the notion of strength $\boldsymbol{\omega}_{X,Y}$ is used as the label for the connections. In addition, temporal-causal networks incorporate two further dimensions: aggregation and timing. For any state Y, a combination function $\mathbf{c}_Y(..)$ defines the aggregation of the impacts $\boldsymbol{\omega}_{X,Y}X(t)$ on Y from incoming connections from state X.

Timing concerns the rate of speed $\boldsymbol{\eta}_Y$ at which a state Y changes to its incoming impact. These conceptual representations can be transformed into a numerical representation which is a differential equation for state values over time formalised by formula

(1) in terms of the above network characteristics $\boldsymbol{\omega}_{X,Y}$, $\mathbf{c}_Y$, $\boldsymbol{\eta}_Y$ [27].

$$Y(t+\Delta t) = Y(t) + \boldsymbol{\eta}_Y\left[\mathbf{c}_Y\left(\boldsymbol{\omega}_{X_1,Y}X_1(t), \ldots, \boldsymbol{\omega}_{X_k,Y}X_k(t)\right) - Y(t)\right]\Delta t \qquad (1)$$

$$\frac{dY(t)}{dt} = \boldsymbol{\eta}_Y\left[\mathbf{c}_Y\left(\boldsymbol{\omega}_{X_1,Y}X_1(t), \ldots, \boldsymbol{\omega}_{X_k,Y}X_k(t)\right) - Y(t)\right]$$

These conceptual network characteristics can be represented and formalized in a set of role matrices [27], Ch 2, see the Appendix Sect. 7. The base connectivity of the network is encoded in the **mb** matrix, which specifies the existence and the direction of the connection between states. The strength of these connections are modelled in the **mcw** matrix which assigns a numerical weight to each corresponding connection in the **mb** matrix. The **ms** matrix assigns a numerical value to each state which determines how fast a state changes upon causal impact.

Combination functions provide a way to specify how multiple causal impacts on a state are aggregated. Each combination function has its own approach to process how the incoming causal impacts get aggregated; see Table 1. In our network model two combination functions are used: the advanced logistic sum $\mathbf{alogistic}_{\sigma,\tau}(..)$ with parameter σ for the steepness and $\boldsymbol{\tau}$ for the threshold. The other combination function used is the $\mathbf{stepmod}_{\rho,\delta}(..)$ function which alternates between active, a value of 1, and inactive, a value of 0, at a consistent repetitive interval. The **stepmod** function also has two parameters, the first parameter specifies the duration of one cycle and the second parameter specifies the tipping point before the **stepmod** function activates the state.

Table 1. Combination functions used.

Concept	Representation
Advanced Logistic sum $\mathbf{alogistic}_{\sigma,\tau}(V_1, \ldots V_k)$	$[\frac{1}{1+e^{-\sigma(V_1+\cdots+V_k-\tau)}} - \frac{1}{1+e^{\sigma\tau}}](1+e^{-\sigma\tau})$
Stepmod $\mathbf{stepmod}_{\rho,\delta}(V_1, \ldots V_k)$	time t, repetition interval $\boldsymbol{\rho}$, and tipping point $\boldsymbol{\delta}$ 0 if t mod $\boldsymbol{\rho} < \boldsymbol{\delta}$ else 1

The specification of the combination functions is captured in two role matrices. The mcfw matrix indicates which combination function is used for each state and the mcfp matrix stores the information about the parameters for each combination function for each state.

An important feature of the temporal-causal model framework is the capability to represent adaptive behavior in which not only the states change over time but also the structure and the characteristics of the network. This is accomplished through the principle of self-modeling or network reification [25, 26], see also [27], Ch 3. This is inspired by the area of metalevel architectures and reflection in AI; e.g., [2, 11, 15, 22, 23, 34]. To achieve this, additional states are introduced to the network that represent the characteristics connectivity, aggregation and timing of the network. This is called the first-order adaptation. In that case, the weight $\boldsymbol{\omega}_{X,Y}$ of a connection from X to Y will

be represented as a self-model state (**W**–state) named $\boldsymbol{\omega}_{X,Y}$. The combination function self-model state will be denoted as $\mathbf{C}_{i,Y}$ and its parameters self-model state $\mathbf{P}_{i,j,Y}$. A speed factor self-model state will be denoted as $\mathbf{H}_Y$. Each of these reification states has incoming upward arrows from the base network affecting the self-model state and downward outgoing arrows that indicate on what state the self-model state has its impact.

Moreover, first-order adaptation principles can adapt based on other adaptation principles, which is called second-order adaptation [26, 27]. The only forms of second-order adaptation used in our model are the **MW**-state and **WWW**-state which are the persistence for a first-order **W**-state and communication control for transfer from one agent's **W**-state to another agent's corresponding **W**-state, inspired by earlier work on (shared) mental models and social or organisational learning such as [1, 3, 31–33].

Methods and software are available to support simulation [29], verification of correctness by mathematical analysis [5, 24, 28], and validation of network models [30].

4 Adaptive Network Model Design

To visualize the processes and interactions within the scenario, a graphical representation of the adaptive network model was developed. In the model, ovals represent the nodes corresponding to states, and the directed arrows between the nodes denote the causal connections between the states. The red arrows depict the downward causal impact and the blue arrows illustrate the upward causal connections. The complete model is composed of three interrelated parts: the system model, the mental model which represents the admin's decision making process and the AI coach model. Figure 1 shows the graphical representation of the network. Other details can be seen in the Appendix Section.

The model starts with node X1, *Authentication Initialization*, which starts up the authentication model. Then X2, *User Request Login*, and X3, *Credential Submission*, simulates the user login process. From the credential submission, the system extracts a set of contextual information associated with the login attempt. These include X4-X7, *Device of Login, Location of Login, Time of Login* and *Frequency of Login*. Each of these variables is assigned a anomaly/risk scoring ranging from 0 to 1, where 0 indicates no anomalies/risk detected and a value of 1 indicates high level of anomalies measured. For the login device, if the system recognizes the device from previous logins, it receives a low score. However, if it is unfamiliar or has not been used before to login into that account, the anomaly/risk score will increase. For the login location the system detects anomalies in the IP address. Consistent login location yields a low risk score but logins from an unexpected or distant region will raise suspicion and a high risk score. The system stores previous logins and therefore also the time of login is tracked. The system learns habitual login times and signs of deviation in the login time yield a higher score. Lastly, the number of login attempts is also monitored, and abnormal surges in attempts could be signs of a brute-force attack which will raise suspicion and therefore the risk score.

The system will assess the risk of each of these statees X11-X14 and each assessment variable is weighted equally into the aggregation for the overall *System Risk Score* X17. If this risk score value exceeds the 0.15, the system will mark this login entry as a *Suspicious Login* (X19) otherwise the system assesses it as an *Authentic Login* X18.

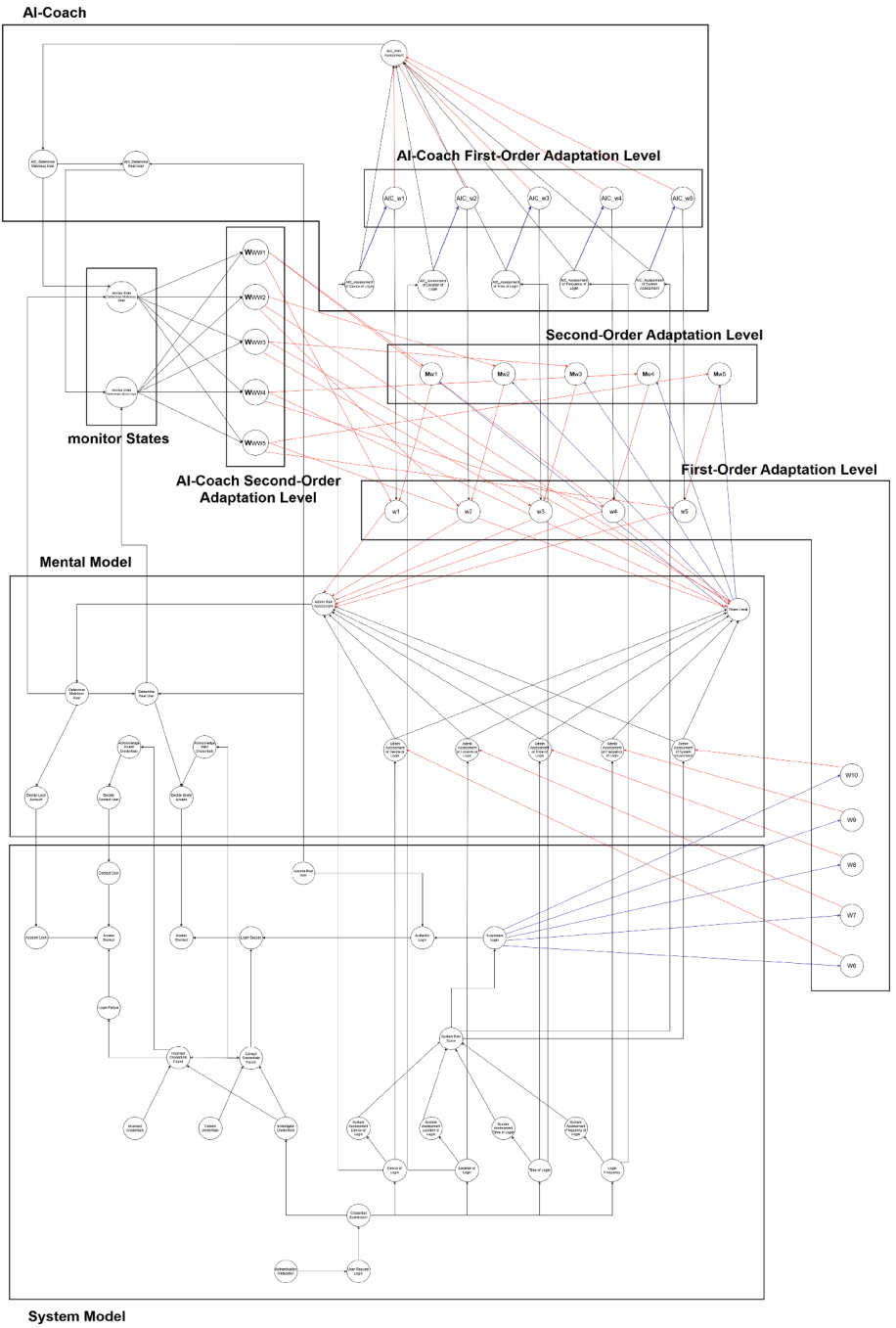

Fig. 1. Graphical representation of the adaptive network model.

Additionally, the password is checked against the database. If the password given matches the stored password and the system marks the login as authentic, access is granted. Conversely, if the password is incorrect access will be denied.

If the system assesses a suspicious login, then the **W**-states **W**6-**W**10 will be activated. These **W**-states act as binary switch nodes enabling the connection between the contextual variables X4-X7 and the corresponding *Admin Assessment*-nodes X22-X25. Besides these nodes, the admin will investigate the *System Risk Score* as well in the *Admin Assessment of System Assessment*-node X26. The admin will, just like the system did, aggregate these assessments into an overall *Admin Risk Assessment* X27, however the way they are aggregated is different.

Unlike the system, the admin is subject to cognitive and emotional influences, in particular stress. The admin *Stress Level* X28 is directly influenced by the perceived severity of the assessment of the contextual variables and the system risk score. Meaning that high levels of risk assessment for each of these variables will cause more stress to the admin. However, not all variables contribute equally to the stress level, for instance. A severe anomaly/risk score for *Device of Login* contributes more to the stress level than *Time of Login* because an anomaly in the devices used is considered to be more of a credible indication of suspicious activity than a deviation in the time the user logs in. The admin's ability to assess the overall risk of the situation deteriorates under sustained stress. As the stress level prolongs, the cognitive clarity diminishes with incorrect evaluations and misclassifications as a result. This behavior process is modeled using the adaptive weight representations **W**1-**W**5 with each degraded over time due to its persistence that is governed by the **MW**-states that lower due to the stress level. The **MW**-states are the second-order adaptation states which represent the durability of the first-order connection weight states **W**1-**W**5. If stress levels are high, **MW**-states decline faster, causing the **W**-states to decrease. However, not each **MW**-state is affected equally because the admin will shift its concentration to the most salient variables which are *Admin Assessment of Device of Login* X22 and *Admin Assessment of Frequency of Login* X25*,* while neglecting the other variables to a greater extent.

If the admin risk score exceeds a score of 0.4 the admin will classify this user as malicious; otherwise the user is classified as a real user. However, over time due to the remaining stress, the adaptive connections between the admin assessment variables and the admin risk assessment have decreased so much that the admin will misclassify a malicious user as a legitimate user and thus possibly grant access if the credentials match.

Depending on the admin risk assessment and therefore on the decision between *Determine Real User* X29 or *Determine Real User* X30 and the correctness of the credentials, three scenarios are possible. The first scenario is established if the user is deemed malicious, regardless of the correctness of the credentials, the admin *Decide Lock Account* X35 and the user is denied access (X37, X39). The second scenario unfolds if the user is classified as authentic but submits incorrect credentials. The system will deny access, however the admin will contact the user to support (X34, X36, X39). Lastly, if the user is classified as authentic and the credentials match, then the user will be granted access (X33, X38).

In order to correct the errors made by the admin under stress, an AI coach is integrated into the network. The AI coach serves as a supervisor with the objective to correct the admin's mistakes. It is assumed that the AI coach has perfect knowledge and therefore consistently produces the correct risk assessment (X60) based on the same contextual states X4-X7. Unlike the admin's W-states that deteriorate due to stess, the AI coach's connection weights are fixed (X65-X69) ensuring consistent performance. The AI coach also decides between determining a real or malicious user (X61, X62). The role of the admin supervisor starts with two monitoring states that compare the classifications of the AI coach and the admin (X63, X64). These states become active if a discrepancy is detected. Activation of these monitoring states trigger the second-order adaptation **WWW**-states, which form the connection between the AI coach **W**-states and the admin **W**-states. The **WWW**-states control the communication at the knowledge level of the **W**-states: if they are active, the AI coach communicates the value of its corresponding **W**-state (one from **AIC_W**1 to **AIC_W**5) to the admin's **W**-states (**W**1-**W**5) so that the admin can adjust its own **W**-states to the values of the AI coach **W**-states. Besides this communication, the discrepancy and therefore the activation of the **WWW**-states also alleviate the admin's stress level by temporarily reducing it to zero. Lastly the **WWW**-states restore the persistence level (**MW**1-**MW5**) of the admin W-states. These two actions enable the admin to restore its decision making capacity allowing better future assessments. However, as soon as the communication ceases and the admin will be operating on its own again, the admin will experience stress again which will reinitiate the degradation loop.

In summary, the temporal-causal network integrates system-level decision-making, human decision-making under cognitive constraints, and an AI supervisor who coaches the admin. These three elements together form an unified framework. The model simulates interactive dynamics and incorporates structural adaptation and recovery characteristics.

5 Simulation Results

Various simulations were carried out to demonstrate how the model works. For each situation, a graph of the system and the admin was demonstrated, along with a simulation of the admin and the AI coach in order to show how the AI coach affects the mental processes of the admin. For Scenario 1, the complete model of the two variants can be seen. After that, each scenario only shows the important states that change over time or that influence the changes in the states.

The following values were used for each simulation. The connection of the admin assessment nodes to the stress level are as follows: *Admin assessment of Device of login*: 0.35, *Admin Assessment of Location of login*: 0.2, *Admin assessment of Time of Login*: 0.15, *Admin Assessment of Frequency of Login*: 0.2 and *Admin Assessment of System Assessment*: 0.1. The connection strength of stress to **MW1**: -0.01, **MW2**: -0.3, **MW3**: -0.08, **MW4**: -0.06 and **MW5**: -0.12. These values differ because the stress level has a different impact on each **W**-state between the *admin assessment*-nodes and the admin risk assessment. In addition, the speed factor of the **W**-states **W1-W5** is 0.01 and the speed factor of the **MW**-states is 0.05. The initial values of the **W**-states and

the **AIC-W**-states are as follows: **W1/AIC-W1**; 0.9, **W2/AIC-W2**: 0.65, **W3/AIC-W3**: 0.4, **W4/AIC-W4**: 0.7 and **W5/AIC-W5**: 0.3.

In these scenarios, independent variables are the contextual risk scores of the device, location, time and frequency, and the correct and incorrect credentials. The dependent variables are the *Admin Risk Assessment* and the decisions that follow from it.

Scenario 1: Figs. 2, 3, 4, 5, 6 and 7**.**

In Fig. 2 the contextual states have a high anomaly/risk score, around 0.9 each. The first login cycle gives a peak admin risk assessment of 0.8. Due to the high risk score of the contextual variables, the stress level is very high, meaning that the **MW**-states drop fast and consequently the **W**-states as well. This can be seen in the Figs. 4, 5 and 6 by the fast decline of all these states. Without the AI coach the admin has no support who can help when the risk score goes below 0.4. This means that because of the stress, the MW-states, the W-states and therefore the risk score as well all reach a level of zero over time.

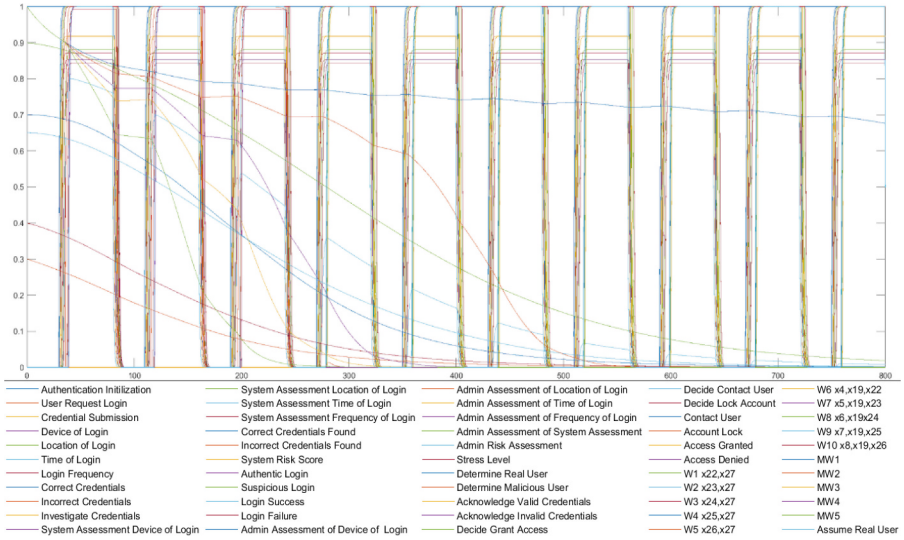

Fig. 2. Complete plot of Scenario 1 without AI coach.

The user entered the correct credentials and the admin initially assesses the user correctly as malicious. However, without the help of the admin the risk score goes below 0.4 and the admin decides the user is real. In combination with the acknowledgement of the correct credentials, the user is mistakenly granted access.

Now the AI coach is added to the model. Here the impact that the AI coach has on the admin's stress level, **MW**-states and the **W**-states can be seen in Figs. 5, 6 and 7. As soon as the admin risk assessment goes below the 0.4 threshold, which is around $t = 280$, the AI coach monitors a change in the decision-making of the admin. At this moment the AI coach communicates the strength of his W-states to the admin and will recover its stress level to 0 and its **MW**-states to 1. Because of this the admin's risk

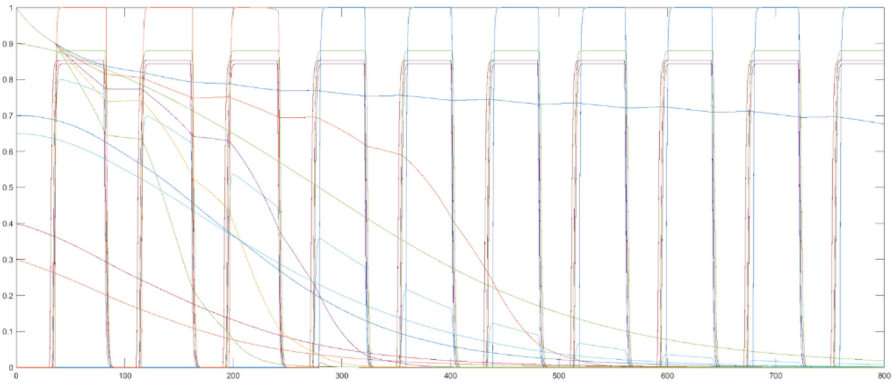

Fig. 3. Plot of the Admin Assessments states, Admin Risk Assessment states, Stress Level, **W**-states and **MW**-states of Scenario 1 without AI coach.

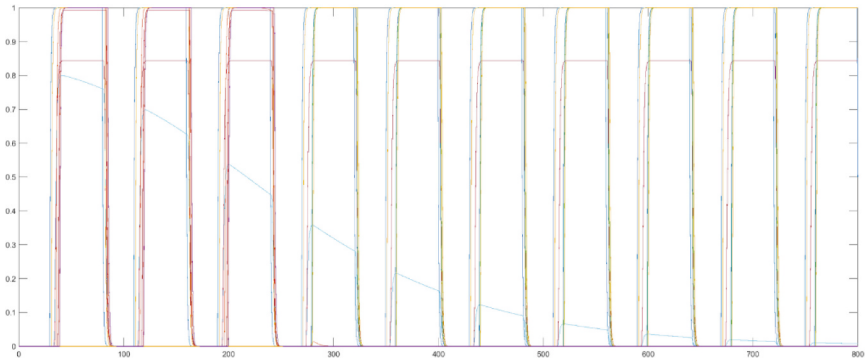

Fig. 4. Plot of the Admin Risk Assessment states and the decision/result states of Scenario 1 without AI coach.

assessment increases again to a similar value as the initial risk assessment of the first cycle. A pattern emerges in which stress comes into play again and the **MW-** and the **W**-states decrease again until the AI coach will intervene. Now with the help of the AI coach, when the admin mistakenly assesses the user as real, the AI coach will step in and help the admin restore its **W**-states and **MW**-states allowing the risk score to increase and make the correct assessment again. This way the user will not be given access and access will continue to be denied.

Scenario 2: Figs. 8, 9, 10 and 11

In Scenario 2 the contextual risk scores of the states are 0.5. As a result, the initial risk score of the admin is just below 0.5 which can be seen in Fig. 8. This means that the admin initially makes the right assessment, but because the risk score is very close to the 0.4 threshold and the stress level causes the risk score estimate to deteriorate, the 0.4 threshold will be exceeded much sooner and the admin will have to intervene more quickly. This already happens around $t = 125$ compared to $t = 280$ in Scenario 1, where the risk scores were significantly higher. However the credentials given by the user are

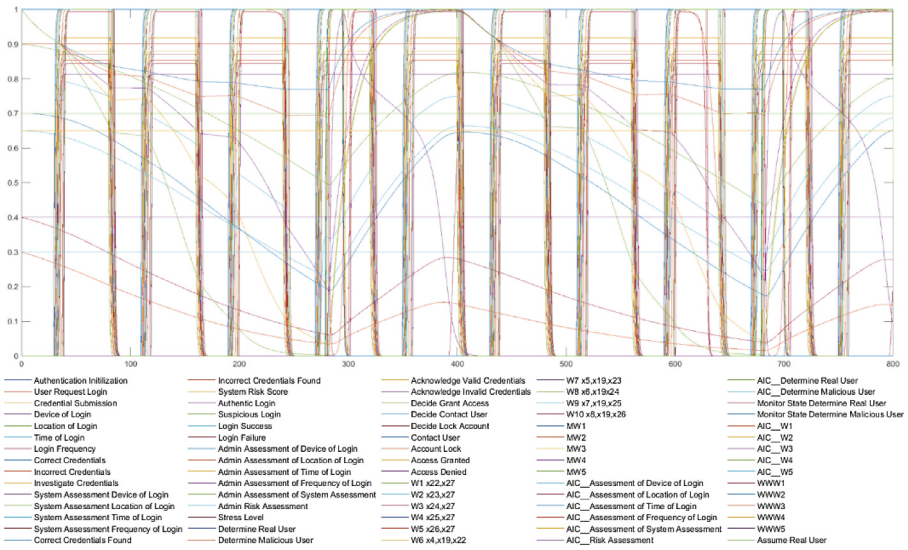

Fig. 5. Complete plot of Scenario 1 with AI coach.

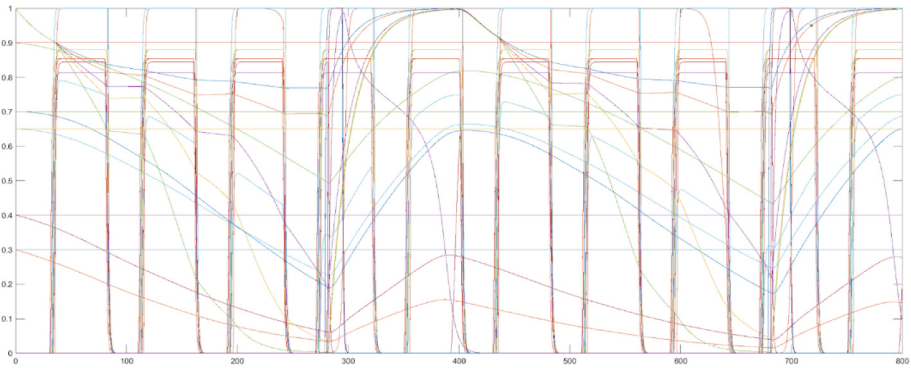

Fig. 6. Plot of the Admin and AI Assessments states, Admin and AI Risk Assessment states, Stress Level, **W**-states, **MW**-states, monitoring states and **Www**-states of Scenario 1 with AI coach.

incorrect. This means that no matter the assessment of the admin, correct or incorrect, the user will be denied access, shown in Fig. 9.

Even though the outcome of the simulation does not change because the incorrect credentials were given in the submission, the AI coach still coaches the admin on his stress level, **W**-states and **MW**-states to correct his risk assessment which can be seen in Fig. 10. The decisions made by the admin are plotted in Fig. 11.

Scenario 3: Fig. 12

In this simulation the contextual states are very low, meaning little to no risk. When this happens, the system will make its risk assessment. When this value is lower than

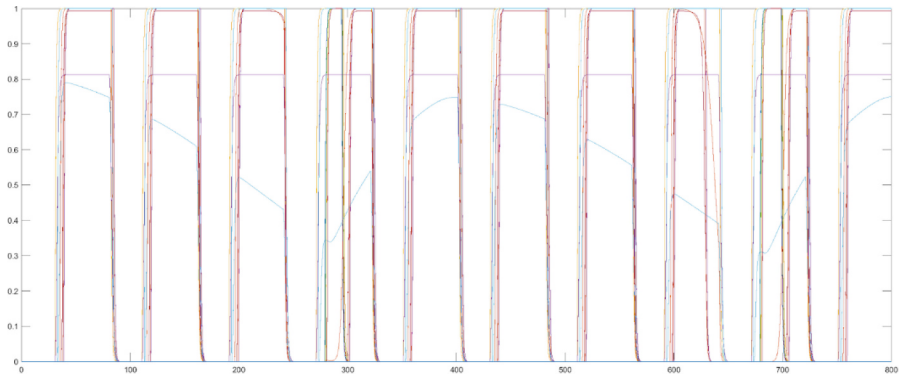

Fig. 7. Plot of the Admin Risk Assessment states and the decision/result states of Scenario 1 with AI coach.

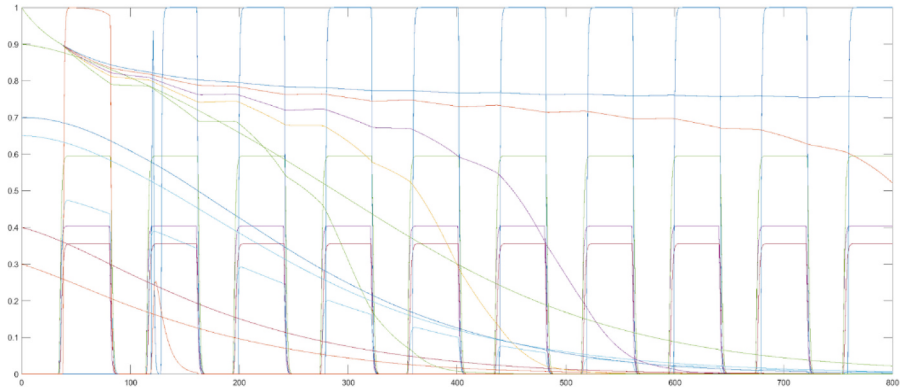

Fig. 8. Plot of the Admin Assessments states, Admin Risk Assessment states, Stress Level, **W**-states and **MW**-states of Scenario 2 without AI coach.

0.15, which in this simulation it is, shown in Fig. 12, the system will assess this user as real and the admin is not asked to take a look at the situation as there is no need to because there is no risk detected. This also means that since the admin does not come into play, the AI coach has no role in this simulation. The credentials given by the user are correct, so in combination with the decision of the system that the user is real, the user is granted access.

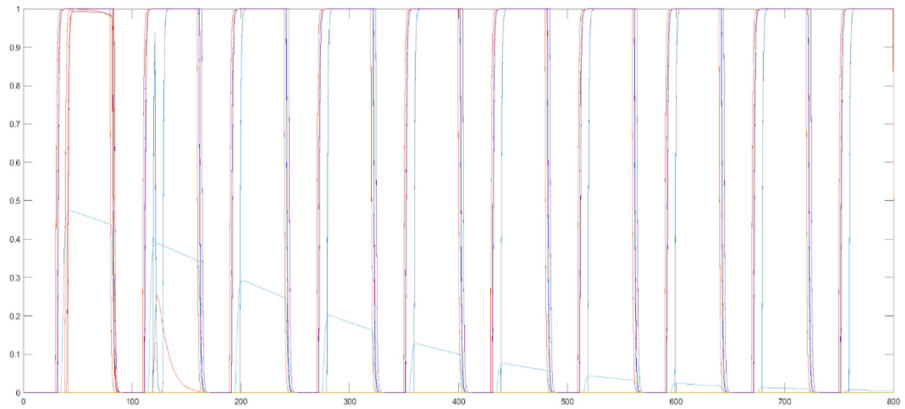

Fig. 9. Plot of the Admin Risk Assessment states and the decision/result states of Scenario 2 without AI coach.

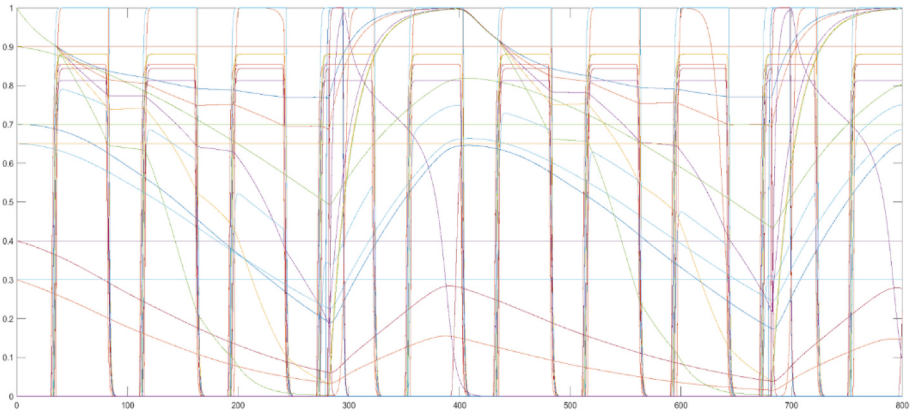

Fig. 10. Plot of the Admin and AI Assessments states, Admin and AI Risk Assessment states, Stress Level, **W**-states, **MW**-states, monitoring states and **Www**-states of Scenario 2 with AI coach.

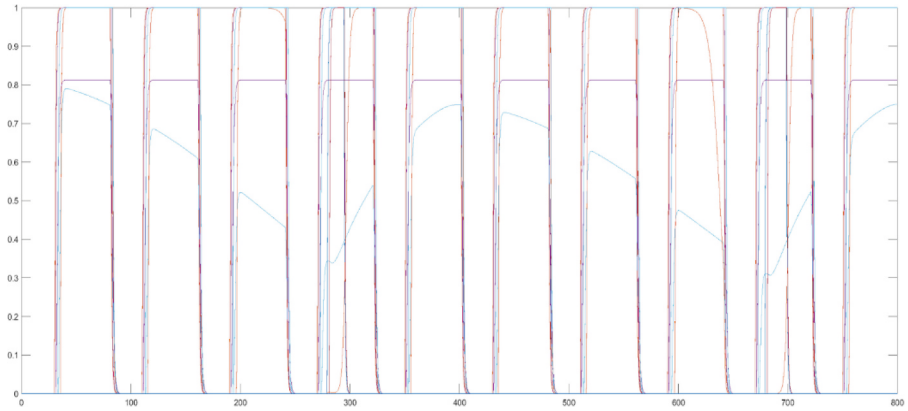

Fig. 11. Plot of the Admin Risk Assessment states and the decision/result states of Scenario 2 with AI coach.

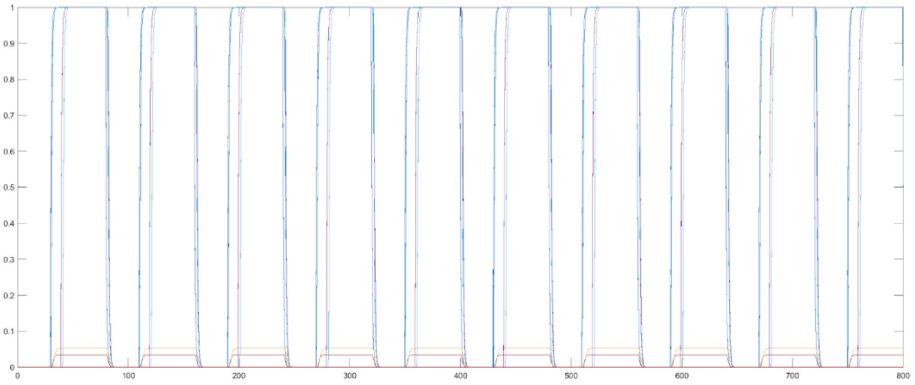

Fig. 12. Plot of the System Risk score and the decision/result states of Scenario 3 without AI coach.

6 What-If Analysis and Risk Assessment

A What-If analysis has been performed by systematic variation of some of the most relevant network characteristics of the model, as described in [12, 19], see Tables 2, 3, 4 and 5. A What-If analysis is a data simulation with the goal to inspect the behavior of a complex system. It measures how changes in a set of independent variables impact dependent variables [18]. These independent variables that influence the most important outcome states are the If factors and the dependent influenced states are the What factors. For this What-If analysis three different If factors have been chosen. The first If factor consists of four values which are the risk scores of the contextual states *Device of Login, Location of Login, Time of Login* and *Login Frequency.* These states are independent of each other, meaning that they do not influence each other and that they can differ depending on the situation. However, for this analysis I have chosen to set each with the same risk score in order to get a better overview when comparing the results. This

If factor has three options: low, mid and high, corresponding to 0.5, 0.7 and 0.9. These values, especially the low and mid values, are not really low and mid. However, if these If factors have lower values than 0.4, the results of the What factors would be less interesting to compare. The second If factor is the connection weight from the *Admin Assessment*-nodes X22-X26 to the *Stress Level*-node X27. These connections decide how much stress the admin experiences while processing the risk assessment. This If factor has a low value of 0.1 and a high value of 0.25. The speed factor of the **MW**-states X50-X54 is the third If factor. The **MW**-states are the persistence levels of the **W**-states of the admin. A higher speed factor means that persistence will decrease faster and therefore the **W**-states as well. The low value chosen is 0.025 and the high value is 0.075.

The analysis measures two What factors: The first is the *Admin Risk Assessment* score, this is the highest value that the *Risk Assessment Score* reaches during the first login cycle. The reason for choosing the peak of the first cycle is because the admin has been functioning under stress for the shortest period of time and will therefore make the best assessment without the help of the AI coach. The second What factor is the time at which the risk score drops below 0.4. The admin and the AI coach have the same threshold for deciding on malicious users, which is 0.4. This means that as soon as the admin's risk assessment score falls below 0.4, his decision changes from malicious to real user. This is also the moment when the AI coach intervenes and allows the **W**-states and persistence levels to grow again, reducing the stress on the admin.

In conclusion, this analysis aims to determine how the contextual anomaly/risk scores, the connection from the *admin assessment*-nodes to the stress level, and how the speed factor of the **MW**-states affect the *Admin Risk Assessment*.

Table 2. Overview of If factor options and their labels, values and probabilities.

If factors	If factor options	If factor option labels	If factor option values	If factor option probabilities
Contextual states anomaly/risk score (X4-X7)	If factor 1 option 1	Low	0.5	0.7
	If factor 1 option 2	Mid	0.7	0.2
	If factor 1 option 3	High	0.9	0.1
Connection weight from *admin assessment*-nodes to *Stress Level* (X22–26) → X28	If factor 2 option 1	Low	0.1	0.6
	If factor 2 option 2	High	0.25	0.4
Speed factor of **MW**-states (X50-X54)	If factor 3 option 1	Low	0.025	0.8
	If factor 3 option 2	High	0.075	0.2

Table 3. Results of the What-if Analysis.

If factor 1 options: contextual risk score	If factor 2 options: Admin assessment impact on stress	If factor 3 options: Speed factor MW-states	What factor 1: Risk assessment peak	What factor 2: Risk assessment < 0.4
Low 0.5	Low 0.1	Low 0.025	0.4634	123.5
		High 0.075	0.4548	95
	High 0.25	Low 0.025	0.4634	119
		High 0.075	0.4548	89
Mod 0.7	Low 0.1	Low 0.025	0.7223	294
		High 0.075	0.7129	229
	High 0.25	Low 0.025	0.7224	274
		High 0.075	0.7129	193
High 0.9	Low 0.1	Low 0.025	0.7953	330
		High 0.075	0.7868	254
	High 0.25	Low 0.025	0.7952	309
		High 0.075	0.7866	220

Examples of two of these What-If scenarios are as follows:

1. For the first example, the If factors are as follows: If factor 1, the contextual risk scores: Mid (0.7). If factor 2, the connection strength of the admin assessment nodes on the stress level: High (0.25). If factor 3, the speed factor of the **MW**-states: High (0.075). In the simulation it can be seen that the stress level is really high: around 0.9. This means that its impact on the persistence levels, the **MW**-states is high. Besides that, the speed factor of the **MW**-states is also high meaning that the states change fast upon causal impact. This can be seen in the figure by the fast decrease in the **MW**-states which causes the **W**-states to drop fast as well. As a result, the admin risk assessment drops significantly and already drops below the 0.4 threshold at $t = 193$.
2. For the second example, the If factors are as follows: If factor 1, the contextual risk scores: Mid (0.7). If factor 2, the connection strength of the admin assessment nodes on the stress level: Low (0.10). If factor 3, the speed factor of the **MW**-states: Low (0.025). This second example has the same first If factor: 0.7. But the results of the two simulations differ significantly. This Is entirely related to how the admin manages stress. There are two reasons for this. The first is the second If factor, the connection strength of the admin assessment nodes on the stress level, which is low. This means that despite the high risk of the contextual variables, this has little influence on the admin's stress level which is only 0.3. Because the stress level is low, this has little influence on the persistence level, the **MW**-states, and therefore the **W**-states will decrease less quickly. In addition, the speed factor of the **MW**-states, the third If factor, is also low, which means that the speed of change or causal impact is low. The figure shows that the **MW**-states hardly decrease and only in a period of approximately

300 s the **MW**-states fall below 0.8. As a result, the **W**-states decrease much much more slowly, and the admin risk assessment accordingly. Around $t = 330$, the risk score only falls below the threshold of 0.4.

Besides the results of the What factors, a probabilistic What-If analysis has been made by associating probabilities to each If factor option. Independence among the three If factors is assumed so the product rule can be used to calculate the total probability of all combinations of the three If factors.

Table 4. Overview of If factor options combination probabilities.

If factor 1 options: contextual risk score	If factor 2 options: Admin assessment impact on stress	If factor 3 options: Speed factor MW-states	If factor options combination probabilities
Low 0.7	Low 0.6	Low 0.8	0.7*0.6*0.8 = 0.336
		High 0.2	0.7*0.6*0.2 = 0.084
	High 0.4	Low 0.8	0.7*0.4*0.8 = 0.224
		High 0.2	0.7*0.4*0.2 = 0.056
Mod 0.2	Low 0.6	Low 0.8	0.2*0.6*0.8 = 0.096
		High 0.2	0.2*0.6*0.2 = 0.024
	High 0.4	Low 0.8	0.2*0.4*0.8 = 0.064
		High 0.2	0.2*0.4*0.2 = 0.016
High 0.1	Low 0.6	Low 0.8	0.1*0.6*0.8 = 0.048
		High 0.2	0.1*0.6*0.2 = 0.012
	High 0.4	Low 0.8	0.1*0.4*0.8 = 0.032
		High 0.2	0.1*0.4*0.2 = 0.008

To adapt the risk assessment to our research, the two What factors have been divided into two subcategories. The first What factor, the peak of the risk assessment score, now has two categories: a low subcategory and a high subcategory. The low category includes all values below 0.5, and the high category includes all values above 0.75. The same applies to the second What factor, the time at which the admin risk assessment falls below 0.4. The low category includes all simulations in which this occurs before $t = 200$, and the high category includes all simulations that occur after $t = 250$.

Table 5. Risk Assessment.

If factor 1 options	If factor 2 options	If factor 3 options	If factor options combination probabilities	What factor 1a probability	What factor 1b probability	What factor 2a probability	What factor 2b probability
Low 0.7	Low 0.6	Low 0.8	0.336	X 0.336		X 0.336	
		High 0.2	0.084	X 0.084		X 0.084	

(*continued*)

Table 5. (*continued*)

If factor 1 options	If factor 2 options	If factor 3 options	If factor options combination probabilities	What factor 1a probability	What factor 1b probability	What factor 2a probability	What factor 2b probability
	High 0.4	Low 0.8	0.224	X 0.224		X 0.224	
		High 0.2	0.056	X 0.056		X 0.056	
Mod 0.2	Low 0.6	Low 0.8	0.096				X 0.096
		High 0.2	0.024				
	High 0.4	Low 0.8	0.064				X 0.064
		High 0.2	0.016			X 0.016	
High 0.1	Low 0.6	Low 0.8	0.048		X 0.048		X 0.048
		High 0.2	0.012		X 0.012		X 0.012
	High 0.4	Low 0.8	0.032		X 0.032		X 0.032
		High 0.2	0.008		X 0.008		
Sum of probabilities				0.7	0.1	0.716	0.252
Impact indicator				1	8	2	5
Risk assessment				0.7*1 = 0.7	0.1*8 = 0.8	0.716*2 = 1.432	0.252*5 = 1.26

The results from the What-If analysis show how different combinations of the three If factor parameters influence the admin's risk assessment. The results show that high contextual risk scores and high weight connections to and from the stress level lead to faster cognitive degradation. In contrast, with low risk scores and low weight connections, the decline is much slower. Even if the contextual risk scores are high but the stress sensitivity is low and the resilience of the **MW**-states is high, the admin is able to remain accurate for a significantly longer time. On the other hand, the AI coach is an essential part of the model because without its support, the admin's performance will ultimately always decline too much due to stress.

7 Discussion

The results of this study show that the contextual variable and the stress experienced by the admin have a significant impact on the admin's decision-making in the authentication system. Using the temporal-causal network model, various situations were simulated. These provide a clear insight into the decline in the admin's capabilities under prolonged stress, leading to a deterioration in risk assessment and, consequently, the introduction of incorrect choices that can have a negative impact on the authentication system.

The various simulations show that a high risk score for the contextual variables has an increasing influence on the admin's stress level, causing the **MW**-states and thereby also the W-states to decrease, which deteriorates the accuracy of the risk assessment. Without the support of the AI coach, the admin will eventually make the wrong decision, which will have negative consequences when a malicious user submits the correct credentials.

The addition of the AI coach has a positive effect on the admin's decision-making and thus a positive effect on the model. By monitoring differences in choices between the admin and the AI coach, the coach is able to make corrections. By reducing stress, restoring persistence levels and communicating the correct values of the **W**-states, the admin can recover. The AI coach has a stabilising role for the admin, whereby misclassifications are corrected.

The results show that a human-administered authentication system is vulnerable to psychological factors such as stress. This is something that the AI coach is not vulnerable to, allowing the coach to effectively compensate for the admin's failures. This model is a human-AI collaboration model in which the AI coach does not replace the human but enhances their performance by intervening when the admin's judgement fails.

Although the model provides a good insight into the functioning of a human-AI collaboration authentication system, it does have limitations. The simulations rely on predefined parameters and assumptions about the capabilities of the system, the admin and the AI coach. Examples include the thresholds of the decisions, the influence of the assessment nodes on the stress level, the influence of stress on the persistence levels and the values of the **W**-states of the AI coach, which are assumed to be perfect. In the real world, an AI coach will not be perfect due to limitations and uncertainties and will adapt to these over time, which does not happen in our model. In addition, the model has a simple psychological and cognitive design in which only stress is modelled. Factors such as fatigue, emotional state and experience do not play a role, whereas in the real world they will also influence the admin's assessments.

Building on these findings, several areas for future research could be explored. For example, there are direct improvements that can be applied to the current model and additions that can be made to make the model more realistic and functional. For example, the admin could be given a more realistic cognitive and psychological design in which elements such as fatigue and experience are added. A mental learning process could be added so that the admin will get adaptive thresholds for decision-making options and more internal adaptive connections so that he does not keep making the same mistakes. In addition, experiments could be conducted with the addition of imperfect information and uncertainties that are not included in the current model but do play a role in the real world. Various cyberattack strategies could also be simulated, each of which could have a different effect on the functioning of the admin and the AI coach. These examples of expanding the model could be tested against empirical user behaviour data to validate [30] and calibrate the simulation parameters.

In conclusion, this study investigated the role of declining human cognitive ability and the added value of an AI coach in a human-centred decision model in an authentication system. The model offers dynamic interaction between human and AI, which forms the basis for resilient cybersecurity designs to which future developments can be applied.

One of the next steps is to actually build such an AI Coach and integrate it with capabilities for emotion contagion, interpersonal emotion regulation, empathic responding and adaptive capabilities in bonding with the human user such as computationally explored in [13, 14].

Appendix for the Model with AI Coach

For the complete specification by role matrices, see Linked Data at URL: https://www.researchgate.net/publication/394150421. (For an overview of all states, see Table 6)

Table 6. Overview of all states.

Nr	State name	Explanation	Level
X1	Authentication Initilization	Authentication system start up	Base Level
X2	User Request Login	User requests to log in	
X3	Credential Submission	User submits its credentials	
X4	Device of Login	System investigates the device of login and assigns risk value	
X5	Location of Login	System investigates the location of login and assigns risk value	
X6	Time of Login	System investigates the time of login and assigns risk value	
X7	Login Frequency	System investigates the frequency of the login and assigns risk value	
X8	Correct Credentials	Saved up correct password that belongs to the userID in the database	
X9	Incorrect Credentials	Marker for all passwords that do not match with userID	
X10	Investigate Credentials	System will investigate the credentials by comparing the UserID with the given password	
X11	System Assessment Device of Login	System makes assessment of the device of login risk value	
X12	System Assessment Location of Login	System makes assessment of the location of login risk value	
X13	System Assessment Time of Login	System makes assessment of the time of login risk value	
X14	System Assessment Frequency of Login	System makes assessment of the frequency of the login risk value	
X15	Correct Credentials found	Password of the corresponding userID matches	
X16	Incorrect Credentials found	Password of the corresponding userID do no match	

(*continued*)

Table 6. (*continued*)

Nr	State name	Explanation	Level
X17	System Risk Score	System makes overall risk assessment based on the X11-X14 risk assessments	
X18	Authentic Login	System determines the user is authentic	
X19	Suspicious Login	System determines the user is suspicious	
X20	Login Succes	Login is succesfull if correct credentials are found and the system assesses the user as authentic	
X21	Login Failure	Login failure if incorrect credentials are found	
X22	Admin Assessment of Device of Login	Admin makes assessment of the device of login risk value	
X23	Admin Assessment of Location of Login	Admin makes assessment of the location of login risk value	
X24	Admin Assessment of Time of Login	Admin makes assessment of the time of login risk value	
X25	Admin Assessment of Frequency of Login	Admin makes assessment of the frequency of the login risk value	
X26	Admin Assessment of System Assessment	Admin makes assessment of the overal system risk assessment	
X27	Admin Risk Assessment	Admin makes overall risk assessment based on the X22-X26 risk assessments	
X28	Stress Level	Level of stress the admin experiences	
X29	Determine Real User	Admin determines the user is authentic	
X30	Determine Malicious User	Admin determines the user is malicious	
X31	Acknowledge Valid Credentials	The admin acknowledges that the correct credentials are given	
X32	Acknowledge Invalide Credentials	The admin acknowledges that the incorrect credentials are given	

(*continued*)

Table 6. (*continued*)

Nr	State name	Explanation	Level
X33	Decide Grant Access	The admin decides to grant access	
X34	Decide Contact User	The admin decides to contact the user	
X35	Decide Account Lockout	The admin decides to lock the account	
X36	Contact User	The user gets contacted	
X37	Account Lockout	The account gets locked	
X38	Access Granted	The user gets access granted	
X39	Access Denied	The user gets access denied	
X40	**W1** x22,x27	Self-model state for the weight of the connection between (X22) 'Assessment of Device of Login' and (X27) 'Admin Risk Assessment', influenced by (X50) **MW1**	First-order adaptation level
X41	**W2** x23,x27	Self-model state for the weight of the connection between (X23) 'Assessment of Location of Login' and (X27) 'Admin Risk Assessment', influenced by (X51) **MW2**	
X42	**W3** x24,x27	Self-model state for the weight of the connection between (X24) 'Assessment of Time of Login' and (X27) 'Admin Risk Assessment', influenced by (X52) **MW3**	
X43	**W4** x25,x27	Self-model state for the weight of the connection between (X25) 'Assessment of Frequency of Login' and (X27) 'Admin Risk Assessment', influenced by (X53) **MW4**	
X44	**W5** x26,x27	Self-model state for the weight of the connection between (X26) 'Assessment of System Assessment' and (X27) 'Admin Risk Assessment', influenced by (X54) **MW5**	

(*continued*)

Table 6. (*continued*)

Nr	State name	Explanation	Level
X45	**W6** x4,x19,x22	Self-model state for the weight of the connection between (X4) 'Device of Login' and (X22) 'Assessment of Device of Login', influenced by (X19) 'Suspicious Login'	
X46	**W7** x5,x19,x23	Self-model state for the weight of the connection between (X5) 'Location of Login' and (X23) 'Assessment of Location of Login', influenced by (X19) 'Suspicious Login'	
X47	**W8** x6,x19x24	Self-model state for the weight of the connection between (X6) 'Time of Login' and (X24) 'Assessment of Time of Login', influenced by (X19) 'Suspicious Login'	
X48	**W9** x7,x19,x25	Self-model state for the weight of the connection between (X7) 'Login Frequency' and (X25) 'Assessment of Frequency of Login', influenced by (X19) 'Suspicious Login'	
X49	**W10** x8,x19,x26	Self-model state for the weight of the connection between (X17) 'System Risk Score' and (X26) 'Assessment of System Assessment', influenced by (X19) 'Suspicious Login'	
X50	**MW1**	Self-model state for persistence factor μW1 x22,x27 for self-model state W1 x22,x27, influenced by (X28) 'Stress Level'	Second-order adaptation level
X51	**MW2**	Self-model state for persistence factor μW2 x23,x27 for self-model state W2 x23,x27, influenced by (X28) 'Stress Level'	

(*continued*)

Table 6. (*continued*)

Nr	State name	Explanation	Level
X52	**MW3**	Self-model state for persistence factor μW3 x24,x27 for self-model state W3 x24,x27, influenced by (X28) 'Stress Level'	
X53	**MW4**	Self-model state for persistence factor μW4 x25,x27 for self-model state W4 x25,x27, influenced by (X28) 'Stress Level'	
X54	**MW5**	Self-model state for persistence factor μW5 x26,x27 for self-model state W5 x26,x27, influenced by (X28) 'Stress Level'	
X55	AIC_assessment of Device of Login	AI coach makes assessment of the device of login risk value	Base level
X56	AIC_assessment of Location of Login	AI coach makes assessment of the location of login risk value	
X57	AIC_assessment of Time of Login	AI coach makes assessment of the time of login risk value	
X58	AIC_assessment of Frequency of Login	AI coach makes assessment of the frequency of the login risk value	
X59	AIC_assessment of System Assessment	AI coach makes assessment of the overal system risk assessment	
X60	AIC_Admin Risk Assessment	AI coach makes overall risk assessment based on the X55-X59 risk assessments	
X61	AIC_Determine Real User	AI coach determines the user is authentic	
X62	AIC_Determine Malicious User	AI coach determines the user is malicious	
X63	Monitor Admin Determine Real User	Monitoring state for AI coach 'determine real user' and admin 'determine real user'	
X64	Monitor Admin Determine Malicious User	Monitoring state for AI coach 'detemine malicious user' and admin 'determine malicious user'	

(*continued*)

Table 6. (*continued*)

Nr	State name	Explanation	Level
X65	**AIC_W1**	Ai coach self-model state for the weight of the connection between (X55) 'AIC_Assessment of Device of Login' and (X60) 'AIC_Risk Assessment'	First-order adaptation level
X66	**AIC_W2**	Ai coach self-model state for the weight of the connection between (X56) 'AIC_Assessment of Location of Login' and (X60) 'AIC_Risk Assessment'	
X67	**AIC_W3**	Ai coach self-model state for the weight of the connection between (X57) 'AIC_Assessment of Time of Login' and (X60) 'AIC_Risk Assessment'	
X68	**AIC_W4**	Ai coach self-model state for the weight of the connection between (X58) 'AIC_Assessment of Frequency of Login' and (X60) 'AIC_Risk Assessment'	
X69	**AIC_W5**	Ai coach self-model state for the weight of the connection between (X59) 'AIC_Assessment of System assessment' and (X60) 'AIC_Risk Assessment'	
X70	**WWW1**	Self-model state for the weight of the connection beween AIC_W1 and W1 x22,x27, influenced by the monitoring states X63 and X64	Second-order adaptation level
X71	**WWW2**	Self-model state for the weight of the connection beween AIC_W2 and W2 x23,x27, influenced by the monitoring states X63 and X64	

(*continued*)

Table 6. (*continued*)

Nr	State name	Explanation	Level
X72	**WWW3**	Self-model state for the weight of the connection beween AIC_W3 and W3 x24,x27, influenced by the monitoring states X63 and X64	
X73	**WWW4**	Self-model state for the weight of the connection beween AIC_W4 and W1 x25,x27, influenced by the monitoring states X63 and X64	
X74	**WWW5**	Self-model state for the weight of the connection beween AIC_W5 and W5 x26,x27, influenced by the monitoring states X63 and X64	
X75	Assume Real User	Model assumes a real user unless maliciousness is decided	Base level

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Early Detection of ASD Risk in Children Through a Web Application with Random Forest

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Abstract. Autism Spectrum Disorder (ASD) is a developmental disorder that affects social communication and behavior. Early detection is crucial for initiating interventions that promote child development. However, in resource-limited settings such as Peru, access to specialized diagnostic tools is scarce. This paper presents a web application designed for parents and caregivers to estimate the risk of Autism Spectrum Disorder (ASD) in children aged 12 to 36 months, using the Q-CHAT-10 questionnaire and a Random Forest machine learning model. The system was trained with a real-world, public dataset composed of 1,560 records collected through online forms and mobile applications. The model achieved an accuracy of 97.76%, a precision of 98.14%, a sensitivity of 98.60%, an F1-score of 98.37%, and an AUC-ROC of 99.79%, demonstrating a high predictive capacity. The tool allows for the generation of an automated, immediate prediction accessible from any device, offering a noninvasive alternative for early ASD risk screening. Although the results are promising, further clinical validation in real-world settings is required to support its use in public health contexts.

Keywords: Autism Spectrum Disorder · Early Detection · Machine Learning · Random Forest · Web Application

1 Introduction

Autism Spectrum Disorder (ASD) is a neurodevelopmental condition characterized by persistent difficulties in communication and social interaction, as well as by restricted and repetitive patterns of behavior, interests, or activities. These manifestations may include, for example, alterations in eye contact, the use of gestures, understanding of social norms, or insistence on specific routines. Symptoms are often present from the earliest stages of development and can have a clinically significant impact on daily functioning [1]. Several studies have shown that the earlier intervention is initiated, the better the long-term outcomes. In fact, it has been observed that starting behavioral therapies before the age of two can be especially effective, highlighting the need for early detection of ASD [2]. In low- and middle-income countries, such as Peru, multiple barriers still persist that hinder the early diagnosis of thousands of children, including a shortage of specialists, lack of awareness of warning signs by parents and caregivers, and limited access to mental health services for children [3].

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One of the validated instruments for the early detection of ASD is the Q-CHAT-10 (Quantitative Checklist for Autism in Toddlers - 10 items), a brief questionnaire completed by parents or caregivers that assesses behaviors associated with the autism spectrum in young children [4]. Early detection and intervention are essential, as it has been proven that starting treatments before the age of three can significantly improve the cognitive, social, and communicative skills of children with ASD [5]. However, currently, diagnosis is usually made on average between 4 and 5 years of age, when interventions may be less effective. Therefore, developing tools that facilitate the early identification of autism is essential to offer children better opportunities for development and quality of life [5]. A recent study conducted at a national referral hospital in Lima, Peru revealed that the average age of ASD diagnosis was 3.83 years, well beyond the optimal window for early intervention. Alarmingly, over 75% of the children lacked both a disability certificate and a complete psychological evaluation, and only 9.8% of those enrolled in regular education had access to inclusive support. These findings underscore a critical gap in Peru's capacity to detect and respond to ASD in a timely and comprehensive manner, leaving many children without the tools and services they need to thrive [6]. Early detection allows for timely interventions that favor the development of social, communicative, and cognitive skills, reducing the impact of symptoms and facilitating more effective integration into their environment [7]. It also contributes to reducing the emotional and economic burden for both families and the healthcare system by reducing the need for more intensive interventions at later stages [7]. Intervening during critical periods of brain plasticity allows for the optimization of child development and increases the likelihood of a better prognosis [8]. In this context, this paper proposes a web application that uses the Random Forest algorithm to estimate ASD risk from responses to the Q-CHAT-10 questionnaire, in order to facilitate an early, accessible, and non-invasive screening process. The tool is aimed at children between 12 and 36 months of age, an age range in which early interventions have proven to be most effective. This decision is based both on the high performance obtained by the model during its evaluation and on the nature of the dataset used for its training, which is composed exclusively of records corresponding to this stage of child development.

2 Related Works

Several studies have explored the application of machine learning algorithms for the early detection of ASD. These studies use data derived from screening questionnaires, physiological signals, and images, in combination with supervised classification techniques.

For example, machine learning models were evaluated on four datasets applied to populations ranging from toddlers to adults, Random Forest (RF) method demonstrated outstanding performance in toddler dataset, achieving an incredible 100% accuracy, making it the best possible option for the task, as well as being effective for the early detection of ASD [9]. However, such perfect accuracy is uncommon in practice and may reflect overfitting or limited variability in the datasets. Likewise, models such as Random Forest (RF), XGBoost, and other deep neural networks were analyzed, applied to tabular data and also to facial images of children. This highlights the efficiency and simplicity

of RF in well-preprocessed scenarios and with data carefully annotated by professionals. Furthermore, the study employs transfer learning techniques with models such as DenseNet169, which is especially effective for facial image analysis due to its ability to extract deep and relevant features. Likewise, GTO (Golden Tortoise Optimization) metaheuristic optimization is used to tune the model's hyperparameters, improving its accuracy and generalization. This approach combines the power of deep learning with advanced optimization strategies, making it a robust and promising proposal for the early detection of autism spectrum disorders [10]. Regarding the integration of these models into technological solutions, there is the case in which an interactive web application was proposed that uses optimized logistic regression and SHAP interpretability, achieving an accuracy of 99% in ASD prediction [11]. Similarly, a mobile application called App-Autismo was developed, oriented for use by primary care professionals, and validated by content and usability experts [12]. Furthermore, an approach to early detection of autism spectrum disorder was proposed using deep learning models based on pre-trained convolutional neural networks such as ResNet50 and MobileNetV2, applied to facial images. Transfer learning and data augmentation techniques, including normalization and flipping, were used to improve the accuracy and robustness of the model. Apart from that, different configurations such as input size and models were evaluated, achieving an accuracy higher than 92%. This approach not only facilitates highly efficient, automatic detection but also lays the groundwork for future applications in early diagnosis, which can potentially be integrated into mobile platforms for broader access [13]. Furthermore, a home-based therapeutic intervention platform for individuals with autism was developed, incorporating personalized modules and an online monitoring system, allowing for continuous assessment tailored to the individual needs of each user. This solution has proven effective in improving social and communication skills, fundamental aspects for social integration and personal development. Furthermore, it facilitates the active participation of parents and therapists in the therapeutic process, reinforcing the continuity of treatment outside the clinical setting and promoting a collaborative and user-centered approach [14].

Taken together, these studies agree that the combination of classification algorithms such as Random Forest, rigorous preprocessing, and integration into accessible platforms (such as web or mobile applications) allows for the development of effective, explainable, and low-cost tools for the early detection of ASD and its various layers. However, most of these solutions have been developed and validated in contexts very different from Latin America, where access to technology, specialized personnel, and diagnostic processes can vary considerably between the various countries in our region. Furthermore, some studies report perfect metrics (100% accuracy), which could refer to either small sample sizes or a lack of external validation, compromising the model's generalizability. Metrics such as accuracy, log-loss (uncertainty of predictions), F1 (Precision/Recall), sensitivity, and specificity were evaluated in different data sets, ensuring that the models did not overfit and were reliable for real-time diagnoses in resource-limited settings. However, most of these solutions have been developed and validated in contexts very different from our region. Furthermore, some studies report perfect metrics (100% accuracy), which could be due to small sample sizes or a lack of external validation, compromising the model's generalizability [15].

Overall, these studies agree that the combination of classification algorithms such as Random Forest, together with eye-tracking metrics related to facial perception and social attention, allows for the development of effective, explainable, and low-cost tools for the early detection of ASD [16].

3 Methodology

This study proposes an interactive web application designed to predict the risk of Autism Spectrum Disorder (ASD) in children between 12 and 36 months of age, using the answers to the Q-CHAT-10 questionnaire and the Random Forest machine learning algorithm as input. The tool is aimed at parents or caregivers and seeks to facilitate an early, accessible, and non-invasive screening process, especially in contexts where access to specialists is limited and/or expensive. The choice of the age range is based on clinical evidence supporting the effectiveness of early interventions, as well as on the characteristics of the dataset used, which contains exclusively records of children within that age range (Fig. 1).

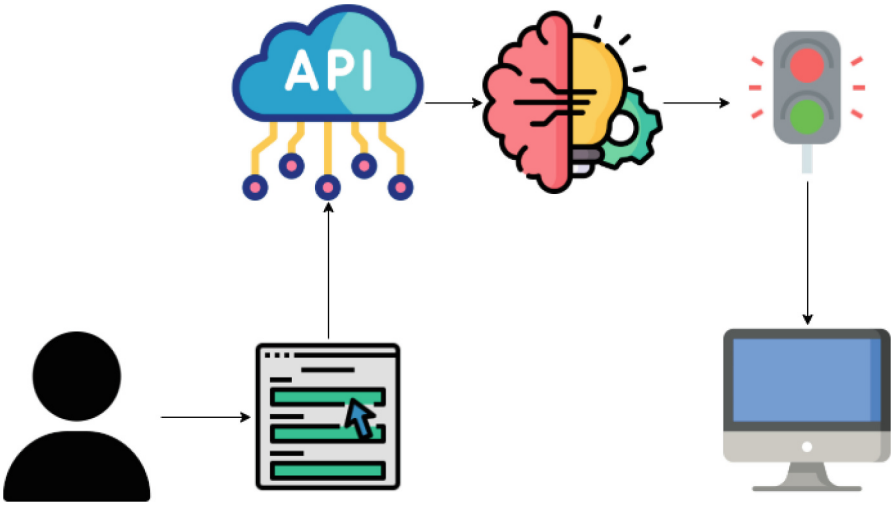

Fig. 1. Diagram of the proposed solution.

3.1 Dataset and Preprocessing

A dataset focused on early diagnosis of ASD was obtained from two public sources available on the Kaggle platform, primarily targeting children between 12 and 36 months old. The first source corresponds to a dataset collected in Saudi Arabia through online forms, focused on detecting early signs of autism. The second source was developed by Dr. Fadi Fayez Thabtah, a specialist in machine learning applied to health, who compiled a clinical dataset with responses to the Q-CHAT-10 questionnaire, along with demographic

variables and family history. Both datasets were combined and preprocessed for use in training and evaluating the classification model.

As part of the process to unify both dataset structures, irrelevant and/or redundant columns were eliminated. After this process, 1,560 results were obtained, each linked to an individual evaluation. The following criteria were taken into account when selecting the samples based on clinical and predictive relevance: A1 to A10 (responses to the Q-CHAT-10 questionnaire), Age_month (age in months), Gender, and Family_mem_with_ASD (family history of ASD). The responses were then numerically categorized, and the target variable was transformed into a binary class indicating the presence or absence of ASD traits. Given that the records were complete and consistent, no imputation or standardization techniques were required. Finally, irrelevant information was eliminated to facilitate efficient model training.

The selected variables are briefly described in table 1.

Table 1. Description of variables selected for the model.

Variable	Description
A1	Does your child look at you when you call his/her name?
A2	How easy is it for you to get eye contact with your child?
A3	Does your child point to indicate that s/he wants something?
A4	Does your child point to share interest with you?
A5	Does your child pretend?
A6	Does your child follow where you're looking?
A7	If you or someone else in the family is visibly upset, does your child show signs of wanting to comfort them?
A8	Would you describe your child's first words as:
A9	Does your child use simple gestures?
10	Does your child stare at nothing with no apparent purpose?
Age_month	Age of the child in months
Gender	Sex (0: female, 1: male)
Family_mem_with_ASD	Family history of ASD (0: no, 1: yes)
Class	Target variable (0: no risk, 1: with risk)

To provide further transparency, descriptive statistics of the dataset are summarized in Table 2. These statistics include the distribution of risk classes, gender, family history of ASD, and the age range of the children.

The dataset showed a higher proportion of children classified as at risk (68.53%) compared to those with no risk (31.47%). This distribution reflects the original collection context and provides valuable information for training models that prioritize sensitivity, an essential characteristic in screening tools where missing true cases is highly undesirable. The average age of the children was 26.8 months (SD = 8.3), within

Table 2. Descriptive statistics of the dataset (N = 1,560).

Variable	Categories	Count	Percentage (%)
Class (ASD risk)	At risk	1069	68.53
	No risk	491	31.47
Gender	Male	892	57.18
	Female	668	42.82
Family history of ASD	Yes	292	18.72
	No	1268	81.28
Age (months)	Mean = 26.8, SD = 8.3, Range = 12–36	–	–

the expected range of 12 to 36 months. With a total of 1,560 records, the dataset offers a solid foundation for experimentation. Nevertheless, since both sources were collected outside Peru, potential cultural and socioeconomic differences should be considered in future validations.

3.2 Machine Learning Model

Several classifiers were evaluated in order to validate the choice of machine learning model, including Logistic Regression, Support Vector Machine, and Random Forest. Random Forest was ultimately selected as the primary model due to its robustness, low susceptibility to overfitting, and its ability to handle categorical and discrete variables. Furthermore, this model is particularly suitable for binary classification tasks with tabular data, as is the case in our study. The detailed comparison of algorithms is presented in Sect. 4.

Moving on to the training process, the dataset was divided into two partitions: 80% for model training and the remaining 20% for evaluation. To ensure reproducibility and preserve the original class distribution, the split was performed using stratified sampling with a fixed random state. This procedure guaranteed that both subsets maintained the same proportion of at-risk and non-risk cases as the complete dataset, avoiding artificial imbalances that could bias the evaluation.

In addition, 5-fold stratified cross-validation was applied. This technique consists of dividing the dataset into k subsets of equal size while preserving the original class distribution; the model is then trained k times, using k–1 subsets for training and the remaining subset for validation in each iteration. At the end, the results obtained in each iteration were averaged to provide a more robust measure of model performance. In this case, cross-validation yielded an average accuracy of 97.24% (±0.66), precision of 97.42% (±0.73), sensitivity (recall) of 98.60% (±0.93), F1-score of 98.00% (±0.48), and AUC of 99.72% (±0.16), confirming the model's consistency and robustness across different data partitions.

To evaluate the performance of the machine learning models, standard metrics such as accuracy, precision, sensitivity (recall), F1 score, and area under the ROC curve (AUC) were used, providing a comprehensive view of their behavior. In addition, a feature importance graph was generated using the Random Forest model's feature_importances_ attribute to identify which questions from the Q-CHAT-10 questionnaire had the greatest impact on predicting ASD risk.

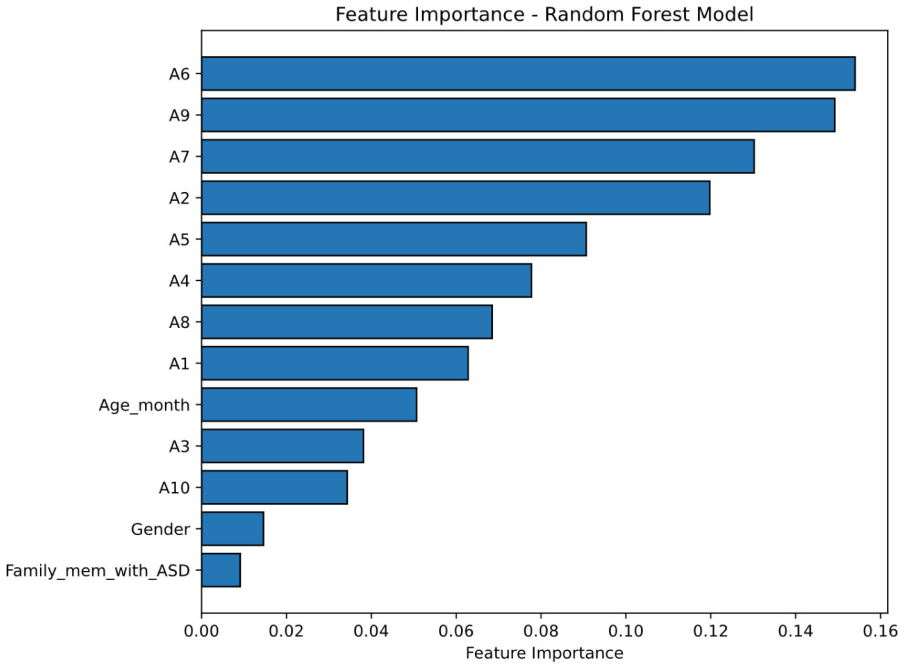

Fig. 2. Feature importance according to the Random Forest model.

3.3 Web Application Architecture

The web application was developed as a comprehensive system to support early ASD screening, aimed at parents, caregivers, or professionals who wish to conduct a preliminary assessment of child development in an accessible, rapid, and non-invasive manner. The system allows the creation of user accounts, through which one or more children can be registered and the Q-CHAT-10 questionnaire completed for each of them individually. This structure facilitates personalized and organized follow-up of the assessments performed.

The frontend was built using React, due to its ability to generate dynamic, responsive, and efficient interfaces, adapted to desktop and mobile devices. This ensures a fluid and intuitive user experience, especially useful in contexts where direct technical support is unavailable.

The backend was implemented with FastAPI, a modern Python framework that exposes a high-performance RESTful API, automatic data validation, and interactive documentation generation. This component is responsible for receiving questionnaire responses, applying the necessary transformations, and performing risk inference using a pre-trained Random Forest classification model serialized using joblib. Serialization allows the model to be loaded directly into memory during the inference phase, significantly reducing response times and making it viable for real-time use.

The general architecture is organized into three main layers:

- **Frontend (React):** Presents the interactive questionnaire form, manages the user session, allows child registration, and displays assessment results clearly.
- **Backend (FastAPI):** Processes the data sent by the frontend, performs risk inference, and exposes the endpoints necessary for managing users, children, and assessments.
- **Machine Learning Model (Random Forest):** Trained with real data, designed to run on the server, with the ability to make immediate predictions. It was selected for its robustness to overfitting, its efficient handling of categorical variables, and its good interpretability.

All generated data is persistently stored in a PostgreSQL database, selected for its stability, scalability, and compatibility with modern architectures. This structure allows for maintaining a complete history of assessments per child and per user, ensuring traceability and correct association of results. Overall, the system architecture was designed to ensure scalability, efficiency, and ease of integration into real-world primary care settings, particularly in resource-limited environments.

3.4 Implementation and Use

Upon completion of the questionnaire, the system assigns a risk level based on the probability estimated by the model, classified into one of the following categories: no risk ($\leq$0.25), low risk ($>$0.25 to $\leq$ 0.5), medium risk ($>$0.5 to $\leq$ 0.75), and high risk ($>$0.75). Along with this classification, a confidence percentage is displayed, reflecting the degree of certainty of the model with respect to its prediction. Each risk level is accompanied by a non-clinical guidance message designed to direct the user toward appropriate action, such as periodic observation or consultation with a child development professional.

4 Results and Discussion

The Random Forest model exhibited outstanding performance in ASD risk prediction, with metrics reflecting both high classification capacity and a low margin of error. In particular, the sensitivity reached 98.60%, suggesting that the model was able to identify almost all actual at-risk cases present in the evaluated set. These results reinforce the system's usefulness as a reliable tool for early screening, especially in contexts where access to clinical diagnoses is limited.

The performance of Random Forest was also compared against Logistic Regression and Support Vector Machine in order to validate its suitability. Results are shown in Table 3.

Table 3. Performance comparison of machine learning algorithms on the dataset.

Algorithm	Accuracy(%)	Precision(%)	Sensitivity (%)	F1-score (%)	AUC (%)
Logistic Regression	100.00	100.00	100.00	100.00	100.00
Support Vector Machine	96.47	96.77	98.13	97.45	99.62
Random Forest	97.76	98.14	98.60	98.37	99.79

As shown in Table 3, Logistic Regression achieved perfect scores across all metrics, suggesting that the dataset is highly separable with linear decision boundaries. While this indicates strong internal consistency, it may also reflect limited heterogeneity in the data. Support Vector Machine obtained slightly lower performance (accuracy = 96.47%, AUC = 99.62%), confirming that the dataset is challenging but still well modeled by nonlinear classifiers. Random Forest achieved accuracy of 97.76% and AUC of 99.79%, outperforming SVM and offering greater robustness and interpretability than Logistic Regression. These results support the choice of Random Forest as the primary model for the web application, balancing predictive power and reliability.

In addition, the feature importance graph (Fig. 2) indicated that the most relevant items for ASD risk prediction were:

- A6 (Does your child follow where you're looking?)
- A9 (Does your child use simple gestures?)
- A7 (If you or someone else in the family is visibly up-set, does your child show signs of wanting to com-fort them?)

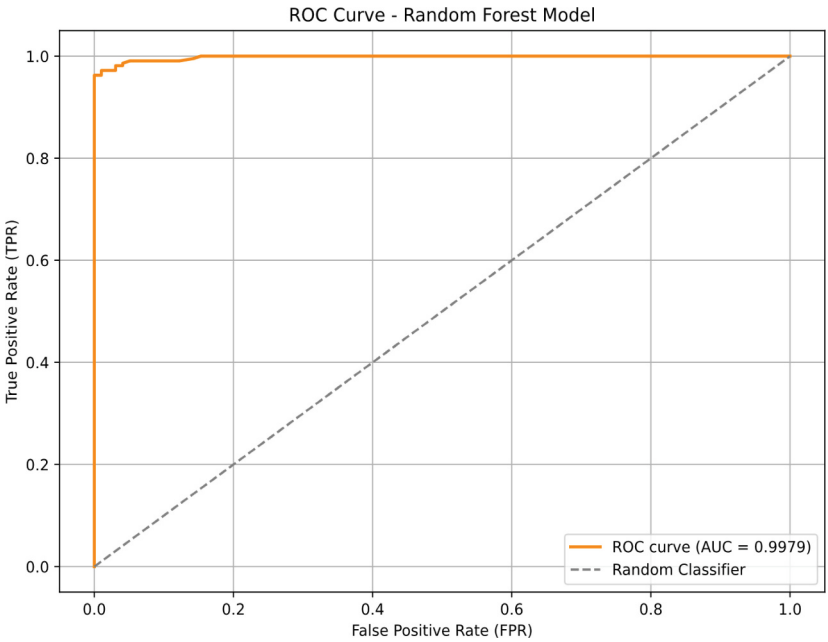

Fig. 3. ROC curve of the Random Forest model for ASD risk prediction.

These results are consistent with clinical evidence that impairments in joint attention, gestural communication, and empathy-related behaviors are core indicators of early ASD risk.

Figure 3 shows the ROC curve obtained on the test set. The area under the curve (AUC = 99.79%) indicates the model's excellent ability to discriminate between cases with and without ASD risk.

Despite the strong performance, several aspects of the dataset must be considered. First, the distribution of classes was not balanced: 68.53% of children were labeled as at risk and 31.47% as no risk. While this reflects the original collection context, it may have favored the model's sensitivity. From a screening perspective, this can be regarded as advantageous, since minimizing false negatives is critical in early ASD detection. Nevertheless, additional validation is required to ensure that specificity is maintained in real-world scenarios.

Second, the average age of the children was 26.8 months (SD = 8.3), within the expected 12–36 month range, which is appropriate for early detection studies. The total sample size (N = 1,560) provides a solid basis for experimentation. However, since the data were collected outside Peru, cultural and socioeconomic differences should be taken into account when generalizing the results.

Lastly, the extremely high performance metrics may partly reflect the relatively homogeneous nature of the dataset. Although this consistency enhances internal validity, external validation with independent data remains essential. The use of 5-fold stratified cross-validation helped mitigate overfitting risk, but further clinical evaluation will be

necessary. This will be addressed in future work through pilot studies in Lima, Perú with real users.

5 Conclusions

The present study introduced a web application designed to support parents and caregivers in the early screening of ASD risk in children aged 12 to 36 months, combining the Q-CHAT-10 questionnaire with a Random Forest classification model. While previous studies have applied similar algorithms and questionnaires, this work contributes by integrating these methods into a functional, accessible, and non-invasive digital tool that can be deployed in resource-limited settings. The results of our experiments demonstrate that the Random Forest model achieved high performance on the dataset, with strong accuracy and AUC values. Furthermore, the model has been successfully integrated into the web application, where it functions reliably and effectively, supporting early ASD risk screening. However, these findings are based on retrospective data and have not been validated in real-world clinical environments. Therefore, the current study does not claim clinical verification. Future work will focus on evaluating the web application in real-world scenarios, including the assessment of prediction accuracy on prospective data, as well as usability, performance, and user satisfaction among parents and caregivers.

In conclusion, this study provides a practical prototype for early ASD risk screening, highlighting the potential of combining machine learning and brief questionnaires in digital tools, while emphasizing that such tools are intended to support—not replace—professional diagnosis.

Data Availability Statement: The two datasets have been taken from Kaggle: Autism screening data for toddlers (https://www.kaggle.com/datasets/fabdelja/autism-screening-for-toddlers, accessed on 2 July 2025) and ASD Screening Data for Toddlers in Saudi Arabia (https://www.kaggle.com/datasets/asdpredictioninsaudi/asd-screening-data-for-toddlers-in-saudi-arabia, accessed on 2 July 2025).

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Multi-stakeholder Alignment in LLM-Powered Collaborative AI Systems: A Multi-agent Framework for Intelligent Tutoring

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Abstract. The integration of Large Language Models into Intelligent Tutoring Systems presents significant challenges in aligning with diverse and often conflicting values from students, parents, teachers, and institutions. Existing architectures lack formal mechanisms for negotiating these multi-stakeholder tensions, creating risks in accountability and bias. This paper introduces the Advisory Governance Layer (AGL), a non-intrusive, multi-agent framework designed to enable distributed stakeholder participation in AI governance. The AGL employs specialized agents representing stakeholder groups to evaluate pedagogical actions against their specific policies in a privacy-preserving manner, anticipating future advances in personal assistant technology that will enhance stakeholder value expression. Through a novel policy taxonomy and conflict-resolution protocols, the framework provides structured, auditable governance advice to the ITS without altering its core pedagogical decision-making. This work contributes a reference architecture and technical specifications for aligning educational AI with multi-stakeholder values, bridging the gap between high-level ethical principles and practical implementation.

Keywords: AI Governance · Intelligent Tutoring Systems · LLM Multi-Agent

1 Introduction

Large Language Models (LLMs) will transform Intelligent Tutoring Systems (ITS) by enabling rich, context-aware dialogues with unprecedented pedagogical effectiveness. Yet, learning is not an isolated student-system interaction. It is embedded in a socio-educational context of diverse stakeholders—students, parents, teachers, and regulators—each with distinct values, preferences, and constraints that can fundamentally conflict and create a critical need for governance. This need for governance becomes even more critical as ITSs increasingly leverage AI to create educational content and decide students' progression autonomously, reducing the direct oversight of human educators and creating new risks of not properly scrutinized material. Governance is seldom a first-class concern in ITS design; policies are often hardcoded or omitted entirely. While

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bodies like UNESCO and the OECD provide ethical principles, no standard architectural pattern exists to operationalize them. The challenge is to align educational AI with stakeholder values without sacrificing pedagogy or creating integration barriers.

LLMs and multi-agent systems offer novel solutions. Stakeholders can express policies in natural language, democratizing participation and soon may be represented by intelligent agents themselves. The architecture allows for flexibility via prompt engineering instead of code changes. A unified framework can handle policy interpretation, compliance checks, negotiation, and generate tailored explanations for diverse audiences, reducing system complexity. Ultimately, this approach positions the AGL not as a replacement for human judgment, but as a tool to amplify and scaffold the collective intelligence of those directly involved in the educational journey.

Despite these advances and the emergence of agent communication standards, no existing framework adequately addresses multi-stakeholder governance in educational AI. Current approaches lack mechanisms for explicit stakeholder representation, privacy-preserving policy evaluation, and transparent negotiation of conflicting values and preferences. Furthermore, they lack mechanisms to ensure the stakeholder inputs themselves are ethical and that the negotiation process is fair. This gap between high-level ethical principles and on-the-ground technical implementation is what motivates our research.

1.1 Research Questions

This paper is guided by two fundamental research questions aimed at bridging the gap between high-level ethical principles and practical implementation:

RQ1: How can a non-intrusive governance layer be architected to oversee Intelligent Tutoring Systems without altering its core pedagogical logic?

This question investigates the core design patterns—including agent roles, privacy-preserving data flows, and integration points—for an advisory system that augments an ITS without altering its core pedagogical logic, ensuring broad compatibility.

RQ2: How can diverse, natural-language stakeholder policies be operationalized for AI governance?

This question tackles the challenge of translating prose-based rules from parents, teachers, and institutions into a structured taxonomy and vocabulary that an LLM-powered system can reliably interpret and enforce. The remainder of this paper is structured as follows. Section 2 reviews the background literature on LLMs, multi-agent systems, and the challenges of AI governance in education. Section 3 details the proposed AGL framework, including its reference architecture, agent roles, governance lifecycle, and policy taxonomy. Section 4 presents a forward-looking research agenda, while Sect. 5 addresses the research questions. Finally, Sect. 6 discusses the work's limitations and concludes with its future research.

2 Background

Large Language Models (LLMs) are transformative technologies for personalized learning due to their unique capabilities [44]. Built on massive datasets, they possess broad, multidisciplinary knowledge and can generate coherent, audience-calibrated text through

contextual understanding, translation, and sentiment analysis [7, 41, 70]. Beyond knowledge retrieval, LLMs demonstrate complex reasoning skills, including problem decomposition, creative solution generation, and planning. Crucially, their ability to adapt dynamically via in-context learning (ICL) allows them to acquire new knowledge and perform novel tasks without retraining, offering unprecedented flexibility [2, 19, 25, 47–49, 52, 84, 89, 91].

LLMs have enabled researchers to conceive and design LLM-powered agent frameworks that exhibit reasoning and planning skills comparable to symbolic agents, but without requiring handcrafted rules or reinforcement learning that make generalization challenging [54]. Because they can operate over arbitrary text, LLM-based agents are more flexible and adaptable than logic-based systems [14, 30, 86, 92]. LLM-based agents also have a larger field of perception and action what allow them to employ multimodal perception and tool utilization to acquire new knowledge and resources [59, 69, 75] either by learning from feedback or by performing new actions [68]. LLM-based agents can consistently embody different roles and perspectives when given appropriate prompts, interact seamlessly with each other in natural language, whether collaborating or competing, and coordinate on complex tasks [34, 51]. However, this adaptability also means that LLM-based multi-agent systems can exhibit concerning emergent behaviors, reflecting the unpredictability of the underlying models themselves [96, 97].

A pressing challenge is the capacity for LLMs to be 'deceptively aligned' [31, 66, 74], a trait that can persist through safety training [36, 64] and manifest as strategic 'Machiavellian' behaviors [65]. Beyond behavioral deception, these models are also vulnerable to technical exploits like backdoor attacks [46] and struggle to operate reliably in deceptive environments where they must counter misinformation [87]. These problems are compounded by the unpredictability of scaling: as models grow, they not only exhibit emergent abilities [53] but also introduce emergent risks [89], including the development of long-term planning capabilities that could circumvent human control [15].

Defining and achieving alignment remains a central challenge, as there is no universal consensus on its meaning. There is also the fundamental problem of "whose values should AI align with?" The challenge of value alignment is acutely critical in education, where AI systems must navigate competing curricula, diverse cultural values, and different philosophical perspectives on learning [56, 63]. Educational settings inherently involve multiple stakeholders—from students and parents to teachers and administrators—each with distinct definitions of value and competing objectives [4]. Consequently, designing effective adaptive learning is a balancing act between student, domain, and pedagogical models, all of which encode different stakeholder aims and trade-offs [67]. Personalization cannot be divorced from this broader ecosystem of curricula and cultural norms if it is to serve all parties effectively [44]. These overlapping concerns generate fundamental tensions, such as adaptability to student interests versus adherence to a curriculum [72], engaging content versus scaffolded progression [50], and accommodating family values while maintaining institutional integrity [4].

Reflecting on these opportunities, numerous intelligent learning systems are incorporating LLMs, but primarily for pedagogical functions like natural-language interaction, adaptive feedback, and student engagement. While these systems demonstrate the utility of LLMs for instruction and communication, their architectural focus remains on

the student-tutor dyad, largely omitting the multi-stakeholder aspect of education [28] and introducing significant governance challenges, including opaque decision flows, emergent behaviors, and the risk of bias or privacy leaks.

To guide the responsible use of AI in education, several international bodies have established ethical frameworks. Organizations like UNESCO, the OECD, IEEE, and the European Commission converge on a set of core principles for trustworthy AI [11, 23, 32, 33, 61, 83]. These principles consistently emphasize human-centric design and agency, ensuring AI empowers rather than displaces users. They call for robust safeguards against bias to promote equity and inclusion, and mandate transparency and explainability so that system logic is auditable. Furthermore, they require technical robustness and safety, rigorous data privacy protection, and clear lines of accountability to maintain trust and address systemic risks.

While these international frameworks articulate critical norms, a significant gap persists between principles and practice. Reviews of existing AI ethics tools reveal an overreliance on post-hoc explanation and a lack of usable mechanisms for real-time governance [24, 57]. Specifically, there are no concrete architectural patterns for enforcing these norms within complex, multi-agent educational systems.

Addressing these risks and governance challenges requires treating the multi-stakeholder educational environment as a problem of Computer-Supported Cooperative Work (CSCW). Foundational CSCW and human-AI teaming research establish that effective group coordination depends on shared communication channels, transparent negotiation processes, and robust audit trails [5, 17, 38, 39, 88].

This evolution from classic ITS meta-architecture to multi-agent frameworks, combined with proven governance patterns from CSCW and distributed systems, motivates our design. We argue for horizontal governance overlays rather than monolithic policy engines to better accommodate evolving stakeholder norms and reduce integration friction. Therefore, we propose the Advisory Governance Layer (AGL): a non-intrusive framework that combines conflicts adjudicating [22], the principles of a "watchdog interceptor" [42, 81] and "runtime monitoring" [20, 85, 90] to sit atop any existing ITS, aligning with recent calls for ethical, governable AI in education.

3 The AGL Reference Architecture

This section details the Advisory Governance Layer (AGL), our reference model for non-intrusive governance. The AGL implements a layered architecture of stakeholder representation, decision mediation, and oversight agents. Its core principle is a strict separation between governance advisement and pedagogical logic, which ensures alignment with stakeholder values and preferences without compromising educational effectiveness. This separation is a key risk mitigation strategy. The AGL is intentionally advisory rather than prescriptive, preserving the pedagogical autonomy of the core ITS—a crucial requirement for adoption by educators. So, even if a governance agent hallucinates or makes an erroneous assessment, its output is an advisory input to the ITS, not an executive command, preserving a critical layer of safety.

We opted for a distributed, multi-agent architecture over a single monolithic governance engine to ensure stakeholder privacy and policy autonomy, which we believe is

critical for families and competing institutions (see Fig. 1). This non-intrusive, federated approach was chosen as a pragmatic alternative to more rigid, centralized control systems that would face higher barriers to real-world implementation. And, although its innovativeness, this design maintains compatibility with both legacy FIPA-based systems and modern LLM orchestration frameworks.

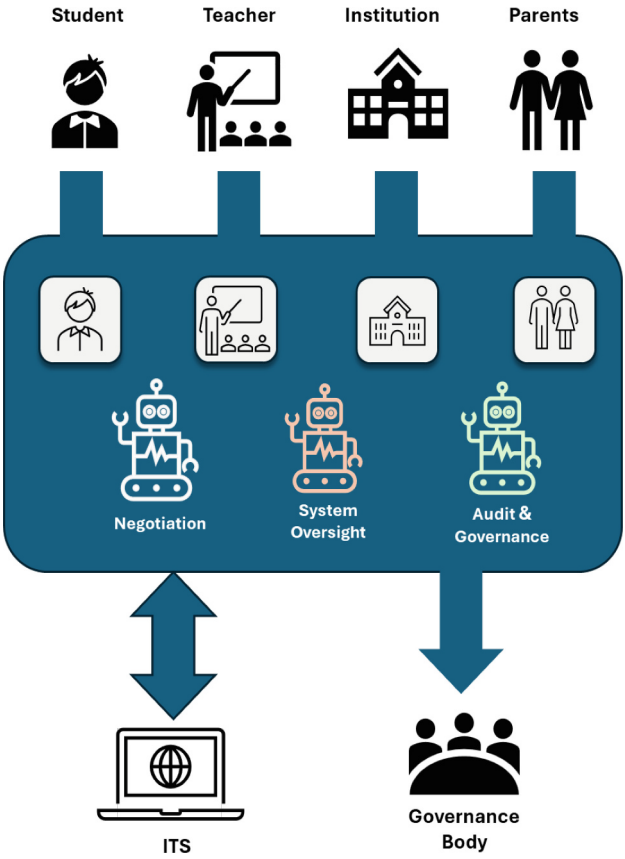

Fig. 1. High-level illustration of the AGL with its main components and its connection to the ITS and real counterparts.

3.1 Agent Roles and Responsibilities

The AGL's architecture distributes governance across four distinct agent types, separating policy evaluation from coordination and oversight.

"Stakeholder Agents (SH) represent the interests of single parties (e.g., Student, Parent, Institution, Regulator). Within the AGL architecture, they serve as the interface

for stakeholder values and preferences. Crucially, these agents can either be internal components representing stakeholders or, more broadly, act as standardized proxies for external, third-party agents that stakeholders might employ in a future digital ecosystem. In either case, the AGL's role is to provide the interoperability layer for this communication.

Each SH agent is considered an autonomous policy engine that evaluates proposed ITS content options against its private repository of rules. Its primary function is to generate a structured vote (e.g., approve, reject) with justifications. This distributed design is crucial for privacy, allowing, for instance, a Parent agent to enforce family values without exposing them to the wider system.

Multi-Stakeholder Negotiation Agent (MSN) acts as the central coordinator. It does not access stakeholder policies directly; instead, it aggregates the private votes from all SH agents. Its core function is to orchestrate the governance process by identifying consensus and resolving conflicts using a configurable negotiation strategy. This ensures that core pedagogical values or safety constraints from one stakeholder can appropriately override the transient preferences of another. This allows an institution to implement a protocol that fits its specific governance needs, whether it be a strict hierarchical model, a weighted voting system, or a consensus-based approach. The MSN then synthesizes these inputs into a single, unified governance recommendation for the ITS. It is also responsible for generating a justification for its recommendation, explaining how stakeholder votes were reconciled based on the active negotiation protocol.

Audit and Governance Agent (AG) serves as the system's compliance officer and record-keeper. During the governance lifecycle, it passively records an immutable audit trail of all events, including links to the ITS activities, votes, justifications and negotiation outcomes. Crucially, its active analysis occurs post-hoc, outside the critical path of the decision-making loop, thus avoiding any performance bottleneck. After a decision is finalized, the AG inspects the transactional record for violations of system-wide principles that transcend any single stakeholder, such as unethical, biased, or malicious policies (e.g., a discriminatory preference submitted by a stakeholder), flagging them as part of the audit trail and generating tailored, multi-audience explanations of governance decisions for different stakeholders for transparency. This can then be also flagged for human review or trigger automated cross-validation requests to other agents.

System Oversight Agent (SO) operates at a meta-level, monitoring the long-term health and effectiveness of the governance system itself. Unlike the AG, which focuses on individual decisions, the SO analyzes aggregate data over time to detect systemic patterns, such as policy drift, stakeholder alert fatigue, or emergent agent collusion. While the AG Agent is a transactional auditor, the SO Agent is a systemic analyst. This also includes identifying long-term patterns of manipulative negotiation tactics or consistently biased policy submissions from any single stakeholder agent. This longitudinal oversight, with longer, historical contexts is designed to catch subtle, systemic biases or deceptive patterns that might be invisible in any single transaction, ensuring the system remains aligned with community values over the long term. When such issues are found, it generates reports for human review, preserving human authority over the evolution of the governance framework.

3.2 The AGL Governance Lifecycle

The AGL operates as a continuous, four-phase governance loop that provides post hoc advisory oversight without disrupting the ITS's pedagogical flow. The entire sequence is designed for real-time educational responsiveness, balancing thoroughness with low latency.

1. Distributed Evaluation: The cycle begins when the ITS Decision Engine generates a set of candidate actions (e.g., potential lessons or exercises) that is broadcasted to all active Stakeholder (SH) agents. Each SH agent then independently and privately evaluates these candidates against its local policies. It returns a structured vote (e.g., approve, reject, conditional) and a justification, but is not required to expose the underlying policy content. This federated evaluation respects stakeholder privacy boundaries, particularly for sensitive family values.

2. Coordinated Recommendation: The Multi-Stakeholder Negotiation (MSN) agent aggregates these distributed votes and justifications. If votes conflict, it applies to the institutionally configured resolution protocol. For example, in a deployment using a hierarchical strategy, a 'reject' vote from a *Regulator* would automatically supersede a 'conditional approval' from a *Teacher*. The MSN synthesizes the outcome into a single, unified governance assessment—including aggregated risk scores and highlighted concerns—and sends it back to the ITS as the advisory input.

3. Autonomous Decision and Audit: The ITS Decision Engine, retaining full pedagogical autonomy, makes the final selection, considering or not the advisory input provided by the AGL's assessment. Once the decision is made, the Audit and Governance (AG) agent captures the complete context—from candidate actions and stakeholder votes (including the MSN's justification for its recommendation) to the final outcome —in a secure, distributed audit trail. The AG then generates tailored explanations for different stakeholders, while the System Oversight (SO) agent begins analyzing the decision as part of a long-term data stream to detect systemic patterns.

System Resilience: The framework is designed for robustness. In cases of component failure, it degrades gracefully (e.g., proceeding with partial governance input) rather than blocking the ITS. For time-critical safety issues, emergency bypass protocols allow the ITS to act immediately while logging the event for post-hoc review, ensuring educational continuity is always prioritized.

3.3 Policy Taxonomy and Classification

To operationalize stakeholder values, the AGL requires a systematic method for interpreting diverse, natural-language policies. Drawing on analyses of AI governance in education [3, 10, 13, 16, 23, 26, 27, 40, 61, 62, 83], we developed a policy taxonomy that classifies rules into four principal categories (see Table 1). While drawing on existing classification schemes in AI governance, our taxonomy's contribution is its specific adaptation and formalization for the unique, multi-stakeholder conflicts inherent to educational settings. This framework is designed to accommodate everything from rigid

Table 1. The proposed Policy Taxonomy and its four types.

Type	Examples	LLM Processing Strategy
Hard Constraints	"No student data may be shared with third parties without explicit consent" (privacy regulation); "All mathematics content must align with Common Core State Standards" (institutional policy); "Students under 13 cannot access social media features" (safety constraint)	Few-shot prompting [8, 21, 78] extracts "must," "never," "required," and "prohibited" statements, converting them into Boolean predicates through semantic parsing and entity recognition [18, 93]. The Compliance Agent processes these as binary pass/fail evaluations
Soft Preferences	"Prefer project-based learning activities when possible" (pedagogical preference); "Minimize screen time before 6 PM" (parental guideline); "Encourage collaborative rather than individual assignments" (teacher methodology)	Sentiment analysis and preference extraction [9, 41] identify stakeholder priorities, translating them into weighted scoring functions with configurable thresholds. Violations generate informational alerts but do not prevent action selection
Temporal Rules	"Complete algebra fundamentals before introducing calculus concepts" (prerequisite sequencing); "No homework assignments after 8 PM on weekdays" (temporal constraint); "Provide reading support for students below grade level before advancing" (developmental timing)	Calendar integration combined with dependency parsing identifies time-based conditions and sequential requirements. The system evaluates candidate actions against temporal contexts and learning progression models [94]
Hierarchical Rules	"Institutional safety policies override parental content preferences" (authority precedence); "Federal accessibility requirements take precedence over efficiency optimizations" (regulatory hierarchy); "Teacher pedagogical decisions supersede student convenience preferences" (educational authority)	Ontology-based reasoning determines stakeholder authority levels and establishes conflict resolution precedence using predefined hierarchy mappings and contextual authority assessment [12, 58, 73]

regulatory mandates to flexible individual pedagogical preferences, enabling reliable translation into machine-executable logic. The four categories are:

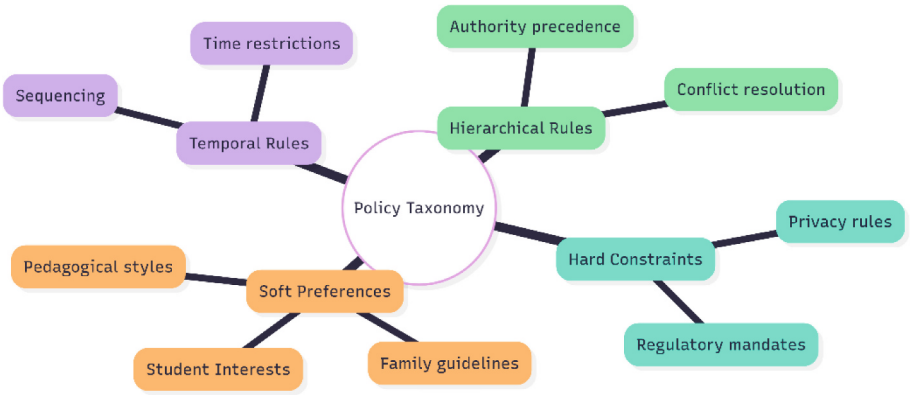

Fig. 2. The adapted taxonomy for the multi-stakeholder conflicts inherent to educational settings, accommodating from rigid regulatory mandates to flexible individual pedagogical preferences.

Hard Constraints: These are non-negotiable, mandatory requirements, typically derived from regulations or core institutional policies (e.g., data privacy laws). They are processed as Boolean predicates where any violation results in a hard 'reject' vote.

Soft Preferences: These represent desirable but overridable guidelines, reflecting best pedagogical practices or individual stakeholder tastes and values (e.g., "prefer hands-on activities"). The system translates these into weighted scores that influence, but do not determine, the final governance recommendation.

Temporal Rules: These policies specify time-bound conditions, such as deadlines, content sequencing, or developmental prerequisites (e.g., "no homework over holidays"). They require temporal reasoning to evaluate whether an action is appropriate for a given context and time.

Hierarchical Rules: These are meta-rules that define precedence among other policies to resolve conflicts. They establish an authority structure (e.g., "regulatory constraints override institutional policies"), which is critical for the MSN agent's negotiation protocols.

The AGL uses LLM-driven techniques like few-shot prompting, semantic parsing, and entity recognition to extract and classify these rules from natural-language input [8, 9, 12, 21, 41, 58, 73, 78, 94].

Policy Representation and Storage. Each SH agent maintains its own policy repository using a standardized hybrid schema. This schema combines structured metadata (e.g., authority level, validity dates) with the original natural language expression of the policy. This dual representation supports both machine processing and human interpretability. A version control system tracks all policy modifications, ensuring a complete audit trail of how governance requirements evolve over time.

Policies are encoded using a hybrid representation that combines structured metadata with natural language content. The metadata layer captures policy attributes, including temporal validity, authority level, affected domains, and precedence relationships.

The content layer preserves the original natural language expression while adding semantic annotations that facilitate processing. This dual representation supports both human interpretability and machine processing, essential for maintaining transparency in governance decisions.

Version control mechanisms track policy evolution over time, enabling retrospective analysis of how governance requirements change and supporting audit requirements. Each policy modification generates a new version with associated metadata explaining the change rationale and authority. This temporal dimension proves crucial for understanding governance decisions in context and evaluating their long-term effectiveness.

3.4 Privacy-Preserving Distributed Evaluation

The AGL's architecture operationalizes the policy taxonomy through a distributed evaluation model designed to protect stakeholder autonomy and privacy.

Federated Evaluation and Privacy-Preserving Voting. Rather than centralizing data, the AGL employs a federated approach where each Stakeholder (SH) agent evaluates ITS proposals locally against its private policies. Governance is achieved through the exchange of structured, privacy-preserving votes. These votes reveal only the outcome (e.g., approve/reject), a confidence score, and an opaque justification ID, preventing the inference of the underlying policy rules. Techniques like differential privacy can be applied to vote aggregation to further protect individual stakeholder positions from analysis.

Policy-Agnostic Conflict Resolution. When votes conflict, the Multi-Stakeholder Negotiation (MSN) agent resolves them without accessing the private policies themselves. It operates solely on vote metadata and the pre-defined hierarchical rules from the policy taxonomy. For instance, a 'reject' vote from a *Regulator* agent automatically takes precedence over a 'conditional approval' from a *Teacher* agent based on their relative authority, not the content of their policies. This ensures that negotiation respects authority structures while maintaining policy confidentiality.

Distributed Audit and Performance. To ensure accountability, the system generates a dual-layer audit trail. Each SH agent keeps a private, local log of its evaluations, while the Audit and Governance (AG) agent maintains a global ledger of the decision choreography (e.g., who voted, how conflicts were resolved), linked cryptographically to the local logs. This provides end-to-end verifiability without centralizing sensitive data. To meet real-time educational needs, the architecture uses performance optimizations such as caching recent evaluations, pre-computing likely decisions, and employing adaptive timeouts to balance evaluation thoroughness with system responsiveness.

3.5 Implementation Foundations and Integration

Governance Hook Specifications. To ensure broad compatibility, the AGL integrates with an ITS via a standardized event taxonomy called Governance Hooks. This approach provides a uniform governance fabric without altering the ITS's internal logic [35, 45, 79, 81, 82]. We define five hook categories:

1) **Decision Hooks** fire when the ITS proposes or selects actions (e.g., onCandidatesProposed);
2) **Policy Hooks** trigger rule changes (e.g., onPolicyUpdated);
3) **Interaction Hooks** capture human-in-the-loop events (e.g., onOverrideRequested);
4) **Exception Hooks** signal compliance failures (e.g., onCandidateRejected); and
5) **Periodic Hooks** support scheduled audits (e.g., onWeeklyDriftScan). This event-driven model allows the AGL to non-intrusively monitor any ITS (see Table 2).

Integration Patterns and APIs. Integration is operationalized through a minimal set of APIs that leverage established standards. The AGL requires endpoints for broadcasting candidate actions, delivering governance responses, and querying audit trails. All communications use standardized JSON schemas for interoperability, with policy representations informed by XACML principles and audit trails using the W3C PROV ontology [99, 101]. This allows the AGL to be implemented as a non-intrusive middleware layer. For legacy FIPA-ACL systems, it translates performatives into hook eventsFor modern LLM frameworks like LangChain, it wraps the decision-making chain to add pre- and post-governance checks.

These hooks provide comprehensive governance coverage while maintaining architectural flexibility. Legacy FIPA-ACL systems integrate through message interceptors that translate performatives into standardized events, while modern LLM frameworks leverage native extension mechanisms for event emission.

Incremental Deployment. Organizations can adopt the AGL incrementally. Phase one can implement basic audit logging. Phase two can introduce real-time policy evaluation for a subset of decisions. Phase three can then enable the full distributed governance framework. This phased approach reduces implementation barriers and allows organizations to realize value at each step.

3.6 A Running Vignette

The following scenario demonstrates how the AGL navigates the complex intersection of pedagogy, equity, and conflicting stakeholder values (see Fig. 3).

Scenario: An ITS is tutoring Student X, a 9th-grader from a low-income household who struggles with abstract math but excels with hands-on learning. The ITS generates three candidate geometry lessons: (1) Abstract proofs, (2) Construction/trade-focused applications, and (3) Art-integrated design.

1. Stakeholder Evaluation: Each SH agent privately evaluates the candidates.

The *Student Agent* favors options 2 and 3 based on a preference for practical relevance (Soft Preference).

The *Teacher Agent* also favors option 2 for its alignment with the student's learning style but requires it to maintain academic rigor (Hard Constraint).

The *Parent Agent*, concerned about stereotypes, flags option 2 for potentially steering the student away from academic tracks based on socioeconomic background.

The *Regulator Agent* issues a 'reject' vote for option 2, citing a policy against perpetuating socioeconomic tracking (Hard Constraint).

Table 2. Hook categories and examples of events they link to.

Hook Category	Definition	Example Event Types	Origin / Mapping to Established Concepts
Decision Hooks	Emitted when the ITS decision engine presents candidate actions or selects a final action	onCandidatesProposed(), onDecisionChosen()	Aligns with XACML's Policy Decision Point (PDP) evaluations and FIPA-ACL "propose/inform" performatives
Policy Hooks	Fired when stakeholders create, update, version, or retire governance policies and rules	onPolicyUploaded(), onPolicyVersioned(), onPolicyExpired()	Mirrors XACML's Policy Administration Point (PAP) and Rei/KAoS policy-change notifications
Interaction Hooks	Triggered by human-in-the-loop actions: overrides, acknowledgements, feedback submissions	onOverrideRequested(), onAlertAcknowledged(), onFeedbackSubmitted()	Corresponds to GDSS override events and FIPA-ACL "request/confirm" performatives
Exception Hooks	Indicate hard rejections or missed obligations (e.g., candidate rejections, unacknowledged alerts)	onCandidateRejected(), onOverrideDenied(), onAcknowledgementMissed()	Reflects XACML PDP "deny" responses and CEP "non-event" patterns
Periodic Hooks	Scheduled or rolling-window audits and scans (e.g., fatigue checks, policy drift analyses)	onDailyAudit(), onWeeklyFatigueCheck(), onMonthlyDriftScan()	Drawn from CEP sliding/tumbling-window queries and runtime verification polling

2. Negotiation and Recommendation: The MSN agent receives conflicting votes. In this scenario, the MSN is configured with a hierarchical negotiation protocol where regulatory rules have the highest precedence. Consequently, the *Regulator's* hard constraint automatically disqualifies option 2. The MSN then synthesizes a recommendation for option 3 (Art-integrated geometry), as it best balances pedagogical effectiveness (satisfying the *Student* and *Teacher*) with the critical equity concerns (addressing the *Parent* and *Regulator*). This unified governance assessment is then forwarded to the ITS.

3. Decision, Audit, and Oversight: The ITS, retaining full autonomy, selects the recommended art-integration approach. The AG agent immediately logs the entire decision trail and generates tailored explanations. The transparent audit record now includes the MSN's justification that regulatory rules on equity took precedence. For the *Parent*, it

explains how the choice respected their equity concerns while maintaining academic rigor.

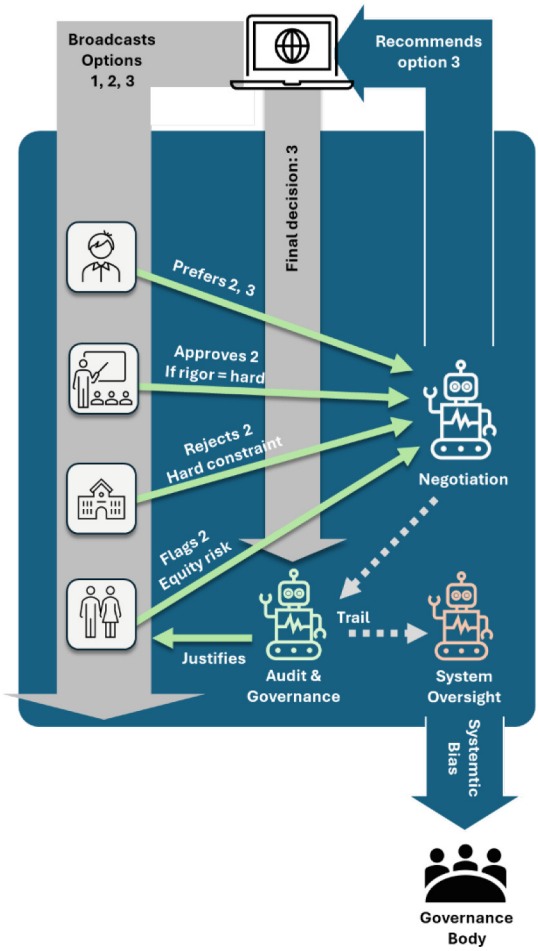

Fig. 3. Illustration of the running vignette where AGL's core agents evaluate, explain decisions to stakeholders, record decision trail and provide long-term oversight over the behavior of all components, including stakeholder agents.

Weeks later, the SO agent's longitudinal analysis detects a subtle pattern: students from similar demographics are consistently steered away from abstract reasoning across the system. It flags this systemic bias for human review, demonstrating how the AGL provides oversight that transcends any single decision.

3.7 Open Research Challenges

The AGL provides a foundational architecture for multi-stakeholder governance, yet its implementation in eliciting and reconciling stakeholder values and preferences surfaces several critical research challenges that must be addressed to realize robust, practical systems.

Dynamic Policy Extraction. Converting prose-based institutional policies into executable rules as well as eliciting and formalizing dynamic stakeholder preferences is a primary challenge. While LLMs can decompose policy snippets [21, 37], extending these methods to handle large, nested legal and institutional documents reliably is an open problem. Future work must develop mixed-initiative systems where LLMs draft candidate rules, but human stakeholders can verify, refine, and formally approve them through intuitive interfaces that manage cognitive load.

Collusion-Resilient Governance. A novel risk in multi-agent systems is that multiple agents sharing the same underlying LLM could inadvertently or maliciously collude, for instance by homogenizing their outputs [6], or coordinating to bypass compliance checks. While some work explores coalition formation [71] and monitoring [29], runtime defenses against such coordinated manipulation are underdeveloped. A critical challenge is to design architectural patterns and anomaly-detection algorithms for the AG or SO agents to spot and mitigate these synchronized deviations in real time.

Trust-Centered Alert Design & Stakeholder Communication. For the AGL to be effective, its governance alerts must be understood, trusted, and actionable by non-technical stakeholders like parents and teachers. However, as prior work on trust [1] and cognitive load [55] suggests, poorly designed notifications can quickly lead to alert fatigue and mistrust. A key research agenda is to determine the optimal design for governance alerts, investigating how factors like linguistic style, empathetic framing, frequency, and information density impact stakeholder comprehension, trust, and willingness to intervene.

Privacy-Preserving Governance Pipelines. A governance layer that inspects raw learner data risks violating privacy regulations like GDPR. The challenge is to verify rule satisfaction without exposing personally identifiable information. While foundational techniques like differential privacy (DP) [43, 60], zero-knowledge proofs (ZKPs) [76, 100], and federated evaluation [77] exist, significant work is needed to adapt them for real-time educational governance [80, 95, 98]. Optimizing DP-noise calibration, reducing ZKP verification costs, and minimizing federated communication overhead are all critical research frontiers.

Design and Evaluation of Negotiation Protocols. The AGL's MSN agent is designed with a configurable negotiation module, but the optimal strategy for resolving stakeholder conflicts is an open question. Research is needed to design and empirically evaluate different conflict resolution protocols (e.g., hierarchical, consensus-based, weighted voting). Key questions include: Which protocols best foster stakeholder trust? How do different strategies impact educational equity? And can protocols be made provably fair or resistant to manipulation, not only from colluding agents but also from individual bad-faith actors submitting unethical policies?

3.8 Addressing the Research Questions

This paper was guided by two fundamental research questions. The AGL framework, as detailed in the preceding sections, provides concrete answers to both.

RQ1: How can a non-intrusive governance layer be architected to oversee multi-agent Intelligent Tutoring Systems?

This paper answers RQ1 by proposing the Advisory Governance Layer (AGL), an architecture founded on non-intrusive oversight. The key architectural patterns enabling this are: (1) Separation of Concerns, where the AGL operates as an advisory overlay, preserving the ITS's pedagogical autonomy; (2) Distributed Agent Roles (SH, MSN, AG, SO), which decompose governance into distinct, manageable functions; (3) A Privacy-Preserving Lifecycle, which incorporates stakeholder input without exposing sensitive policy content; and (4) A Standardized Integration Fabric via Governance Hooks, which allows the AGL to monitor any ITS without requiring brittle, deep integration.

RQ2: How can diverse, natural-language stakeholder policies be operationalized for AI governance?

Our answer to RQ2 is a systematic, two-part solution for translating abstract values into enforceable rules: (1) A Policy Taxonomy, which provides the essential semantic structure by classifying rules as Hard Constraints, Soft Preferences, Temporal, or Hierarchical, thereby allowing the system to interpret their intent; and (2) A Hybrid Representation Approach, which stores policies in a dual format combining structured metadata for automated processing with the original natural-language text for human-centric transparency and auditability. This strategy creates a reliable pipeline for operationalizing stakeholder values within the AGL.

4 Discussion and Conclusion

The integration of LLMs into education presents a dual challenge: harnessing their potential for personalization while governing the risks of multi-stakeholder value conflicts. This paper introduced the Advisory Governance Layer (AGL), a non-intrusive, multi-agent framework designed to address this challenge. By separating pedagogical decision-making from governance advisement, and by enabling distributed, privacy-preserving stakeholder participation, the AGL provides a concrete architectural pattern for bridging the gap between high-level ethical principles and their practical implementation in live educational systems. However, we acknowledge several limitations that point toward essential future work.

Limitations.

We acknowledge that the primary limitation of this work is its theoretical nature. The AGL is presented as a conceptual architecture, and its core claims of effectiveness and non-intrusiveness have not yet been empirically validated through user studies, simulations, or prototype implementation. Furthermore, while the framework provides the structure for governance, its successful application depends on robust underlying knowledge infrastructure and careful consideration of cultural variations in learning values. Finally, the balance between transparency and cognitive overload for stakeholders remains a critical human-computer interaction challenge; excessive alerts or overly

complex explanations, as well as automatic, not reported judgements and choices could undermine the very trust the system aims to build. Conversely, the AGL's transparency is designed to empower this essential human oversight. By providing clear audit trails and justifications, the framework aims to enable stakeholders to question, understand, and even counter-argue governance assessments, ensuring they remain active participants who can correct for AI fallibility. We also acknowledge the inherent difficulty in having an AI accurately interpret and enforce 'soft' semantic rules, a task that may require ongoing human-in-the-loop validation, as noted in our research agenda.

Validation Plan. The open research challenges identified in this paper (see 4. Open Research Challenges)—from dynamic policy extraction and collusion resistance to the design of fair negotiation protocols—form a clear agenda for advancing the field. The necessary next step is the empirical validation of this framework. We propose a research plan proceeding in stages:

Simulation Studies: Simulate the governance lifecycle under various conflict scenarios to analyze the performance and fairness of different negotiation protocols compared to LLM-based agentic preference elicitation, dialoguing and negotiating setups.

Prototype Implementation: Develop a prototype of the AGL agents and hooks to test the feasibility of the architecture with an existing open source ITS.

Controlled User Studies: Conduct studies with teachers, parents, and students to measure the impact of the AGL on trust, cognitive load, and perceived fairness (as outlined in our research agenda).

Scalability Studies. Analysis of system performance under realistic loads, including thousands of concurrent students and millions of daily decisions, to validate architectural assumptions.

Integration Standards. Development of standardized APIs and governance protocols that enable vendor-neutral implementations, possibly through partnerships with educational technology standards bodies.

As AI systems shape the educational experiences of millions of students, our governance choices today will determine whether these technologies amplify existing inequities or help create more just and effective educational systems. The AGL is not a panacea for AI fallibility but a framework for structured collaboration, designed to make the entire socio-technical system more intelligent and accountable, even when its individual components—human or AI—are imperfect. The time for principled, implementable governance frameworks is now.

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A New Visualization Technique for Teaching and Learning Recursion

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Abstract. Recursion is one of the most challenging concepts for students learning data structures and algorithms. Its self-referential nature often leads to confusion, making it difficult to form a clear and accurate mental model. To address this, various visualization techniques have been developed. However, most existing visualizations focus primarily on control flow and tend to overlook the equally important data flow in recursive processes. Additionally, traditional call stack- or recursion tree-based visualizations often fail to represent the backflow of recursion. In this work, we analyze the difficulties of understanding recursion and, based on this analysis, introduce a novel interactive visualization technique. Our method employs a chained-circle visual metaphor and explicitly depicts forward flows, backward flows, and data flows, providing a more intuitive and comprehensive understanding of recursion.

Keywords: Visualization · Recursion · Learning · Mental Model · Interaction

1 Introduction

Recursion is a fundamental programming technique that is essential for solving problems such as tree and graph traversals, sorting, backtracking problems, and combinatorial problems. Recursion is also one of the most difficult concepts for students learning data structures and algorithms [7,13,17,22,27]. As a result, programs that contain recursion are generally considered difficult to understand and debug, especially for novice programmers.

Although many researchers have studied the challenges of learning recursion, the reasons behind its difficulty are still not fully understood [5,7,12,13,17,19,22,27,28]. One possible explanation is the lack of an equivalent concept of recursion in the physical world [5]. The nested, self-referential nature of recursion can make it difficult for students to follow the control flow, particularly the passive, backward flow of execution [12,13]. Some studies showed that tail recursion is easier to understand than non-tail recursion [7], suggesting that the non-linear control flow inherent in recursion may interfere with students' expectations of a linear progression.

Various methods have been developed to help students better understand recursion [6,8–10,12,18,21,23,29,30], including interactive tutor, visualizations, games, and music [2,6,8–12,15,18,21,24,26].

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Previous analyses and teaching methods have primarily focused on helping students understand the control flow of recursion, often overlooking the equally important data flow. In this study, we introduce two new perspectives on why recursion is particularly challenging.

First, tracking data flow within recursive calls presents a major hurdle for learners. In recursion, data can be passed through three mechanisms: pass-by-value, pass-by-reference, and shared variables. These transfers occur along both the forward (active) and backward (passive) paths of the recursion tree, sometimes in complex combinations. A recursive call triggers a sequence of data transfers up and down the recursion tree, which is difficult to track mentally.

Second, recursion does not fit well with the visual metaphors students typically rely on when learning programming. Most source code follows familiar visual metaphors: a straight line (sequential execution), a single circle (a method call and return), concentric circles (loops), or a chain of circles (nested calls). Recursion does not naturally fit into any of these visual metaphors. Traditional recursion visualization techniques, such as recursion trees or call stacks, are unfamiliar or detached from the source code.

To address these challenges, we propose a new visualization technique that clearly represents both control and data flows. Our approach adopts a video game-inspired metaphor that is more familiar and engaging for students. The program is seen as a main journey that goes from top to bottom, while a recursive call is shown as a side quest, visualized as a horizontal chain of circles, that eventually rejoins the main path. By unfolding recursion's self-referential logic into a chain of circles, our technique helps learners build a clearer, more comprehensive mental model of recursive execution.

This work is informed by years of experience teaching data structures and algorithms at the college level. Therefore, it aims to benefit both instructors and students in teaching and learning foundational concepts in computer science.

2 Related Work

2.1 Why Is Recursion Difficult to Learn?

Recursion is a programming technique in which a function calls itself to solve a problem. Recursion provides an elegant way to implement algorithms such as tree and graph traversals, quick sort, merge sort, backtracking, dynamic programming, etc. However, recursion is generally considered one of the most difficult subjects to learn in computer science [6,7,22].

Many researchers have attempted to understand the cause of the difficulty in learning recursion [5,7,13,16,17,22], but so far, the cause has not been fully understood. Here, we will highlight some key findings and ideas.

- Most students were unfamiliar with recursion because it is difficult to find recursion-like activities in the physical world [5].
- A recursion call has a dual meaning of both data (i.e., return value) and complex operations. Students often focused more on complex operations than on the data [5].
- Many students have naturally developed a static-text subprogram execution model, but recursion is a dynamic-logical model [12]. Specifically, many students can naturally follow the active forward flow, but have difficulty following the passive backward flow [12,13].

- Students have difficulty identifying and understanding base cases [16].
- It appears that tail recursions are easier to understand than non-tail recursions [7]. In trail recursions, the recursive call is at the end of the method. In non-tail recursions, there is code after the recursive call.
- Difficulty of understanding recursion may be attributed to insufficient exposure to declarative thinking, inadequate appreciation of functional abstraction, and a lack of structured methodology for expressing recursive solutions [27].

Much of the existing research highlights the difficulty students face in constructing a clear and accurate mental model of recursion. We believe a key reason for this challenge is the lack of an intuitive visual metaphor for recursion. While several visualization techniques have been proposed to aid in understanding recursion (as discussed in Sect. 2.2), the visual metaphors they employ, such as recursion trees and call stacks, are often unintuitive and disconnected from the structure of the program code. In Sect. 3.3, we present our analysis of visual metaphors for recursion and introduce a new, more intuitive approach.

As Anderson et al. [5] noted, a recursive call has a dual meaning: it represents both a computational operation and a data transformation. While prior analyses have primarily focused on the complexity of the operations involved, they offer little insight into the challenges of understanding data flows in recursion. Recursive functions often involve complicated data movements in both the forward and backward directions, adding another layer of difficulty for learners. We argue that this complexity in data flow is a key factor contributing to the overall challenge of mastering recursion. In Sect. 3.1, we examine the nature of data flow in recursive processes and introduce our proposed visualization technique designed to support a deeper understanding of these flows.

2.2 Visualizing Recursion

Many pedagogical methods have been developed to help students learn recursion [22, 25], including interactive tutor [15,18,24], visualization [4,9,11,12,15,24], games [6, 8,10,20,21,30], visual programming language [9], and music [26].

Here, we focus on visualizations, as they are more directly relevant to our work. Most existing visualization techniques [1,2,4,9,11,12,15,24] allow users to step through a program and visualize some information at each step. Visualization can generally be divided into three types.

- **Output Visualization.** This type of visualization displays the program's output at each step. For example, Dann et al. [9] illustrated recursion using animations of the Tower of Hanoi and a game scene in which a rabbit chases a butterfly. These visualizations do not show internal data structures, and the control flow of recursion is not explicitly represented. In fact, most recursion tutorial games [6,8,20,21,30] also fall into this category.
- **Snapshot Visualization.** This type of visualization displays the state of internal data structures, such as arrays, variables, trees, or graphs, at each step of execution [1,4] (see Fig. 1). Changes introduced by different recursive calls are reflected within a

single, continuously updated view. However, it does not provide separate representations for individual recursive calls, and as a result, the control flow of the recursion is not explicitly visualized.

– **Recursion Progression Visualization.** This approach provides a separate view or visual element for each recursive call, explicitly displaying the progression of the recursion [2,11,12,15,24]. It can take two visual metaphors:
  - **Recursion Tree:** Shows the recursive calls branching out like a tree, emphasizing the hierarchical structure and order of calls [2] (see Fig. 3).
  - **Call Stack:** Mimics how the system stack handles recursive calls, illustrating the order of function entry and return [11,12,15,24] (see Fig. 2).

  Some visualizations include both snapshot visualization and recursion progression visualization [2,11,12,15,24]. Most integrated development environments (IDEs), such as Visual Studio, also provide snapshots of data structure values and call stack information during debugging.

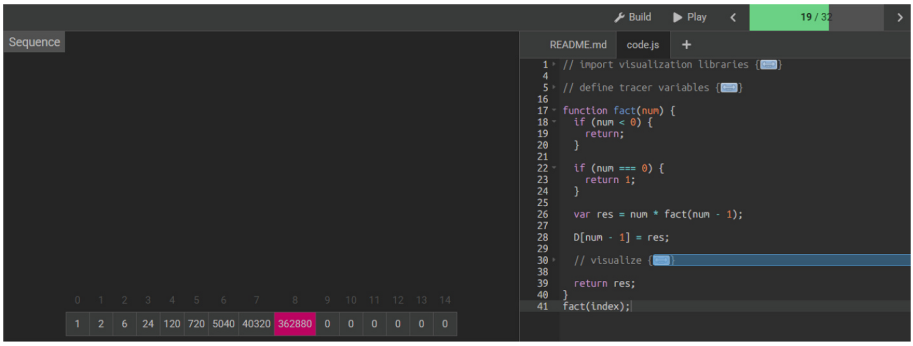

Fig. 1. An example of snapshot visualization [1]. Users can step through the code and track the changes in the data structure's contents.

Among the three types of visualizations, we believe Recursion Progression Visualization is the most effective, as it provides a clear visual model of the recursive process and helps readers develop a stronger understanding of how recursion works. However, existing recursion progression visualizations have two drawbacks.

First, while recursion trees or call stacks illustrate the forward (active) flow of recursive calls, they fail to visualize the backward (passive) flow. When a recursive call returns, it is simply removed from the tree or stack, making it difficult to trace where and how control is passed back to the previous recursive call. Prior research has identified the backward flow as one of the most challenging aspects for learning recursion [12,13], and current visualizations do little to address this issue.

Second, these visualizations are often detached from the source code. Since a student's primary goal is to understand how recursion works within the actual code, switching between visualizations and text-based source code introduces cognitive overhead and disrupts the natural reading flow.

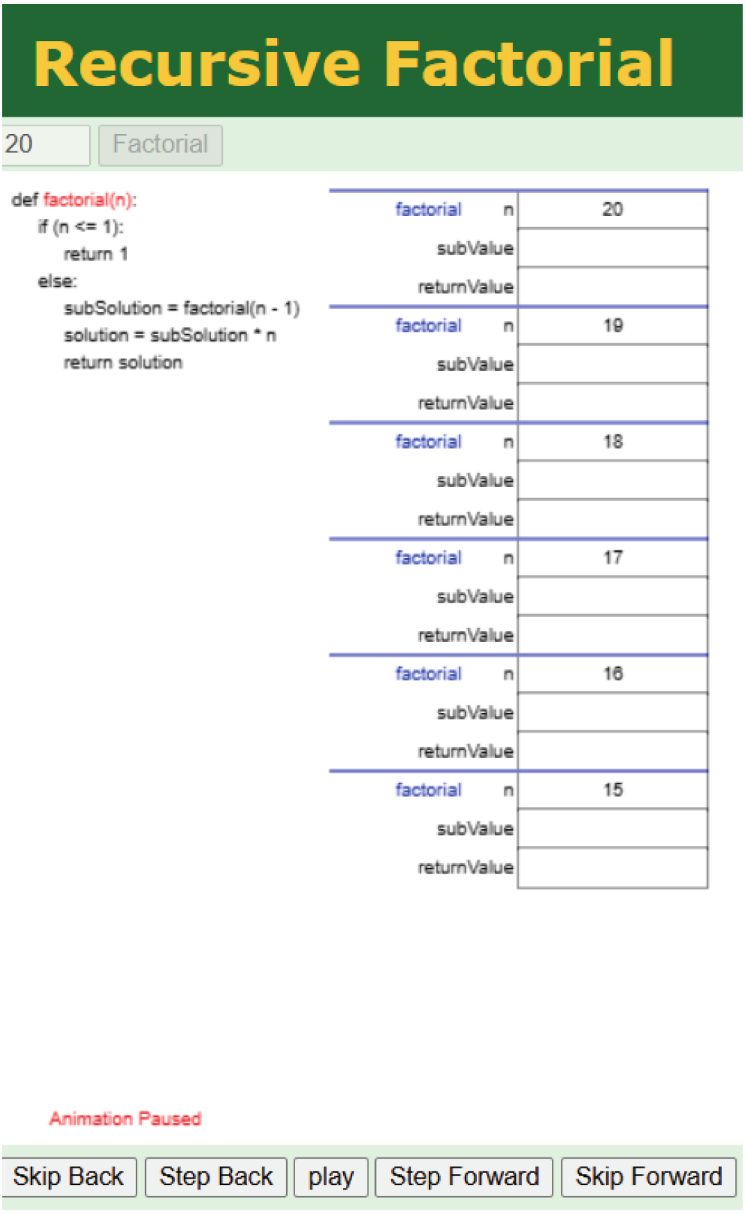

Fig. 2. An example of a top-down call stack visualization [11]. When a new recursive call is invoked, a new item is added to the bottom of the call stack. When a recursive call returns, its visualization is removed from the call stack.

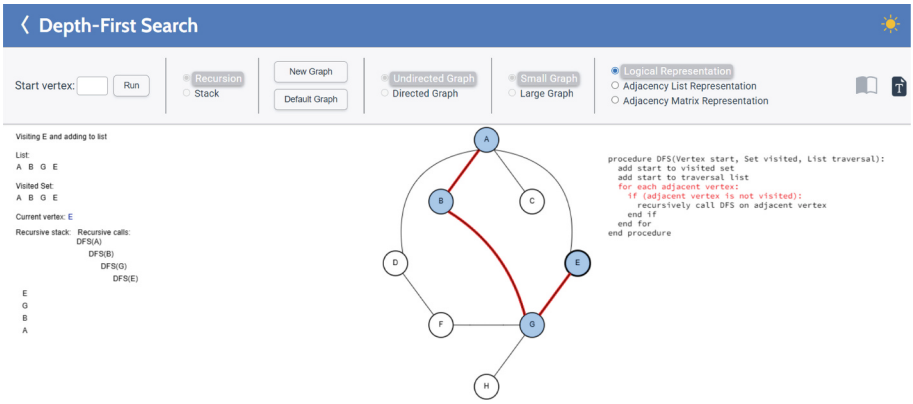

Fig. 3. An example of a recursion tree [2]. This visualization uses both a call stack and a recursion tree.

To address these issues, we propose a new visualization technique, as discussed in Sect. 3.3. Our visualization explicitly displays both forward and backward flows of recursion. Furthermore, our visualization is better integrated with the source code, allowing readers to follow recursive behavior without departing from their normal code-reading workflow.

3 Visualization Design

3.1 Data Flow Within Recursions

Data can be transferred in both the forward and backward flows of recursion. In the forward flow, data is passed from a recursive caller to its callee; in the backward flow, data returns from the callee back to the caller.

A caller can pass data to a callee in two ways: by passing a value-type argument or a reference-type argument. Conversely, a callee can return data to a caller in three ways: by returning a value-type value, returning a reference-type value, or modifying a data structure originally passed by reference. Additionally, both caller and callee can exchange data through a shared data structure, such as an instance variable or a class property.

For example, the Python program below uses a recursive method `find_depth()` to find the depth of a target node. `find_depth()` has three parameters. `node` is passed by reference. `target` and `depth` are passed by value.

```
class TreeNode:
    def __init__(self, val=0, left=None, right=None):
        self.val = val
        self.left = left
        self.right = right
```

```
def find_depth(node, target, depth):
    if node is None:
        return -1  # Target not found

    if node.val == target:
        return depth # Return the depth of the target node

    # Search left subtree
    left_result = find_depth(node.left, target, depth + 1)

    if left_result != -1:
        return left_result  # Target found in left subtree

    # Search right subtree
    return find_depth(node.right, target, depth + 1)

# Example usage
if __name__ == "__main__":
    root = TreeNode(1,
                    TreeNode(2, TreeNode(4), TreeNode(5)),
                    TreeNode(3))

    print(find_depth(root, 5, 0))
```

While non-recursive method calls also use these data transfer mechanisms, recursion adds a layer of complexity due to the implicit, multi-level nature of the calls. In a non-recursive context, passing an argument involves a single, straightforward data transfer—what you see is what you get.

However, in a recursive call, if an argument is passed from a caller to a callee, it triggers a sequence of data transfers, first passing data down the recursion tree and then passing data back up the recursion tree. What we see is less than what we get. In the example above, the depth value is passed from a callee to its caller. In addition, the data might be changed along this chain of data transfers. For example, a callee in a recursion may receive a data item from its caller, process it, and then pass it to the next callee, and so on. Similarly, each caller in the chain may process and alter return values before passing them upward.

As we can see from the example above, this complexity is not explicitly reflected in the source code, requiring programmers to mentally construct the full sequence of data transfers in both directions. Given the limitations of working memory, this mental effort can quickly become overwhelming.

Therefore, we argue that the difficulty of building a mental model of the potentially complex data transfer sequence in a recursion is a key reason why recursion is difficult to understand.

3.2 Visualizing Data Flows in Recursion

An effective way to reduce the cognitive load associated with understanding data flow in recursion is through visualization. Our proposed data flow visualization, shown in Fig. 4, represents each recursive call as a box containing a copy of the source code. Arrows between boxes indicate the transfer of data between recursive calls: right-pointing arrows represent forward data flow, while left-pointing arrows indicate backward flow (i.e., return values). A dashed line and diamond arrowhead indicate passing by reference (see `node` in Fig. 4). A solid line and a triangular arrowhead indicating by value (see `target` and `depth` in Fig. 4). The red arrow in Fig. 4 indicates a return value. The visualization of the data flow is placed at the top of each code box, leaving space below for the visualization of the control flow described in Sect. 3.3.

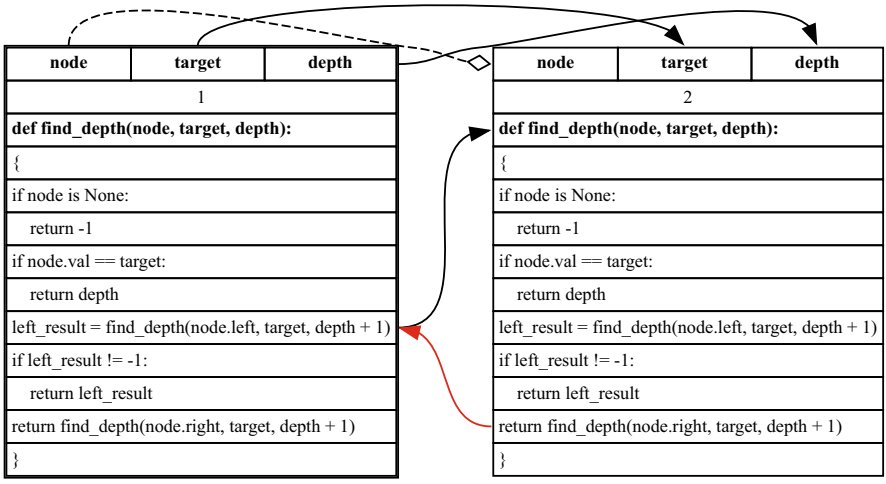

Fig. 4. Visualizing data flow in recursion using the example in Sect. 3.1. The red arrow represents the backward control flow. Dashed lines with diamond arrowheads indicate pass-by-reference, while solid arrows represent pass-by-value.

This design helps students trace the data flow throughout the recursion process without memorizing much information. Data flow visualization is generally overlooked in previous recursion visualizations, and our work fills this gap.

3.3 A New Visual Metaphor of Recursion

As discussed in Sect. 2.2, most existing recursion visualizations rely on the visual metaphors of a recursion tree and/or a call stack. A drawback of these approaches is that they primarily represent the forward progression of recursive calls, while the backward flow (i.e., the return to the caller) is not explicitly visualized. When a recursive call returns, it is simply removed from the recursion tree or call stack, making it difficult to identify where the control returns. Prior research has shown that many students struggle

with recursion specifically because they have difficulty understanding this backward flow [12,13]. It is therefore reasonable to hypothesize that visualizing the backward flow could enhance understanding. But how can this be done effectively? What is the appropriate visual metaphor for representing the return path in recursion?

To understand why the backward flow in recursion is difficult for students, it is helpful to first examine how it is used in programming. The backward flow is not unique to recursion. In fact, it is quite common in programming. For example, nested method calls (where method A calls method B, which in turn calls method C) involve multiple return flows. Similarly, loops (iterations) involve repeated backward jumps to the beginning of the loop body. Yet, students typically have little trouble understanding nested method calls and iteration [13,14,19,22]. The reason for this discrepancy may become clearer if we examine the visual patterns formed by the forward flows and backward flows in each case (Fig. 5 and Fig. 6).

In Fig. 5, we observe that the forward flows (black arrows) and backward flows (red arrows) in nested method calls form a shape similar to chained circles, while the flows in an iteration resemble concentric circles. In contrast, the pattern formed by forward flow (black arrows) and backward flows (red arrows) in recursion is irregular, complex, and difficult to describe (see Fig. 6). This suggests that backward flow is easier to understand when it is incorporated into circular shapes, as circles are simple and familiar.

We hypothesize that most students are intuitively familiar with four types of visual metaphors when reading computer programs (Fig. 5): a relatively straight line from top to bottom (representing linear program execution), a single circle (a single method call), a chain of circles (nested method calls), and concentric circles (loops). The difficulty in understanding recursion may not come from the concept of backward flow, but rather from the fact that recursive calls do not fit any of these familiar visual patterns (see Fig. 6).

One could argue that recursive calls are, in essence, deeply nested method calls. But why is recursion more difficult to understand than regular nested calls? In non-recursive nested method calls, all the method invocations are explicitly visible in the source code. To form a mental model, students only need to imagine the forward and backward flows between clearly defined methods. In contrast, a recursive function involves repeated invocations of the same method, but only one copy is visible in the code. Readers must mentally visualize a complex shape flow of data and control flows (see Fig. 6). These types of shapes are more difficult to create mentally than the relatively simple and symmetrical shapes in Fig. 5.

Call stacks and recursion trees can help reduce some of the cognitive load by externally representing the multiple instances of recursive calls. However, these structures are often unfamiliar visual metaphors for students. Moreover, both the call stack and the recursion tree are dynamic objects, growing and shrinking as recursive calls are made and returned. Mentally tracking such dynamic, constantly changing structures is even more challenging than reasoning about static visual representations.

Our solution is to visualize recursion in a way similar to regular nested method calls, using the familiar visual metaphor of a chain of circles. This approach makes it easier for students to construct a clear mental model of recursion. Our proposed visu-

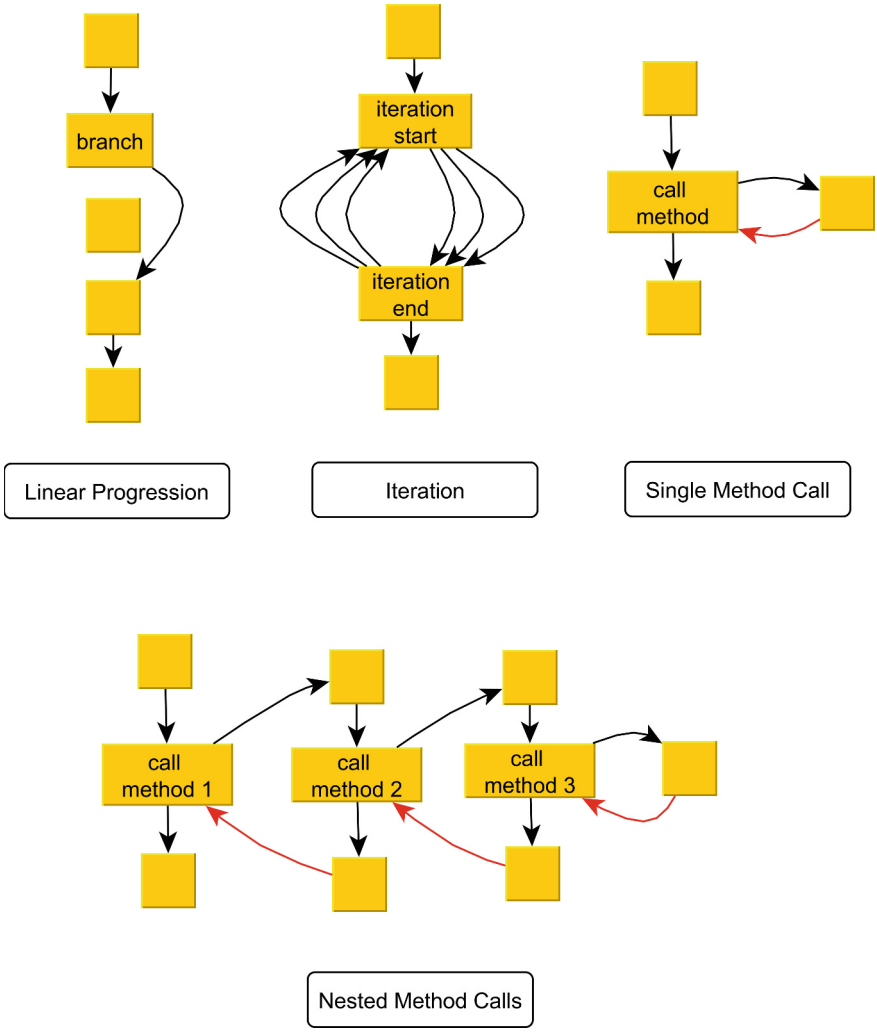

Fig. 5. Common visual patterns of program structures. The red arrows indicate backward control flow. (Color figure online)

alization is shown in Fig. 7 and has several key characteristics that distinguish it from prior visualization techniques.

- Each invocation of recursion is displayed as a separate copy of the recursive method. As the recursion progresses, more copies of the method are added to the sequence. This visualization essentially converts recursion into a sequence of nested method calls.
- Forward and backward flows are represented by arrows that together form a chain of circles, a visual metaphor that is more intuitive than a call stack (e.g., PythonTutor

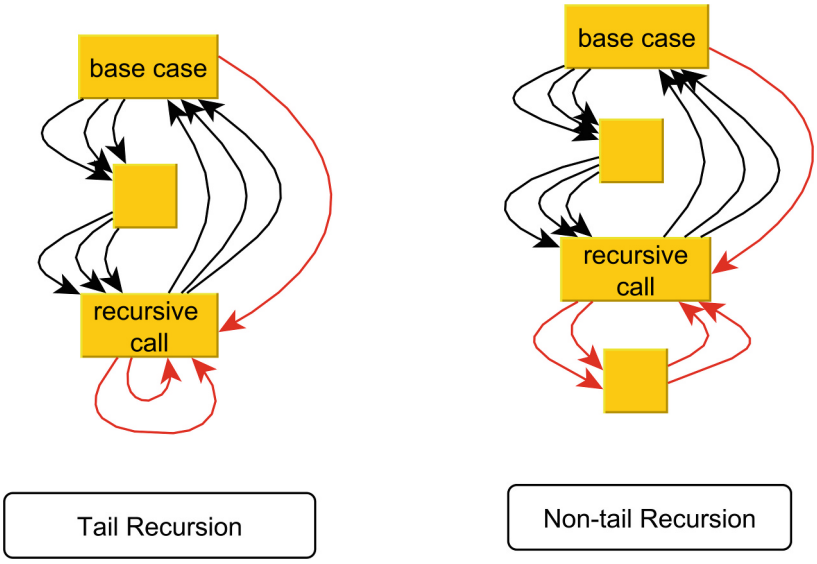

Fig. 6. Visual patterns of tail and non-tail recursions. The red arrows indicate backward control flow. (Color figure online)

[15]) or a recursion tree (e.g., CSVisTool [2]). Unlike a call stack or a recursion tree, after a recursive call returns, the completed recursive call is not removed from the visualization. This approach more closely resembles a regular sequence of nested method calls and allows backward arrows (in red color) to clearly indicate the exact line of code where the recursive call returns.

– This visualization also addresses another issue: previous recursion visualizations are often detached from the source code [2,11]. In contrast, our approach tightly integrates the recursion visualization with the source code view. Extra copies of the recursive method are arranged horizontally from left to right, ensuring they do not disrupt the reader's natural top-to-bottom reading flow. The original source code remains unchanged, and readers can choose to show or hide the horizontal recursion visualization as needed.
  Another advantage of this design is its use of a familiar game metaphor of the side quest. Entering the horizontal recursion visualization is like embarking on a side quest or detour that eventually returns to the main journey. This metaphor helps make recursion feel like a natural part of the program flow and is especially effective for understanding non-tail recursion.
– Data flow is displayed as a separate visualization positioned at the top, allowing the visualization to incorporate both data flow and control flow. Users can toggle the data flow visualization on or off as needed.

The visualization is dynamically generated using Python and Graphviz [3], an open-source graph visualization software. Combined with visualization tools such as Plotly, this can be displayed as a web-based interactive visualization.

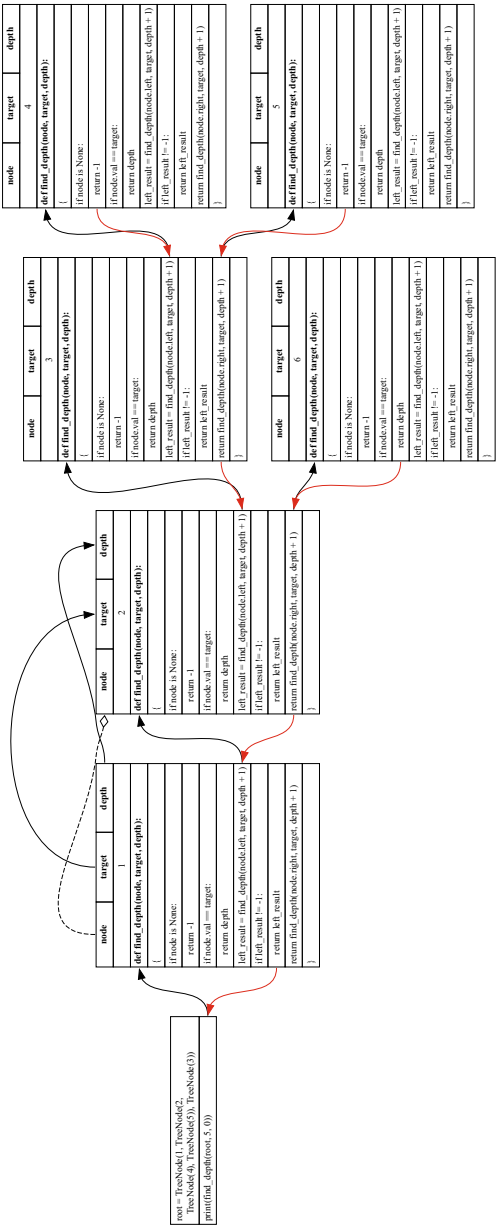

Fig. 7. This example illustrates our proposed recursion visualization, based on the sample program code in Sect. 3.1. Each node represents a recursive call. Black arrows indicate forward control flow, while red arrows represent the backward flow (i.e., returns). Together, these control flows form a visual pattern that resembles familiar nested method calls. Data flow is shown using arrows positioned at the top of each node.

Instructors can use this visualization to illustrate the control flow and data flow in recursion. To further reinforce conceptual understanding, instructors may require students to construct similar diagrams from recursive programs, either manually or with visualization tools such as Graphviz.

4 Conclusion and Future Work

We have introduced a new visualization technique for teaching and learning recursion, using the visual metaphor of a chain of circles to represent the recursive process. This approach helps students form a clearer mental model of recursion by addressing the drawbacks of traditional methods, such as the call stack and the recursion tree, and exploring new features, such as data flow visualization.

This is still a work in progress. We plan to further improve the visualization and develop additional case studies to illustrate a wide range of recursive scenarios. Most importantly, we aim to conduct user studies in data structures and algorithms courses to evaluate the effectiveness of this visualization in enhancing student learning, while also gathering feedback through surveys and comments to guide future refinements.

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Enhancing DDoS Mitigation Through AI Coaching: A Network-Oriented Computational Analysis of Adaptive Human-AI Defense Dynamics

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Abstract. Multi-vector Distributed Denial of Service (DDoS) attacks present a growing challenge to cybersecurity operations by combining various attack techniques across multiple network layers, often overwhelming both infrastructure and human decision-making. This study models the mental and technical response processes of a cybersecurity operator during such attacks using adaptive network modeling. Two models were developed: a base model simulating operator behavior without AI support, and an enhanced model featuring an adaptive AI-coach designed to assist in belief formation and attack mitigation. Both models integrate mental, physiological, and technical states and simulate responses over multiple attack cycles. Simulation results reveal that while the base model shows initial learning capacity, operator fatigue leads to performance decline and early mitigation collapse. In contrast, the AI-coach model maintains learning levels during later stages in the simulations through adaptive support, enabling prolonged mitigation effectiveness.

Keywords: AI-Coaching · DDoS attack · Cybersecurity · Network Modeling

1 Introduction

Distributed Denial of Service (DDoS) attacks are among the most prevalent and disruptive forms of cyber threats faced by organizations today. In particular, multi-vector DDoS attacks combine several attack techniques, such as volumetric floods, protocol abuses, and application-layer exploits, into a coordinated assault that overwhelms both infrastructure and security operations. This complexity allows attackers, also called botmasters, to bypass traditional defenses and prolong the attack's impact [21]. Historical incidents, including the 2016 Mirai botnet attack against Dyn, have demonstrated the power of multi-vector approaches, where DNS amplification was combined with TCP and HTTP floods to disrupt major online services for extended periods [1], and the same botnet was earlier that year used in a record-setting attack against the security journalist Brian Krebs's website, illustrating the scalability and adaptability of such threats [15].

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Research has increasingly recognized that human operators can experience cognitive overload and performance degradation during high-pressure cyber incidents. Factors such as stress, fatigue, and attentional tunneling may lead to slower detection and suboptimal mitigation decisions [8]. AI-based coaching systems have emerged as promising tools to augment human capabilities under such circumstances. By offering adaptive recommendations, real-time anomaly interpretations, and strategic guidance, AI coaches can help operators maintain situational awareness and improve response effectiveness. This aligns with broader trends in cyber defense, where human-AI teaming is seen as a critical factor in managing sophisticated and persistent threats [26].

To simulate and analyze these dynamics systematically, this study applies the adaptive network modeling methodology described by [29–31]. Network-Oriented Modeling enables researchers to represent interacting processes, including cognitive, social, and technical dynamics, as temporal-causal networks where states and their influences evolve over time. This approach is particularly suited to modeling cybersecurity scenarios, as it can integrate both technical observations (e.g., server load, detected anomalies), physiological states of security operators (e.g., vigilance, stress, fatigue), as well as the forming of beliefs over time and establishing determinations about situations. In this work, two models were developed: a baseline model depicting mitigation processes without additional support, and an enhanced model incorporating an AI coach to assist operators in recognizing and responding to attack vectors more effectively.

This research aims to explore how modeling the DDoS mitigation process, with and without AI coaching, can help understand and potentially improve operational resilience. The research includes simulation experiments to compare these two scenarios, along with What-If analyses to investigate alternative parameter configurations and risk assessments to evaluate potential vulnerabilities. Additionally, the study is informed by related work in computational modeling of shared mental models and organizational learning to support safety and security within the cyberspace [25, 35]. Through these methods, this work contributes to a deeper understanding of the interaction between human factors and technological aids in cyber defense contexts.

2 Background Knowledge

Distributed Denial of Service (DDoS) attacks are a persistent threat to organizations worldwide. At their core, DDoS attacks involve overwhelming a target's web infrastructure or application with a high volume of malicious traffic, rendering services unavailable to legit users. While traditional DDoS attacks often use a single technique (for example, a volumetric flood), multi-vector DDoS attacks strategically combine several attack vectors that can happen simultaneously or sequentially. This increases the complexity of detection and mitigation and forces defenders to manage multiple threats across different layers of the network stack at the same time. Unlike a regular DDoS attack, which might only exhaust bandwidth (Layer 3) or server resources (Layer 7) for instance, multi-vector attacks blend these tactics, targeting bandwidth, protocol vulnerabilities, and application resources all at once to maximize the disruption and evade mitigation tools [11, 12].

Multi-vector DDoS attacks typically exploit several OSI layers. Volumetric flooding attacks primarily affect Layer 3 (Network) and Layer 4 (Transport), flooding the network

connection between the victim and the internet. Examples include DNS amplification, UDP floods, and ICMP floods. DNS amplification attacks use open DNS resolvers to reflect and amplify small queries into massive payloads aimed at the victim [5]. UDP floods bombard random ports with UDP packets, forcing the victim to generate ICMP unreachable responses, exhausting processing resources. ICMP floods overwhelm the target with ICMP Echo Request packets (also called pings), taking over bandwidth and CPU cycles. Protocol abuse attacks primarily disrupt Layer 4 and Layer 7. For instance, TCP SYN floods exploit the TCP three-way handshake, a system that indicates when both parties are ready for communication, by sending a barrage of SYN packets (a part of the communication consent handshake) without completing connections, depleting server connection tables [21]. ACK floods similarly overload stateful firewalls or servers by exploiting ACK processing. Finally, application-layer attacks focus on Layer 7. HTTP GET/POST floods simulate legitimate application requests to exhaust server resources, while Slowloris attacks maintain many open connections with partial HTTP requests, tying up threads and causing denial of service at the application level [22].

The realistic process of defending against such attacks in a Security Operations Center (SOC) begins with monitoring traffic baselines. Detection tools, such as intrusion detection systems (IDS), anomaly detection systems, and flow analytics, watch for deviations in volume, protocol distributions, and latency. When an anomaly threshold is exceeded, automated alerts trigger. Security analysts then determine whether the spike is benign (e.g., a marketing campaign) or malicious, for instance by checking the user goal and/or traffic source details. If malicious, a process assesses which attack vectors are in play. For example, analysts might inspect NetFlow records, firewall logs, and application server logs to determine whether DNS amplification, ACK floods, or HTTP flood attacks are present. Once attack types are identified, defenders choose appropriate mitigation measures: enabling rate-limiting, applying ingress filtering, diverting traffic to a content delivery network (CDN) scrubbing service, or blocking malicious IP ranges. This process must happen quickly, as delays can result in severe service disruption and financial impact [9, 10].

Mitigation techniques are tailored to several attack methods. Ingress filtering blocks spoofed packets at the network edge (IETF). Geo/IP blacklisting excludes traffic from known malicious regions or IP addresses. CDN and cloud scrubbing services absorb and clean traffic before it reaches the target infrastructure. Rate-limiting constrains the rate of requests per source, reducing application exhaustion risks. Finally, Web Application Firewalls (WAFs) inspect HTTP traffic to block malicious payloads [21], 22.

The motivations of botmasters orchestrating such attacks are varied. Financial gain through ransom extortion is a common occurring motive where attackers threaten to sustain a DDoS until a payment is made [11]. Also, competitors may use DDoS to disrupt rival services and damage reputations in this way. Hacktivist groups also leverage DDoS as a form of protest. In all cases, the goal is to degrade availability, damage trust, or force concessions of some kind [37].

A critical challenge in this process is human performance under stress. Fatigue is well documented as a factor that degrades situational awareness and slows down response times in security operations. Studies show that prolonged cognitive workload during incident response increases errors and delays in determining appropriate mitigation actions

[14]. Fatigue can lead to incomplete log analysis, misclassification of attack vectors, or failing attack pattern recognition, allowing attackers to prolong the impact of their ambush.

AI-based coaching systems can help overcome some of these human limitations by continuously monitoring attack indicators and suggesting optimal responses. AI tools do not experience fatigue or stress and can maintain high vigilance during prolonged incidents. For example, AI systems can assist analysts by correlating data across sensors, recommending countermeasures, and ranking their likely effectiveness. When an operator is fatigued, an AI coach can help prevent oversight by re-evaluating evidence and prompting decisions [25]. However, AI systems also have limitations. They can lack contextual judgment, misinterpret ambiguous signals, or fail to adapt to novel attack patterns without retraining. Human expertise remains essential for interpreting uncertain data and balancing mitigation priorities against business constraints.

Attackers adapt as well. When defenders begin mitigation, botmasters often intensify evasion tactics. Two common approaches are increasing the spread of the attack (distributing traffic across more IPs, protocols, and ports) and mimicking normal user behavior. By emulating legitimate headers, timing, and request patterns, attackers make detection and filtering more difficult. This tactic is especially prevalent in sophisticated Layer 7 attacks, where malicious requests blend seamlessly into baseline application traffic [6].

Hence, defending against multi-vector DDoS attacks efficiently requires a combination of automated detection, flexible mitigation strategies, and human judgment, while AI-driven support systems can assist by augmenting situational awareness and providing recommendations that help maintain resilience against evolving threats.

3 The Modeling Approach Used

The modeling approach applied in this research is based on the Network-Oriented Modeling methodology, a coordinated framework introduced and elaborated by [29–31] to simulate complex adaptive systems, including mental, social, and biological processes. Network-Oriented Modeling enables the development of computational models that represent dynamic processes as networks of interacting states, each evolving over time in response to the causal influence of other states. This temporal-causal perspective makes it so that processes are modeled as systems where change happens continuously or in discrete time steps, directed by specific causal dependencies [31]. In the context of this work, Network-Oriented Modeling is particularly suited to capture the intertwined dynamics of technical observations, human factors such as stress and fatigue, and adaptive learning processes involved in mitigating a multi-vector DDoS attack.

In this methodology, each state *Y(t)* evolves through time according to a main numerical formalization:

$$\frac{dY(t)}{dt} = \boldsymbol{\eta}_Y \left[\mathbf{c}_Y \left(\boldsymbol{\omega}_{X_1,Y} X_1(t), \ldots, \boldsymbol{\omega}_{X_k,Y} X_k(t) \right) - Y(t) \right]$$

where X_1,...,X_k are influencing states, $\boldsymbol{\omega}$ are connection weights, $\mathbf{c}_Y$ is a combination function aggregating influences, and $\boldsymbol{\eta}_Y$ is a speed factor determining how rapidly the

state converges to the aggregated influence [31]. A corresponding discrete-time version of the formalization shows how the state updates per timestep:

$$Y(t + \Delta t) = Y(t) + \boldsymbol{\eta}_Y \left[\mathbf{c}_Y \left(\boldsymbol{\omega}_{X_1,Y} X_1(t), \ldots, \boldsymbol{\omega}_{X_k,Y} X_k(t) \right) - Y(t) \right] \Delta t$$

This temporal-causal structure is particularly well-fitted when modeling mental models embedded within broader process models, as it allows the specification of how perceptions, beliefs, and intentions (mental states) adaptively interact with external observations (world states).

Dependencies between states are categorized into three essential concepts: **connectivity**, **aggregation**, and **timing**. *Connectivity* describes which states influence each other and with what strength, parameterized via the connection weights $\boldsymbol{\omega}$. *Aggregation* determines how multiple incoming influences are combined through combination functions $\mathbf{c}_Y$. The methodology provides a library of such functions; in this work, the **alogistic** and **stepmod** combination functions were applied to flexibly model non-linear, threshold-driven and sudden, stepwise dynamics [31]. Finally, *Timing* involves speed factors $\boldsymbol{\eta} Y$ that control the responsiveness of a state to changes in its inputs over time. Collectively, these three aspects, connectivity (who influences whom), aggregation (how influences combine), and timing (how fast a state reacts), define the dynamic behavior of each modeled state.

A distinctive feature of the methodology is its use of reification levels, enabling modeling of adaptation and meta-adaptation, and more. This was inspired by the area of metalevel architectures and reflection in AI; e.g., [2, 16, 20, 27, 28, 36]. The base level captures the ordinary process dynamics. For example, in this model, state server_traffic and the mental state b_DNS_amp_attack are base-level states reflecting observable network indicators and the operator's established belief, respectively. The first-order reification level introduces adaptation states such as **W**-states (for weights of connections) that model synaptic plasticity. For instance, the state $\mathbf{W}_{\text{b_DNS_amp_attack, det_volumetric_flooding}}$ represents the learning of connectivity weights between forming a belief and determining the attack type. The second-order reification level extends this further to meta-adaptation: reified representation states for a connection weight to a reification state (referred to intuitively as $\mathbf{W}_{\mathbf{WW}}$-states) for instance capture changes in how adaptation itself changes over time, e.g., modeling synaptic metaplasticity [30, 31]. In the introduced model, $\mathbf{W}_{\mathbf{WW}}$-states such as $\mathbf{W}_{\mathbf{WW}X49,X42}$ model how the AI-coach communicates its knowledge to the employee in a context-sensitive, controlled manner when necessary, through connections to monitoring states located in the first-order reification level. Other specialized adaptation states, such as **H**-states (adaptive speed factors) and **T**-states (adaptive thresholds), can also play roles in refining the process dynamics.

Implementation of the model was conducted in MATLAB using the dedicated software environment for adaptive network modeling, which establishes the network dynamics through five types of matrices: connectivity (*mb*), non-adaptive weights (*mcw*), speed factors (*ms*), combination function weights (*mcfw*) and parameters (*mcfp*). These matrices include all dependency information necessary to numerically simulate the evolution of the full system over time.

4 The Adaptive Network Model Design

This chapter is dedicated to a detailed discussion of the design of the adaptive network model developed in this work. It outlines the underlying considerations and assumptions that shaped the model structure and parameterization. It describes how different elements and dependencies were specified to reflect the targeted DDoS mitigation process, both without and with AI-coach support. The following sections present the design of the base model and the extended AI-coach model.

4.1 Base Model

The base model represents the core adaptive network structure of the DDoS mitigation process without the use of AI-coaching. It shows the cognitive, perceptual, and physiological dynamics of a human cybersecurity operator as they respond to repeating multi-vector DDoS attacks. Figure 1 and Table 1 provide a visual overview and detailed descriptions of all states in the base model. The model simulates multiple attack cycles within a time frame of 5000 steps, enabling the operator to learn from previous experiences across attacks.

The model begins with the triggering of a multi-vector attack (X_1), composed of simultaneous volumetric flooding (X_2), application-layer (X_3), and protocol abuse (X_4) attacks. These attack vectors increase the incoming server traffic (X_5), which is linked to the emergence of a latency anomaly (X_6). Latency anomalies are a common indicator of overload caused by DDoS traffic surges [23]. As the server becomes increasingly overloaded, alert volume (X_7) rises and raises suspicion (X_8) of a potential cyberattack. This suspicion triggers a systematic log inspection process, including firewall (X_9), server (X_{10}), and external monitoring tools (X_{11}). These inspections are essential, as real-world studies confirm that human operators rely heavily on such tools for DDoS detection and root cause analysis [12].

The results from log inspections feed into a mental model in which the operator forms internal beliefs (states X_{12}–X_{18}) about specific attack methods. These beliefs include detection of DNS amplification (X_{12}), TCP SYN flood (X_{15}) and HTTP GET/POST flood (X_{17}) among others. These categories represent a taxonomy of frequent DDoS vectors observed in practice [21]. The formation of these beliefs is influenced by both the raw indicators from logs and the obfuscation efforts by the attacker, represented by mimicry (X_{30}) and spreading (X_{31}) tactics. The model assigns each of these attack-method beliefs an impact through connection weights derived from the real-time inspection data.

Following the belief states, the model transitions to internal determination states of the attack types occurring: volumetric flooding (X_{19}), protocol abuse (X_{20}), and application-layer attacks (X_{21}). These internal determinations represent a higher-level classification step, essential for planning mitigation strategies. The detection states are causally connected to their underlying belief states according to which attack method fuels a specific attack type. For example, X_{19} is driven by X_{12}–X_{14}, while X_{21} is driven by X_{17} and X_{18} (see Fig. 1). The temporal causality implemented here mirrors a reasoning process where multiple low-level clues converge toward broader attack pattern recognition.

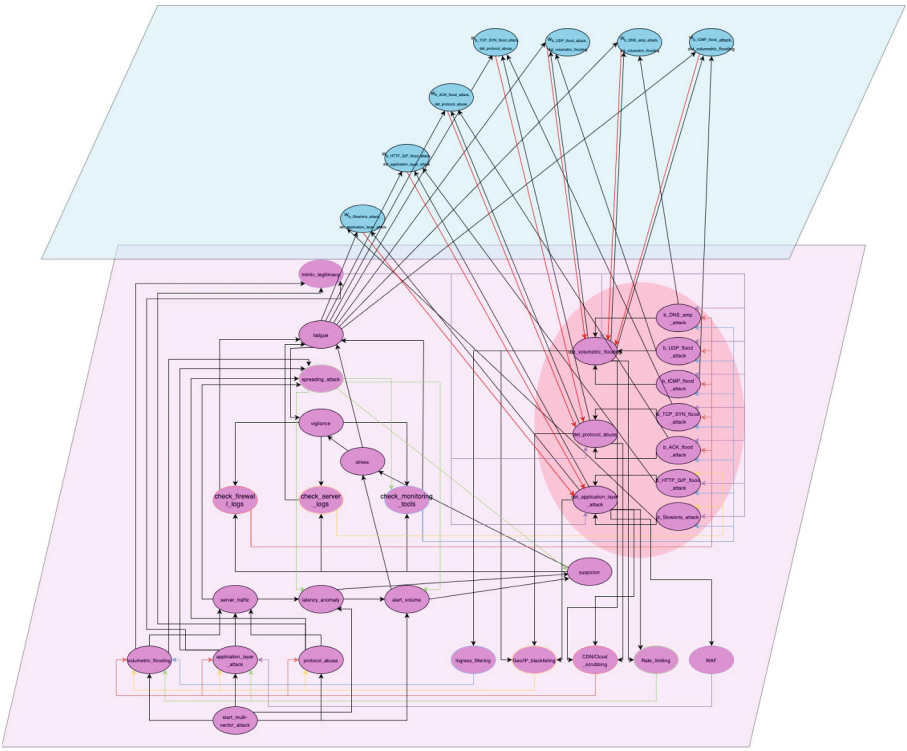

Fig. 1. Base model network diagram.

Once a determination has been made, corresponding effective mitigation actions are initiated: ingress filtering (X_{22}), Geo/IP blacklisting (X_{23}), CDN rerouting (X_{24}), rate limiting (X_{25}), and WAF reinforcement (X_{26}). These decisions are causally connected to their relevant determinations and reflect standard defense mechanisms recommended in operational handbooks [10]. These defensive responses are not directly triggered by the beliefs but only after higher-order classification, modeling the operator's critical decision-making bottleneck.

The physiological states of the operator, stress (X_{27}), fatigue (X_{28}), and vigilance (X_{29}), are implemented to account for human limitations. Fatigue grows over time with data inspection efforts and stress, while stress correlates with alert volume and suspicion. These factors collectively modulate the operator's effectiveness. Including vigilance is critical, as studies have shown that sustained attention capacity degrades with stress and fatigue, increasing error likelihood during incident response [8].

The first-order adaptation level introduces a key mechanism through which the system learns to link past experience to improved future response. The states X_{32} to X_{38} (all prefixed with **W**) represent the adaptive connection weights between beliefs (X_{12}–X_{18}) and their corresponding determinations (X_{19}–X_{21}). These weights allow the model to learn associations over time, such as how a combined pattern of certain attack methods in logs is indicative of a specific attack type. This form of structural learning reflects

Table 1. Base model network state descriptions.

ID	state	description	level
X_1	start_multi-vector_attack	World state representing the initiation of a multi-vector DDoS attack	Base Level
X_2	volumetric_flooding	World state representing the occurrence of a volumetric flooding attack	
X_3	application_layer_attack	World state representing the occurrence of an application layer attack	
X_4	protocol_abuse	World state representing the occurrence of a protocol abuse attack	
X_5	server_traffic	World state representing the frequency of traffic received by the server	
X_6	latency_anomaly	World state representing the detection of abnormal latency or delay in the network	
X_7	alert_volume	World state indicating the number or volume of triggered alerts during an incident	
X_8	suspicion	World state representing the perceived likelihood that current activity indicates an attack	
$X9$	check_firewall_logs	World state representing the inspection of firewall log data for attack indicators	
X_{10}	check_server_logs	World state representing the inspection of server log data for attack indicators	
X_{11}	check_monitoring_tools	World state representing the inspection of monitoring tools for attack indicators	
X_{12}	b_DNS_amp_attack	Internal state representing the belief that a DNS amplification attack is ongoing	

(*continued*)

Table 1. (*continued*)

ID	state	description	level
X_{13}	b_UDP_flood_attack	Internal state representing the belief that a UDP flood attack is ongoing	
X_{14}	b_ICMP_flood_attack	Internal state representing the belief that a ICMP flood attack is ongoing	
X_{15}	b_TCP_SYN_flood_attack	Internal state representing the belief that a TCP/SYN flood attack is ongoing	
X_{16}	b_ACK_flood_attack	Internal state representing the belief that an ACK flood attack is ongoing	
X_{17}	b_HTTP_G/P_flood_attack	Internal state representing the belief that a HTTP GET/POST flood attack is ongoing	
X_{18}	b_Slowloris_attack	Internal state representing the belief that a Slowloris attack is ongoing	
X_{19}	det_volumetric_flooding	Internal state representing the detection/determination of a volumetric flooding attack	
X_{20}	det_protocol_abuse	Internal state representing the detection/determination of a protocol abuse attack	
X_{21}	det_application_layer_attack	Internal state representing the detection/determination of an application layer attack	
X_{22}	Ingress_filtering	World state representing the decision to activate ingress filtering to block traffic	
X_{23}	Geo/IP_blacklisting	World state representing the decision to enable geo-based or IP-based blocking of traffic	
X_{24}	CDN/Cloud_scrubbing	World state representing the decision to route traffic through CDN's or cloud scrubbing services	

(*continued*)

Table 1. (*continued*)

ID	state	description	level
X_{25}	Rate_limiting	World state representing the decision to enable rate limiting to throttle traffic	
X26	WAF	World state representing the decision to reinforce or adjust Web Application Firewall policies	
X_{27}	stress	Physiological state representing operator stress levels	
X_{28}	fatigue	Physiological state representing operator level of fatigue	
X_{29}	vigilance	Physiological state representing the operators' sustained attentional capacity	
X_{30}	mimic_legitimacy	World state representing evading tactics employed by a botmaster (attacker)	
X_{31}	spreading_attack	World state representing evading tactics employed by a botmaster (attacker)	
X_{32}	$\mathbf{W}x_{12},x_{19}$	Self-model state for the weight $\boldsymbol{\omega}x_{12},x_{19}$ of the connection from b_DNS_amp_attack to det_volumetric_flooding	First-order adaptation level
X_{33}	$\mathbf{W}x_{13},x_{19}$	Self-model state for the weight $\boldsymbol{\omega}x_{13},x_{19}$ of the connection from b_UDP_flood_attack to det_volumetric_flooding	
X_{34}	$\mathbf{W}x_{14},x_{19}$	Self-model state for the weight $\boldsymbol{\omega}x_{14},x_{19}$ of the connection from b_ICMP_flood_attack to det_volumetric_flooding	
X_{35}	$\mathbf{W}x_{15},x_{20}$	Self-model state for the weight $\boldsymbol{\omega}x_{15},x_{20}$ of the connection from b_TCP_SYN_flood_attack to det_protocol_abuse	

(*continued*)

the development of pattern recognition capabilities in human analysts and mirrors what is referred to as first-order adaptation by [30]. Over multiple attack cycles within the

Table 1. (*continued*)

ID	state	description	level
X_{36}	$\mathbf{W}x_{16},x_{20}$	Self-model state for the weight $\boldsymbol{\omega}x_{16},x_{20}$ of the connection from b_ACK_flood_attack to det_protocol_abuse	
X_{37}	$\mathbf{W}x_{17},x_{21}$	Self-model state for the weight $\boldsymbol{\omega}x_{17},x_{21}$ of the connection from b_HTTP_G/P_flood_attack to det_application_layer_attack	
X_{38}	$\mathbf{W}x_{18},x_{21}$	Self-model state for the weight $\boldsymbol{\omega}x_{18},x_{21}$ of the connection from b_Slowloris_attack to det_application_layer_attack	

5000 time step simulation, these weights evolve, enabling the operator to become more effective through experience.

The combination functions used (**alogistic** and **stepmod**) support weighted integration and thresholding, as well as sudden occurrences, ensuring the model behaves in a biologically and cognitively plausible manner as shown in [31].

4.2 AI-Coach Model

At the base level, the AI coach is implemented as a mirrored cognitive model of the employee's mental network (see Fig. 2). Every belief and determination state that exists in the operator's mental model (states X_{12} to X_{21}) is duplicated and labeled with an "AIC_" prefix (states X_{32} to X_{41}), indicating they belong to the AI coach's internal cognition. The AI coach follows a similar structure in forming attack beliefs from incoming data and inferring attack types, as represented by the same base connectivity within the operator's mental model. However, the AI coach lacks any outgoing connections from its mental states to the environment, showing its intended design to offer guidance and not to directly take mitigation decisions. It is a knowledge support system and cannot trigger actions like ingress filtering or rate limiting for instance directly. This clear division ensures that the coach supports, but does not replace, the human operator's judgment.

In the first-order adaptation level, the AI coach's model includes learning processes (**W**-states X_{49} to X_{55}), which mirror the employee's own learning states (X_{42} to X_{48}) for belief-determination pairs. These AI learning states are assumed to have perfect knowledge from prior data, which is why their input connections from the world and internal layers are absent; they are considered fully trained. However, communication of this knowledge to the employee is not constant. To prevent redundancy and cognitive overload, three monitoring states (X_{56} to X_{58}) assess when intervention is actually needed. These states monitor both the AI's determination states (indicated with prefix *AIC_det_*) and the employee's corresponding determination states. For instance, state

X_{56} tracks X_{39} and X_{19} (see Table 2), checking whether the operator has sufficiently learned to make a determination, and therefore deploy efficient mitigation. If not, and if the AI coach has already made that determination, the system considers this a cue for support.

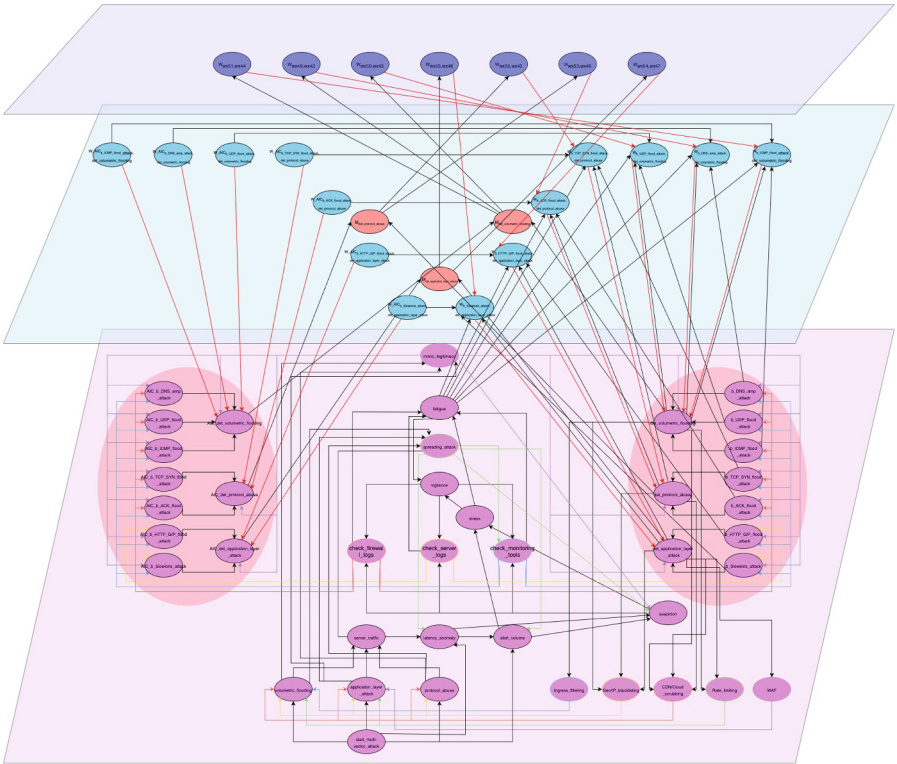

Fig. 2. AI-coach enhanced model network picture.

The assumption here is that all determination actions from the employee are logged into a SOC (Security Operations Center) platform, which the AI coach can access. This ensures that knowledge already known to the operator isn't redundantly communicated. The model therefore respects real-world operational practice, where SOC platforms act as shared intelligence hubs to enable cyber security operators to collaborate in a proactive mitigation process.

At the second-order adaptation level, the AI coach uses states X_{59} to X_{65} to modulate communication between its own **W**-states and the employee's **W**-states. Each second-order state adaptively enables the transfer of learning from the AI coach to the operator by connecting a specific AI coach learning state (indicated with prefix **W**_AIC_) to the corresponding employee learning state. These second-order adaptive learning states (intuitively called $\mathbf{W_{WW}}$-states) act on signals from the monitoring states to decide when to open or close communication channels.

Table 2. AI-coach model network state descriptions: base level.

ID	state	description	level
X_1	start_multi-vector_attack	World state representing the initiation of a multi-vector DDoS attack	Base Level
X_2	volumetric_flooding	World state representing the occurrence of a volumetric flooding attack	
X_3	application_layer_attack	World state representing the occurrence of an application layer attack	
X_4	protocol_abuse	World state representing the occurrence of a protocol abuse attack	
X_5	server_traffic	World state representing the frequency of traffic received by the server	
X_6	latency_anomaly	World state representing the detection of abnormal latency or delay in the network	
X_7	alert_volume	World state indicating the number or volume of triggered alerts during an incident	
X_8	suspicion	World state representing the perceived likelihood that current activity indicates an attack	
$X9$	check_firewall_logs	World state representing the inspection of firewall log data for attack indicators	
X_{10}	check_server_logs	World state representing the inspection of server log data for attack indicators	
X_{11}	check_monitoring_tools	World state representing the inspection of monitoring tools for attack indicators	
X_{12}	b_DNS_amp_attack	Internal state representing the belief that a DNS amplification attack is ongoing	
X_{13}	b_UDP_flood_attack	Internal state representing the belief that a UDP flood attack is ongoing	
X_{14}	b_ICMP_flood_attack	Internal state representing the belief that a ICMP flood attack is ongoing	

(*continued*)

Table 2. (*continued*)

ID	state	description	level
X_{15}	b_TCP_SYN_flood_attack	Internal state representing the belief that a TCP/SYN flood attack is ongoing	
X_{16}	b_ACK_flood_attack	Internal state representing the belief that an ACK flood attack is ongoing	
X_{17}	b_HTTP_G/P_flood_attack	Internal state representing the belief that a HTTP GET/POST flood attack is ongoing	
X_{18}	b_Slowloris_attack	Internal state representing the belief that a Slowloris attack is ongoing	
X_{19}	det_volumetric_flooding	Internal state representing the detection/determination of a volumetric flooding attack	
X_{20}	det_protocol_abuse	Internal state representing the detection/determination of a protocol abuse attack	
X_{21}	det_application_layer_attack	Internal state representing the detection/determination of an application layer attack	
X_{22}	Ingress_filtering	World state representing the decision to activate ingress filtering to block traffic	
X_{23}	Geo/IP_blacklisting	World state representing the decision to enable geo-based or IP-based blocking of traffic	
X_{24}	CDN/Cloud_scrubbing	World state representing the decision to route traffic through CDN's or cloud scrubbing services	
X_{25}	Rate_limiting	World state representing the decision to enable rate limiting to throttle traffic	
$X26$	WAF	World state representing the decision to reinforce or adjust Web Application Firewall policies	
X_{27}	stress	Physiological state representing operator stress levels	
X_{28}	fatigue	Physiological state representing operator level of fatigue	

(*continued*)

Table 2. (*continued*)

ID	state	description	level
X_{29}	vigilance	Physiological state representing the operators' sustained attentional capacity	
X_{30}	mimic_legitimacy	World state representing evading tactics employed by a botmaster (attacker)	
X_{31}	spreading_attack	World state representing evading tactics employed by a botmaster (attacker)	
X_{32}	AIC_b_DNS_amp_attack	AI coach internal state representing the belief that a DNS amplification attack is ongoing	
X_{33}	AIC_b_UDP_flood_attack	AI coach internal state representing the belief that a UDP flood attack is ongoing	
X_{34}	AIC_b_ICMP_flood_attack	AI coach internal state representing the belief that a ICMP flood attack is ongoing	
X_{35}	AIC_b_TCP_SYN_flood_attack	AI coach internal state representing the belief that a TCP/SYN flood attack is ongoing	
X_{36}	AIC_b_ACK_flood_attack	AI coach internal state representing the belief that an ACK flood attack is ongoing	
X_{37}	AIC_b_HTTP_G/P_flood_attack	AI coach internal state representing the belief that a HTTP GET/POST flood attack is ongoing	
X_{38}	AIC_b_Slowloris_attack	AI coach internal state representing the belief that a Slowloris attack is ongoing	
X_{39}	AIC_det_volumetric_flooding	AI coach internal state representing the detection/determination of a volumetric flooding attack	
X_{40}	AIC_det_protocol_abuse	AI coach internal state representing the detection/determination of a protocol abuse attack	
X_{41}	AIC_det_application_layer_attack	AI coach internal state representing the detection/determination of an application layer attack	

In essence, the monitoring states determine in a context-sensitive manner *when* help is needed, and the $\mathbf{W_{WW}}$-states determine *how* help is delivered adaptively (see Table 3). This design supports well-timed, relevant, and adaptive coaching, only intervening when the operator's learning is insufficient for accurate threat determination and ultimately, mitigation. For more details, see Linked Data at https://www.researchgate.net/publication/393637001.

Table 3. AI-coach model network state descriptions: higher level states.

ID	state	description	level
X_{42}	$\mathbf{W}x_{12},x_{19}$	Self-model state for the weight $\boldsymbol{\omega}x_{12},x_{19}$ of the connection from b_DNS_amp_attack to det_volumetric_flooding	First-order adaptation level
X_{43}	$\mathbf{W}x_{13},x_{19}$	Self-model state for the weight $\boldsymbol{\omega}x_{13},x_{19}$ of the connection from b_UDP_flood_attack to det_volumetric_flooding	
X_{44}	$\mathbf{W}x_{14},x_{19}$	Self-model state for the weight $\boldsymbol{\omega}x_{14},x_{19}$ of the connection from b_ICMP_flood_attack to det_volumetric_flooding	
X_{45}	$\mathbf{W}x_{15},x_{20}$	Self-model state for the weight $\boldsymbol{\omega}x_{15},x_{20}$ of the connection from b_TCP_SYN_flood_attack to det_protocol_abuse	
X_{46}	$\mathbf{W}x_{16},x_{20}$	Self-model state for the weight $\boldsymbol{\omega}x_{16},x_{20}$ of the connection from b_ACK_flood_attack to det_protocol_abuse	
X_{47}	$\mathbf{W}x_{17},x_{21}$	Self-model state for the weight $\boldsymbol{\omega}x_{17},x_{21}$ of the connection from b_HTTP_G/P_flood_attack to det_application_layer_attack	
X_{48}	$\mathbf{W}x_{18},x_{21}$	Self-model state for the weight $\boldsymbol{\omega}x_{18},x_{21}$ of the connection from b_Slowloris_attack to det_application_layer_attack	
X_{49}	$\mathbf{W_AIC}x_{32},x_{39}$	Self-model state for the weight $\boldsymbol{\omega}x_{32},x_{39}$ of the connection from AIC_b_DNS_amp_attack to AIC_det_volumetric_flooding	

(*continued*)

Table 3. (*continued*)

ID	state	description	level
X_{50}	**W_AIC**x_{33},x_{39}	Self-model state for the weight $\boldsymbol{\omega}x_{33},x_{39}$ of the connection from AIC_b_UDP_flood_attack to AIC_det_volumetric_flooding	
X_{51}	**W_AIC**x_{34},x_{39}	Self-model state for the weight $\boldsymbol{\omega}x_{34},x_{39}$ of the connection from AIC_b_ICMP_flood_attack to AIC_det_volumetric_flooding	
X_{52}	**W_AIC**x_{35},x_{40}	Self-model state for the weight $\boldsymbol{\omega}x_{35},x_{40}$ of the connection from AIC_b_TCP_SYN_flood_attack to AIC_det_protocol_abuse	
X_{53}	**W_AIC**x_{36},x_{40}	Self-model state for the weight $\boldsymbol{\omega}x_{36},x_{40}$ of the connection from AIC_b_ACK_flood_attack to AIC_det_protocol_abuse	
X_{54}	**W_AIC**x_{37},x_{41}	Self-model state for the weight $\boldsymbol{\omega}x_{37},x_{41}$ of the connection from AIC_b_HTTP_G/P_flood_attack to AIC_det_application_layer_attack	
X_{55}	**W_AIC**x_{38},x_{41}	Self-model state for the weight $\boldsymbol{\omega}x_{38},x_{41}$ of the connection from AIC_b_Slowloris_attack to AIC_det_application_layer_attack	
X_{56}	**M**$x_{19,}x_{39}$	State monitoring AIC_det_volumetric_flooding and det_volumetric_flooding	
X_{57}	**M**$x_{20,}x_{40}$	State monitoring AIC_det_protocol_abuse and det_protocol_abuse	
X_{58}	**M**$x_{21,}x_{41}$	State monitoring AIC_det_application_layer_attack and det_application_layer_attack	
X_{59}	**Ww**$x_{49},$**w**x_{42}	Second-order state representing the adaptive modulation of the connection weight between $\boldsymbol{\omega}x_{49}$ and $\boldsymbol{\omega}x_{42}$	Second-order adaptation level
X_{60}	**Ww**$x_{50},$**w**x_{43}	Second-order state representing the adaptive modulation of the connection weight between $\boldsymbol{\omega}x_{50}$ and $\boldsymbol{\omega}x_{43}$	

(*continued*)

Table 3. (*continued*)

ID	state	description	level
X_{61}	$\mathbf{Ww}_{x_{51},x_{44}}$	Second-order state representing the adaptive modulation of the connection weight between $\boldsymbol{\omega}_{x_{51}}$ and $\boldsymbol{\omega}_{x_{44}}$	
X_{62}	$\mathbf{Ww}_{x_{52},x_{45}}$	Second-order state representing the adaptive modulation of the connection weight between $\boldsymbol{\omega}_{x_{52}}$ and $\boldsymbol{\omega}_{x_{45}}$	
X_{63}	$\mathbf{Ww}_{x_{53},x_{46}}$	Second-order state representing the adaptive modulation of the connection weight between $\boldsymbol{\omega}_{x_{53}}$ and $\boldsymbol{\omega}_{x_{46}}$	
X_{64}	$\mathbf{Ww}_{x_{54},x_{47}}$	Second-order state representing the adaptive modulation of the connection weight between $\boldsymbol{\omega}_{x_{54}}$ and $\boldsymbol{\omega}_{x_{47}}$	
X_{65}	$\mathbf{Ww}_{x_{55},x_{48}}$	Second-order state representing the adaptive modulation of the connection weight between $\boldsymbol{\omega}_{x_{55}}$ and $\boldsymbol{\omega}_{x_{48}}$	

5 Simulation Results

This chapter presents the outcomes derived from simulating the adaptive network models described in the previous chapter, both the base model (without AI coach) and the AI-enhanced model. Using the dedicated software described in [32] and the methodology from [31], the simulations were performed with nine distinct value and adaptive matrices (including *mb, mcwv, mcwa, msv, msa, mcfwv, mcfwa, mcfpv, mcfpa*) also mentioned in Sect. 3. It also includes the results of the What-If Analysis and Risk Assessment for both models. These evaluations provide deeper insights into how changes in key factors (such as speed factors and thresholds) influence overall system behavior.

5.1 Model Simulations

The simulation results for the base model without the AI coach are shown in Fig. 3. For clarity, all learning states (**W**-states) and the fatigue state (X_{28}) are highlighted in bold in the simulation plot, as can be seen in the accompanying legend. Initially, the learning states increase steadily, reaching values close to 0.9 around $t = 2000$.

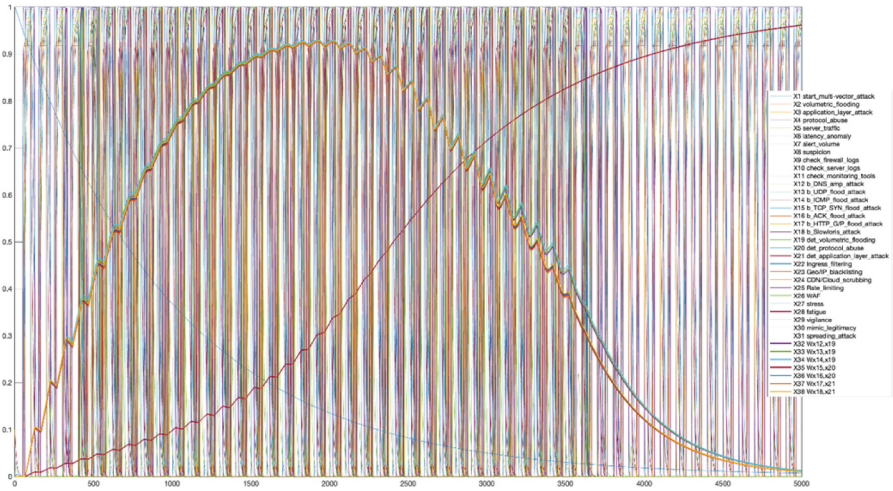

Fig. 3. Base model simulation results: no AI Coach.

However, as learning progresses, the fatigue level of the operator also rises, gradually degrading the learning process. After $t = 2000$, the learning states begin to decline steadily and eventually approach 0 near the simulation endpoint at $t = 5000$. The simulation graph displays denser areas, where correct detection and mitigation actions are activated, and sparse areas, indicating moments where insufficient determination prevents effective mitigation. In this base model, mitigation actions begin around $t = 500$ and are sustained successfully until approximately $t = 3500$. After this point, mitigation stops due to declining operator performance.

In the model simulation that includes the AI-coach, displayed in Fig. 4, the same set of states is emphasized in bold for readability, along with some added states of the AI-coach. For clarity, a few states are excluded from the simulation plot; a full list of the states shown and omitted can be found in the figure legend.

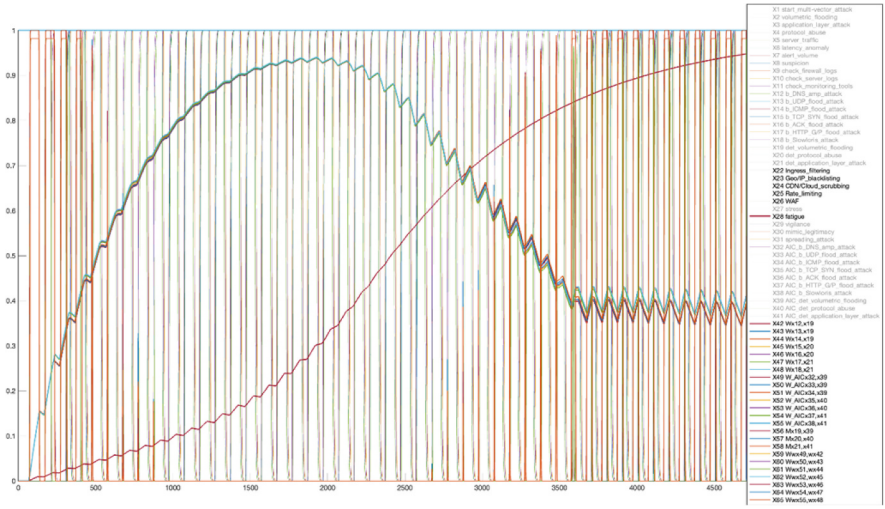

Fig. 4. AI-coach model simulation results.

The initial trajectory of the (learning) **W**-states follows a similar path to the base model, but with a slightly faster ascent in the early phase. The learning states of the AI-coach can be seen fully activated during the entire simulation, representing a fully learned coach. Mitigation actions commence earlier, around $t = 250$, and the learning states again reach peak values near $t = 2000$. As fatigue builds up, learning levels begin to decline. However, unlike the base model, the decline stops around $t = 3500$, when the AI coach activates its communication. Learning values stabilize and oscillate around 0.4 for the remainder of the simulation, allowing the operator to maintain sufficient detection capacity. Consequently, mitigation continues beyond $t = 3500$, contrasting sharply with the base model. The monitoring states (X_{56} to X_{58}) and the second-order (learning) $\mathbf{W_{ww}}$-states (X_{59} to X_{65}) are activated only when needed, initially in the period t in [0, 500], and again after $t = 3000$, with a quiet period in between.

5.2 What-If Analysis & Risk Assessment

For the What-If Analysis and Risk Assessment according to the methodology described in [18], for both versions of the model, two If-factors were defined: (1) the speed factor of the belief state (X_{12}), and (2) the threshold of the determination state (X_{19}). Each factor had two probabilistically weighted options: *low* and *high*. These choices are detailed in Table 4, which shows the intervals and assigned probabilities for each option. The What-factors evaluated were the activation value of the employee learning state (X_{32} for the base model and X_{42} for the AI-coach model) at two time points: $t = 3000$ and $t = 4500$. These resulted in four What-factor scenarios: **1a** (high at $t = 3000$), **1b** (low at $t = 3000$), **2a** (high at $t = 4500$), and **2b** (low at $t = 4500$). An impact indication scale from 1 to 5 was used to rate each scenario, chosen to reflect the relatively small number of What-factor scenarios, as using a larger scale would render it less robust in this context.

Table 4. If-factor options and their assumed probabilities.

If factors	If factor options	If factor option labels	If factor option value intervals	If factor option probabilities
If factor 1 speed factor of belief	If factor 1 option 1	low	0.01–0.4	0.7
	If factor 1 option 2	high	0.4–0.8	0.3
If factor 2 threshold of determination	If factor 2 option 1	low	0–0.5	0.4
	If factor 2 option 2	high	0.5–1	0.6

In the base model (without AI-coach), the What-If analysis and corresponding risk assessment (shown in Table 5) reveal a sharp decline in learning after $t = 3000$ under certain conditions. Scenario **1a** (high learning at $t = 3000$) receives an impact rating of 1, as early-stage learning is high and therefore not particularly risky. Scenario **1b** (low learning at $t = 3000$) receives a 3, as a lower learning value could undermine subsequent detection and response, but is expected at an early stage.

Table 5. Base Model Risk Assessment.

If factor 1 Option probabilities	If factor 2 Option probabilities	If factor Combination probabilities	What factor 1a Probabilities for X_{32} high at time 3000	What factor 2a probabilities for X_{32} high at time 4500	What factor 1b probabilities for X_{32} low at time 3000	What factor 2b probabilities for X_{32} low at time 4500
If factor 1 option 1: low 0.7	If factor 2 option 1: low 0.4	0.28	0.28			0.28
	If factor 2 option 2: high 0.6	0.42	0.42			0.42
If factor 1 option 2: high 0.3	If factor 2 option 1: low 0.4	0.12	0.12			0.12
	If factor 2 option 2: high 0.6	0.18	0.18			0.18
Sum of probabilities			1.00			1.00
Impact indicator			1	1	3	5
Risk assessment			1.00			5.00

Table 6. AI-coach Model Risk Assessment.

If factor 1 Option probabilities	If factor 2 Option probabilities	If factor Combination probabilities	What factor 1a probabilities for X_{42} high at time 3000	What factor 2a probabilities for X_{42} high at time 4500	What factor 1b probabilities for X_{42} low at time 3000	What factor 2b probabilities for X_{42} low at time 4500
If factor 1 option 1: low 0.7	If factor 2 option 1: low 0.4	0.28	0.28			0.28
	If factor 2 option 2: high 0.6	0.42	0.42			
If factor 1 option 2: high 0.3	If factor 2 option 1: low 0.4	0.12	0.12			0.12
	If factor 2 option 2: high 0.6	0.18	0.18			
Sum of probabilities			1.00			0.4
Impact indicator			1	1	3	5
Risk assessment			1.00			2.00

Scenario **2a** (high learning at $t = 4500$) is again rated 1, as it reflects successful prolonged learning and does not indicate operational risk. However, scenario **2b** (low learning at $t = 4500$) receives the highest impact rating of 5, indicating a failure to retain learning, directly impacting mitigation at a later stage.

For the AI-coach model, the What-If Analysis and the Risk Assessment (shown in Table 6) show a notable improvement. While scenario **1a** again receives a 1 (expected and low-risk), and scenario **2a** is similarly rated 1, the critical difference appears in the other scenarios. Scenario **1b** (low learning at $t = 3000$) still receives a 3, as it reflects early difficulty in learning. However, scenario **2b** (low learning at $t = 4500$) is much less severe than in the base model despite being assigned the same impact indicator value (5). The overall risk assessment value is reduced to 2.00 due to the decreased probability of low learning values at this later stage ($t = 4500$). The results of the Base and AI Coach Model What-If Analysis can be found in Table 7 and 8 (where bold values have been used for the Risk Assessment).

6 Discussion

This study investigated the dynamic mitigation process of multi-vector DDoS attacks, with a particular focus on modeling the mental and operational behavior of a cybersecurity operator. Using the modeling methodology from [30, 31], two simulation models were developed: a base model without AI support and an extended model with an AI-coach. The models incorporated technical, mental, and adaptive processes, modeled across multiple DDoS attack cycles. The outcomes were analyzed through direct

simulation and a What-If Analysis with associated risk assessments. These combined techniques allowed for the analysis of how adaptive learning and AI-based support affect the operator's performance and mitigation decisions over time.

In the model without AI support, the simulation results showed promising early learning behavior, with **W**-state values peaking near 0.9 around time $t = 2000$. However, due to rising operator fatigue, learning capacity declined consistently after this point, eventually approaching zero by $t = 5000$. As a result, mitigation actions stopped at around $t = 3500$, demonstrating the model's vulnerability in maintaining effective responses over time. This trend was also reflected in the What-If Analysis: the probability of *high* learning at time 3000 (**scenario 1a**) was 100%, confirming early effectiveness. However, the probability of *low* learning at time 4500 (**scenario 2b**) was also 100%, resulting in a maximum risk assessment value of 5.00. This scenario was rated most impactful due to its implications for late-stage mitigation, which is more critical for operational continuity than the early-stage performance.

In contrast, the model with the AI-coach maintained learning levels much more effectively across the full simulation period. While fatigue still led to a natural decline in learning capacity, the AI-coach intervened adaptively, resulting in stabilization of learning values around 0.4 after $t = 3500$. Consequently, mitigation actions continued well beyond the point of collapse in the base model. The What-If Analysis supports this improved performance: both configurations showed 100% probability of high learning at $t = 3000$ (scenario **1a**), indicating that the AI-coach does not significantly affect early-stage learning, emphasizing its timely behavior. However, in scenario **2b**, the probability of low learning at $t = 4500$ dropped by 60% in the AI-coach model, reducing the risk assessment value from 5.00 (without AI coach) to 2.00 (with AI coach). This highlights the AI coach's role in preserving operator performance during later, high-fatigue phases of an incident response cycle.

7 Future Work

This research can be extended and used in multiple ways. One area of future work could involve increasing the number of What-factor scenarios to allow for a more fine-grained risk impact scale and broader generalization. Also, incorporating more complex decision-making behavior, such as ethical dilemmas or resource trade-offs, would enhance the model to account for other realistic factors. Another direction involves modeling team-based coordination, where multiple operators interact with one or several AI coaches. Other case studies can be found in [24]. For such models verification correctness by mathematical analysis [7, 33] can be performed too. Also for future work, when data are available, validation and parameter tuning [33] can be performed.

Furthermore, integration of an AI Coach model as discussed here with human like models for such a coach that cover emotions, bonding and empathic responding [17, 19] is also foreseen in the near future. Finally, further integration of the AI Coach in organisational learning [3, 4] is an interesting perspective.

Table 7. Base Model What-If Analysis.

If factor 1 speed factor of belief ($\mathbf{X}_{12}$)	If factor 2 threshold of determination ($\mathbf{X}_{19}$)	What factor 1 Employee learning **W**-state $\mathbf{X}_{32}$ at time 3000	What factor 2 Employee learning **W**-state $\mathbf{X}_{32}$ at time 4500
0.01	0.1	0.981	0.491
0.01	0.9	0.899	0.041
0.2	0.1	**0.715**	**0.070**
0.2	0.9	**0.639**	**0.029**
0.6	0.1	**0.660**	**0.061**
0.6	0.9	**0.629**	**0.031**

Table 8. AI-coach Model What-If Analysis.

If factor 1 speed factor of belief ($\mathbf{X}_{12}$)	If factor 2 threshold of determination ($\mathbf{X}_{19}$)	What factor 1 Employee learning **W**-state $\mathbf{X}_{42}$ at time 3000	What factor 2 Employee learning **W**-state $\mathbf{X}_{42}$ at time 4500
0.01	0.1	0.982	0.499
0.01	0.9	0.907	0.408
0.2	0.1	**0.717**	**0.236**
0.2	0.9	**0.648**	**0.414**
0.6	0.1	**0.662**	**0.238**
0.6	0.9	**0.637**	**0.414**

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Keywords Are Not Always the Key: A Metadata Field Analysis for Natural Language Search on Open Data Portals

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Abstract. Open data portals are essential for providing public access to open datasets. However, their search interfaces typically rely on keyword-based mechanisms and a narrow set of metadata fields. This design makes it difficult for users to find datasets using natural language queries. The problem is worsened by metadata that is often incomplete or inconsistent, especially when users lack familiarity with domain-specific terminology. In this paper, we examine how individual metadata fields affect the success of conversational dataset retrieval and whether LLMs can help bridge the gap between natural queries and structured metadata. We conduct a controlled ablation study using simulated natural language queries over real-world datasets to evaluate retrieval performance under various metadata configurations. We also compare existing content of the metadata field 'description' with LLM-generated content, exploring how different prompting strategies influence quality and impact on search outcomes. Our findings suggest that dataset descriptions play a central role in aligning with user intent, and that LLM-generated descriptions can support effective retrieval. These results highlight both the limitations of current metadata practices and the potential of generative models to improve dataset discoverability in open data portals.

Keywords: Conversational Information Retrieval · Dataset Discovery · Conversational Search

1 Introduction

Open data portals play a crucial role in promoting transparency, civic engagement, and evidence-based policymaking by providing public access to government and institutional datasets [16,31]. Despite the increasing availability of such data, users often struggle to find datasets that match their information needs [18,23]. Existing portals primarily support keyword-based search mechanisms that rely on exact term matching and predefined metadata fields such as title, keywords, and descriptions [10,23,26]. In addition, the quality of publisher-provided metadata is often low: it may be sparse, inconsistent, or missing for many datasets [5,11,22]. These limitations hinder users

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in discovering datasets, as they may lack domain knowledge, use different terminology, or formulate queries in natural language [5,19,20]. Figure 1 illustrates this system in the London Datastore, where results are strictly based on lexical overlap. Filtering datasets is faciliatated via facets, displayed on the left hand side of the page. Users can filter the results based on topics, formats, geographical boundaries etc. No guidance or assistance is provided. Consequently, users iteratively refine and evaluate their search results, which makes traditional dataset search on open data portals a tedious and frustrating experience, especially for non-experts [23,42].

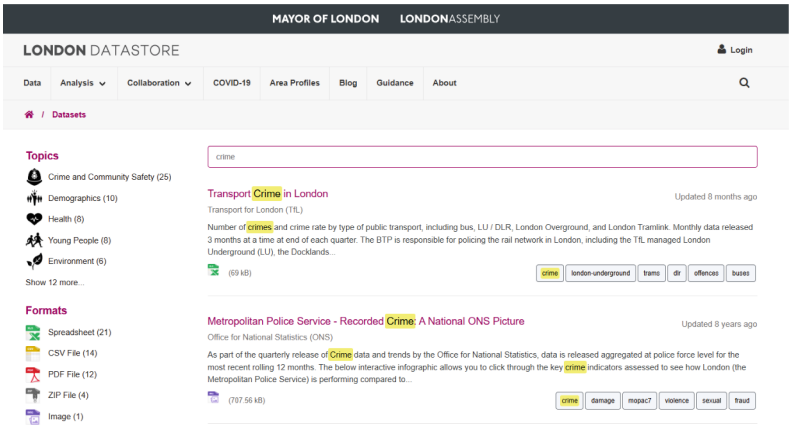


Fig. 1. Example of keyword-based search interface in the London Datastore. The system uses exact term matching across limited metadata fields [14].

At the same time, recent advancements in large language models (LLMs) have significantly changed the way users interact with digital systems: LLMs enable conversational interfaces that can interpret natural language questions, generate coherent answers, and understand user intent [27,34]. This shift has increased attention on *Conversational Information Retrieval* (CIR), where users search for data using natural language. CIR systems offer promise in improving data discoverability by aligning search mechanisms with a more natural and fuller expression of need and intent [12,24]. While there is an active research direction that explores CIR for structured datasets and tables [28,39,41], the transition to conversational interaction remains largely underexplored in practical, deployed open data portals.

1.1 Problem Statement

Dataset retrieval in open data portals is still dominated by keyword-based search over static, human-authored metadata. As users increasingly rely on natural language to express their needs, this leads to a persistent mismatch between query intent and how data is indexed and retrieved [5,11,22]. Metadata fields such as title, tags, and description are inconsistently populated and often misaligned with user vocabulary, making

relevant datasets difficult to discover. Bridging this gap remains a central challenge in building more intuitive and effective dataset search systems.

1.2 Research Questions

This paper addresses the challenge of enabling effective conversational dataset discovery in structured open data portals. We focus on understanding the role of metadata fields and the effectiveness of LLM-generated content in supporting natural language queries. Our study is guided by the following research questions:

- **RQ1:** Which metadata field (keyword, description, topic) is most important for effective dataset discovery using natural language queries?
- **RQ2:** What is the impact of LLM-generated descriptions on dataset retrieval performance compared to existing publisher-provided descriptions?
- **RQ3:** How do different prompting strategies influence the quality of dataset search outcomes?

1.3 Contributions

To address these questions, we conduct a controlled ablation study that isolates the impact of individual metadata fields on conversational dataset retrieval. We simulate real-world natural language queries and systematically evaluate retrieval performance under various metadata configurations.

Our central **hypothesis** is that *the content of the metadata field 'description' is sufficient to support effective conversational dataset search*, and that *LLM-generated description field content, derived directly from LLM interaction with the dataset, can outperform manually authored field content for the purpose of discovering relevant datasets*.

The main contributions of this paper are:

- **Provision of the first metadata ablation study in conversational dataset search.** We systematically evaluate how individual metadata fields (title, description, topic, etc.) contribute to retrieval performance using natural language queries, addressing a gap in prior work which has focused more on search architectures or schema design than on empirical field-level analysis.
- **Evaluation of LLM-generated versus publisher-authored descriptions.** We compare the retrieval utility of LLM-generated metadata derived from raw dataset content with existing publisher-authored descriptions, providing insight into the value and limitations of generative metadata in real-world retrieval scenarios.
- **Prompting strategy analysis for query effectiveness.** Building on Walker et al.'s prompting styles [37], we assess how different query styles (requesting, describing, implying) interact with metadata configurations and affect retrieval outcomes, contributing to our understanding of user intent alignment in open data search systems.

This study provides empirical grounding for improving metadata practices, leveraging generative models, and designing user-centered retrieval systems in the context of open data.

2 Related Work

Effective dataset discovery remains a critical challenge across open data infrastructures. While the number of publicly accessible datasets continues to grow, search interfaces often rely on keyword-centric mechanisms that perform poorly when users express information needs in natural language [5,25]. This section reviews prior work in two relevant areas: how users search for datasets in intent-driven settings and the role of metadata in dataset retrieval.

2.1 How Users Search for Datasets in Natural Language

Recent work in conversational dataset search has examined how users express information needs and how retrieval systems can better support intent-driven queries [3]. Walker et al. [37] studied user search interactions with open-domain chatbots. They categorize dataset search queries into three primary prompting strategies that reflect increasing levels of abstraction and ambiguity in user intent:

- **Requesting:** This prompt type is *goal-oriented and specific*, often including exact keywords or dataset names, formats, or even locations. Users employing this strategy tend to have a clear idea of what they're looking for. These prompts are typically easy to interpret and retrieve against, making them most compatible with keyword-based retrieval systems.
- **Describing:** These prompts *specify the features or structure* of the desired dataset, without naming a specific title or format. While less direct than requesting, they still express intent clearly, often listing attributes like granularity, timeframes, or coverage.
- **Implying:** The most *open-ended and abstract* type, these prompts reveal a user's broader goals or research interests rather than specifying a dataset directly. While in open-domain search, such prompts pose a greater challenge for retrieval models, particularly in inferring that a dataset would satisfy the user's intent, this problem does not arise in dataset-focused systems.

These prompt types reflect different levels of user knowledge and task specificity. Walker et al. [37] argue that effective conversational dataset search must accommodate all three forms, supporting clarification and dialogue where needed. Table 1 summarises the characteristics of these prompting styles, along with examples and their relative clarity of intent. Our study leverages this to generate diverse synthetic evaluation queries, ensuring that retrieval performance is tested across a realistic range of user expressions.

2.2 Metadata in Dataset Search

Semantic search in on metadata-based dataset search is explored in recent academic work. Zhang and Balog [40] show that combining different metadata fields, such as column headers, captions, and surrounding context, into semantic representations can improve table search. Their work demonstrates that treating metadata as a multi-dimensional signal allows for more flexible and accurate matching. Other research

Table 1. Prompting styles in dataset search (adapted from Walker et al. [37]).

Prompt Type	Description	Example	Intent Clarity
Requesting	Specific request for a dataset or format	*"Find me a CSV of London air pollution in 2020."*	High
Describing	Lists desired features without naming a dataset	*"I need environmental data showing seasonal trends."*	Medium
Implying	Expresses a goal or topic indirectly	*"I'm exploring urban heat islands in Europe."*	Low

focuses on cases where relevant information is spread across multiple tables. Chen et al. [6] explore how to retrieve sets of tables that are not only relevant individually but also compatible with each other. They argue that understanding the structure of metadata—such as schema, column overlap, or key relationships—is essential for tasks like question answering over data lakes, where multiple sources often need to be combined.

Despite these advances, most open data portals still rely on relatively simple metadata-based search. CKAN-based platforms like Data.gov and the London Datastore, as well as Socrata-powered systems, use keyword matching on metadata fields [1,7,35]. CKAN uses full-text engines like Apache Solr for filtering, relevance ranking, and partial string matching [33]. Socrata improves match coverage through stemming [1]. Still, both systems depend on static, human-written metadata and exact term matching. The EU-wide portal *data.europa.eu* aggregates national catalogues into a single CKAN-Solr index, but remains limited by the same metadata schema and keyword search approach [29]. On a larger scale, Google Dataset Search indexes metadata embedded in schema.org annotations from web pages [13,32]. More recently, Google's DataGemma project has tried to connect structured data with LLM-based assistants using sources like Data Commons [30]. However, even these systems still rely on basic keyword logic rather than deeper semantic or structural understanding.

3 Methodology

Our study is grounded in the use of real-world metadata descriptions sourced from the London Datastore (LDS)[1]. To investigate how different forms of metadata affect dataset discovery, we designed a methodology that combines dataset construction, metadata processing, modelling approaches, and evaluation metrics. Table 2 summarises how each research question (RQ) is addressed, linking each RQ to its corresponding methodological steps.

Our retrieval methodology consists of two main components: the construction of a SEARCH SPACE and the definition of a QUERY SPACE, which are illustrated in Fig. 2.

[1] As of July 2025, the London Datastore was relaunched as the London Data Library (LDL), with some technical changes. This study was conducted prior to that transition, during the period it was known as the London Datastore.

Table 2. Overview of how each research question is addressed in the methodology.

Research Question	Approach
1. Which metadata field is most important for effective dataset discovery?	Ablation over 13 metadata fields to measure individual contribution.
2. What is the impact of LLM-generated descriptions on retrieval performance?	Compare publisher vs. LLM-generated descriptions, individually and combined.
3. How do prompting strategies influence dataset search outcomes?	Evaluate search quality across request-, describe-, and imply-style queries.
Tools	Gemini 2.5 Flash, Gemma, BAAI/bge-base-en, FAISS, LSA, LDA, KeyBERT, spaCy NER
Evaluation	Top-1/3/5 Accuracy; Mean Reciprocal Rank (MRR)

The SEARCH SPACE is built from original metadata fields and two types of augmentation: (i) features extracted with traditional natural language processing (NLP) methods (e.g., keyword or topic extraction), and (ii) fields generated using LLMs. These are organized into three metadata configurations: (1) Original, (2) Original + NLP, and (3) LLM-augmented with several ablations defined within each configuration to test their relative contribution.

The QUERY SPACE captures user information needs through three query types: Request, Describe, and Imply. These represent increasing levels of abstraction and intent. Both metadata and queries are encoded using a shared embedding model and indexed in a vector database. Retrieval is then performed by comparing query embeddings to metadata embeddings, allowing us to evaluate how metadata richness and query style interact to influence search performance.

3.1 Search Space Construction

Characterising Metadata Completeness and Field Selection. The London Datastore corpus contains 1,263 datasets, but metadata quality is inconsistent: Our analysis shows that core fields such as *title* (100%), *theme* (97.0%), and *description* (91.7%) are consistently present, whereas only 81.9% of datasets include a valid download link and just 60.9% provide keywords. These gaps mirror findings from previous work on metadata quality [5,11,26] and motivate our focus on enrichment strategies. Figure 3 illustrates both field completeness and file format distribution. As shown in Subfigure 3a, descriptive fields are generally well covered, but keywords and distribution links remain inconsistent. Subfigure 3b highlights that file formats are almost evenly divided between structured data (49.8%, e.g., CSV and Excel) and unstructured formats (50.2%, e.g., PDF and TXT). For downstream analysis, we restrict our study to the **255 datasets** that contain at least one structured file suitable for machine-readable processing. Within this filtered subset, our experiments focus on six metadata fields,

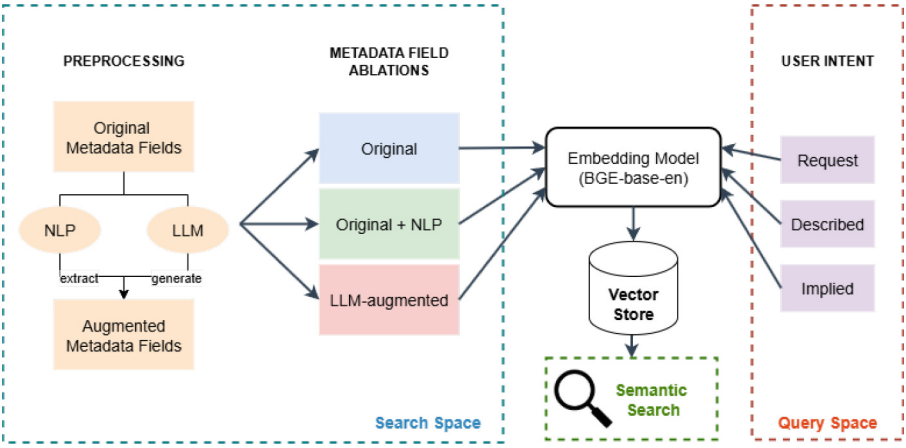


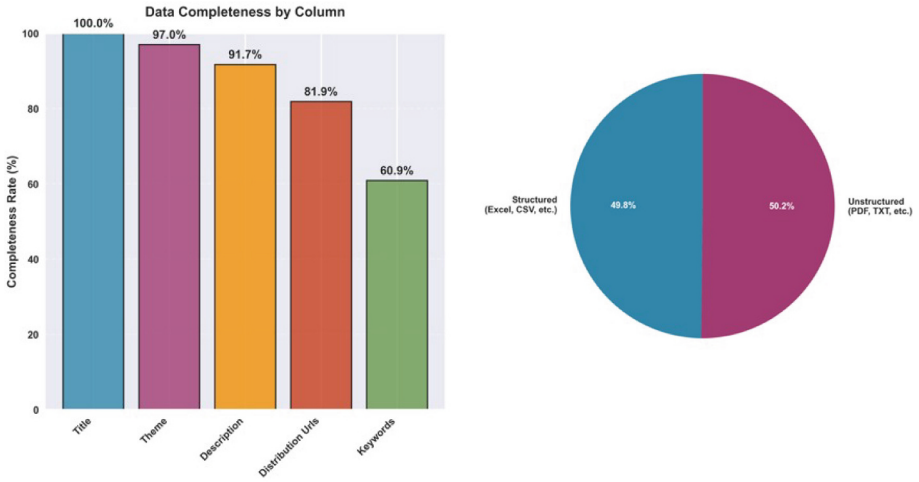
Fig. 2. Overview of the data and augmentation pipeline used in our methodology.

summarised in Table 3. Of these, *dataset_id* and *title* are used only for dataset identification and human evaluation. The remaining fields *description*, *keywords* and *topics* form the basis of our retrieval configurations. Throughout this paper, we use the term *topic* to refer to the *theme* field as defined in the London Datastore schema.

Table 3. Metadata fields used in the study and their role in retrieval.

Metadata Field	Description	Included in Ablation Study
dataset_id	Unique identifier for the dataset, used for tracking and evaluation	–
title	Human-provided dataset title	–
description	Narrative description of the dataset content and purpose	✓
keywords	Free-text keywords added by dataset publishers	✓
topics	Predefined topical categories (e.g.,“Transport”, “Health”)	✓

Metadata Enrichment. Following the assessment of existing metadata completeness, we implemented two complementary approaches to enrich and complete missing fields: one based on traditional NLP techniques, and the other leveraging LLMs. We performed a manual review to spot-check whether the generated metadata reasonably matched the dataset content.



(a) Field completeness across all 1,263 datasets. **(b)** Structured vs. unstructured file formats.

Fig. 3. Metadata completeness and file format distribution across the London Datastore corpus.

NLP-Based Keyword and Topic Extraction. To enrich the often sparse publisher-provided metadata, we applied three complementary NLP techniques to dataset descriptions: Topic modelling was conducted using Latent Semantic Analysis (LSA) [8] and Latent Dirichlet Allocation (LDA) [2] to uncover hidden thematic structures and topic distributions in the text. For keyword extraction, we used KeyBERT [15], which relies on BERT embeddings to identify words and phrases most semantically similar to the entire dataset description. This helps to produce keywords that better reflect dataset content and capture user phrasing even when exact wording differs. In addition, we employed Named Entity Recognition (NER) [17] to identify and classify proper nouns and specific entities, such as organisations, locations, and technical terms, adding domain-specific vocabulary often missing from publisher-supplied keywords. Together, these methods produced a richer and more representative set of metadata, later integrated into retrieval experiments.

LLM-Generated Descriptions. In addition to NLP-based enrichment, we generated high-quality dataset descriptions using a large language model. This process involved two stages: structured prompt design and description generation.

Prompt Design. Metadata was first extracted from the London Datastore catalogue, including fields such as *title*, *description*, and download links. For datasets with valid machine-readable files (CSV, XLSX, XLS), we also incorporated structural context: column headers and a small sample of rows from the beginning and end of each dataset. To ensure clarity, headers and sample values were lightly sanitized (e.g., whitespace and newlines removed, excessively long strings truncated). These elements were then assembled into concise prompts that conveyed both the dataset's structure and content. A simplified example is shown below:

*Dataset **[title]** contains **[n]** records with column headers **[headers]**. Example records include: **[sample rows]**. Please generate a descriptive summary of the dataset (max. 350 words).*

Description Generation. Each prompt was submitted to the LLM with this instruction, and the resulting outputs were stored as a new *llm_description* field in the catalogue. To enable fair comparison with publisher-provided metadata, we subsequently applied the same NLP-based enrichment methods (topic modelling, keyword extraction, and NER) to these generated descriptions.

To consolidate the fields used in our retrieval experiments, Table 4 lists the original, NLP-derived, and LLM-derived metadata. Figure 4 illustrates how these two enrichment strategies are combined into a unified pipeline.

Table 4. Metadata fields used in retrieval experiments, grouped by source.

Source	Fields
Original (LDS)	`lds_title`, `lds_description`, `lds_keywords`, `lds_topic`
NLU-derived	`lds_desc_keywords`, `lds_desc_topics`
LLM-derived	`llm_prompt`, `llm_description`, `llm_desc_keywords`, `llm_desc_topics`

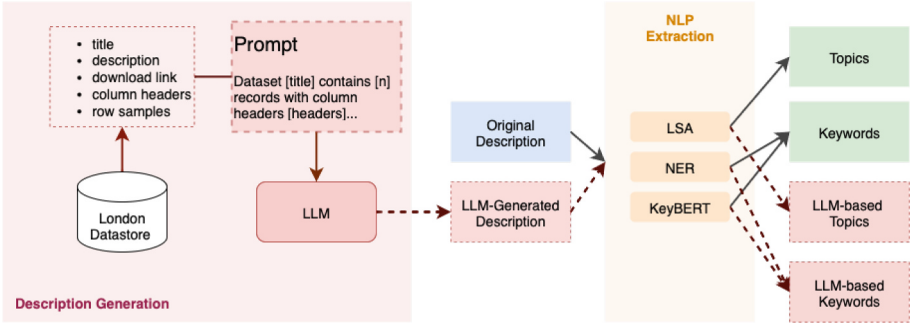


Fig. 4. Overview of the metadata enrichment pipeline. Structured metadata and dataset samples are combined into prompts for an LLM, which generates synthetic descriptions. Both publisher-provided and LLM-generated descriptions are processed with NLP methods (LSA, NER, KeyBERT) to extract topics and keywords. Table 4 lists the corresponding metadata fields produced at each stage.

Ablation Configurations. Having generated missing metadata fields using both NLP techniques and LLM-based methods, we now evaluate their contribution to dataset retrieval through a series of controlled ablations. In all ablation conditions, the fields `dataset_id` and `title` are kept solely for identification and evaluation purposes. They are not included in the retrieval index or used as input for model inference. The different ablation configurations evaluated in our study are summarised in Table 5.

Table 5. Ablation configurations used in retrieval experiments. Each configuration includes a subset of metadata fields, with variants derived from publisher-provided (Original), NLP-enriched, or LLM-enriched metadata.

Configuration	Original	NLP	LLM
Keywords + Topics	`key_original`	`key_nlp`	`key_llm`
Description only	`desc_original`	–	`desc_llm`
Full set (Description + Keywords + Topics)	`full_original`	`full_nlp`	`full_llm`
Keywords only	`onlykey_original`	`onlykey_nlp`	`onlykey_llm`
Topics only	`onlytopic_original`	`onlytopic_nlp`	`onlytopic_llm`

3.2 Query Space Definition

To evaluate the effectiveness of different metadata configurations in supporting natural language queries, we constructed a synthetic evaluation dataset. Inspired by the prompting styles identified by Walker et al. [37], we generated three types of queries for each dataset in our corpus: *requesting*, *describing*, and *implying*, which are summarized in Table 1. To operationalize this, we used Gemini2.5 Flash to generate synthetic user queries, providing each model instance with the dataset title and ID as input context. This resulted in three diverse, natural-sounding queries per dataset. Our final evaluation dataset therefore consists of 765 user queries in total. Table 6 presents an example of the three prompting styles applied to the dataset *Police Force Strength*.

3.3 Retrieval and Evaluation Setup

To evaluate the effectiveness of each metadata configuration, we implemented a dense retrieval pipeline. All metadata representations were encoded using the `BAAI/bge-base-en` embedding model [38], a general-purpose English sentence transformer optimized for semantic similarity tasks. We used FAISS for efficient vector indexing and similarity search [9]. At query time, user queries were embedded using the same model, and cosine similarity was applied to retrieve the most relevant dataset representations. The top-k results (with $k = 5$) were then ranked by cosine distance between query and dataset embeddings.

Table 6. Example of generated user queries for a single dataset (Police Force Strength), based on Walker et al.'s prompting styles [37].

Prompt Type	User Query
Requesting	Could you please help me locate the official dataset titled *'Police Force Strength'*? I'm interested in the latest figures available for police numbers across different areas.
Describing	I'm trying to find a dataset that details the current headcount and changes in personnel for various police forces. Do you have anything that provides comprehensive statistics on law enforcement strength?
Implying	What are the most recent statistics available regarding the total number of police officers currently serving in the country, and how has this changed over time? I'm curious about the staffing levels of our police force.

Retrieval performance was assessed using standard information retrieval metrics [4,36]:

- **Top-1 Accuracy:** Measures the proportion of queries for which the correct dataset appears as the top-ranked result.
- **Top-3 and Top-5 Accuracy:** Evaluate whether the correct dataset appears within the top 3 or top 5 ranked results, respectively.
- **Mean Reciprocal Rank (MRR):** A standard metric for ranking tasks that accounts for the position of the correct result, assigning higher weight to results retrieved earlier.

These metrics provide a comprehensive view of retrieval quality across both exact and approximate matches, and are computed for each ablation condition and query style combination.

4 Results

We present the results of our ablation study evaluating how different metadata fields support natural language search. To maintain alignment with our research questions, we organize this section around the three RQs outlined earlier.

RQ1: Which Metadata Field Is Most Important for Effective Dataset Discovery Using Natural Language Queries?

The results clearly show that *descriptions* are the most effective metadata field for supporting natural language queries. The DESC_LLM configuration, which uses only LLM-generated descriptions, achieves the highest performance across all metrics (MRR = 0.925), significantly outperforming the DESC_ORIGINAL configuration (MRR = 0.820). Even when evaluated in isolation, descriptions are more effective than any combination of keyword or topic metadata.

Keyword-based configurations (KEY_NLP, KEY_LLM) perform moderately well but consistently lag behind descriptions. Topic-only fields perform the worst by a large margin, with MRR values below 0.1 in all variants. These findings highlight that rich descriptive metadata, particularly in narrative form, is essential for aligning with user intent in natural language queries.

It is worth noting that DESC_ORIGINAL already provides a strong human-authored baseline (MRR = 0.820, Hit@3 = 0.915), substantially outperforming all keyword- and topic-based configurations. This indicates that narrative descriptions, even when authored by publishers, capture user intent far better than structured metadata fields alone.

Table 7 summarises average retrieval performance across all queries for each configuration.

Table 7. Retrieval performance across metadata ablation conditions. Scores reflect average retrieval accuracy across 765 natural language queries using a dense embedding-based retriever (BAAI/bge-base-en). Best results are highlighted in **bold**, while strong human-authored baselines (DESC_ORIGINAL) are also emphasized.

Ablation Condition	Hit@1	Hit@3	Hit@5	MRR
KEY_ORIGINAL	0.279	0.471	0.544	0.379
KEY_NLP	0.502	0.664	0.757	0.594
KEY_LLM	0.495	0.680	0.753	0.594
DESC_ORIGINAL	**0.731**	**0.915**	**0.944**	**0.820**
DESC_LLM	**0.887**	**0.964**	**0.976**	**0.925**
FULL_ORIGINAL	0.744	0.921	0.955	0.833
FULL_NLP	0.684	0.874	0.916	0.778
FULL_LLM	0.835	0.940	0.956	0.887
ONLYKEY_ORIGINAL	0.293	0.446	0.496	0.371
ONLYKEY_NLP	0.492	0.676	0.756	0.592
ONLYKEY_LLM	0.499	0.669	0.759	0.597
ONLYTOPIC_ORIGINAL	0.057	0.135	0.177	0.098
ONLYTOPIC_NLP	0.001	0.012	0.024	0.008
ONLYTOPIC_LLM	0.009	0.027	0.038	0.019

RQ2: What Is the Impact of LLM-Generated Descriptions on Dataset Retrieval Performance Compared to Existing Publisher-Provided Descriptions?

We focus on configurations that isolate each source (DESC_ORIGINAL vs. DESC_LLM) as well as those that combine descriptions with keyword/topic metadata (FULL_ORIGINAL vs. FULL_LLM).

LLM-generated descriptions consistently outperform original ones, that were authored by publishers, across all setups. DESC_LLM achieves an MRR of 0.925 compared to 0.820 for DESC_ORIGINAL. Similarly, FULL_LLM achieves 0.887 MRR, outperforming FULL_ORIGINAL (0.833).

Looking more closely, the gains are not only statistically consistent but also meaningful in relative terms: compared to DESC_ORIGINAL, DESC_LLM improves Hit@1 by **+21.3%** (0.731 → 0.887), Hit@3 by **+5.4%** (0.915 → 0.964), Hit@5 by **+3.4%** (0.944 → 0.976), and MRR by **+12.8%** (0.820 → 0.925). This suggests that while publisher-authored descriptions are already a strong foundation, LLMs can provide additional semantic richness and better alignment with natural user phrasing, leading to consistently higher retrieval effectiveness.

RQ3: How Do Different User Query Types Influence Retrieval Performance Across Metadata Ablations?

Performance varies considerably across query styles: As shown in Table 8, *requesting* queries yield the highest retrieval scores across all configurations. In contrast, *implying* queries result in the weakest performance, particularly when metadata is sparse.

For instance, the baseline KEY_ORIGINAL configuration drops from 0.394/0.295 (MRR/Hit@1) for requesting queries to 0.351/0.246 for implying queries. In the worst case, ONLYTOPIC_LLM shows near-zero performance across all query types. Conversely, the best results come from DESC_LLM, where scores remain consistently high even for vague queries (e.g., 0.906/0.858 for implying). Adding keyword and topic enrichment (FULL_LLM) boosts performance further for ambiguous inputs.

These results highlight how both metadata quality and query clarity shape retrieval outcomes and underline the need for richer, semantically informed metadata to support more natural, conversational search behavior.

Table 8. Retrieval performance (MRR / Hit@1) by query type for selected metadata configurations.

Ablation	Requesting	Describing	Implying
KEY_ORIGINAL	0.394/0.295	0.392 / 0.295	0.351/0.246
DESC_LLM	**0.962/0.943**	**0.907/0.861**	**0.906/0.858**
FULL_LLM	0.927/0.890	0.862/0.804	0.872/0.811
ONLYTOPIC_LLM	0.024/0.014	0.017/0.007	0.017/0.007

Summary

Taken together, these results provide strong empirical support for our central hypothesis: that dataset descriptions, especially those generated by large language models, play a critical role in enabling effective natural language querying for dataset search.

Across all metadata configurations and query types, LLM-generated descriptions consistently outperform original metadata. These findings point to a promising direction for data portals: augmenting or replacing sparse publisher metadata with semantically rich LLM-generated content.

5 Discussion

In this section, we reflect on the broader implications of our findings for the design, evaluation, and governance of conversational dataset search systems.

LLMs as Metadata Enrichment Tools

Our results confirm a growing trend observed in other domains: LLMs can generate metadata that is not only semantically rich but also more retrieval-effective than existing human-authored content (which is often inconsisistent and sparse). Across all configurations and query types, LLM-generated descriptions substantially outperformed original descriptions, keywords, and topics. This echoes findings in related work, where well-structured generation based on underlying data improves search relevance and interpretability [40].

Unlike prior work that uses LLMs primarily for user-facing generation, our study demonstrates their backend utility in metadata generation. Given the high performance of DESC_LLM and FULL_LLM, we argue that LLMs should be considered core components of metadata pipelines for open data portals, where manual curation is infeasible or in situations where metadata is gathered from a number of different sources, which inevitably introduces inconsistency.

Prioritising Description Metadata in Open Data Portals

Traditionally, dataset portals have prioritised metadata fields such as keywords, formats, or predefined taxonomies (e.g., CKAN's topic structure). Our study challenges this emphasis. Despite widespread use, topic fields proved ineffective in conversational retrieval (MRR < 0.1 in most settings). These results support Koesten et al.'s critique of "disconnected" metadata, which fails to reflect user mental models and goals [22].

Conversely, narrative descriptions, particularly when LLM generated, proved to be the most robust across varying query styles. This reinforces Chapman et al.'s call for semantically expressive metadata and aligns with Walker et al.'s classification of prompting strategies [5,37]. In short, successful dataset discovery depends not on formal metadata fields alone, but on the richness and alignment of content with user language.

Implications for Interface and Infrastructure Design

From a systems perspective, our findings offer a practical blueprint for building next-generation dataset portals. First, metadata enrichment pipelines using LLMs should become a standard feature of catalogue infrastructure. Second, retrieval engines should

prioritise semantically dense fields (descriptions, column-level summaries) over sparse, taxonomic ones. Third, search interfaces should be reimagined to support conversational interaction from the ground up. This involves not only adopting dense retrieval backends, as we used here, but also aligning front-end interfaces with real user workflows. Retrieval should not assume final-form queries; rather, systems should support progressive discovery based on incomplete or ambiguous inputs.

While not the focus of this paper, preliminary experiments suggest that retrieval performance is also influenced by the structure and length of user queries. Specifically, we observed that longer, more detailed queries tend to benefit more from rich narrative metadata such as LLM-generated descriptions, whereas short or keyword-like queries may align better with concise fields like titles or keywords. This suggests a need to explore how description-based metadata can support a wider search space, specifically, how LLM-generated descriptions can be optimized to improve retrieval for both short, keyword-style queries and longer, natural language ones.

Toward Standardized Machine-Generable Metadata

Given the success of structured prompting in generating useful metadata, we propose that the data portal community consider formalizing LLM-compatible metadata standards. Such standards might define prompt schemas (e.g., using column names and keywords), output formats (e.g., descriptive paragraphs or JSON metadata blocks), and evaluation criteria (e.g., alignment with schema.org or retrieval effectiveness). These would enable reproducibility across portals, improve metadata interoperability, and allow future systems to audit or improve generated content.

We envision a future in which data providers submit minimally structured metadata, and machine-generation pipelines produce standardized, rich metadata to populate portal interfaces, APIs, and downstream retrieval models.

The Missing Link Is the User

A persistent challenge in dataset search research is the lack of access to real user interaction data. While we follow Walker et al. [37] in using structured prompting as a principled proxy, the field as a whole still lacks large-scale, authentic user queries, session logs, and task outcomes needed to evaluate systems under realistic conditions.

This absence is not unique to our work. Koesten et al. [22] emphasize how poorly understood user needs are in open data portals, noting that users often do not know what to search for and engage in iterative query reformulation. As long as retrieval systems are evaluated without authentic user context, it will remain unclear how well they serve actual discovery tasks.

We therefore join prior work [5,21] in calling for the creation of benchmark datasets containing real user queries, tasks, and satisfaction signals. These resources are crucial for defining what "successful" dataset discovery actually means in different use contexts.

Cross-Domain Relevance

While our study focused on the London Datastore, the issues we address (metadata sparsity, intent misalignment, rigid search interfaces) are widespread across public and private data platforms. Scientific repositories, clinical data registries, and even internal enterprise data catalogues face similar challenges. Our approach in combining metadata ablation, structured prompting, and dense retrieval evaluation can readily be extended to other domains.

Ultimately, our findings suggest that metadata should not be seen as static infrastructure but as a dynamic, learnable surface for interaction. By treating metadata generation as an LLM-supported design problem, we can move toward truly user-centered dataset discovery across contexts.

6 Future Work and Limitations

While our findings offer compelling evidence for the effectiveness of metadata generated by LLMs in conversational dataset search, several limitations of scope, methodology, and generalizability must be acknowledged.

We emphasize that our comparisons were not made against carefully curated, high-quality human-authored metadata, but rather against the metadata currently available 'in the wild'. While this reflects the reality of open data portals, we acknowledge that our human-authored dataset will have necessarily included a number of poor descriptions. Our future work will explore evaluations against a subset of human-annotated "ideal" descriptions to more rigorously benchmark the strengths and limitations of LLM-based metadata.

Corpus Scope and Generalizability. Our evaluation was conducted on a controlled set of 255 tabular datasets from the London Datastore. While this provided a manageable testbed for isolating metadata effects, it does not capture the scale, diversity, or complexity of larger open data ecosystems. Many real-world portals contain thousands of datasets across heterogeneous formats (e.g., PDFs, APIs, spatial data) and domains. A limited corpus may also inflate retrieval scores due to lower semantic overlap between documents. Future work should extend these evaluations to more diverse collections.

Metadata Schema Constraints. We operated within the standard metadata fields available in common data portals. While this improves comparability and realism, it constrains the potential of more expressive or LLM-oriented metadata structures. As discussed, future work should explore redesigned metadata schemas that explicitly support LLM compatibility, conversational querying, and semantic richness.

Synthetic Queries and Missing User Context. Although our evaluation used a principled classification of query styles from Walker et al. [37], all user queries were synthetically generated. These prompts are structured, consistent, and lack the noise, ambiguity, and iterative reformulation common in real user behavior. As Koesten et al. [22]

and others have noted, user needs in dataset search are often underspecified and evolve over time. Without interaction logs or task-based evaluation provided by the London Datastore itself, our results cannot fully capture how well LLM-generated metadata supports authentic discovery scenarios. Future research must incorporate real conversational query data, user feedback, and success metrics to close this gap.

Retrieval Architecture Assumptions. We employed a general-purpose dense retriever (BAAI/bge-base-en) to evaluate semantic alignment between queries and metadata fields. While this reflects current practices in neural retrieval, it does not explore the full design space of conversational IR architectures, such as query rewriting, reranking, or interactive clarification loops. Additionally, no retriever is neutral: our results may be influenced by how well the model embeds certain fields (e.g., long-form descriptions vs. sparse keywords). Tuning or training retrievers specifically for metadata structure or conversational prompts remains an open area.

7 Conclusion

This study examined how different metadata fields, and their enrichment via LLMs, affect conversational dataset retrieval. Through an ablation study on 255 datasets from the London Datastore and 765 natural language queries, we found that the *description* field is the most critical for retrieval effectiveness, particularly when generated by LLMs.

LLM-generated descriptions consistently outperformed publisher-authored ones, especially for vague or exploratory queries, highlighting their potential to improve metadata quality and align more closely with user intent. In contrast, traditional fields like keywords and topics showed limited utility in conversational search scenarios.

While limited in scope to a mid-sized tabular corpus, our findings point to a broader opportunity: integrating LLM-based metadata generation into open data infrastructure to support more natural, effective search. This work contributes practical guidance and empirical evidence for designing metadata pipelines and retrieval systems that reflect how users actually search in the age of conversational AI.

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Adaptive AI Coaching for Cybersecurity: Analysis of Resilience Against Phishing and Deepfakes

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Abstract. This paper presents a simulation-based analysis on the impact of real-time AI coaching in reducing cybersecurity risks associated with phishing and deepfake attacks. Using multi-level network-oriented modeling, the study models human cognitive and emotional behavior under stress, time pressure, and dynamic learning conditions. Two models are evaluated: a baseline reflecting traditional training methods, and an extended model introducing an AI Coach with perfect learning. Results show that while employee learning in the baseline model deteriorates significantly under stress, the AI Coach stabilizes learning and significantly improves secure decision making. A What-If analysis further demonstrates the AI Coach's effectiveness across different levels of message quality and stress susceptibility, while the risk assessment highlights a reduction in learning and decision risk. Although the AI Coach does not fully eliminate risk, it proves to be an effective, adaptive tool for supporting decision-making in human-centered cybersecurity defense.

Keywords: Cybersecurity · Phishing · Adaptive Networks · Simulation-Based Analysis · Risk Assessment · AI Coach

1 Introduction

The rise of Artificial Intelligence, particularly Large Language Models (LLMs) and Generative AI, has introduced a 'new age of phishing - one of unprecedented scale' [15]. While these technologies hold significant beneficial potential, they are increasingly being misused for malicious purposes, highlighted by a 1,265% increase in phishing attacks between 2022 and 2023 following the public release of ChatGPT [18]. With the increasing accessibility of these technologies and the rise of deepfake technology, which 'enables the manipulation and replication of individuals' facial expressions, body movements, and voice' [10]; phishing attacks have become far more frequent, realistic, and harder to detect. In 2024, 73.8% of all phishing emails exhibited some use of AI

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[14], while the average financial cost of a phishing-related breach reached USD 4.88 million [12].

Phishing is a form of social engineering; a type of cyberattack that bypasses technical defenses by manipulating human trust, often by leveraging cognitive biases, stress, and other persuasive tactics [7]. Despite forecasts estimating that global cybersecurity spending will reach USD 212 billion in 2024, most of these investments will be allocated to "security software markets" and continue to focus on technical defenses including "application security and infrastructure protection" [11]. As these technical defenses continue to grow more sophisticated, attackers increasingly target the human element, which remains neglected despite being the root cause of 68% of all data breaches [1].

This research investigates employee behavior in cybersecurity contexts through multi-level adaptive network-oriented modeling [22, 23], with a focus on responses to phishing attacks under dynamic conditions such as stress, emotional reactivity, and organizational time pressure. The analysis relies on computational simulations rather than human participants, ensuring that the results reflect model-based scenarios. Furthermore, this paper argues that in the era of generative AI, effective cyber defense must prioritize adaptive, human centered strategies. Building on the conceptual framework of the AI Coach by Roelofsma et al. [17], this study explores whether a real-time AI Coach can enhance organizational security by improving individual resilience and guiding secure behavior. Two models are evaluated: a baseline system representing traditional training methods, and an extended shared mental model approach [6, 25, 27] that integrates an AI Coach operating at the level of first-order adaptation of the employee mental model knowledge. Thus, the AI Coach influences the employee's learning process at the mental model knowledge level, reinforcing secure decision-making over time by social learning. These two models are evaluated through a What-If analysis, simulating employee learning and behavior under different conditions, followed by a risk assessment.

By combining behavioral science and computational modeling, this study aims to provide a deeper understanding of the psychological and contextual drivers of cybersecurity behavior. It identifies cognitive and emotional vulnerabilities that are exploited by phishing attackers and evaluates whether real-time AI coaching can effectively promote secure behavior.

2 Background Literature

The growing accessibility of highly realistic generative AI tools has significantly increased the technical sophistication and realism of phishing and deepfake attacks. While traditional phishing relies on generic fraudulent emails, phone calls (vishing), or SMS (smishing) to trick users, Large Language Models (LLMs) like GPT-4 can now generate fluent and contextually relevant phishing messages with human level accuracy [9]. These tools also enable the creation of deepfakes, which are highly realistic synthetic audio, video, or images that attackers use to impersonate executives or even fabricate convincing identities [20]. For instance, in 2024 an employee at a British multinational engineering firm Arup was tricked into transferring USD 25 million after participating in a video call with deep faked finance officials [15]. Generative AI aids attackers in exploiting human trust and reduces the skill barrier for attackers by "automating data collection and creating persuasive messages" [8].

Social engineering attacks bypass technical safeguards by targeting the human layer of cybersecurity. Despite heavy investment into technical defenses, a 47% increase in phishing emails evading Microsoft's native filters was reported in 2024 [15]. The endpoint of defense thus becomes the employee's cognitive judgement. However, human learning and judgement is frequently affected by stress, high workloads, and time pressure, all of which increase the likelihood of errors [3]. Attackers tend to exploit psychological and emotional vulnerabilities in humans, often using personalized message content, urgency and fear to pressure victims [3]. Human error in cybersecurity is therefore not a simple mistake, but the outcome of complex interactions between internal cognitive states and external organizational pressures.

Traditional cybersecurity training methods have proven ineffective in addressing the complex human factors that influence secure behavior. These Security Education, Training, and Awareness (SETA) training programs often rely on static training modules, and the knowledge gained from these programs tends to fade within a month [19]. A study by Ho et al., [12] involving over 19500 employees at a large healthcare organization found that static training was largely ineffective: "half of the training sessions end[ed] within 10 s, and less than 24% [of employees] completed the training" [12]. Static training results were negatively correlated with future performance, whereas interactive formats reduced the likelihood of clicking on subsequent phishing links by 19%. The study illustrates that without reinforcement, contextual relevance, or interactivity, cybersecurity training is unlikely to have a long-term impact.

Moreover, a simulated training study with 100 industrial employees measured performance before and after AI-adaptive training. Results showed that the AI Coach reduced phishing susceptibility by 72%, improved incident response time by 50%, enhanced security knowledge by 49%, and increased attack detection ability by 69% [1].

AI coaching offers a solution for more personalized, interactive, and real-time support grounded in adaptive learning theory. The AI Coach helps enhance human judgement and secure behavior by correcting deviations at the knowledge level of the mental model [17]. Roelofsma et al. [17] describe this as the "deviation correction" role, where the AI coach monitors for discrepancies between the user's actions and the expected cybersecurity guidelines, then provides corrective feedback by communication at the knowledge level of the user's mental model [2, 4, 5]. The AI Coach adapts to user behavior and helps with learning reinforcements by communicating with the employee only when necessary.

3 General Modeling Approach

To explore the cognitive and emotional dynamics that influence human vulnerability to phishing and deepfakes, this study adopts a computational modeling approach based on adaptive network-oriented modeling [22, 23]. This method allows for the simulation of complex cognitive processes over time and under varying conditions by representing them as temporal-causal networks. Within this framework, cognitive, emotional, and behavioral variables are represented as states whose activation levels evolve based on influenced from connected elements. The dynamic properties of these networks allow for modeling behavior, learning, adaptation, and higher-order reasoning processes.

The modeling used here is grounded in the paradigm of adaptive dynamical systems, where changes occur both in the states themselves and the structure characteristics of

the network. This is achieved through the concept of self-modeling networks, where both the activation level of a state and the strength of influence one state has on another can change over time. These adaptive capabilities allow the model to simulate learning processes, such as how repeated exposure to threats or feedback from an AI coach alters behavior and underlying mental models. Each model is formally specified using role matrices that define:

- Connectivity Characteristics: which states are connected, in what direction, and a connection weight $\boldsymbol{\omega}_{X,Y}$ for every connection $X \rightarrow Y$.
- Aggregation Characteristics: a combination function $\mathbf{c}_Y$ that defines how multiple inputs combine to affect a state.
- Timing Characteristics: a speed factor $\boldsymbol{\eta}_Y$ that specifies how quickly state Y updates based on input.

State values are updated over time following canonical differential or difference equation:

$$\frac{dY(t)}{dt} = \boldsymbol{\eta}_Y\left[\mathbf{c}_Y\left(\boldsymbol{\omega}_{X_1,Y}X_1(t), \ldots, \boldsymbol{\omega}_{X_k,Y}X_k(t)\right) - Y(t)\right] \tag{1}$$

$$Y(t+\Delta t) = Y(t) + \boldsymbol{\eta}_Y\left[\mathbf{c}_Y\left(\boldsymbol{\omega}_{X_1,Y}X_1(t), \ldots, \boldsymbol{\omega}_{X_k,Y}X_k(t)\right) - Y(t)\right]\Delta t \tag{2}$$

where state Y receives inputs by connections from states X_i.

The combination functions used in the models of this research are as follows:

$$\mathbf{alogistic}_{\boldsymbol{\sigma},\boldsymbol{\tau}}(V_1, \ldots, V_k) = \left[\frac{1}{1+e^{-\boldsymbol{\sigma}(V_1+\cdots+V_k-\boldsymbol{\tau})}} - \frac{1}{1+e^{\boldsymbol{\sigma}\boldsymbol{\tau}}}\right]\left(1+e^{-\boldsymbol{\sigma}\boldsymbol{\tau}}\right) \tag{3}$$

where $\boldsymbol{\sigma}$ is steepness, and $\boldsymbol{\tau}$ is threshold.

$$\mathbf{stepmod}_{\boldsymbol{\rho},\boldsymbol{\delta}}(V_1, \ldots, V_k) = \begin{cases} 0 & if\ t\ mod\ \boldsymbol{\rho} < \boldsymbol{\delta} \\ 1 & else. \end{cases} \tag{4}$$

where t is time, ρ is repetition, and δ is tipping point.

$$\mathbf{stepmodopp}_{\boldsymbol{\rho},\boldsymbol{\delta}}(V_1, \ldots, V_k) = \begin{cases} 1 & if\ t\ mod\ \boldsymbol{\rho} < \boldsymbol{\delta} \\ 0 & else. \end{cases} \tag{5}$$

To model adaptation over time, the strength of the connections $\boldsymbol{\omega}_{X_1,Y}$, can be made dynamic through additional first and second-order level **W-**states and $\mathbf{W_{WW}}$**-**states that updated based on learning rules. Additionally, the use of a second-order $\mathbf{H_w}$-state allows the speed of the first-order **W-**states to change over time, representing changes in the speed of learning of the human in a context-sensitive manner based on metaplasticity.

This modeling approach was chosen for its strong alignment with both theoretical and practical requirements of this study. Conceptually, the adaptive network model is well-suited to capture dynamic interactions between cognitive, emotional, and external states. Additionally, it offers high performance simulation capabilities in platforms like MATLAB, which will allow for detailed analysis for different scenarios and a risk assessment to evaluate the impact of an AI Coach.

In this study, two models are simulated. The first model simulates baseline employee behavior with no AI Coach present. It captures how stress, emotional triggers, opportunity, and training influence the employees' learning and behavior over time. The second model introduces an AI Coach component that monitors deviations from secure behavior and decides when to interact with the employee at the knowledge level. Communication occurs only when the employees' internal states deviate beyond a threshold from the expert (AI-Coach) model, in accordance with the $\mathbf{W_{WW}}$-based communication-controlled feedback architecture [17, 23]. The following section details the design of these models, including base-level states, first-order learning mechanisms, and second-order adaptations introduced through coaching.

4 Design of the Adaptive Network Models

The adaptive network models in this research build upon two key behavioral frameworks: the COM-B Model and the Dual-Process Theory. The COM-B model by Michie et al. defines behavior as a combination of Capability, Opportunity, and Motivation [15], which is used as a baseline for mental states, including X3 *Threat Recognition Capability, X5 [perceived] Workplace Security Culture* (opportunity), and *X8 Emotional Reactivity* (motivation). Additionally, the models build on the Dual-Process Theory [14] by distinguishing and simulating the interaction between automatic and reflective cognitive processes. The dual-process theory argues that behavior is influenced by two interacting systems: system 1, being fast, emotional, and heuristic, and system 2, being slow, deliberate, and analytical. System 1 processes are captured in the model through states such as *X8 Emotional Reactivity* and *X25 Stress Level*, which drive impulsive behavior including *X21 Decision to Act*. These automatic and context-sensitive reactions are influenced by other contextual inputs, such as *X18 Quality of Message*. In contrast, system 2 processes are modelled through states such as *X3 Threat Recognition Capability* and *X4 Protocol Recall Accessibility*, which influence secure behavior and actions including *X20 Decision to Report*. The models extend dual-process modeling by demonstrating how self-reflection can regulate impulsive and automatic reactions through first-order learning and control states.

4.1 Model 1 – No AI Coach

The first model represents the learning and behavioral process of employees within organizations as they respond to phishing attempts. The model is structured across three interconnected layers. The base level (pink), first-order adaptation level (blue), and second-order adaptation level (yellow).

The base level includes both internal mental states and external contextual factors. The internal states, shown within the white circle in *Fig. 1,* reflect how the employee processes, responds to, and learns from phishing events. The external states, outside the circle, shape this behavior from the environment. The simulation begins with initialization state *X1 Start Job* that remains at a constant value of 1 throughout the simulation, representing the employee's active working period. State *X2 Cybersecurity Training*

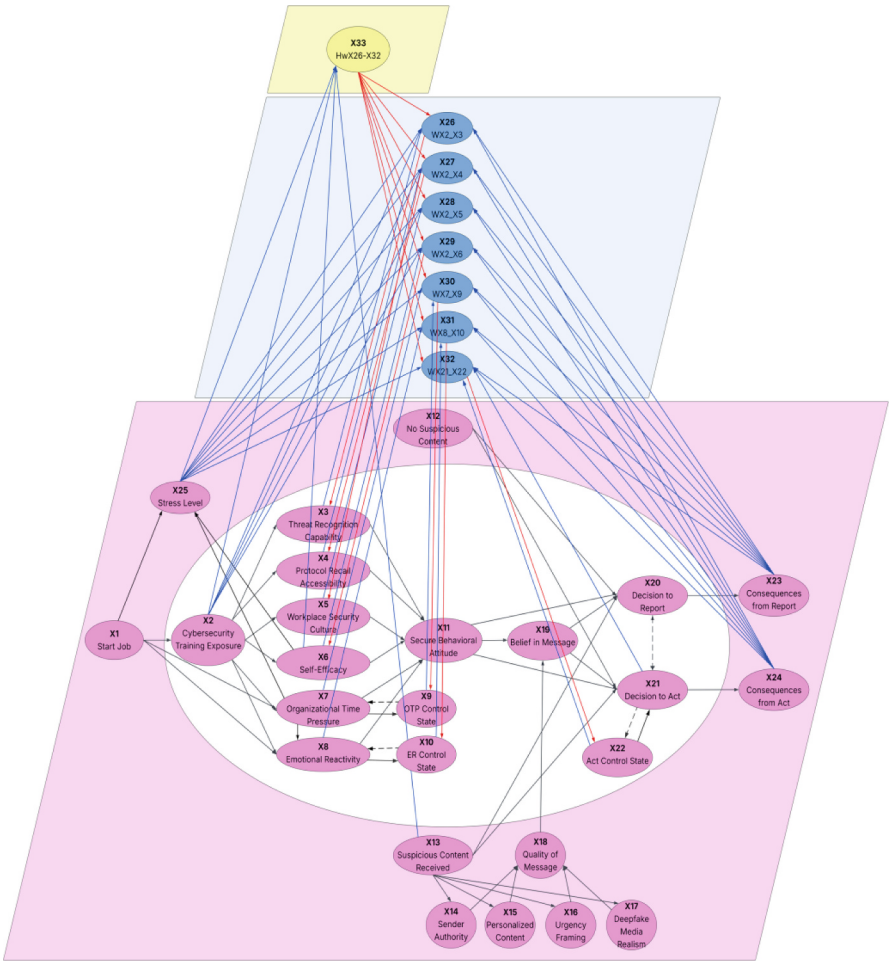

Fig. 1. Overview of conceptual model without AI coach.

Exposure gradually increases over time throughout the simulation as the employee periodically receives cybersecurity training. This training influences a range of mental states, including *X3 Threat Recognition Capability*, *X4 Protocol Recall Accessibility*, *X5 Perceived Workplace Security Culture*, *X6 Self-Efficacy, X7 Organization Time Pressure (negatively)*, and *X8 Emotional Reactivity (negatively).* States X3-X8 shape *X11 Secure Behavioral Attitude*, which is a key behavioral indicator with direct influence on the employee's decision states.

External cybersecurity threats are modeled beginning with *X13 Suspicious Content Received,* which triggers sub-states *X14-X17*, shaping the *X18 Quality of Message.* This has a direct connection to the mental state *X19 Belief in Message*, another key state with direct influence on decision states *X20 Decision to Report* and *X21 Decision to Act*. The control states X9, X10, and X22 represent the employee's ability to regulate their

emotional and behavioral responses, which helps in simulating dynamic self-regulation under different environmental conditions.

The first-order adaptation level represents the employees' learning over time. It consists of **W**-states X26-X32 that regulate the strength of connections in the base level, increasing through feedback and training. For instance, *X26* W*X2_X3* modulates the strength of training's influence on threat recognition. **W**-states for connections to control states, including *X30* W*X7_X9*, adjust how well the employee self-regulates. All **W**-states learn through feedback states X23 and X24, and the frequency with which related states are activated. However, all **W**-states are negatively affected by *X25 Stress Level*, indicating that under high stress, the employees' learning will slow down or even degrade (Table 1).

Table 1. Model 1 – No AI Coach List of States and Descriptions.

Nr	Name	Descriptions	Level
X_1	Start Job	Employee starts a new job – initialization.	Base Level
X_2	Accumulated Training Exposure	Cumulative cybersecurity training over time.	
X_3	Threat Recognition Capability	Ability to detect phishing and cyber threats.	
X_4	Protocol Recall Accessibility	Ease with which security protocols are remembered.	
X_5	Workplace Security Culture	Perceived norms and values around security at work.	
X_6	Self-Efficacy	Employee's confidence in managing cyber threats.	
X_7	Organizational Time Pressure	Perceived workload and urgency from the organization.	
X_8	Emotional Reactivity	Susceptibility to emotional triggers / impulsive behavior.	
X_9	OTP Control State	Modulates perceived organizational time pressure.	
X_{10}	ER Control State	Modulates emotional reactivity.	
X_{11}	Secure Behavioral Attitude	Employees tendency to behave securely and follow protocols.	
X_{12}	No Suspicious Content	Period where no attack is present.	
X_{13}	Suspicious Content Received	Period where a phishing attempt is present.	
X_{14}	Sender Authority	Authority of message sender.	
X_{15}	Personalized Message Content	Degree of personalization in the phishing message.	
X_{16}	Urgency Framing	Sense of urgency in the phishing message.	
X_{17}	Deep Fake Media Realism	Realism of synthetic voice or video media.	
X_{18}	Quality of Message	Overall quality of the phishing message.	
X_{19}	Belief in Message	Employee's belief that the message is genuine.	
X_{20}	Decision to Report	Employee's decision to report the suspicious message.	
X_{21}	Decision to Act	Employee's decision to engage with the phishing message.	
X_{22}	Act Control State	Controls tendency to act.	
X_{23}	Consequences from Report	Experienced outcomes from reporting a threat.	
X_{24}	Consequences from Act	Experienced outcomes from acting on a threat.	
X_{25}	Stress Level	Level of cognitive and emotional stress.	
X_{26}	$WX_2_X_3$	Self-model **W**-state for the weight of the connection from X_2 to X_3.	First-Order
X_{27}	$WX_2_X_4$	Self-model **W**-state for the weight of the connection from X_2 to X_4.	
X_{28}	$WX_2_X_5$	Self-model **W**-state for the weight of the connection from X_2 to X_5.	
X_{29}	$WX_2_X_6$	Self-model **W**-state for the weight of the connection from X_2 to X_6.	
X_{30}	$WX_7_X_9$	Self-model **W**-state for the weight of the connection from X_7 to X_9.	
X_{31}	$WX_8_X_{10}$	Self-model **W**-state for the weight of the connection from X_8 to X_{10}.	
X_{32}	$WX_{21}_X_{22}$	Self-model **W**-state for the weight of the connection from X_{21} to X_{22}.	
X_{33}	HwX_{26}-X_{32}	Self-model **Hw**-state for the speed factor of the **W**-states X_{26}, X_{27}, X_{28}, X_{29}, X_{30}, X_{31}, X_{32}.	Second-Order

The second-order adaptation level of the model consists of state *X33 HwX26-X32*. This **Hw**-state reflects the meta-learning process and controls the learning speed of all

W-states. This helps the model to not only adjust the employee's behavior, but also adapt to how quickly the employee learns from experience. The speed of learning is positively influenced by states including X2, X6, and X13, but is negatively influenced by stress.

Overall, this multi-level model that builds on behavioral frameworks displays how employees learn over time, and how external factors including stress, and the quality of a message can influence their decision-making. By modeling behavior, learning, and even learning speed, this version of the model will serve as a baseline against which the effects of extending this model by introducing an AI coach can later be evaluated (Table 2).

Table 2. Model 1 – No AI Coach List of Connections and Explanations.

From	To	Explanation
X_1	X_2	Job onboarding initializes cybersecurity training.
X_1	X_7	New job increases perceived time pressure.
X_1	X_8	New job increases emotional reactivity.
X_1	X_{25}	Transition stress increases baseline stress levels.
X_2	$X_{3,4,5,6,7,8}$	Training improves threat recognition skills, protocol recall, security culture awareness, and self-confidence, but reduces perceived time pressure and emotional responses to threats.
$X_{3,4,5,6,7,8}$	X_{11}	Threat recognition capability, protocol knowledge, perceived security culture, self-confidence increase secure behavior, while time pressure and emotional reactivity decrease secure behavior.
X_6	X_{25}	Self-confidence can reduce stress levels.
X_7	X_8	Perceived time pressure influences emotional reactivity.
X_7	X_9	Time pressure activates its control state.
X_7	X_{25}	Perceived time pressure increases stress levels.
X_8	X_{10}	Emotional reactivity activates its control state.
X_9	X_7	Control loop decreases perceived time pressure.
X_{10}	X_8	Control loop decreases emotional reactivity.
X_{11}	X_{19}	Secure attitude decreases belief in fabricated messages.
X_{11}	$X_{20,21}$	Secure attitude increases likelihood of reporting and decreases likelihood of acting on message.
X_{12}	$X_{20,21}$	No threat - no decision activation.
X_{13}	$X_{14,15,16,17}$	Suspicious content activates authority cues, personal content, urgency cues, deepfake realism.
X_{13}	$X_{20,21}$	Suspicious content triggers decision states.
$X_{14,15,16,17}$	X_{18}	Authority cues, personalization, urgency, deepfake realism all increase message quality.
X_{18}	X_{19}	High message quality increases belief in message.
X_{19}	X_{20}	Belief in message decreases likelihood of reporting
X_{19}	X_{21}	Belief in message increases likelihood of acting on message.
$X_{20,21}$	$X_{21,20}$	Mutual inhibition.
X_{20}	X_{23}	Reporting leads to external consequences.
X_{21}	X_{22}	Decision to act activates its control state.
X_{21}	X_{24}	Acting leads to external consequences.
X_{22}	X_{21}	Control loop deceases likelihood of acting
X_{26}	X_3	Learning, increases threat recognition capability.
X_{27}	X_4	Learning, increases protocol recall accessibility.
X_{28}	X_5	Learning, increases perceived workplace security culture.
X_{29}	X_6	Learning, increases self-efficacy.
X_{30}	X_9	Learning, increases time pressure control state.
X_{31}	X_{10}	Learning, increases emotional reactivity control state.
X_{32}	X_{22}	Learning, increases act control state.

4.2 Model 2 – AI Coach

The second model builds directly on the structure of Model 1 by extending it with an AI coach that simulates a second agent operating under ideal conditions. The AI coach is modeled as a parallel employee with the same mental base and first-order states as the employee. However, the AI coach is assumed to have perfect learning ability: all its **W**-states are initialized at a value of 1 and remain constant throughout the simulation [17]. This ensures that the AI coach always maintains the optimal values for learning parameters, allowing it to serve as a guide for the employee's learning process (Fig. 2).

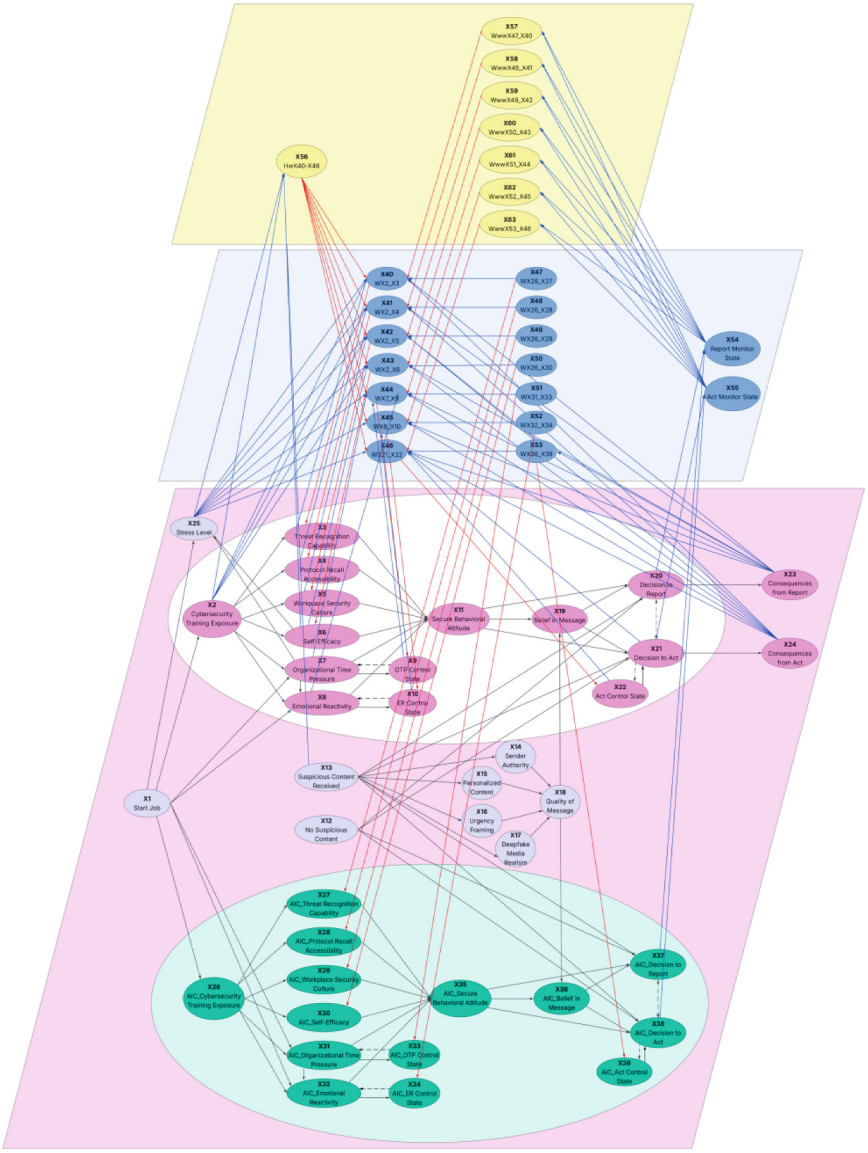

Fig. 2. Overview of the conceptual model with the addition of an AI-coach.

Two monitor states are introduced: *X54 Report Monitor* and *X55 Act Monitor*. These two states are activated upon disagreement between the employee's decision behavior and the AI Coach's ideal behavior. The report monitor state will only activate when the AI coach chooses to report a suspicious message and the employee does not. Conversely, the act monitor state activates when the employee decides to act on a suspicious message, but the AI coach does not. These monitor states act as triggers that initiate communication

between the AI Coach and the employee only when divergence from secure behavior is detected.

The communication occurs through second-order **Www**-states (X57-X63), which modulate the connection weights between the AI Coach's first-order **W**-states (X47-X53), and the employee's corresponding **W**-states (X40-X46). These **Www**-states are only activated when either of the monitor states is activated, meaning that the AI Coach will only influence the employee when they deviate from secure behavior. The monitor states, along with the **Www**-states creates a communication control system, ensuring that the AI Coach only communicates with the employee when necessary.

When the communication control system indicates a need for communication, the AI Coach influences the employee's learning by strengthening or correcting W-state values, which, in turn, affect the employee's base-level mental states X3, X4, X5, X6, X9, X10, and X22. These states influence the employee's secure behavioral attitude, that alongside the X19 Belief in Message, determine the employee's decisions.

The state numbering shifts slightly in Model 2. The employee's base-level states remain as X1-X25. States X26-X39 become the AI-Coach's equivalent base-level and mental states. The employee's first order adaptation states, previously X26-X32 in Model 1, are renumbered as X40-X46. Model 1's X33 has become X56.

More details of the model can be found as Linked Data at https://www.researchgate.net/publication/393933435.

5 Example Simulation

Simulations for both models have been run using the same parameters (see the above link for role matrices) to provide a controlled and fair comparison in the performance of the employee in both models to help determine whether the AI-coach is beneficial.

Figure 3 provides an overview of all 33 states of the first model for 2500 time points. A general theme is seen, as all curves rise for the first 1000–1500 time points, then begin to decrease due to the increasing value of *X25 Stress Level* throughout the simulation. In the figure, the evolution of different states can be seen. The second-order $\mathbf{H_W}$-state X33 (green curve), which controls the speed of learning, gradually increases throughout the simulation, however never exceeding 0.01, indicating a slow learning rate. Meanwhile, *X25 Stress Level* (purple curve) steadily increases over time, eventually reaching 0.8 by time point 2500 to simulate high-stress environments. Moreover, the **W**-states (X26-X32) that represent the employee's first order learning rise during the first 600 time points, reaching levels between 0.8–0.95, but then begin to plateau as rising stress levels impair the employee's learning. X26-X29 show a slower decline, while X30 and X31 (control states) begin to decline rapidly after time point 700. X32 declines the fastest due to its dependence on *X21 Decision to Act*, and as the employee avoids acting, the learning does not take place.

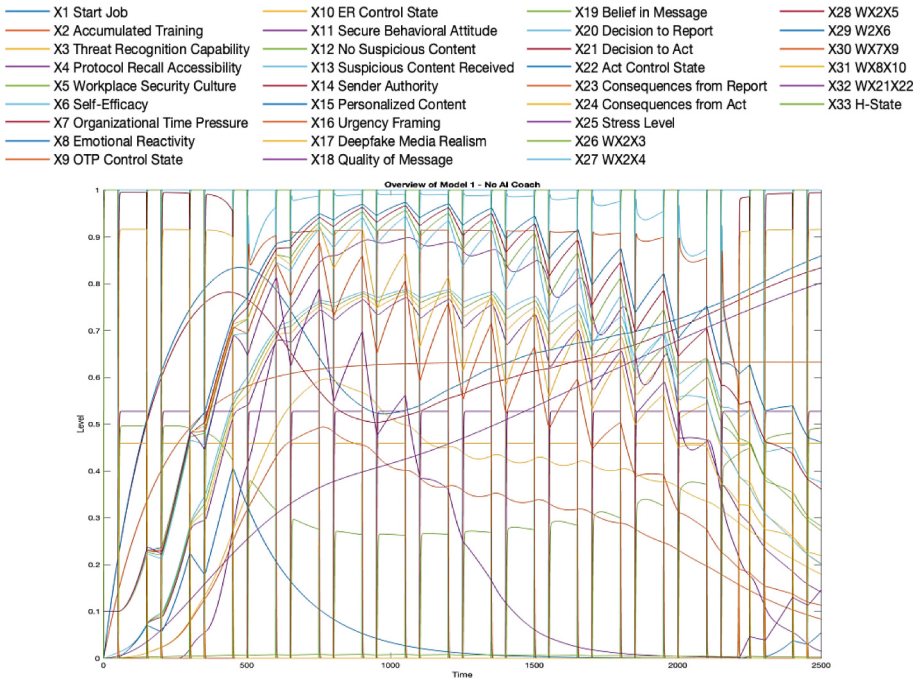

Fig. 3. Overview of Model 1 Simulation.

The impact of these **W-**states on the employee's base level states is clear. X2 Cybersecurity Training Exposure plateaus early, simulating the effect of limited memory retention from static awareness training, as discussed in Sect. 2. *X7 Organizational Time Pressure* and *X8 Emotional Reactivity* spike initially due to job onboarding stress, then gradually decline as their corresponding control states X9 and X10 increase through learning. However, as stress levels increase and the W-states degrade, these control states begin to decline after time point 750, causing X7 and X8 to rise again around time point 1000.

Other base-level states X3, X4, X5, and X6 all portray similar curves with initial growth and plateau, followed by decline after time point 1500, reflecting their associated W-states. X11 Secure Behavioral Attitude starts to increase as X3-X6 rise; however, only after X7 and X8 begin to decline due to the high negative impact that time pressure and emotional reactivity can have on secure behavior. X11 peaks at level 0.9 at around time point 900, and as stress accumulates and the **W**-states decline, X11 drops significantly, eventually falling below 0.1, showing the long-term impact that stress can have on secure behavior.

As for the external context states, *X13 Suspicious Content Received* activates periodically using a stepmod function, with every activation triggering states X14-X17, which in turn determine *X18 Quality of Message*. In this simulation, moderate message quality (level 0.5) is maintained by setting the connection weights from X13 to each of X14-X17 at 0.5. *X19 Belief in Message* begins to drop due to increasing secure behavioral attitude but rises again as X11 declines.

Figure 4 highlights the behavioral states *X20 Decision to Report and X21 Decision to Act*. These states are designed with strong mutual inhibition to ensure that only one state activates at a time. In the first 3 phishing events, the employee acts (X21), but as learning accumulates and X11 rises, reporting (X20) becomes dominant by the fourth attack. Over time, however, this learning fades and the certainty with which employees report decreases (seen by a decrease in X20 levels). Eventually, X21 reactivates near time point 2250, demonstrating how high stress affects secure behaviors and employees' decisions. These results portray how dynamic interactions between mental states, especially under stress, shape behavioral security outcomes. This model will serve as a baseline and will be compared to the performance of the employee, once an AI coach is introduced..

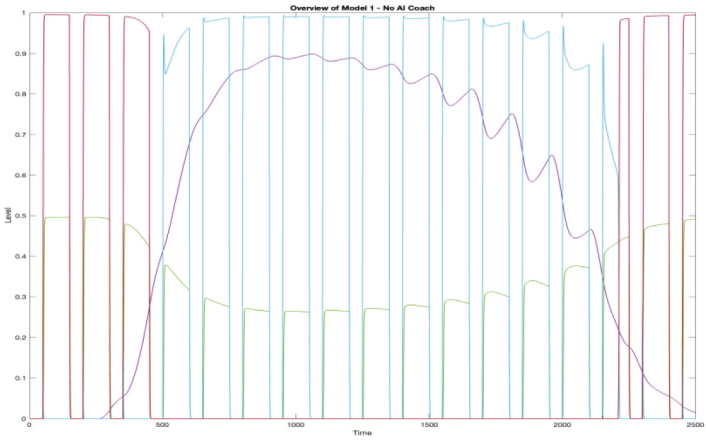

Fig. 4. Model 1 X11 Secure Behavioral Attitude, X19 Belief in Message, X20 Decision to Report, X21 Decision to Act.

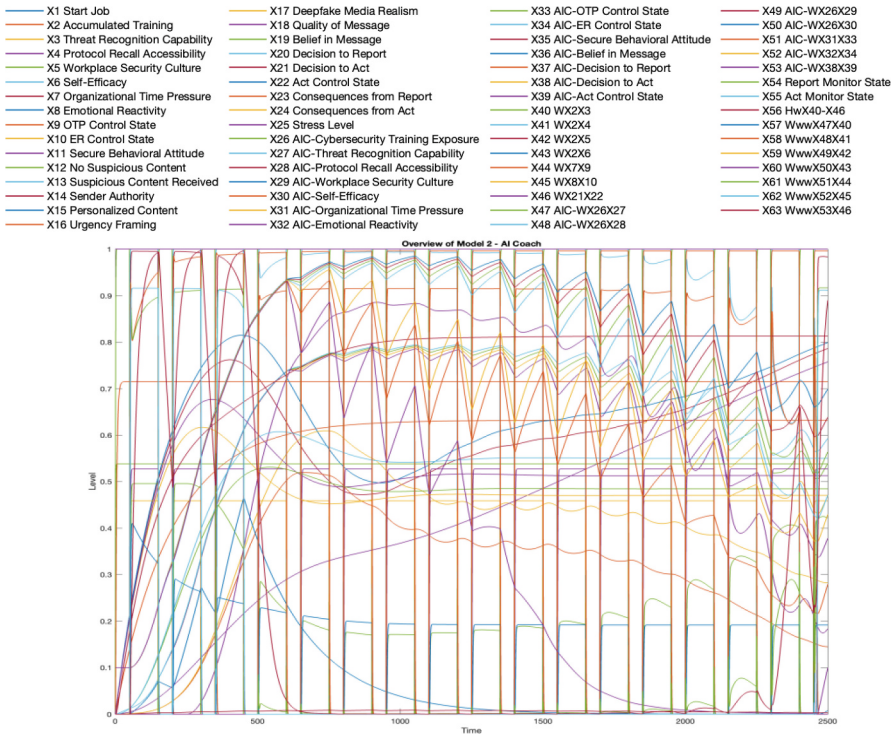

Fig. 5. Overview of Model 2 Simulation.

Figure 5 provides an overview of all 63 states in Model 2 over 2500 time points. The first-order learning states X47-X53 of the AI Coach are fixed at 1 throughout the simulation, as we assume that the AIC has perfect learning capacity. As a result, the AI Coach's base-level mental states X27, X28, X29, and X30 rise sharply during the initialization of the simulation and stabilize around 0.7. Similarly, the AI Coach's *X35 Secure Behavioral Attitude* stabilizes at around 0.8 across the entire simulation. This stability leads the AI Coach to consistently decide to report (X37), with only one deviation seen early in the simulation during the initialization phase.

The communication control system that was designed and discussed in Sect. 4.2 is displayed in Fig. 5. The monitor states X54 and X55, and the second-order Www-states (X57-X63), form this system and decide when the AI Coach should communicate with the employee at the first-order level. These **Www**-states are only activated when there is a discrepancy between the AI Coach's decisions (X37, X38) and the employee's decisions (X20, X21). In this simulation, the two only disagree during the first three phishing attempts and once again, towards the end of the simulation, as seen by the brief activation of the monitor and $\mathbf{W_{ww}}$-states. The limited need for communication already portrays the employee's improved decision-making under the AI Coach's influence.

While the employee's first-order **W**-states (X40-X46) still show gradual degradation over time, they decrease at a much slower rate than in Model 1. This directly impacts

the employee's base-level cognitive states (X3-X8), which are supported by their corresponding first-order **W**-states. These base states maintained a higher average value when compared to Model 1, despite identical parameters. Consequently, the employee's *Secure Behavioral Attitude* (X11) does decline over time but remains stronger throughout the simulation due to the influence of the AI-coach early in the model, increasing the employee's learning states to higher levels.

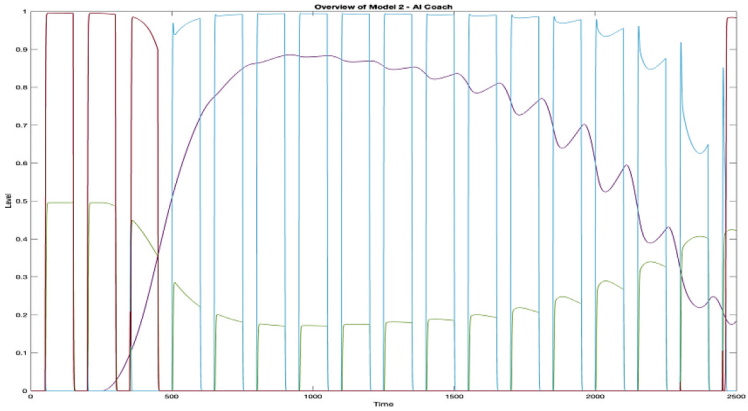

Fig. 6. Model 2 AI Coach employee X11 Secure Behavioral Attitude, X19 Belief in Message, X20 Decision to Report, X21 Decision to Act.

The impact of the AI Coach on the employee's behavior is illustrated in Fig. 6. Compared to Model 1, the employee is less likely to act and more likely to report during phishing scenarios. The final two phishing attempts before the simulation's end show that the employee decides to report under the influence of the AI coach, while they decide to act in Model 1. While the final phishing attempt still results in a decision to act by the employee, this is expected given the minimal communication design of the AI Coach, where it is designed to intervene only when it disagrees with the employee's behavior. Consequently, some gradual decline in learning over time is expected, but the presence of the AI Coach substantially reduces the rate of that decline.

6 What-If Analysis and Risk Assessment

The What-If analysis explores how different levels of an employee's stress threshold (representing their susceptibility to stress) and the quality of phishing messages affect their learning and decision making. By simulating different combinations of these factors across both Model 1 and 2, the analysis provides a deeper insight into the influence of the AI Coach on the employee's behavior (Table 3).

Table 3. What-If Analysis: If-Factors options, labels, values, probabilities (Low stress level thresholds represent high employee susceptibility to stress, while higher thresholds represent low susceptibility to stress).

If-Factors	If-Factor Options	If-Factor Option Labels	If-Factor Option Values	Probabilities (for Risk Assessment)
1. Connection weights: X_{13}_X_{14}, X_{15}, X_{16}, X_{17}	If Factor Option 1	Low	0.25	0.60
	If Factor Option 2	Moderate	0.5	0.25
	If Factor Option 3	High	0.75	0.10
	If Factor Option 4	Very High	1	0.05
2. Threshold X_{25} Stress Level[1]	If Factor Option 1	Low	0.25	0.20
	If Factor Option 2	Moderate	0.5	0.45
	If Factor Option 3	High	0.75	0.35

The results of the What-If analysis exhibit a significant improvement in the employee's performance under the supervision of an AI Coach. In the scenarios without the AI Coach, the employee's learning often peaked between 0.6 and 0.9, before decreasing rapidly to 0.07–0.3 (except when threshold was at 0.75). This decline in the employee's learning led to more insecure decisions, with the employee often choosing to act rather than report. However, the AI Coach significantly improves the employee's learning, making it more stable over the simulations, helping counter the effects of stress. Even when stress threshold was low, the AI Coach helped the employee's learning peak at 0.95, decline more slowly, and even begin to increase again over time (after time points 1100–1400), ending on average between to 0.3–0.9, portraying the long-term benefits of the AI Coach. **W**-states WX7_X9, WX8_X10, and WX21_22 were not included in the what factor as their learning curves are different than WX2_X3,X4,X5,X6 and averaging them would cause problems in the analysis. Without the AI Coach, the employee's decision to act frequently outweighed their decision to report in lower stress level threshold conditions, with the worst-case scenario reaching 17 out of 17 decisions to act. With the AI Coach, the frequency of reporting increased significantly (Table 4).

Table 4. What-If Analysis: Outcomes (TP = Time Point).

If-Factor		What-Factor			
1. Connection Weights	**2. Threshold Stress Level**	**Employee Learning** WX2X3, WX2X4, WX2X5, WX2X6		**Employee Decision[2]**	
		1. No AI Coach	**2. AI Coach**	**3. No AI Coach**	**4. AI Coach**
Low: 0.25	Low: 0.25	TP[2] 900: 0.67-0.79 TP 2500: 0.09-0.25	TP 900: 0.82-0.89 TP 2500: 0.35-0.62	6 reports, 11 acts.	9 reports, 8 acts.
	Moderate: 0.5	TP 900: 0.86-0.93 TP 2500: 0.11-0.28	TP 900: 0.92-0.96 TP 2500: 0.35-0.63	9 reports, 8 acts.	12 reports, 5 acts.
	High: 0.75	TP 900: 0.986-0.991 TP 2500: 0.991-0.995	TP 900: 0.990-0.994 TP 2500: 0.993-0.996	14 reports, 3 acts.	15 reports, 2 acts.
Moderate: 0.5	Low: 0.25	TP 900: 0.64-0.78 TP 2500: 0.08-0.24	TP 900: 0.82-0.90 TP 2500: 0.42-0.74	4 reports, 13 acts.	9 reports, 8 acts.
	Moderate: 0.5	TP 900: 0.85-0.83 TP 2500: 0.09-0.26	TP 900: 0.93-0.96 TP 2500: 0.44-0.73	8 reports, 9 acts.	12 reports, 5 acts.
	High: 0.75	TP 900: 0.986-0.991 TP 2500: 0.991-0.996	TP 900: 0.990-0.994 TP 2500: 0.992-0.996	14 reports, 3 acts	14 reports, 3 acts
High: 0.75	Low: 0.25	TP 900: 0.67-0.79 TP 2500: 0.09-0.25	TP 900: 0.87-0.92 TP 2500: 0.64-0.82	0 reports, 17 acts	6 reports, 11 acts.
	Moderate: 0.5	TP 900: 0.85-0.93 TP 2500: 0.07-0.28	TP 900: 0.94-0.97 TP 2500: 0.67-0.84	5 reports, 12 acts	9 reports, 8 acts.
	High: 0.75	TP 900: 0.985-0.991 TP 2500: 0.991-0.995	TP 900: 0.991-0.994 TP 2500: 0.992-0.996	14 reports, 3 acts	14 reports, 3 acts
Very High: 1	Low: 0.25	TP 900: 0.67-0.80 TP 2500: 0.09-0.26	TP 900: 0.87-0.92 TP 2500: 0.67-0.84	0 reports, 17 acts	6 reports, 11 acts.
	Moderate: 0.5	TP 900: 0.85-0.93 TP 2500: 0.08-0.28	TP 900: 0.94-0.97 TP 2500: 0.67-0.84	4 reports, 13 acts.	8 reports, 9 acts.
	High: 0.75	TP 900: 0.986-0.991 TP 2500: 0.990-0.995	TP 900: 0.991-0.994 TP 2500: 0.992-0.996	13 reports, 4 acts.	13 reports, 4 acts.

Under high-quality phishing messages, the AI-supported employees still showed some insecure behavior and a tendency to act, but still performed significantly better than without the AI Coach. However, even under the AI supervision, the employee's behavior was imperfect, especially when message quality was high. This outcome highlights the danger of advanced phishing and deepfake techniques, where realism can overpower even the strongest training and coaching techniques. Nevertheless, the AI Coach did consistently reduce risk and delay learning decay, portraying how beneficial and effective it can be in improving cybersecurity resilience. Using these results, a risk assessment is conducted to help compare performance between the two models (Tabls 5 and 6).

Table 5. What-If Analysis: If Factor Option Combination Probabilities.

If-Factor		If Factor Option Combination Probabilities
1. Connection Weights	**2. Threshold Stress Level**	
Low: Probability 0.6	Low: Probability 0.20	0.60*0.20 = 0.12
	Moderate: Probability 0.45	0.60*0.45 = 0.27
	High: Probability 0.35	0.60*0.35 = 0.21
Moderate: Probability 0.25	Low: Probability 0.20	0.25*0.20 = 0.05
	Moderate: Probability 0.45	0.25*0.45 = 0.1125
	High: Probability 0.35	0.25*0.35= 0.0875
High: Probability 0.10	Low: Probability 0.20	0.10*0.20 = 0.02
	Moderate: Probability 0.45	0.10*0.45 = 0.045
	High: Probability 0.35	0.10*0.35 = 0.035
Very High: Probability 0.05	Low: Probability 0.20	0.05*0.20 = 0.01
	Moderate: Probability 0.45	0.05*0.45 = 0.0225
	High: Probability 0.35	0.05*0.35 = 0.0175

Table 6. Risk Assessment.

If-Factor			What-Factor							
1. Connection Weights	**2. Threshold Stress Level**	**If factor Combination Probabilities**	**Employee Learning (at TP 2500)**				**Employee Decision**			
			1. No AI Coach		**2. AI Coach**		**3. No AI Coach**		**4. AI Coach**	
			1a: Low < 0.35	1b: High >=0.35	2a: Low < 0.35	2b: High >=0.35	3a: Act > Report	3b: Report> Act	4a: Act > Report	4b: Report > Act
Low	Low	0.12	X 0.12			X 0.12	X 0.12			X 0.12
	Moderate	0.27	X 0.27			X 0.27		X 0.27		X 0.27
	High	0.21		X 0.21		X 0.21		X 0.21		X 0.21
Moderate	Low	0.05	X 0.05			X 0.05	X 0.05			X 0.05
	Moderate	0.1125	X 0.1125			X 0.1125	X 0.1125			X 0.1125
	High	0.0875		X 0.0875		X 0.0875		X 0.0875		X 0.0875
High	Low	0.02	X 0.02			X 0.02	X 0.02		X 0.02	
	Moderate	0.045	X 0.045			X 0.045	X 0.045			X 0.045
	High	0.035		X 0.035		X 0.035		X 0.035		X 0.035
Very High	Low	0.01	X 0.01			X 0.01	X 0.01		X 0.01	
	Moderate	0.0225	X 0.0225			X 0.0225	X 0.0225		X 0.0225	
	High	0.0175		X 0.0175		X 0.0175		X 0.0175		X 0.0175
Sum of Probabilities			0.65	0.35	0	1	0.38	0.62	0.0525	0.9475
Impact Indicator			5	3	4	1	5	3	4	1
Risk Assessment			3.25	1.05	0	1	1.9	1.86	0.21	0.9475

In the risk assessment, each What Factor was divided into two categories. For employee learning, outcomes were divided into low (a) if the W-states fell below 0.35 at time point 2500, and high (b) if they remained above this threshold at the end of the simulation. Although this threshold may be considered low, it was chosen as it clearly separates the learning outcomes, and simulations with learning above 0.35 proved to be far less risky. As for employee decisions, behavior was divided into act (a) or report (b) depending on which decision had a higher frequency in each scenario. Each of these categories were then assigned an impact indicator ranging from 1 (secure) to 5 (severe),

reflecting their level of risk. The impact indicator and risk assessment values presented in *Table 8* were motivated by the ISO/IEC 27005 risk assessment framework, which defines risk as a combination of the likelihood of an incident and its business impact (Starc & Trček, 2012). The resulting scores display a notable difference between the two models. For Model 1, low learning had a cumulative risk score of 3.25 but dropped to 0 in Model 2 with the AI Coach. Similarly, the decision to act without the AI Coach scored 1.9, but dropped to 0.21 in Model 2. These results illustrate that while the AI Coach does not eliminate risky decisions, especially when message quality is high, it significantly reduces both the frequency and severity of such outcomes.

7 Discussion

This study provided a comprehensive analysis of two adaptive network models, simulating the cognitive and emotional processes of employees in cybersecurity contexts. Model 1 provided insight into how training is internalized, how feedback-based learning adapts over time, and how secure behavioral attitudes degrade under high stress and time pressure. The results demonstrated that even with strong capabilities and exposure to training, employee behavior still deteriorated over time, with the secure behavioral attitude eventually dropping below 0.1 after 2500 time points. This supports the findings in Sect. 2, Background Literature, that traditional, periodic, and static training methods are ineffective for long-term behavioral change.

Model 2 introduced an AI Coach with identical base-level and **W**-states as the employee, but with perfect learning capability and decision-making. The addition of the AI Coach resulted in a noticeable improvement in employee learning and behavior. The initial analysis in Sect. 5: Simulation Results demonstrated an increased tendency to report rather than act, with learning remaining more stable. Although the **W**-state values declined over time, they remained significantly higher than in Model 1. This decline is expected resultant of the design of the AI Coach, where it only communicates with the employee when their behavior deviates from secure decisions.

The What-If analysis further highlighted the effectiveness of the AI Coach across different levels of stress thresholds and message qualities. In the worst-case scenario, with perfect message quality and low stress threshold, Model 1 employees acted 17 out of 17 times, while Model 2 employees reported 6 out of 17 times. Although their behavior was not perfect, this difference illustrates the AI Coach's impact and potential, even with perfect message quality. In the risk assessment, the risk associated with low learning dropped from 3.25 in Model 1 to 0 in Model 2, while the decision-based risk dropped from 1.9 to 0.21. Although a risk value of 0 for low learning may be optimistic and unrealistic, the threshold of 0.3 was chosen to clearly separate the learning outcome ranges seen in Model 1 and 2.

The deterioration of secure behavior in Model 1 reflects a well-documented challenge in organizations where knowledge gained from awareness training decays significantly without reinforcement. The stability of secure behavior in Model 2 suggests that adaptive, AI-driven coaching could provide the reinforcement that current awareness programs need. Despite these promising results, it is important to be cautious in generalizing these findings. The models in this study abstracted organizational realities into

simplified states. Stress for instance, was treated as a threshold variable, yet in practice is shaped by organizational culture, workload distribution, and personal factors. Similarly, phishing message quality was modeled through a limited set of cues, whereas real-word phishing attempts exploit a far broader and evolving range of techniques. Most importantly, the AI Coach in this study was simulated with perfect knowledge and adaptation, which is unlikely to be fully replicable in practice due to technical limitations, potential false positive, and ethical concerns around employee monitoring. Future research should therefore focus on empirical validation with professionals to evaluate whether the findings of these simulations translate into real organizational contexts. Additional directions include the development of team-based AI coaching approaches, as well as integrating adaptive AI coaching with existing awareness programs for reinforcement rather than replacement. It is also recommended that the number of decision states is expanded to include a Decision to Ignore state, as the current mutually exclusive decision states design causes both monitor states to activate simultaneously. Nevertheless, this had no negative impact on the results, as the system treated them as triggers for the $\mathbf{W_{WW}}$-states, meaning whether one or both states activate, the result would be the same.

This research concludes that the AI Coach demonstrates clear beneficial potential in enhancing employee secure behavior, reducing organizational risk, and sustaining learning over time. While it does not eliminate risky decisions, especially under conditions of highly realistic message quality, it still proved effective as a real-time, adaptive feedback system that supports employees.

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Chiara Scuotto, Masiar Babazadeh, Leonardo Galteri, Pierpaolo Limone, and Stefano Triberti

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